Course Guide

IBM Planning Analytics: Design and Develop Models in Architect (v2.0)

Course code P8361 ERC 1.0



IBM Training

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Course overview

Preface overview

This course is designed to teach modelers how to build a complete model in IBM Planning Analytics using the Architect tool. Through a series of lectures and hands-on exercises, students will learn how to set up and verify dimensions, cubes, and views, manually enter data into these structures, and define the data that users can see. Students will also learn how to transfer data into the IBM Planning Analytics model, including the use of TurboIntegrator scripts to perform data transfer. In addition, the course outlines how to customize drill paths, convert currencies, model for different fiscal requirements, and deploy IBM Planning Analytics applications to the web.

Intended audience

This course is recommended for Modelers.

Topics covered

Topics covered in this course include:

- Examine the components of an IBM Planning Analytics model
- Create dimensions
- Build cubes and views
- Load and maintain data
- Add business rules
- Optimize business rules
- Transfer data into the model
- Customize drill paths
- Use rules for advanced modeling
- Convert currencies
- Model for different fiscal requirements
- Deploy IBM Planning Analytics applications to the web

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Course prerequisites

Participants should have:

- Understanding of the metrics and drivers of your business
- Significant experience with Microsoft Excel spreadsheets (functions, macros, etc.)
- Basic knowledge of OLAP and IBM Planning Analytics

To gain this knowledge, you can visit the following websites:

- **IBM Planning Analytics Learning Center** at http://ibmtvdemo.edgesuite.net/software/analytics/learning-centers/planning-analytics/index.html
- **IBM Planning Analytics YouTube Channel** at https://www.youtube.com/channel/UCGYcFhVXoA29v8kSbspLudA/featured
- **IBM Planning Analytics Community** at https://www.ibm.com/communities/analytics/planning-analytics/

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Document conventions

Conventions used in this guide follow Microsoft Windows application standards, where applicable. As well, the following conventions are observed:

- **Bold**: Bold style is used in demonstration and exercise step-by-step solutions to indicate a user interface element that is actively selected or text that must be typed by the participant.
- *Italic*: Used to reference book titles.
- CAPITALIZATION: All file names, table names, column names, and folder names appear in this guide exactly as they appear in the application.
  To keep capitalization consistent with this guide, type text exactly as shown.

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Exercises

Exercise format

Exercises are designed to allow you to work according to your own pace. Content contained in an exercise is not fully scripted out to provide an additional challenge. Refer back to demonstrations if you need assistance with a particular task. The exercises are structured as follows:

The business question section

This section presents a business-type question followed by a series of tasks. These tasks provide additional information to help guide you through the exercise. Within each task, there may be numbered questions relating to the task. Complete the tasks by using the skills you learned in the unit. If you need more assistance, you can refer to the Task and Results section for more detailed instruction.

The task and results section

This section provides a task based set of instructions that presents the question as a series of numbered tasks to be accomplished. The information in the tasks expands on the business case, providing more details on how to accomplish a task. Screen captures are also provided at the end of some tasks and at the end of the exercise to show the expected results.

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Additional training resources

Visit the IBM Skills Gateway (www.ibm.com/training/) for details on:

- Instructor-led training in a classroom or online
- Self-paced training that fits your needs and schedule
- Comprehensive curricula, learning journeys, and training paths that help you identify the courses that are right for you
- IBM Professional Certification Program (http://www-03.ibm.com/certify/)
- For other resources that will enhance your success, bookmark the IBM Analytics Skills Gateway https://www-03.ibm.com/services/learning/ites.wss/zz-en?pageType=page&c=C067650S63836C42

You can find more videos and information about IBM Planning Analytics at the websites listed below:

- **IBM Planning Analytics Learning Center** at https://developer.ibm.com/clouddataservices/docs/ibm-planning-analytics/how-to
- **IBM Planning Analytics YouTube Channel** at https://www.youtube.com/channel/UCGYcFhVXoA29v8kSbspLudA/featured
- **IBM Planning Analytics Community** at https://www.ibm.com/communities/analytics/planning-analytics/

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IBM product help

Help type	When to use	Location
Task-oriented	You are working in the product and you need specific task-oriented help.	*IBM Product* - Help link
Books for Printing (.pdf)	You want to use search engines to find information. You can then print out selected pages, a section, or the whole book. Use Step-by-Step online books (.pdf) if you want to know how to complete a task but prefer to read about it in a book. The Step-by-Step online books contain the same information as the online help, but the method of presentation is different.	Start/Programs/*IBM Product*/Documentation
IBM on the Web	You want to access any of the following:	
	• IBM - Training and Certification	• http://www-01.ibm.com/software/analytics/training-and-certification/
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IBM Training

Overview of IBM Planning Analytics

IBM Planning Analytics (v2.0)

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Unit objectives

- Review financial performance management
- Identify the IBM Planning Analytics position in a performance management system
- Describe the IBM Planning Analytics components and architecture
- Describe TM1 server fundamentals
- Describe fundamental modeling concepts

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Unit objectives

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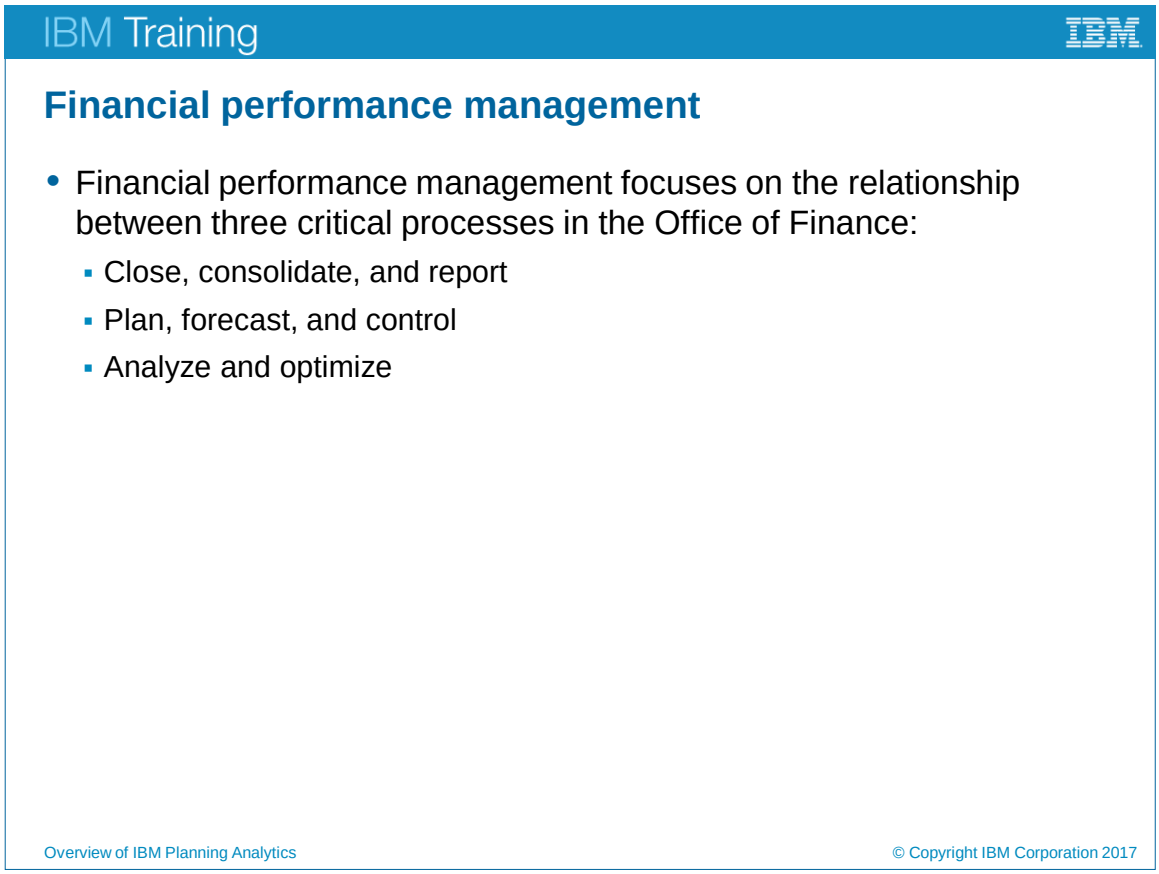
Review IBM Planning Analytics

- IBM Planning Analytics helps you answer three critical business questions from a single platform:
  - Flexible and timely planning answers "What should we be doing?"
  - World-class scorecarding and dashboarding answers "How are we doing?"
  - Accurate reporting and analysis answers "Why are we doing it?"

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Review IBM Planning Analytics

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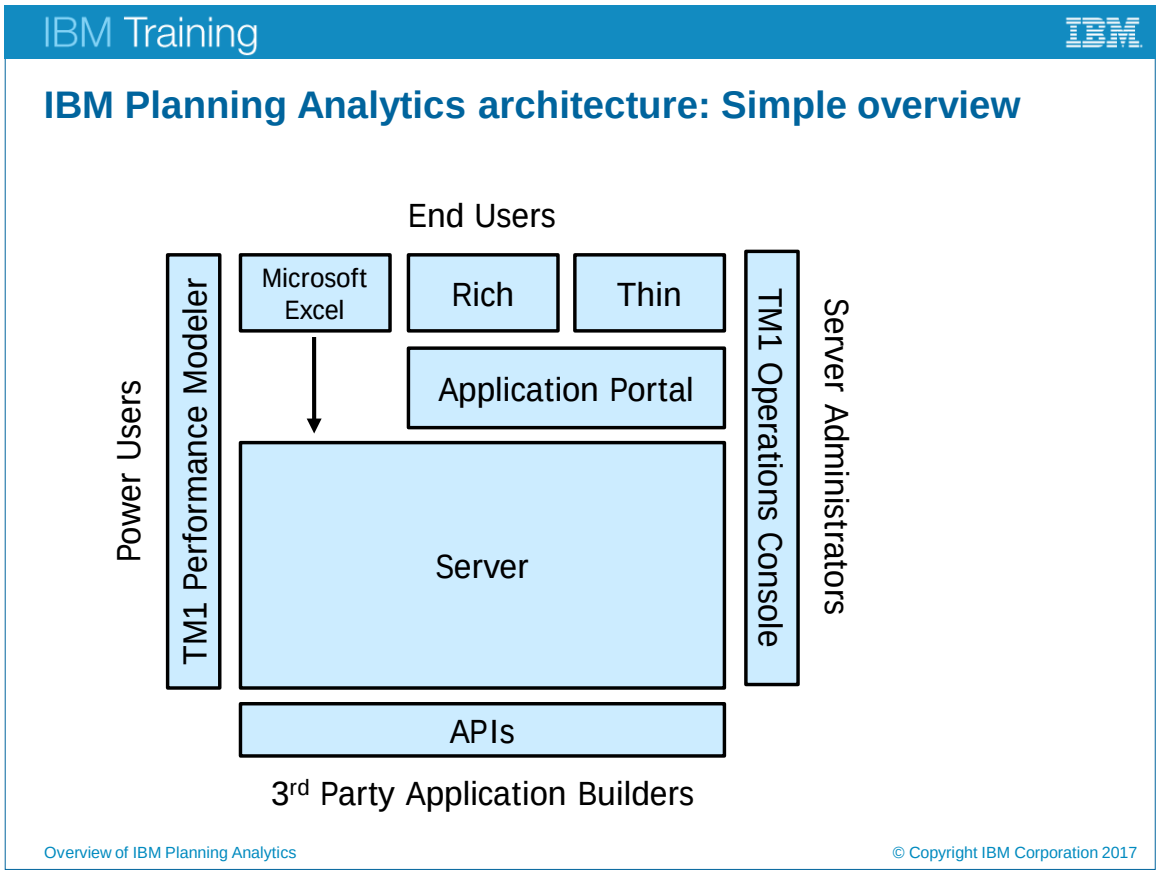


Financial performance management

Finance organizations around the world face the same challenges. They struggle to drive growth and performance while simultaneously ensuring compliance with a myriad of legal, tax, and accounting requirements. Financial performance management solutions from IBM give Finance the tools to win that struggle.

Built on the powerful OLAP engine of IBM TM1, IBM Planning Analytics automates your essential planning, budgeting and forecasting processes. But it also provides self-service analytics for business users and flexible, multidimensional modeling for your finance team. IBM Planning Analytics discovers insights automatically – directly from your data – and empowers users with the speed, agility, and foresight they need to deliver better business results.

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IBM Planning Analytics architecture: Simple overview

Use TM1 Architect and TM1 Performance Modeler to build models that use dimensions, cubes, and rules. You can create applications from cube views, assign workflow, and set up security. You can then deploy, administer, and maintain your applications.

End users have many options to select from, depending on their requirements. IBM Planning Analytics clients include two Microsoft Excel offerings: TM1 Perspectives and IBM Planning Analytics for Microsoft Excel.

IBM Planning Analytics Workspace is a Web-based interface that provides you with exciting ways to plan, create, and analyze your IBM Planning Analytics content.

IBM Cognos Insight is a personal analysis tool that you can use to analyze almost any set of data, including data in your IBM Planning Analytics system.

Thin clients are TM1 Web and TM1 Application Web.

TM1 Performance Modeler, IBM Cognos Insight, and TM1 Application Web work with the TM1 Applications portal to interact with the underlying TM1 server.

TM1 Operations Console is a Web-based operations tool that is designed to facilitate the monitoring, support, and management of TM1 servers, providing greater insight into day-to-day server operations.

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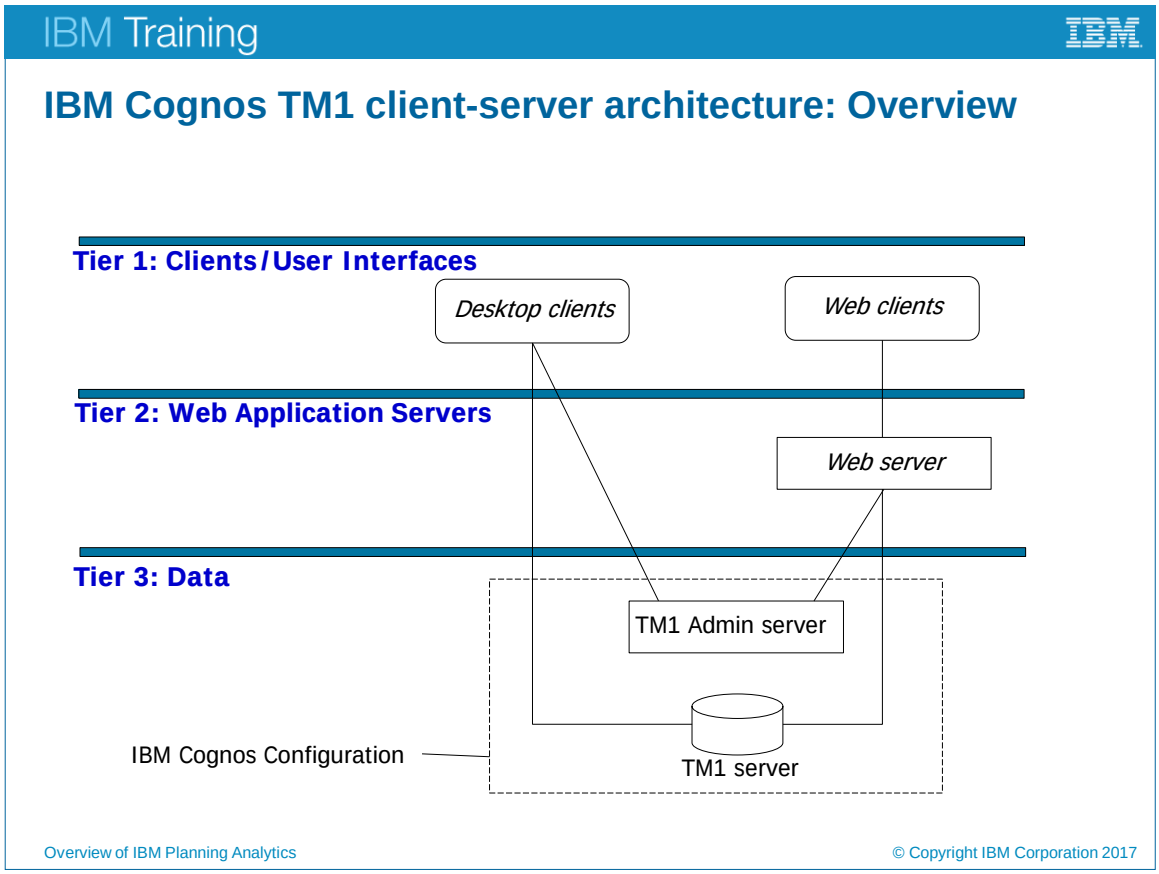
IBM Planning Analytics components

- IBM Planning Analytics Workspace
- IBM Planning Analytics for Microsoft Excel
- IBM Cognos TM1 Architect
- IBM Cognos Insight
- IBM Cognos TM1 Performance Modeler
- IBM Cognos TM1 Server
- IBM Cognos TM1 Admin Server
- IBM Cognos TM1 Application Server
- IBM Cognos TM1 Operations Console
- IBM Cognos TM1 Perspectives
- Web-based planning and analysis
  - IBM Cognos TM1 Applications
  - IBM Cognos TM1 Web

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IBM Planning Analytics components

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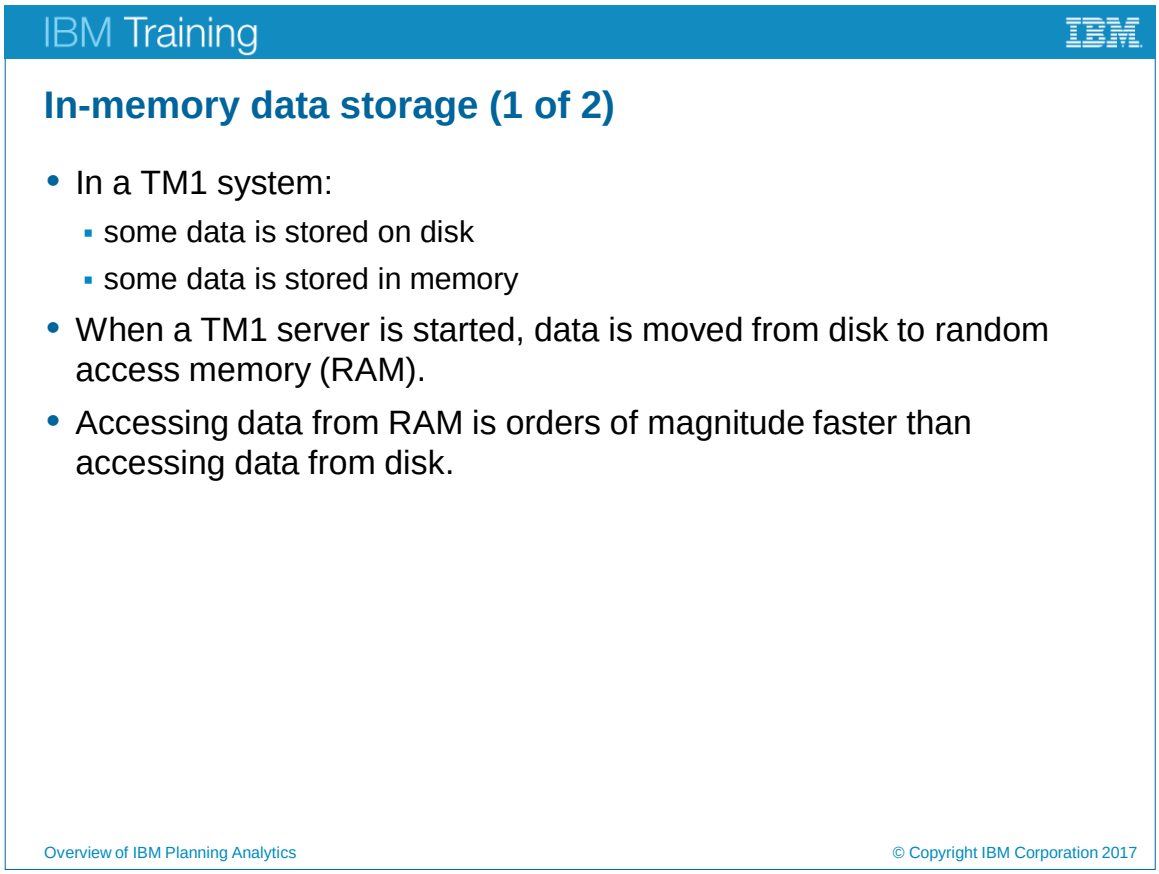
IBM Cognos TM1 clients-server architecture: Overview

IBM Cognos TM1 uses a distributed client-server architecture that consists of the IBM Cognos TM1 server to which a combination of different clients can connect.

IBM Planning Analytics includes the following clients and user interfaces:

- IBM Cognos TM1 Perspectives
- IBM Cognos TM1 Architect
- IBM Cognos TM1 Web
- IBM Cognos TM1 Application portal and workflow (using the TM1 Application Server)
- IBM Cognos TM1 Performance Modeler
- IBM Cognos Insight

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In-memory data storage

Most software and databases store data on hard drives or "disks."

With hard drives, the time it takes to retrieve data is impacted by "seek" time.

The access time to data in RAM is on the order of 50 nanoseconds (5 x 10 **-8 seconds).

The access time to data on disk is on the order of 5 milliseconds (5 x 10 **-3 seconds).

The difference is 10**5 or a factor of 100 thousand increase in speed.

RAM can only hold values while power is going through it. If the power is interrupted, the RAM will lose all the values in it. Therefore, RAM is volatile.

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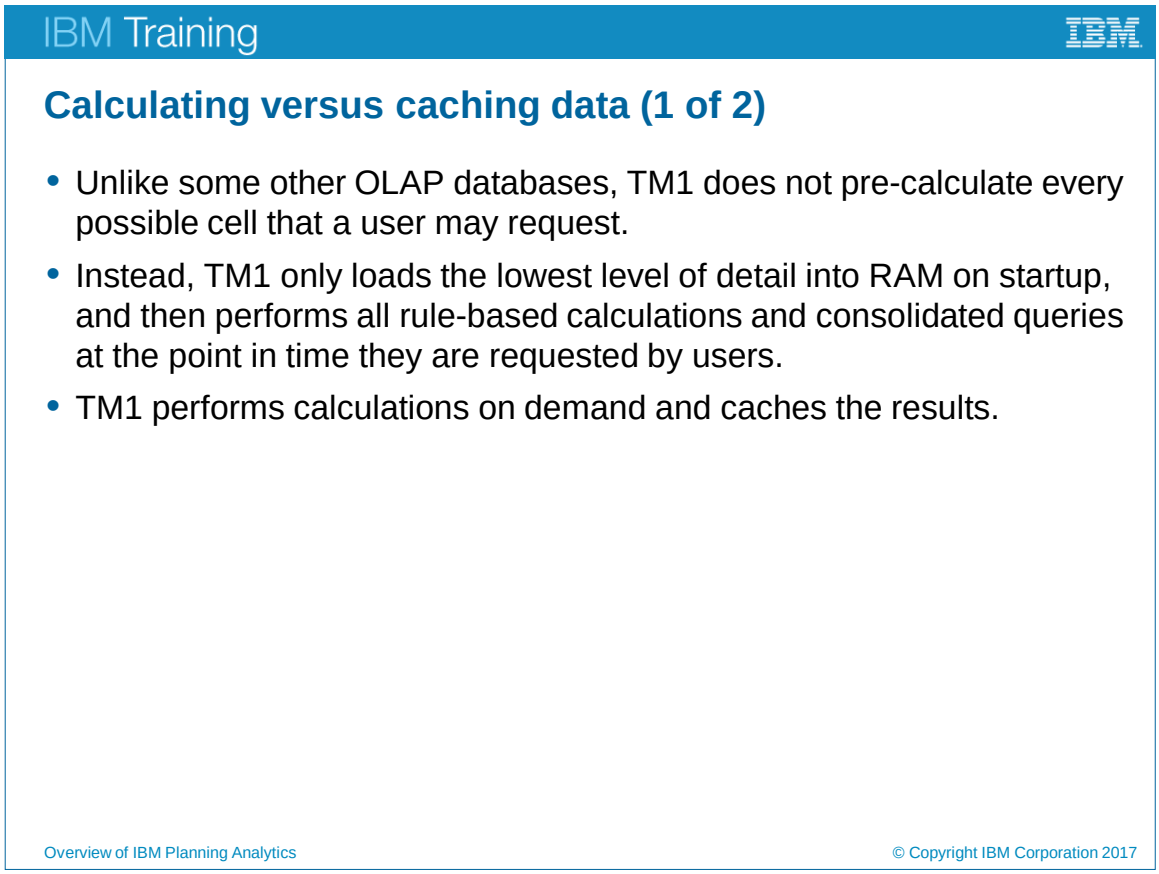
In-memory data storage (2 of 2)

- When a TM1 server is started:
  - all dimensions are loaded into RAM
  - all cube data is loaded into RAM
  - all feeder statements in rules files are processed
  - configuration settings are evaluated
  - log settings are evaluated

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Dimensions, cubes, rules, and feeders will be discussed later in this course.

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Calculating versus caching data

TM1 calculates both aggregate and rule-defined calculations when cells are requested by users. This functionality is often referred to as "real-time" or "on-demand" processing.

These "requests" can come in numerous formats, such as when a user retrieves a cube view from the TM1 Web client.

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Calculating versus caching data (2 of 2)

- "Cache:" computer memory with very short access time used for storage of frequently or recently used instructions or data.
- Also called "cache memory."
- When users query data that has not already been calculated, TM1 will calculate the data and cache it.
- All future queries, for all users, for this same data, will be immediately available until the cache is cleared.

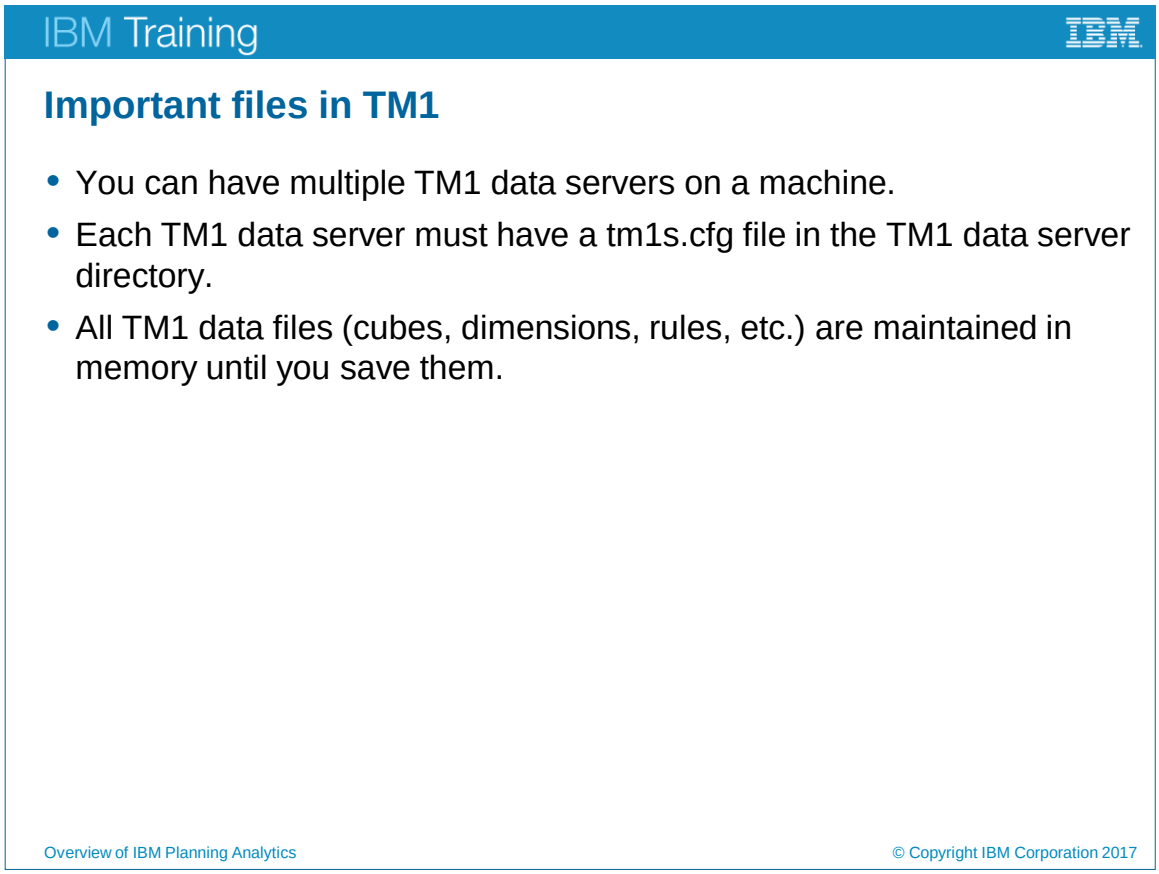
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Two types of cache:

- aggregation cache
  - associated with calculating data based on predefined consolidated dimension elements
- calculation cache
  - associated with calculating data based on cube rules

Writing data to any cell in the cube will eliminate all cached data for the entire cube, as will writing data to any cell in any cube linked via rules. Other changes, such as reordering dimensions in a cube or stopping the TM1 server, will clear the cache.

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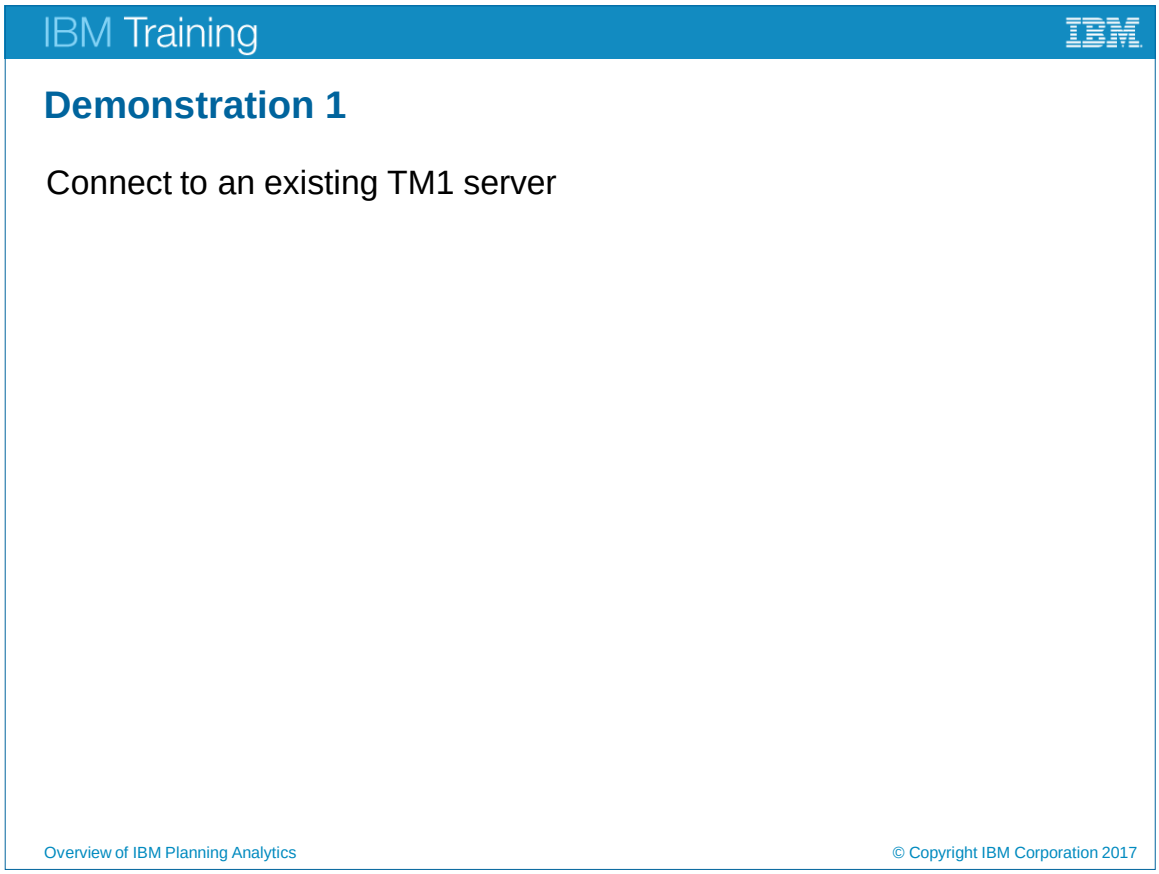
Important files in TM1

Because all objects reside in memory, processing time is very fast.

When objects are saved, only the leaf-level element is saved. Consolidations and calculations are not stored; they are performed as required. They do remain in memory on the server until the server is recycled. If another user requests the same calculations, they will receive the results much faster than the first time because TM1 will give them the value that resides in memory.

You are prompted to save files when you stop the TM1 data server, or you can choose to save data manually or using TurboIntegrator. You will see this later in the course.

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Demonstration 1: Connect to an existing TM1 server

Solution files

(do not complete these steps as part of Demonstration 1)

Solution files are available for each unit in a subfolder located in the C:\Training\P8361 directory.

A solution folder was created for each unit in which the TM1 model is being built (units 1 to 11). Each solution folder reflects the state of the model at the *end* of that unit. To restore your environment to the end of a particular unit:

1. Stop the **IBM Cognos TM1 Server - SampleOutdoors** service from the Services window.
2. Go to **C:\Program Files\IBM\cognos\tm1\samples\tm1\SampleOutdoors**.
3. Rename the **DataFiles** subfolder to **DataFiles_OLD** or something similar.
4. Copy the **DataFiles** subfolder from the solution folder for the unit to which you want to restore your environment (for example, the end of unit 5).
5. Paste this copied **DataFiles** subfolder to the **C:\Program Files\IBM\cognos\tm1\samples\tm1\SampleOutdoors** folder.
6. Start the **IBM Cognos TM1 Server - SampleOutdoors** service.

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Demonstration 1: Connect to an existing TM1 server

Purpose:
You will create a Desktop shortcut that points to a TM1 server. You will then launch the TM1 server as a Windows application and log on to it using Architect.

Task 1. Browse through the different TM1 services that are installed.

1. From the **Start** menu, click **Control Panel** and **Administrative Tools**, and then double-click **Services**.
2. Scroll down to **IBM Cognos TM1 Admin Server x64**.
3. Ensure that the **IBM Cognos TM1 Admin Server x64** service is **Started**.

   The Admin Server runs as a service and keeps track of all TM1 servers running on a network. When a TM1 server starts, the server registers itself with an Admin Server that is running on a specified Admin Host.

   A TM1 server will not run if the Admin Server is not started.
4. Right-click the **IBM Cognos TM1 Server - greatoutdoors** service and click **Stop**.

   The service is stopped (this will take a few moments). You will run this TM1 server as a Windows application for now. You cannot run a TM1 server as both a service and a Windows application at the same time.
5. Close the **Services** window, and then close **Administrative Tools** and the **Control Panel**.

Task 2. Add a shortcut to the Great Outdoors TM1 server.

1. Open **Windows Explorer** and navigate to **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors**.
2. Copy the **greatoutdoors** shortcut to your **Desktop**.
3. Close **Windows Explorer**.
4. On the **Desktop**, right-click the **greatoutdoors** shortcut and click **Properties**.

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5. On the **Shortcut** tab of the **greatoutdoors Properties** dialog box, change the value in the **Target** box to **"C:\Program Files\IBM\cognos\tm1_64\bin\tm1s.exe" -z "C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors"**

   The Target box shows that tm1s.exe is run, and indicates the location of the tm1s.cfg configuration file. This file must be in the same directory as the server's data files.
6. Change the value in the **Start in** box to **"C:\Program Files\IBM\cognos\tm1_64\bin"**
7. On the **General** tab, change the value from **greatoutdoors** to **Great Outdoors**, and then click **OK**.

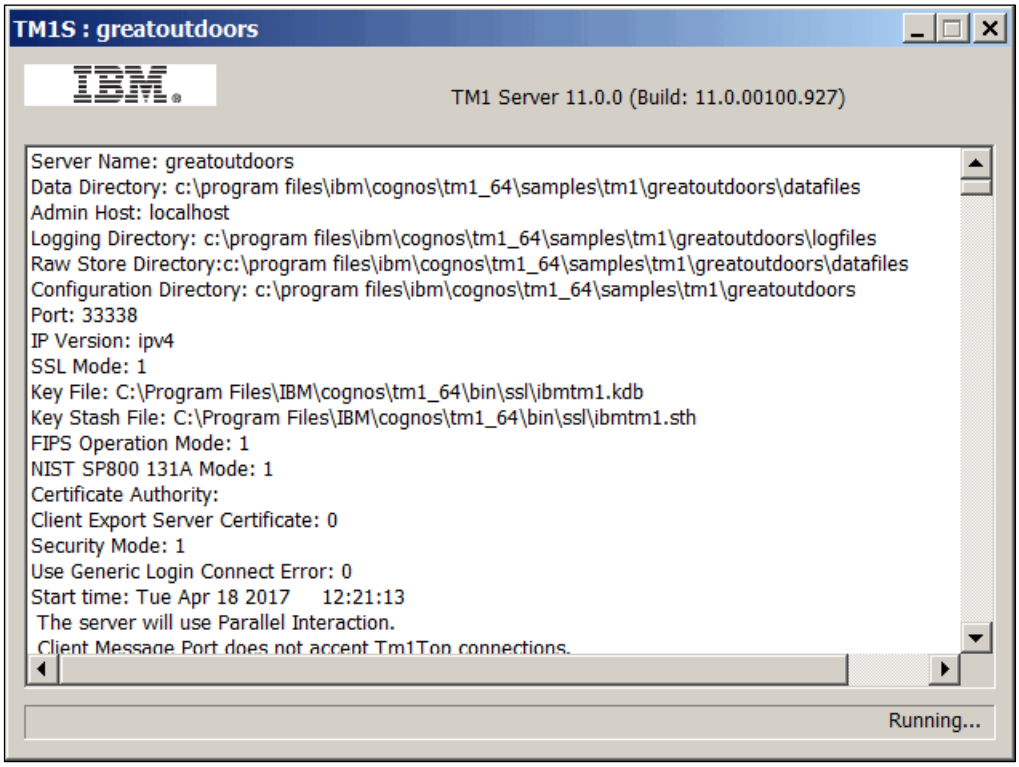
Task 3. Start the Great Outdoors TM1 server as a Windows application.

1. On the **Desktop**, double-click the **Great Outdoors** shortcut.

   If you receive a Windows Firewall-related message, click **Allow access**.

   The Great Outdoors TM1 server is started as a Windows application. If you are prompted to recover changes, click **No**.

   The result appears similar to that shown below:



   When the TM1 server is running, the TM1S: greatoutdoors dialog box is minimized on the toolbar.

   You can now start working with this server using the Architect tool.

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Task 4. Connect to TM1 using Architect.

1. From the **Start** menu, click **All Programs** and **IBM Cognos TM1 - 64**, and then click **Architect**.
2. Maximize the window, expand **TM1**, double-click **greatoutdoors**, and then log on with:

   UserName: **admin**

   Password: **apple**

   You can now access the TM1 server. As the admin user you can add new objects (such as cubes) to the TM1 server and modify existing objects. You can also enter data into cubes.
3. Close **Architect**.
4. On the **Taskbar**, click the **TM1S: greatoutdoors** dialog box, and then close it.
5. Click **No** when prompted to save changes.

   The TM1 server is shut down. As shown previously, this server is set to start automatically as a Windows service when the operating system starts. If you run a TM1 server as a service, you cannot also run it as a Windows application, and vice versa.

Results:
You created a Desktop shortcut that points to a TM1 server. You then launched the TM1 server as a Windows application and logged on to it using Architect.

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IBM Training

Demonstration 2

Create a new TM1 server

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Demonstration 2: Create a new TM1 server

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Demonstration 2: Create a new TM1 server

Purpose:
You want to create a new TM1 server to use for the course model. You will copy and modify a tm1s.cfg file for the course server. The server will be configured in IBM Cognos Configuration in order for the TM1 Admin server to find it. You also need to change the admin password for the new server.

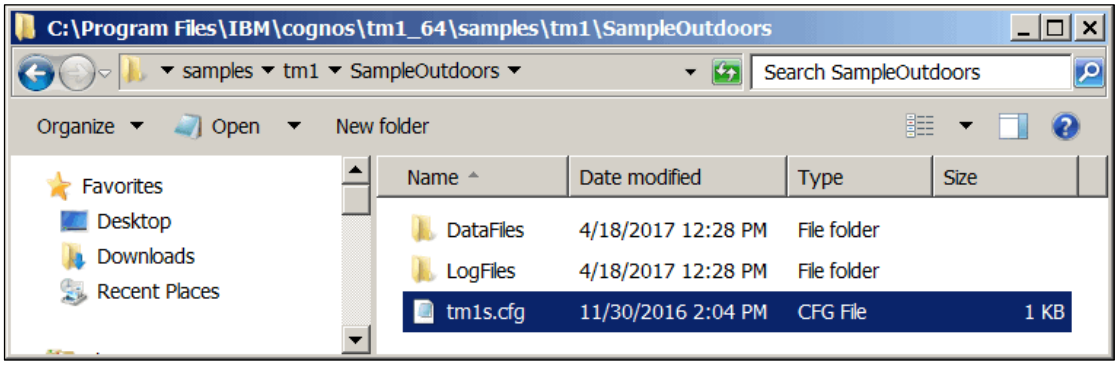
Task 1. Create a folder for the server files.

1. In the **Taskbar**, click the **Services** button  , stop the **IBM Cognos** service (it will take a few moments to fully stop), and then close the **Services** window.
   Having this service stopped will prevent any encryption-related errors from appearing when you save and start your TM1 configuration.
2. Right-click the **Start** menu and click **Open Windows Explorer**.
3. Navigate to the following path:
   C:\Program Files\IBM\cognos\tm1_64\samples\tm1\GO_New_Stores
   This is the path to one of the sample TM1 servers that comes installed with the software. Each TM1 server must have a location for the data files and a configuration file that points to this location. It is recommended that a separate folder is also specified to store any log files.
4. Right-click the **tm1s.cfg** file, and then click **Copy**.
5. Navigate to the **C:\Program Files\IBM\cognos\tm1_64\samples\tm1** folder.
6. Right-click in the workspace, point to **New**, and then click **Folder**.
7. Type **SampleOutdoors**, and then press **Enter**.
8. Double-click **SampleOutdoors**.
9. Repeat steps **6** and **7** to create two subfolders called **DataFiles** and **LogFiles**.

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10. Paste the **tm1s.cfg** file to the **SampleOutdoors** folder.

    The results appear similar to the following:



    The files could be located anywhere as long as it they are accessible to the TM1 Admin server.

11. Open **tm1s.cfg** in Notepad.

    Most text editors are fine to edit this file.

12. Next to **ServerName=**, delete **GO_New_Stores**, and then type **SampleOutdoors**.

13. Next to **PortNumber=**, delete **45557**, and then type **45454**.

14. Net to **HTTPPortNumber=**, change **5010** to **50100**.

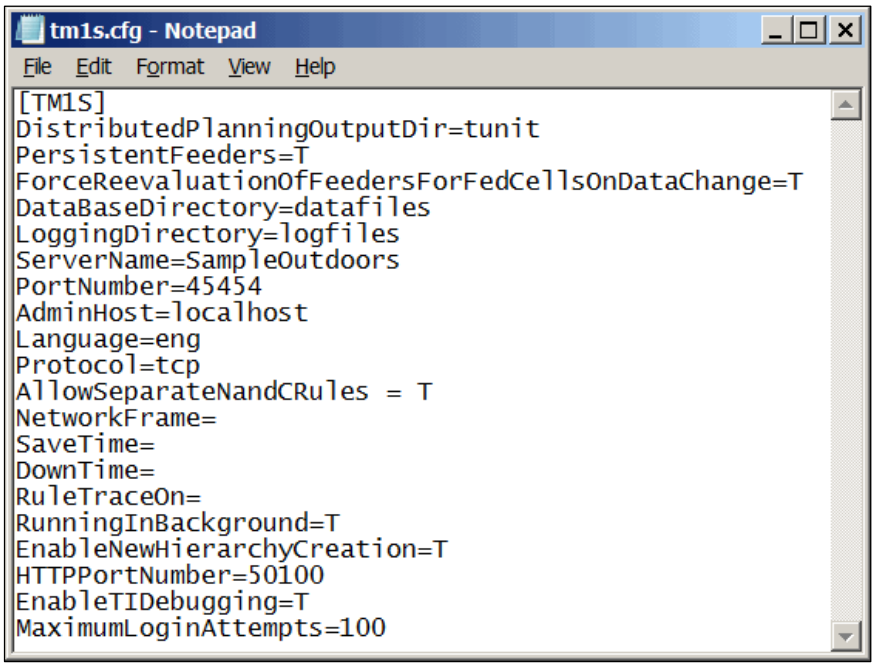
    The ServerName, PortNumber, and HTTPPortNumber values must be unique. If they are not, the TM1 server will not start.

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15. At the bottom of the file, type **MaximumLoginAttempts=100**.

    This parameter sets the maximum number of failed user login attempts permissible on the TM1 server. If you do not include the MaximumLoginAttempts parameter in tm1s.cfg, by default, the server allows three login attempts. If a user does not successfully log in to the TM1 server within the specified number of attempts, the server issues an error.

    The results appear as follows:



16. From the **File** menu, click **Save**, and then close **Notepad**.

    You know that the GO_New_Stores TM1 server starts and runs correctly. It is often easier to start with an existing configuration file and just modify it.

17. In **Windows Explorer**, navigate to **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\SData**, and then open t**m1s.cfg** in **Notepad**.

18. Scroll to the bottom of the file, on the last blank line, type **MaximumLoginAttempts=100**, and then save and close the file.

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19. Repeat steps **16** and **17** for the following files:
    - **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\GO_New_Stores\tm1s.cfg**
    - **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\go_contributor\tm1s.cfg**
    - **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\tm1s.cfg**
    - **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\GO_scorecards\tm1s.cfg**
    - **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\PlanSamp\Tm1s.cfg**
    - **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\Proven_Techniques\tm1s.cfg**
20. Close **Windows Explorer**.

Task 2. Use IBM Cognos Configuration to add a new TM1 server.

1. From the **Start** menu, point to **All Programs\IBM Cognos TM1 - 64**, and then click **IBM Cognos Configuration**.
   IBM Cognos Configuration opens (this may take a few moments).
2. Maximize the window.
   Any TM1 servers that are already running are shown in the Explorer pane on the left side. When you select a line in the Explorer pane, properties are displayed for the selected component on the right side.
3. In the **Explorer** pane, right-click **TM1 Server**, point to **New resource**, and then click **TM1 Server instances**.
4. In the **Name** box, type **SampleOutdoors**, and then click **OK**.
   Spaces are permitted but not recommended unless necessary.
5. In the right pane, click the **Value** box for **TM1 Server configuration path**, and then click the **edit** button  .
   You are prompted to select a folder.
6. Navigate to the following path and then click **Select**:
   C:\Program Files\IBM\cognos\tm1_64\samples\tm1\SampleOutdoors
   Do not select the tm1s.cfg file, just the path to where it is located.
7. In the **Explorer** pane, right-click the **SampleOutdoors** node and click **Test**.
   The server is tested successfully.

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8. Click **Close**.
9. In the **Explorer** pane, click **Environment**, and then in the right pane, ensure that the values appear as follows (modify them if necessary, but do not change the value for the Dispatcher password property):

Environment - Group Properties		
	Name	Value
	Data files location	../data
	Temporary files location	../temp
	Encrypt temporary files?	False
*	Format specification file location	../configuration/cogformat.xml
*	IP Version for Host Name Resolution	Use IPv4 addresses
	Gateway Settings	
*	Gateway URI	http://vclassbase:9300/bi/v1/disp
	Dispatcher Settings	
*	External dispatcher URI	http://vclassbase:9300/p2pd/servlet/dispatch
*	Internal dispatcher URI	http://vclassbase:9300/p2pd/servlet/dispatch
	Dispatcher password	**********
	Other URI Settings	
*	Dispatcher URI for external applications	http://vclassbase:9300/p2pd/servlet/dispatch
*	Content Manager URIs	http://vclassbase:9300/p2pd/servlet

10. In the **Explorer** pane, click **TM1 Applications**, and then in the right pane, ensure that the values appear as follows (modify them if necessary, but do not change the value for the User ID and password property):

TM1 Applications - Component Properties		
	Name	Value
*	Maximum memory in MB	768
	Shutdown port number	9513
*	TM1 Application Server Gateway URI	http://vclassbase:9510/pmpsvc
*	External server URI	http://vclassbase:9510
*	Session timeout (min)	1440
*	Force qualified paths	True
	Notifications provider	HTS
*	Enable Business Viewpoint	False
*	Business Viewpoint URI	http://localhost:9510/bv
*	Enable Scorecarding	True
*	Cognos Connection Folder Name	IBM Cognos TM1 Application - My Applications
	Application definition path	
	User preference path	
	Service configuration file path	
*	TM1 Application Server Dispatcher URI	http://vclassbase:9510/pmpsvc/dispatcher/servlet
	User ID and password	**********
	Namespace	

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11. In the **Explorer** pane, click **Local Configuration**, and then from the **File** menu, click **Save**.

    The configuration is saved (this will take a few moments).

    If you see an error message during the save process (such as **[ERROR] null**), or if it appears that IBM Cognos Configuration has stopped working, do the following:

    a) Right-click the **Taskbar**.
    b) Click **Start Task Manager**.
    c) On the **Applications** tab of Windows Task Manager, click the **IBM Cognos Configuration** entry.
    d) Click **End Task** and then click the **No** and/or **End Now** buttons (if necessary) to close IBM Cognos Configuration.
    e) Close **Windows Task Manager**.
    f) Open **Windows Explorer**.
    g) Navigate to **C:\Program Files\IBM\cognos\tm1_64\configuration**.
    h) Delete the **cogstartup.lock** file.
    i) Repeat steps **1** to **11** of **Task 2** in this demonstration. If the SampleOutdoors TM1 server is included in the list of TM1 servers in the Explorer pane on the left side (under the TM1 Server node), do not add it again.

12. After the configuration is saved, click **Close**.
13. In the **Explorer** pane, right-click **Local Configuration**, and then click **Restart**.

    All of the TM1 services and servers are restarted (this will take a few moments). The SampleOutdoors TM1 server that you added is started for the first time.
14. After the restart process is finished, click **Close**, and then close **IBM Cognos Configuration**.

Task 3. Log on to the SampleOutdoors TM1 server in Architect and change the admin password.

1. From the **Start** menu, point to **All Programs\IBM Cognos TM1**, and then click **Architect**.
2. Expand the **TM1** node and double-click the **SampleOutdoors** TM1 server.
3. In the **User name** box, type **admin**, leave the **Password** box blank, and then click **OK**.

   You are logged on to the SampleOutdoors TM1 server. Notice that this server does not yet contain any objects (such as cubes).
4. Right-click the **SampleOutdoors** server, point to **Security**, and then click **Change Password**.

   The Password change for admin dialog box appears.

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5. In the **Password** box, type **apple**, and then click **OK**.
   You are prompted to reenter your password.
6. Type **apple**, and then click **OK**.
   The password is successfully changed.
7. Right-click the **SampleOutdoors** server and click **Log Out**.
8. Double-click the **SampleOutdoors** server and log on with a user ID of **admin** and a password of **apple**.
   This TM1 server does not currently contain any user-created objects. You will add TM1 objects to this server and modify them throughout this course.
9. Close all open windows.

Results:
You customized an IBM Cognos TM1 server path and tm1s.cfg file. Next you added the TM1 server to the IBM Cognos Configuration tool and started the new server. Then you changed the admin password for this TM1 server.

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IBM Training

Unit summary

- Review financial performance management
- Identify the IBM Planning Analytics position in a performance management system
- Describe the IBM Planning Analytics components and architecture
- Describe TM1 server fundamentals
- Describe fundamental modeling concepts

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Unit summary

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IBM Training

Create dimensions

IBM Planning Analytics (v2.0)

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IBM Training

Unit objectives

- Describe cubes and dimensions
- Create dimensions manually
- Edit dimensions
- Create dimensions using TurboIntegrator

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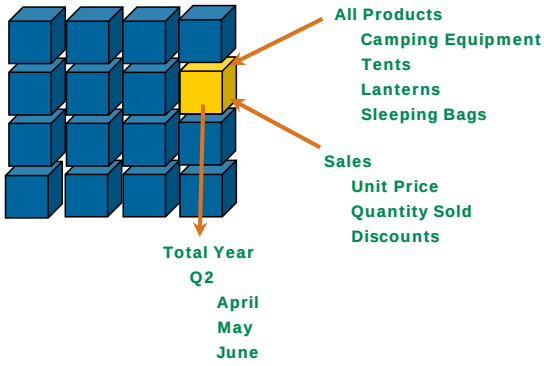
Unit objectives

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IBM Training

What is a dimension?

- Fundamental building block of TM1 cubes.
- Contain one or more hierarchies.



Create dimensions

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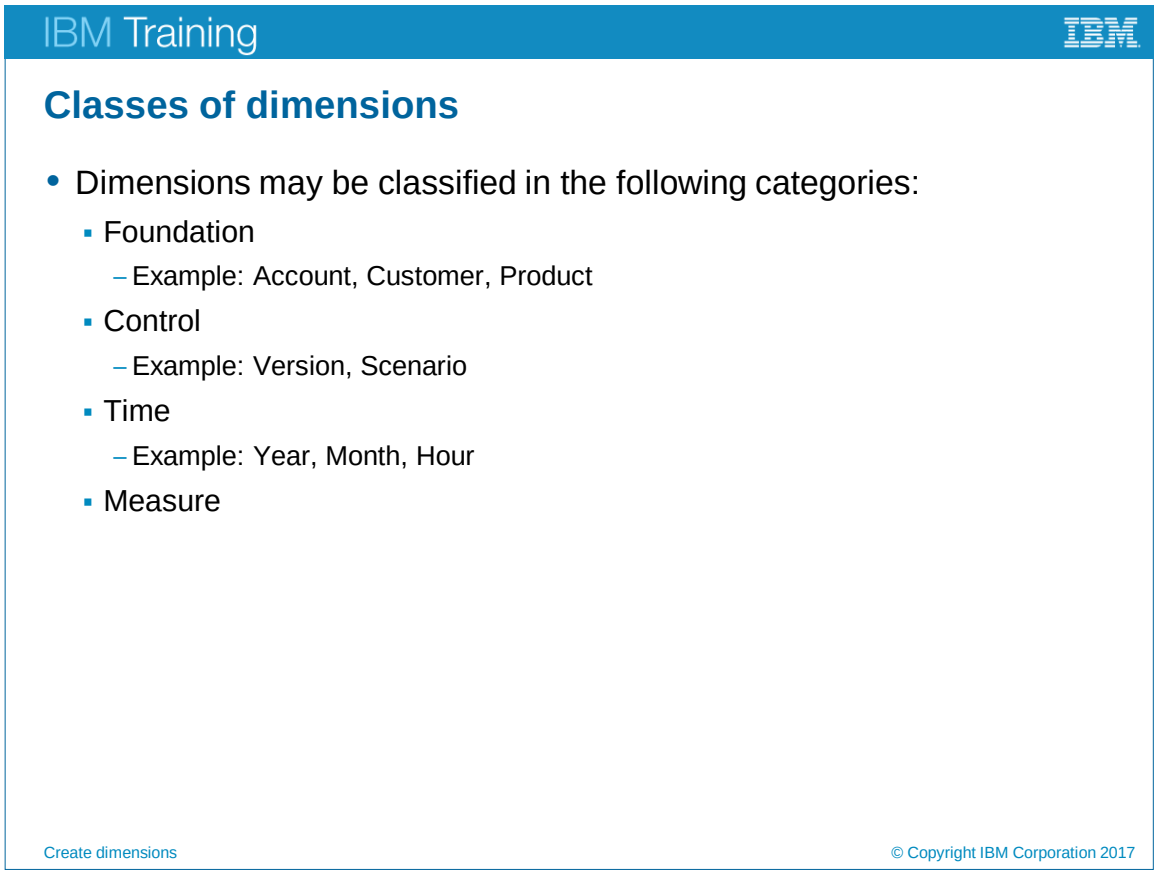
What is a dimension?

Dimensions are made up of elements (sometimes also called "members"), which are structured lists of related items that define the data stored in cubes.

The slide shows that the Time dimension contains hierarchies, where months roll up into quarters, and quarters into the year. The Product dimension also contains hierarchies, where product types roll into a product line, and product lines into all product lines.

TM1's aggregation is implicit. You do not need to specify the children who roll up to a parent. The consolidation is assumed and does not need to be defined by calculations.

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Classes of dimensions

Foundation dimensions describe the data.

Control dimensions are most often used to implement business processes.

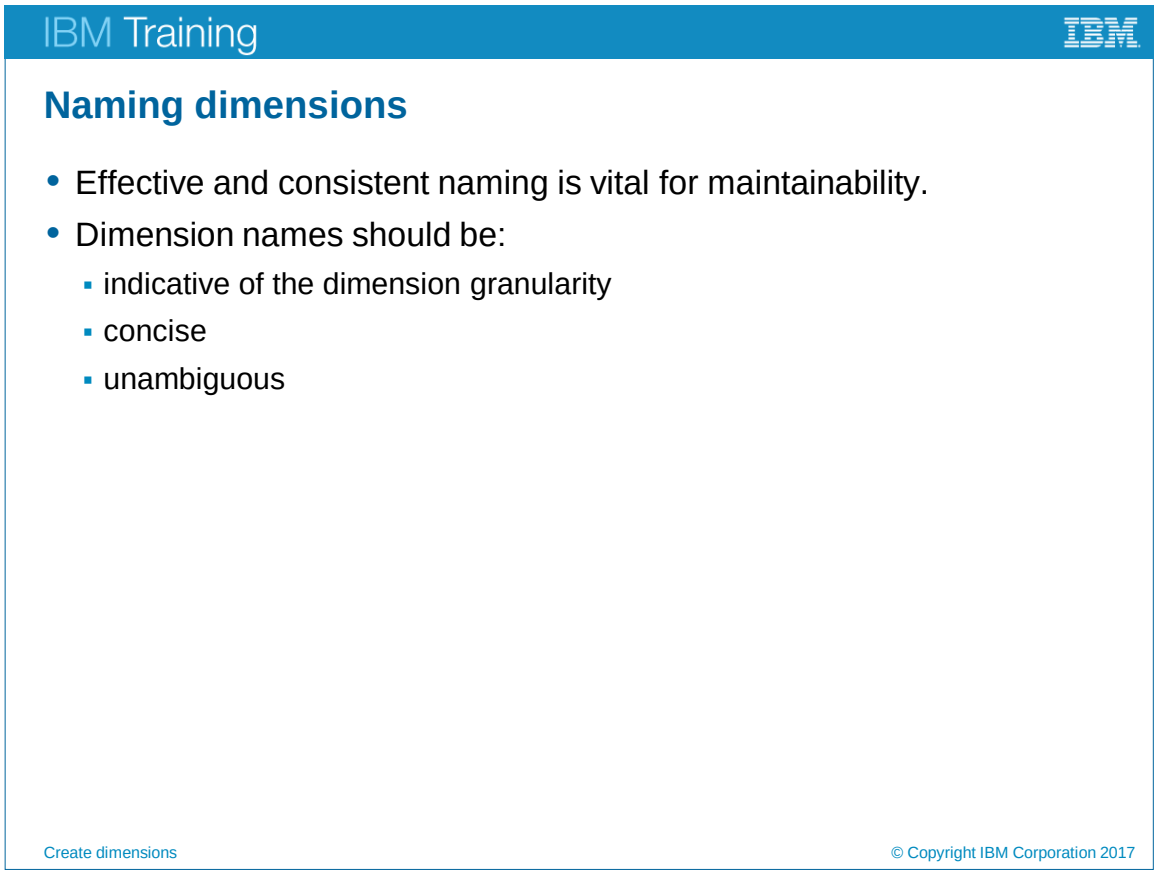
Time dimensions represent the critical time of the data.

The measures dimension represents the cardinal values of interest. There should be a measures dimension for almost every cube. It may be a good idea to name it after the cube for which it will be used, along with a prefix or suffix marking it as the measures.

Cubename_m or cubename_Measures may be used. In this course, we will use cubename_Measures to identify the measures dimensions.

Note: These dimension classifications are theoretical and are not enforced or marked in TM1.

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Naming conventions

Dimensions identify the location of data points in a cube. Consistent naming conventions should be established and enforced in the model.

Objects in TM1 may not be renamed. Instead they must be deleted and recreated. It is recommended that the data be exported to text before the object is deleted.

This concept is true for all objects in TM1, including dimensions, elements, cubes, processes, and so forth.

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IBM Training

What is an element?

- Elements are used to define data in cubes.
- Sometimes also referred to as "members".
- Every element has a type associated with it:
  - simple (number)
  - string
  - consolidated element
- As TM1 modelers, you will create, edit and format the attributes of the elements (precision, format).

Create dimensions

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What is an element?

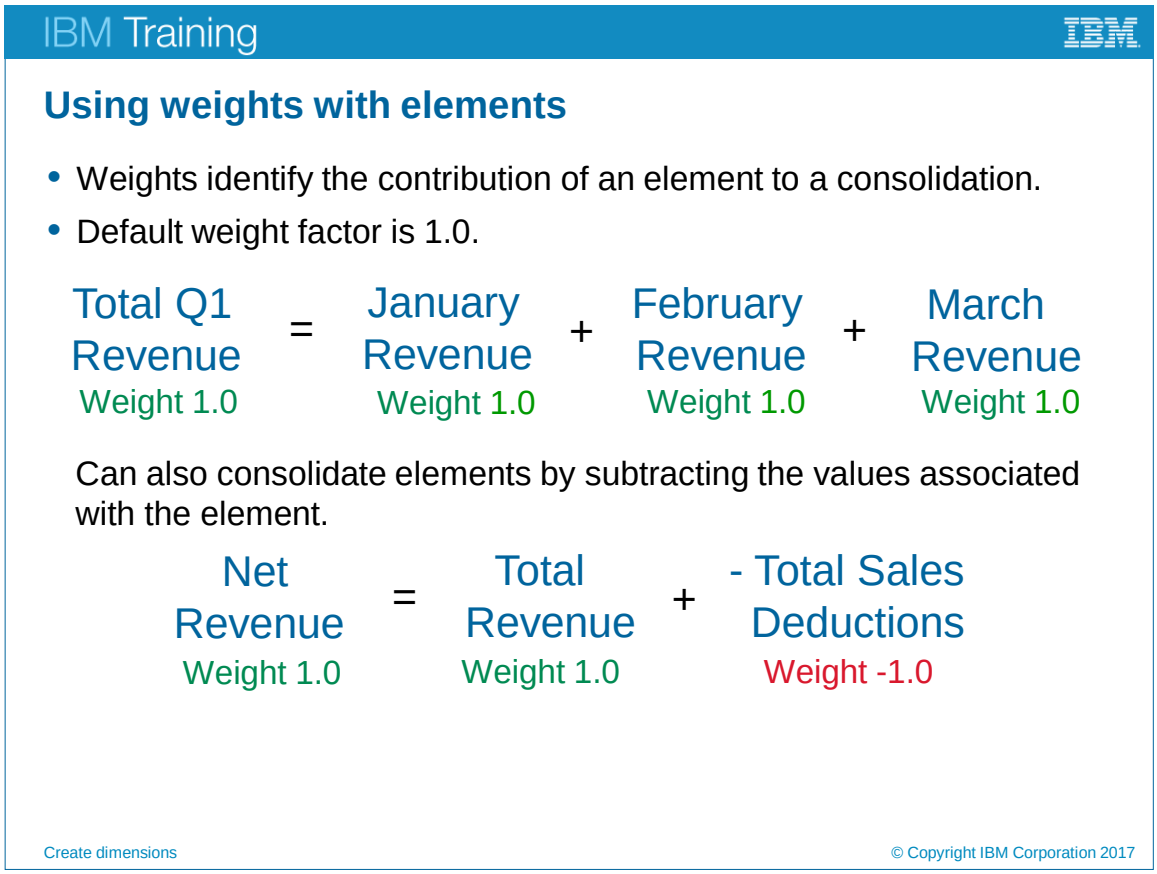
Symbol	Element Type	Description
	Simple	An element without any children. Also known as a leaf element. It is the lowest level in the hierarchy.
	String	An element that stores text strings in cells.
	Consolidated	An element that has aggregated data in a cube. Two or more elements roll up to a consolidated element.

String elements are not aggregated and are typically used for descriptive purposes. To include a string in a cell in a cube, the element from the last dimension defining the cell must be a string element. TM1 treats string elements that occur in any dimension other than the last one as numeric elements.

It is best practice to make the measures dimension the last dimension. In a later exercise, you will add a string element to a measures dimension that is in a cube.

TM1 only stores data at the leaf level element, and consolidations are done in RAM. This allows TM1 to quickly compute calculations.

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Using weights with elements

TM1 assumes all children will be aggregated (addition).

Credit balances may be stored. Unsigned and negative weights allow you to apply a reduction.

Weighting is a property of a child's relationship to a parent. It allows the value of the child element to be subtracted from the parent value (instead of aggregated).

Net Revenue = Total Revenue + (Total Sales Deductions * (-1))

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IBM Training

Using attributes

- Element attributes describe an element.
- Use attributes to:
  - describe features of elements
  - provide alternative names, or aliases
  - format the display for numeric data

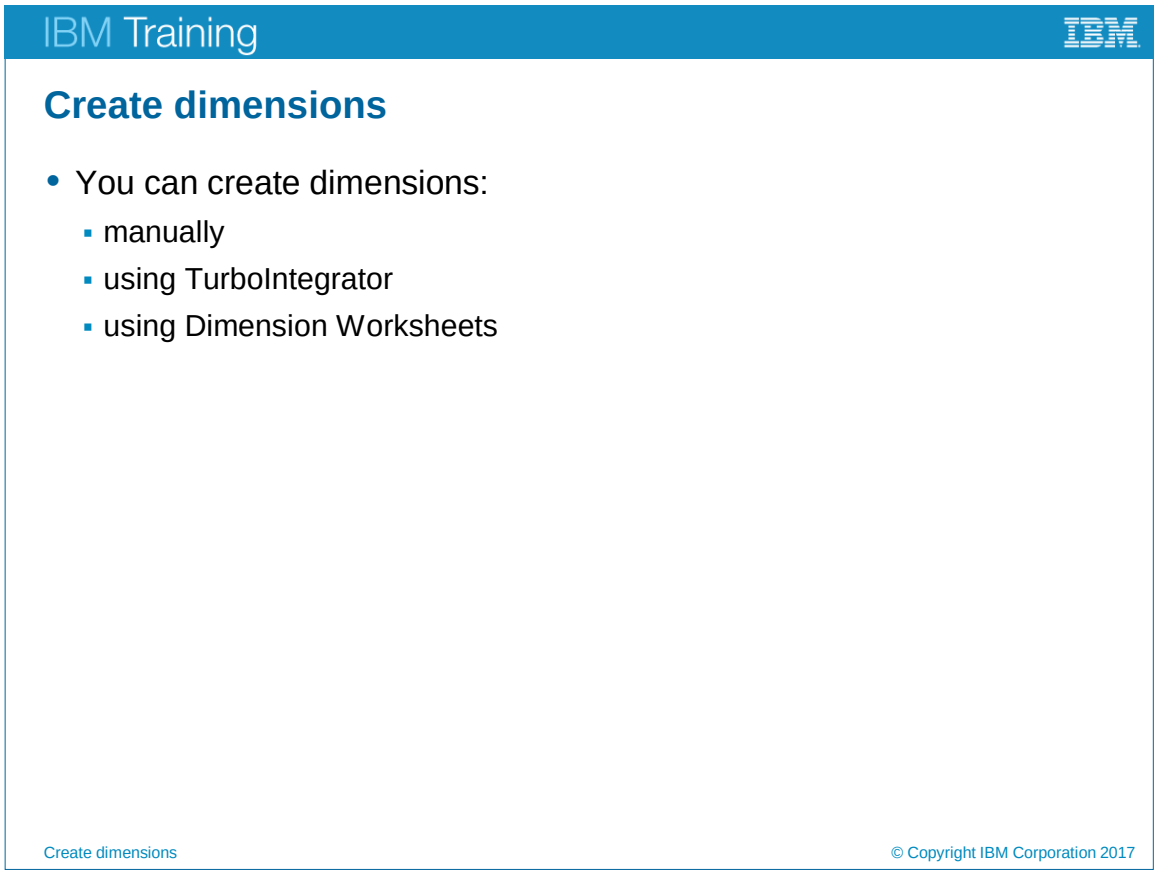
Create dimensions

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Using attributes

If you want to:	Attribute Type:	Example:
describe features of an element	text or numeric	• square footage of a store • the engine size of a car model
provide alternative names, or aliases	alias	• descriptive names of general ledger accounts • element name in another language
format the display for numeric data	format	• number of decimal places • currency or percentage • how negatives appear

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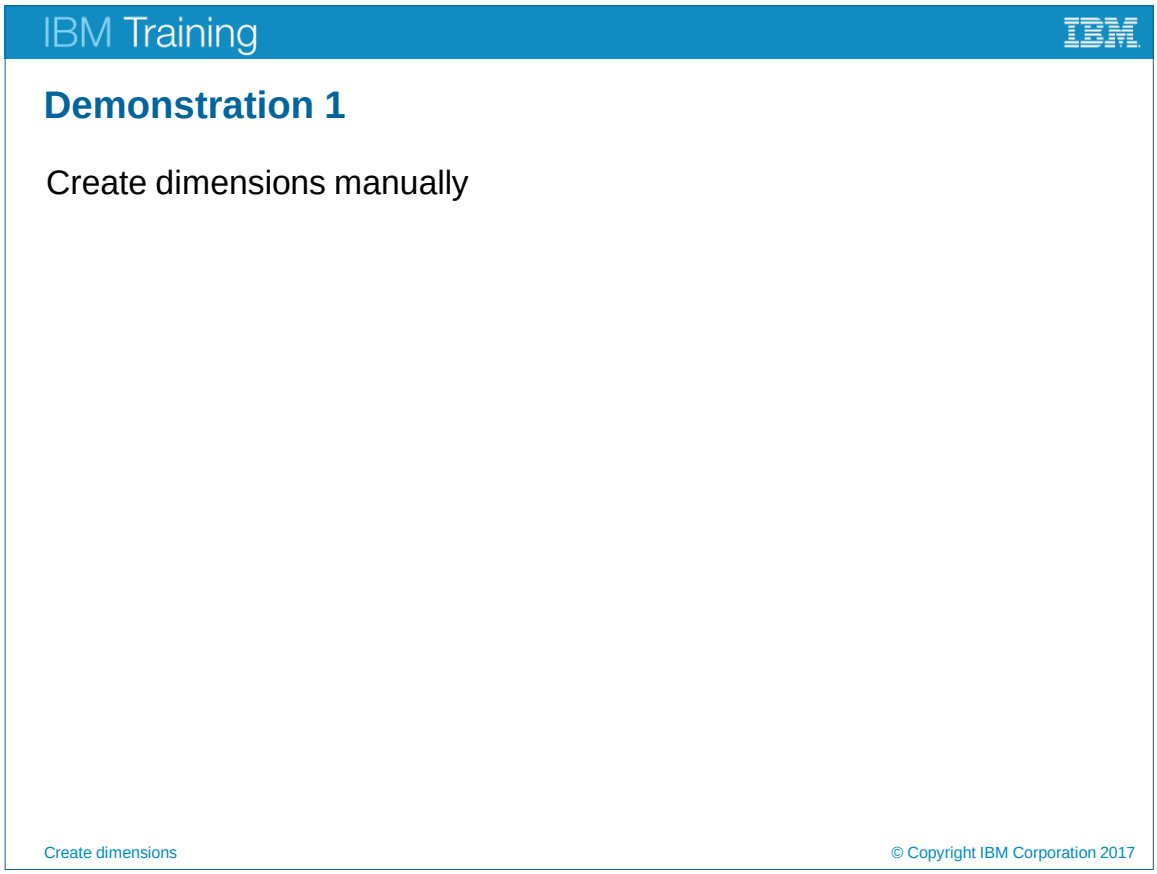


Create dimensions

It is best practice to name the dimension with an intuitive noun that indicates it is a leaf-level element.

TM1 supports the use of spaces and some special characters in names. However, it is better to use underscores (_) instead of spaces in order to minimize conflict with other applications.

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Demonstration 1: Create dimensions manually

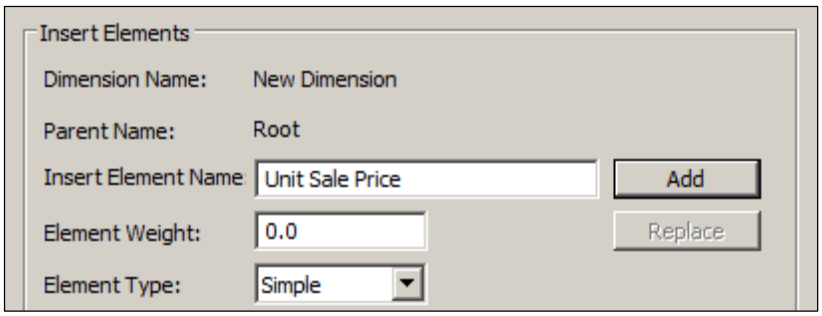
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Demonstration 1: Create dimensions manually

Purpose:
You want to create the Price and Cost, Months, and Versions dimensions manually.

Task 1. Create the Price_and_Cost_Measures dimension.

1. From the **Start** menu, point to **All Programs\IBM Cognos TM1 - 64**, and then click **Architect**.
2. Maximize the window, expand **TM1**, and then double-click **SampleOutdoors**.
3. Log on with a user ID of **admin** and a password of **apple**.
   Notice that this TM1 server does not currently contain any objects.
4. From the **View** menu, deselect the **Display Control Objects** menu item if it is currently selected.
5. Right-click **Dimensions**, and then click **Create New Dimension**.
6. From the **Edit** menu, click **Insert Element**.
7. In the **Insert Element Name** box, type **Unit Sale Price**.
   The result appears as shown below:

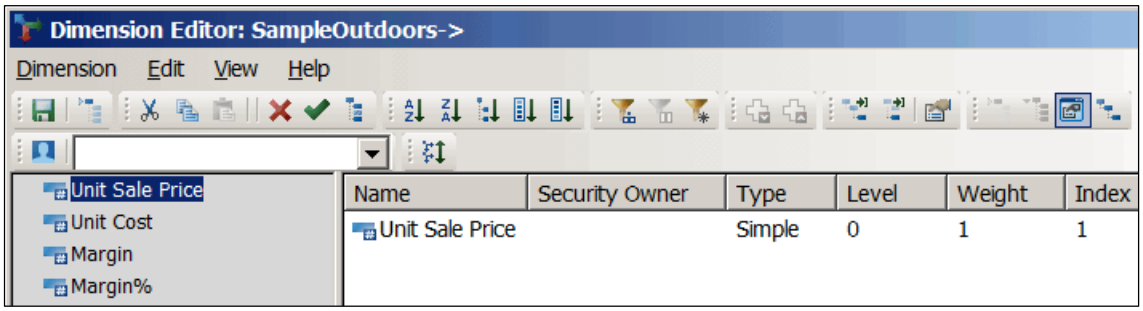


8. Leave the default **Element Weight** and **Element Type**, and then click **Add**.
   The new element appears in the Inserted Elements pane.
   You can also press <Enter> to add elements to the Inserted Elements pane.
9. Repeat steps **6** and **7** to add **Unit Cost**, **Margin** and **Margin%**.
10. Click **OK**.

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11. From the **View** menu, click **Properties Window**.
12. Select **Unit Sale Price** on the left.

    The result appears as shown below:



    If the Dimension Editor does not appear with two panes as shown you can also click the **Display Properties Window**  button on the toolbar.
13. From the **Dimension** menu, click **Save**.
14. In the **Name** box, type **Price_and_Cost_Measures**, and then click **OK**.
15. Click **OK** to close the **Dimension Editor**.

Task 2. Create the Months dimension.

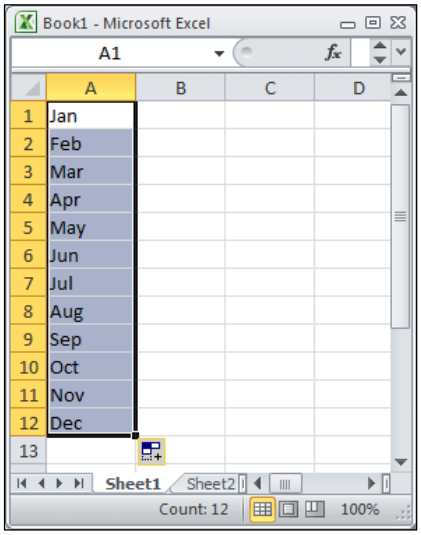
Next you will create the Months dimension. Instead of typing each month, you will use Microsoft Excel to populate the months.

1. Right-click **Dimensions**, and then click **Create New Dimension**.
2. Launch **Microsoft Excel**, create a new blank workbook (if necessary), and then in cell **A1**, type **Jan**.

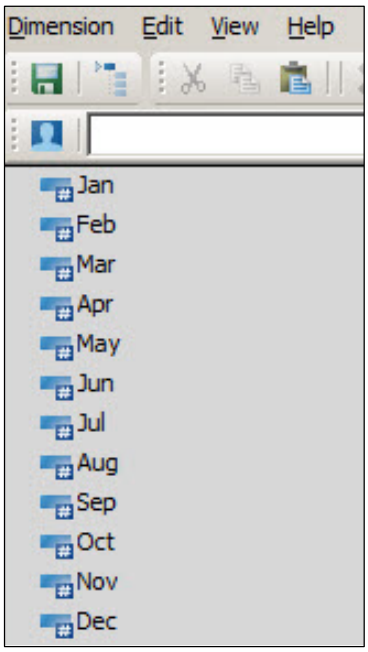
   You will use Autofill functionality to populate the remaining months of the year.
3. Click cell **A1**, and then place the pointer over the **lower right corner** of cell **A1**, until you see a plus sign.

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4. Click and drag the cursor to cell **A12**, and then release the mouse button.
   A list of all 12 months appears:



5. Copy the 12 months from **Microsoft Excel**, and paste them in the grey pane in the **Dimension Editor**.
6. Close **Microsoft Excel** without saving.
7. In the **Dimension Editor** window, from the **Dimension** menu, save the dimension with the name **Months**.
   The result appears as shown below:



The icon to the left of each element indicates that these are leaf-level numeric elements, at the lowest level of detail in the dimension. In a cube that contains only numbers, all the lowest level elements are numeric.

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8. Click **OK** to close the Dimension Editor.

   It is a proven practice to include the measures dimension as the last dimension in the cube. However, dimensions can be created in the order of your choice.

   The measures dimension is usually named something like <cubename>_m or <cubename>_measures.

   In the next unit, you will create the Price and Cost cube.

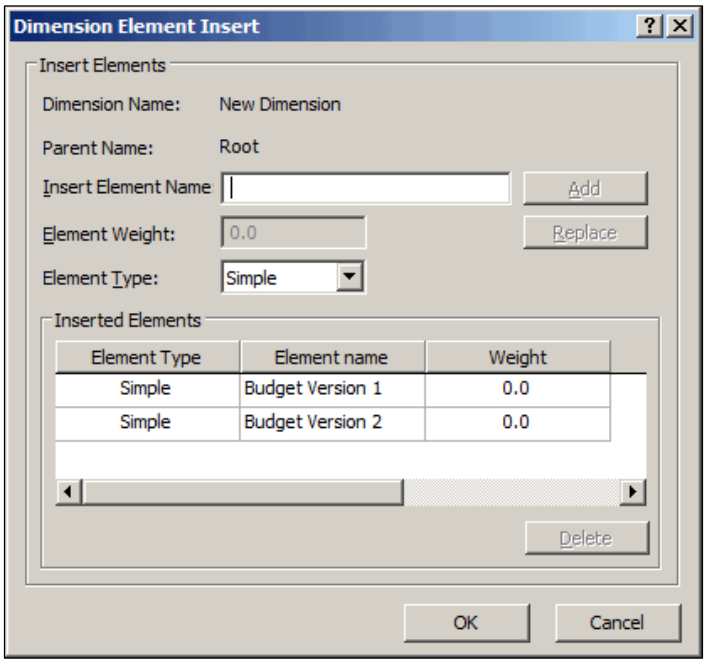
   TM1 supports string elements. They are not aggregated and are usually used for descriptive purposes. If string elements exist, they must be defined in the last dimension in the cube (which can also be the measures dimension).

   A dimension does not actually contain anything. It maps to data points when used in a cube. The text string "Jan" does not point to text. It points to a value that "Jan" describes.

Task 3. Create the Versions dimension.

1. Right-click **Dimensions**, and then click **Create New Dimension**.
2. In the **Dimension Editor**, insert the following elements:
   - **Budget Version 1**
   - **Budget Version 2**

   The results appear as follows:



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3. Save the dimension with the name **Versions**, and then close the **Dimension Editor**.
4. Leave Architect open for the next demonstration.

Results:
You have created the Price_and_Cost_Measures, Months, and Versions dimensions.

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IBM Training

Demonstration 2

Modify existing dimensions

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Demonstration 2: Modify existing dimensions

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Demonstration 2: Modify existing dimensions

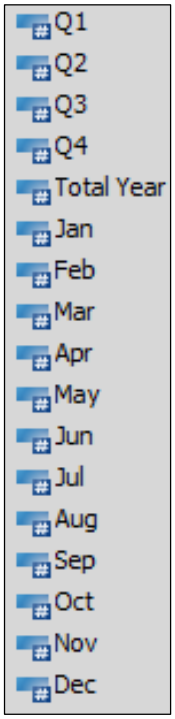
Purpose:
You want to modify the Months dimension to include the elements for quarters and year, and then sort the Months dimension. You will add descriptive attributes for each month to identify the next and previous months, as well as an alias for each month in order to refer to each month in either short or long form. You will also edit the attribute elements to format the Price_and_Cost_Measures dimension.

Task 1. Add Quarters and Total Year elements to the Months dimension and then sort the dimension.

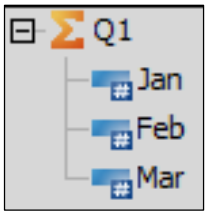
1. In **Server Explorer**, right-click **Months**, and then click **Edit Dimension Structure**.
2. From the **Edit** menu, click **Insert Element**, and then insert the following elements with the default **Element Weight** and **Element Type**:
   - **Q1**
   - **Q2**
   - **Q3**
   - **Q4**
   - **Total Year**

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3. Click **OK**.
   The result appears as shown below:



4. Click **Jan**, Shift+click **Mar** to select Jan to Mar, and then drag and drop the elements onto **Q1**  .
   The icon indicates that the selected elements will be dropped as child elements.
   The result appears as shown below:

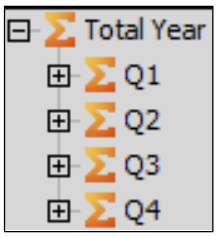


   The "Σ" indicates consolidated items. Jan, Feb, and Mar are summed into Q1.
5. Right-click **Q1**, and then click **Element Properties**.
   Notice that the element type is "Consolidated".
6. Under **Element Type**, click the arrow beside **Consolidated**.
   Notice that that the other available element types are Simple and String.
7. Click **Cancel**, and then view the **Element Properties** for **Jan**.
   Notice that the element type is Simple.
8. Click **Cancel**.

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9. Select **Apr**, **May**, and **Jun**, and then drop them onto **Q2**.
10. Select **Jul**, **Aug**, and **Sep**, and then drop them onto **Q3**.
11. Select **Oct**, **Nov**, and **Dec**, and then drop them onto **Q4**.
12. Collapse and select **Q1**, **Q2**, **Q3**, and **Q4**, and then drop the selection onto **Total Year**.

    The result appears as shown below:



    Next, you want to make sure that these items will maintain this element order when you open the dimension again.

13. On the toolbar, click **Hierarchy Sort**  .
14. On the toolbar, click **Set Dimension order**  .

    This button will reset the index number of each element. Each time you add a new element, the next available index number is assigned to it. This number is often used in ordering the dimension. If you moved the elements around, you may wish to reset the index number to reflect the new order.

    However, index numbers are sometimes used in TM1 functions so you want to be sure you will not alter the result of these functions if you reorder the dimension.

15. Save the **Months** dimension, and then click **OK** to close the **Dimension Editor**.

Task 2. Add element attributes for Next and Previous Months.

Next you will add descriptions that identify next and previous months.

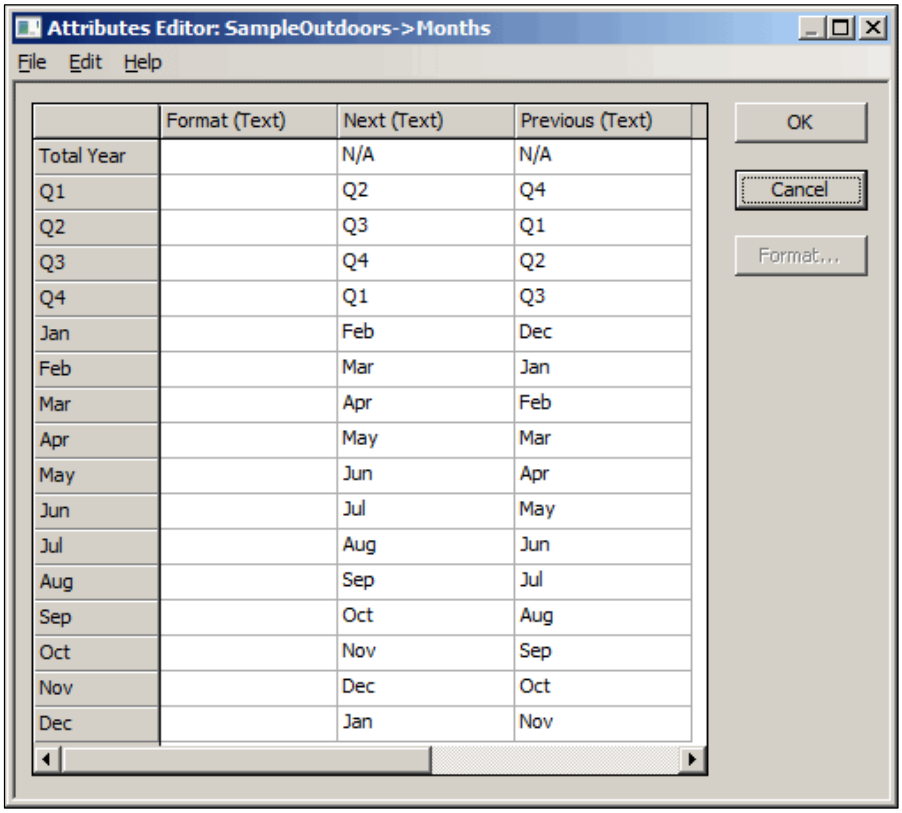
1. In **Server Explorer**, right-click the **Months** dimension and click **Edit Element Attributes**.
2. From the **Edit** menu, click **Add New Attribute**.
3. In the **Name** box, type **Next**, under **Type**, ensure that **Text** is selected, and then click **OK**.
4. Repeat steps **2** and **3**, and in the **Name** box, type **Previous**.

   You will need to identify the Next and Previous month.

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5. Enter the values as shown below:
   Note: You can paste Autofill values from Microsoft Excel.



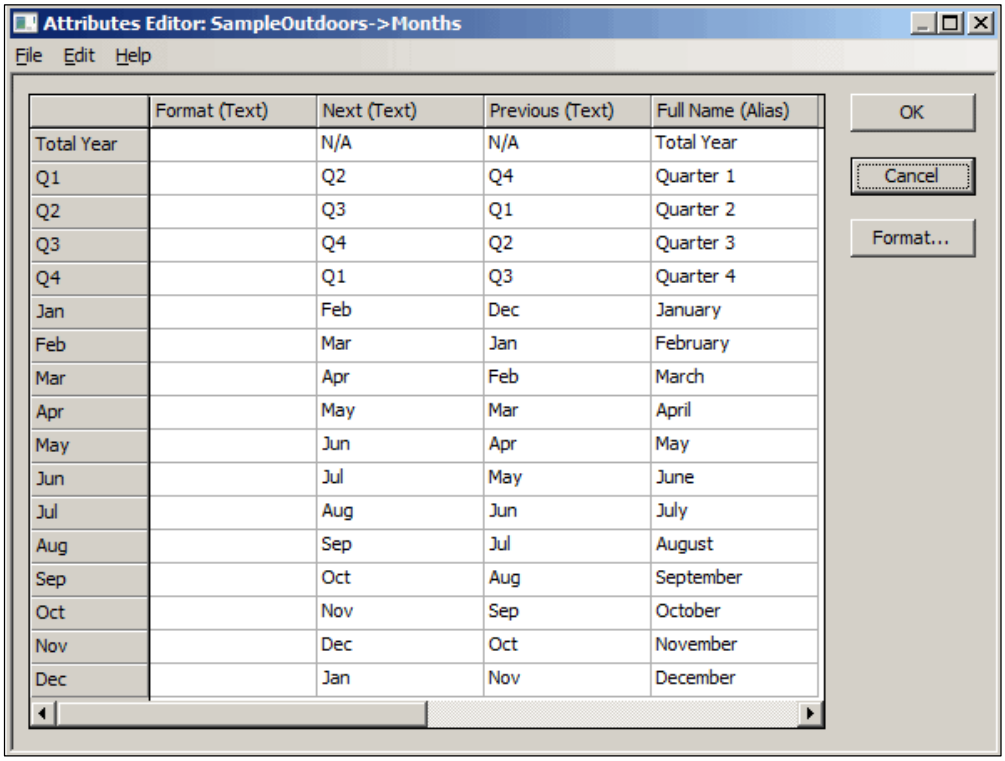
	Format (Text)	Next (Text)	Previous (Text)
Total Year		N/A	N/A
Q1		Q2	Q4
Q2		Q3	Q1
Q3		Q4	Q2
Q4		Q1	Q3
Jan		Feb	Dec
Feb		Mar	Jan
Mar		Apr	Feb
Apr		May	Mar
May		Jun	Apr
Jun		Jul	May
Jul		Aug	Jun
Aug		Sep	Jul
Sep		Oct	Aug
Oct		Nov	Sep
Nov		Dec	Oct
Dec		Jan	Nov

Task 3. Add attributes to alias the month name in full name form.

1. From the **Edit** menu, click **Add New Attribute**.
2. In the **Name** box, type **Full Name**, select **Alias**, and then click **OK**.
   The warning that TM1 will set initial values for all objects for an alias attribute appears.
3. Click **OK**.
   You will manually change the values in the column to use full names.

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4. Modify the values in the **Full Name (Alias)** column to the full names. The result appears as shown below:

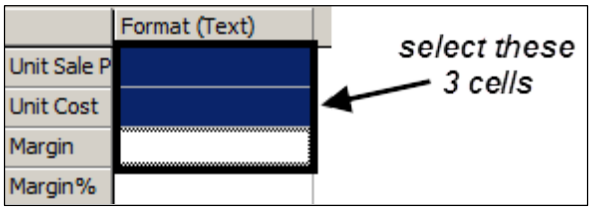


	Format (Text)	Next (Text)	Previous (Text)	Full Name (Alias)
Total Year		N/A	N/A	Total Year
Q1		Q2	Q4	Quarter 1
Q2		Q3	Q1	Quarter 2
Q3		Q4	Q2	Quarter 3
Q4		Q1	Q3	Quarter 4
Jan		Feb	Dec	January
Feb		Mar	Jan	February
Mar		Apr	Feb	March
Apr		May	Mar	April
May		Jun	Apr	May
Jun		Jul	May	June
Jul		Aug	Jun	July
Aug		Sep	Jul	August
Sep		Oct	Aug	September
Oct		Nov	Sep	October
Nov		Dec	Oct	November
Dec		Jan	Nov	December

5. Click **OK**.

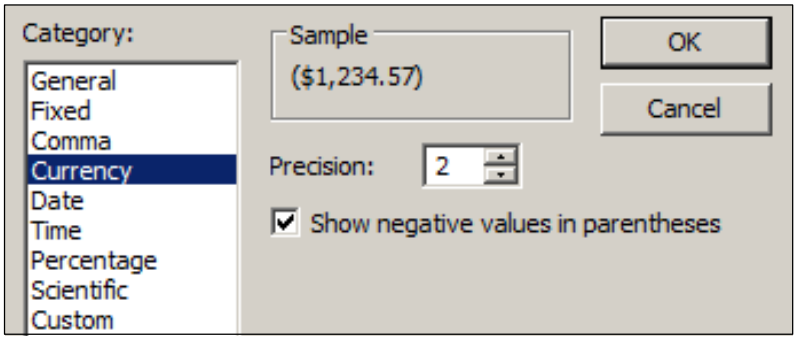
Task 4. Format the elements in the Price_and_Cost_Measures dimension.

1. In the **Server Explorer** window, right click **Price_and_Cost_Measures**, and then click **Edit Element Attributes**.
2. Click and drag - from the **Unit Sale Price** cell to the **Margin** cell - to highlight all three cells.



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3. Click **Format**.



4. Click **Currency**, and then click **OK**.
5. Click the **Margin%** cell, and then click **Format**.
6. Click **Comma**, and then click **OK**.

   The results appear as follows:

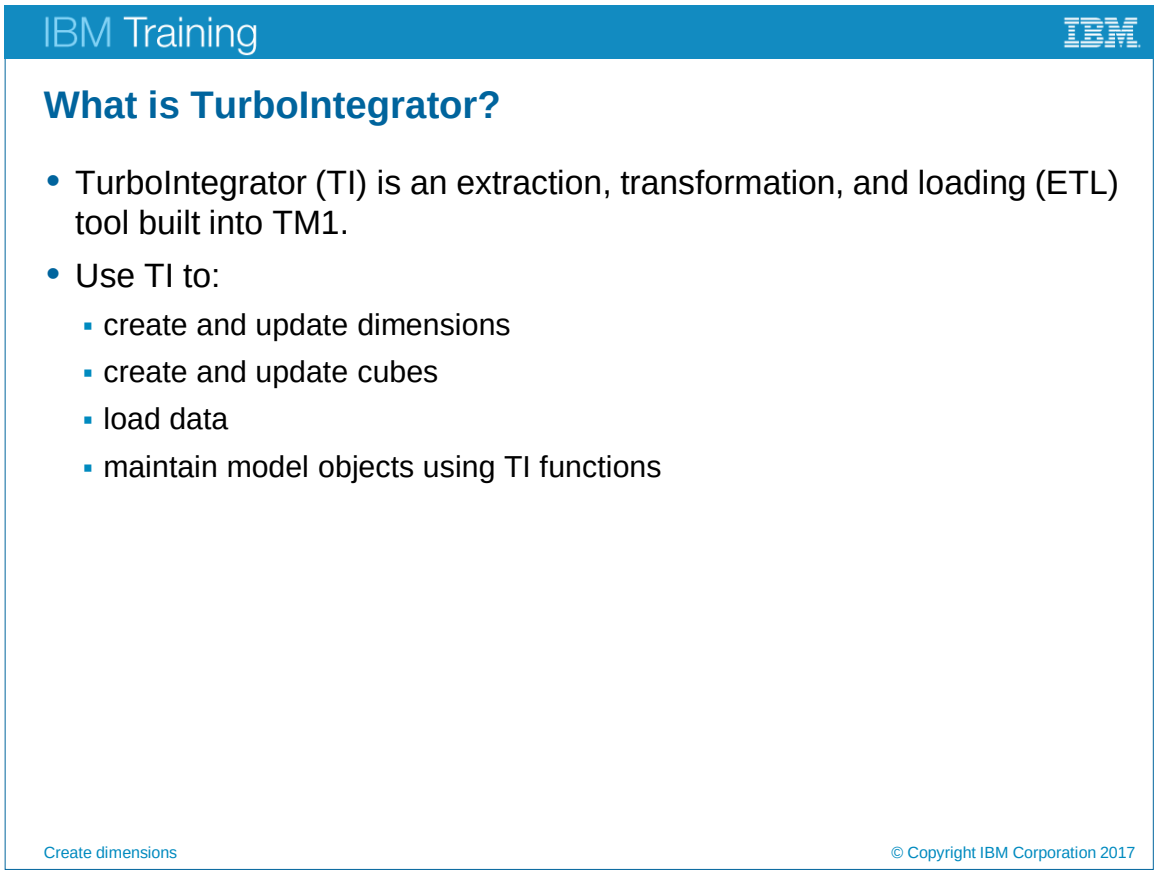
	Format (Text)
Unit Sale Price	b:\$#,###0.00;(\$#,###.00)ϞC|2|Y|$
Unit Cost	b:\$#,###0.00;(\$#,###.00)ϞC|2|Y|$
Margin	b:\$#,###0.00;(\$#,###.00)ϞC|2|Y|$
Margin%	b:#,##0.00;(#,##0.00)ϞCO|2|Y

7. Click **OK**.
8. Leave Architect open for the next demonstration.

Results:
You modified the Months dimension to include the elements for quarters and year, and then sorted the Months dimension. Next you added descriptive attributes for each month to identify the next and previous months, as well as an alias for each month to be able to refer to it in either short or long form. Lastly, you edited the attribute elements to format the Price_and_Cost_Measures dimension.

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What is TurboIntegrator?

TurboIntegrator (TI) is used to create a process that recognizes the data structure of the source and transforms it into the appropriate structure for the model. Once the TI process is developed, you can rerun it or schedule it to be used when importing data from a dynamic source.

TI may import data from a variety of sources including TM1, text files, ODBC, and more.

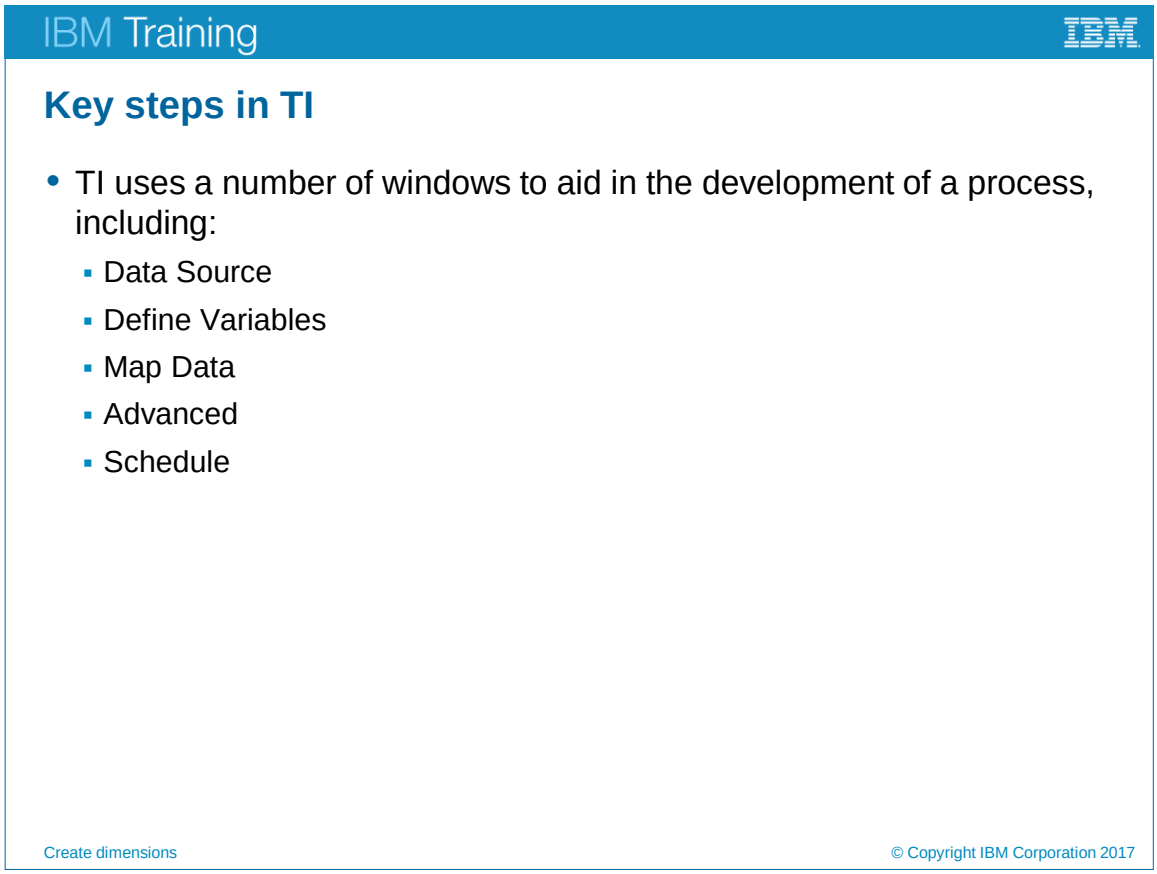
In this unit, the focus is on creating dimensions. However, additional uses will be discussed in the following units.

When creating an object using TI, be sure to review the resulting object for accuracy.

It is possible for a TI process to run without an error, while the resulting structure is not what was intended. This may be due to bad or unexpected data (garbage in, garbage out, or GIGO), or a number of other reasons. It is recommended that you always review the resulting objects for accuracy.

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Key steps in TI

The Data Source tab allows you to select and preview the data from the data source as it appears to TI.

Define Variables is used to identify how the incoming data will be transformed for use in the TM1 structures.

Map Data allows for more detailed mapping of data structures and objects.

Scripts used by a TI process are displayed and edited from the Advanced tab.

The Schedule tab is used to schedule the process to run repetitively, or off hours.

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IBM Training

Demonstration 3

- Create a dimension using TurboIntegrator

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Demonstration 3: Create a dimension using TurboIntegrator

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Demonstration 3: Create a dimension using TurboIntegrator

Purpose:
You will create the Products dimension using TurboIntegrator. You will identify text data as the source, create variables to transform the data, map the variables to TM1 objects, review the scripts generated, and run the process. Finally, you will review the resulting dimension for accuracy.

Task 1. Add data source connection information.

1. In the **Server Explorer** window, under **SampleOutdoors**, right-click **Processes**, and then click **Create New Process**.

   TurboIntegrator opens.
2. On the **Data Source** tab, click **Text**.
3. Next to **Data Source Name**, click **Browse**.

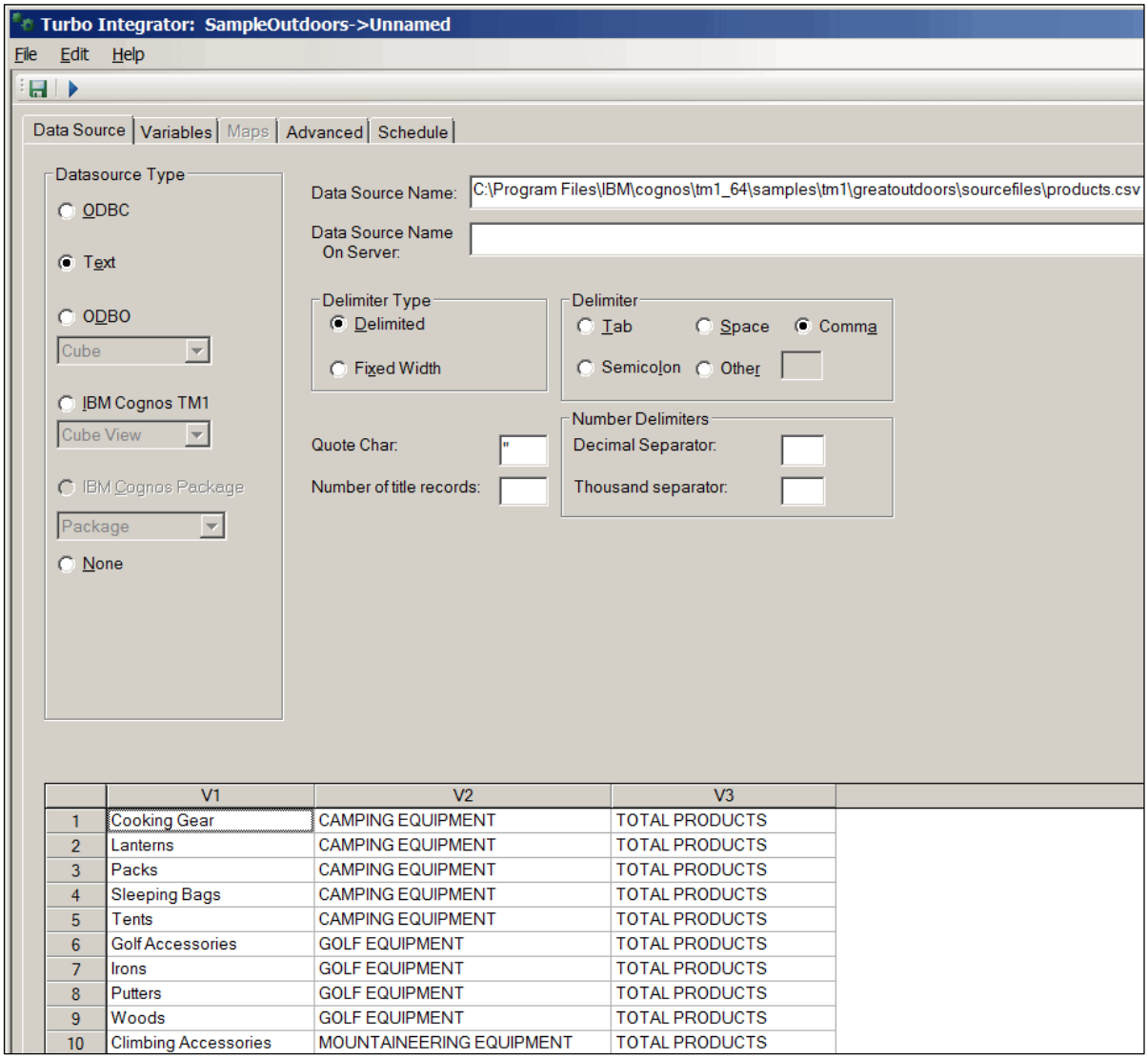
   The Select Input File dialog box appears.
4. Navigate to **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles**, and then double-click **products.csv** to open the file.
5. Click **OK** to dismiss the warning.

   If you were working on a remote server, you would use a Universal Naming Convention (UNC) path for files. The UNC path is of the form \\ComputerName\SharedFolder\Resource.
6. Ensure that the **Delimiter Type** is **Delimited** and that the **Delimiter** is **Comma**.

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7. Click **Preview**.

   You can see the first 10 rows of data in your source:



	V1	V2	V3
1	Cooking Gear	CAMPING EQUIPMENT	TOTAL PRODUCTS
2	Lanterns	CAMPING EQUIPMENT	TOTAL PRODUCTS
3	Packs	CAMPING EQUIPMENT	TOTAL PRODUCTS
4	Sleeping Bags	CAMPING EQUIPMENT	TOTAL PRODUCTS
5	Tents	CAMPING EQUIPMENT	TOTAL PRODUCTS
6	Golf Accessories	GOLF EQUIPMENT	TOTAL PRODUCTS
7	Irons	GOLF EQUIPMENT	TOTAL PRODUCTS
8	Putters	GOLF EQUIPMENT	TOTAL PRODUCTS
9	Woods	GOLF EQUIPMENT	TOTAL PRODUCTS
10	Climbing Accessories	MOUNTAINEERING EQUIPMENT	TOTAL PRODUCTS

Task 2. Modify variable names.

1. At the top of the **TurboIntegrator** window, click the **Variables** tab.
2. In the first row, double-click **V1** and rename it to **vProducts**. Leave the Variable Type column as **String**.

   The Variable column is used to identify to TI what type of data is coming from the source. You define formulas to transform this variable type into the final data structure required for the TM1 object being created/updated in the TI process.

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3. In the **Contents** column, in the **vProducts** row, click the down arrow  , and then click **Element**.

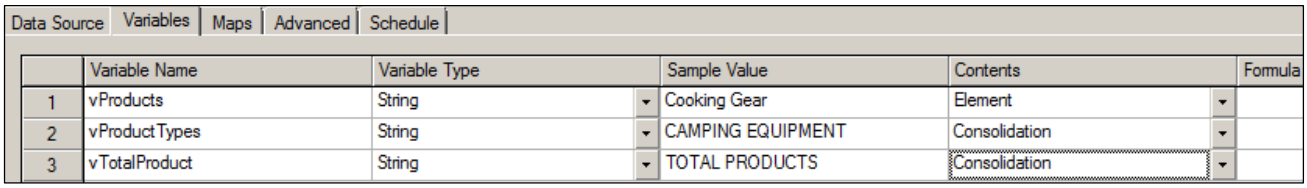
   The Contents column identifies how the incoming data will be used or transformed in the TI process. It is defined by selecting one of the following settings:

Contents	Description
Ignore	Ignore the contents of the column when processing the data source.
Element	The column contains simple elements.
Consolidation	The column contains consolidated elements.
Data	The column contains data values in a cube.
Attribute	The column contains element attributes for the dimension you want to create or update.
Other	The column contains data that does not fall into any of the previous four categories. Typically, this setting is used for columns containing data that will be processed through custom variables and formulas.

4. Repeat steps **2** and **3** with the following information.

Old Variable Name	New Variable Name	Contents
V2	vProductTypes	Consolidation
V3	vTotalProduct	Consolidation

The result appears as shown below:



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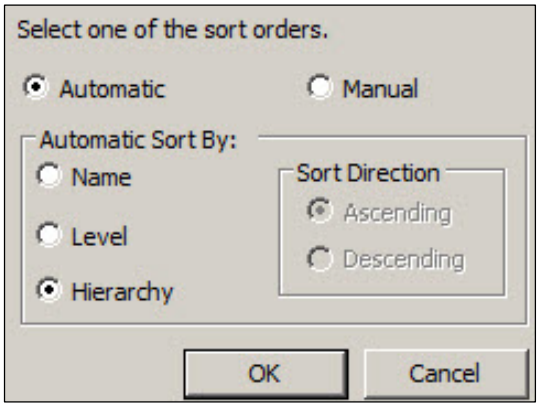
Task 3. Identify how data is mapped.

1. Click the **Maps** tab, and then click the **Dimensions** sub-tab.
2. For the **vProducts** element variable, in the **Dimension** column, type **Products**.
3. In the **Action** column, click the down arrow  , and then click **Create** (if necessary).
4. In the **Element Type** column, ensure that **Numeric** is selected.

   Numeric is the default and should only be changed if the data will be used for a String element. When you review the elements in a dimension in the Subset or Dimension Editors, their icon includes either a # or *ab* (at the lowest level) identifying a numeric or a string element.
5. In the **Element Order** column, click **By Input**.

   The Dimension Element Ordering dialog box appears.
6. Select **Automatic**, and then select **Hierarchy**.

   The result appears as shown below:



7. Click **OK**.
8. Click the **Consolidations** tab.

   This tab identifies the parent-child relationship(s) in the dimension. You can map more than one hierarchy here if necessary.

   It is often useful to diagram the parent-child relationships on paper or in a diagram tool prior to working in this screen:

   vTotalProduct

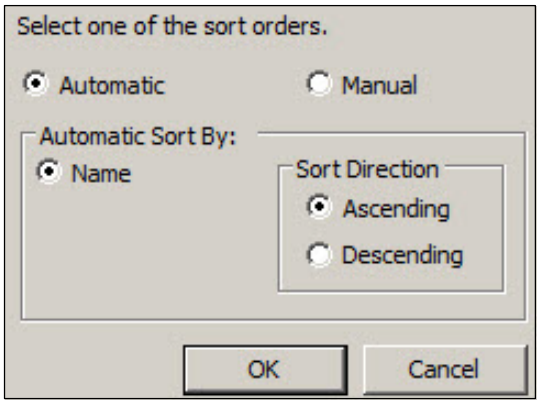
   vProductTypes

   vProducts
9. For the **vProductTypes** consolidation variable, click the arrow to the right of the **Child Variable** cell, select **vProducts**, and then click **OK**.
10. For the **vTotalProduct** consolidation variable, click the arrow to the right of the **Child Variable** cell, select **vProductTypes**, and then click **OK**.

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11. Under **Component Order** for **vProductTypes**, click **By Input**.
    The Component Element Ordering dialog box appears.
12. Click **Automatic**.
    The result appears as shown below:



13. Click **OK**.
    The Server Explorer window may move to the front. Click the TurboIntegrator window to see it.
    The results appear as follows:

Cons. Variable	Dimension	Child Variable	Weight	Sample Value	Component Order
vProductTypes	Products	vProducts	1.000000	CAMPING EQUIP	By Name
vTotalProduct	Products	vProductTypes	1.000000	TOTAL PRODUC	By Input

14. Repeat steps **11** to **13** for the **vTotalProduct** variable.
    The results appear as follows:

Cons. Variable	Dimension	Child Variable	Weight	Sample Value	Component Orde
vProductTypes	Products	vProducts	1.000000	CAMPING EQUIPMENT	By Name
vTotalProduct	Products	vProductTypes	1.000000	TOTAL PRODUCTS	By Name

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Task 4. Save and execute the process.

1. Click the **Advanced** tab.
2. Click the **Prolog**, **Metadata**, **Data**, and **Epilog** tabs.

   There are four scripts that are run when executing a TI process. There is a tab for each of them here (along with a tab to identify runtime variables) for the given process.

   It is a good practice to review each script and ensure they have been updated, especially if you are modifying an existing process.

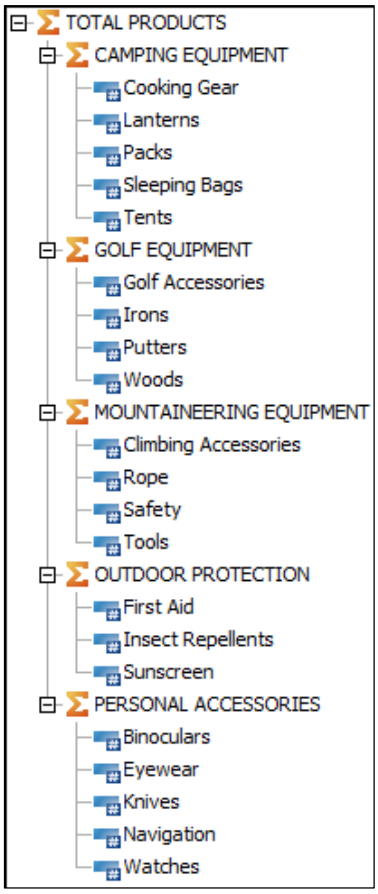
   Clicking each tab will force the scripts on each tab to be updated and compiled.
3. From the **File** menu, click **Save**.
4. In the **Name** box, type **CreateProductsCSV**, and then click **OK**.
5. Click **Run**  .
6. Click **OK** when the process has completed.
7. Close **TurboIntegrator**.

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8. In the **Server Explorer** window, double-click the **Products** dimension.

   The Subset Editor opens with all the elements that were loaded from the CSV file.



   You ordered your elements in the TI process. If you had not initially chosen to order by hierarchy and elements by name, you could reorder them using the Dimension Editor or in the Server Explorer window.

   If you want to order a dimension by hierarchy after running the TI process, in Server Explorer, right-click the dimension name, click **Set Elements Order**, select **Automatic, by Hierarchy**, and then click **OK**. Open the dimension in the Subset Editor and it should now appear sorted by hierarchy.

9. Close the **Subset Editor**.
10. Leave Architect open for the upcoming exercise.

Results:
You created a dimension using TurboIntegrator. You identified the variables, mapped the data, reviewed the scripts, executed the process, and confirmed the resulting Products dimension.

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IBM Training

Unit summary

- Describe cubes and dimensions
- Create dimensions manually
- Edit dimensions
- Create dimensions using TurboIntegrator

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Unit summary

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IBM Training

Exercise 1

- Create a dimension using TurboIntegrator

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Exercise 1: Create a dimension using TurboIntegrator

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Exercise 1: Create a dimension using TurboIntegrator

Analysts have mentioned that they need to be able to analyze sales revenue by distribution channel, or type of store. To satisfy this requirement, create the Channels dimension using TM1.

To accomplish this:

- Add data source connection information, to connect to the channels.csv file in the C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles folder.
- Modify variable names.
- Identify how data is mapped.
- Save the process as CreateChannelsCSV and execute the process.

For more information about where to work and the exercise results, refer to the Tasks and results section that follows. If you need more information to complete a task, refer to earlier demonstrations for detailed steps.

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Exercise 1: Tasks and results

Task 1. Add data source connection information.

- In the **Server Explorer** window, right-click **Processes**, and then click **Create New Process**.
- On the **Data Source** tab, click **Text**.
- Click **Browse**, navigate to **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles**, and then double-click **channels.csv** to open the file.
- Click **OK** to dismiss the warning.
- Ensure that the **Delimiter Type** is **Delimited** and the **Delimiter** is **Comma**.
- Click **Preview**.

  You can see all 8 rows of data in your source.

Task 2. Modify variable names.

- Click the **Variables** tab.
- In the first row, click **V1**, rename it **vChannels**, and then leave the **Variable Type** column as **String**.
- In the **Contents** column, in the **vChannels** row, click the down arrow  , and then click **Element**.
- Click **V2**, and rename it **vTotalChannels**.
- In the **Contents** column, in the **vTotalChannels** row, click the down arrow, and then click **Consolidation**.

Task 3. Identify how data is mapped.

- Click the **Maps** tab, and then click the **Dimensions** sub-tab.
- For the **vChannels** element variable, in the **Dimension** column, type **Channels**.
- In the **Action** column, select **Create** (if necessary), and in the **Element Type** column, ensure that **Numeric** is selected.
- Click **By Input**, select **Automatic**, select **Hierarchy** in the **Automatic Sort By** section, and then click **OK**.

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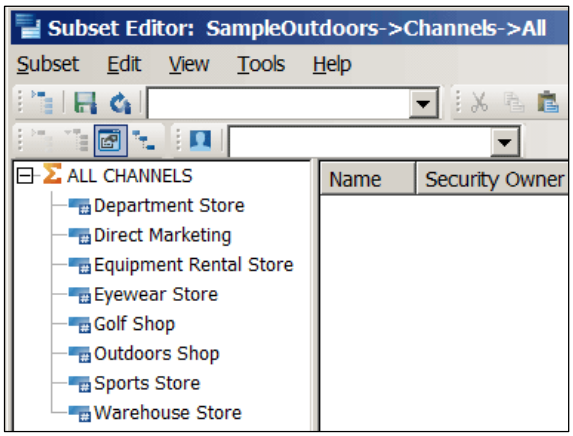
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- Click the **Consolidations** sub-tab.
- For the **vTotalChannels** consolidation variable, ensure that **vChannels** is selected in the **Child Variable** column.
- Click **By Input**, select **Automatic**, select **Name** in the **Automatic Sort By** section, and then click **OK**.

Task 4. Save and execute process.

- Click the **Advanced** tab.
- Click the **Prolog**, **Metadata**, **Data**, and **Epilog** tabs.
- From the **File** menu, click **Save**, in the **Name** box, type **CreateChannelsCSV**, and then click **OK**.
- Click **Run**  , and then click **OK** when the process has completed.
- Close **TurboIntegrator**.
- In the **Server Explorer** window, double-click the **Channels** dimension.

  The Subset Editor opens with the elements that were loaded from the CSV file:



- Close the **Subset Editor**.
- Close all open windows.

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IBM Training

Build cubes and views

IBM Planning Analytics (v2.0)

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IBM Training

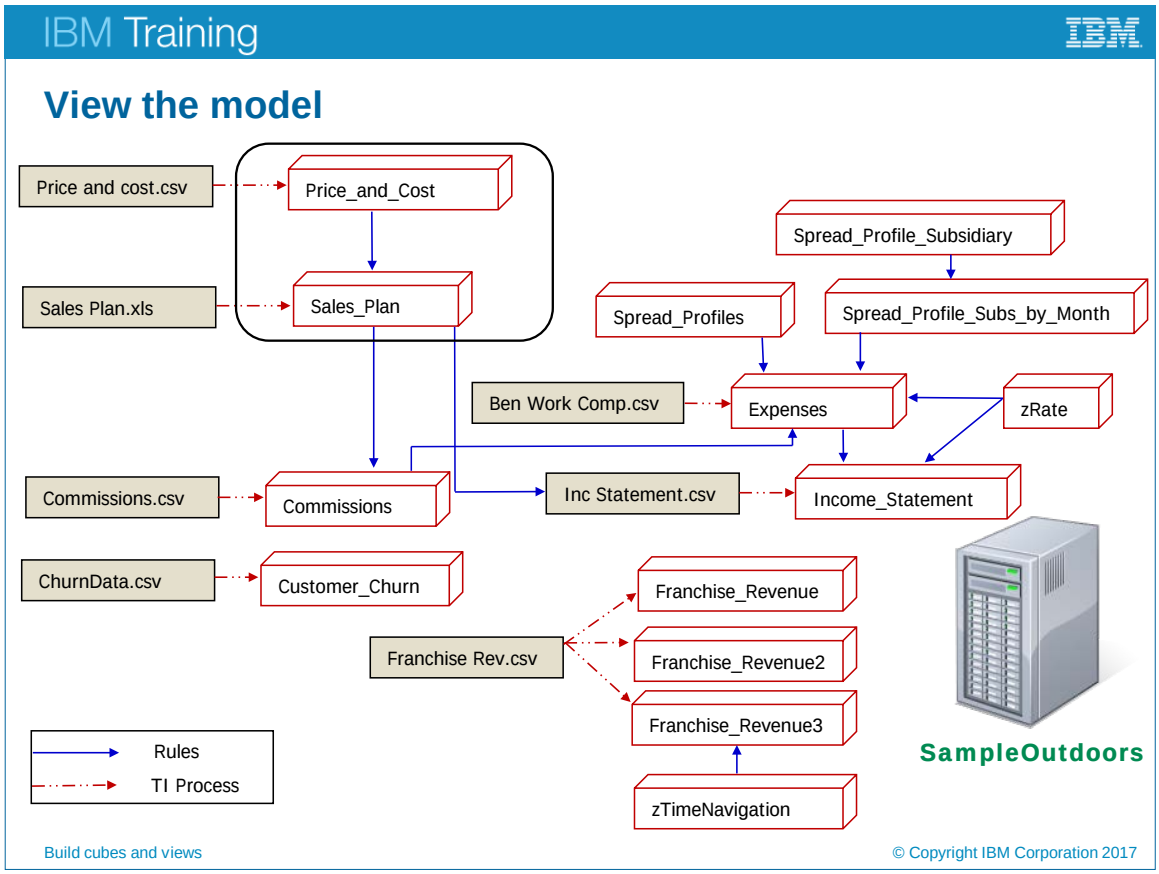
Unit objectives

- Describe cubes and data points
- Create cubes
- Construct views of data in cubes
- Create and use a pick list

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Unit objectives

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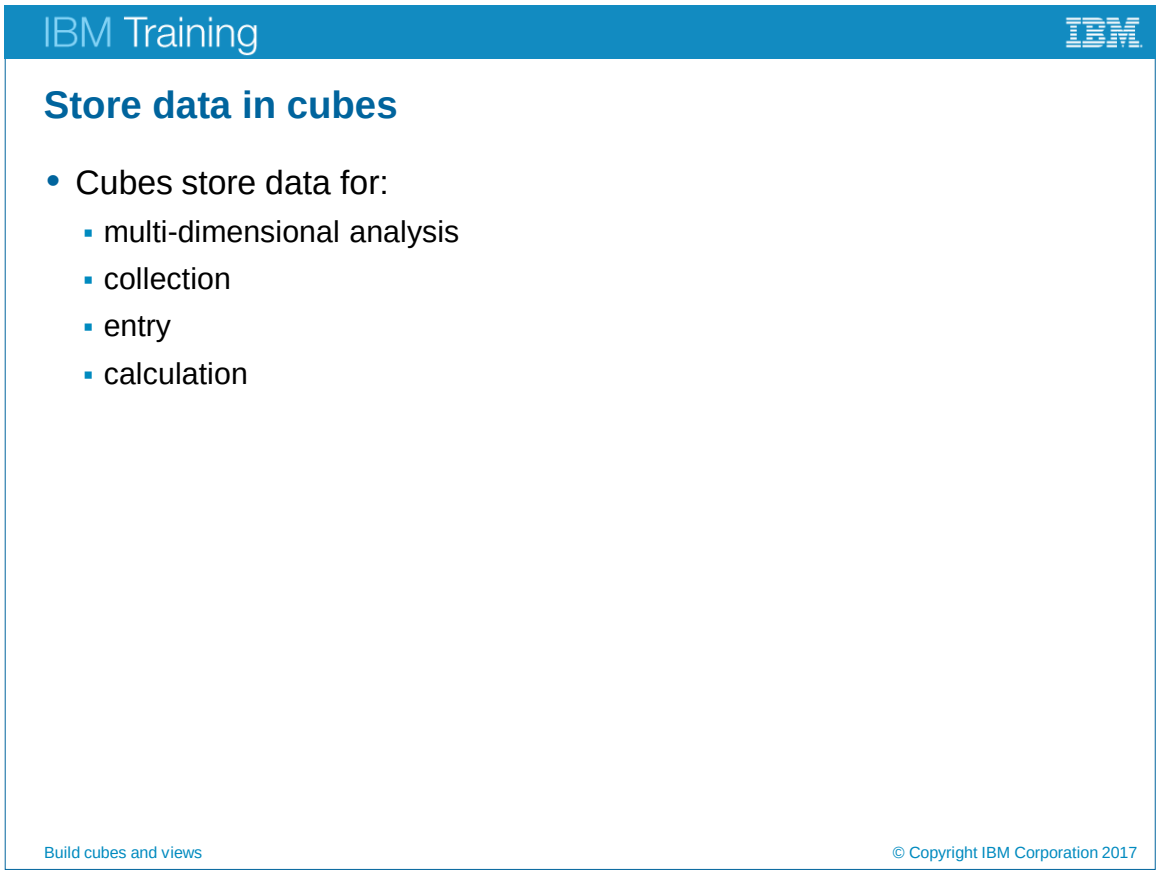


View the model

In this unit, you will create the following cubes:

- Price_and_Cost
- Sales_Plan

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Store data in cubes

You may have any number of cubes in your TM1 model.

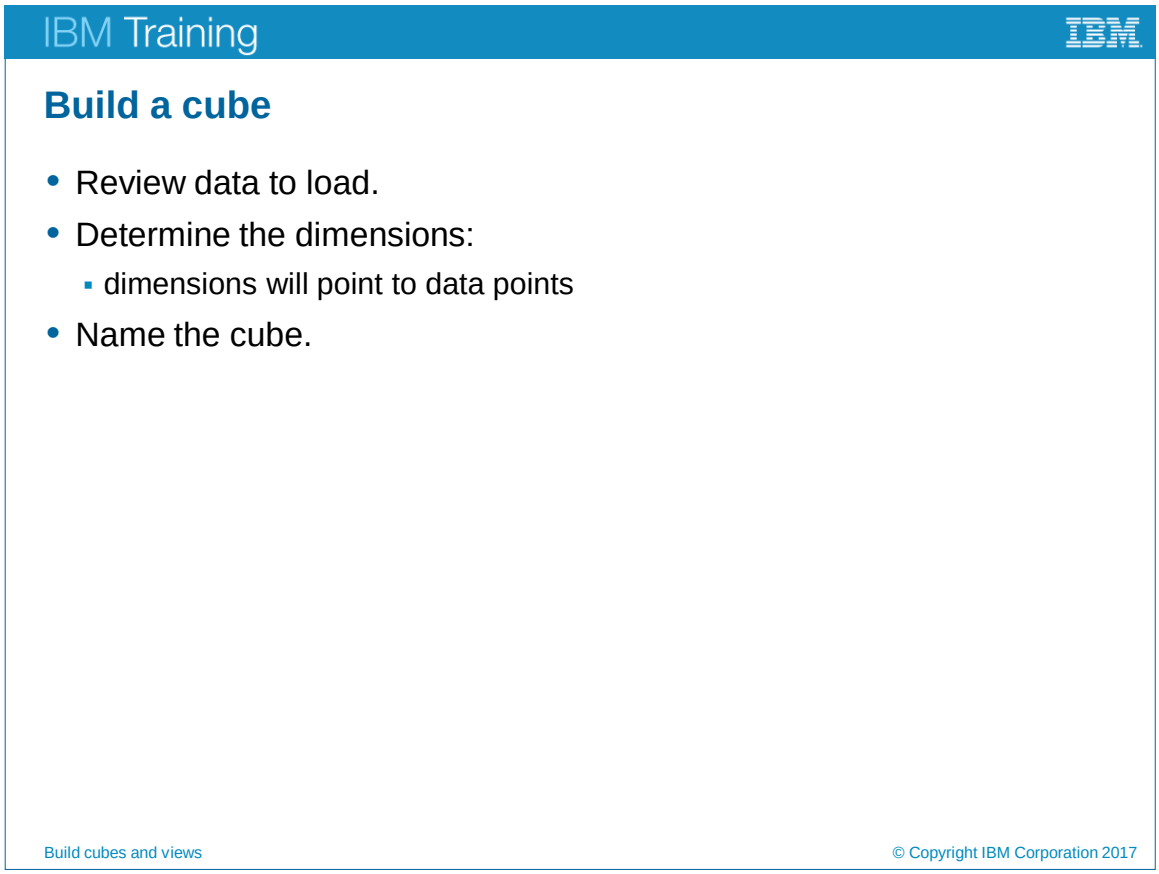
Cubes store data for reports, analysis, or staging to be used in further calculations.

Data may come from a number of sources and be combined in a single cube or in multiple cubes using business logic.

TM1 can be its own 'data mart', pulling data from many sources into a cube for further analysis.

For example, HR has one system, and GL has another. They share a common time dimension and can be combined into a single cube or multiple cubes.

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Build a cube

Cubes are made up of two or more dimensions.

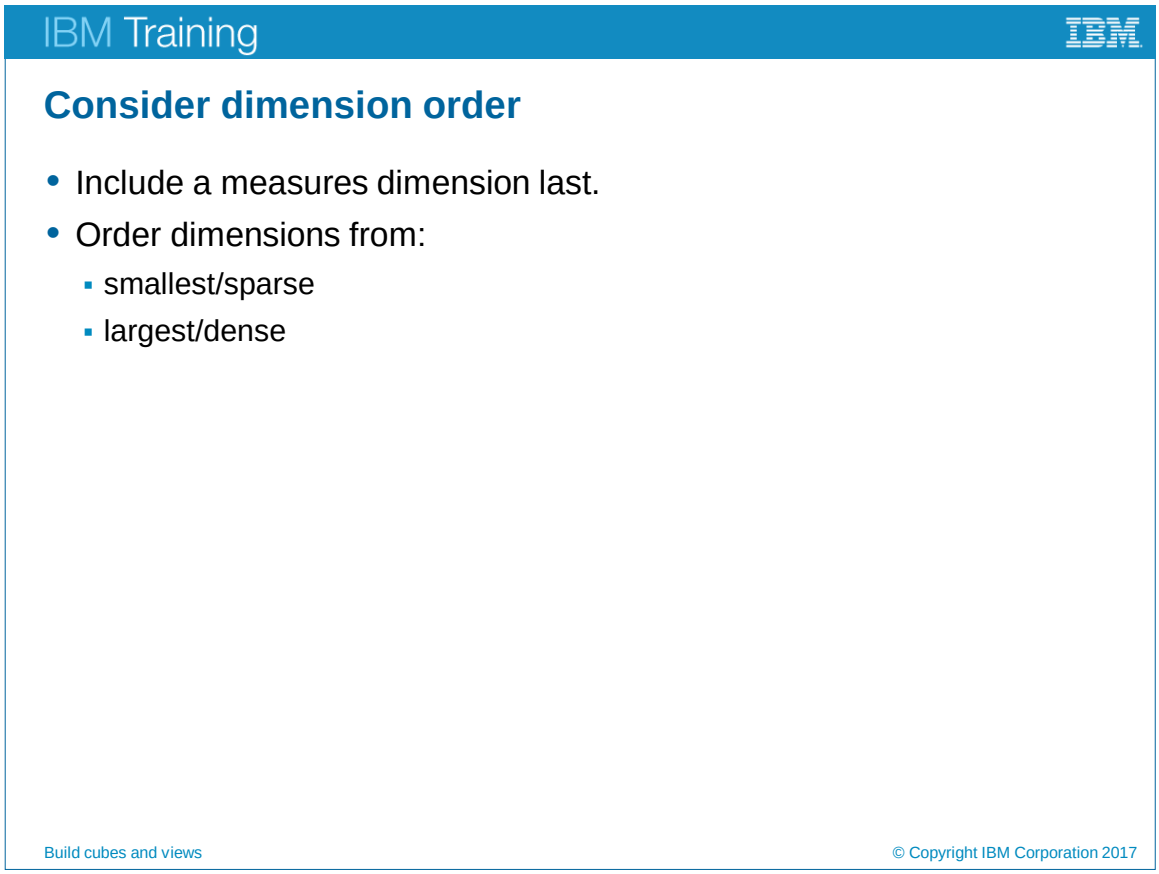
Dimensions are made up of one or more elements. Available element types are consolidations, numeric, or string. Only numeric elements may be consolidated.

One element from each dimension identifies the location of a specific data point in a cube.

TM1 will support spaces and some other characters in object names. However, it is better to use _ instead of spaces and avoid using special characters. This will prevent conflicts with other applications and will make names easier to read.

Establishing a naming convention for your model will make maintenance and expansion easier.

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Consider dimension order

Every cube should have its own measures dimension, even if it only contains a single element (dollar, amount, or value).

The measures dimension should always be the last dimension specified when defining the cube.

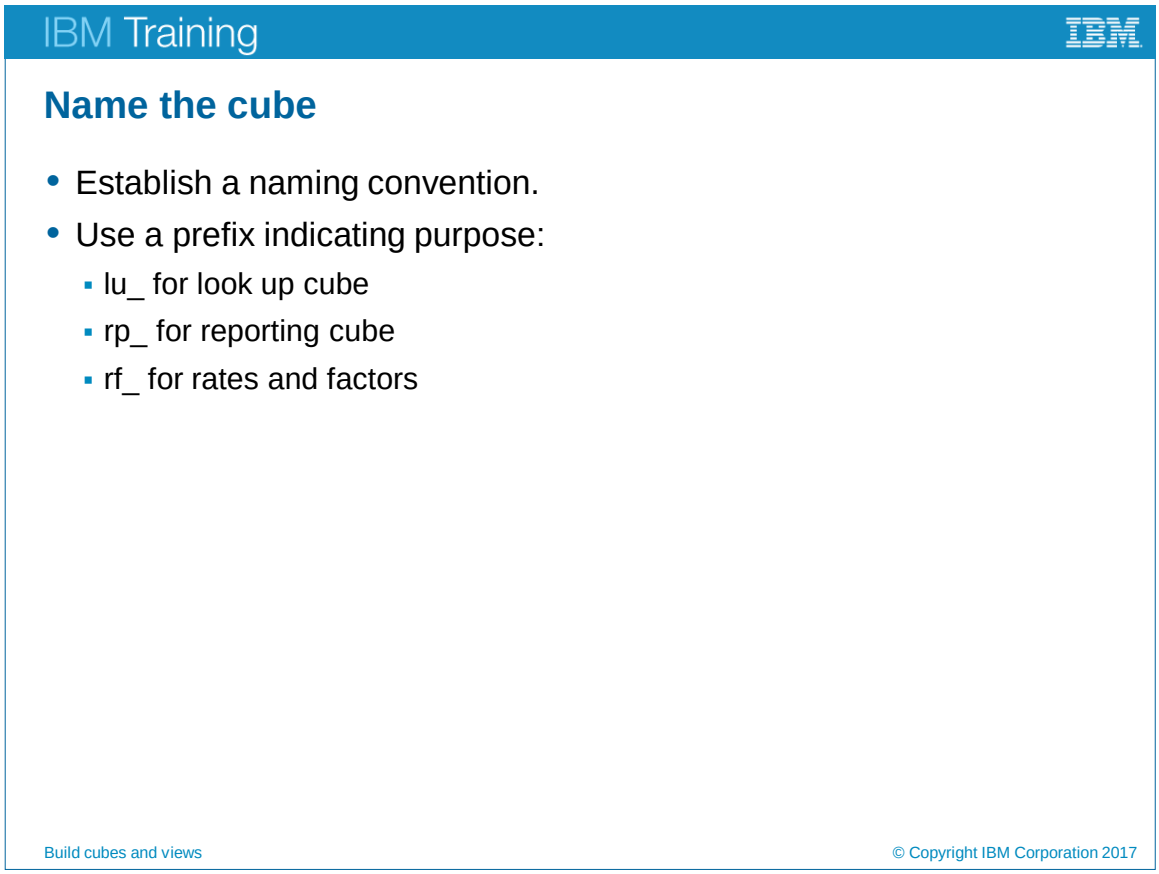
Following these recommendations will:

- make every cube extensible
- make cube data easier to locate
- ensure string measures will always work
- simplify rule and feeder statements

A sparse dimension has a low percentage of available data positions filled (for example, Products, Markets, or Regions). A dense dimension has a high percentage of probability that one or more data points are occupied in every combination of dimensions (such as Version or Scenario).

Sparsity and density are usually estimated, so the statement above is a guideline and not a set rule. After loading data, use the Cube Optimizer to improve calculation speed.

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Name the cube

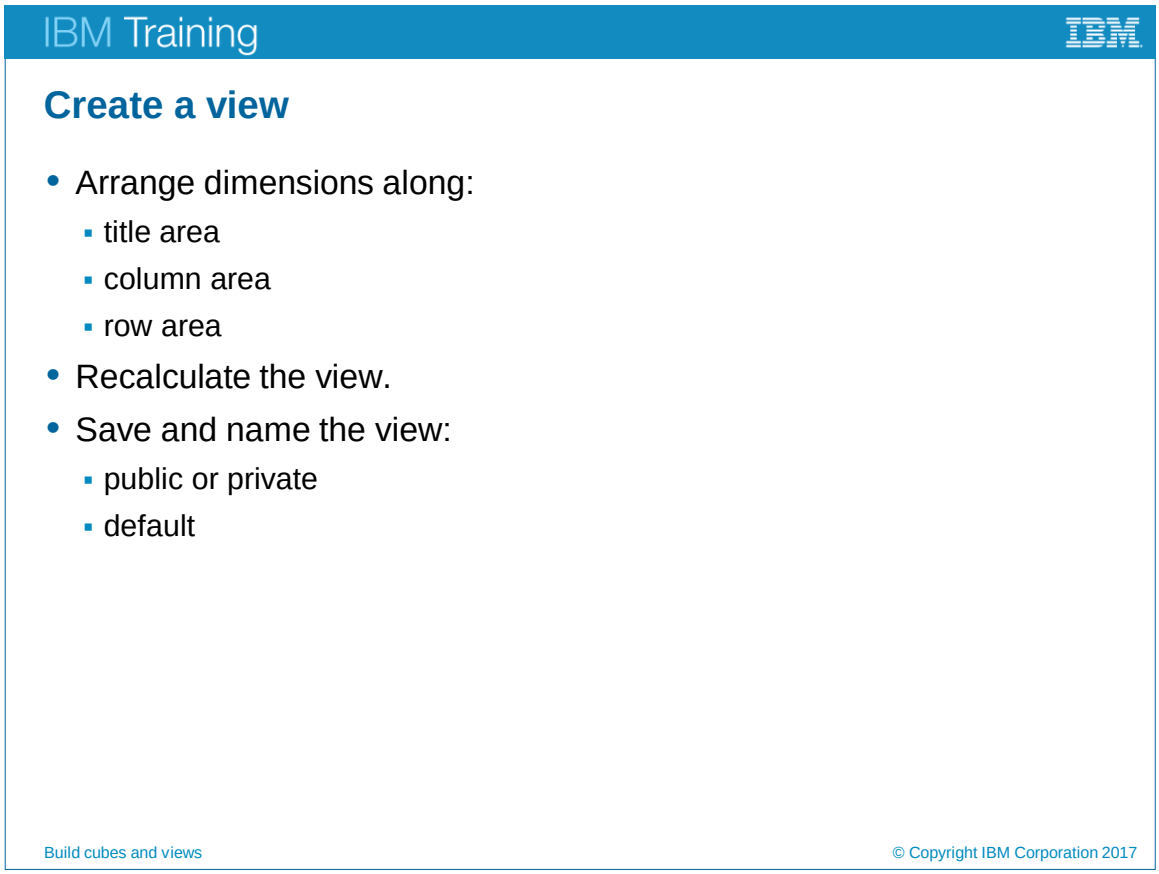
Create a consistent naming standard for cubes.

When naming cubes, consider the following:

- Is it readable (use lowercase, and underscores for spaces)?
- Should it contain a prefix or suffix to identify its purpose?

The Server Explorer window sorts objects in alphabetical order. If cube names have prefixes, similarly named cubes will be sorted together.

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Create a view

A view saves the arrangement of dimensions for a cube.

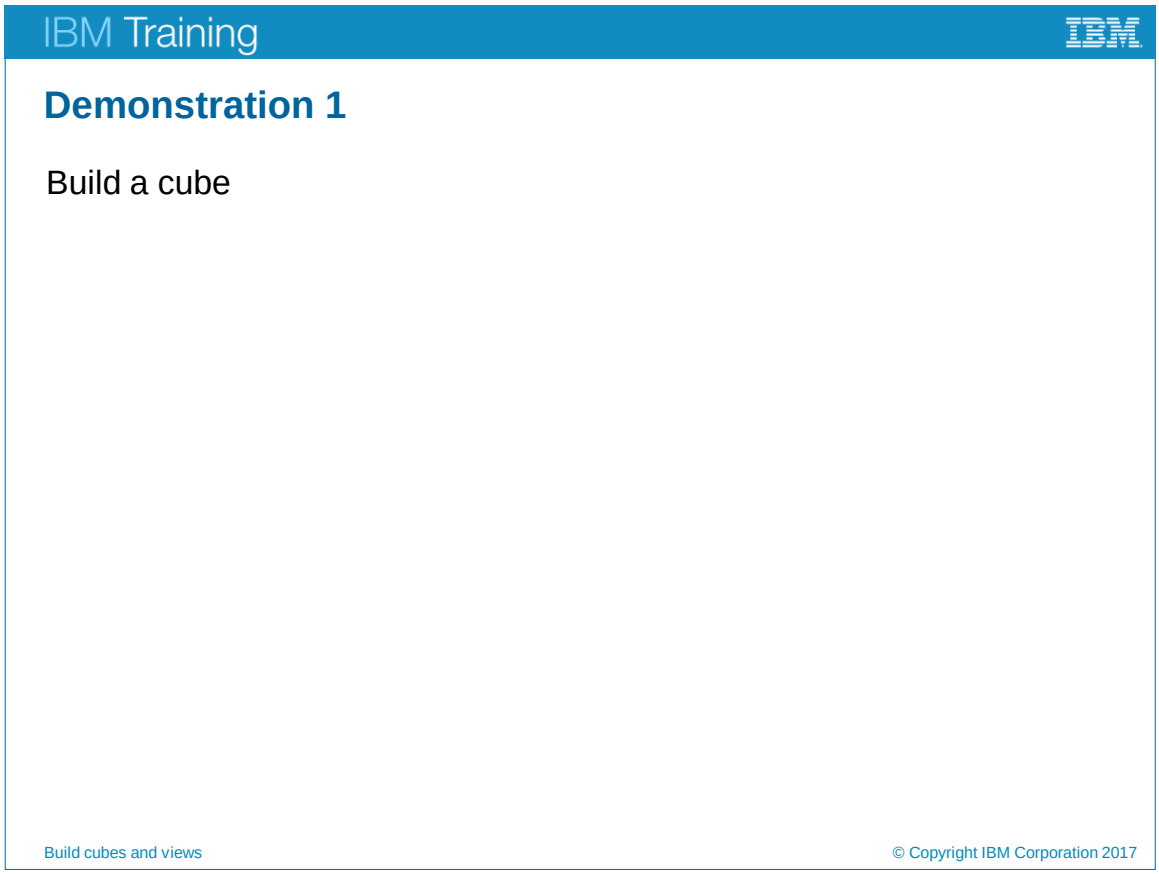
Views of a cube are not calculated by default. After stacking or swapping dimensions click the Recalculate button (or press F9) to see the resulting data.

Only administrators may save public views for everyone to see.

An administrator may also publish a private view for public viewing.

The title area allows the selection of a single element. The row and column areas allow for display of multiple elements.

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Demonstration 1: Build a cube

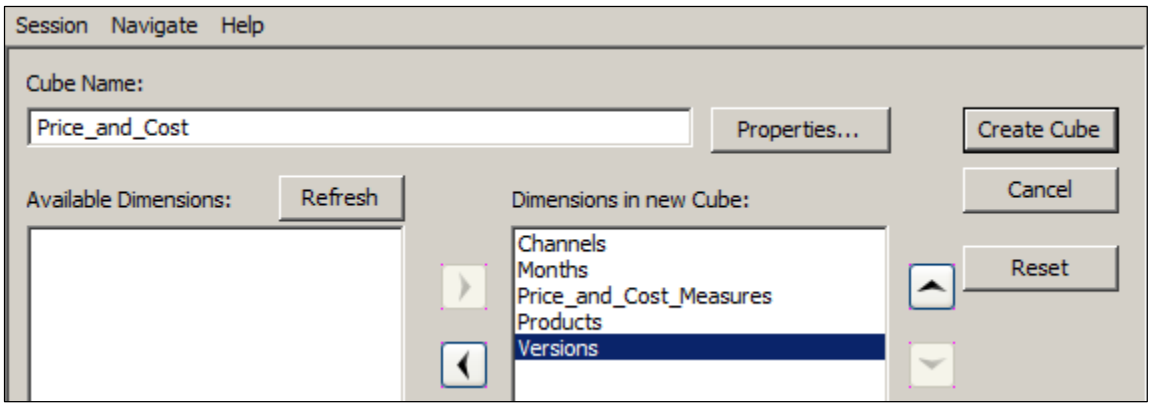
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Demonstration 1: Build a cube.

Purpose:
The Sample Outdoors Company sales plans rely on current price and cost information to forecast profit and margin. The price and cost information will be stored for all products for the two versions of the budget. You need to create a cube to hold this data so it can be referenced by other cubes.

Task 1. Create the Price_and_Cost cube.

1. From the **Start** menu, navigate to **All Programs\IBM Cognos TM1 - 64**, and then click **Architect**.
2. Maximize the window.
3. Expand **TM1**, double-click **SampleOutdoors**, and then log on with a user ID of **admin** and a password of **apple**.
4. Right-click **Cubes** and then click **Create new cube**.
5. In the **Cube Name** box, type **Price_and_Cost**.
6. In the **Available Dimensions** list, double-click all the available dimensions to move them to the **Dimensions in new Cube** list.



You can use the Right and Left arrows to select and deselect dimensions to appear in the cube. You can also use the Up and Down arrows to change the order of the dimensions.

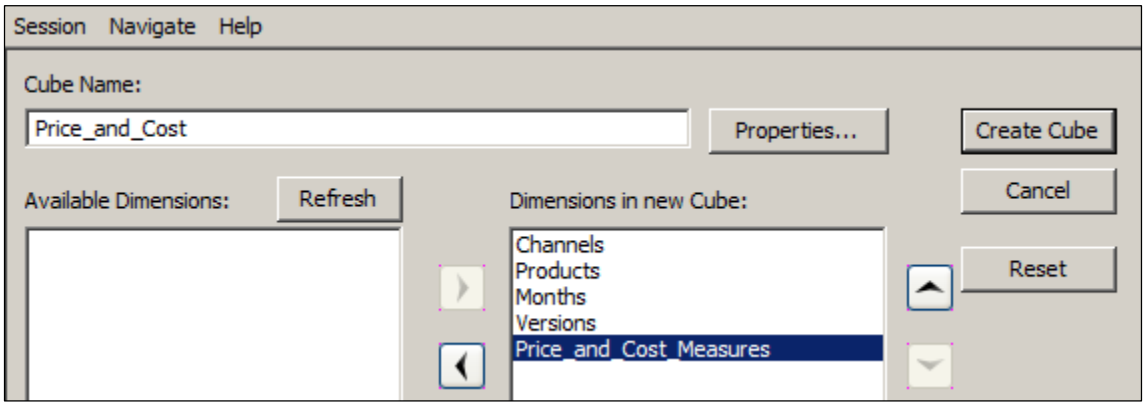
You need to make the measures dimension last. You will also make a guess at ordering the dimensions from smallest/sparse to largest/dense.

7. Click **Products** and then click the **Up** button twice to move **Products** below **Channels**.

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8. Click **Price_and_Cost_Measures** and then click the **Down** button to move **Price_and_Cost_Measures** to the bottom.

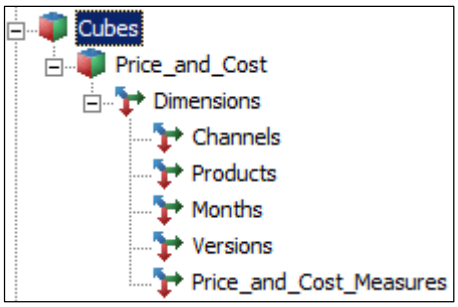
   The results appear as follows:



9. Click **Create Cube**.

10. Under **Cubes**, expand **Price_and_Cost**, and then expand **Dimensions**.

    The results appear as follows:



Once a cube has been created, you may NOT add, delete or rename dimensions. You must delete and recreate the cube if you wish to change the dimensions.

You can use the Cube Optimizer after loading data into a cube to help determine the most efficient dimension order. You right-click the cube name and click Optimize. The lowest percentage is the most efficient. This will reorganize the dimensions in TM1 but it will not physically reorder them (so functions and rules are not impacted).

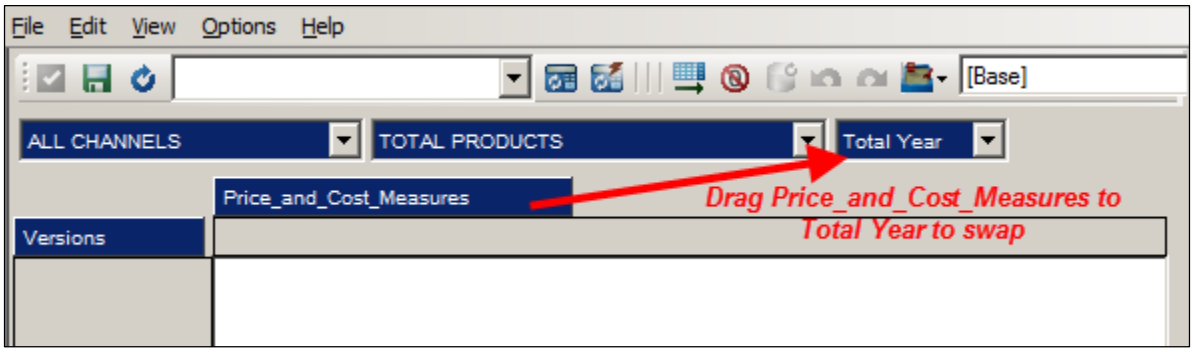
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Task 2. Arrange the dimensions in the Price_and_Cost cube.

1. Double-click the **Price_and_Cost** cube.

   The Cube Viewer opens, displaying the dimensions in the title, row and column areas.

2. Drag **Price_and_Cost_Measures** on top of the **Months** dimension (displaying Total Year) in the title area to swap them.



   The Price_and_Cost_Measures dimension should display the Unit Sale Price element in the drop down list.

3. Click the **Months** dimension to open the Subset Editor.
4. Click **All**  .
5. Click **Filter by Level**  , select **0**, and then click **OK**.

   Now only the lowest level elements should appear.

6. From the **Subset** menu, click **Save As**.
7. In the **Select or Enter Subset Name** box, type **Months_Only**, clear the **Private** check box, and then click **OK**.

   This saves a public subset named Months_Only that can be used in other cubes and views.

8. Click **OK** to close the **Subset Editor**.
9. Click and drag the **Versions** dimension onto the **Products** dimension (currently showing the TOTAL PRODUCTS element) to swap them.
10. Click **Recalculate**  .

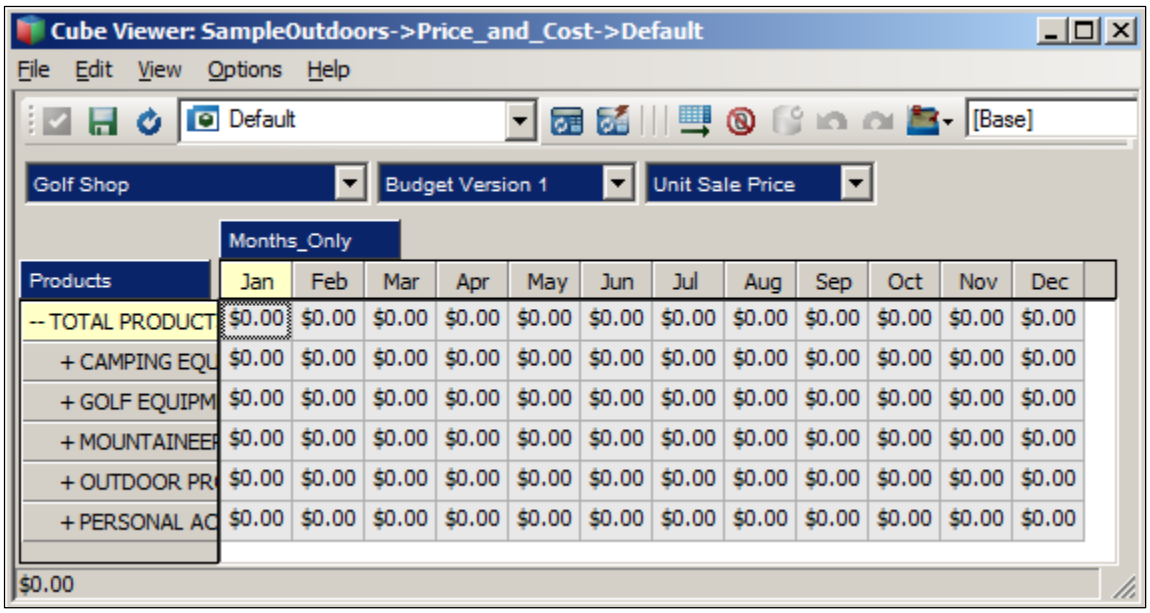
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Task 3. Save the default view.

1. Click the **down** arrow next to **ALL CHANNELS**, and then click **Golf Shop**.
2. Click **Recalculate**.
3. Click **File** and then **Save As**.
4. Select **Default**.

   This will disable the Select or Enter Named View list because TM1 will name this view Default. The Default view is automatically opened when you double-click on the cube name.
5. Clear the **Private** check box and click **OK**.

   The results appear as follows:



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6. Close the **Cube Viewer**.

   You may create both a public and a private default view. When administrators create a public, default view, it is available to all users. Users may also create their own private, default view that will override the public one.

   By default, the Cube Viewer displays a view with the last dimension in the column area, the second to last dimension in the row area and all the other dimensions in the title area.

   The Subset Editor will return either the selected element or all displayed elements (if none are selected in the display). Be sure not to click on any one element in the display if you wish to return them all.

   Data for this cube will be loaded later in the course.

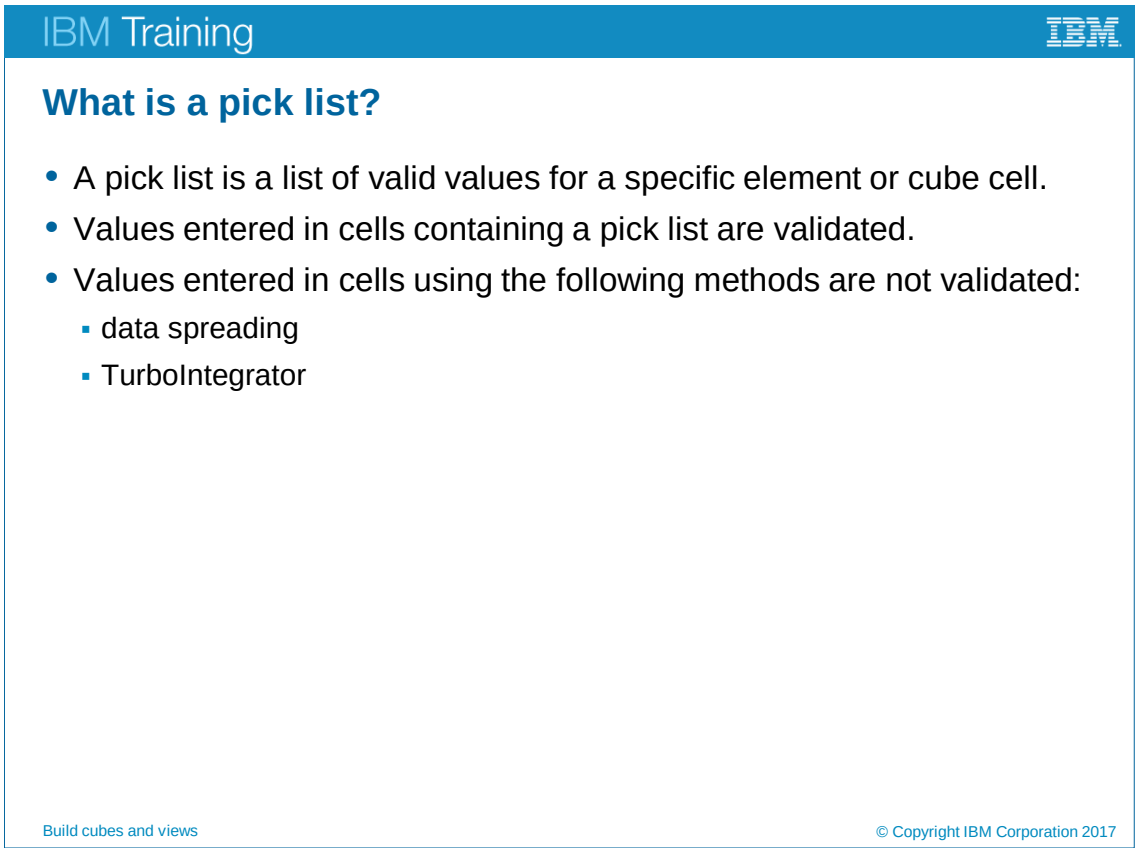
   You may also want to right-click the cube name in the Server Explorer and select Properties. You can specify which dimension represents the time dimension and which dimension represents the measures dimension. This has no impact on TM1, but if your cubes will be sourcing other applications (such as Framework Manager), they may require this setting.

7. Leave Architect open for the next demonstration.

Results:
The Price_and_Cost cube is now available to store product price and cost information. It contains a default view for all users to see. You are now ready to load data from an external data source, and use it to forecast margin ratios.

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What is a pick list?

When an administrator defines a pick list for an element or a cell, a drop-down menu containing the defined values is available for data entry into the specified cell when browsing a cube in any of the TM1 clients.

The data format of the pick list must match the data type of the element for which it is defined. If the list contains strings, it can only be used for a string element. If it is applied to a string element, it must reside in the last dimension of the cube.

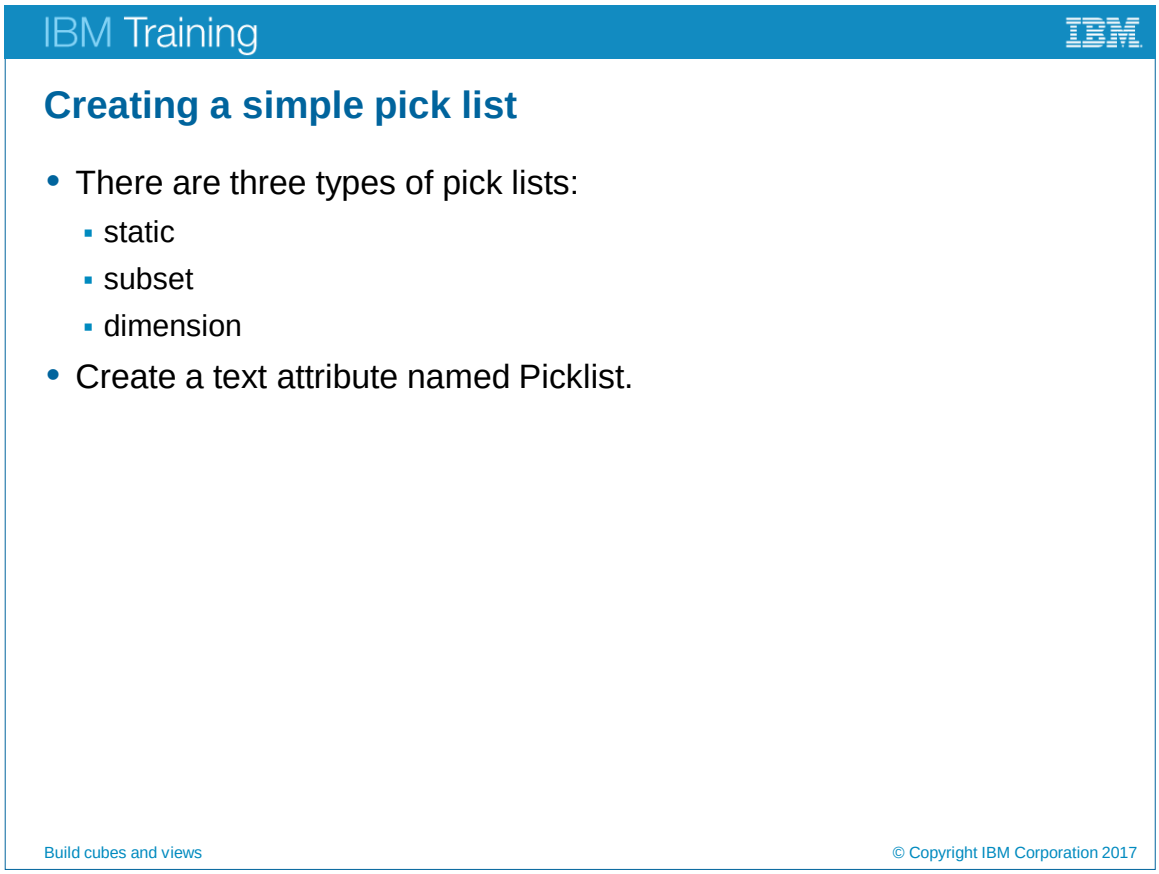
Values in cells containing a pick list are validated, meaning a user must select one of the predefined values for the cell. If a user attempts to enter a value that is not valid for the cell, an error appears indicating that only values from the pick list can be entered in the cell.

It is a good idea to apply pick lists to the measures dimension (especially if it is a string).

Pick lists are available to end users when they browse a cube containing the dimension with the Picklist attribute.

Data entered via TI, spreading, or after the pick list is defined will not be 'validated' by the pick list. This means it is possible to see values in cells that do not match the ones in the pick list.

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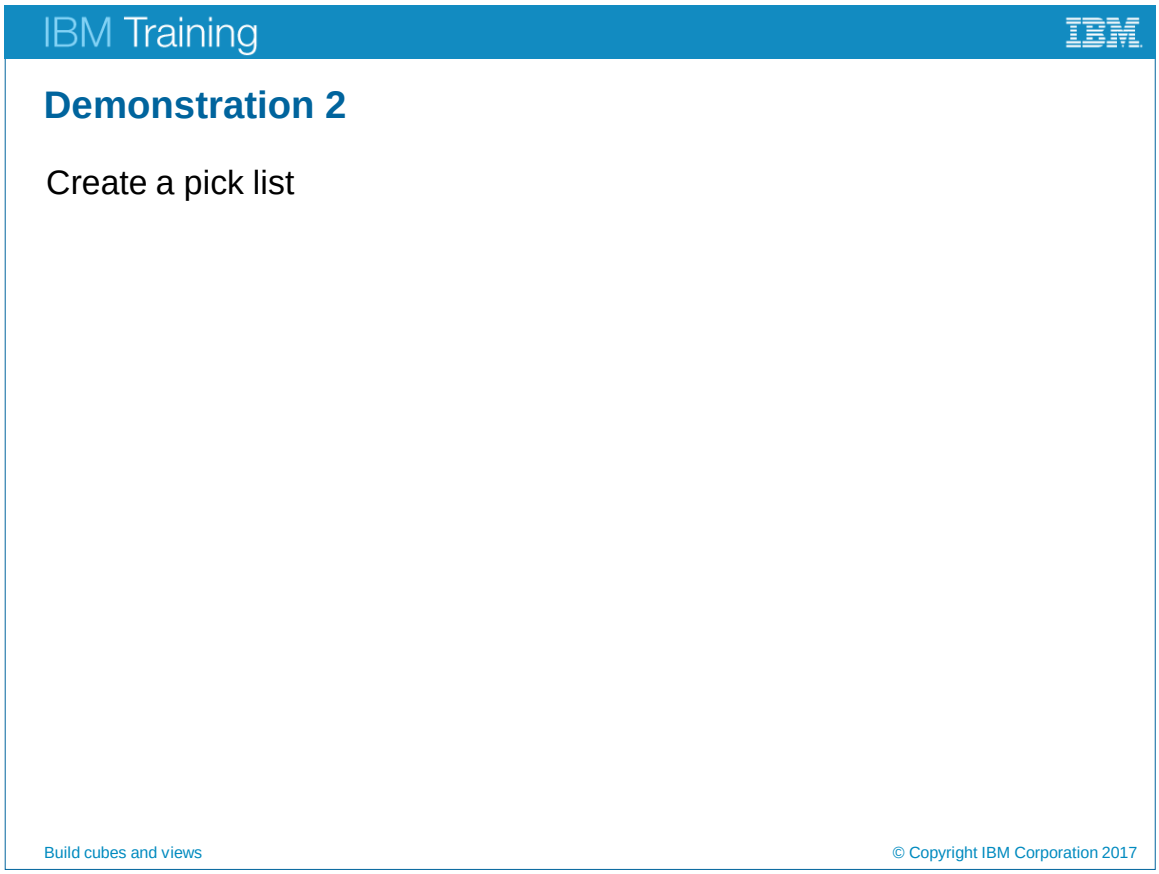
Creating a simple pick list

There are two ways to create a pick list. A simple pick list is created by creating a new attribute called picklist. A simple picklist will appear in every cube containing that dimension. Another option is to use a rule to create an advanced pick list. A rule may apply to specific elements, or can use logic to determine when the pick list should appear.

A static pick list is composed of a colon-delimited list of values using the syntax, **static:value1:value2:value3:value4**.

A subset pick list contains values corresponding to all elements of a named subset. If the members of the subset change, the values available in the pick list change.

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Demonstration 2: Create a pick list

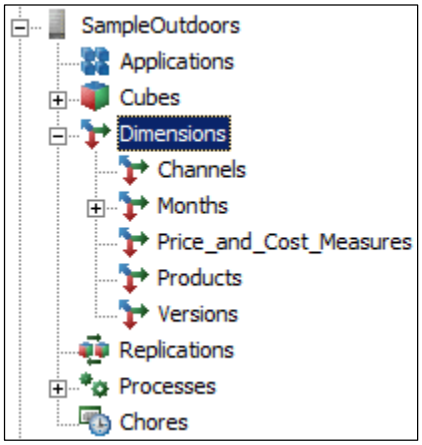
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Demonstration 2: Create a pick list.

Purpose:
Create a pick list for the Price_and_Cost cube so that users can enter a comment indicating the price level for different products. Users have asked to be able to select standard comments so you will create a pick list of options they can choose from.

Task 1. Create a string element.

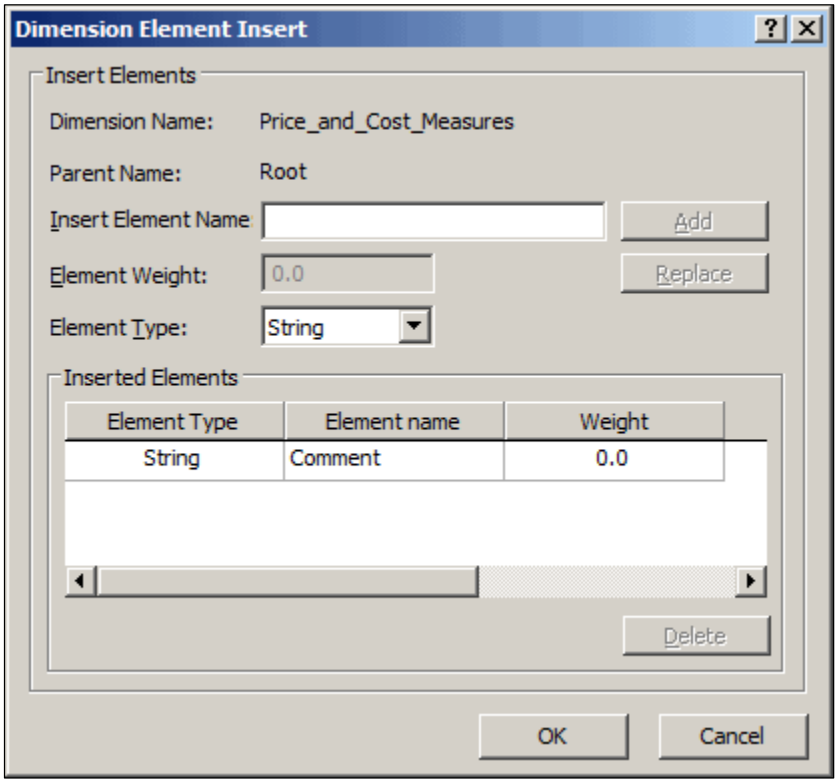
1. In the Server Explorer window, under **SampleOutdoors**, expand **Dimensions**. The results appear as follows:



2. Right-click **Price_and_Cost_Measures**, and then click **Edit Dimension Structure**.
3. Click the last element in the list (**Margin%**).
4. From the **Edit** menu, click **Insert Element**.
5. In the **Insert Element Name** box, type **Comment**.
6. In the **Element Type** list, select **String**.

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7. Click **Add**.
   The results appear as follows:



8. Click **OK**.
9. Click **OK** to close the **Dimension Editor** window and then click **Yes** to save changes to the dimension.

Task 2. Create a static pick list.

1. Right-click **Price_and_Cost_Measures**, and then click **Edit Element Attributes**.
2. From the **Edit** menu, click **Add New Attribute**.
3. In the **Name** box, type **Picklist**, ensure **Text** is selected, and then click **OK**.

	Format (Text)	Picklist (Text)
Unit Sale Price	b:\$#,###0.00;(\$#,###.00)ŞC|2|Y|$	
Unit Cost	b:\$#,###0.00;(\$#,###.00)ŞC|2|Y|$	
Margin	b:\$#,###0.00;(\$#,###.00)ŞC|2|Y|$	
Margin %	b:#,##0.00;(#,##0.00)ŞCO|2|Y	
Comment		

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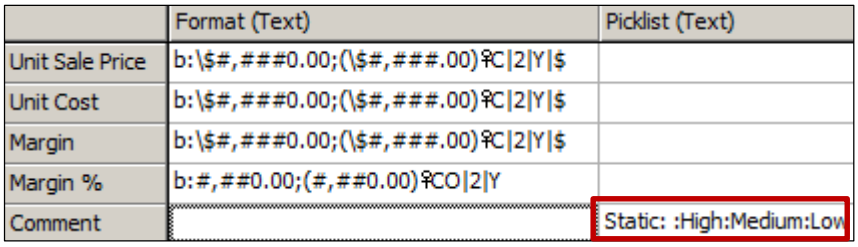
4. In the cell next to **Comment**, under **Picklist** type the following:

   Static: :High:Medium:Low

   There is a space in before **:High**. This defines a blank string at the top of the picklist, before the **High** text string.

   The syntax to define a subset pick list is:

   static: value1: value2:…valuen



	Format (Text)	Picklist (Text)
Unit Sale Price	b:\$#,###0.00;(\$#,###.00)%C|2|Y|$	
Unit Cost	b:\$#,###0.00;(\$#,###.00)%C|2|Y|$	
Margin	b:\$#,###0.00;(\$#,###.00)%C|2|Y|$	
Margin %	b:#,##0.00;(#,##0.00)%CO|2|Y	
Comment		Static: :High:Medium:Low

5. Click **OK**.

   The pick list contains a list of text strings ("High," "Medium," and "Low"). You created a string element to hold the text once it is selected from the pick list.

   String elements must reside in the last dimension in a cube. If the string element is not defined in the last dimension, it is ignored by the cube. If you expand the list of dimensions under the Price_and_Cost cube, the Price_and_Cost_Measures dimension appears last.

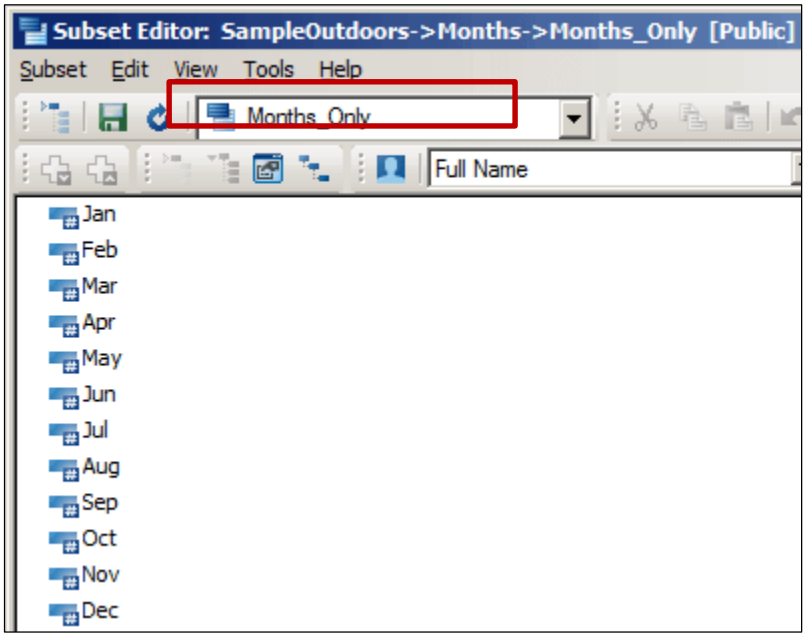
Task 3. Create a view to use the pick list.

1. In Server Explorer, expand the **Cubes** node (if necessary) and double-click the **Price_and_Cost** cube to open it in the **Cube Viewer**.
2. Double-click the **Price_and_Cost_Measures** dimension (currently displaying the **Unit Sale Price** element) to open this dimension in the Subset Editor.
3. Click **All**, and then click **OK** to close the **Subset Editor**.
4. In the **Price_and_Cost_Measures** drop down list, click **Comment** and then click **Recalculate**.

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5. Click **Months_Only**.

   The Months dimension currently has the **Months_Only** subset selected:



   If you want to show all the elements in this dimension, you would click the **All** button, click **OK**, and then click **Recalculate**.

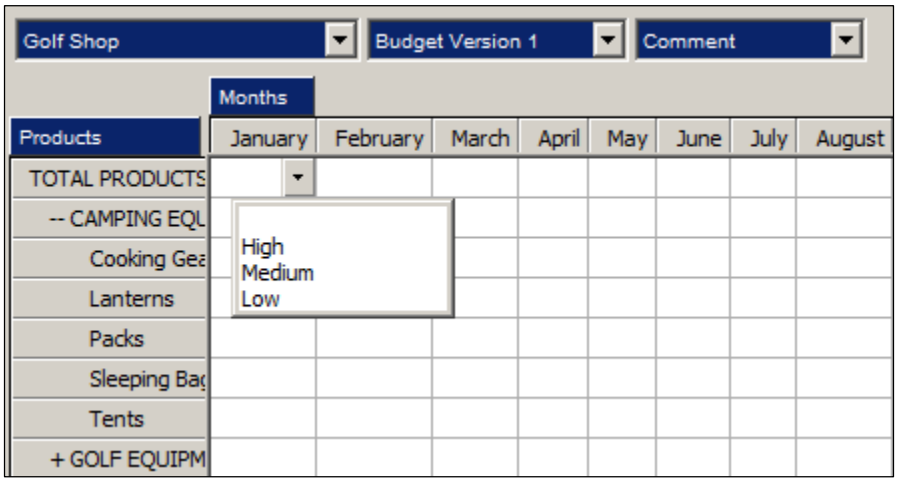
6. Click **Use Aliases(currently off)**  (be sure the Select Alias box displays **Full Name**), and then click **OK**.
7. Click **Recalculate**.

   The columns area no longer shows the Months_Only subset because this subset does not use aliases. Instead, the columns area shows the full names of the twelve months due to the Full Name alias being applied. You could save this selection of elements as another subset of the Months dimension.
8. From the **File** menu, click **Save As**.
9. Ensure that **Private** is selected, in the **Select or Enter Named View** box, type **Price Comment** and then click **OK**.

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10. Expand **TOTAL PRODUCTS** and **CAMPING EQUIPMENT** and then click in the first cell under **January**.

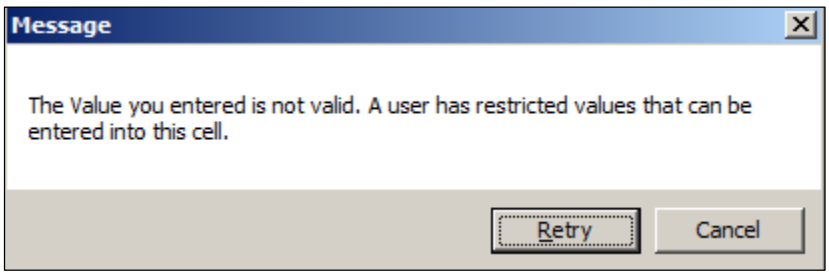
    The results appear as follows:



    A drop-down list appears in the cell. Click on any of the selections to populate the cell.

11. Click the cell again, type **Jan**, and then press **Enter**.

    An error is generated because the value typed did not match any of the values from the pick list. The pick list validates items that are entered into a cell containing a pick list.



12. Click **Retry**, click the pick list again and then select **High** from the list.

    The pick list was defined for the Comment element in the Price_and_Cost_Measures dimension. This is the only element that will generate the picklist.

13. Close the **Price Comment** view without saving changes.
14. Leave Architect open for the upcoming exercise.

Results:
You have created a pick list to enter the price level for different products. You created a string element in which to display the result of the pick list.

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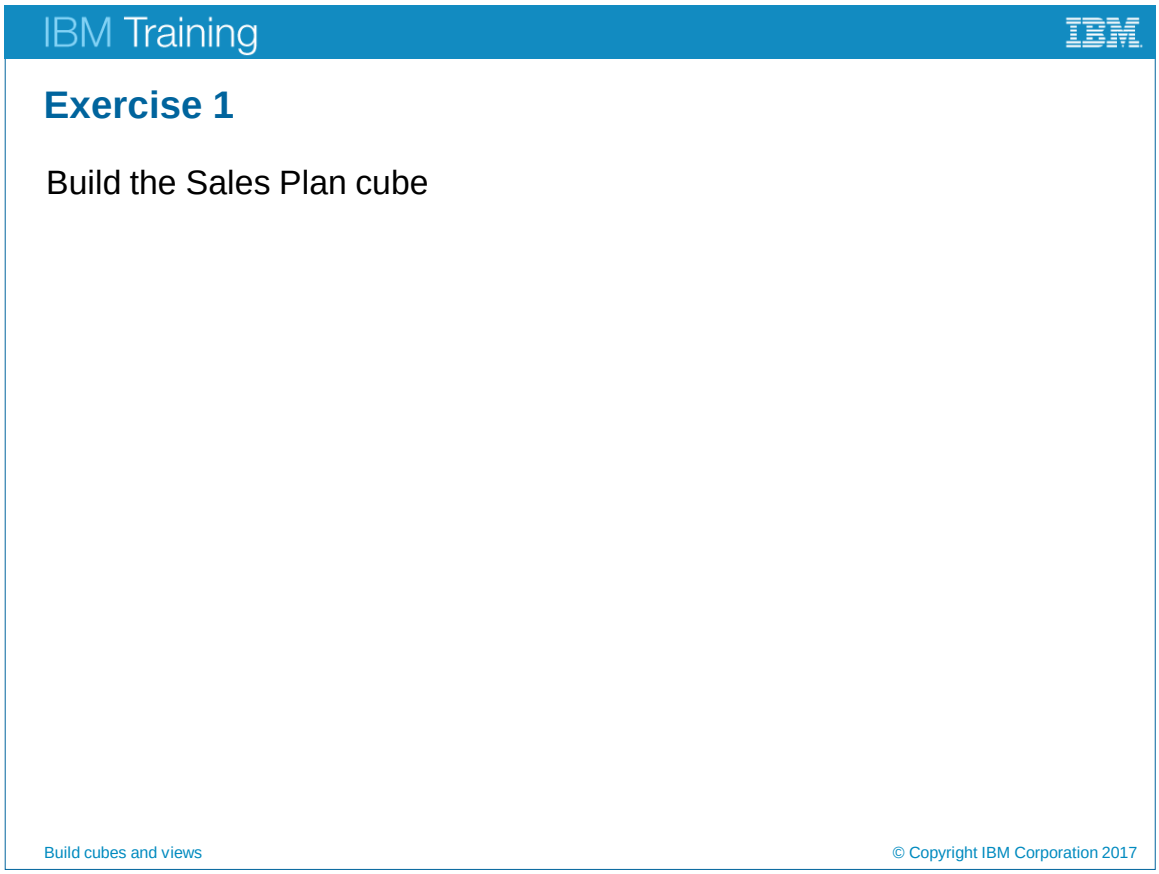
Unit summary

- Describe cubes and data points
- Create cubes
- Construct views of data in cubes
- Create and use a pick list

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Unit summary

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Exercise 1: Build the Sales Plan cube

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Exercise 1: Build the Sales Plan cube

To forecast future sales based on past performance, the Sample Outdoors model requires a cube that will store sales data based on several dimensions. Currently, Sample Outdoors products are sold through several channels, in several worldwide subsidiaries. The cube must present sales over time for two separate versions of the budget, and should include details about unit cost, price, and quantity sold. Users would also like to add some information about the peak month for each product.

To accomplish this:

- Create two new dimensions using the comma-delimited text files provided.
  - One contains information on subsidiaries.
  - One contains sales metrics or measures such as Quantity and Unit Sale Price.
- Create the Sales_Plan cube so it includes subsidiaries, channels, products, months, versions, and Sales_Plan_Measures.
- Re-orient the cube so that Sales_Plan metrics are on rows, periods are on the columns, and the title shows data for lanterns, all channels, and all subsidiaries. Save a default view.
- Create a subset pick list for the Sales_Plan_Measures that identifies the peak month for each product using the Months_Only subset for the Months dimension. Format values as necessary.

For more information about where to work and the exercise results, refer to the Tasks and results section that follows. If you need more information to complete a task, refer to earlier demonstrations for detailed steps.

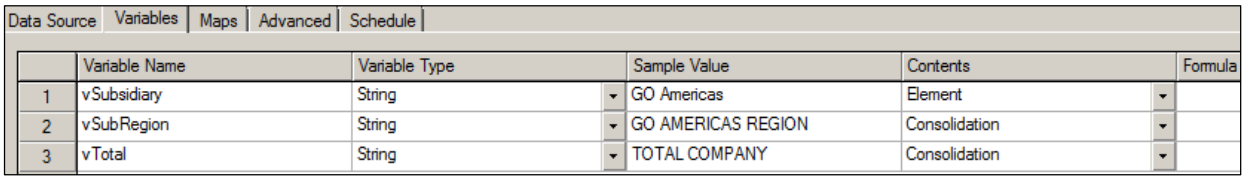
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Exercise 1: Tasks and results

Task 1. Create the Subsidiaries dimension.

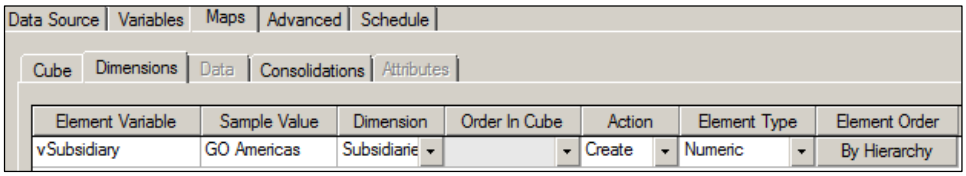
- In the **Server Explorer** window, under **SampleOutdoors**, right-click **Processes**, click **Create New Process**, and then click **Text**.
- Click **Browse**, navigate to **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles** and then double-click **subsidiaries.csv** to select it.
- Click **OK** to dismiss the warning, click **Preview**, and then click the **Variables** tab.
- Click **V1**, type **vSubsidiary**, and then under **Contents**, click **Element**.
- Click **V2**, type **vSubRegion**, and then under **Contents**, click **Consolidation**.
- Click **V3**, type **vTotal**, and then under **Contents**, click **Consolidation**.

  The results appear as follows:



- Click the **Maps** tab, and then click the **Dimensions** sub-tab.
- In the **Dimension** box, type **Subsidiaries**.
- Click **By Input**, select **Automatic** and **Hierarchy**, and then click **OK**.

  The results appear as follows:

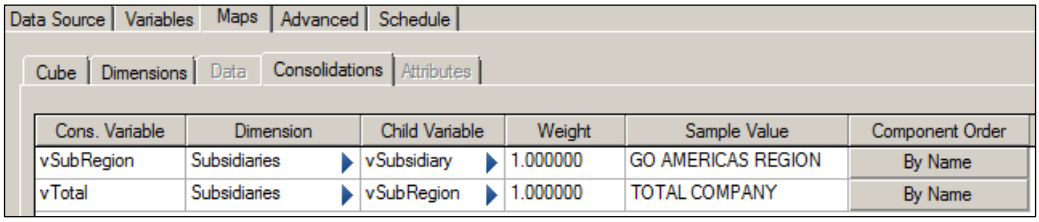


- Click the **Consolidations** sub-tab, and then in the **vSubRegion** row, click the arrow under **Child Variable**.
- Click **vSubsidiary**, and then click **OK**.
- In the **vTotal** row, click the arrow under **Child Variable**, click **vSubRegion**, and then click **OK**.

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- For each variable, click **By Input**, select **Automatic** and **Name**, and then click **OK**.

  The results appear as follows (if the TurboIntegrator window is minimized to the Windows taskbar, maximize it again):



- Click the **Advanced** tab, and then click the **Prolog**, **Metadata**, **Data** and **Epilog** tabs.
- Save the process with the name **CreateSubsidiariesDimCSV** and then click **OK**.
- Click **Run**, click **OK** to dismiss the confirmation message, and then close **TurboIntegrator**.

Task 2. Create the Sales_Plan_Measures dimension.

- In **Server Explorer**, right-click **Dimensions** then select **Create New Dimension**.

  Since this dimension is flat, it is easier to create it manually.

- Open **Windows Explorer**, navigate to **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles**, and then open the **sales _metrics.csv** file in Microsoft Excel.
- Select cells **A1-A13**, and then press **Ctrl+C** to copy.
- In the **Dimension Editor**, click **Edit**, and then click **Paste**.
- Close **Microsoft Excel** without saving.
- In the **Dimension Editor**, click **GROSS MARGIN %**, and then from the **Edit** menu, click **Insert Element**.
- Name the element **Peak Month**, select **String**, click **Add**, and then click **OK**.
- From the **Dimension** menu, click **Save As**.
- In the **Name** box, type **Sales_Plan_Measures**, and then click **OK**.
- Click **OK** again to close the **Dimension Editor**.

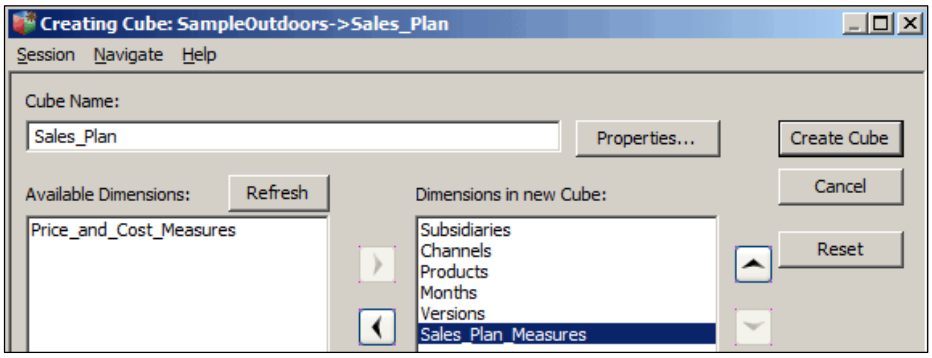
  Sales_Plan_Measures should appear under the Dimensions icon.

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Task 3. Create the Sales_Plan cube.

- In the **Server Explorer** window, right-click **Cubes** and then click **Create new cube**.
- In the **Cube Name** box, type **Sales_Plan**.
- Double-click each of the following dimensions in the **Available Dimensions** list:
  - **Subsidiaries**
  - **Channels**
  - **Products**
  - **Months**
  - **Versions**
  - **Sales_Plan_Measures**

  Use the Up and Down arrows to arrange the dimensions in this order if needed. The results appear as follows:



- Click **Create Cube**.

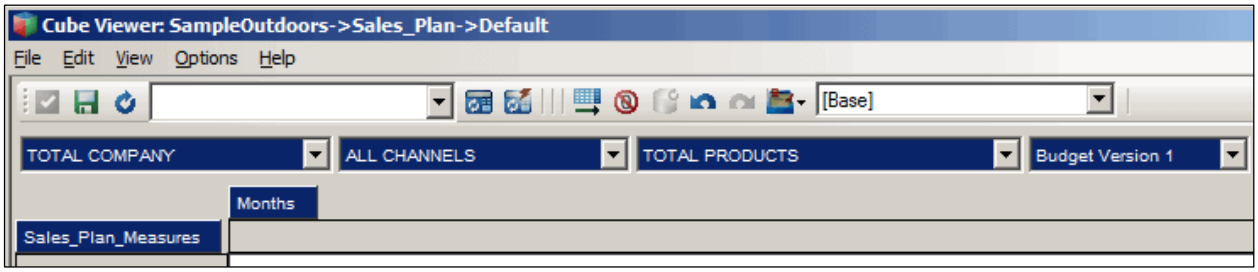
  The Sales_Plan cube will appear below the Cubes icon.

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Task 4. Create a default view of the Sales_Plan cube.

- In **Server Explorer**, double-click the **Sales_Plan** cube.
- Click **Sales_Plan_Measures** (in the column area) and drag it on to **Versions** (in the row area) to swap dimensions.
- Click **Versions** (in the column area) and drag it on top of the **Months** dimension (in the title area, currently displaying the Total Year element) to swap them.

  The results appear as follows:



- Click **Months** (in the column area) to open the **Subset Editor**.
- Click the **down** arrow beside the **Select Subset** list at the top and then click **Months_Only**.

  You should now see a list of months.
- Click **OK** to close the Subset Editor.
- Click **Recalculate**  .

  The months should appear across the columns in the view. If not, repeat the previous steps and do not click on any single month before clicking OK.
- Double-click **TOTAL PRODUCTS** (in the title area) to open the Subset Editor for the Products dimension.
- Expand **CAMPING EQUIPMENT**, click **Lanterns**, and then click **OK**.
- Click **Recalculate**.
- From the **File** menu, click **Save As**.
- Select **Default** and then click **OK**.

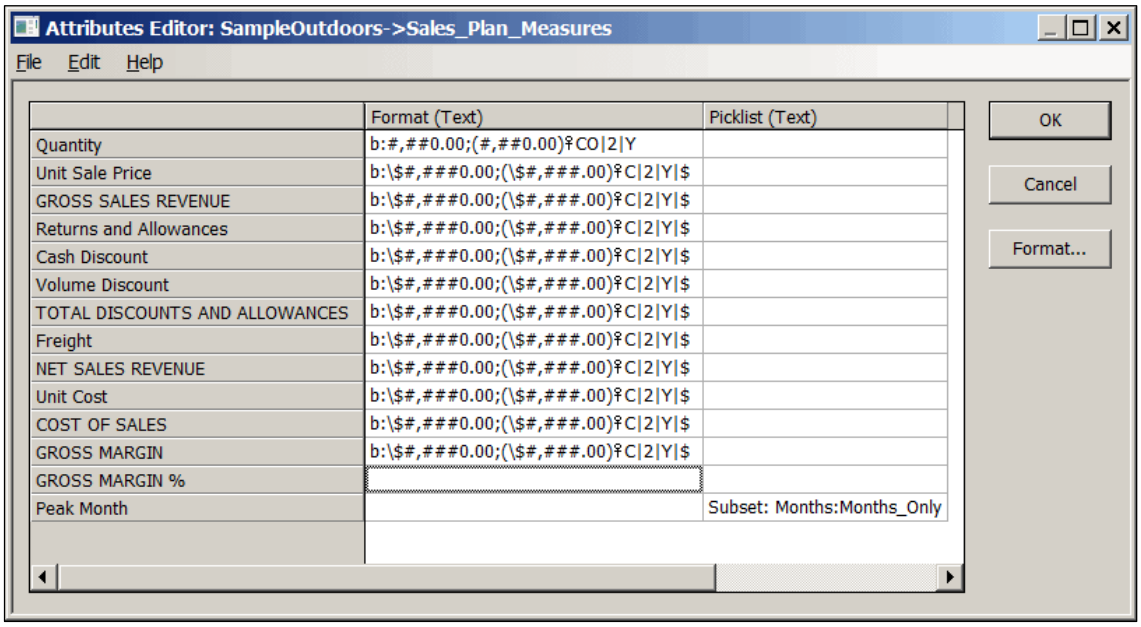
  Leave the default view open.

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Task 5. Create a subset pick list for the Sales_Plan_Measures dimension.

- In Server Explorer, right-click the **Sales_Plan_Measures** dimension and then click **Edit Element Attributes**.
- From the **Edit** menu, click **Add New Attribute**.
- In the **Name** box type **Picklist** and then click **OK** (the Text button should be selected by default).
- In the **Picklist** column, next to **Peak Month**, type:
  - **Subset: Months:Months_Only**
- Under the **Format** column, click the **Quantity** cell, click the **Format** button, click **Comma** (if necessary), and then click **OK**.
- Under the **Format** column, click the **Unit Sale Price** cell and drag down to **GROSS MARGIN**.
- Click **Format**.
- Click **Currency** and then click **OK**.

  The results appear as follows:

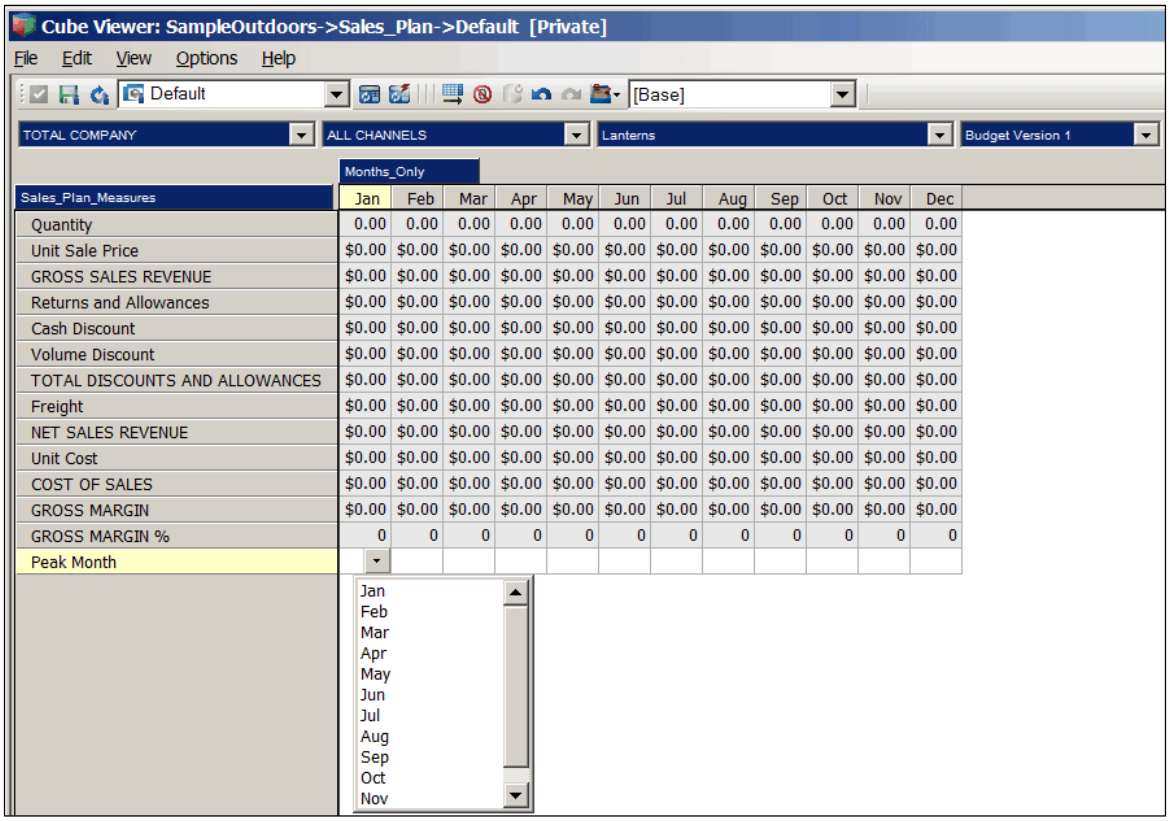


- Click **OK** to close the **Attributes Editor**.

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- Return to the **default** view of the Sales_Plan cube and then click **Recalculate**.

  The results appear as follows:



- Click the **down** arrow in the **Jan** cell for **Peak Month** to see the pick list of months.
- Close the **Cube Viewer** (saving changes and overwriting the existing view if prompted), and then close all other open windows.

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IBM Training

Load and maintain data

IBM Planning Analytics (v2.0)

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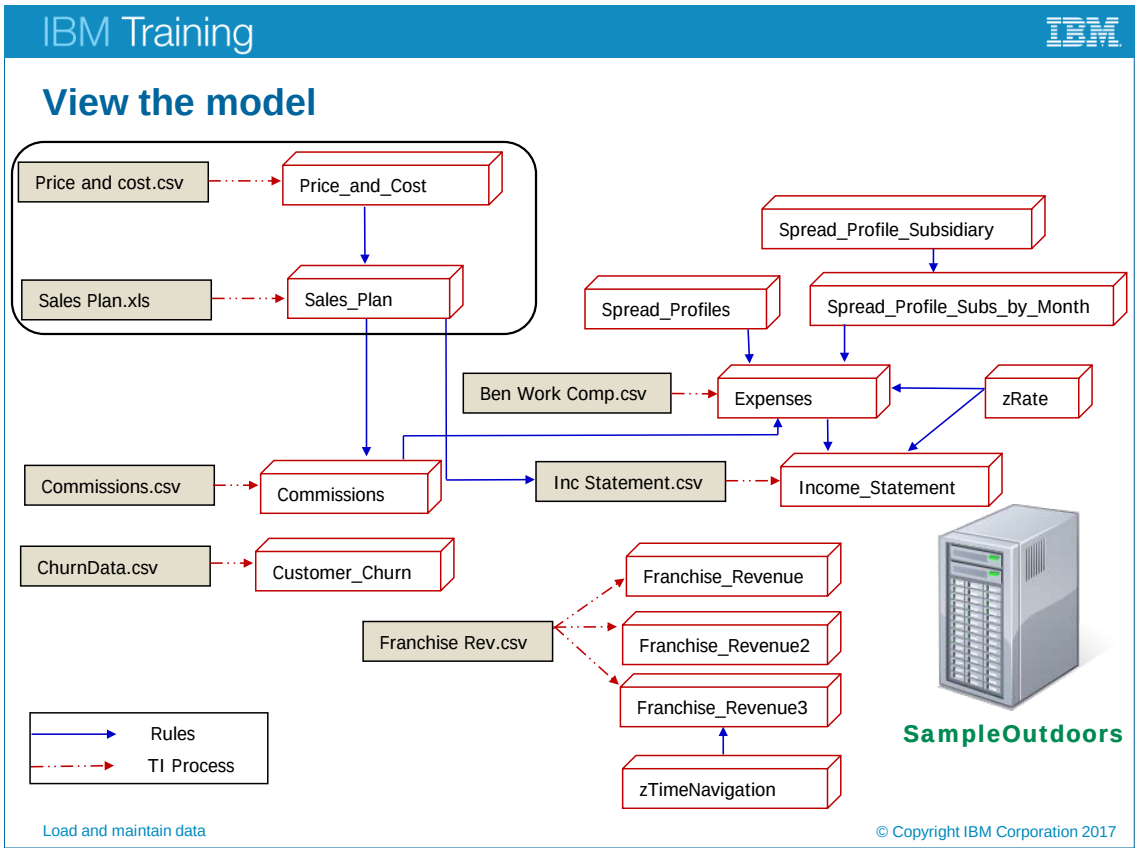
IBM Training

Unit objectives

- Identify data sources
- Create processes to load data
- Create processes to delete data in a cube
- Create processes to update and maintain the model

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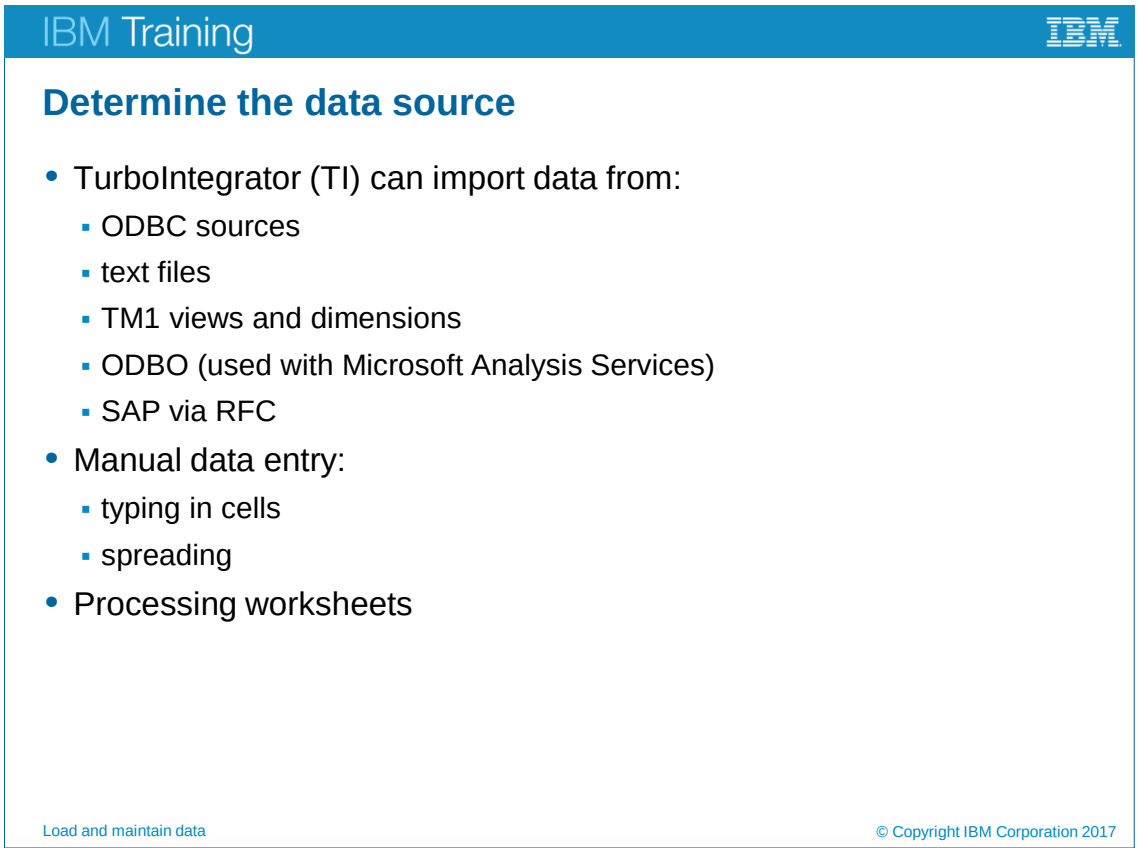


View the model

In this unit, you will use TurboIntegrator to load data into the following cubes:

- Price_and_Cost
- Sales_Plan
- Franchise_Revenue

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Determine the data source

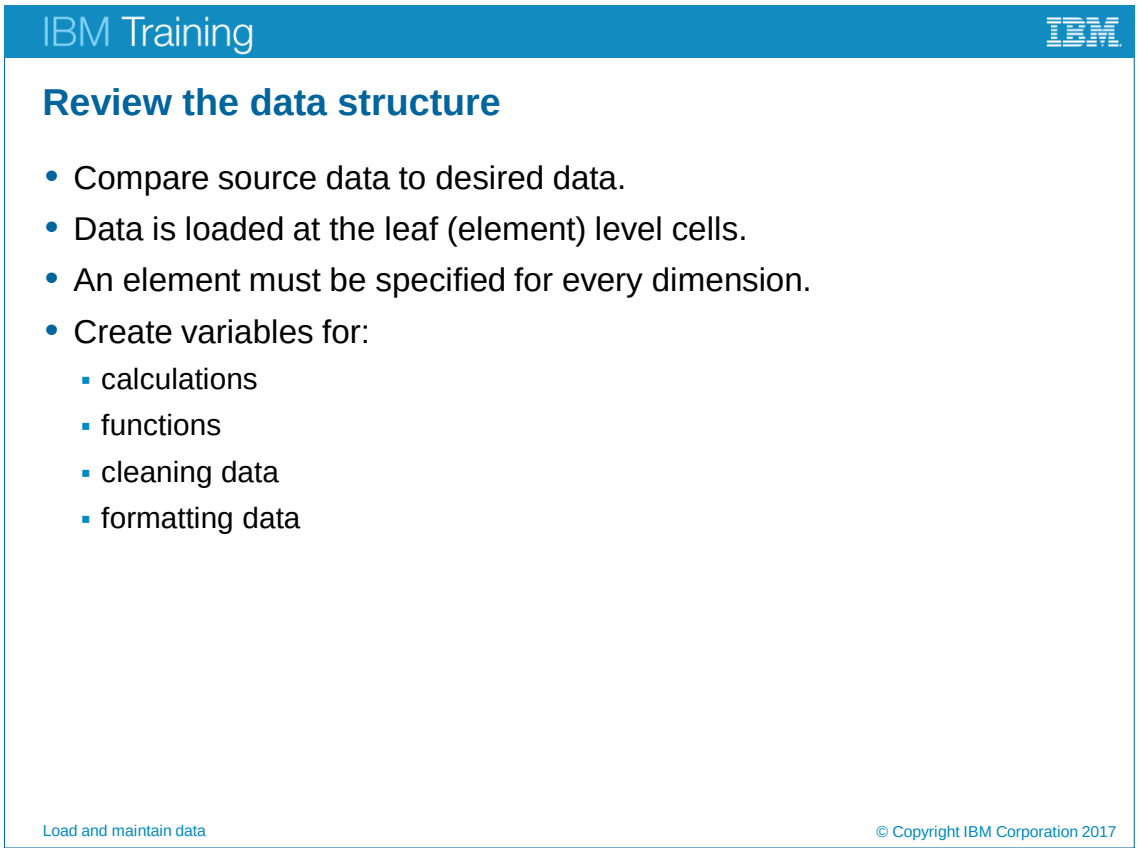
You must establish connections to data sources such as:

- ODBC
- ODBO
- SAP

Contact your database administrator for connections to other databases. You may need additional software or passwords in order to access these sources.

A processing worksheet is a specially formatted Microsoft Excel spreadsheet that contains data to load into a cube.

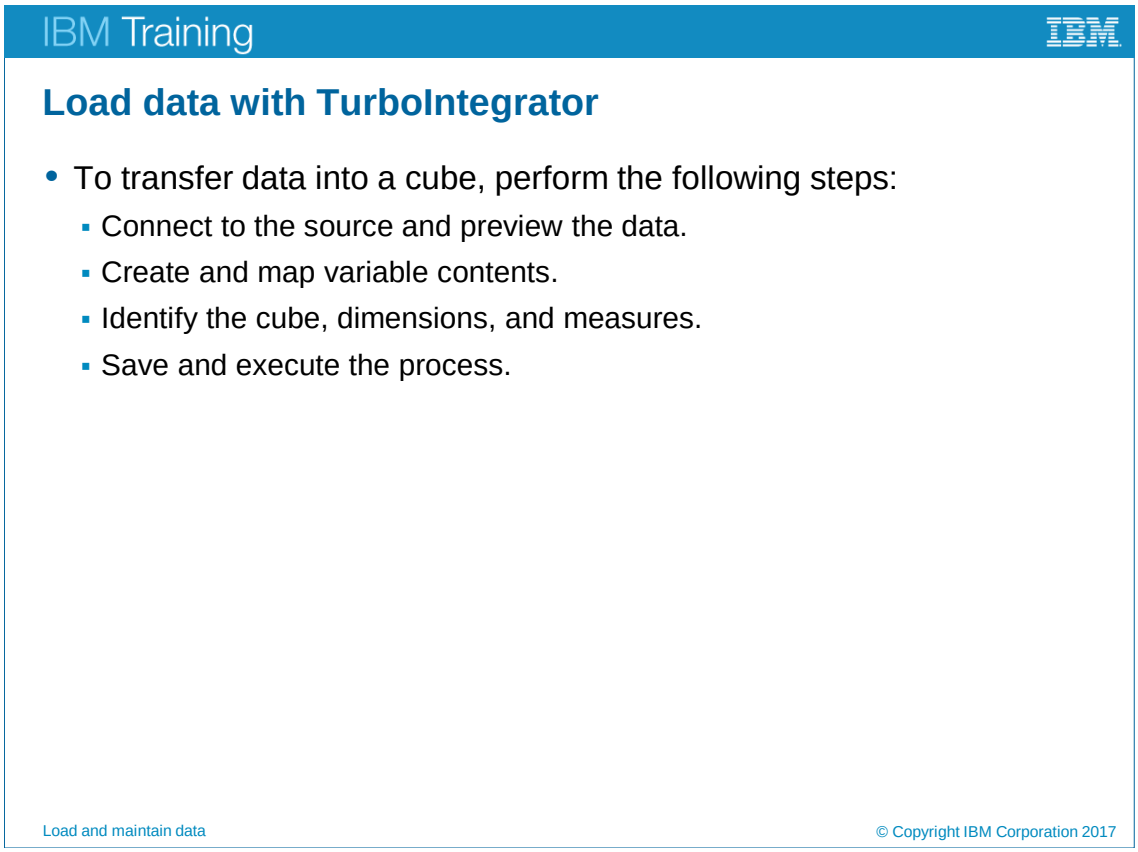
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Review the data structure

Data is loaded by specifying the leaf elements' names or aliases.

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Load data with TurboIntegrator

TurboIntegrator is neither case sensitive nor space sensitive.

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IBM Training

Demonstration 1

Import data from an ASCII file

Load and maintain data

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Demonstration 1: Import data from an ASCII file

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Demonstration 1: Import data from an ASCII file

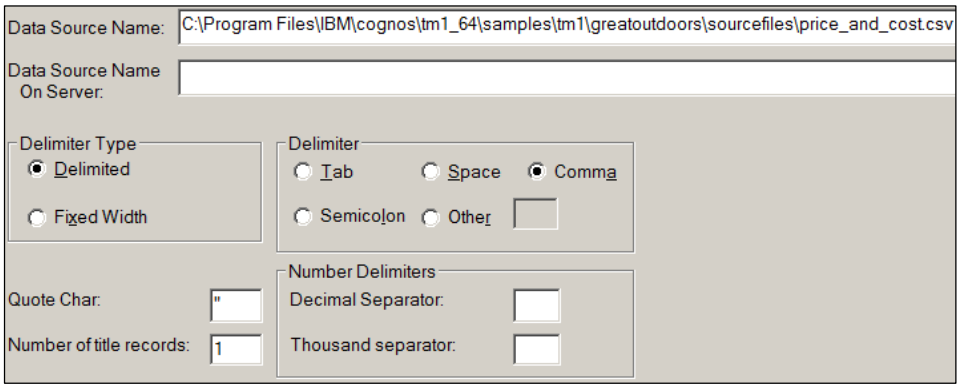
Purpose:
You have just received a file that contains the price and cost information for each of your products. You will need to populate the Price_and_Cost cube with this data so that it can be used to determine Gross Sales Revenue in another cube.

Task 1. Specify the data source.

1. From the **Start** menu, navigate to **All Programs\IBM Cognos TM1 - 64**, and then click **Architect**.
2. Maximize the window, expand the **TM1** node, and then log on to the **SampleOutdoors** TM1 server with a user ID of **admin** and a password of **apple**.
3. Right-click **Processes**, click **Create New Process**, and then click **Text**.
4. Beside the **Data Source Name** box, click **Browse**, navigate to **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles**, double-click **price_and_cost.csv** to select the file, and then click **OK** to dismiss the warning.

   If you were working on a remote server, you would use a Universal Naming Convention (UNC) path for files.
5. In the **Number of title records** box, type 1.

   This indicates that the first row in the text file contains descriptive information about the data:



6. Click **Preview**.

   You can see the first 10 rows of data in your source.

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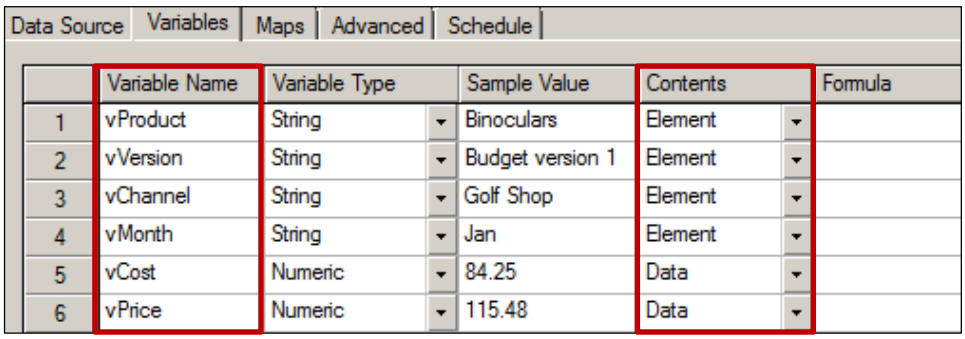
Task 2. Create and map variables.

1. Click the **Variables** tab.
2. Double-click **V1**, type **vProduct**, and then in the **Contents** column, click **Element**.

   If the column name is a reserved word to TI or contains a space, it will be replaced by Vx.

   It is good practice to place a v next to custom variable names to identify them as such in TI scripts. It is also recommended to use names that will help identify the data contents. This will help when mapping the variables to dimensions and elements.
3. In the **Variable Name** column, change **Version** to **vVersion** and **Channel** to **vChannel**.
4. Double-click **V4**, type **vMonth**, and then in the **Contents** column, click **Element**.
5. Double-click **V5**, type **vCost**, and then in the **Contents** column, click **Data**.
6. Double-click **V6**, type **vPrice**, and then in the **Contents** column, click **Data**.
7. Next to **vVersion**, under **Content**, click **Element**.
8. Next to **vChannel**, under **Content**, click **Element**.

   The results appear as follows:



	Variable Name	Variable Type	Sample Value	Contents	Formula
1	vProduct	String	Binoculars	Element	
2	vVersion	String	Budget version 1	Element	
3	vChannel	String	Golf Shop	Element	
4	vMonth	String	Jan	Element	
5	vCost	Numeric	84.25	Data	
6	vPrice	Numeric	115.48	Data	

   This cube contains five dimensions including the measures dimension. You should have an element from each dimension in your variable list. In this case, Data is a type of element, so you do not need to map an element to the measures dimension.

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Task 3. Identify the cube, dimensions, and measures.

1. Click the **Maps** tab.
2. Select **Update Cube**.

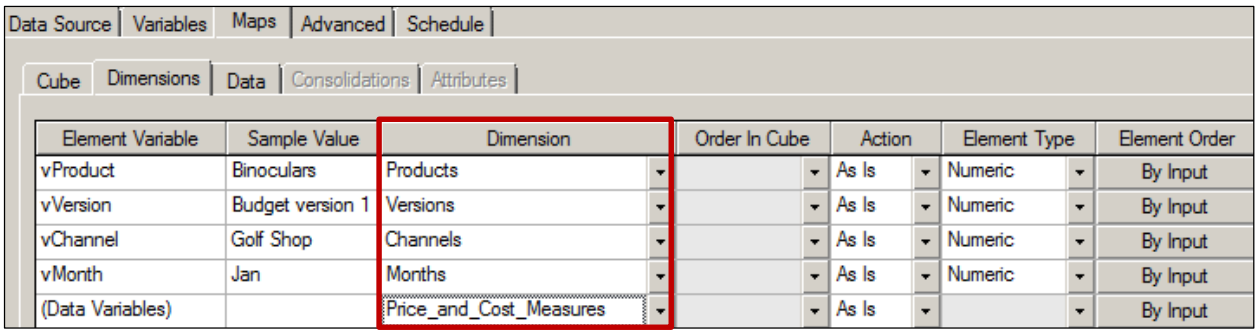
   Cube Action determines what TI will do to a cube. It can:

   - Create a cube
   - Recreate a cube (destroy it and then create it)
   - Update the cube
   - Take no action

   The Price_and_Cost cube already exists so you are going to update it.
3. In the list next to **CubeName**, select **Price_and_Cost**.
4. Click the **Dimensions** tab.
5. In the **Dimension** column, click the dimension that corresponds to each variable in the list:

   - **vProduct: Products**
   - **vVersion: Versions**
   - **vChannel: Channels**
   - **vMonth: Months**
   - **(Data Variables): Price_and_Cost_Measures**

   The results appear as follows:



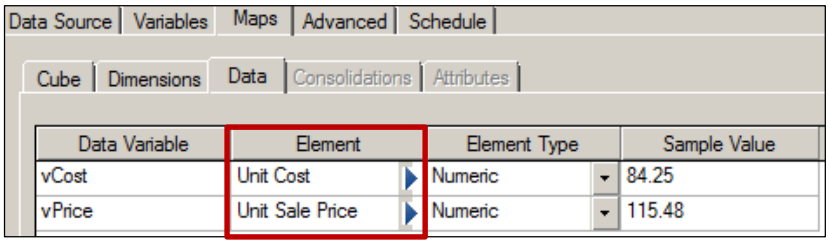
   Only change the Element Type if you are loading String data into the cube. Otherwise, leave it as Numeric.

   You are not updating or creating any dimensions so you can leave the Action column set to As Is for all of the element variables.
6. Click the **Data** tab.
7. Next to **vCost**, under **Element**, click the arrow to open the Subset Editor.
8. Select **Unit Cost**, and then click **OK** to close.

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9. Next to **vPrice**, under **Element**, click the arrow to open the Subset Editor.
10. Select **Unit Sale Price**, and then click **OK** to close.

    The results appear as follows:



Task 4. Save and execute the process.

1. Click the **Advanced** tab.
2. Click the **Prolog**, **Metadata**, **Data**, and **Epilog** tabs.

   When you click on each of the tabs, you regenerate the scripts that are associated with the tab.

   You will notice that there are statements regarding logging changes on the Prolog and Epilog tabs. When a cube is created, by default, any changes made to it are written to a log file. However, when data is loaded to a cube, there may be many records going into a single cube. TM1 will write each record to the cube (one or more times depending on the script), and if logging is on, it will also write each record to the log file. This may cause the data load to take a long time, and it could use a lot of space on your server (especially if the log files are not removed).

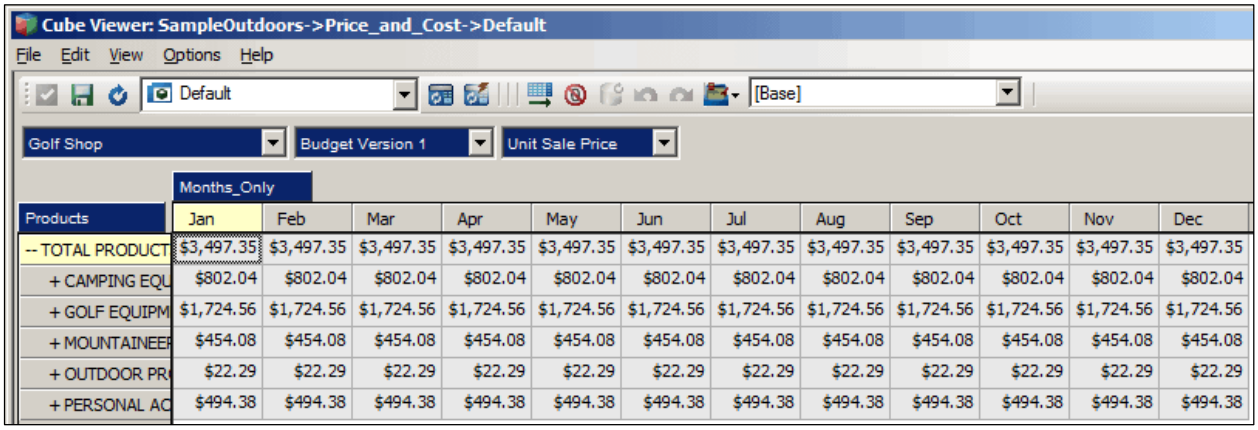
   For these reasons, the default when loading data into a cube is for logging to be turned off. It can be turned on (usually for troubleshooting purposes) by selecting the Enable Cube Logging check box on the Maps > Cube sub-tab.

   The script on the Prolog tab looks at the current Log changes setting, records it, and then turns logging off. On the Epilog tab, the Log changes setting is set back to the state it was in prior to the Prolog changes.
3. From the **File** menu, click **Save**.
4. In the **Name** box, type **LoadPriceAndCostCSV** and then click **OK**.
5. Click **Run**  .
6. Click **OK** when the process has completed, and then close **TurboIntegrator**.

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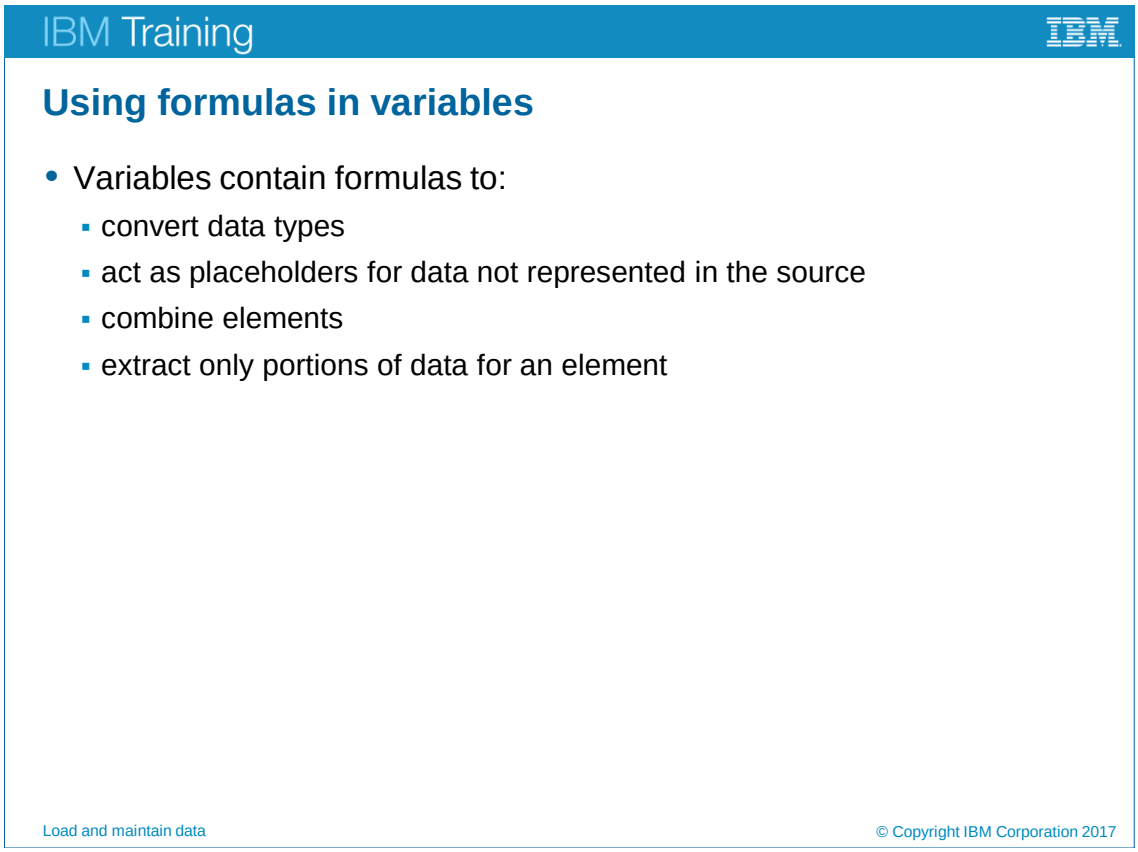
7. In the **Server Explorer** window, double-click the **Price_and_Cost** cube.
   The cube appears in the Cube Viewer as shown below:



8. Close **Cube Viewer** and then leave Architect open for the next demonstration.

Results:
You created the process to load data from a comma separated values file. During process creation, you specified which columns contained dimension elements and which contained data elements. You then saved and executed the process.

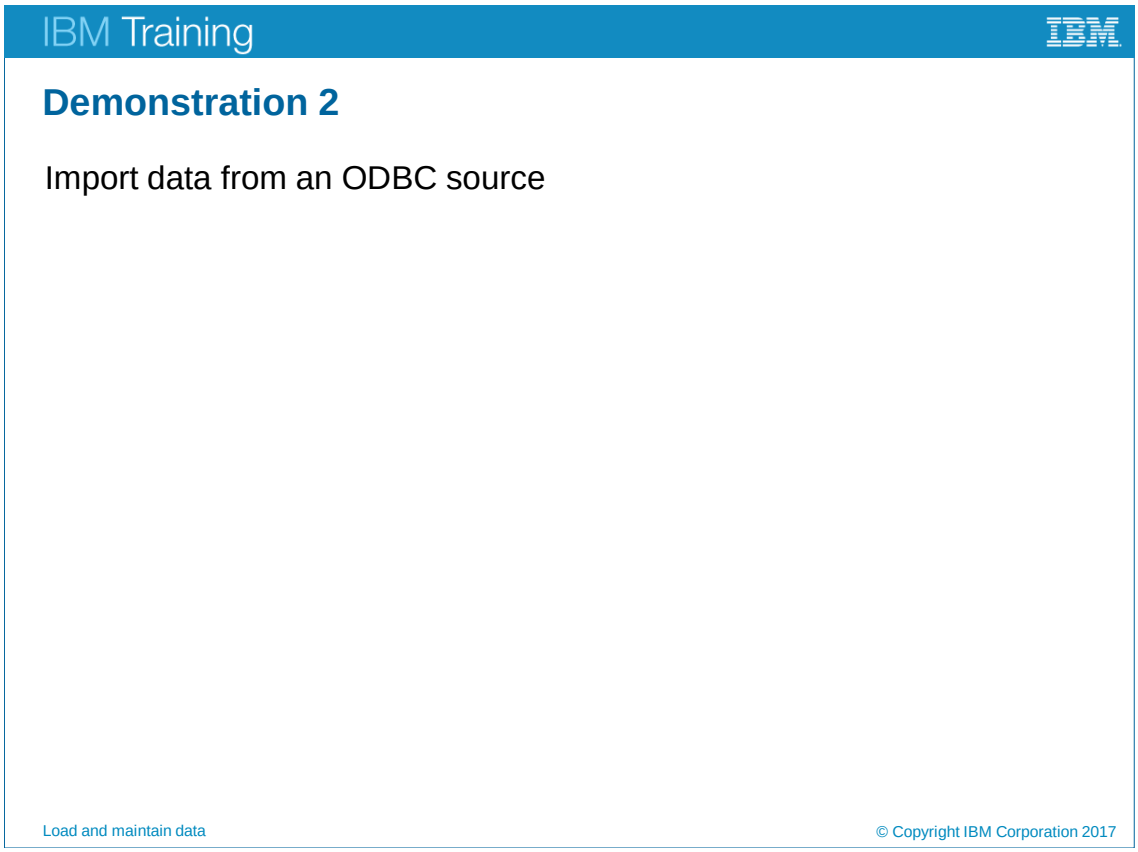
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Using formulas in variables

Many functions may be used in both TurboIntegrator and rules. Rules are discussed later in this course.

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Demonstration 2: Import data from an ODBC source

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Demonstration 2: Import data from an ODBC source

Purpose:
You want to import data from a Microsoft Excel spreadsheet used as an ODBC source. There is no Month column in the spreadsheet, but there is a Date column.

You will create an ODBC data source name that identifies the spreadsheet. Next you will create an alias in the Months dimension that refers to each month by number. Then you will create a process that uses a function to convert the Date into the month number, as well as load the data from the spreadsheet. Finally, you will run the process and confirm that the data has been loaded successfully.

Task 1. View data and decide how to proceed.

1. Open **Microsoft Excel**.
2. Click the **File** tab, click **Open**, browse to **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles**, and then double-click **sales_plan2.xls**.

   The first row contains descriptive information (labels) about the data. There are 11 columns in the Excel file.

   Notice the format of the Order Date column. The values in this column are formatted as dates (for example, 7-Jul). Click any cell in the Order Date column to see the underlying data in the formula (*fx*) bar.

   The underlying data uses numbers for the months' values (for example, 7/7). When importing into TM1, it is the underlying values that are imported, not the formatted display values.

   Within TM1, you need to add an alias to the Months dimension to map numbered months to the month element names. You will do this in the following task. You do not want the information related to years.

   Data may be loaded if it matches either the element name or an alias.

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3. In the **Name** box, to the left of the Function bar, click the down arrow, and then click **Sales**.

   The named range, Sales, refers to the contents of the entire spreadsheet. You will use it as the database name in TurboIntegrator.

4. Close **sales_plan2.xls**.

   If you do not close the .xls file you will get an error when you try to preview the data in TurboIntegrator later in this demonstration.

Task 2. Create aliases for month numbers.

1. In **Server Explorer**, right-click the **Months** dimension and then select **Edit Element Attributes**.

   You will create an attribute for the number of each month. This will allow you to use a TI function to extract the month number from the Order Date and then map it to the alias instead of the element name. TM1 allows you to use either the element name or the element alias to load data.

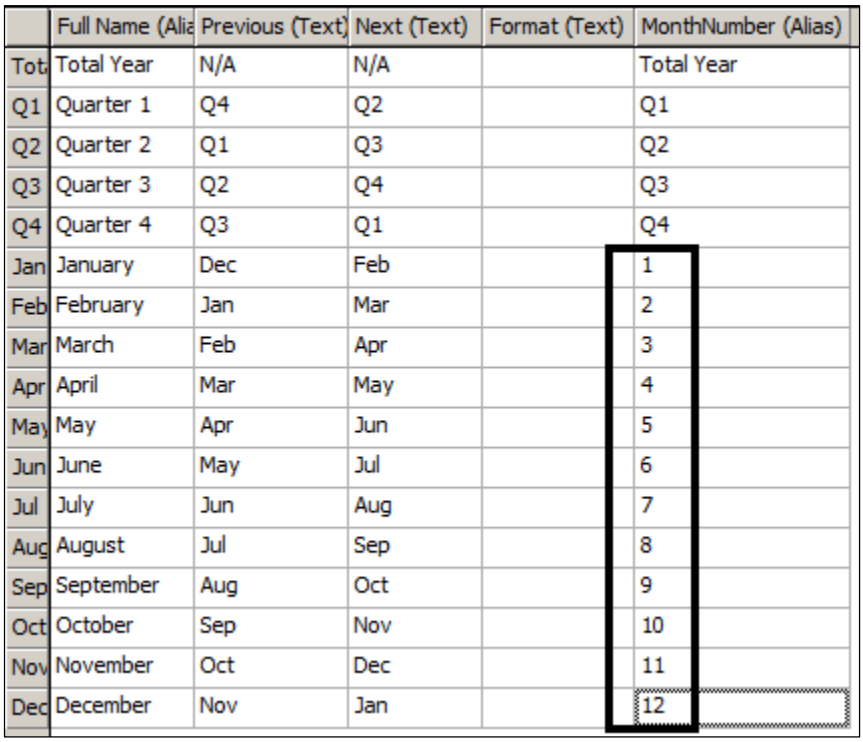
2. From the **Edit** menu, click **Add New Attribute**.
3. In the **Name** box, type **MonthNumber**, select the **Alias** button, and then click **OK**.
4. Click **OK** to dismiss the warning.

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5. In the **MonthNumber (Alias)** column, type the numbers from **1** to **12** for the corresponding months.

   The result appears as shown below:



	Full Name (Alia	Previous (Text)	Next (Text)	Format (Text)	MonthNumber (Alias)
Tot	Total Year	N/A	N/A		Total Year
Q1	Quarter 1	Q4	Q2		Q1
Q2	Quarter 2	Q1	Q3		Q2
Q3	Quarter 3	Q2	Q4		Q3
Q4	Quarter 4	Q3	Q1		Q4
Jan	January	Dec	Feb		1
Feb	February	Jan	Mar		2
Mar	March	Feb	Apr		3
Apr	April	Mar	May		4
May	May	Apr	Jun		5
Jun	June	May	Jul		6
Jul	July	Jun	Aug		7
Aug	August	Jul	Sep		8
Sep	September	Aug	Oct		9
Oct	October	Sep	Nov		10
Nov	November	Oct	Dec		11
Dec	December	Nov	Jan		12

6. Click **OK**.

Task 3. Add a connection to an ODBC driver.

1. From the Start menu, click **Control Panel**, click **Administrative Tools**, and then double-click **Data Sources (ODBC)**.

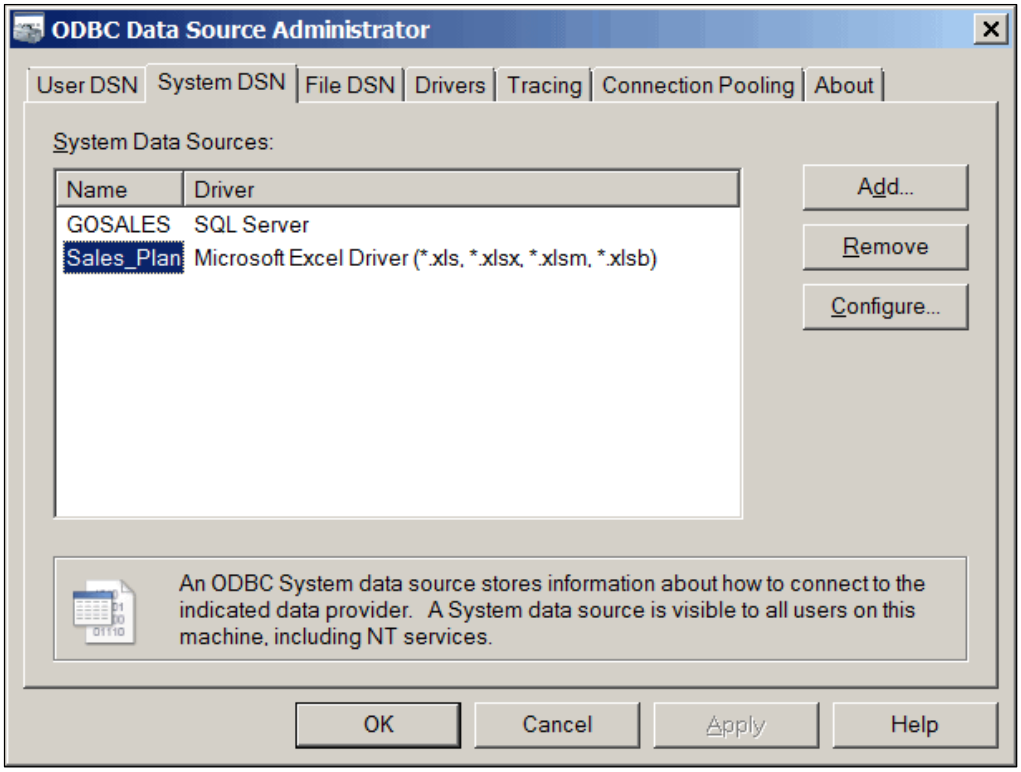
   The ODBC Data Source Administrator dialog box appears.

2. Click the **System DSN** tab.
3. Click **Add**, scroll down if necessary, and then double-click **Microsoft Excel Driver (*.xls, *.xlsx, *.xlsm, *.xlsb)**.
4. On the ODBC Microsoft Excel Setup page, in the **Data Source Name** box, type **Sales_Plan**.
5. Click **Select Workbook**, browse to **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles**, and then click **sales_plan2.xls**.

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6. Click **OK** to select the workbook, and then click **OK** to close the ODBC Microsoft Excel Setup window.

   The Sales_Plan ODBC connection appears in the System Data Sources box, as shown below:



7. Click **OK** to close the **ODBC Data Source Administrator** dialog box, and then close the **Administrative Tools** and **Control Panel** windows.

Task 4. Specify the ODBC data source.

1. In the Server Explorer window, under the **SampleOutdoors** server, right-click **Processes** and click **Create New Process**.
2. Select the **ODBC** button.
3. Beside the **Data Source Name** box, click **Browse**, and then double-click **Sales_Plan**.

   This is the data source name that you created using the ODBC Administrator.

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4. In the **Query** box, type **select * from sales**.

   The query is not case sensitive, so you can type the select statement in lower case. 'Sales' refers to the named range in Microsoft Excel.

5. Click **Preview**.

   The result appears as shown below:

	Subsidiaries	Channel	Product Type	Version	Sales Staff	Order Date	Quantity	Cash Discount
1	GO Americas	Department Store	Binoculars	Budget version 1	Barros, Viviam	2008-01-07 00:00:00	1969.000000	7706.610000
2	GO Americas	Department Store	Binoculars	Budget version 1	Barros, Viviam	2008-10-07 00:00:00	4623.000000	18097.700000
3	GO Americas	Department Store	Binoculars	Budget version 1	Barros, Viviam	2008-09-07 00:00:00	2522.000000	9871.490000
4	GO Americas	Department Store	Binoculars	Budget version 1	Bretton, Frank	2008-12-07 00:00:00	509.000000	1991.570000

Task 5. Create variables.

1. Click the **Variables** tab.

   You need to rename the variables to make them more intuitive. You also need to identify how the content will be used.

2. Rename **V3** as **vProducts**, and then in the **Contents** column, click **Element**.

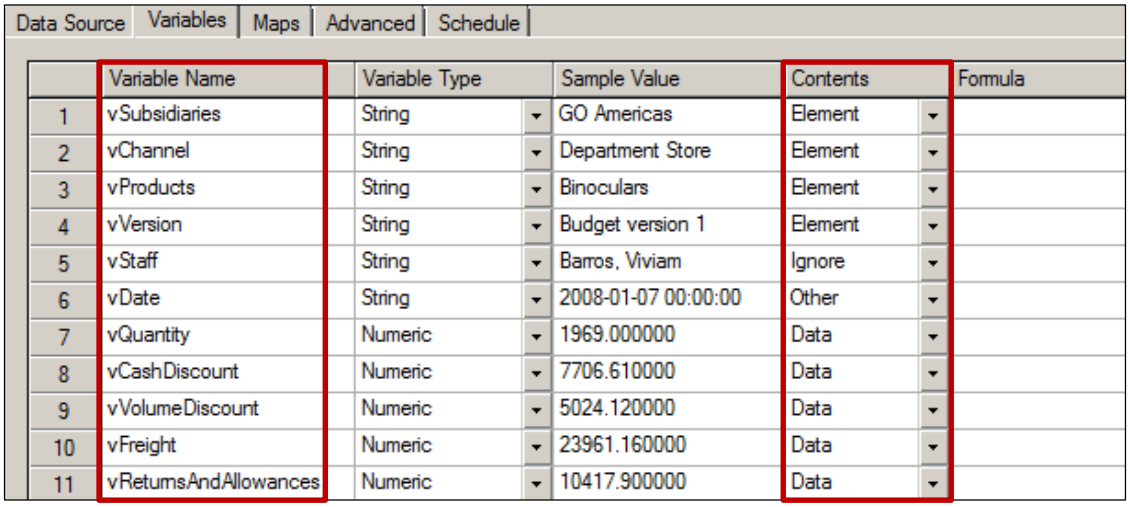
3. Repeat step **2**, with the following information:

Old Variable Name	New Variable Name	Contents
Subsidiaries	vSubsidiaries	**Element**
Channel	vChannel	**Element**
Version	vVersion	**Element**
V5	**vStaff**	**Ignore**
V6	**vDate**	**Other**
Quantity	vQuantity	**Data**
V8	**vCashDiscount**	**Data**
V9	**vVolumeDiscount**	**Data**
Freight	vFreight	**Data**
V11	**vReturnsAndAllowances**	**Data**

A variable whose Contents is set to Other is available to a TI process but not required to be mapped to an Element, Consolidation, or Attribute. Variables set to Ignore are not available to the process at all.

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The results appear as follows:



Data Source | Variables | Maps | Advanced | Schedule

	Variable Name	Variable Type	Sample Value	Contents	Formula
1	vSubsidiaries	String	GO Americas	Element	
2	vChannel	String	Department Store	Element	
3	vProducts	String	Binoculars	Element	
4	vVersion	String	Budget version 1	Element	
5	vStaff	String	Barros, Viviam	Ignore	
6	vDate	String	2008-01-07 00:00:00	Other	
7	vQuantity	Numeric	1969.000000	Data	
8	vCashDiscount	Numeric	7706.610000	Data	
9	vVolumeDiscount	Numeric	5024.120000	Data	
10	vFreight	Numeric	23961.160000	Data	
11	vReturnsAndAllowances	Numeric	10417.900000	Data	

Next you want to create a new variable called vMonth.

4. On the right side of the **TurboIntegrator** window, click **New Variable**, and rename it **vMonth**.

   You want to supply the vMonth variable with the number that corresponds to the MonthNumber alias. You will use the MONTH function to retrieve the month from the vDate variable and the NUMBERTOSTRING function to convert the number to text (again, to match the MonthNumber alias).

   Syntax:

   MONTH (date); - where date is in the format YY-MM-DD or YYYY-MM-DD

   NUMBERTOSTRING(value); - where value is a real value

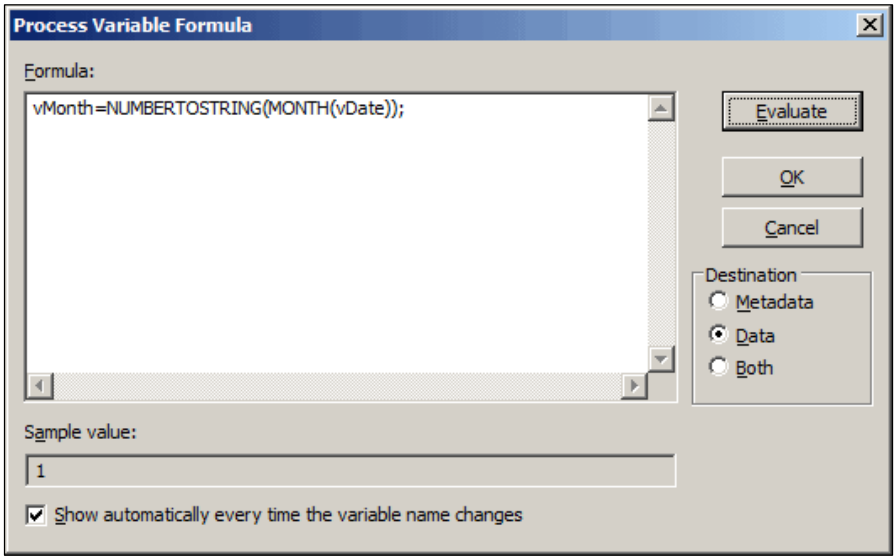
   It is recommended that you use capital letters for functions. This way you immediately know that MONTH is a function and not a variable or data.

5. Click the **Formula** button for the **vMonth** variable to open the **Process Variable Formula** dialog box, and then in the **Formula** box type: **vMonth=NUMBERTOSTRING(MONTH(vDate));**

   Remember the **;** (semicolon) at the end.

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6. Below **Destination**, select the **Data** button, and then click **Evaluate**.
   The result appears as shown below:



7. Click **OK**.
8. For **vMonth**, in the **Variable Type** column, click **String**, and then in the **Contents** column, click **Element**.
   The results appear as follows:

Data Source | Variables | Maps | Advanced | Schedule

	Variable Name	Variable Type	Sample Value	Contents	Formula
1	vSubsidiaries	String	GO Americas	Element	
2	vChannel	String	Department Store	Element	
3	vProducts	String	Binoculars	Element	
4	vVersion	String	Budget version 1	Element	
5	vStaff	String	Barros, Viviam	Ignore	
6	vDate	String	2008-01-07 00:00:00	Other	
7	vQuantity	Numeric	1969.000000	Data	
8	vCashDiscount	Numeric	7706.610000	Data	
9	vVolumeDiscount	Numeric	5024.120000	Data	
10	vFreight	Numeric	23961.160000	Data	
11	vReturnsAndAllowances	Numeric	10417.900000	Data	
12	vMonth	String	1	Element	Formula

Variables that are mapped to Element or Consolidation must be of type String.

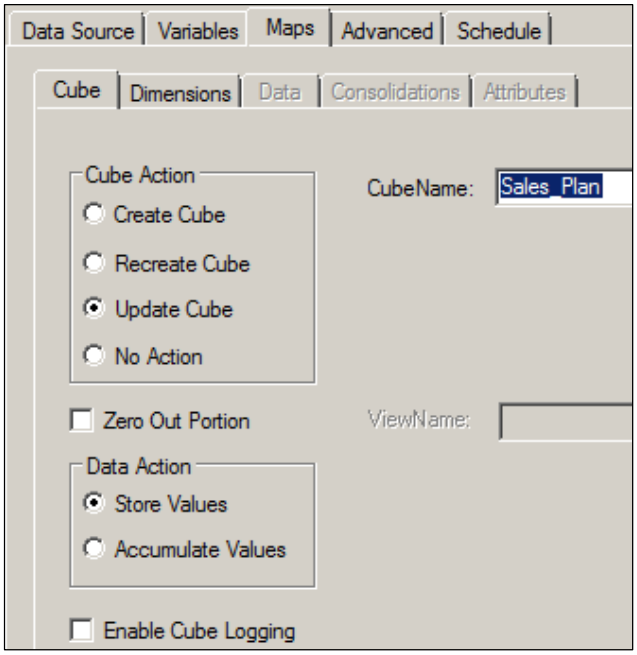
Next you will map the variables to TM1 objects like cubes, elements, consolidations, and attributes. You will also determine what will be done with objects (for example, creating, updating, or recreating).

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Task 6. Map cube, dimension, and data information.

1. Click the **Maps** tab, and then on the **Cube** sub-tab, select the **Update Cube** button.
2. Click the down arrow beside **CubeName**, and then select **Sales_Plan**.
3. Below **Data Action**, click the **Store Values** button (if necessary).

   The result appears as shown below:



You chose Update Cube because the cube was created previously.

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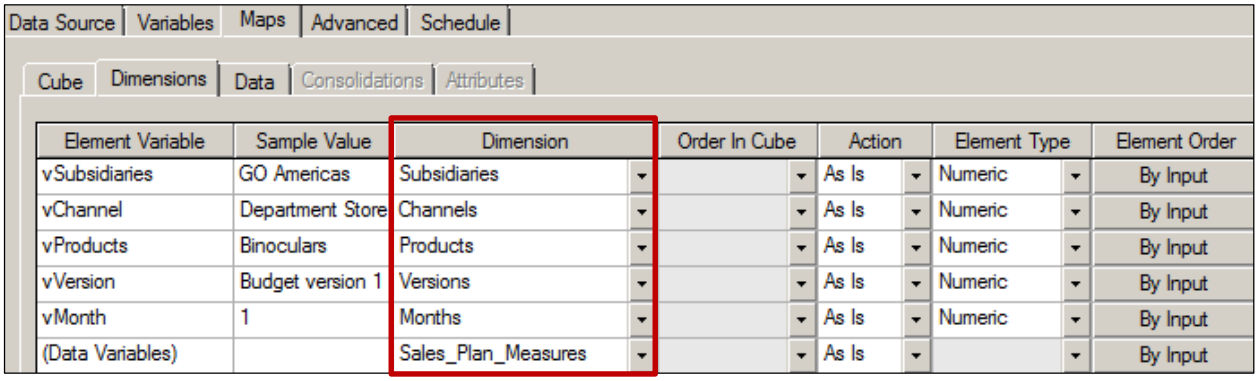
4. Click the **Dimensions** sub-tab.

   On this tab, you map the variables to the dimensions in TM1. Since the dimensions have already been created, the Action column remains "As Is".

   Beside each element variable, click the appropriate dimension.

Element Variable	Dimension
vSubsidiaries	**Subsidiaries**
vChannel	**Channels**
vProducts	**Products**
vVersion	**Versions**
vMonth	**Months**
(Data Variables)	**Sales_Plan_Measures**

   The results appear as follows:



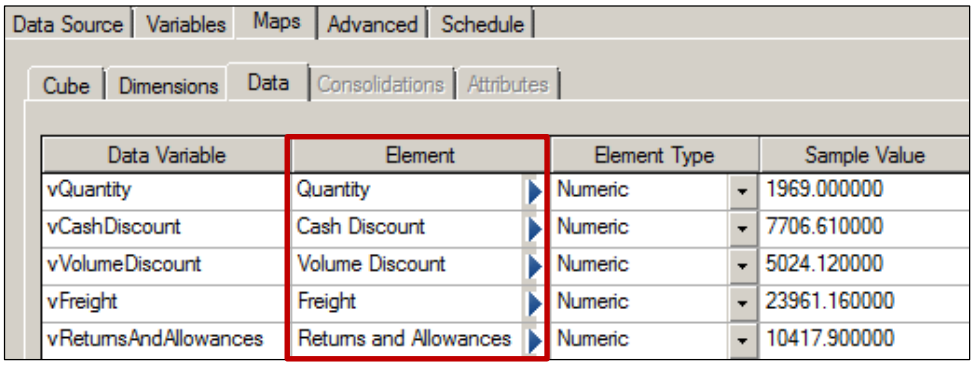
5. Click the **Data** tab.

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6. Beside each data variable, click the appropriate element:

Data Variable	Element
vQuantity	**Quantity**
vCashDiscount	**Cash Discount**
vVolumeDiscount	**Volume Discount**
vFreight	**Freight**
vReturnsAndAllowances	**Returns and Allowances**

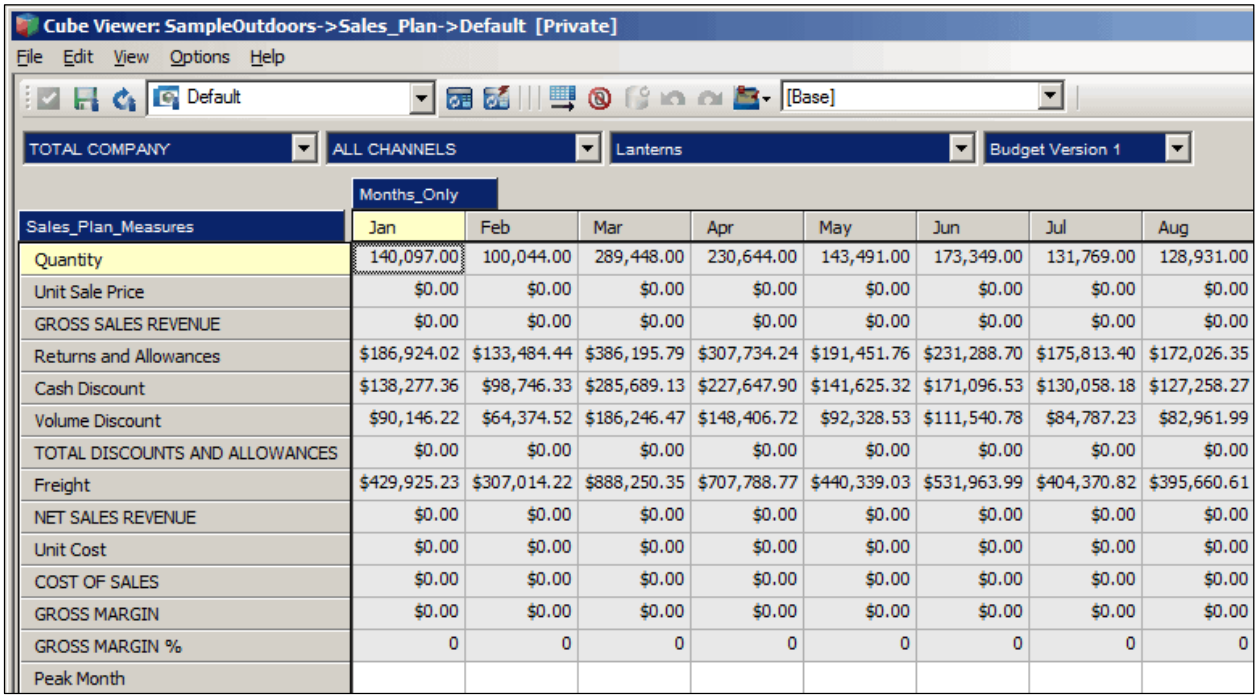
The results appear as follows:



7. Click the **Advanced** tab, and then click on the **Prolog**, **Metadata**, **Data**, and **Epilog** tabs.
8. Save the process as **LoadSalesPlanODBC** and then run it.
9. Click **OK** to dismiss the confirmation message, and then close **TurboIntegrator**.

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10. Double-click the **Sales_Plan** cube.
    The result appears as shown below:

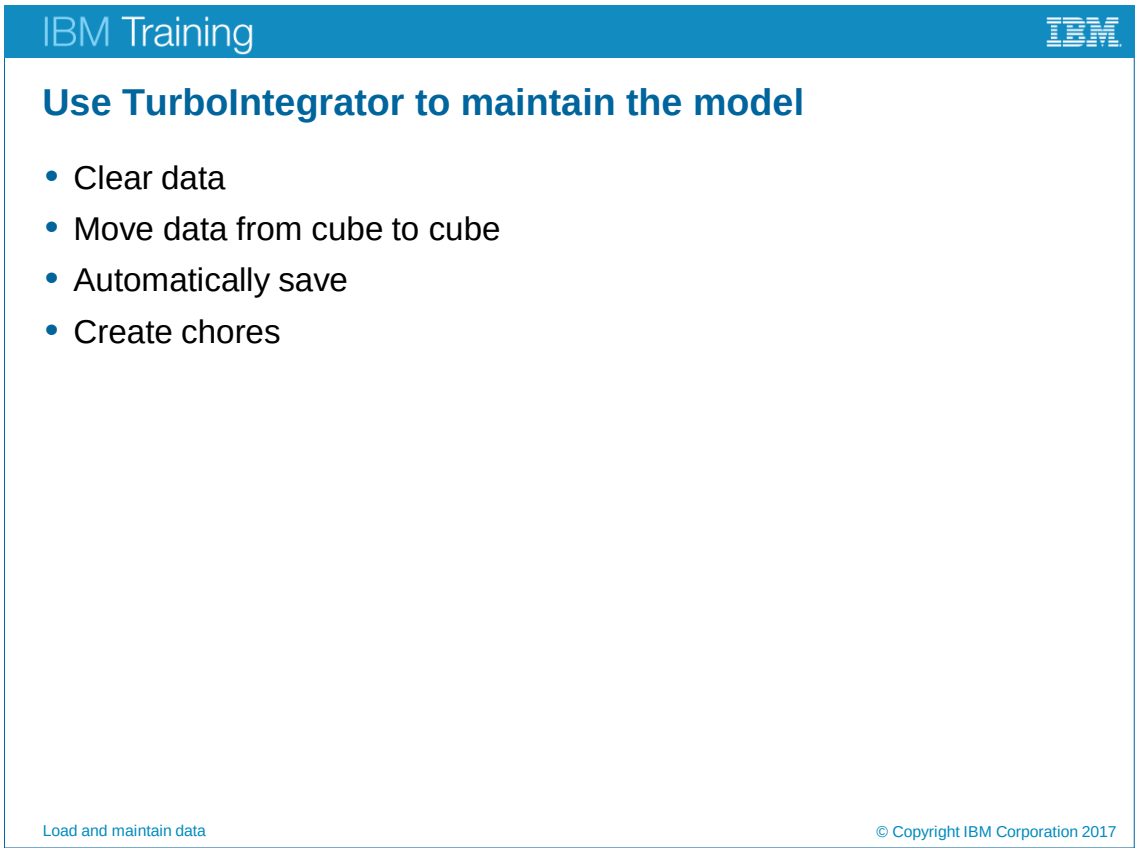


Sales_Plan_Measures	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug
Quantity	140,097.00	100,044.00	289,448.00	230,644.00	143,491.00	173,349.00	131,769.00	128,931.00
Unit Sale Price	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
GROSS SALES REVENUE	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
Returns and Allowances	$186,924.02	$133,484.44	$386,195.79	$307,734.24	$191,451.76	$231,288.70	$175,813.40	$172,026.35
Cash Discount	$138,277.36	$98,746.33	$285,689.13	$227,647.90	$141,625.32	$171,096.53	$130,058.18	$127,258.27
Volume Discount	$90,146.22	$64,374.52	$186,246.47	$148,406.72	$92,328.53	$111,540.78	$84,787.23	$82,961.99
TOTAL DISCOUNTS AND ALLOWANCES	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
Freight	$429,925.23	$307,014.22	$888,250.35	$707,788.77	$440,339.03	$531,963.99	$404,370.82	$395,660.61
NET SALES REVENUE	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
Unit Cost	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
COST OF SALES	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
GROSS MARGIN	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
GROSS MARGIN %	0	0	0	0	0	0	0	0
Peak Month								

    The default view was generated in a previous exercise. If your default view does not look like the above, arrange the dimensions and save the default view now.
11. Right-click the **SampleOutdoors** node and click **Save Data**.
12. Close **Cube Viewer** and leave Architect open for the next demonstration.

Results:
You created an ODBC data source name that identified the spreadsheet. You then created an alias in the Months dimension that refers to each month by number. Next, you created a process that uses a function to convert the Date into the month number, as well as load the data from the spreadsheet. Finally, you ran the process and confirmed that the data was loaded successfully.

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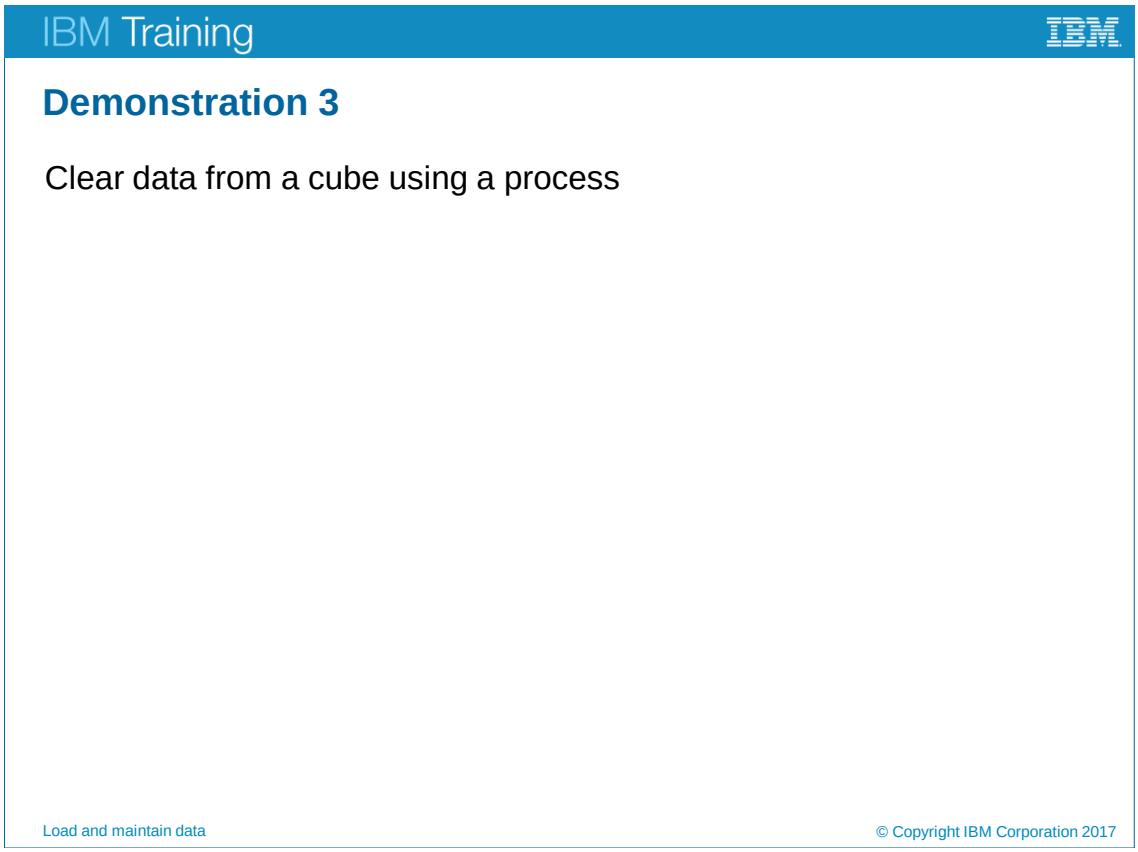
Use TurboIntegrator to maintain the model

A chore is a TM1 object that executes one or more processes at a user-defined frequency. You can schedule a process for automatic execution as a chore from within TurboIntegrator.

When scheduling a TI process, TI automatically creates a chore that will be scheduled. Either a chore or a process can be run on-demand from the menu. Users may run TI processes (if granted security access). However, only admin group members may edit processes.

In order to prevent a chore from trying to execute while it is being edited, it must be de-activated before you can open and edit it. After editing a chore, it must be re-activated in order to run at its scheduled time.

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Demonstration 3: Clear data from a cube using a process

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Demonstration 3: Clear data from a cube using a process

Purpose:
You want to reload data into the Sales_Plan cube using a modified process. Before you reload the data, you want all the existing data from the cube to be removed, so that the data you view later will be correct. You will create a new process that clears all the data from the cube.

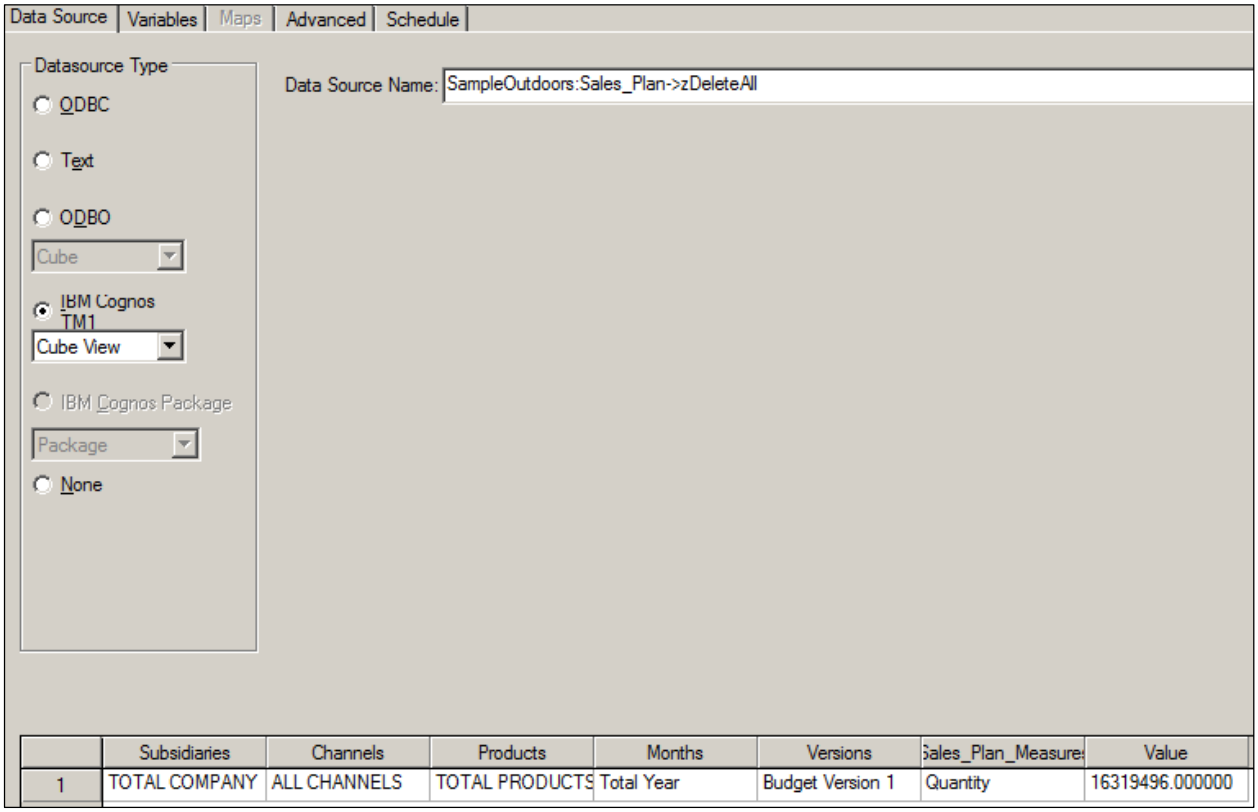
Task 1. Identify the view to be cleared.

1. In the Server Explorer window, right-click **Processes**, and then click **Create New Process**.
2. On the **Data Source** tab, click **IBM Cognos TM1**.
   Cube View is enabled automatically.
3. Click **Browse**.
   All the cubes that are available on your server appear.
4. Click the **SampleOutdoors: Sales_Plan** cube, and then click **Create View**.
5. In the **View Extract** dialog box, click **OK**.
   You do not need to make any selections as the default is to select All elements.
6. In the **Select or Enter Named** box, type **zDeleteAll** and then click **OK**.
   This view should only be used in a TI process. It will try to display all the cells in the cube which may result in a very large view that may be too large in memory. TM1 displays views in alphabetical order so placing a 'z' at the beginning will make it more difficult to accidentally select this view.
   Private is disabled because TI processes may only use Public views.
7. In the **Select View** column, click **zDeleteAll**, and then click **OK**.

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8. Click **Preview**.

   The result appears as shown below:



9. Save the process as **SalesPlanDeleteData**.

Task 2. Add a function to the Prolog tab.

1. Click the **Variables** tab to ensure that all the variables have **Ignore** selected in the **Contents** column.
2. Click the **Advanced** tab, and then click the **Prolog** tab.

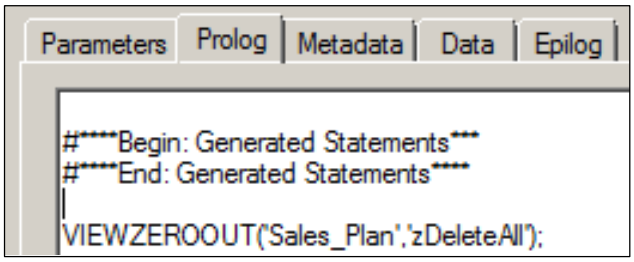
   You want to add a function that sets all data values to zero in the Sales_Plan cube.

   Syntax:

   VIEWZEROOUT('Cube', 'ViewName');

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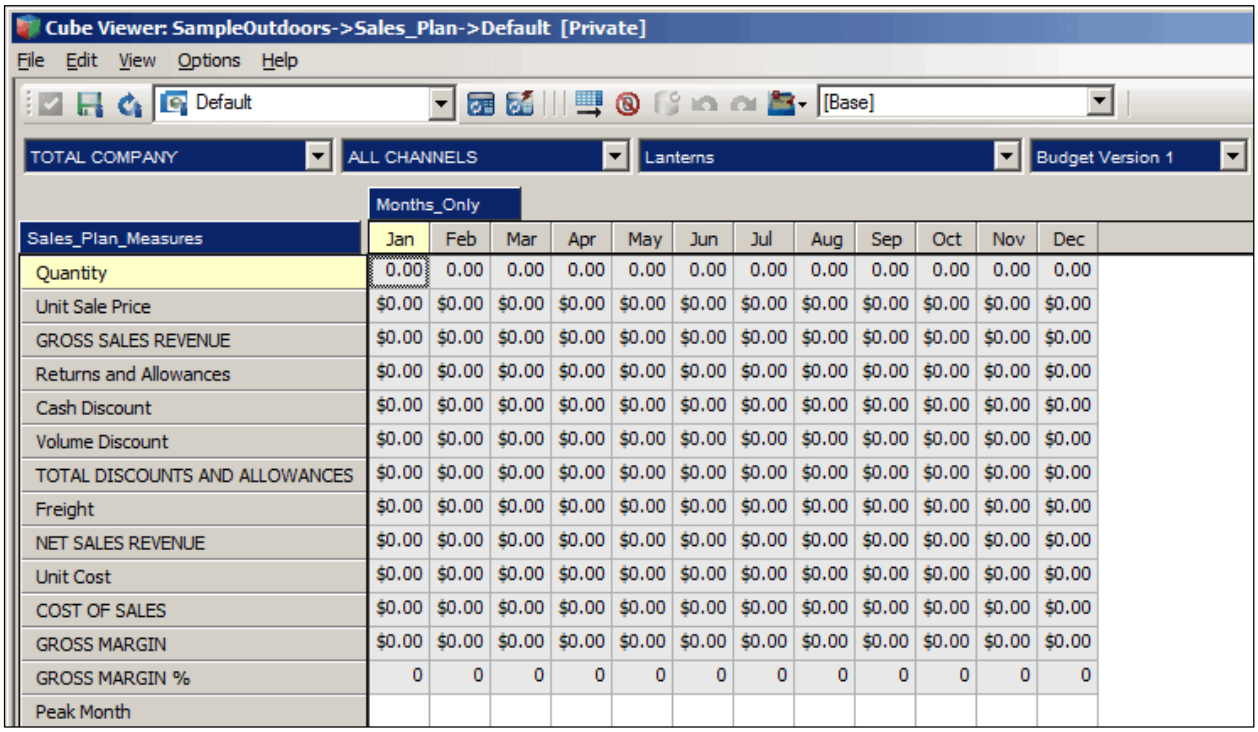
3. Below the two lines already there, type **VIEWZEROOUT('Sales_Plan','zDeleteAll');**
   The results appear as follows:



4. Save and run the process, and then click **OK**.
5. Close **TurboIntegrator**.

Task 3. View the cube.

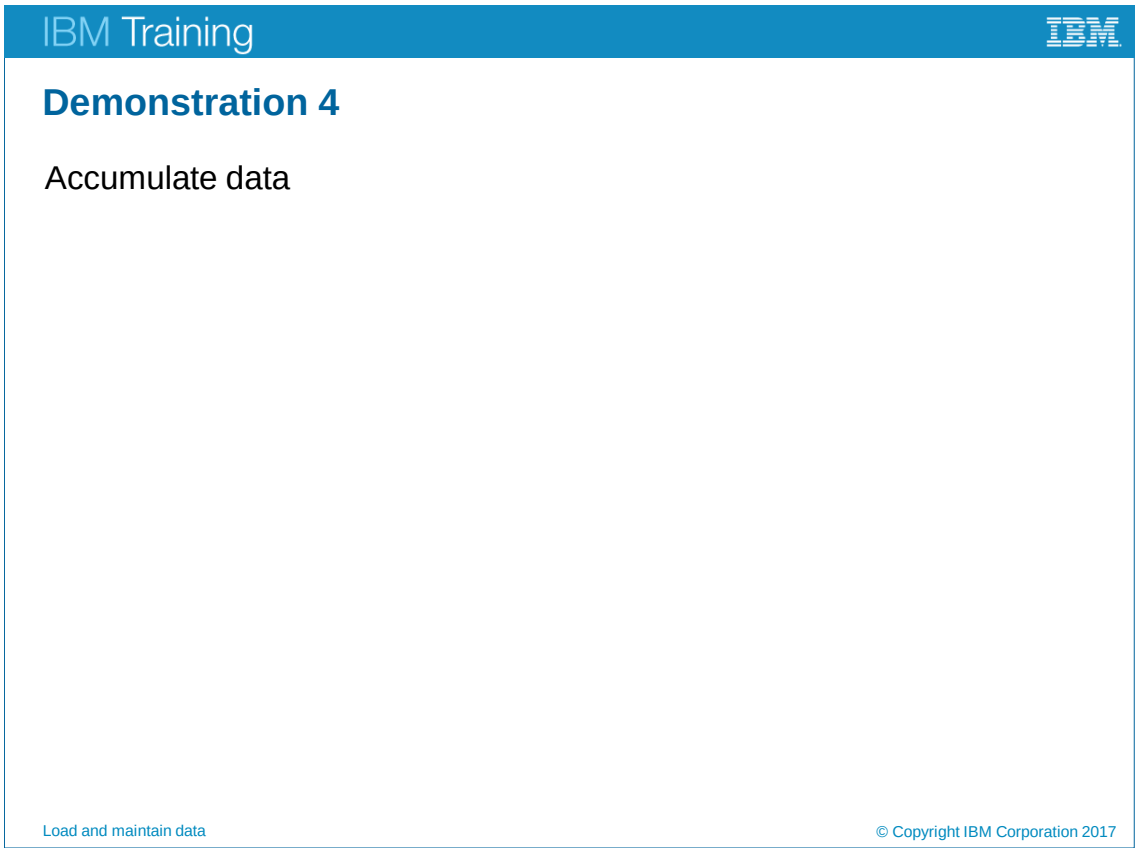
1. Double-click the **Sales_Plan** cube.
   The result appears as shown below:



2. Close the **Cube Viewer** without saving.
3. Right-click the **SampleOutdoors** node, and then click **Save Data**.
4. Leave Architect open for the next demonstration.

Results:
You created a process that clears all the data from a cube.

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Demonstration 4: Accumulate data

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Demonstration 4: Accumulate data

Purpose:
It has been brought to your attention that since you previously ignored the data for staff, and you did not accumulate data going into the Sales_Plan cube, the cube only contains the last data for each targeted cell.

Your cube is empty again after running the delete process so now you can reload data from the ODBC source and accumulate the data.

You will edit the process to accumulate the data, rerun the process, and then compare the results with the results from Demonstration 2.

Task 1. Specify the data action.

1. In the Server Explorer window, expand **Processes** (if necessary), and then double-click **LoadSalesPlanODBC**.
2. Click the **Maps** tab.
3. In the **Data Action** box, click **Accumulate Values**.

   In some cases, such as this one, you may have a number of records in your data source that need to be summarized and put into a cell. This is very common when you have more columns in the data source than you have dimensions in your cube.

   In your example, you do not have a "Staff" dimension, so you ignored the names of the staff members. TI will need to ignore the "staff" column and keep adding data into the appropriate cells. If the data were not accumulated, then each record would overwrite the previous one and the data would be incorrect (too low).

Task 2. Create variables.

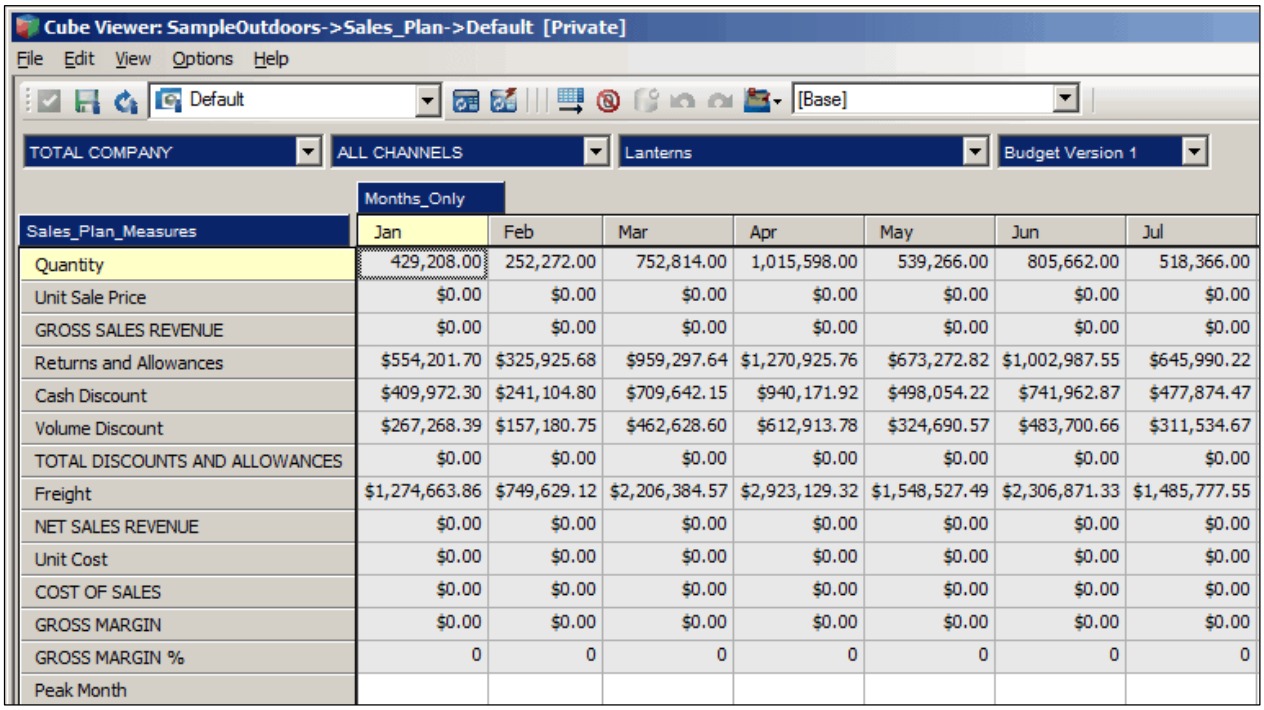
1. Click the **Advanced** tab, and then click on the **Prolog**, **Metadata**, **Data**, and **Epilog** tabs.

   Because you made a change to the process, you need to be sure the underlying scripts reflect this change.
2. Click **Save**, and then click **Run**.

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3. Click **OK** to dismiss the confirmation message and then close **TurboIntegrator**.
4. Double-click the **Sales_Plan** cube.

   The result appears as shown below:



Sales_Plan_Measures	Jan	Feb	Mar	Apr	May	Jun	Jul
Quantity	429,208.00	252,272.00	752,814.00	1,015,598.00	539,266.00	805,662.00	518,366.00
Unit Sale Price	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
GROSS SALES REVENUE	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
Returns and Allowances	$554,201.70	$325,925.68	$959,297.64	$1,270,925.76	$673,272.82	$1,002,987.55	$645,990.22
Cash Discount	$409,972.30	$241,104.80	$709,642.15	$940,171.92	$498,054.22	$741,962.87	$477,874.47
Volume Discount	$267,268.39	$157,180.75	$462,628.60	$612,913.78	$324,690.57	$483,700.66	$311,534.67
TOTAL DISCOUNTS AND ALLOWANCES	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
Freight	$1,274,663.86	$749,629.12	$2,206,384.57	$2,923,129.32	$1,548,527.49	$2,306,871.33	$1,485,777.55
NET SALES REVENUE	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
Unit Cost	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
COST OF SALES	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
GROSS MARGIN	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
GROSS MARGIN %	0	0	0	0	0	0	0
Peak Month							

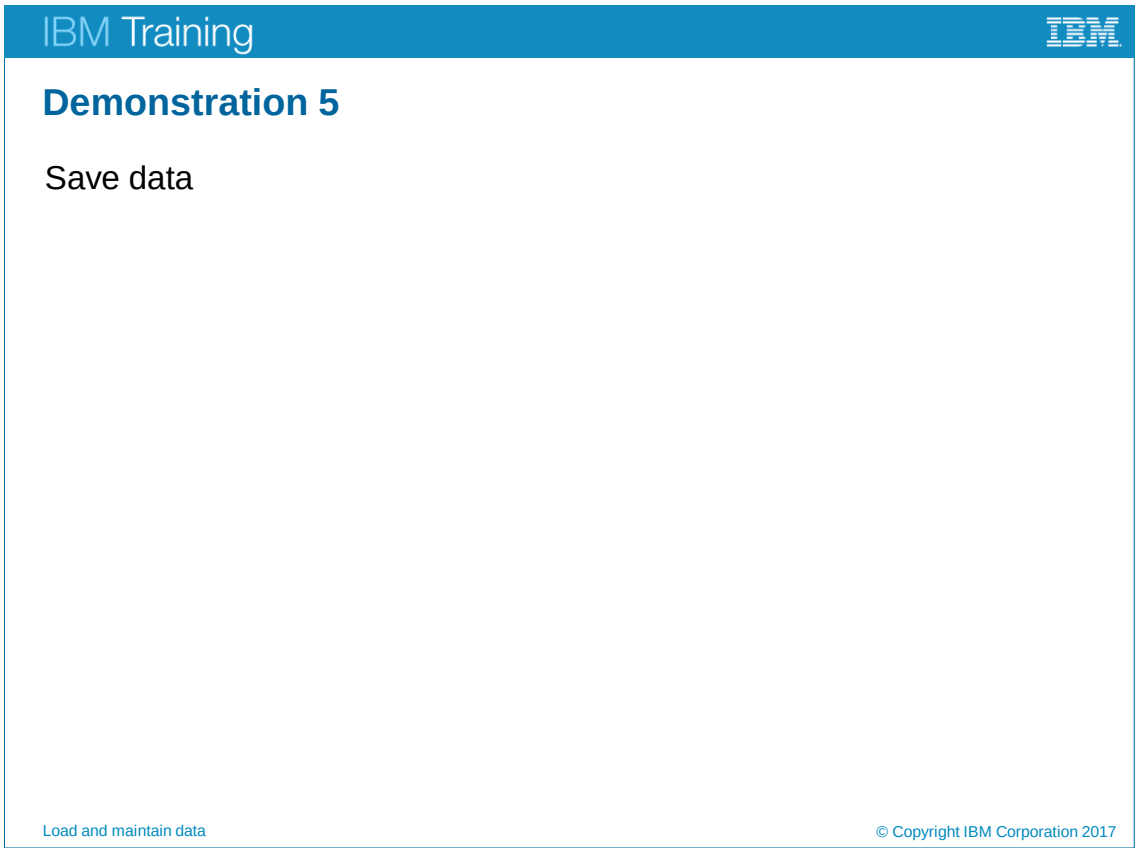
5. Compare your results with the results in the screen capture found in **Demonstration 2**, **Task 6**, **Step 11** of this unit.

   In that view, the Quantity for Jan was only **140,097**. Now you can see it is **429,208**. The records are now being accumulated where they were previously being overwritten.

   As expected, the values are significantly higher than earlier, because data is now aggregated.
6. Right-click the **SampleOutdoors** node and click **Save Data**.
7. Close the **Cube Viewer** and leave Architect open for the next demonstration.

Results:
You edited a process to accumulate data from an ODBC source, re-ran the process, and then compared the results with the results from the previous Demonstration 2.

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Demonstration 5: Save data

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Demonstration 5: Save data

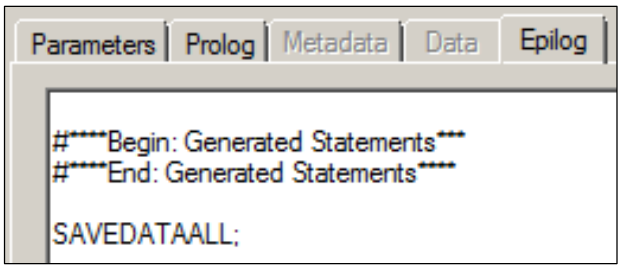
Purpose:
All objects and data in TM1 are in memory until you save them to the server. You are normally prompted to do this whenever you shut down the server when it is running as a Windows application, but you want to ensure that data is saved periodically during the day. You will create a process that saves all your changes, test it, and then schedule it to run every two hours

Task 1. Create a process.

1. In the Server Explorer window, right-click **Processes**, and then click **Create New Process**.

   You want to run a TurboIntegrator function that affects all data, and not just a specific cube or view. You do not need to make any changes to the Data Source or Variables tabs and can start working on the Advanced tab.
2. Click the **Advanced** tab.
3. Click the **Epilog** tab.
4. After the **End: Generated Statements** line, type **SAVEDATAALL;**

   The result appears as shown below:

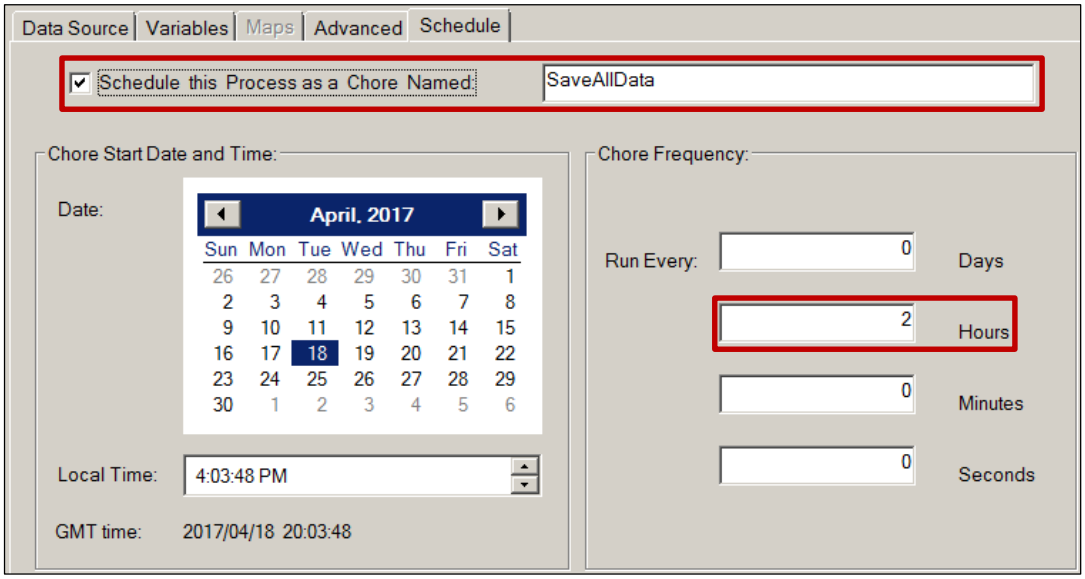


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Task 2. Schedule the process to run every two hours.

1. Click the **Schedule** tab, select the **Schedule this Process as a Chore Named** check box, and then beside it, type **SaveAllData**.
2. Below **Chore Frequency**, beside **Hours**, type **2**.

   The result appears similar to that shown below:



3. Click **Save**, and then save the process as **SaveAllData**.

   This chore will run every two hours starting immediately.
4. Close **TurboIntegrator**.

   Below Processes, the new process SaveAllData appears.
5. Expand **Chores**.

   Below Chores, the new chore SaveAllData appears.

   You can run either the chore or the process by right-clicking it and then clicking Run.
6. Leave Architect open for the upcoming exercise.

Results:
You created a new process to save all data.

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IBM Training

Unit summary

- Identify data sources
- Create processes to load data
- Create a process to delete data in a cube
- Create processes to update and maintain the model

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Unit summary

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IBM Training

Exercise 1

Create the Franchise_Revenue cube

Load and maintain data

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Exercise 1: Create the Franchise_Revenue cube

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Exercise 1: Create the Franchise_Revenue cube

The budget data for franchise revenue is now available. The data is currently in a comma separated values file called franchise_rev.csv and needs to go into a cube that holds franchise revenue information for all subsidiaries and months for both budget versions.

To accomplish this:

- Create the Franchise_Revenue cube containing the following dimensions:
- Subsidiaries
- Months
- Versions
- Create a TI process to import the data into the Franchise_Revenue cube.
  - Use the SUBST () function to extract specific characters from a column of data in order to match the format of the dimension.
  - This cube will also NOT contain a specified measures dimension. In a later example we will see the difficulty this presents.
- Run the process to load the data.
- Browse the cube to verify that the data loaded correctly.

For more information about where to work and the exercise results, refer to the Tasks and results section that follows. If you need more information to complete a task, refer to earlier demonstrations for detailed steps.

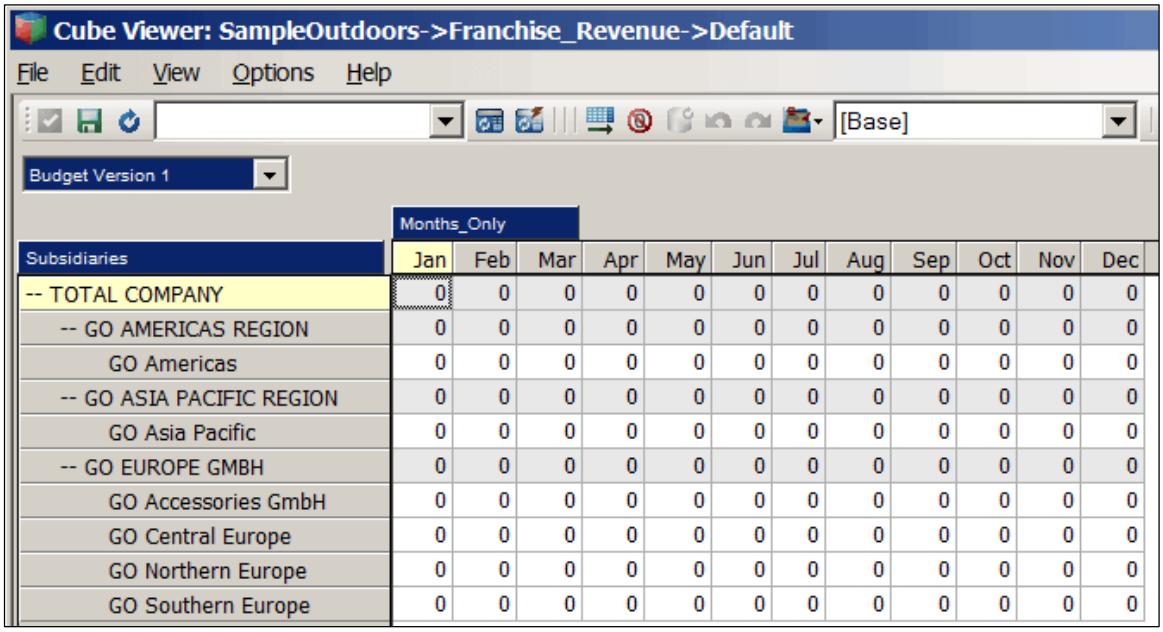
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Exercise 1: Tasks and results

Task 1. Create the Franchise_Revenue cube.

- In Server Explorer, right-click **Cubes** and then click **Create new cube**.
- Name the cube **Franchise_Revenue**.
- Select the **Subsidiaries**, **Months**, and **Versions** dimensions (in that order).
- Click **Create Cube**.
- In Server Explorer, double-click **Franchise_Revenue**.
- Put **Subsidiaries** in the rows, **Months** in the columns and **Versions** in the title area and then click **Recalculate**.
- Select the **Months_Only** subset from the **Subset Editor** and then **Recalculate**.
- Drill-down on the elements in the **Subsidiaries** dimension.

  The result appears as follows:



Subsidiaries	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
-- TOTAL COMPANY	0	0	0	0	0	0	0	0	0	0	0	0
-- GO AMERICAS REGION	0	0	0	0	0	0	0	0	0	0	0	0
GO Americas	0	0	0	0	0	0	0	0	0	0	0	0
-- GO ASIA PACIFIC REGION	0	0	0	0	0	0	0	0	0	0	0	0
GO Asia Pacific	0	0	0	0	0	0	0	0	0	0	0	0
-- GO EUROPE GMBH	0	0	0	0	0	0	0	0	0	0	0	0
GO Accessories GmbH	0	0	0	0	0	0	0	0	0	0	0	0
GO Central Europe	0	0	0	0	0	0	0	0	0	0	0	0
GO Northern Europe	0	0	0	0	0	0	0	0	0	0	0	0
GO Southern Europe	0	0	0	0	0	0	0	0	0	0	0	0

- Save the view as the public **Default** and close the **Cube Viewer**.

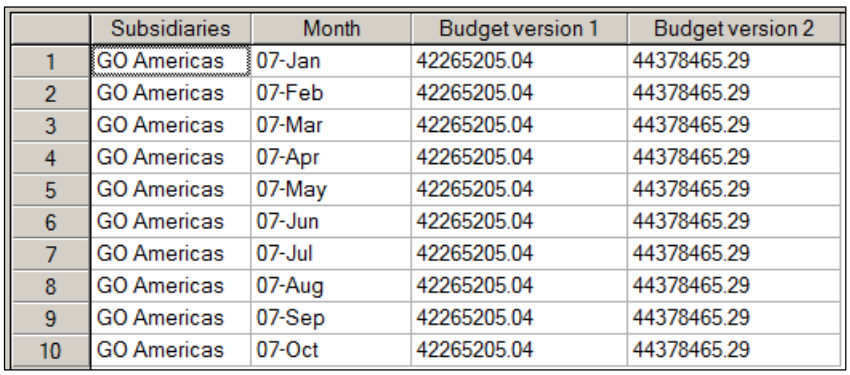
Task 2. Create a process to load data from a .csv file.

- In Server Explorer, right-click **Processes** and then click **Create New Process**.
- On the **Data Source** tab, click **Text**.

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- Browse to **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles\franchise_rev.csv** and then click **Open**.
- Click **OK** to dismiss the warning message.
- In **Number of title records**, type **1**, and then click **Preview**.

  The result appears as follows:



	Subsidiaries	Month	Budget version 1	Budget version 2
1	GO Americas	07-Jan	42265205.04	44378465.29
2	GO Americas	07-Feb	42265205.04	44378465.29
3	GO Americas	07-Mar	42265205.04	44378465.29
4	GO Americas	07-Apr	42265205.04	44378465.29
5	GO Americas	07-May	42265205.04	44378465.29
6	GO Americas	07-Jun	42265205.04	44378465.29
7	GO Americas	07-Jul	42265205.04	44378465.29
8	GO Americas	07-Aug	42265205.04	44378465.29
9	GO Americas	07-Sep	42265205.04	44378465.29
10	GO Americas	07-Oct	42265205.04	44378465.29

Task 3. Define variables.

- Click the **Variables** tab.
- Name and map the variables as follows:

Name	Variable Type	Contents
vSubsidiaries	String	**Element**
vMonth	String	**Other**
vBudget1	Numeric	**Data**
vBudget2	Numeric	**Data**

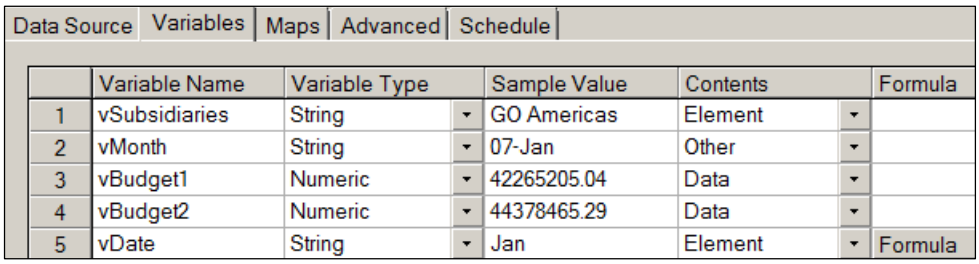
- Click **New Variable**, and then rename the **V5** variable to **vDate**.
- Click the **Formula** button for the **vDate** variable.
- In the **Process Variable Formula** box type, **vDate=SUBST(vMonth, 4, 3);**

  This will return just the 3 characters in the vMonth variable starting with character number 4 (inclusive).
- In the **Destination** section, select the **Data** button.
- Click **Evaluate**, and then click **OK**.

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- For **vDate**, under **Variable Type** select **String**, and under **Contents** select **Element**.

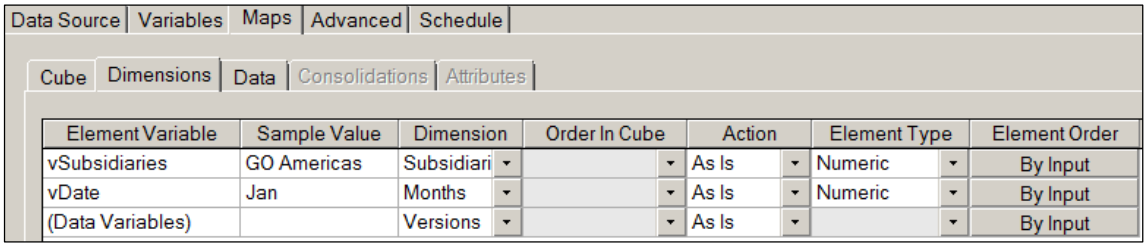
  The result appears as follows:



Task 4. Map variables to the Franchise_Revenue cube.

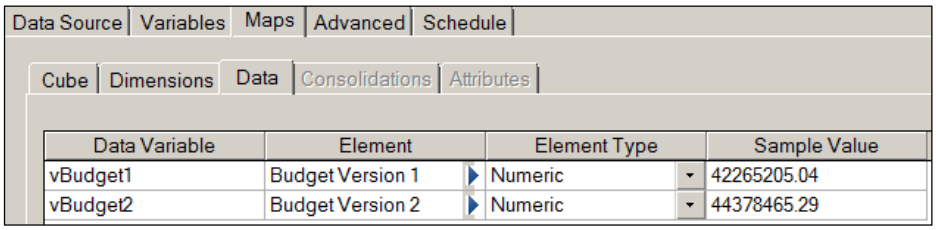
- Click the **Maps** tab.
- Click **Update Cube** and then, in the **CubeName** list, select **Franchise_Revenue**.
- Click the **Dimensions** sub-tab.
- Next to **vSubsidiaries**, in the **Dimension** column, click **Subsidiaries**.
- Next to **vDate**, in the **Dimension** column, click **Months**.
- Next to **(Data Variables)**, in the **Dimension** column, click **Versions**.

  The result appears as follows:



- Click the **Data** sub-tab.
- Under **Element**, map **vBudget1** to **Budget Version 1**.
- Under **Element**, map **vBudget2** to **Budget Version 2**.

  The result appears as follows:



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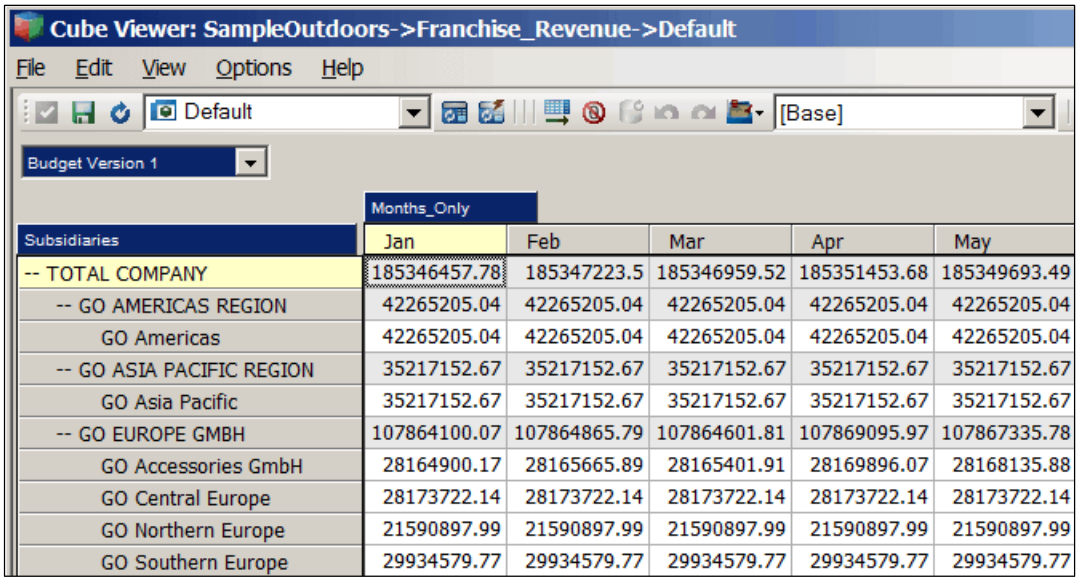
Task 5. Save and run the process.

- Click the **Advanced** tab and all tabs below.
- Save the process as **LoadFranchiseRevenueCSV**.
- Run the process, click **OK** to close the confirmation message, and then close **TurboIntegrator**.

Task 6. Browse the Franchise_Revenue cube.

- Double-click the **Franchise_Revenue** cube.
- Recalculate the view (if necessary).

  The result appears as follows:



Subsidiaries	Jan	Feb	Mar	Apr	May
-- TOTAL COMPANY	185346457.78	185347223.5	185346959.52	185351453.68	185349693.49
-- GO AMERICAS REGION	42265205.04	42265205.04	42265205.04	42265205.04	42265205.04
GO Americas	42265205.04	42265205.04	42265205.04	42265205.04	42265205.04
-- GO ASIA PACIFIC REGION	35217152.67	35217152.67	35217152.67	35217152.67	35217152.67
GO Asia Pacific	35217152.67	35217152.67	35217152.67	35217152.67	35217152.67
-- GO EUROPE GMBH	107864100.07	107864865.79	107864601.81	107869095.97	107867335.78
GO Accessories GmbH	28164900.17	28165665.89	28165401.91	28169896.07	28168135.88
GO Central Europe	28173722.14	28173722.14	28173722.14	28173722.14	28173722.14
GO Northern Europe	21590897.99	21590897.99	21590897.99	21590897.99	21590897.99
GO Southern Europe	29934579.77	29934579.77	29934579.77	29934579.77	29934579.77

- Close the **Cube Viewer** (saving the view and overwriting the existing view if prompted).
- Right-click the **SampleOutdoors** node and click **Save Data**.
- Close all open windows.

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IBM Training

Add business rules

IBM Planning Analytics (v2.0)

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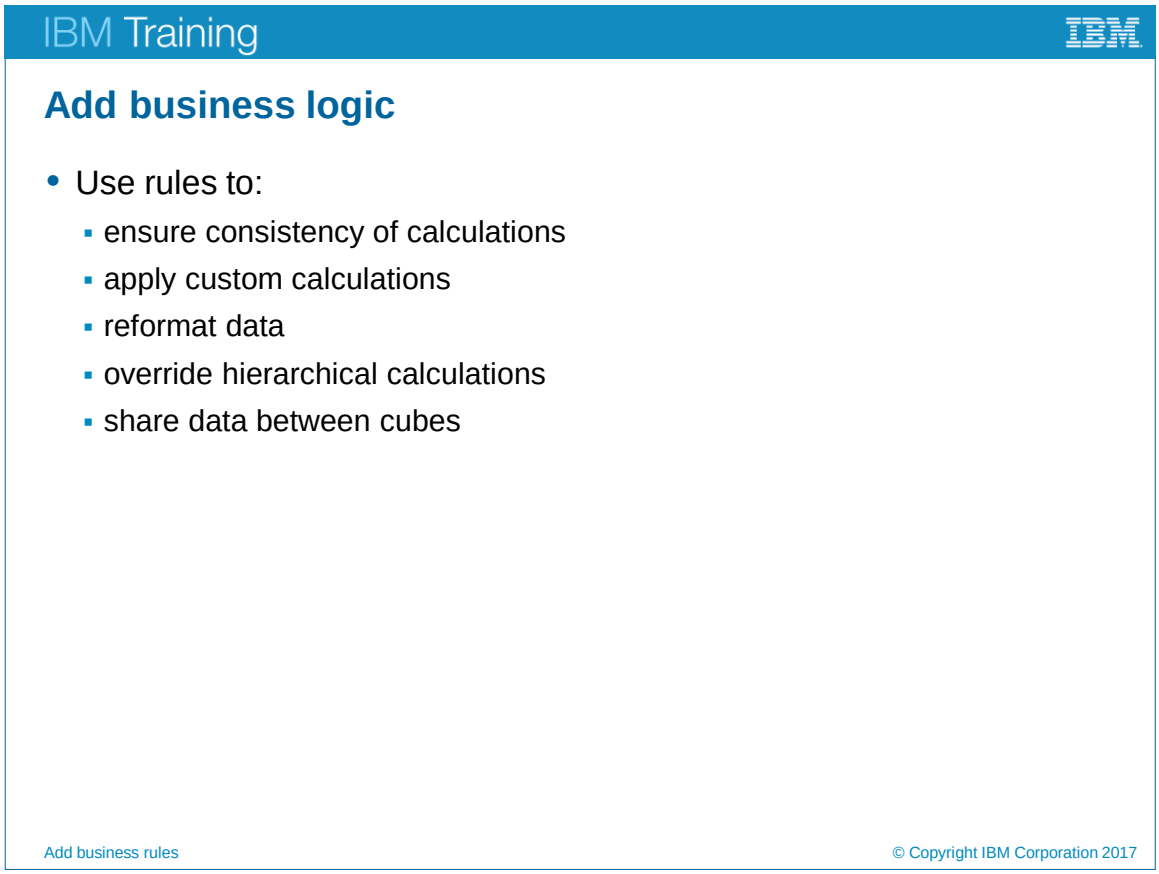
Unit objectives

- Describe a rule
- Construct rules for elements or consolidations
- Use functions in rules
- Optimize rule performance
- Describe rules in a rule worksheet

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Unit objectives

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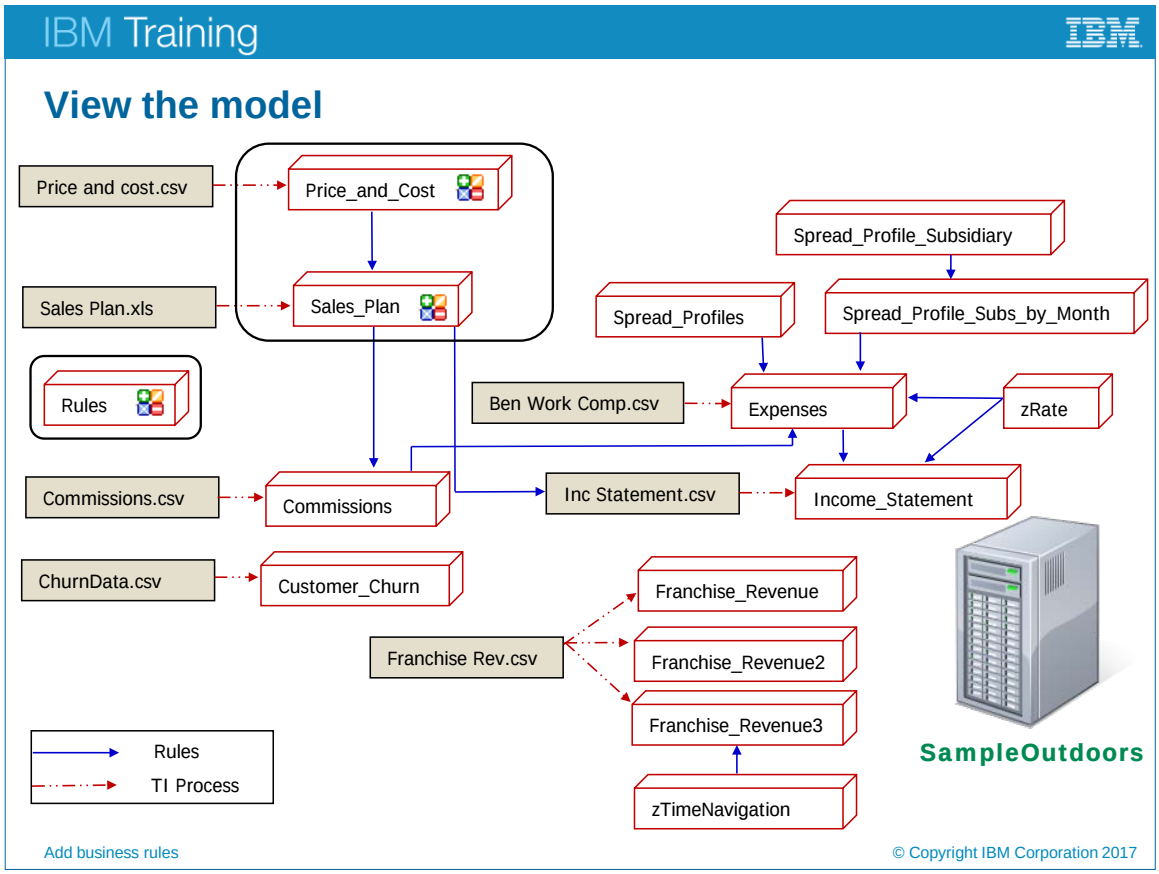


Add business logic

Business logic can be created and stored with the cube on a server by using rules. This ensures that all users have the same calculations in their analysis.

Rules are stored in text files with the extension *.rux. In this unit you will use the TM1 Rules Editor to create and modify rules for your cubes.

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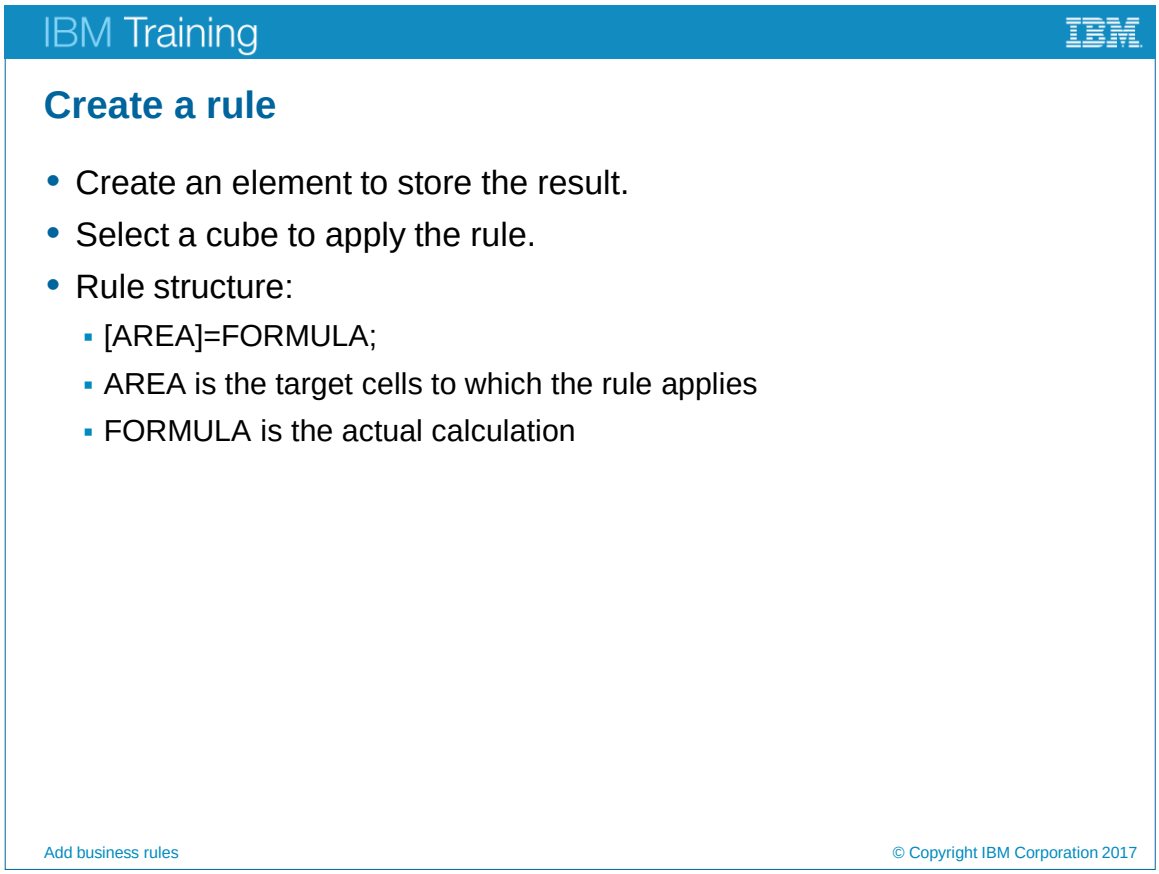
View the model

In this unit you will create a new cube called Rules. It will be used to practice techniques and syntax for business rules attached to cubes.

You will add rules to the following cubes:

- Rules
- Price_and_Cost
- Sales_Plan

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Create a rule

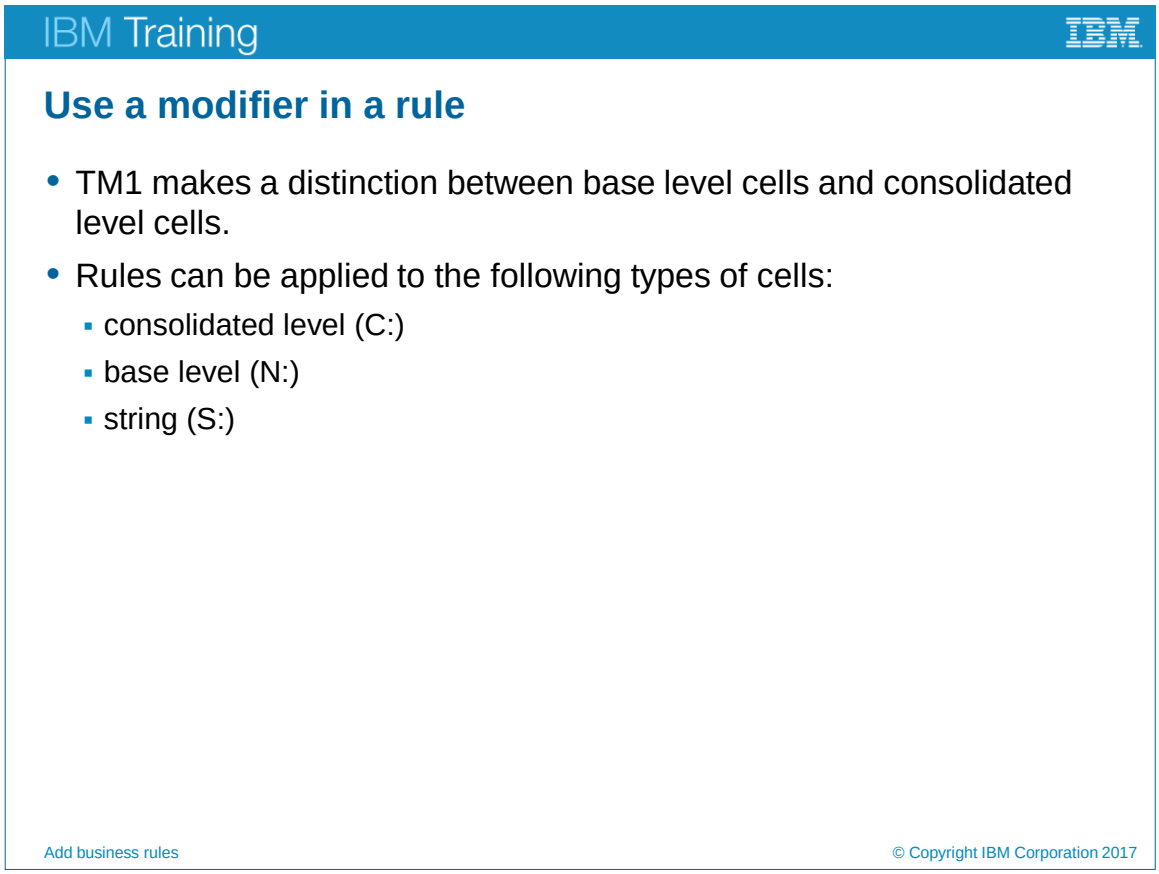
The area definition tells TM1 what values should be calculated by this rule. It restricts the calculation so that it only applies to certain areas of the cube. The calculation can be restricted so that it applies to just one cell in the cube, but more typically we will restrict the calculation to members in one or two of the dimensions. TM1 will then automatically apply the calculation to all the members of the other dimensions.

Using areas, we can specify different calculations for different parts of the cube. Specify elements using the following guides:

- A list: [{'January', 'February', 'March'}]
- All elements: []
- A specific cell: ['May', 'London', 'Price']

While area definitions are powerful, they do not allow you to identify a subset of members from a dimension. If you want a rule to apply to four types of fish, you will need to write four separate rules.

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Use a modifier in a rule

If the N: modifier is used, the rule will only be applied at the base level. The values of cells at higher levels will be calculated according to the consolidation hierarchies defined in the cube's dimensions. In most cases, if a calculation can be performed just at the base level, it is more efficient to allow TM1 to simply consolidate the results, rather than performing the calculation at every level of the hierarchies.

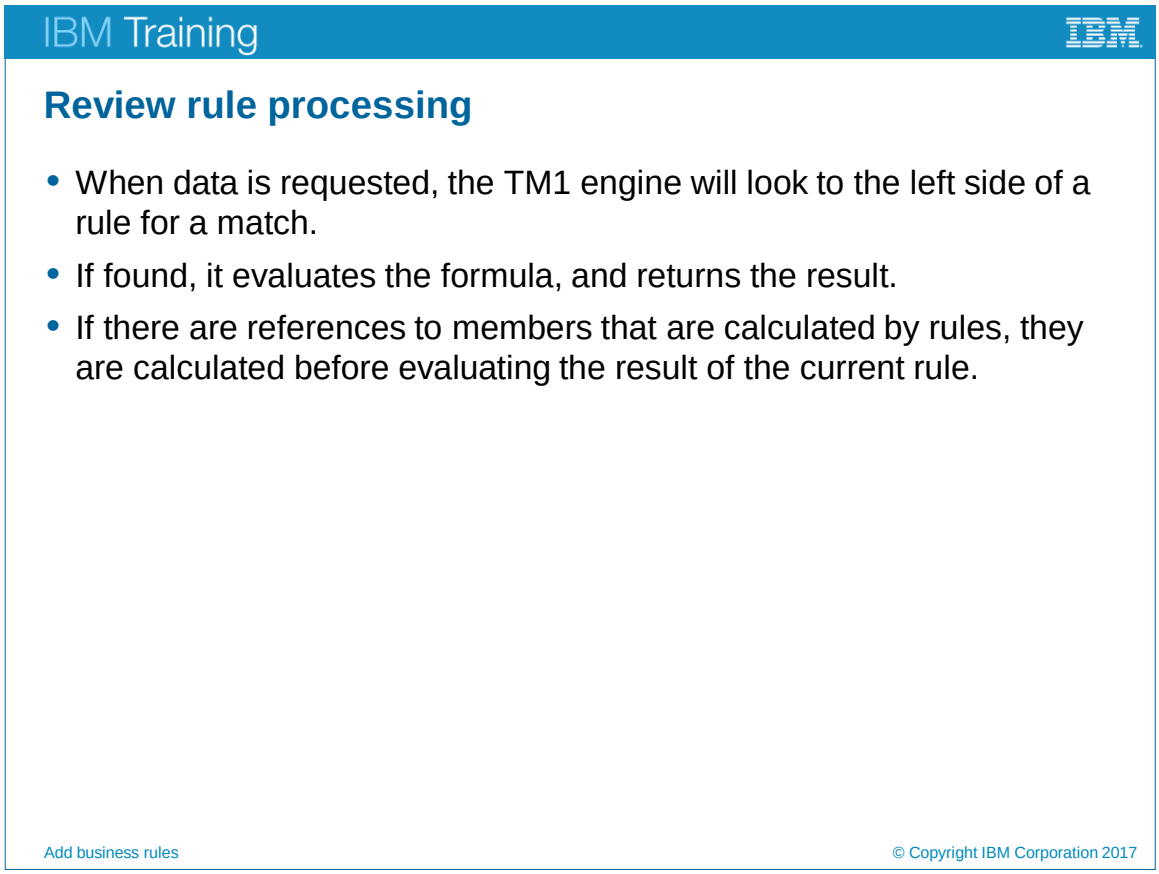
If the C: modifier is used, the calculation will be applied to consolidated cells only. This can be useful to override the normal consolidation that the dimension hierarchies provide. However, this should be used with care to avoid over complicating the model. A suitable use of the C: modifier in our example might be to prevent the price from being consolidated:

['Price']=C:0;

The lack of a modifier indicates that the calculation is to be applied at all levels. This is necessary for calculations such as ratios, percentages and averages, which usually cannot be consolidated and must be performed at each level of the dimensions' hierarchies.

There is another modifier, S:, which is used when the rule is being used to 'calculate' a string (text) value instead of a numeric value.

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Review rule processing

The first rule with an area definition that matches the requested data will be used. Any subsequent rules with area definitions that also match the requested data will not be evaluated. You should take advantage of this by placing rules with more specific area definitions ahead of rules with more general area definitions. In this way, you can use a more specific rule to handle an exceptional case, while the other cases drop through to the more general rule.

For example, if Gross Profits for New York were calculated differently from the rest, you would need to put the New York rule first, and the general rule for Gross profit last.

['New York','Sales'] = ['Units Sold']*['New York','Price'];

['Sales'] = ['Units Sold']*['Price'];

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IBM Training

Demonstration 1

Create simple rules

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Demonstration 1: Create simple rules

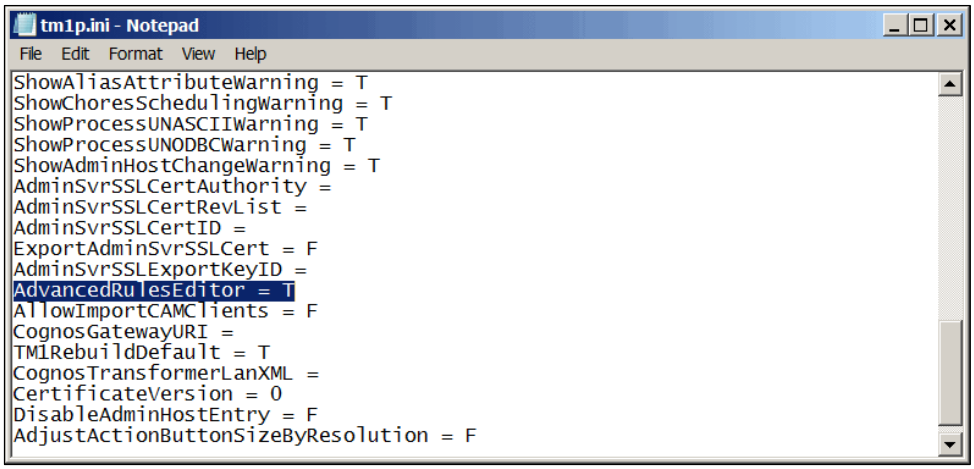
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Demonstration 1: Create simple rules

Purpose:
You recognize that the rules are performed in the order they are written, and that a cell has a rule applied to it only once. To see the impact of rules on a cube, you will create a cube called Rules. The cube will contain rules for leaf level elements, consolidations, and string elements.

Task 1. Enable the Advanced Rules Editor.

1. Open **Windows Explorer**.
2. In the top left corner, click **Organize**, and then click **Folder and search options**.
3. Click the **View** tab and select the **Show hidden files, folders, and drives** button - if it is not already selected, and then click **OK**.
4. Navigate to **C:\Users\Student\AppData\Roaming\Applix\TM1** and open **tm1p.ini** in Notepad.
5. Change the value of the **AdvancedRulesEditor** property from F to **T**:



   You can use the Simple Rules Editor by leaving this parameter set to F. However, the steps to create rules will be different than those outlined below, since the Rules Editor will not have as much functionality in "simple" mode.

   Although the Advanced Rules Editor is no longer officially supported after version 10.1.1 of TM1, it is functional in your environment. For more information, see the following Technote: http://www-01.ibm.com/support/docview.wss?uid=swg21615331.
6. Save and close the file and then close **Windows Explorer**.

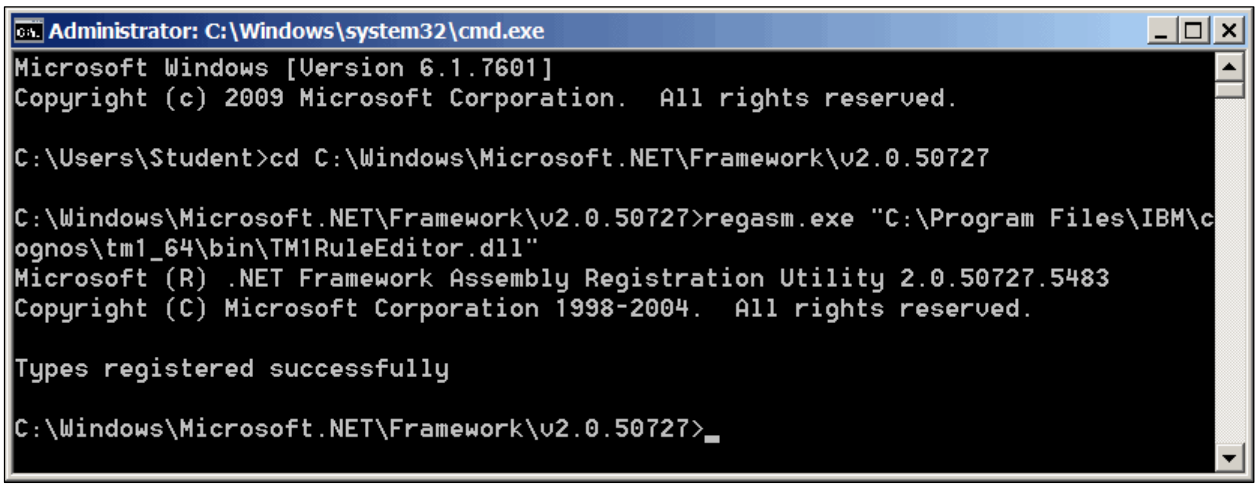
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7. In the **taskbar**, click the **Services** button  .
8. Right-click **IBM Cognos TM1 Admin Server x64**, click **Restart**, and then click **Yes** when prompted to restart the other services.
9. After the services have restarted, close the **Services** window.

Task 2. Create a cube to contain the rules.

To work with the advanced rules editor, you must first open the 32-bit version of Architect. You will also need to register a .dll file.

1. In the bottom left corner, click the **Start** button, in the **Search programs and files** box, type **cmd**, and then press **Enter**.
2. At the command prompt, type the following:
3. **cd C:\Windows\Microsoft.NET\Framework\v2.0.50727**
4. Press **Enter**.
5. At the command prompt, type the following:
   regasm.exe "C:\Program Files\IBM\cognos\tm1_64\bin\TM1RuleEditor.dll"
6. Press **Enter**.
   The result appears as follows:



7. Close the command window.
8. Open **Windows Explorer** and navigate to **C:\Program Files\IBM\cognos\tm1_64\bin**.

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9. Double-click **tm1a.exe**.
10. Maximize the window.
11. Expand **TM1**, double-click **SampleOutdoors**, and then log on with a user ID of **admin** and a password of **apple**.
12. In Server Explorer, right-click **Cubes** and then click **Create new cube**.
13. In the **Cube Name** box, type **Rules**.
14. Under Available Dimensions, double-click **Months** and **Price_and_Cost_Measures**.

    Typically you would create a new measures dimension for the new cube. However, this cube is only being used for demonstrations and is not part of the overall model.
15. Click **Create Cube**.
16. Double-click the new **Rules** cube to open it in the Cube Viewer.
17. Click **Recalculate**.

    The result appears as follows:

	Price_and_Cost_Measures				
Months	Unit Sale Price	Unit Cost	Margin	Margin%	Comment
+ Total Year	$0.00	$0.00	$0.00	0.00	

    It is often helpful to open a view of the cube so you may immediately see the result of the rule. You can arrange the windows to display the view and the rule at the same time.

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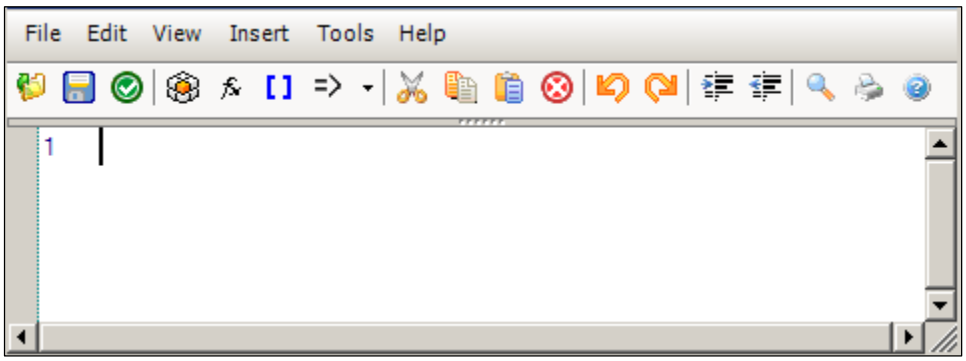
Task 3. Create a rule to set Unit Sale Price to 1 for all values including consolidations.

1. Expand **Months** to display **Total Year**, **Q1**, **Jan**, **Feb**, and **Mar**.

	Price_and_Cost_Measures				
Months	Unit Sale Price	Unit Cost	Margin	Margin%	Comment
-- Total Year	$0.00	$0.00	$0.00	0.00	
-- Q1	$0.00	$0.00	$0.00	0.00	
Jan	$0.00	$0.00	$0.00	0.00	
Feb	$0.00	$0.00	$0.00	0.00	
Mar	$0.00	$0.00	$0.00	0.00	
+ Q2	$0.00	$0.00	$0.00	0.00	
+ Q3	$0.00	$0.00	$0.00	0.00	
+ Q4	$0.00	$0.00	$0.00	0.00	

2. In the Server Explorer window, right-click **Rules** and then click **Create Rule**.

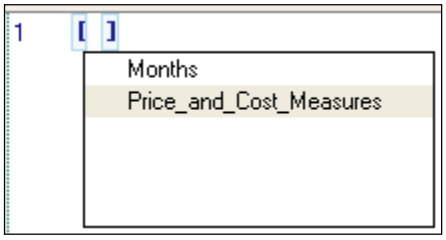
   The Advanced Rules Editor should now be open (this may take a few moments). The cursor should be blinking on the first row of the editor next to the number **1**. This is where you will create your rules statements.



   In the Advanced Rules Editor, line numbers are displayed for each row. You may use  to indent lines for better organization. You may also Undo and Redo using  . Other tools will be reviewed as they are needed.

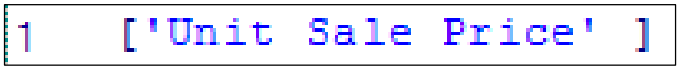
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3. Click **Brackets** [] and then double-click **Price_and_Cost_Measures**.



4. Click **Unit Sale Price**, and then click **OK**.

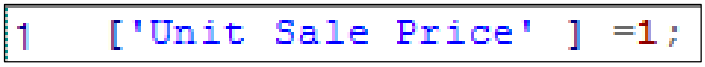
   The result appears as shown below:



   This is the area of the cube to populate with our rule.

5. At the end of **line 1**, type **= 1;**.

   All rules statements must end with a semicolon (;).



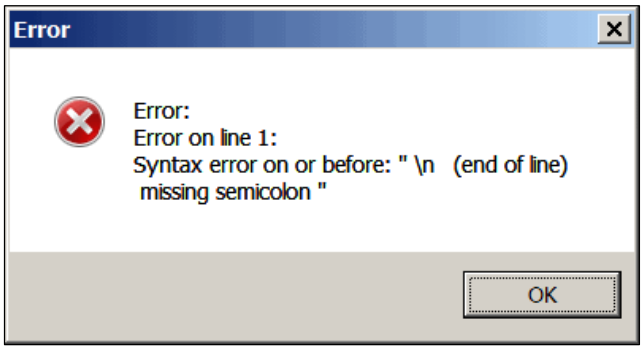
   Unit Sale Price is blue indicating that it is a string. Strings must be enclosed in single quotes ('). In the Advanced Rules Editor, numbers appear in red.

6. Click **Check Syntax**  .

   Check Syntax will check your rule for any syntax errors and indicate the row number of an error. It is common to forget to place a semicolon after a rule. This would generate a syntax error.

7. Delete the semicolon (**;**) at the end of **line 1**.

8. Click **Check Syntax**  .



   This is an example of a syntax error message generated in the Rules Editor.

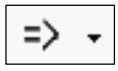
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9. Click **OK** to dismiss the error message.
10. Click at the end of **line 1**, and then type a semicolon (**;**).
11. Click **Check Syntax** and then click **Save**  .
12. Navigate back to the **Cube Viewer** and click **Recalculate**.
    The result appears as shown below:

	Price_and_Cost_Measures				
Months	Unit Sale Price	Unit Cost	Margin	Margin%	Comment
-- Total Year	$1.00	$0.00	$0.00	0.00	
-- Q1	$1.00	$0.00	$0.00	0.00	
Jan	$1.00	$0.00	$0.00	0.00	
Feb	$1.00	$0.00	$0.00	0.00	
Mar	$1.00	$0.00	$0.00	0.00	
+ Q2	$1.00	$0.00	$0.00	0.00	
+ Q3	$1.00	$0.00	$0.00	0.00	
+ Q4	$1.00	$0.00	$0.00	0.00	

All cells in the Unit Sale Price column contain $1.00. This rule applies to all levels of the Unit Sale Price, overriding any consolidations.

Task 4. Apply the rule to only the leaf level cells.

1. Navigate to the **Rules Editor**.
2. Double-click the equals sign and then click **Delete**  to remove the equals sign.
3. Click the down arrow of **Insert Qualifier or Operator**  , and then click **=N:**.

```
1   ['Unit Sale Price' ]  = N: 1;
```

4. Click **Check Syntax** and then click **Save**.

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5. Click the **Cube Viewer** and then click **Recalculate**.

   The result appears as shown below:

	Price_and_Cost_Measures				
Months	Unit Sale Price	Unit Cost	Margin	Margin%	Comment
-- Total Year	$12.00	$0.00	$0.00	0.00	
-- Q1	$3.00	$0.00	$0.00	0.00	
Jan	$1.00	$0.00	$0.00	0.00	
Feb	$1.00	$0.00	$0.00	0.00	
Mar	$1.00	$0.00	$0.00	0.00	
+ Q2	$3.00	$0.00	$0.00	0.00	
+ Q3	$3.00	$0.00	$0.00	0.00	
+ Q4	$3.00	$0.00	$0.00	0.00	

   Now you see that $1.00 appears only in the leaf (**n**) level cells and that the consolidations are performed in the hierarchy by the aggregation engine. By using the N: after the equal sign, you have told TM1 to apply the rule only to the lowest level in the hierarchy and to use the consolidation engine for the consolidations.

   Also notice the calculated cells are gray, indicating that data may not be entered into these cells. Rules-driven cells will not accept data entry.

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Task 5. Apply a rule to both the leaf and consolidation levels.

1. In the **Rules Editor**, click after the semicolon and type **C: 20;**.
   The result appears as shown below:

```
1   ['Unit Sale Price' ]  = N: 1; C: 20;
```

2. Click **Save**, and then recalculate the view.
   The result appears as shown below:

	Price_and_Cost_Measures				
Months	Unit Sale Price	Unit Cost	Margin	Margin%	Comment
-- Total Year	$20.00	$0.00	$0.00	0.00	
-- Q1	$20.00	$0.00	$0.00	0.00	
Jan	$1.00	$0.00	$0.00	0.00	
Feb	$1.00	$0.00	$0.00	0.00	
Mar	$1.00	$0.00	$0.00	0.00	
+ Q2	$20.00	$0.00	$0.00	0.00	
+ Q3	$20.00	$0.00	$0.00	0.00	
+ Q4	$20.00	$0.00	$0.00	0.00	

   All consolidated values are $20.00 and the leaf level cells are $1.00. The rule overrides the consolidation engine.

3. In the **Rules Editor**, on the **second line**, type the following rule:
   ['Q1', 'Unit Sale Price'] = 7;
   The result appears as follows:

```
1   ['Unit Sale Price' ]  = N: 1; C: 20;
2   ['Q1', 'Unit Sale Price'] = 7;
```

   This rule is more specific because it identifies only the Q1 cells for Unit Sale Price.

4. Click **Check Syntax**, and the click **Save** in the Rules Editor.

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5. In the **Cube Viewer**, click **Recalculate**.

   The result appears as follows:

	Price_and_Cost_Measures				
Months	Unit Sale Price	Unit Cost	Margin	Margin%	Comment
-- Total Year	$20.00	$0.00	$0.00	0.00	
-- Q1	$20.00	$0.00	$0.00	0.00	
Jan	$1.00	$0.00	$0.00	0.00	
Feb	$1.00	$0.00	$0.00	0.00	
Mar	$1.00	$0.00	$0.00	0.00	
+ Q2	$20.00	$0.00	$0.00	0.00	
+ Q3	$20.00	$0.00	$0.00	0.00	
+ Q4	$20.00	$0.00	$0.00	0.00	

   The result did not change because of the order in which rules are calculated. Once a rule applies to a cell, TM1 does not look for any other calculation for that cell. Instead, it calculates other cells. This allows cells to be calculated quicker. It also means you should put more specific rules before generic rules in the editor.

6. In the **Rules Editor**, insert an extra row at the top.
7. Select the **['Q1'. 'Unit Sale Price']** line, click **Edit** and then click **Cut**.
8. Click on the first row, click **Edit** and then click **Paste**.

   The result appears as shown below:

```
1   ['Q1', 'Unit Sale Price'] = 7;
2   ['Unit Sale Price' ]  = N: 1; C: 20;
```

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9. Save the rule, and then recalculate the view.

   The result appears as shown below:

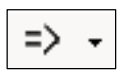
	Price_and_Cost_Measures				
Months	Unit Sale Price	Unit Cost	Margin	Margin%	Comment
-- Total Year	$20.00	$0.00	$0.00	0.00	
-- Q1	$7.00	$0.00	$0.00	0.00	
Jan	$1.00	$0.00	$0.00	0.00	
Feb	$1.00	$0.00	$0.00	0.00	
Mar	$1.00	$0.00	$0.00	0.00	
+ Q2	$20.00	$0.00	$0.00	0.00	
+ Q3	$20.00	$0.00	$0.00	0.00	
+ Q4	$20.00	$0.00	$0.00	0.00	

   By changing the order of the rules, the more specific rule is calculated first and then the more generic rule is applied to all the "other" cells. Since Q1 for Unit Sale Price was calculated first, it is not recalculated as a part of the second rule. This allows TM1 to be more efficient in the calculations and saves resources.

10. Save the view as **Default**.

Task 6. Populate a string element with a rule.

Previously you populated the Comment element using a Pick List attribute on the Products dimension. The Rules cube does not contain the Products dimension so that pick list is not available in this cube. Instead you will create a rule to populate the Comment.

1. Expand the **Rules** cube and double-click the **Rules** icon  to open the **Rules Editor** (if it is not already open).
2. Press **Enter** after the last rule to create a new line (if necessary).
3. On **line 3**, click **Brackets**  , and then double-click **Price_and_Cost_Measures**.
4. Click **Comment**, and then click **OK**.
5. Click outside the brackets on **line 3**, click **Insert Qualifier or Operator**  and then click **= S:**.

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6. Type **'Winter';** (including quotes).

   The result appears as follows:

```
1   ['Q1', 'Unit Sale Price'] = 7;
2   ['Unit Sale Price' ]  = N: 1; C: 20;
3   ['Comment' ]  = S: 'Winter';
```

   If you are comparing two string elements ('A'='B'), you must use @= instead of =.

7. Click **Check Syntax** and if there are no errors, click **Save** and then recalculate the view.

   The result appears as follows:

	Price_and_Cost_Measures				
Months	Unit Sale Price	Unit Cost	Margin	Margin%	Comment
-- Total Year	$20.00	$0.00	$0.00	0.00	Winter
-- Q1	$7.00	$0.00	$0.00	0.00	Winter
Jan	$1.00	$0.00	$0.00	0.00	Winter
Feb	$1.00	$0.00	$0.00	0.00	Winter
Mar	$1.00	$0.00	$0.00	0.00	Winter
+ Q2	$20.00	$0.00	$0.00	0.00	Winter
+ Q3	$20.00	$0.00	$0.00	0.00	Winter
+ Q4	$20.00	$0.00	$0.00	0.00	Winter

   All cells now show a Comment of Winter. When the result of a rule is a string, it must be a string element. You must also specify a string rule using S:. Remember, a string element must reside in the last dimension in a cube in order to be stored properly. If you make your measures dimension the last dimension, you will always know where to put the string elements.

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Task 7. Add logic to a rule.

1. In the **Rules Editor**, click the end of the string rule and delete **'Winter';**.
   You will add logic to this rule that will evaluate the Unit Sale Price and display 'Summer' when the price is over 5 and 'Winter' when the price is under 5.
2. After the S: type, **IF (['Unit Sale Price'] > 5, 'Summer', 'Winter');**
   Syntax for IF: **IF (test, true_condition, false_condition)**
   The result appears as follows:

```
1   ['Q1', 'Unit Sale Price'] = 7;
2   ['Unit Sale Price' ]  = N: 1; C: 20;
3   ['Comment' ]  = S: IF ( ['Unit Sale Price'] > 5, 'Summer', 'Winter');
```

3. Save the rule and recalculate the view.
   The result appears as follows:

	Price_and_Cost_Measures				
Months	Unit Sale Price	Unit Cost	Margin	Margin%	Comment
-- Total Year	$20.00	$0.00	$0.00	0.00	Summer
-- Q1	$7.00	$0.00	$0.00	0.00	Summer
Jan	$1.00	$0.00	$0.00	0.00	Winter
Feb	$1.00	$0.00	$0.00	0.00	Winter
Mar	$1.00	$0.00	$0.00	0.00	Winter
+ Q2	$20.00	$0.00	$0.00	0.00	Summer
+ Q3	$20.00	$0.00	$0.00	0.00	Summer
+ Q4	$20.00	$0.00	$0.00	0.00	Summer

   You will now exclude the Q1 cell from this rule. To exclude an area from a rule, use the STET function immediately before the rule.
   Syntax for STET is: **STET;** (it has no parameters or brackets).
4. Click the beginning of **line 3** and press **Enter** to insert a blank line.
5. On the blank line type, **['Q1'] = STET;**
   The result appears as follows:

```
1   ['Q1', 'Unit Sale Price'] = 7;
2   ['Unit Sale Price' ]  = N: 1; C: 20;
3   ['Q1'] = STET;
4   ['Comment' ]  = S: IF ( ['Unit Sale Price'] > 5, 'Summer', 'Winter');
```

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6. Save the rule and recalculate the view.

   The result appears as follows:

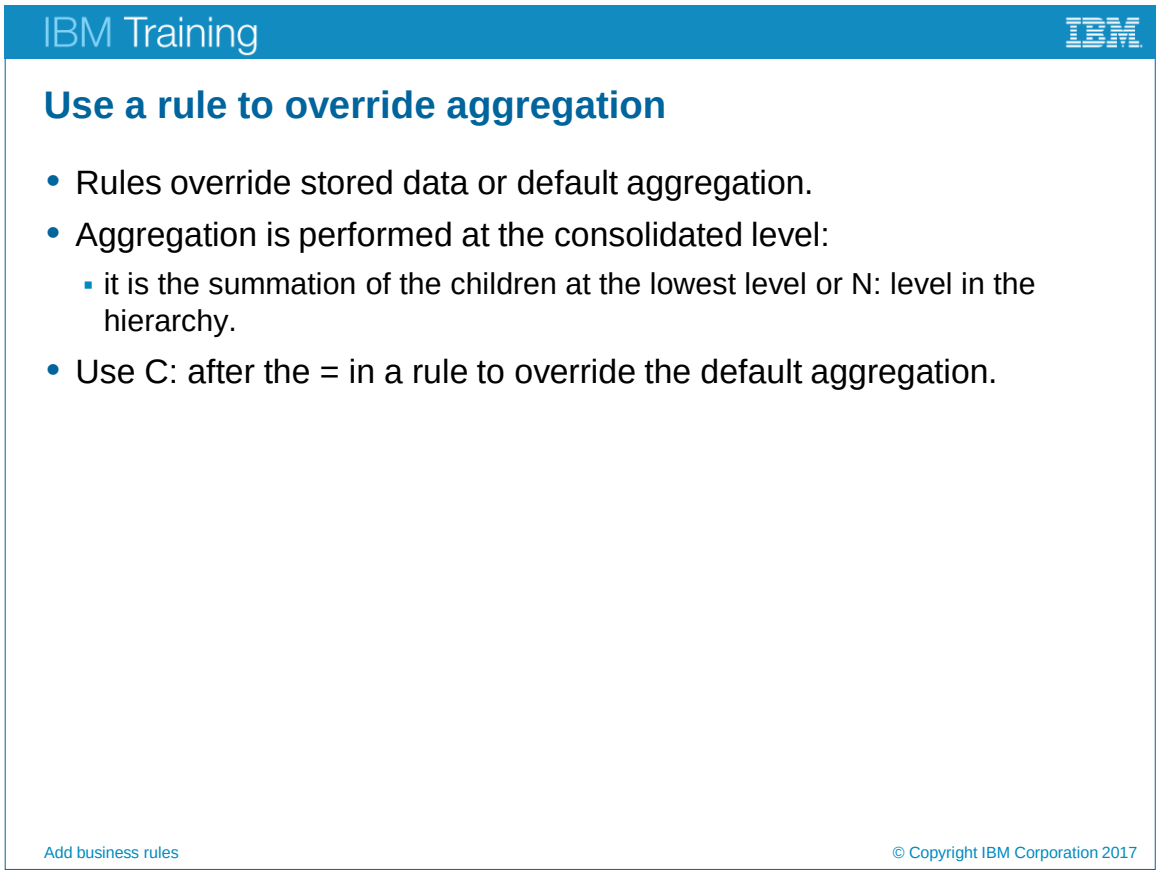
	Price_and_Cost_Measures				
Months	Unit Sale Price	Unit Cost	Margin	Margin%	Comment
-- Total Year	$20.00	$0.00	$0.00	0.00	Summer
-- Q1	$7.00	$0.00	$0.00	0.00	
Jan	$1.00	$0.00	$0.00	0.00	Winter
Feb	$1.00	$0.00	$0.00	0.00	Winter
Mar	$1.00	$0.00	$0.00	0.00	Winter
+ Q2	$20.00	$0.00	$0.00	0.00	Summer
+ Q3	$20.00	$0.00	$0.00	0.00	Summer
+ Q4	$20.00	$0.00	$0.00	0.00	Summer

   No rule was applied to the Q1 cell for Comment. String cells cannot be aggregated so no value is displayed.

7. Close the **Rules Editor** and then close the **Cube Viewer**, clicking **No** if asked to save changes to the view.
8. In **Server Explorer**, right-click the **SampleOutdoors** server and click **Save Data**.
9. Leave Architect open for the next demonstration.

Results:
To see the impact of rules on a cube, you created a cube called Rules. The cube contained rules for leaf level elements, consolidations, and string elements. Next you will use cubes that make up the SampleOutdoors model to display the results of your business rules.

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Use a rule to override aggregation

When a dimension is created, TM1 automatically aggregates the lowest levels according to the dimension hierarchy. Sometimes it is necessary to override the default aggregation with custom calculations or no calculations.

One example is a Price of a Product that should not be summed along the dimension, but instead display a zero or an average calculation.

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IBM Training

Demonstration 2

Create a rule to suppress a consolidation

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Demonstration 2: Create a rule to suppress a consolidation

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Demonstration 2: Create a rule to suppress a consolidation

Purpose:
You do not want unit price values to be summed or calculated at the consolidation level. You will add elements to contain the values for calculating average price at the leaf level. Next, at the consolidation level, you will create the average calculation using your leaf-level calculations.

Task 1. Create elements to hold calculations.

1. In Server Explorer, under the **Dimensions** node of the **SampleOutdoors** server, right-click **Price_and_Cost_Measures** and select **Edit Dimension Structure**.

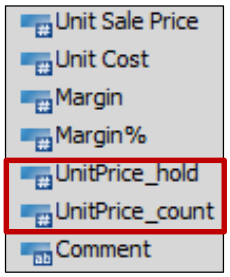
   The calculation for average price at the consolidated level = (total price for elements)/(total number of prices). You have the total price for elements in the consolidated elements for Unit Sale Price, so you need to retain that information in a holder element. You will override the consolidations for Unit Sale Price with the result of the rule.

   You do not currently have an element that indicates how many elements have prices so you will create a count element and assign the value of 1 where a price is found.

   You will create two new elements to hold data for your calculation.
2. Click **Margin%**, and then from the **Edit** menu, click **Insert Element**.
3. Add two new simple elements:

   - **UnitPrice_hold**
   - **UnitPrice_count**

   The results appear as follows:

4. Click **Save**.
5. Close the **Dimension Editor**.

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Task 2. Create the rule.

1. Expand the **Cubes** node (if necessary), and then expand the **Price_and_Cost** and **Views** nodes.
2. Open the **Default** view for the **Price_and_Cost** cube.
3. Swap **Months_Only** and **Price_and_Cost_Measures** (currently displaying Unit Sale Price).
4. Click **Price_and_Cost_Measures** (in the column area) to open the Subset Editor.
5. In the **Subset Editor**, click **All**  , Ctrl+click **Unit Sale Price**, **UnitPrice_count** and **UnitPrice_hold**, click **Keep**  , and then click **OK**.
6. Click **Recalculate**.
7. Click **CAMPING EQUIMPMENT** to expand this element.
8. Ensure that **Budget Version 1** is selected, and then click **Recalculate** (if necessary).

   The result appears as shown below:

Golf Shop	Budget Version 1	Jan	
	Price_and_Cost_Measures		
Products	Unit Sale Price	UnitPrice_hold	UnitPrice_count
-- TOTAL PRODUCT	$3,497.35	0	0
-- CAMPING EQU	$802.04	0	0
Cooking Gea	$40.31	0	0
Lanterns	$29.12	0	0
Packs	$175.37	0	0
Sleeping Bag	$103.83	0	0
Tents	$453.41	0	0
+ GOLF EQUIPM	$1,724.56	0	0
+ MOUNTAINEEI	$454.08	0	0
+ OUTDOOR PR	$22.29	0	0
+ PERSONAL AC	$494.38	0	0

9. Save the view as **AverageRule**.
10. In the **Server Explorer** window, right-click the **Price_and_Cost** cube and then click **Create Rule**.

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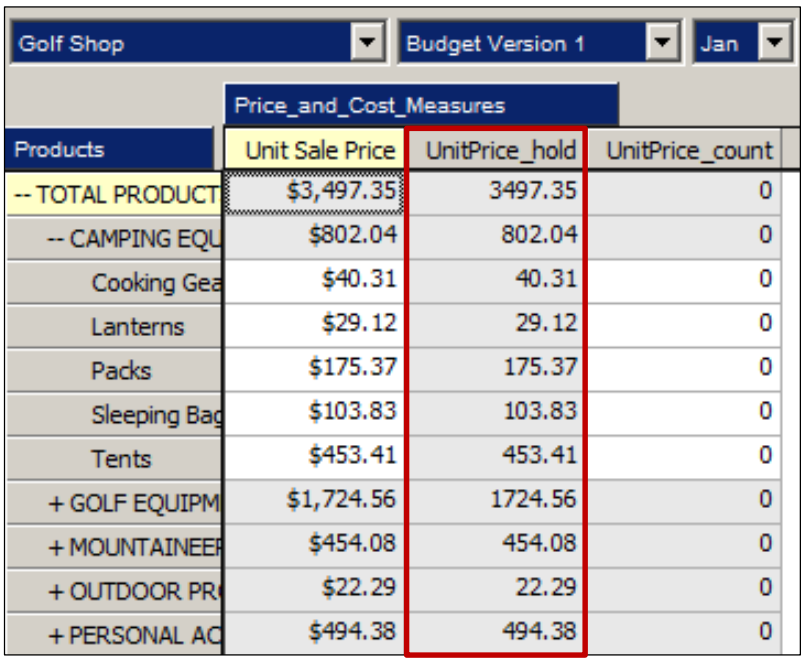
11. On **line 1** type, **# hold the Unit Sale Price for the average price**.

    The # indicates that the rest of the line is a comment. It will be ignored even if it contains a calculation. This is useful for documenting the following lines or to temporarily disable a rule.
12. Press **Enter** to add another line, click **Brackets**, and then double-click **Price_and_Cost_Measures**.
13. Click **UnitPrice_hold**, and then click **OK** to return to the Rules Editor.
14. Click to the right of the **]**, click the down arrow beside the **Insert Qualifier or Operator** icon, and then click **=N:**.
15. Click **Brackets**, and then double-click **Price_and_Cost_Measures**.
16. Click **Unit Sale Price** and then click **OK**.
17. Click to the right of the last bracket and then type a semicolon.

```
1   # hold the Unit Sale Price for the average price
2   ['UnitPrice_hold' ]   = N: ['Unit Sale Price' ] ;
```

18. Click **Check Syntax** and correct any syntax errors.
19. Click **Save** and recalculate the view.

    The result appears as follows:



Golf Shop	Budget Version 1	Jan	
	Price_and_Cost_Measures		
Products	Unit Sale Price	UnitPrice_hold	UnitPrice_count
-- TOTAL PRODUCT	$3,497.35	3497.35	0
-- CAMPING EQU	$802.04	802.04	0
Cooking Gea	$40.31	40.31	0
Lanterns	$29.12	29.12	0
Packs	$175.37	175.37	0
Sleeping Bag	$103.83	103.83	0
Tents	$453.41	453.41	0
+ GOLF EQUIPM	$1,724.56	1724.56	0
+ MOUNTAINEEI	$454.08	454.08	0
+ OUTDOOR PR	$22.29	22.29	0
+ PERSONAL AC	$494.38	494.38	0

The UnitPrice_hold column has the same values as the Unit Sale Price column (notice that you have not applied formatting to the UnitPrice_hold element). Next you want to create a counter that indicates the number of items that are being counted in the Unit Sale Price.

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Task 3. Create a counter.

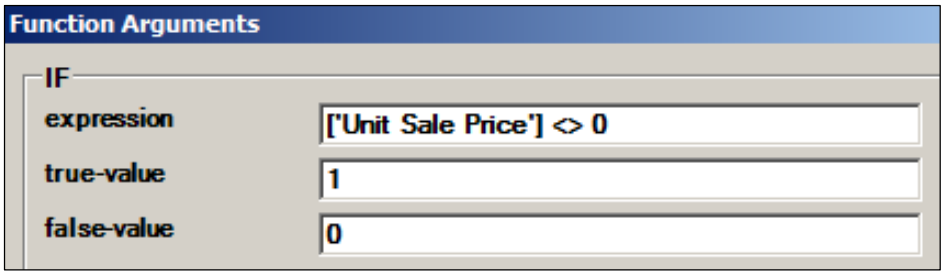
1. On **line 3** of the **Rules Editor**, click **Brackets**, and then double-click **Price_and_Cost_Measures**.
2. Click **UnitPrice_count** and then click **OK**.
3. After the brackets type **=N:**
4. Click **Insert Function**  .

   You will create a function that evaluates the Unit Sale Price. When there is a value, the function enters the number 1 in the corresponding cell; if there is no value, it enters 0 in the cell. You will use this function in another calculation to summarize how many cells contain prices, and also use it in your average price calculation. To do this you will use an IF statement.

   SYNTAX for IF: **IF(condition, true_condition, false_condition**)
5. In the **Select a Category** list, click **Logical**.
6. In the **Select a Function** list, click **IF** and then click **OK**.
7. In the **expression** box type **['Unit Sale Price'] <> 0**.
8. In the **true-value** box type, **1**.
9. In the **false-value** box type, **0**.

   This expression will look at the Unit Sale Price. If there is a value other than 0, it will return a 1 to the counter. If there is a 0, it will place a 0 in the counter.

   The results appear as follows:



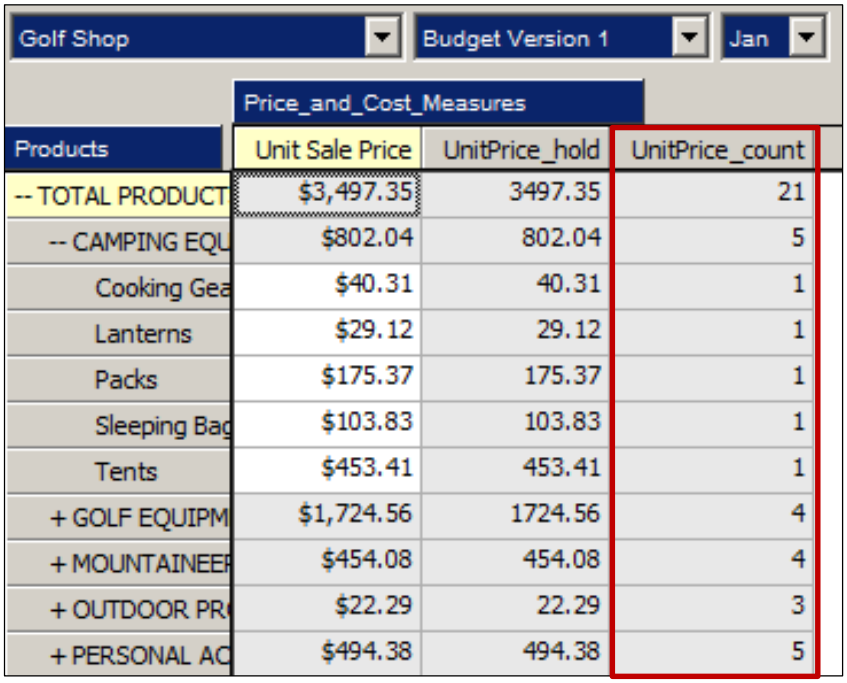
10. Click **OK**.
11. Type a **semicolon**, click **Check Syntax**, and then click **Save**.

```
1   # hold the Unit Sale Price for the average price
2   ['UnitPrice_hold' ]   = N: ['Unit Sale Price' ] ;
3   ['UnitPrice_count' ]   = N: IF(['Unit Sale Price'] <> 0, 1, 0);
```

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12. Recalculate the view.

    The result appears as follows:



Golf Shop	Budget Version 1	Jan	
	Price_and_Cost_Measures		
Products	Unit Sale Price	UnitPrice_hold	UnitPrice_count
-- TOTAL PRODUCT	$3,497.35	3497.35	21
-- CAMPING EQU	$802.04	802.04	5
Cooking Gea	$40.31	40.31	1
Lanterns	$29.12	29.12	1
Packs	$175.37	175.37	1
Sleeping Bag	$103.83	103.83	1
Tents	$453.41	453.41	1
+ GOLF EQUIPM	$1,724.56	1724.56	4
+ MOUNTAINEEI	$454.08	454.08	4
+ OUTDOOR PR	$22.29	22.29	3
+ PERSONAL AC	$494.38	494.38	5

    This confirms that for the leaf-level elements, the Unit Sale Price is really the price for one unit. For consolidated items, you want to look at the average unit price.

Task 4. Calculate the average unit price.

1. On **line 4** of the **Rules Editor**, click the **Brackets**, and then double-click **Price_and_Cost_Measures**.
2. Click **Unit Sale Price**, and then click **OK**.
3. Click outside the brackets and type **= C:**.
4. Click **Brackets**, and then double-click **Price_and_Cost_Measures.**
5. Click **UnitPrice_hold** and then click **OK**.
6. Click to the right of the last bracket, and then type ****.

   In TM1, \ indicates division, where a number divided by zero is replaced by 0. If you use / (the normal division sign), when you divide by zero you get 'NA'.
7. Click **Brackets**, and then double-click **Price_and_Cost_Measures**.

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8. Click **UnitPrice_count**, click **OK**, and then add a **;** to the end of the line.
   The result appears as follows:

```
1   # hold the Unit Sale Price for the average price
2   ['UnitPrice_hold' ]   = N: ['Unit Sale Price' ] ;
3   ['UnitPrice_count' ]   = N: IF(['Unit Sale Price'] <> 0, 1, 0);
4   ['Unit Sale Price' ]   = C: ['UnitPrice_hold' ] \ ['UnitPrice_count' ] ;
```

9. Save the rule, and then recalculate the view.
   The result appears as follows:

Golf Shop	Budget Version 1	Jan	
	Price_and_Cost_Measures		
Products	Unit Sale Price	UnitPrice_hold	UnitPrice_count
-- TOTAL PRODUCT	$166.54	3497.35	21
-- CAMPING EQU	$160.41	802.04	5
Cooking Gea	$40.31	40.31	1
Lanterns	$29.12	29.12	1
Packs	$175.37	175.37	1
Sleeping Bag	$103.83	103.83	1
Tents	$453.41	453.41	1
+ GOLF EQUIPM	$431.14	1724.56	4
+ MOUNTAINEER	$113.52	454.08	4
+ OUTDOOR PR	$7.43	22.29	3
+ PERSONAL AC	$98.88	494.38	5

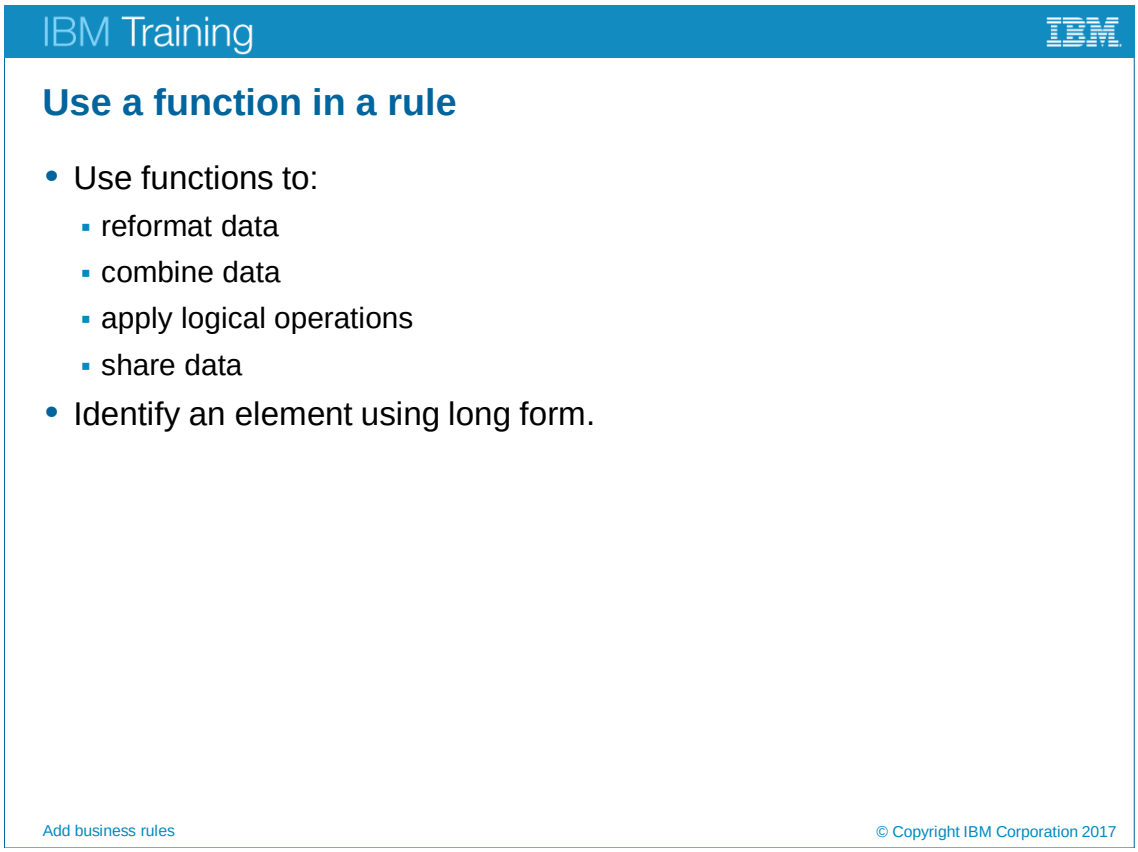
   In Task 3, Step 12 it shows the Unit Sale Price for CAMPING EQUIPMENT as $802.04. Now you have added a rule to use the average price of the products for the consolidated elements which reflects the average Unit Sale Price for CAMPING EQUIPMENT ($160.41).

   There is no need to format UnitPrice_hold or UnitPrice_count because these will probably not be used in reports or analyses, only in the calculation.

10. Close the **Rules Editor** and the **Price_and_Cost** cube and click **Yes** to save changes to the Average Rule view if prompted.
11. In Server Explorer, right-click **SampleOutdoors** and click **Save Data**.
12. Leave Architect open for the next demonstration.

Results:
You added elements to contain the values for calculating average price at the leaf level. Next, at the consolidation level, you created the average calculation using your leaf-level calculations.

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Use a function in a rule

A cell in the current cube may be referenced by the element name in brackets. You may also reference cells in the current cube or across cubes using the DB function:

- Syntax: DB('cube_name', 'dimension1', 'dimension2', … 'dimensionN');
- An element_name or !dimension must be referenced for each 'dimension' in the source cube in the order listed in the Server Explorer tree.
- DB('Price_and_Cost', !Channels, 'TOTAL PRODUCTS', !Months, !Versions, 'Unit Sale Price')
- Enclose the cube name or any element name in single quotes.
- Use an exclamation mark (!) preceding a dimension name to reference the current element.

In the example displayed, !Channels means the function will look to the target cell for a match to the Channels dimension in the source cube, 'TOTAL PRODUCTS' is an element from the Products dimension, and 'Unit Sale Price' is in the Measures dimension in the source.

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IBM Training

Demonstration 3

Create a rule to share data between cubes

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Demonstration 3: Create a rule to share data between cubes

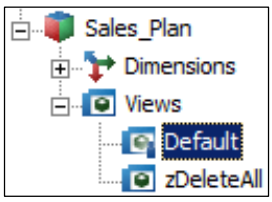
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Demonstration 3: Create a rule to share data between cubes

Purpose:
In the Sales_Plan cube, the Quantity has been loaded but you need the Unit Sale Price in order to calculate Gross Revenue. The Unit Sale Price exists in the Price_and_Cost cube so you will use a rule to pull that value into this cube. This way, if the price in the Price_and_Cost cube is updated, the change will automatically be reflected in the Sales_Plan cube.

Task 1. Create a rule to bring the Unit Sale Price to the Sales_Plan cube.

1. In the Server Explorer window, open the **Default** view of the **Sales_Plan** cube:

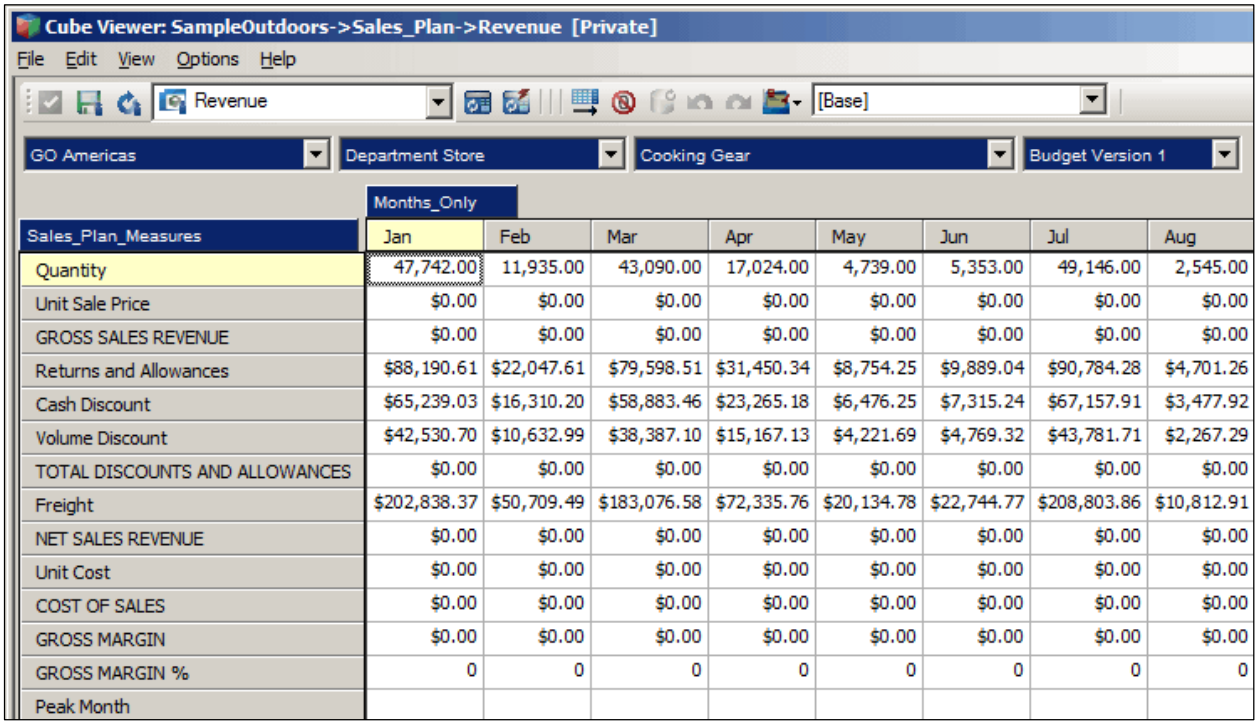


2. Pick a leaf element from each dimension:
   - Subsidiaries: **GO Americas** (open the Subset Editor to select this element from within the GO AMERICAS REGION element)
   - Channels: **Department Store**
   - Products: **Cooking Gear**
   - Versions: **Budget Version 1**

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3. Click **Recalculate**, and then save the view as **Revenue**.

   The result appears as follows:



Sales_Plan_Measures	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug
Quantity	47,742.00	11,935.00	43,090.00	17,024.00	4,739.00	5,353.00	49,146.00	2,545.00
Unit Sale Price	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
GROSS SALES REVENUE	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
Returns and Allowances	$88,190.61	$22,047.61	$79,598.51	$31,450.34	$8,754.25	$9,889.04	$90,784.28	$4,701.26
Cash Discount	$65,239.03	$16,310.20	$58,883.46	$23,265.18	$6,476.25	$7,315.24	$67,157.91	$3,477.92
Volume Discount	$42,530.70	$10,632.99	$38,387.10	$15,167.13	$4,221.69	$4,769.32	$43,781.71	$2,267.29
TOTAL DISCOUNTS AND ALLOWANCES	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
Freight	$202,838.37	$50,709.49	$183,076.58	$72,335.76	$20,134.78	$22,744.77	$208,803.86	$10,812.91
NET SALES REVENUE	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
Unit Cost	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
COST OF SALES	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
GROSS MARGIN	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
GROSS MARGIN %	0	0	0	0	0	0	0	0
Peak Month								

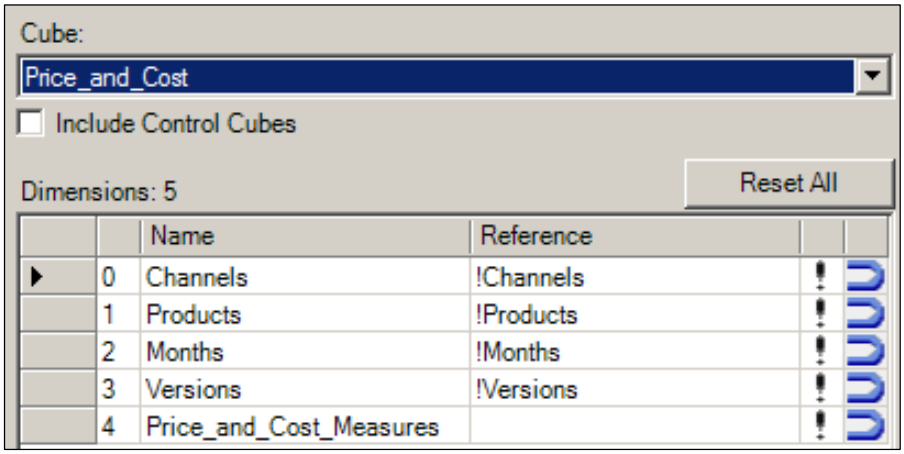
4. In the Server Explorer window, right-click the **Sales_Plan** cube and then click **Create Rule**.
5. Click **Brackets**, and then double-click **Sales_Plan_Measures**.
6. Click **Unit Sale Price** and then click **OK**.
7. Click to the right of the last bracket, click the **Insert Qualifier or Operator** icon, and then click **=N:**
8. Click the **Insert Function** icon, scroll down in the **Select a Function** list, click **DB**, and then click **OK**.

   The Insert Cube Reference dialog box appears.

   The DB() function uses the long form name for any element. It can be used to refer to elements in the current cube or across cubes. In this example, you will use it to pull the Unit Sale Price from the Price_and_Cost cube into the Sales_Plan cube. This ensures that when a price is added or updated in the Price_and_Cost cube, the change will be reflected in the Sales_Plan.

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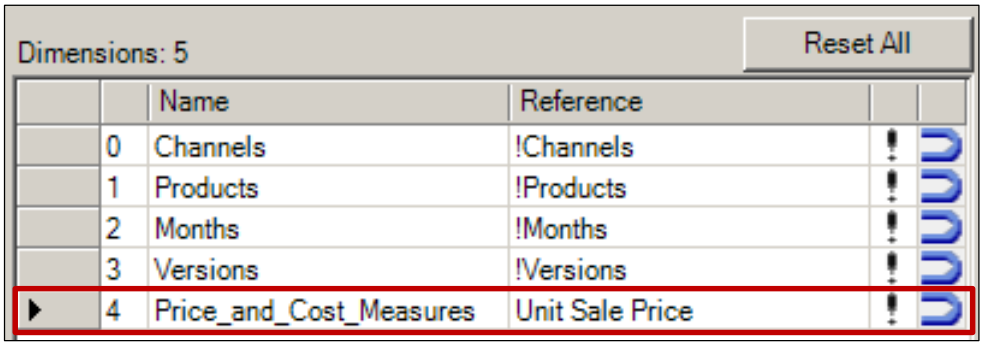
9. In the **Cube** list, click **Price_and_Cost**.



All the dimensions in the Price_and_Cost cube are listed. The reference to be used in the DB() function is displayed in the Reference column. You can select a specific element from a dimension by clicking the Subset icon.

The ! in front of the Reference dimensions indicates that TM1 will use the current value for that dimension in the function instead of using a specific element in the function. This allows the function to be more dynamic.

10. In the **Price_and_Cost_Measures** row, click the **Subset** icon  .
11. In the **Subset Editor**, click **Unit Sale Price**, and then click **OK**.



You used the element name for Unit Sale Price because you only want that value brought into the Sales_Plan cube and no other elements.

12. Click **OK** to close the **Insert Cube Reference** dialog box.
13. Add a **;** to the end of the line.

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14. Press **Enter**, and then type the following:

    C:['GROSS SALES REVENUE']\['Quantity'];

    The result appears as follows:

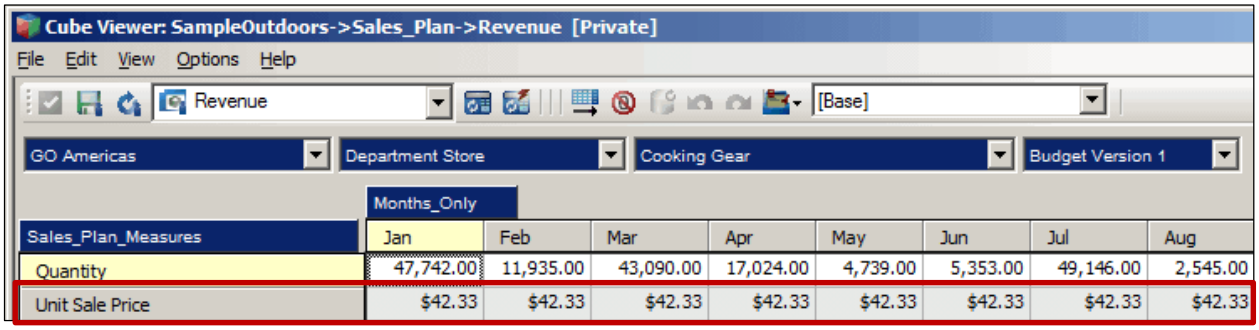
```
1   ['Unit Sale Price' ]  = N: DB('Price_and_Cost',
2      !Channels, !Products, !Months, !Versions, 'Unit Sale Price');
3   C:['GROSS SALES REVENUE']\['Quantity'];
```

    The first rule pulls the Unit Sale Price from the Price_and_Cost cube at the N: level. The consolidated levels instead will compute the Unit Sale Price based on the Gross Sales Revenue divided by Quantity so it will not be aggregated at the consolidated levels.

    Spaces and returns may be inserted into rules to make them more readable.

15. Click **Save** in the Rules Editor, and then recalculate the view.

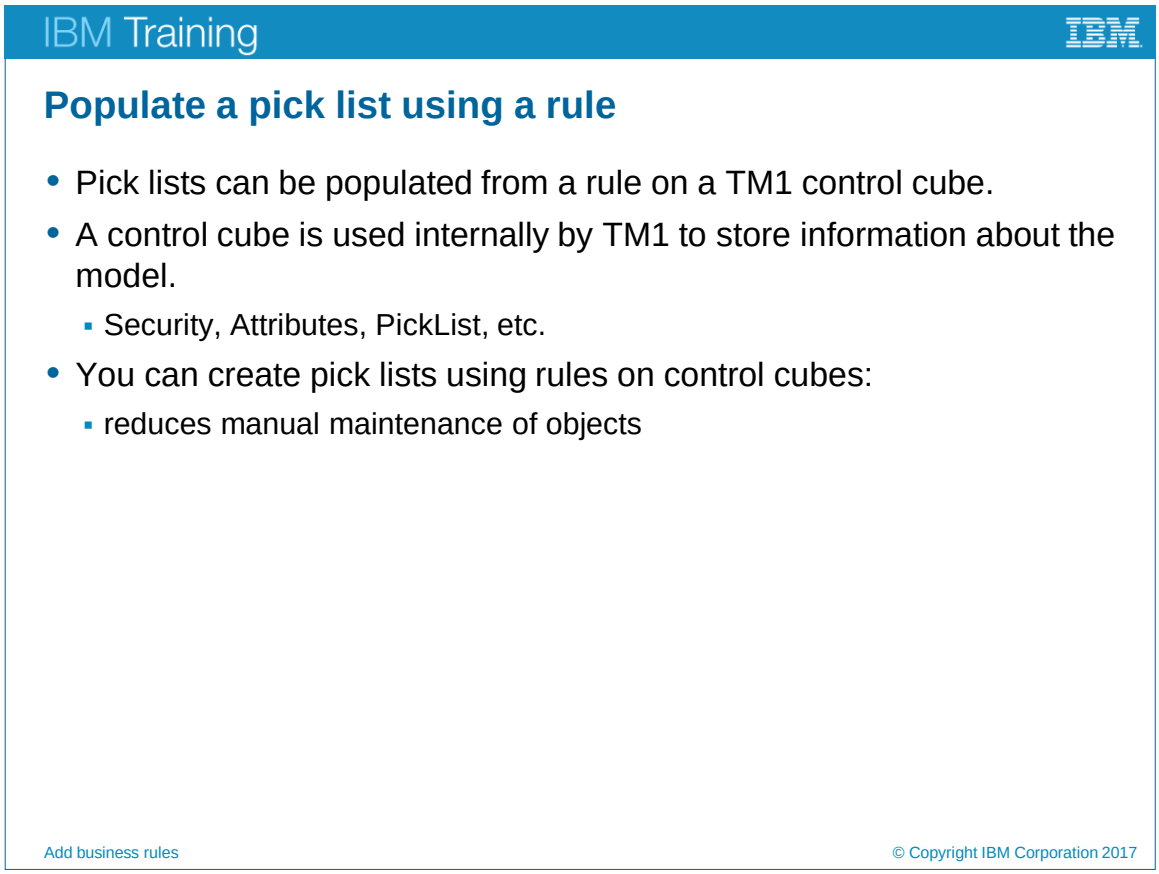
    The result appears as follows:



16. Close the **Rules Editor**, and then close the **Cube Viewer** without saving.
17. In the Server Explorer, right-click **SampleOutdoors** and click **Save Data**.
18. Leave Architect open for the next demonstration.

Results:
You shared the Unit Sale Price in the Price_and_Cost cube with the Sales_Plan cube by using a DB() function in a rule.

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Populate a pick list using a rule

You can also create rules for the pick list control cube, which allows you to define pick lists for any section of a cube, from a single cell to the entire cube.

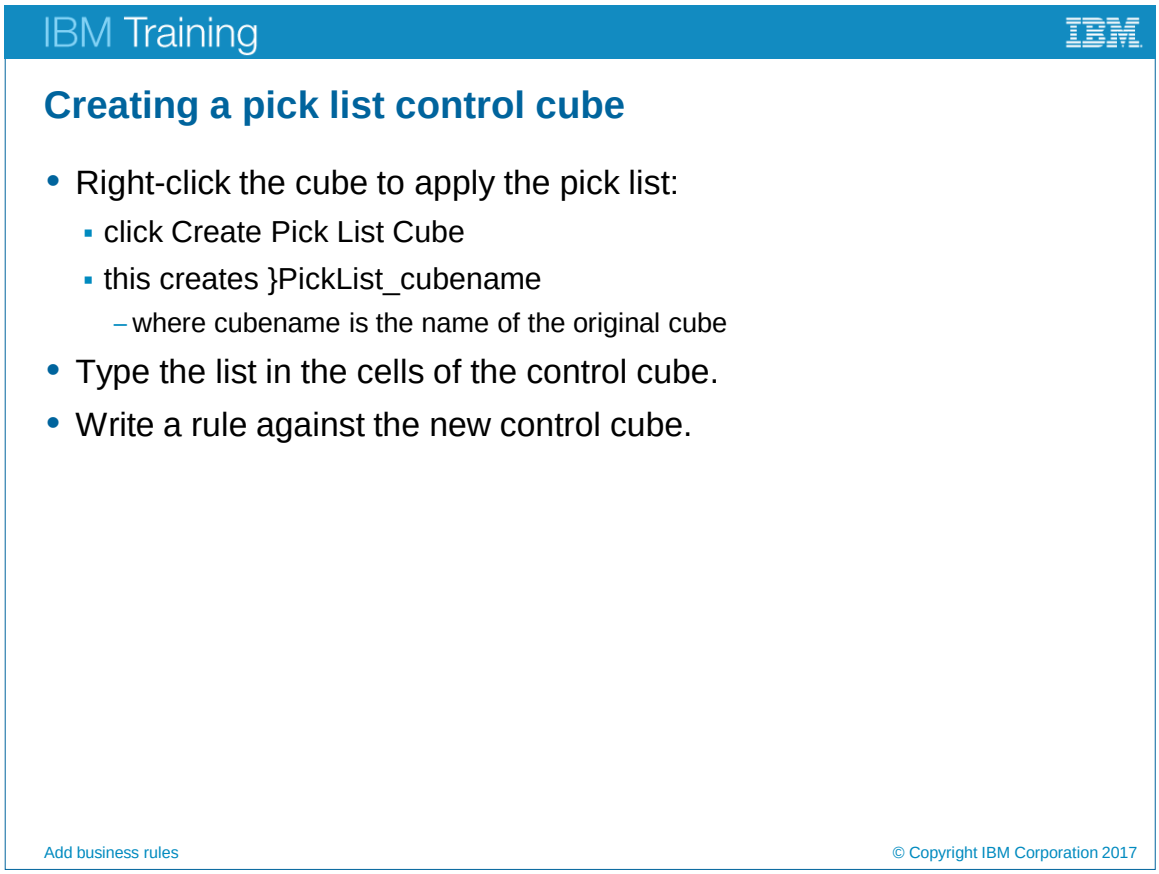
This gives you greater control over which cube cells should contain pick lists, and allows greater flexibility in defining pick lists for individual cells.

A pick list control cube is composed of the same dimensions as the regular cube it is associated with, along with an additional dimension named }Picklist.

The }Picklist dimension contains a single string element, named Value.

Control cubes and dimensions are hidden by default. They are preceded with a right-curly bracket (}).

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Creating a pick list control cube

When typing into the cell of the control cube, use the same syntax as the attribute:

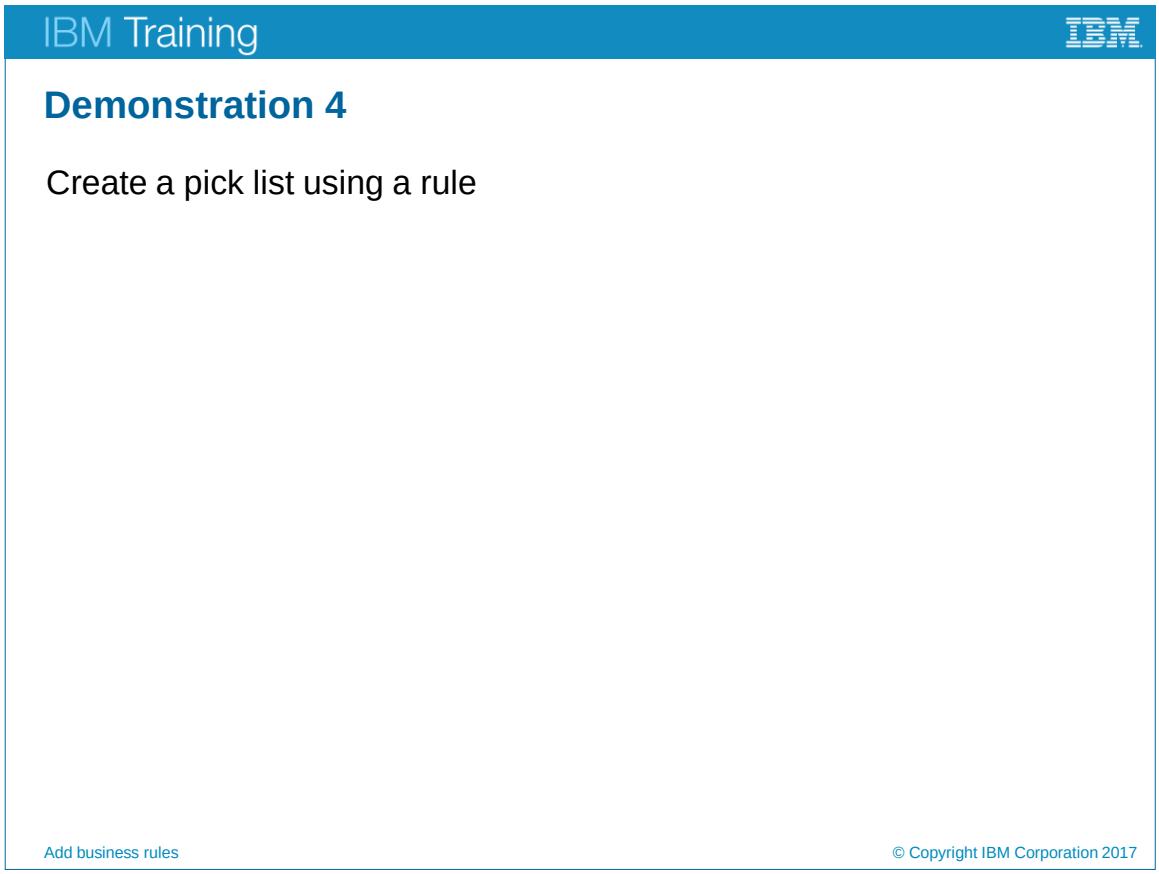
- Static: value1: value2…valueN
- Subset:Dimension_name:Subset_name
- Dimension:Dimension_name

Rules that define pick lists follow the same conventions as all other TM1 rules.

- Pick lists are strings so the string qualifier must be used (=S:)

It is possible to have a numeric pick list, but not probable. You would only do this if you wanted the pick list to aggregate.

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Demonstration 4: Create a pick list using a rule

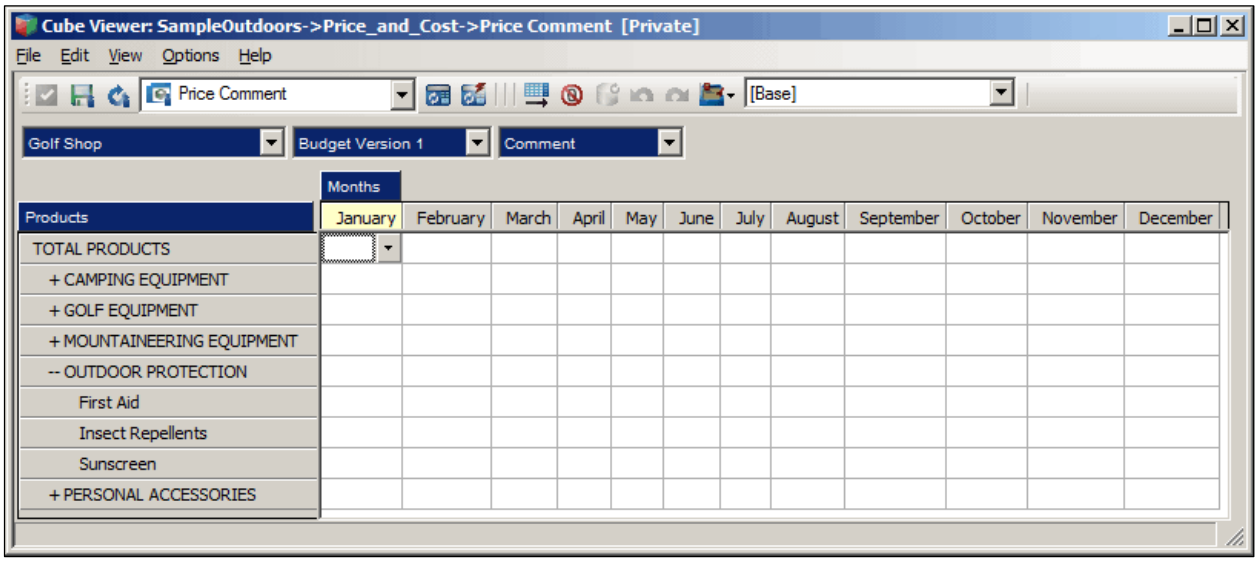
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Demonstration 4: Create a pick list using a rule

Purpose:
You need to add more pick lists to the Comment element in the Price_and_Cost cube. You would like to use rules to populate the lists so you will not have to type them all manually in the attributes.

Task 1. Display control cubes.

1. Open the **Price Comment** view of the **Price_and_Cost** cube and then click to expand **OUTDOOR PROTECTION**.
2. If necessary, select **Comment** from the **Price_and_Cost_Measures** dimension, and then collapse **Camping Equipment** (if necessary).

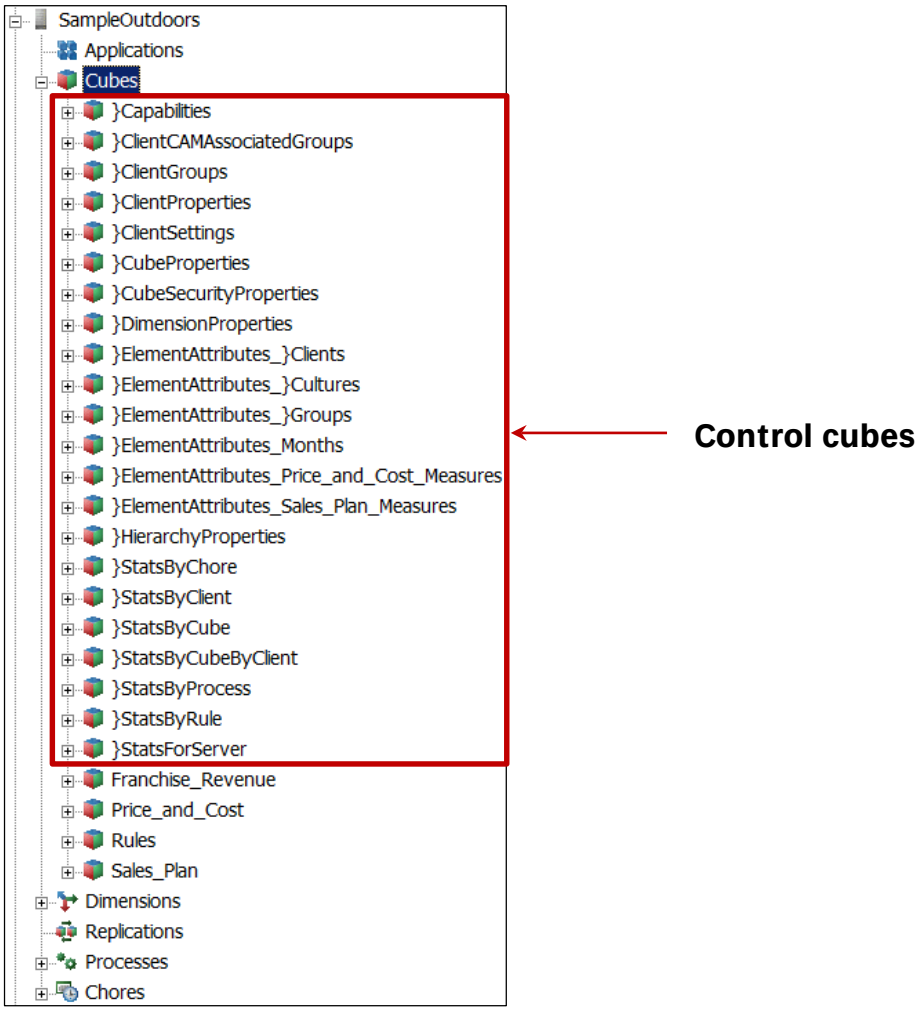


Move this view to the side so that you can see the Server Explorer tree.

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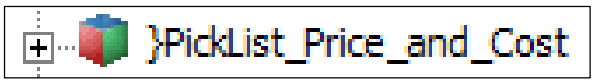
3. In the Server Explorer window, from the **View** menu, click **Display Control Objects**.

   The results appear similar to the following:



   Control objects (cubes and dimensions) are identified with the curly-bracket (}).

4. Right-click the **Price_and_Cost** cube, and then click **Create Pick List Cube**.



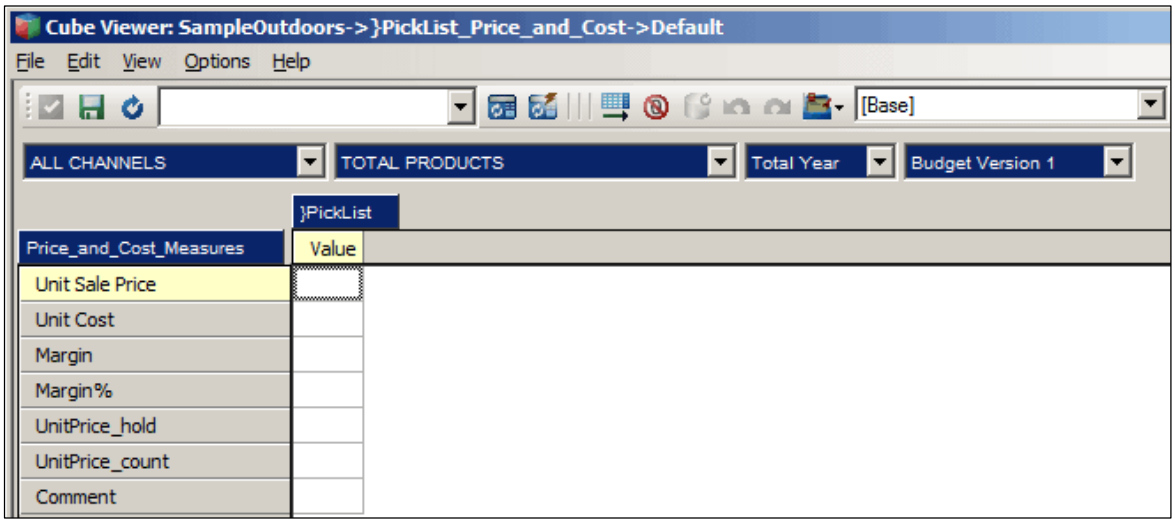
   A control cube is added and named }PickList_*cubename*.

   This cube contains the same dimensions as the Price_and_Cost cube with one more called }PickList. You can apply rules to this cube just as you can apply rules to a data cube in TM1. This allows for greater flexibility and easier maintenance of your pick lists.

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Task 2. Create and use a static pick list.

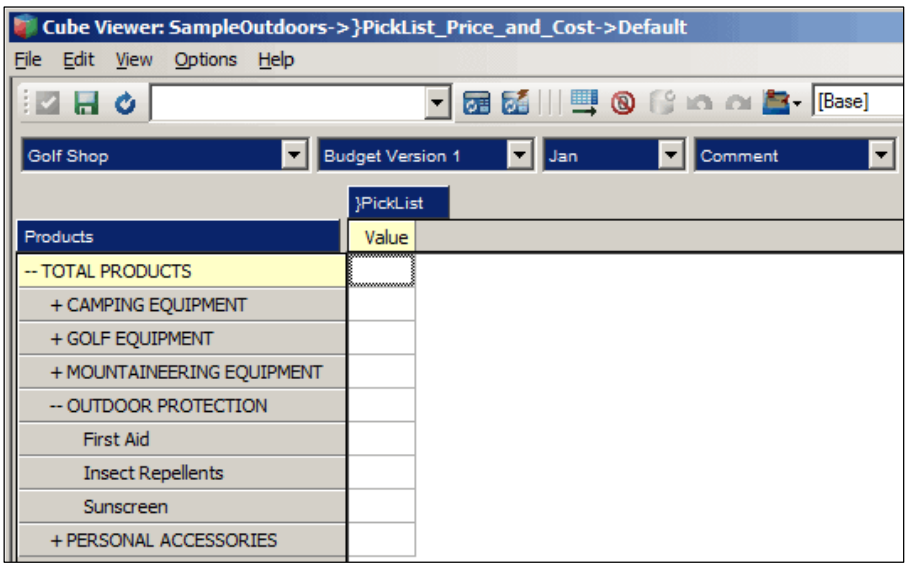
1. Double-click the **}PickList_Price_and_Cost** control cube, and then click **Recalculate**.



The cube contains all the dimensions of the Price_and_Cost cube and another dimension named }PickList.

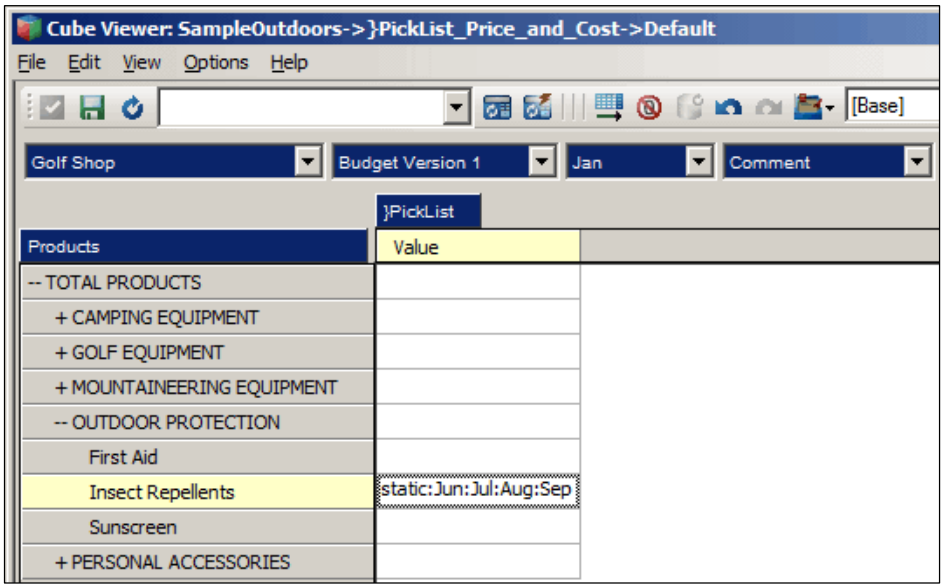
2. Arrange the }Picklist cube to look like following:

- Title elements: **Golf Shop**, **Budget Version 1**, **Jan**, **Comment** (from the **Price_and_Cost_Measures** dimension)
- Columns: **}Picklist**
- Rows: **Products** and drill down on **OUTDOOR PROTECTION** (you may need to click Recalculate first)

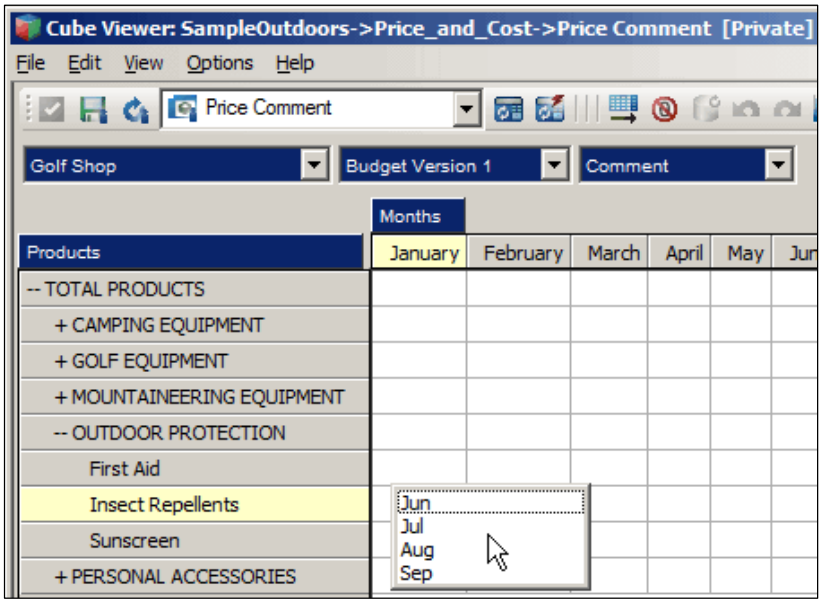


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3. Click the **Value** for **Insect Repellents** and type the following: **static:Jun:Jul:Aug:Sep**
4. Click **Recalculate**.



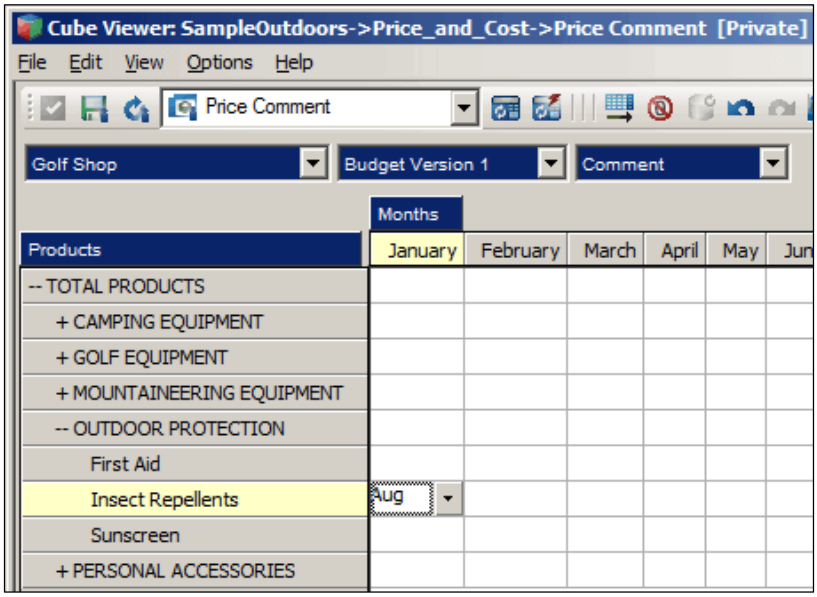
5. Click (or open) the **Price Comment** view of the **Price_and_Cost** cube and delete any existing Comment values.
6. If necessary, click **OUTDOOR PROTECTION** to expand.
7. Click **Comment** for **Insect Repellents** in **January**.



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8. Click **Aug** to select.

   The results appear as follows:



   This pick list was defined only for the intersection of Golf Shop, Budget Version 1, Comment, January, and Insect Repellents. When you created a Pick List using the attribute, it was for all elements in the dimension. Defining a pick list in the control cube allows the definition to be much more specific.

   You can leave this view open for the next task.

   Click in the other cells and you will see that the original Picklist attribute (with values of blank, "High," "Medium," and "Low") is still in effect. However, using a rule to populate a pick list will override the attribute.

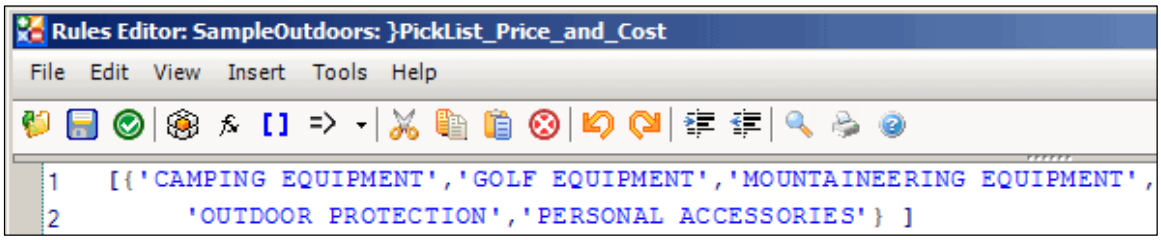
Task 3. Create a pick list using a rule.

In the previous example, if we needed to expand the range for the pick list, we could do it manually. However, it is much easier to maintain if we write a rule to define and populate the pick list.

1. In Server Explorer, right-click the **}PickList_Price_and_Cost** cube and then click **Create Rule**.
2. On **line 1**, click the **Brackets** and then double-click **Products**.

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3. Ctrl-click the following elements from the **Products** dimension, and then click **OK**:
   - **CAMPING EQUIPMENT**
   - **GOLF EQUIPMENT**
   - **MOUNTAINEERING EQUIPMENT**
   - **OUTDOOR PROTECTION**
   - **PERSONAL ACCESSORIES**

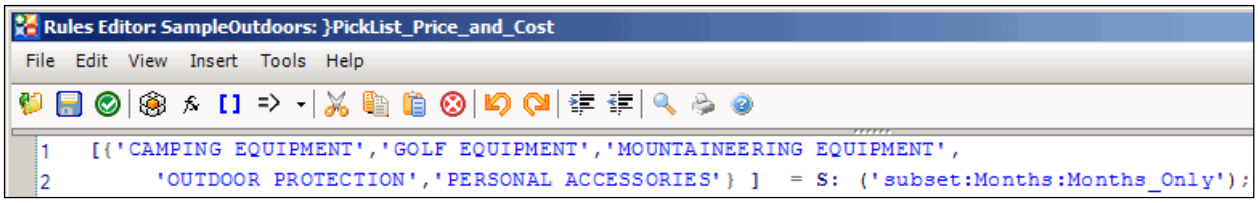


   You can insert returns to make the line easier to read.

4. Click to the right of the bracket, click **Insert Qualifier or Operator** and then click **=S:**
5. Type the following:

   ('subset:Months:Months_Only');

   The first Months reference is the dimension name and the second one is the name of a subset in that dimension.
6. Click **Check Syntax**, fix any errors and then click **Save**.

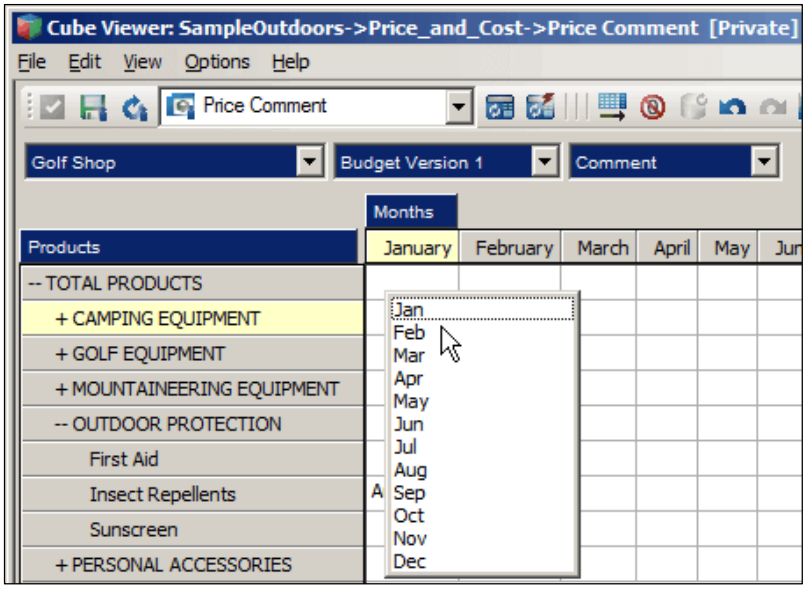


7. Recalculate the **Price Comment** view of the **Price_and_Cost** cube.

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8. Click **Comment** for any of the consolidations (such as CAMPING EQUIPMENT) and you should now see a drop down list of the 12 months in the Months subset.

   The results appear as follows:



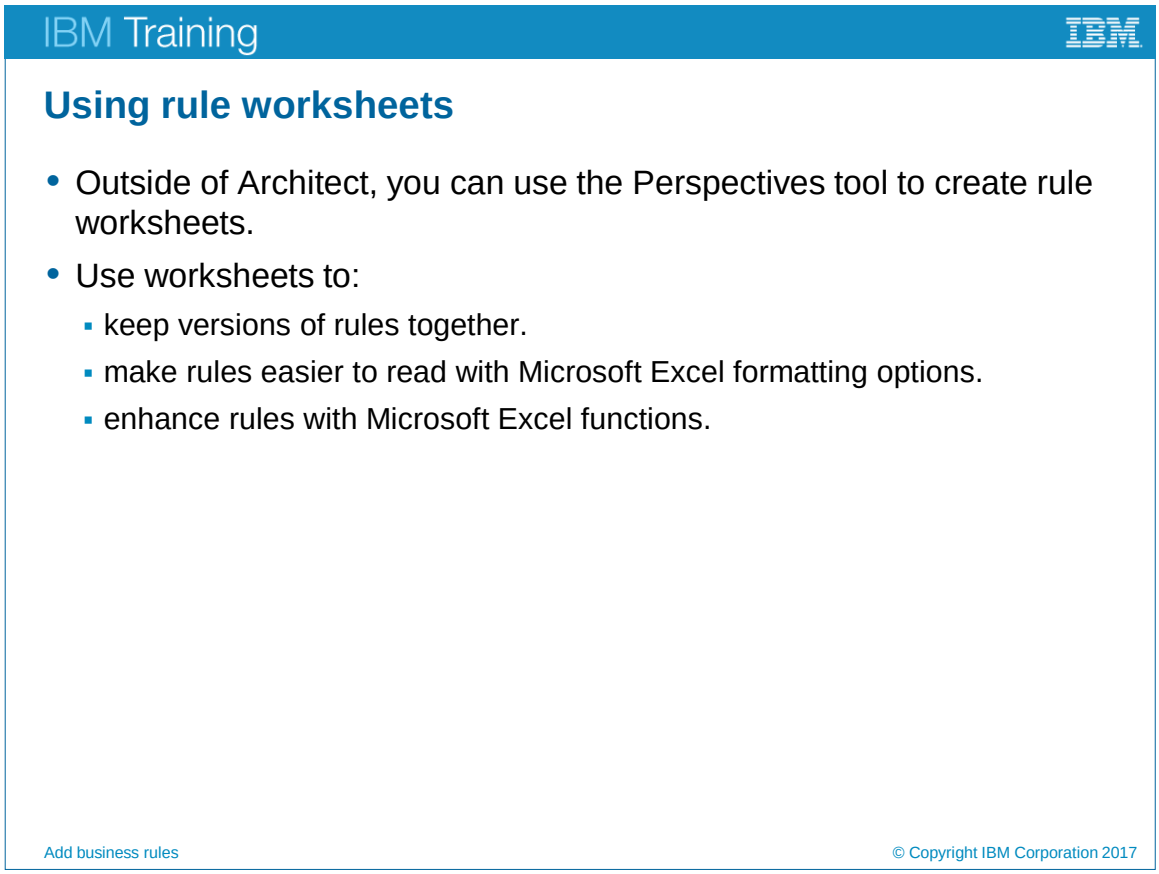
9. Close the **Price Comment** view, saving changes and overwriting the existing view when prompted.
10. Close the rules for the **}PickList_Price_and_Cost** cube.
11. Close the **}PickList_Price_and_Cost** cube and do not save changes.
12. In Server Explorer, click the **View** menu and then deselect **Display Control Objects**.

    This will turn off the display of control objects.
13. Right-click **SampleOutdoors** and click **Save Data**.
14. Leave Architect open for a later demonstration.

    The next instructor-only demonstration will use the Perspectives tool instead of Architect.

Results:
You added more pick lists to the Comment element in the Price_and_Cost cube using a control cube and rules. You added both a static and a subset pick list.

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Using rule worksheets

IBM Cognos TM1 Perspectives is a TM1 add-in for Microsoft Excel. It can be used for both TM1 model development and for analyzing data via Microsoft Excel capabilities. Like Architect, Perspectives supports the creation and maintenance of all TM1 objects (other than those that are specific to the Performance Modeler tool, such as diagrams).

Do not use both the rule worksheets and the Rules Editor for the same cube as this may cause data to be lost. Write rules with either the Rules Editor or a rule worksheet.

When saving rules in a worksheet, you must use Save Rule from the TM1 menu to save the file. Using the TM1 menu saves the *.xru file and generates the *.rux file from column A. The Rules Editor only reads/writes to the *.rux (compiled) file.

Using File - Save from the Microsoft Excel menu will save only the *.xru file without compiling the *.rux file. If the rules are then opened in the Rules Editor, changes in the *.xru file will not be reflected and business logic may be lost.

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IBM Training

Demonstration 5

Review a rule worksheet (instructor only)

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Demonstration 5: Review a rule worksheet (instructor only)

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Demonstration 5: Review a rule worksheet (instructor only)

Purpose:
You want to look at rules in rule worksheets. You will open the rule worksheet for the Rules cube and review the format in Microsoft Excel.

Task 1. Open a rule worksheet.

1. On the **Desktop**, double-click **Perspectives for MS Excel** and click **Close** if you are prompted to recover any documents.
2. Click **Enable Macros**, if prompted.
3. In **Microsoft Excel**, on the **TM1** tab, click **Connect**.
4. Enter the following:
   - Server Id: **SampleOutdoors**
   - Client Id: **admin**
   - Password: **apple**
5. Click **OK** to close the **Connect to TM1 Server** dialog box, and then click **OK** to close the confirmation dialog box.
6. On the **TM1** tab, click the **Developer** button in the top-right corner, and then click **Rule New**.

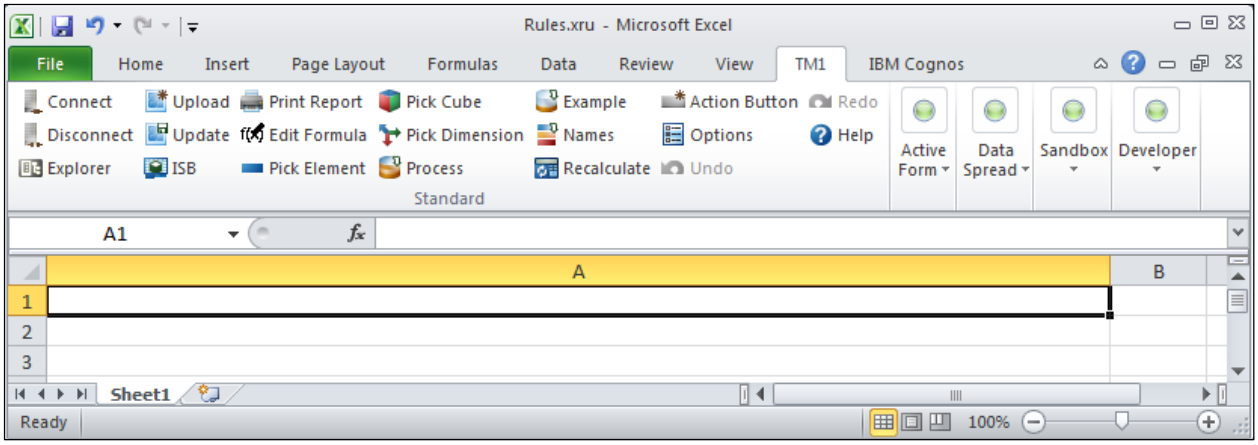
   You are prompted to select a cube from the SampleOutdoors server.

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7. Click **SampleOutdoors:Rules**, and then click **OK**.

   Rules.xru opens in another Microsoft Excel window. Notice that column A is very wide.

   Only statements placed in column A are compiled. You may place comments or older versions of rules in other columns to document them.



   Rules created in the Rules Editor will not appear. Rule worksheets read in and write to the worksheet (*.xru) files. This is why you should only create or edit rules using either the Rules Editor or rule worksheets for a cube. You should never use both editors for a single cube.

   You write rules in column A using the same format as with the Rules Editor.

   Since only column A is compiled, it can be useful to use the additional columns for comments or versioning of your rules.

8. Close the rule worksheet (**Rules.xru**) and do not save it.
9. Close **Microsoft Excel**.

Results:
You reviewed steps to create a rule worksheet for the Rules cube.

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Determine location of logic

Location	Use for:
Hierarchies & Weighting	Core calculations
Rules	Formal modeling with comments
TI	ETL, reports & KPIs, performance optimization
Spreading	Spontaneous modeling
Microsoft Excel	Ad-hoc, specialized
VBA	Usability (not business logic)
VBA and API	Specialized solutions

Add business rules

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Determine location of logic

You have looked at a number of options for calculating data. Understanding these options will allow you to decide where to add business logic to best suit the needs of your users and your data. Calculations may be performed within the dimension hierarchy, when data is pulled into a cube via TI, or in TM1 rules.

In order to optimize performance of the model, anything that can be processed on the server, instead of in spreadsheets and local files, should be processed there.

- dimension hierarchies, cube rules, TI processes
- define complex SQL queries and views in the RDBMS

When looking at the model, ask: "does this really need to be computed dynamically or can it be stored?" It is also a good idea to review the model periodically, asking this question.

The dimension hierarchy can perform calculations using weighting. To have a consolidation do subtraction, you can assign an element a weight of -1. This will be multiplied by the value associated with this element (giving it a negative value) upon calculating the consolidation.

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IBM Training

Demonstration 6

Calculate gross margin

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Demonstration 6: Calculate gross margin

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Demonstration 6: Calculate gross margin

Purpose:
You need to calculate GROSS MARGIN in the Sales_Plan cube. You know you can use rules to do this but the calculation is simple addition and subtraction. Instead of writing a rule, you determined that the best place to calculate GROSS MARGIN is within the dimension hierarchy (using weighting).

Task 1. Edit the dimension structure.

1. In Server Explorer, under the **SampleOutdoors** server, expand the **Dimensions** node, right-click the **Sales_Plan_Measures** dimension and click **Edit Dimension Structure**.

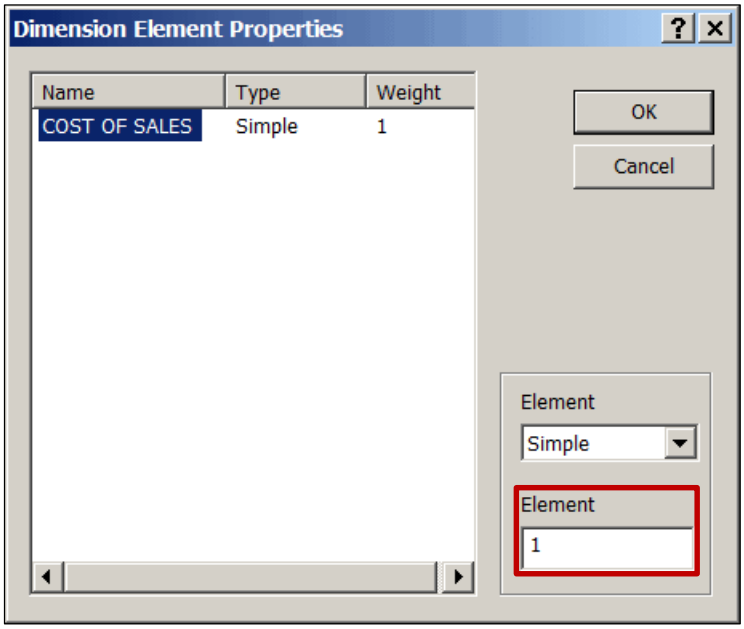
   The Dimension Editor window opens.
2. Click **NET SALES REVENUE**, and then from the **Edit** menu, click **Cut**.

   You could also copy the element if you wished to see the value both within GROSS MARGIN and as a stand- alone element. Both would display the same value.
3. Click **OK** to dismiss the confirmation message.
4. Click **GROSS MARGIN**, and then from the **Edit** menu, click **Paste as Child**.

   The sigma (summation) icon appears in front of GROSS MARGIN, and NET SALES REVENUE appears below it.
5. Repeat steps **2** to **4** with **COST OF SALES**, pasting it as a child of **GROSS MARGIN**.

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6. Right-click **COST OF SALES**, and then click **Element Properties**.

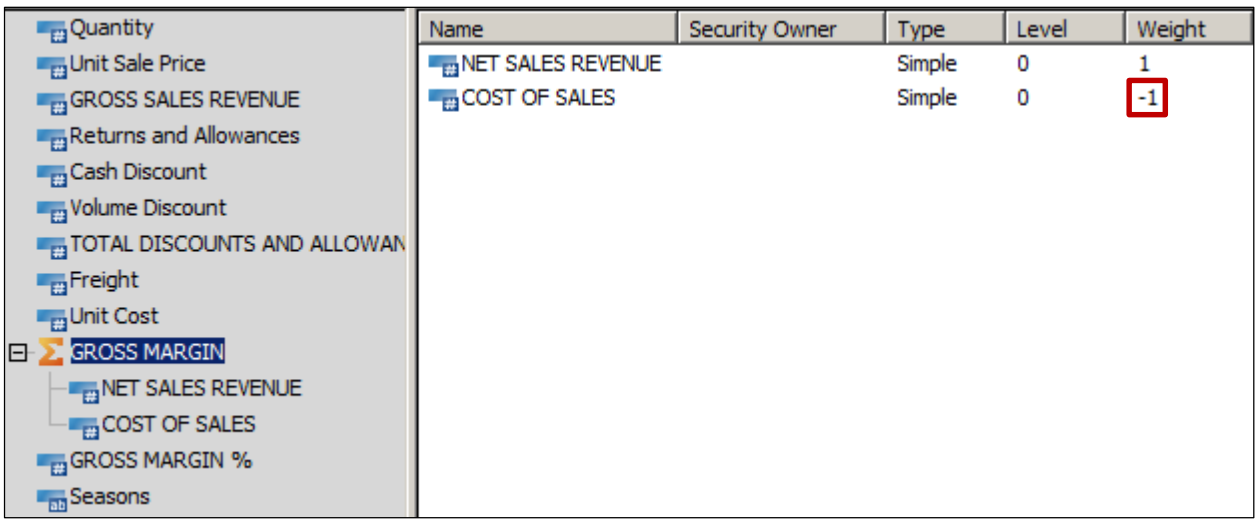


Here you may change the type of element or its weight. Consolidations are summations by default. Giving a child element a negative weight is how subtraction may be achieved in the consolidation.

7. In the **Element** box type, **-1**, and then click **OK** to close the Dimension Element Properties dialog box.

8. Click **GROSS MARGIN**, and then from the **View** menu, click **Properties Window** (if it is not already selected).

The element list shows COST OF SALES with a weight of -1.



The consolidation GROSS MARGIN is a summation of NET SALES REVENUE – COST OF SALES.

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9. Click **Save**, and then close the **Dimension Editor**.
10. Right-click **SampleOutdoors**, and then click **Save Data**.
11. Leave Architect open for the upcoming exercise.

Results:
You calculated GROSS MARGIN in the dimension hierarchy using weightings in the consolidation algorithm.

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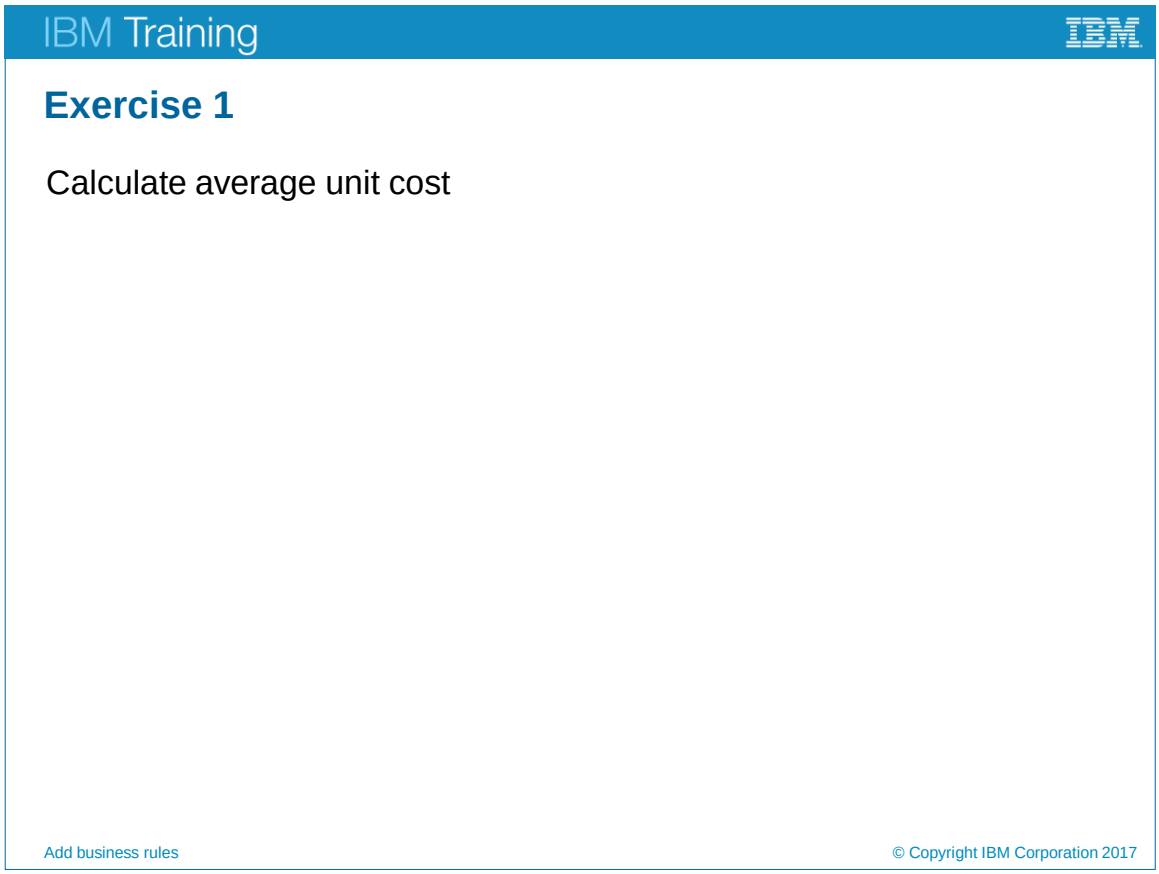
Unit summary

- Describe a rule
- Construct rules for elements or consolidations
- Use functions in rules
- Optimize rule performance
- Describe rules in a rule worksheet

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Unit summary

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Exercise 1: Calculate average unit cost

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Exercise 1: Calculate average unit cost

Now that you have computed average unit price for consolidations in the Price_and_Cost cube, you will need to compute the average unit cost for consolidations. Create the rules and review the consolidated Unit Cost values to ensure they are the average of the base level elements.

To accomplish this:

- Create new elements in the Price_and_Cost_Measures dimension:
  - UnitCost_hold
  - UnitCost_count
- Add a rule to hold the Unit Cost data.
- Add a rule to count the number of non-empty cells in Unit Cost.
- Add a rule to compute the average unit cost at the consolidated level.

For more information about where to work and the exercise results, refer to the Tasks and results section that follows. If you need more information to complete a task, refer to earlier demonstrations for detailed steps.

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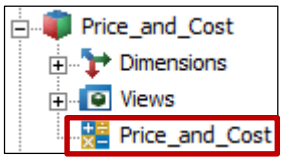
Exercise 1: Tasks and results

Task 1. Create new elements in the Price_and_Cost_Measures dimension.

1. In Server Explorer, right-click the **Price_and_Cost_Measures** dimension, click **Edit Dimension Structure**, and then click **All**.
2. Click **UnitPrice_count**, and then from the **Edit** menu, click **Insert Element**.
3. In **Insert Element Name**, type **UnitCost_hold** and then click **Add**.
4. In **Insert Element Name**, type **UnitCost_count** and then click **Add**.
5. Click **OK**, click **OK** again and then click **Yes** to save the changes to the dimension.

Task 2. Create a rule for UnitCost_hold.

1. Expand the **Price_and_Cost** cube (if necessary), and then double-click the rules file for the **Price_and_Cost** cube to open the **Rules Editor**.



2. Place your cursor on **line 5** and add the following comment:
3. **#Calculate Average Unit Cost for consolidated levels.**
4. On **line 6**, type the following:
   ['UnitCost_hold'] = N: ['Unit Cost'];
5. Press **Enter**.

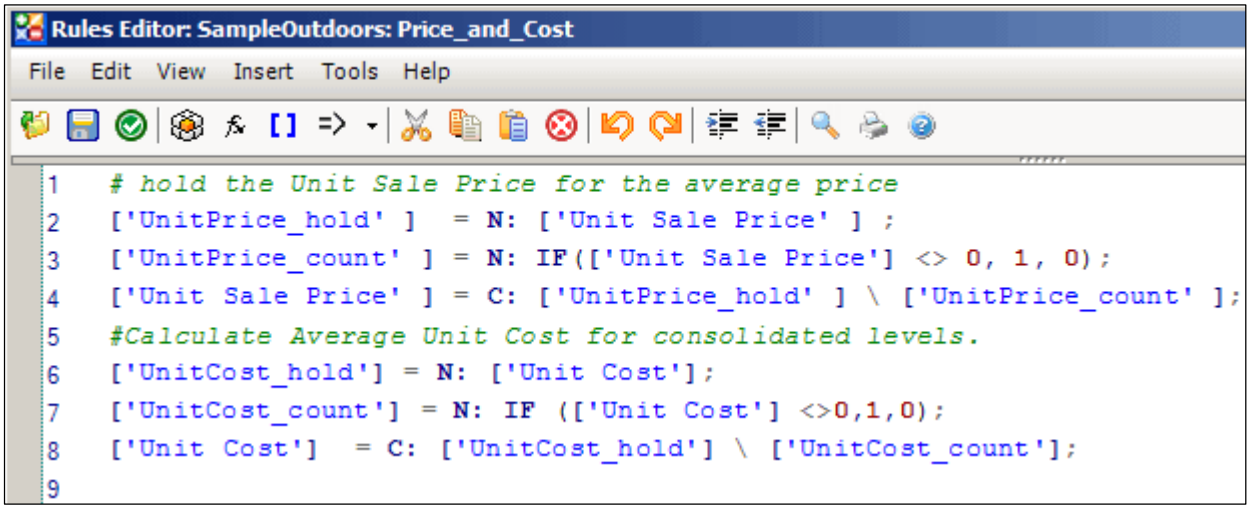
Task 3. Create a rule for UnitCost_count.

1. On **line 7**, type the following:
   ['UnitCost_count'] = N: IF (['Unit Cost'] <>0,1,0);
2. Click Check Syntax.
3. Press Enter.

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Task 4. Create a rule for the Average Unit Cost for consolidated cells.

1. On **line 8**, type the following:
   ['Unit Cost'] = C: ['UnitCost_hold'] \ ['UnitCost_count'];
2. Press **Enter**.
   The result appears as follows:



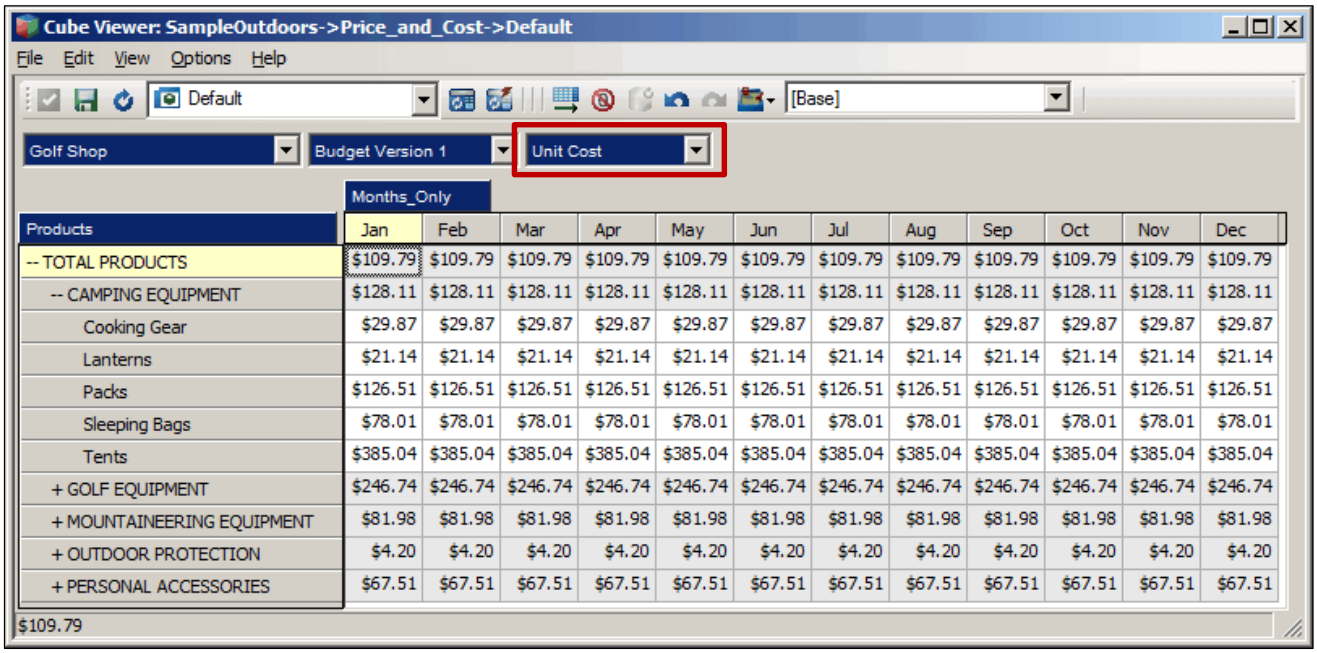
3. Click **Save**.
4. Close the **Rules Editor**.

Task 5. Review the default view.

1. In Server Explorer, double-click the **Price_and_Cost** cube.
2. In the **Price_and_Cost_Measures** drop down list, click **Unit Cost**.

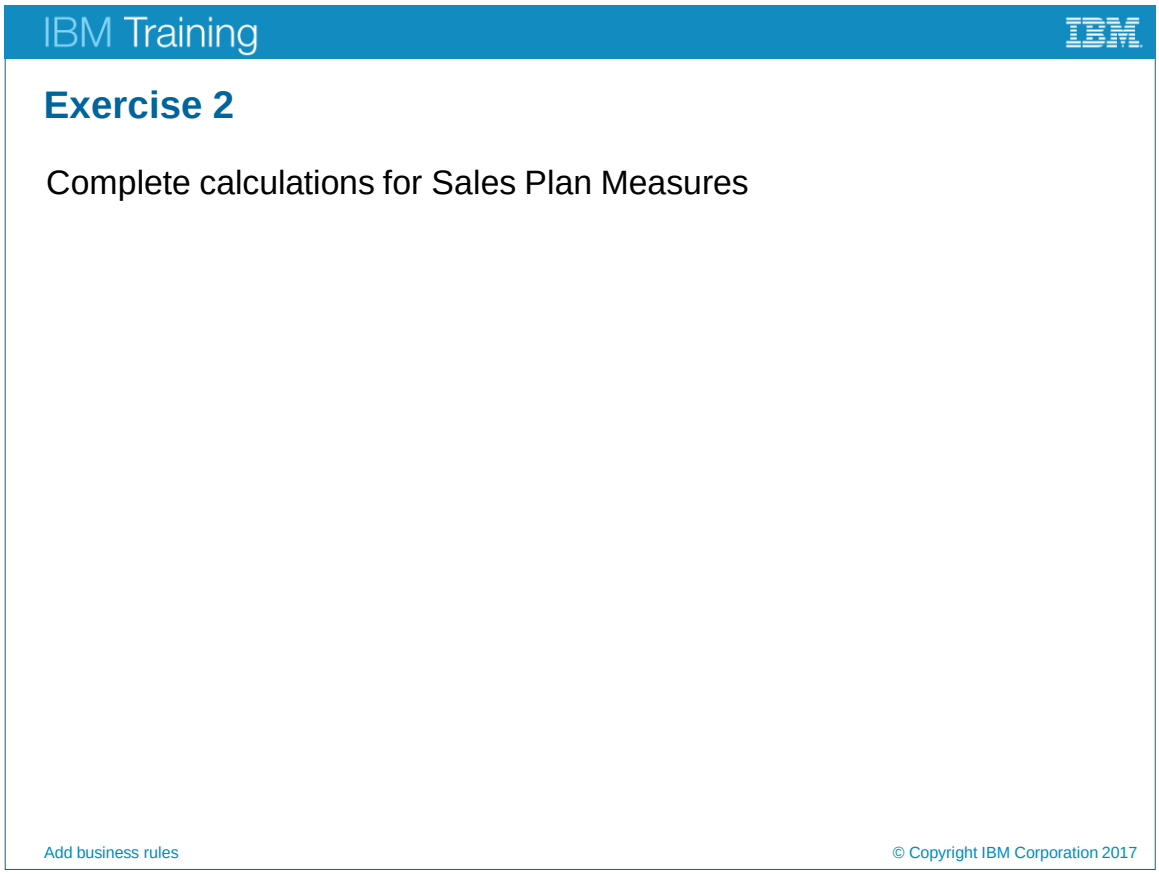
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3. Recalculate the view, expand **CAMPING EQUIPMENT**, and then review the results:



4. Close the **Cube Viewer** without saving.
5. Right-click **SampleOutdoors** and click **Save Data**.
6. Leave Architect open for the next exercise.

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Exercise 2: Complete calculations for Sales Plan Measures

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Exercise 2: Complete calculations for Sales Plan Measures

The Sales_Plan_Measures dimension is not complete. Now that you have brought the Unit Sale Price and Unit Cost measures into the cube, you want to finish calculating the rest of the measures. You will take advantage of the different calculation options TM1 has to offer.

To accomplish this:

- Change the dimension structure to:
  - Net Sales Revenue will aggregate:
    - Gross Sales Revenue
    - Total Discounts and Allowances (weight of -1)
    - Freight (weight of -1)
  - Total Discounts and Allowances will aggregate:
    - Volume Discount
    - Cash Discount
    - Returns and Allowances
- Create a rule to bring Unit Cost to the Sales_Plan cube.
- Create a rule to calculate Cost of Sales:
  - Quantity*Unit Cost
- Calculate a view of the Sales_Plan cube.

For more information about where to work and the exercise results, refer to the Tasks and results section that follows. If you need more information to complete a task, refer to earlier demonstrations for detailed steps.

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Exercise 2: Tasks and results

Task 1. Edit the Sales_Plan_Measures dimension.

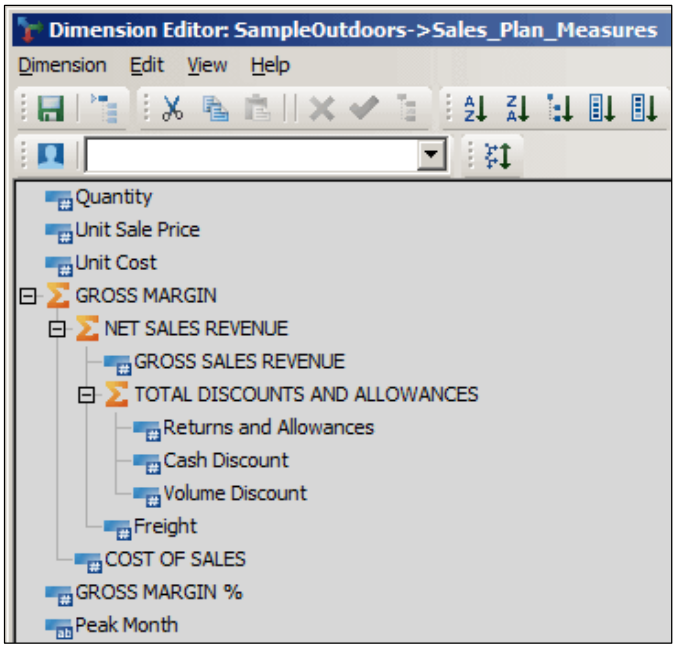
- In Server Explorer, under the **Dimensions** node, right-click **Sales_Plan_Measures** and then click **Edit Dimension Structure**.
- Ctrl+click **GROSS SALES REVENUE**, **TOTAL DISCOUNTS AND ALLOWANCES**, and **Freight** to select.
- Press **Ctrl+X** to cut, click **OK** to dismiss the confirmation message, and then click **NET SALES REVENUE**.
- From the **Edit** menu, click **Paste as Child**.
- Right-click **TOTAL DISCOUNTS AND ALLOWANCES** (under NET SALES REVENUE) and then click **Element Properties**.
- In the **Element** box type **-1**, and then click **OK**.
- Repeat steps **5** - **6** for **Freight**.
- Ctrl+click **Volume Discount**, **Cash Discount** and **Returns and Allowances** to select.
- Press **Ctrl+X** to cut, click **OK** to dismiss the confirmation message, and then click **TOTAL DISCOUNTS AND ALLOWANCES** (under NET SALES REVENUE).
- From the **Edit** menu, click **Paste as Child**.

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- Click **Set Dimension order**.

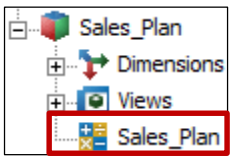
  The result appears as follows:



- Click **OK** to close the Dimension Editor, and then click **Yes** to save changes to the dimension.

Task 2. Create a rule for Unit Cost.

- Open the rules file for the **Sales_Plan** cube.



- Start a new line, and then type the following:

  ['Unit Cost'] = N:

- Click **Insert Cube Reference**.
- In the **Cube** list, click the **Price_and_Cost** cube.

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- Click the **Subset** icon next to **Price_and_Cost_Measures**.
- Click the **Unit Cost** element, click **OK** to close the **Subset Editor** window, and then click **OK** to close the **Insert Cube Reference** dialog box.
- Type a **;** at the end of the line, and then press **Enter**.
- On the new line, type the following:

  C:['COST OF SALES'] \ ['Quantity'];
- Press **Enter**.

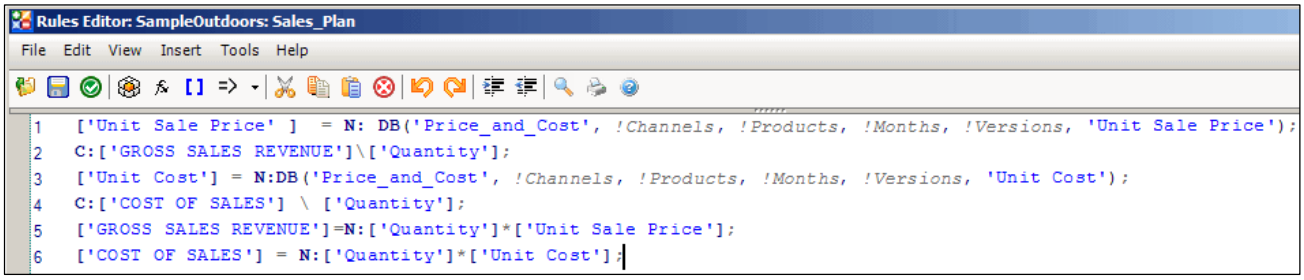
Task 3. Create rules for Cost of Sales and Gross Sales Revenue.

- On the new line, type the following rule:

  ['GROSS SALES REVENUE']=N:['Quantity']*['Unit Sale Price'];
- Press **Enter**.
- On the new line, type the following rule:

  ['COST OF SALES'] = N:['Quantity']*['Unit Cost'];
- Click **Save**.

  The result appears as follows (your line indenting may be different, but that is acceptable):



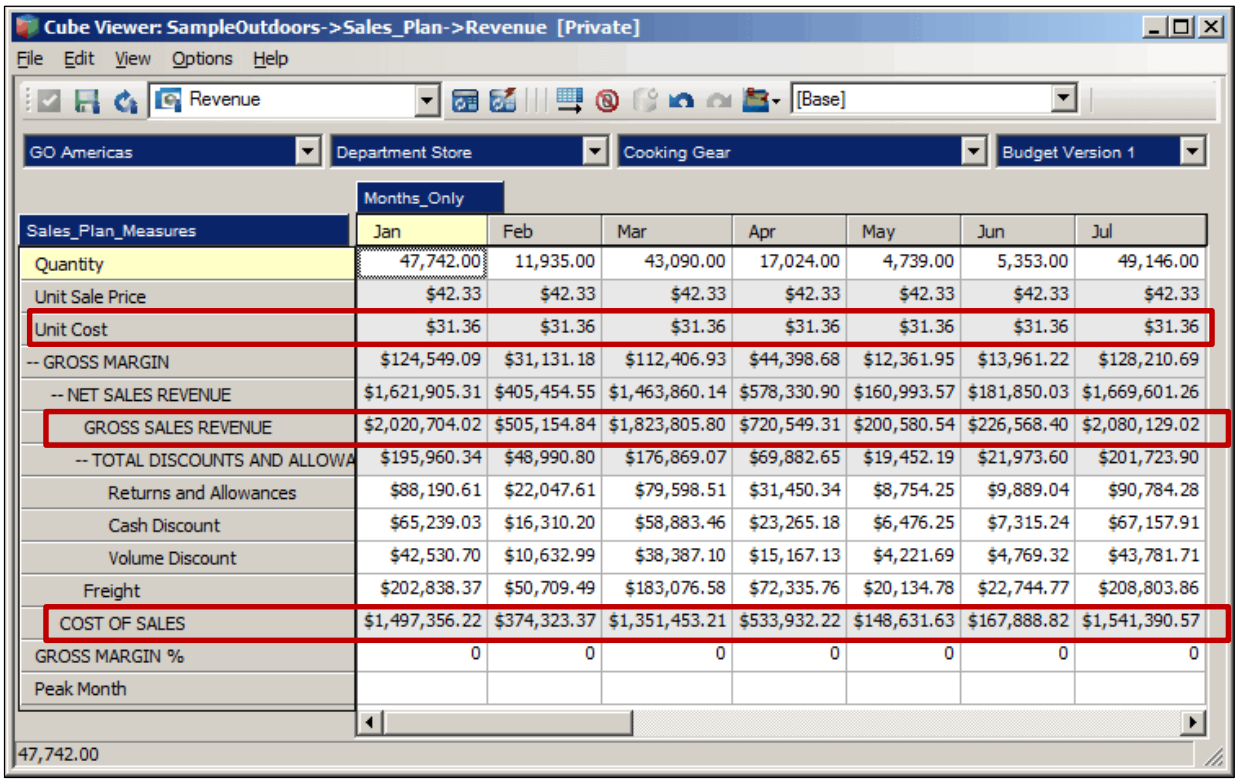
Task 4. Calculate Revenue view of the Sales_Plan cube.

- Open the **Sales_Plan** cube.
- In the **Select View** list, select **Revenue**.
- Click **Sales_Plan_Measures**, click **All** and then click **OK**.
- Select **GO Americas** from the **Subsidiaries** dimension (if necessary) by double-clicking the dimension and expanding **GO AMERICAS REGION** in the **Subset Editor**.

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- Click **Recalculate**.

  Unit Cost should now contain values and the Sales_Plan_Measures should be reorganized:



- Close the **Rules Editor**.
- Close the **Cube Viewer**, saving the view and overwriting the existing view if prompted.
- Right-click **SampleOutdoors** and click **Save Data**.
- Close **Architect** and all other open windows.

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IBM Training

Optimize rule performance

IBM Planning Analytics (v2.0)

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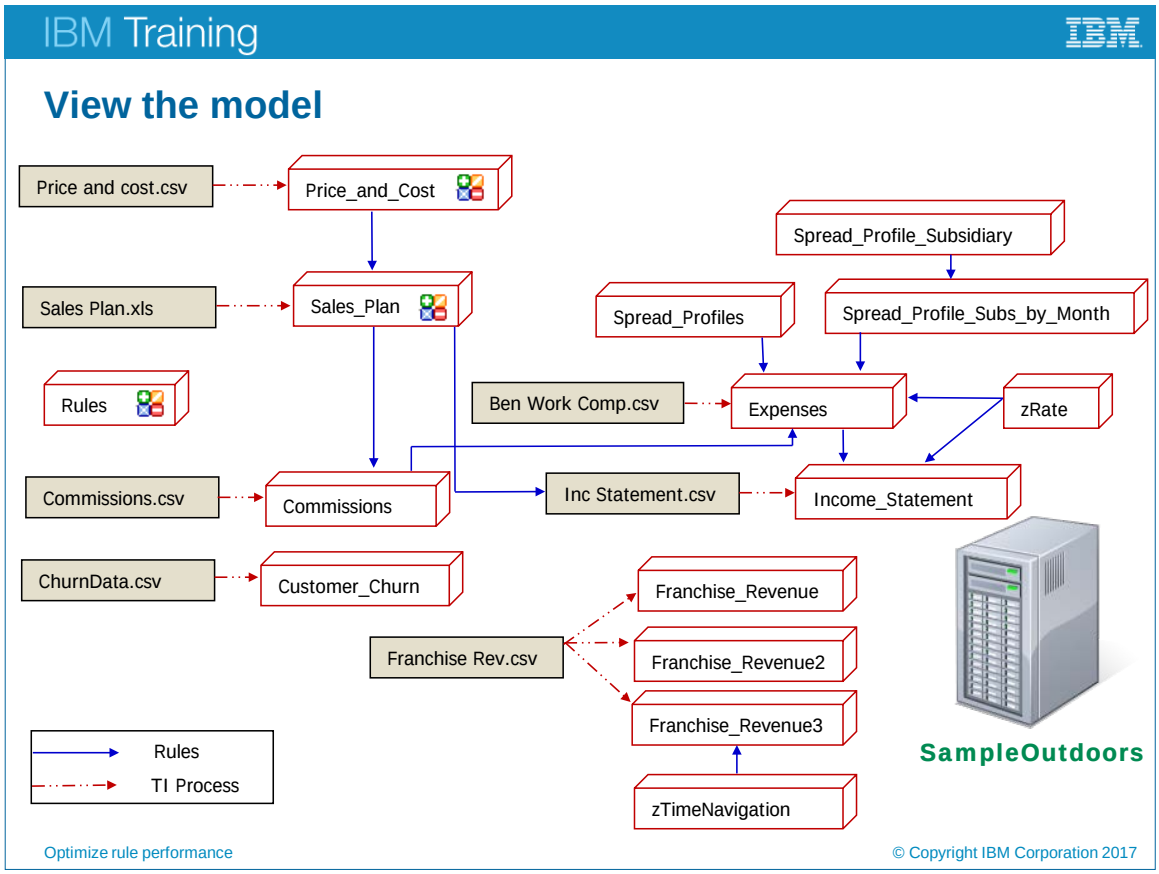
Unit objectives

- Describe consolidations and sparsity
- Enhance consolidation performance using SKIPCHECK
- Use FEEDERS to optimize aggregations
- Check the accuracy of rules and feeders

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Unit objectives

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View the model

In this unit you will not create any new cubes. Instead, you will modify existing rules and cubes.

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Understanding consolidations and sparsity

- The consolidation engine skips empty cells or zeros.
- TM1 generates a zero when a stored or calculated value is not found.
- Example: not all products are sold all year:

	Total Year	Q1	Q2	Q3	Q4
Total Products	130	10	20	40	70
Product1	40	10	0	10	20
Product2	20	0	20	0	0
Product3	40	0	0	30	10
Product4	40	0	0	0	40

Optimize rule performance

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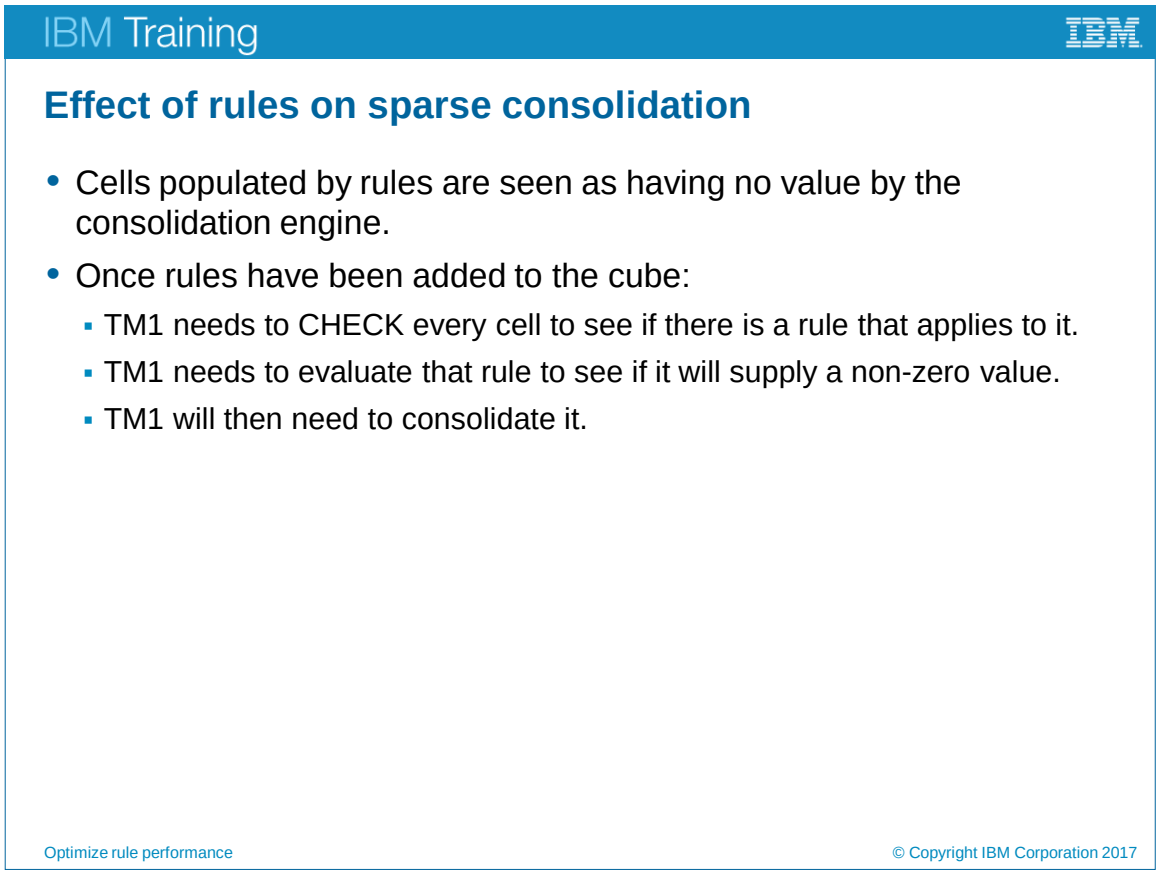
Understanding consolidations and sparsity

In the above example there are four quarters in the Time dimension and four products in the Products dimension. There are a total of 16 values (4*4=16) in the cube. However, TM1 only uses 7 of the 16 cells when aggregating the consolidated values (for example, Total Year or Total Products).

If you have 100 products, 60 time periods, 20 markets, and 5 measures you need to track, you would have 100*60*20*5=600,000 potential values in your cube. In a multidimensional model, this is a small example and it is very likely that most of these values will be zero. It is much faster to look at 1% of the values (6,000) instead of evaluating 600,000 values.

By avoiding the consolidation of zero values, TM1 also avoids the explosion in the size of the cube that can arise if zero values at the base level are consolidated to create even more zero values at consolidated levels.

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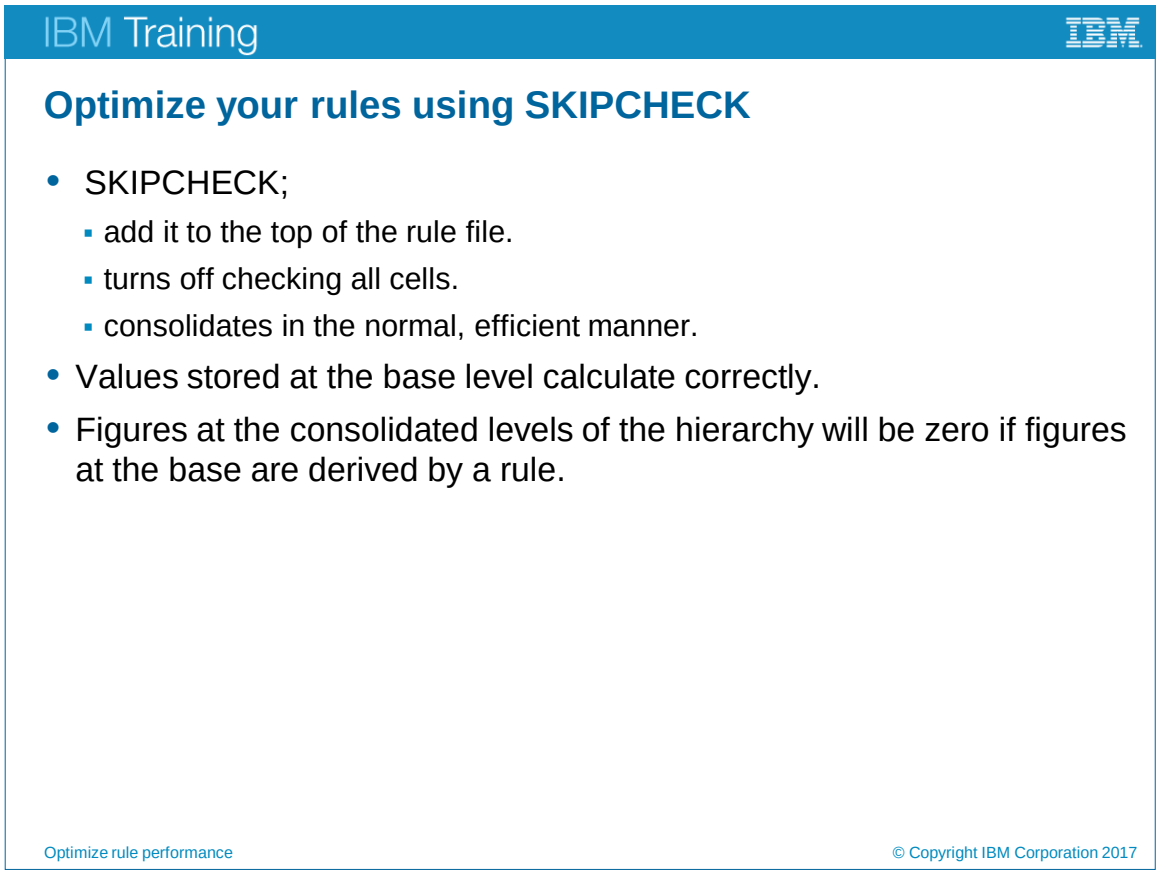
Effect of rules on sparse consolidation

Rule checking occurs as soon as a rule file is associated with a cube, even if the file contains no rules.

This will significantly increase the time it takes to aggregate data for consolidations.

You can turn off this cell CHECKing by using the SKIPCHECK statement in the rule file. Adding the SKIPCHECK statement restores TM1's sparse consolidation algorithm and improves performance. In many models, less than one in a million cells is actually populated.

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Optimize your rules using SKIPCHECK

In this course we are working with a limited amount of data. In your model, you may have much more data to calculate and consolidate. This may cause the aggregation engine to slow significantly.

In a large cube the number of zero values will typically be very high. The ratio between the number of non-zero cells in a cube and the total number of cells in a cube is called sparsity. Often times, there are more zero values in a cube than non-zero values.

Zeroes are a fact of life in multi-dimensional cubes. Managing those zeroes correctly is very important in multi-dimensional modeling tools. To avoid wasting space, TM1 never stores a zero value. TM1 simply generates a zero when someone requests a value for a combination of elements that does not have a figure. As TM1 does not store zero values, this reduces the storage requirements for large cubes.

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SKIPCHECK with consolidations

- The last two columns contain calculations.
- The Total Products consolidation ignores the values in them:

	Total Year	Q1	Q2	Q1+Q2	Q3+Q4
Total Products	30	10	20	0	0
Product1	10	10	0	10	20
Product2	20	0	20	20	0
Product3	0	0	0	0	10
Product4	0	0	0	0	40

Optimize rule performance © Copyright IBM Corporation 2017

SKIPCHECK with consolidations

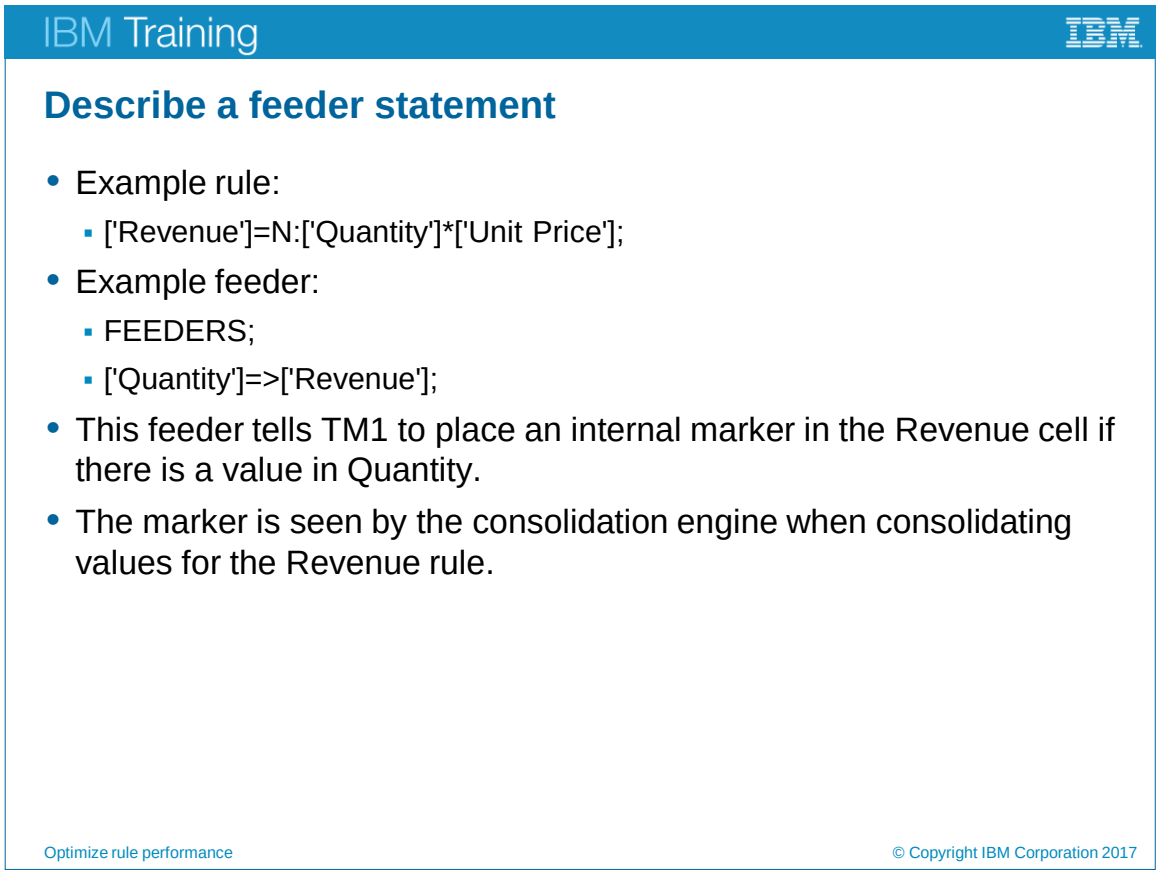
To correct the problem where consolidations are displayed as zeros, FEEDERS are used.

Feeders are a way of telling TM1 which calculated cells will receive non-zero values via rules, so TM1 knows where there are non-zero values to be consolidated. This improves the speed of consolidations while preserving accuracy.

Use the FEEDERS; statement in a rule file below all rules statements, and then follow it with statements indicating exactly which cells will contain values used in rules.

For example, assume Total Year consolidates Q1 and Q2 separately and not Q1+Q2 and Q3+Q4. FYI: Q1+Q2 is not consolidated, but calculated by a rule.

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Describe a feeder statement

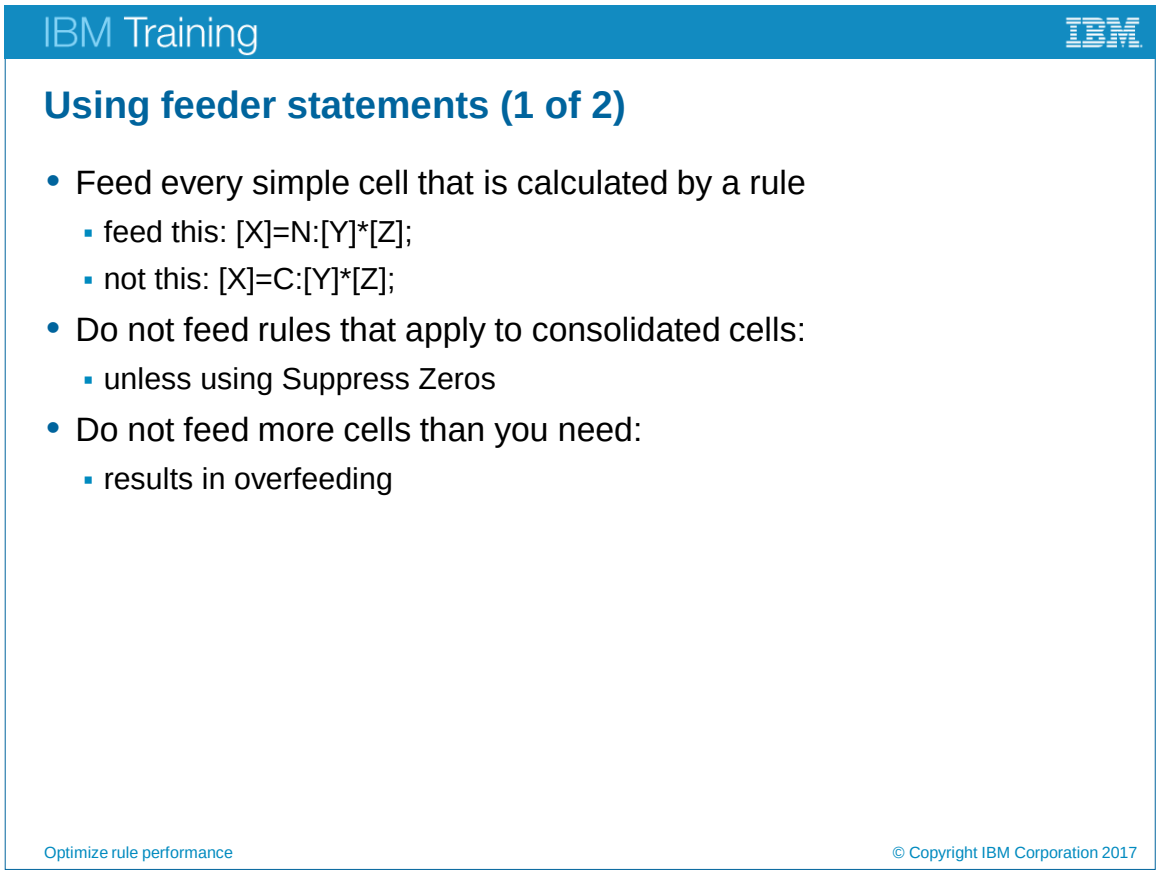
Feeders are activated when the cube is first loaded into memory, and thereafter, whenever a value is changed in the cube. This is the opposite of rules, which are only activated when a value is requested by a user. You can think of rules as PULLING values in, while feeders PUSH markers out. Once a feeder is fired, it does not fire again until the server is restarted or the cube is reloaded.

When writing feeders you need to ask the question "Which element or combination of elements on the right hand side of the rule determines whether there will be a value on the left hand side of the rule?"

Writing feeders for rules is known as "feeding" cells.

In this example, you would feed 'Revenue' based on 'Quantity' because the Quantity would only contain a value if there were actually sales made. Unit Price is more likely to contain values even if there were no quantities sold. Quantity is more likely to contain zeros and allow TM1 to look at fewer cells so it is a better choice.

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Using feeder statements

We need to be careful about over-feeding: for example, feeding more cells than is necessary. In other words, we do not want to feed cells whose result will be zero anyway.

Remember that when you feed a cell, a small marker is stored in that cell, which takes up a small amount of memory. Feeding a cell also means that TM1 will evaluate any rule associated with this cell when it is consolidating, even if that rule is going to return a zero value.

Feeders take up processor time, particularly when the cube is first loaded, since at that time the processor has to evaluate all the feeders. Thereafter, the burden is lower, since TM1 only needs to evaluate feeders when a value changes, and even then, only those feeders related to the value that was changed.

Chronic over-feeding will cause performance problems since it effectively returns us to the state that we were in, before we added the SKIPCHECK statement.

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Using feeder statements (2 of 2)

- In multiplication or division, feed using the least data:
  - [Revenue]=[Quantity]*[Price];
  - [Quantity]=>[Revenue];
- In addition, subtraction, or logic, feed using each section:
  - [HalfYear]=[Q1]+[Q2];
  - [Q1]=>[HalfYear]; [Q2]=>[HalfYear];

Optimize rule performance

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A little over-feeding is not normally a problem. When in doubt it is best to over-feed. Under-feeding (i.e., failing to feed cells for calculated results that will be non-zero) will result in incorrect consolidation of results.

You may need to feed consolidations as well as base level cells when users turn on Suppress Zeros for a view. Suppress Zeros hides all values displaying only zeros in the entire row or column. If both the base level and consolidated level calculation is not fed, Suppress Zeros may hide the columns or rows containing calculations.

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IBM Training

Inter-cube feeders

Feeder Cube (Source)			
	Jan	Feb	Mar
P1	1 ⇨	2 ⇨	8 ⇨
P2	2 ⇨	4 ⇨	6 ⇨
P3	3 ⇨	6 ⇨	4 ⇨
P4	4 ⇨	8 ⇨	2 ⇨

Rules Cube				
	Q1	Jan	Feb	Mar
P1	11	1 **DB()**	2 **DB()**	8 **DB()**
P2	12	2 **DB()**	4 **DB()**	6 **DB()**
P3	13	3 **DB()**	6 **DB()**	4 **DB()**
P4	14	4 **DB()**	8 **DB()**	2 **DB()**

- Rules can 'pull' data into cells from other cubes.
- Feeders put a marker is the source cells and 'push' it to the calculated cell in the target cube.

Optimize rule performance

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Inter-cube feeders

While rules reside in the target cube, feeders generally reside in the source cube.

The rule is attached to the cube that will display the result.

Use the DB() function to reference the cell(s) containing the data and pull in the data.

This is because of the "push" nature of feeders. The feeder statement tells the system to feed cells when a value changes in the source cells. If the source cells are in another cube, the feeder statement needs to be there too.

The feeder statements are attached to the cube that contains the source data and push the marker to the target calculation.

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IBM Training

Demonstration 1

Add SKIPCHECK and FEEDERS

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Demonstration 1: Add SKIPCHECK and FEEDERS

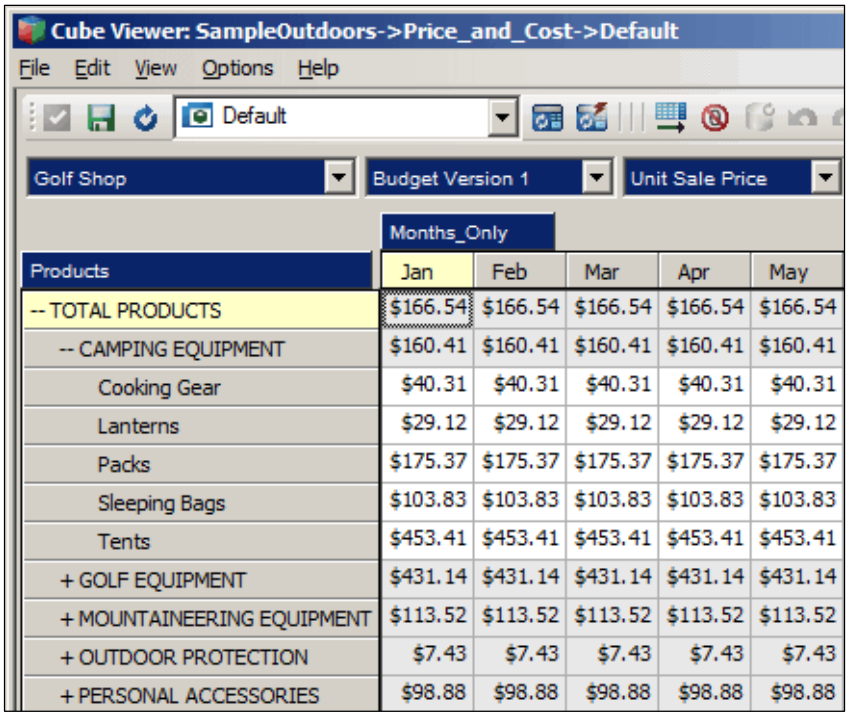
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Demonstration 1: Add SKIPCHECK and FEEDERS

Purpose:
You want to improve the performance of calculations in the Price_and_Cost cube. To do this you will add SKIPCHECK and FEEDERS to your rules in this cube.

Task 1. Determine which rules must be fed.

1. Open **Windows Explorer** and go to **C:\Program Files\IBM\cognos\tm1_64\bin**.
2. Double-click **tm1a.exe** and maximize the Architect window.
   You will be using the 32-bit version of Architect so that you can work with the Advanced Rules Editor.
3. Expand **TM1**, double-click **SampleOutdoors**, and then log on with a user ID of **admin** and a password of **apple**.
4. In the Server Explorer window, open the **Default** view of the **Price_and_Cost** cube.
5. Drill down on **CAMPING EQUIPMENT**.
   The result appears as shown below:



Products	Jan	Feb	Mar	Apr	May
-- TOTAL PRODUCTS	$166.54	$166.54	$166.54	$166.54	$166.54
-- CAMPING EQUIPMENT	$160.41	$160.41	$160.41	$160.41	$160.41
Cooking Gear	$40.31	$40.31	$40.31	$40.31	$40.31
Lanterns	$29.12	$29.12	$29.12	$29.12	$29.12
Packs	$175.37	$175.37	$175.37	$175.37	$175.37
Sleeping Bags	$103.83	$103.83	$103.83	$103.83	$103.83
Tents	$453.41	$453.41	$453.41	$453.41	$453.41
+ GOLF EQUIPMENT	$431.14	$431.14	$431.14	$431.14	$431.14
+ MOUNTAINEERING EQUIPMENT	$113.52	$113.52	$113.52	$113.52	$113.52
+ OUTDOOR PROTECTION	$7.43	$7.43	$7.43	$7.43	$7.43
+ PERSONAL ACCESSORIES	$98.88	$98.88	$98.88	$98.88	$98.88

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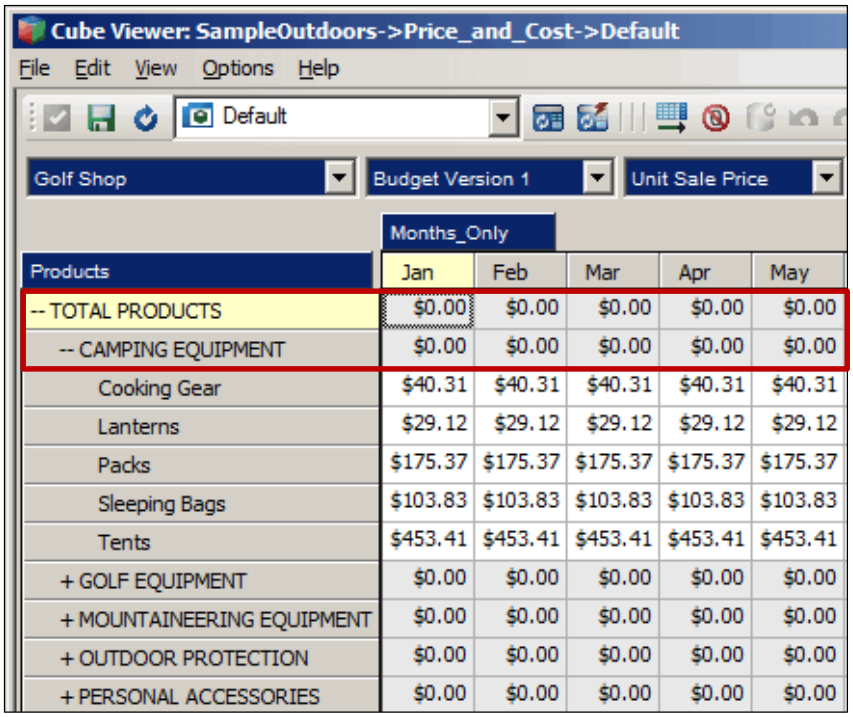
6. Open the **Rules Editor** for **Price_and_Cost**.



7. Insert a new line at the beginning of row 1 and type **SKIPCHECK;**.

```
SKIPCHECK;
# hold the Unit Sale Price for the average price
['UnitPrice_hold' ]  = N: ['Unit Sale Price' ] ;
['UnitPrice_count' ]  = N: IF(['Unit Sale Price'] <> 0, 1, 0);
['Unit Sale Price' ]  = C: ['UnitPrice_hold' ] \ ['UnitPrice_count' ] ;
#Calculate Average Unit Cost for consolidated levels.
['UnitCost_hold'] = N: ['Unit Cost'];
['UnitCost_count'] = N: IF (['Unit Cost'] <>0,1,0);
['Unit Cost']  = C: ['UnitCost_hold'] \ ['UnitCost_count'] ;
```

8. Save the rule and recalculate the view.



The consolidation no longer calculates the average because TM1 does not see the calculated values in the N: level elements UnitPrice_hold and UnitPrice_count. They are calculations stored only in memory. TM1 does not know they contain values to compute, and they are seen as 0 when aggregating consolidated values.

You will need to feed those expressions so the values for these cells can be used for other calculations (like the average Unit Sale Price).

You will feed the N: level calculations for UnitPrice_hold and UnitPrice_count using Unit Sale Price.

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Task 2. Add the feeder statement.

1. Create a blank row at the bottom of the rules, on **line 11** type **FEEDERS;** and then press **Enter**.
2. Click **Brackets** and then double-click **Price_and_Cost_Measures**.
3. Click **Unit Sale Price** and then click **OK** to return to the rules.
4. Click to the right of the bracket, click **Insert Qualifier or Operator** and then click **=>**.
5. Click **Brackets** and then double-click **Price_and_Cost_Measures**.
6. Click **UnitPrice_hold** and then click **OK** to return to the rules.
7. Click to the right of the bracket and type a **,** (comma)
8. Click **Brackets**, double-click **Price_and_Cost_Measures**, and then select **UnitPrice_count**.
9. Click **OK** to return to the rules, click to the right of the bracket, and then add a **;** to the end of the line.

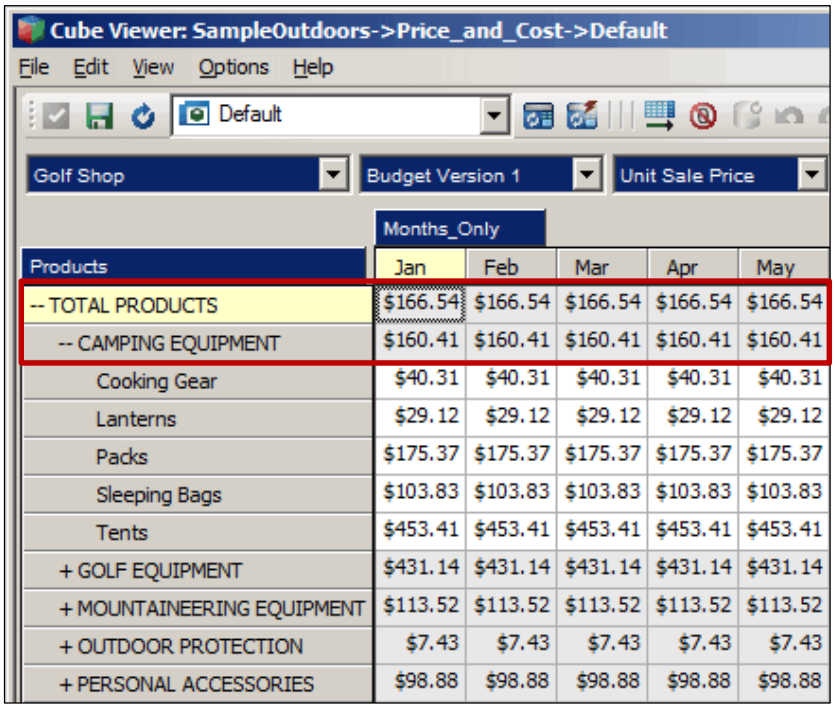
   In this example you would feed both of the rules with the same element so you can reference them both in a single line by separating them with commas.

   The result appears as follows:

```
1   SKIPCHECK;
2   # hold the Unit Sale Price for the average price
3   ['UnitPrice_hold' ]  = N: ['Unit Sale Price' ] ;
4   ['UnitPrice_count' ]  = N: IF(['Unit Sale Price'] <> 0, 1, 0);
5   ['Unit Sale Price' ]  = C: ['UnitPrice_hold' ] \ ['UnitPrice_count' ] ;
6   #Calculate Average Unit Cost for consolidated levels.
7   ['UnitCost_hold'] = N: ['Unit Cost'];
8   ['UnitCost_count'] = N: IF (['Unit Cost'] <>0,1,0);
9   ['Unit Cost']  = C: ['UnitCost_hold'] \ ['UnitCost_count'] ;
10
11  FEEDERS;
12  ['Unit Sale Price' ]  => ['UnitPrice_hold' ] , ['UnitPrice_count' ] ;
```

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10. Click **Save** and then recalculate the view.
    The view should now consolidate properly.
    The result appears as shown below:

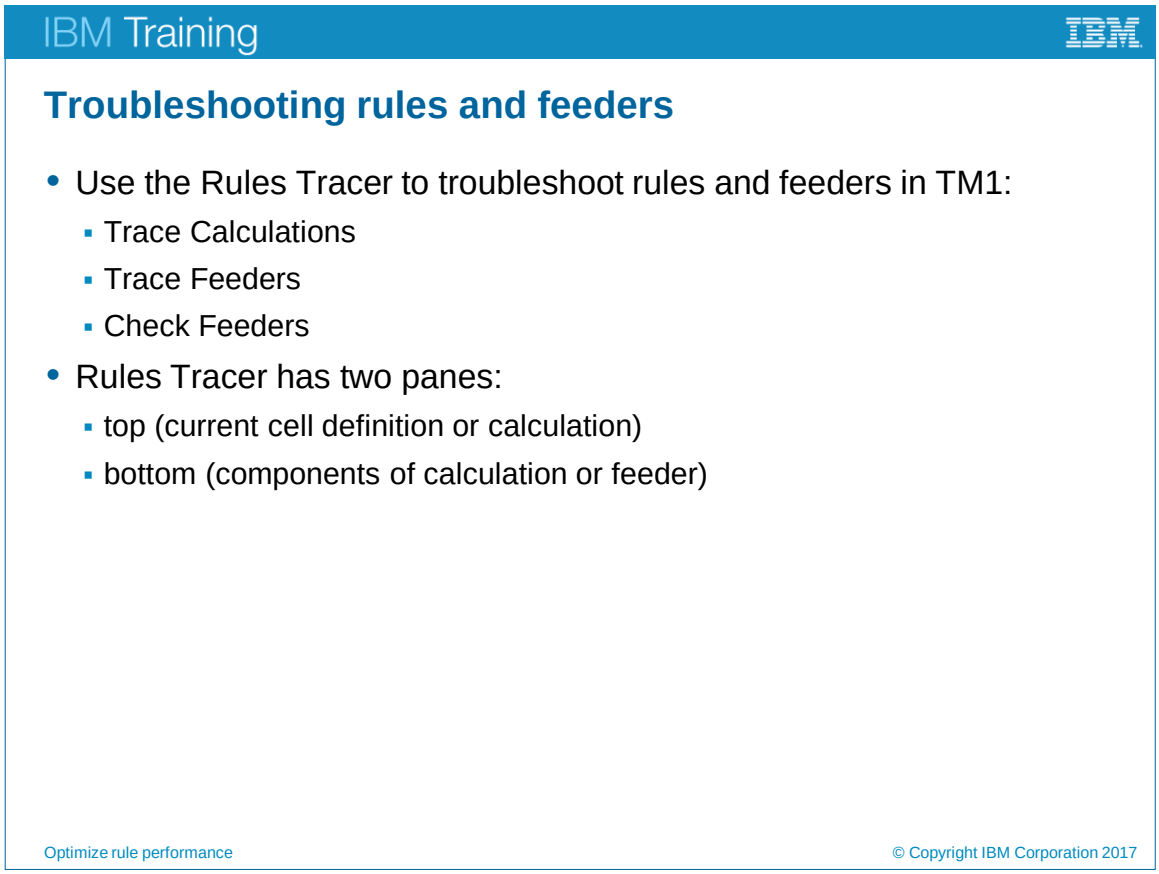


    As you add more data to your cube, it will calculate more efficiently with the feeders that you added, than if you did not add the feeder statements.
11. Close the Rules Editor (saving if prompted).
12. Close the Cube Viewer without saving changes.
13. Right-click **SampleOutdoors** and click **Save Data**.
14. Leave Architect open for the next demonstration.

Note: Our sample data is very small so we did not see a large performance impact without proper feeders. However, it is much more common to see a large increase in calculation time when opening views of a cube that have just added rules. This has caused many people to avoid using rules. Rules are a very powerful addition to a cube. It is recommended that you consider feeders while you write your business rules.

Results:
You improved the performance of calculations in the Price_and_Cost cube by adding SKIPCHECK and FEEDERS to your rules in the cube.

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Troubleshooting rules and feeders

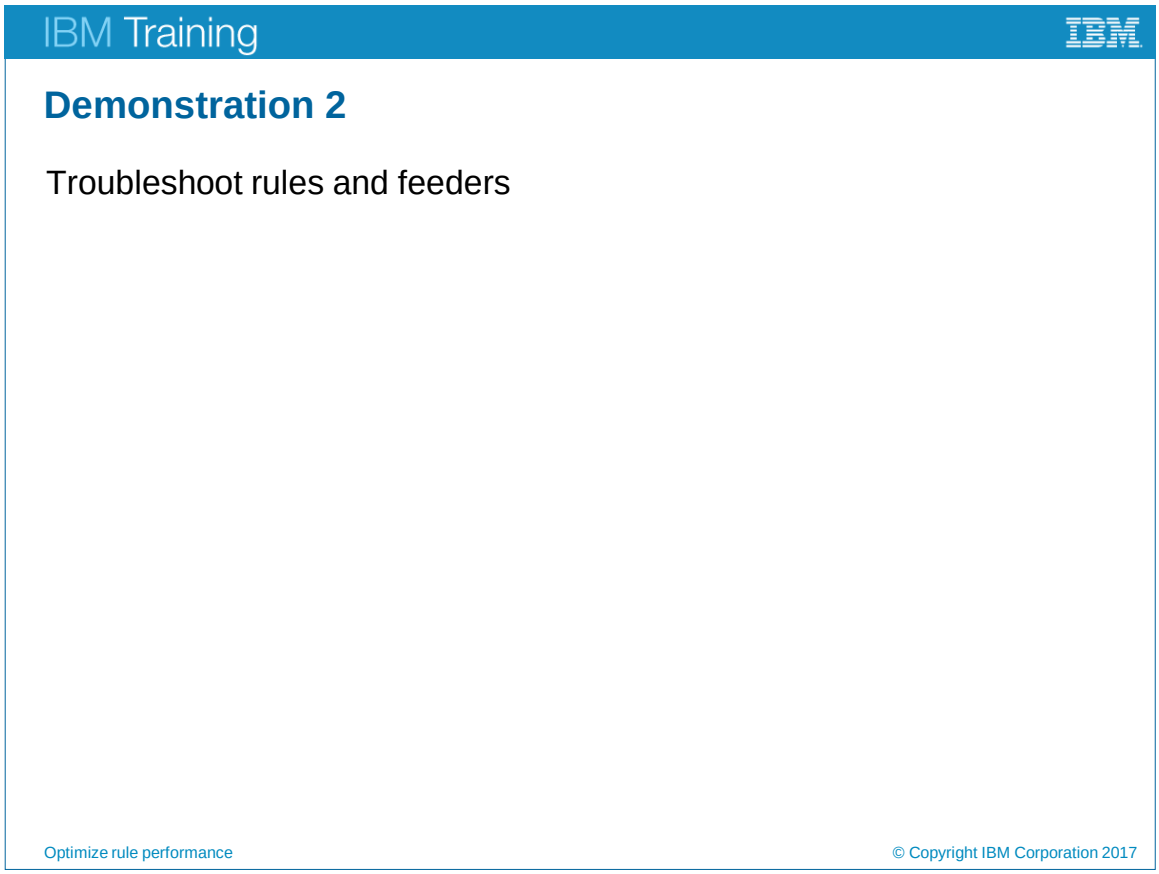
Right-click any cell and choose Trace Calculation to trace it to the base or leaf level elements.

Trace Feeders checks the way the selected cell feeds other cells. Trace Feeders is only enabled for base or leaf level cells. Trace Feeders is not enabled for consolidated cells.

If a cube contains a rule with SKIPCHECK and FEEDERS statements, you can use the Rules Tracer to Check Feeders to check that TM1 properly feeds the components of the consolidation.

If the Trace pane is empty, it means the consolidation is fed. Otherwise you will see 'not fed' displayed.

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Demonstration 2: Troubleshoot rules and feeders

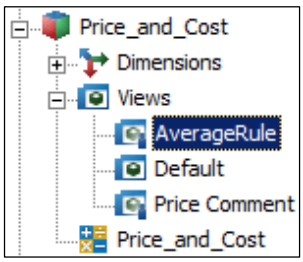
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Demonstration 2: Troubleshoot rules and feeders

Purpose:
You want to verify that your calculations are consolidating properly. In addition, you want to ensure that your feeders are working as expected. You will use the Rules Tracer to review your calculations and the associated feeder statements.

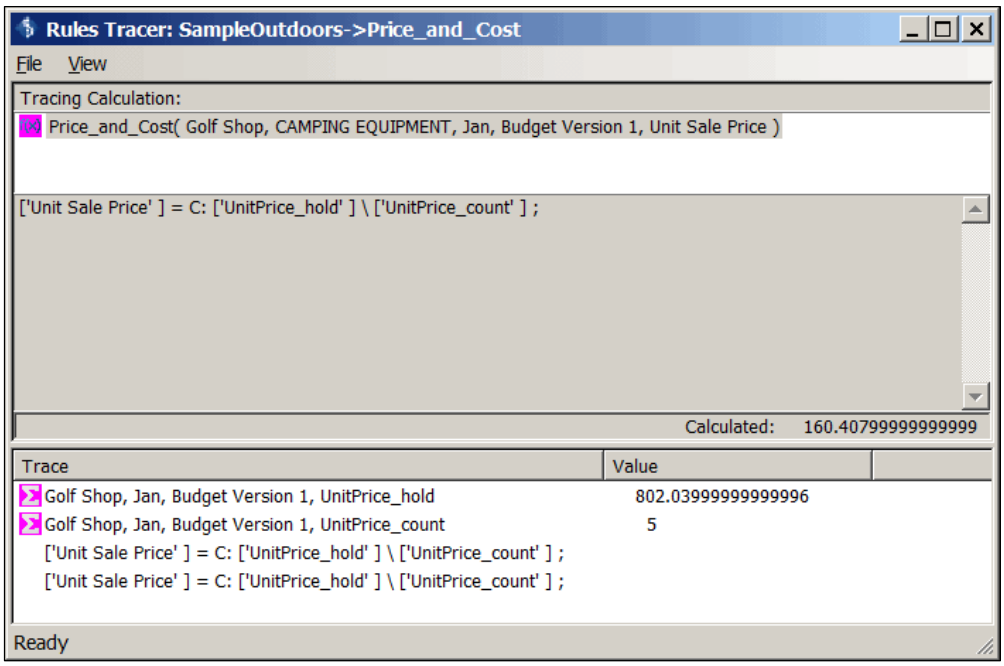
Task 1. Trace your calculations.

1. Open the **AverageRule** view of the **Price_and_Cost** cube:



2. Right-click the **Unit Sale Price** for **CAMPING EQUIPMENT** cell (containing $160.41) and then click **Trace Calculation**.

   The results appear similar to the following (the contents of your Trace pane may appear slightly different):



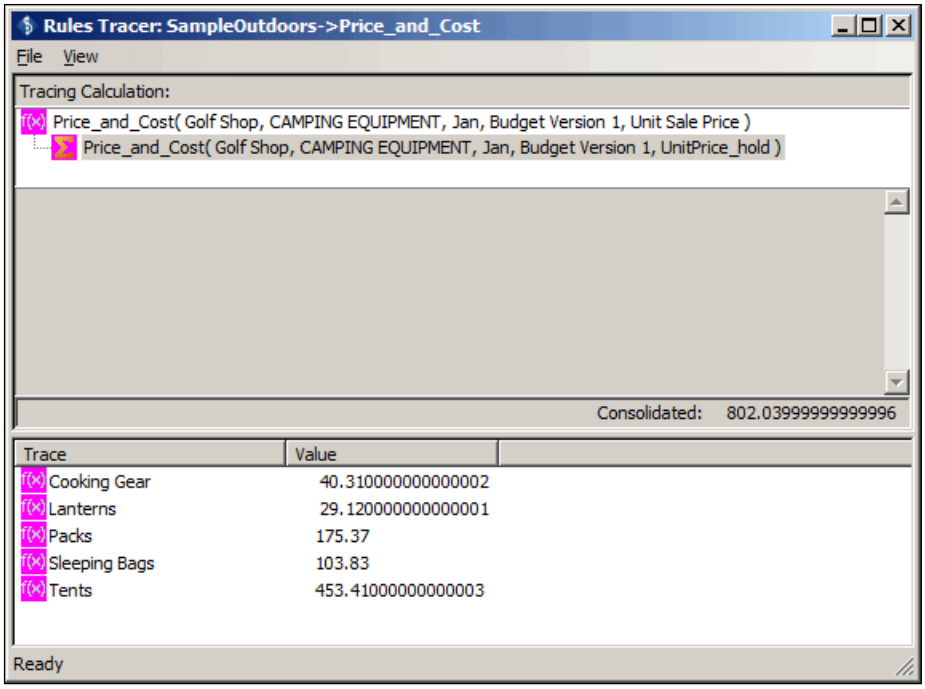
   The top pane shows the calculation (consolidation) and the bottom pane shows the values that make up the calculation above.

   You can double click a value in the bottom pane to trace it.

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3. Double click the first line in the bottom pane (it should say something like **Golf Shop, Jan, Budget Version 1, UnitPrice_hold**).

   The results appear similar to the following (the contents of your Trace pane may appear slightly different):



   Under Tracing Calculation, it displays the path that you are tracing. You can double-click on an expression here to return to a previously traced expression.

   The bottom Trace will show the base level elements with a gray bullet  , indicating there is nothing further to trace.

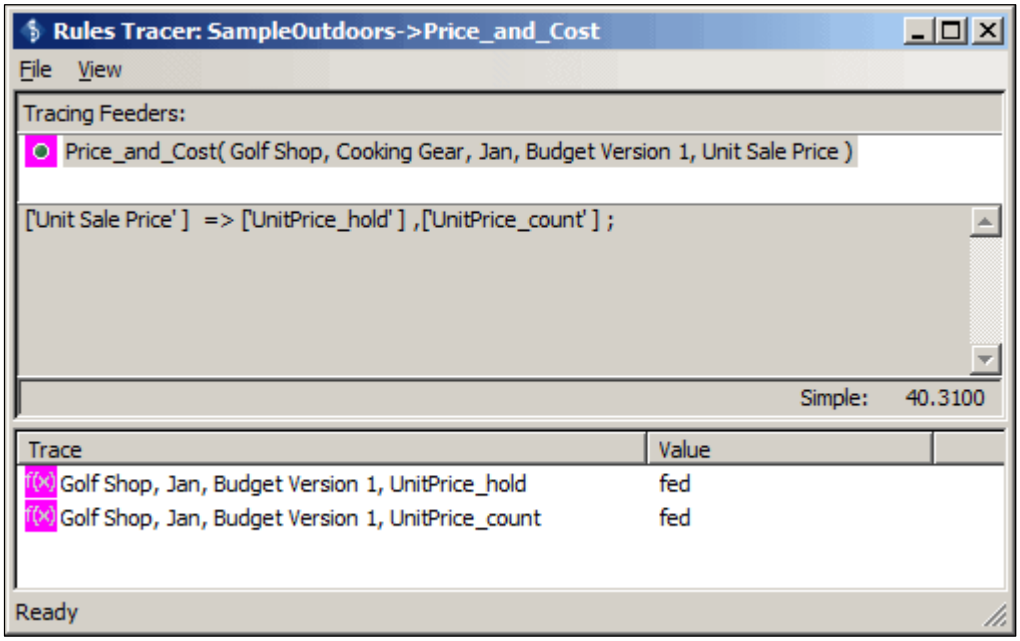
4. Click **File** and then click **Exit** to close the window.
5. Leave the **AverageRule** view open for the next task.

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Task 2. Use the Rules Tracer to trace a feeder.

1. Right-click **Cooking Gear/Unit Sale Price** (containing 40.31) and then click **Trace Feeders**.

   The results appear similar to the following (the contents of your Trace pane may appear slightly different):



   In the bottom pane it shows the exact cells that are being fed by the feeder displayed in the top pane.

   You see 'fed' in the Value column. The window displays the feeder using this cell in the gray area and the expressions being fed at the bottom.

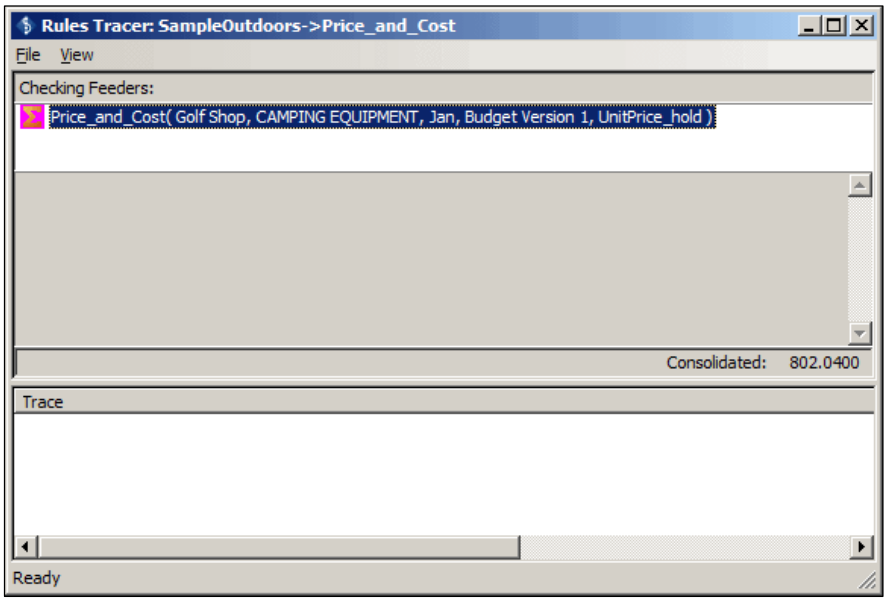
   If you had selected a consolidated element, such as CAMPING EQUIPMENT, Trace Feeders would be disabled on the shortcut menu because Trace Feeders is only available at the base or n level elements.

2. Close the **Rules Tracer** window.

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Task 3. Check feeders.

1. Right-click **CAMPING EQUIPMENT/UnitPrice_hold** (containing 802.04) and then click **Check Feeders**.



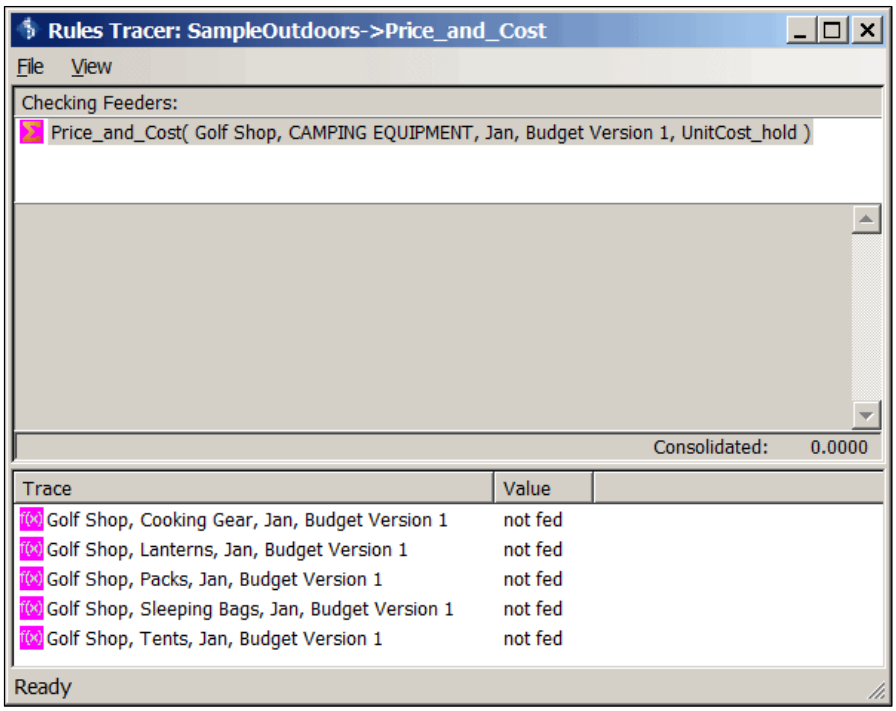
   Nothing is displayed in the Rules Tracer except for the calculation. This is the desired result. This is a consolidation that is fed properly.

2. Close the **Rules Tracer**.
3. In the **Cube Viewer**, click **Price_and_Cost_Measures** to open the **Subset Editor**, click **All**, and then click **OK**.
4. Click **Recalculate**.

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5. Right-click **CAMPING EQUIPMENT/UnitCost_hold** (containing 0) and then click **Check Feeders**.

   The results appear similar to the following (the contents of your Trace pane may appear slightly different):



   Here we see the elements making up this consolidation and none of them are being used to feed it. You will fix this in a later exercise.

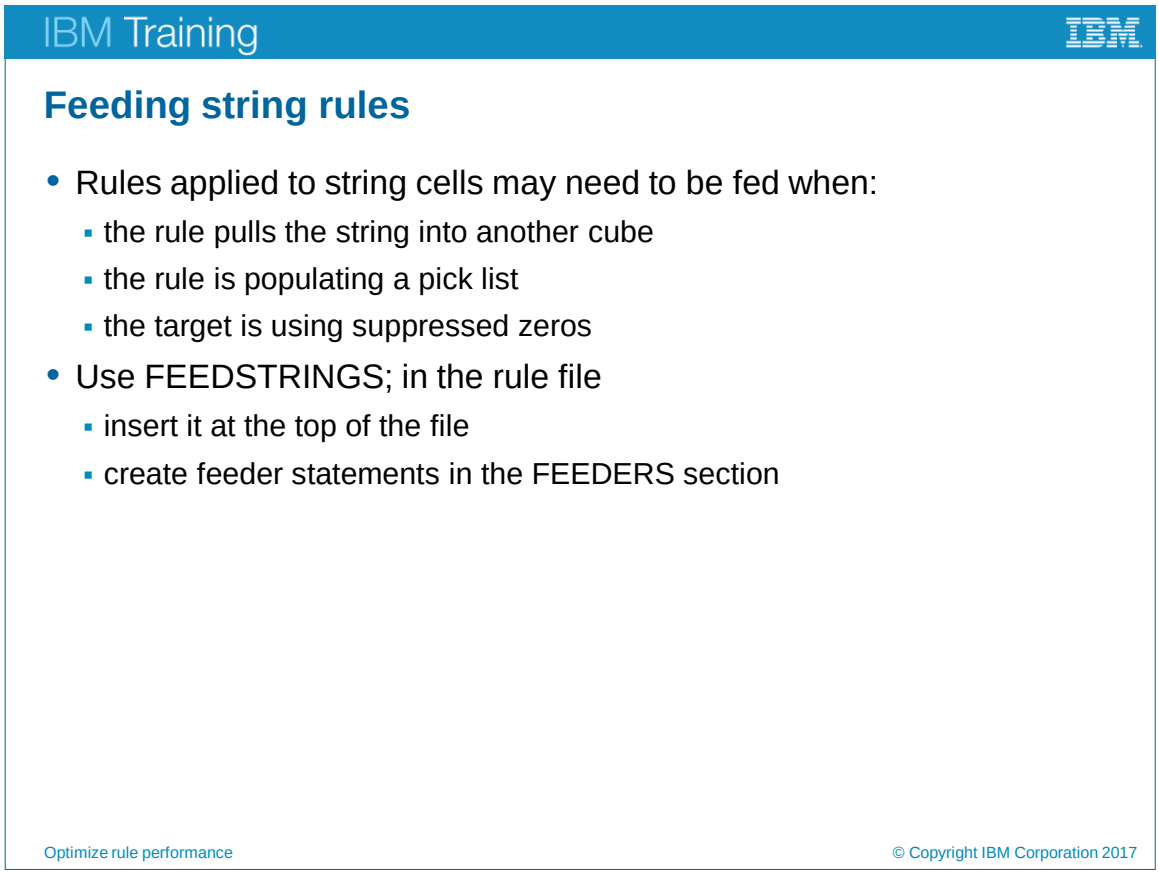
6. Close the **Rules Tracer** and the **Cube Viewer** window without saving changes.

   Note: You can only check the FEEDERS for a cell once per TM1 session. The action of checking the FEEDERS actually feeds the components of the consolidation. Any subsequent checking of the FEEDERS does not yield accurate results. If you want to check the FEEDERS for a cell more than once, you must recycle (shut down and restart) the TM1 server before every check.

7. Leave **Architect** open for the upcoming exercise.

Results:
You used the Rules Tracer to review your calculations. You also verified the leaf level cells feeding other calculations and that the calculations are being fed by the correct base cells.

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Feeding string rules

FEEDSTRINGS;

Once this declaration is in place, you can set up feeders for string cells in a cube view, and rely on the string to be available to other rules even if the view is zero-suppressed. Statements that define feeders for string cells should be created below the FEEDERS; declaration in your rule.

To enable feeding of string cells, insert the FEEDSTRINGS; declaration as the first line of your rule.

Feeder statements are placed under the FEEDERS; declaration, after the rule statements.

Feeders from string cells fire whenever a value changes.

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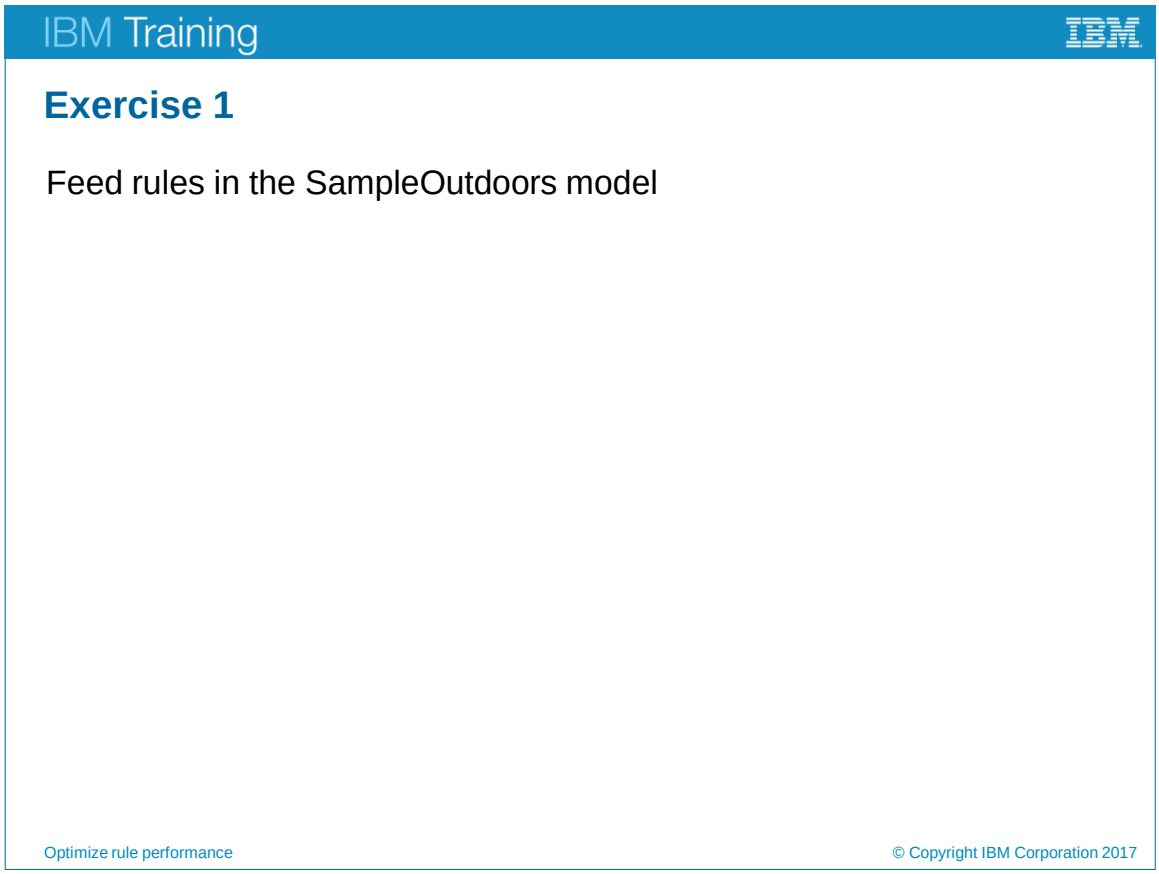
Unit summary

- Describe consolidations and sparsity
- Enhance consolidation performance using SKIPCHECK
- Use FEEDERS to optimize aggregations
- Check the accuracy of rules and feeders

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Unit summary

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Exercise 1: Feed rules in the SampleOutdoors model

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Exercise 1: Feed rules in the SampleOutdoors model

Now that you have computed the average unit cost and average unit sale price for the Price_and_Cost cube, you will need to feed it to ensure the consolidations work properly and efficiently. In addition, you need to feed rules in the Sales_Plan cube.

- Add feeders to the rules for Price_and_Cost.
- Verify the correct data in the Price_and_Cost cube.
- Add SKIPCHECK and FEEDERS to the Sales_Plan rules.
- Verify the correct data in the Sales_Plan cube.

For more information about where to work and the exercise results, refer to the Tasks and results section that follows. If you need more information to complete a task, refer to earlier demonstrations for detailed steps.

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Exercise 1: Tasks and results

Task 1. Add feeders to Price_and_Cost rules.

1. Right-click the **Price_and_Cost** cube, and then click **Edit Rule**.
2. On the last line type the following:
   ['Unit Cost'] => ['UnitCost_hold'] , ['UnitCost_count'] ;
3. Click **Check Syntax** and correct any errors.
   The result appears as follows:

```
10
11   FEEDERS;
12   ['Unit Sale Price' ]  => ['UnitPrice_hold' ] , ['UnitPrice_count' ] ;
13   ['Unit Cost' ]  => ['UnitCost_hold' ] , ['UnitCost_count' ] ;
```

4. Click **Save** and then close the Rules Editor.

Task 2. Verify that correct data is calculated in the Price_and_Cost cube.

1. Double-click the **Price_and_Cost** cube.
2. From the **Select View** list, click **AverageRule**.
3. Click **Price_and_Cost_Measures** to open the Subset Editor, click **All**, and then click **OK**.
4. Click **Recalculate**.
5. Verify that the **CAMPING EQUIPMENT** cells for **UnitCost_hold** and **UnitCost_count** are now displaying values instead of zeros:

Golf Shop | Budget Version 1 | Jan

Products	Price_and_Cost_Measures: Unit Sale Price	Unit Cost	Margin	Margin%	UnitPrice_hold	UnitPrice_count	UnitCost_hold	UnitCost_count
-- TOTAL PRODUCT	$166.54	$109.79	$0.00	0.00	3497.35	21	2305.6	21
-- CAMPING EQU	$160.41	$128.11	$0.00	0.00	802.04	5	640.57	5
Cooking Gea	$40.31	$29.87	$0.00	0.00	40.31	1	29.87	1
Lanterns	$29.12	$21.14	$0.00	0.00	29.12	1	21.14	1
Packs	$175.37	$126.51	$0.00	0.00	175.37	1	126.51	1
Sleeping Bag	$103.83	$78.01	$0.00	0.00	103.83	1	78.01	1
Tents	$453.41	$385.04	$0.00	0.00	453.41	1	385.04	1

6. Close the **Cube Viewer** without saving changes.

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Task 3. Add SKIPCHECK and FEEDERS to Sales_Plan rules.

1. Right-click the **Sales_Plan** cube and then click **Edit Rule**.
2. At the beginning of the first line, press **Enter**.
3. On **line 1** type:
   SKIPCHECK;
4. At the end of the last line, press **Enter** to add a blank line, and then type:
   FEEDERS;
5. Press **Enter**, and then type:
   ['Quantity'] => ['GROSS SALES REVENUE'], ['COST OF SALES'];
6. Click **Check Syntax** and correct any errors.
   The result appears as follows (your indenting may be different, but that is acceptable):

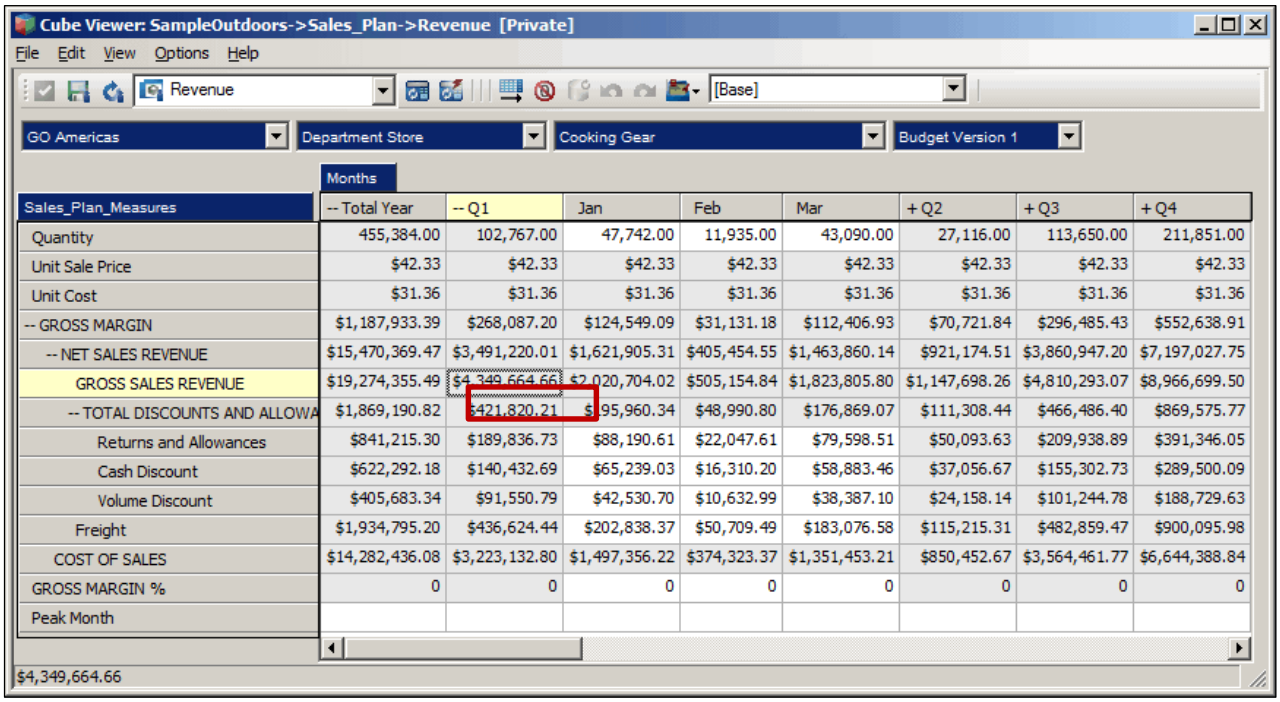
```
1    SKIPCHECK;
2    ['Unit Sale Price' ]  = N: DB('Price_and_Cost',
3       !Channels, !Products, !Months, !Versions, 'Unit Sale Price');
4       C:['GROSS SALES REVENUE']\['Quantity'];
5    ['Unit Cost'] = N:DB('Price_and_Cost',
6       !Channels, !Products, !Months, !Versions, 'Unit Cost');
7       C:['COST OF SALES'] \ ['Quantity'];
8    ['GROSS SALES REVENUE']= N:['Quantity']*['Unit Sale Price'];
9    ['COST OF SALES'] = N:['Quantity']*['Unit Cost'];
10
11   FEEDERS;
12   ['Quantity' ] => ['GROSS SALES REVENUE' ], ['COST OF SALES' ];
```

7. Click **Save** and close the **Rules Editor**.

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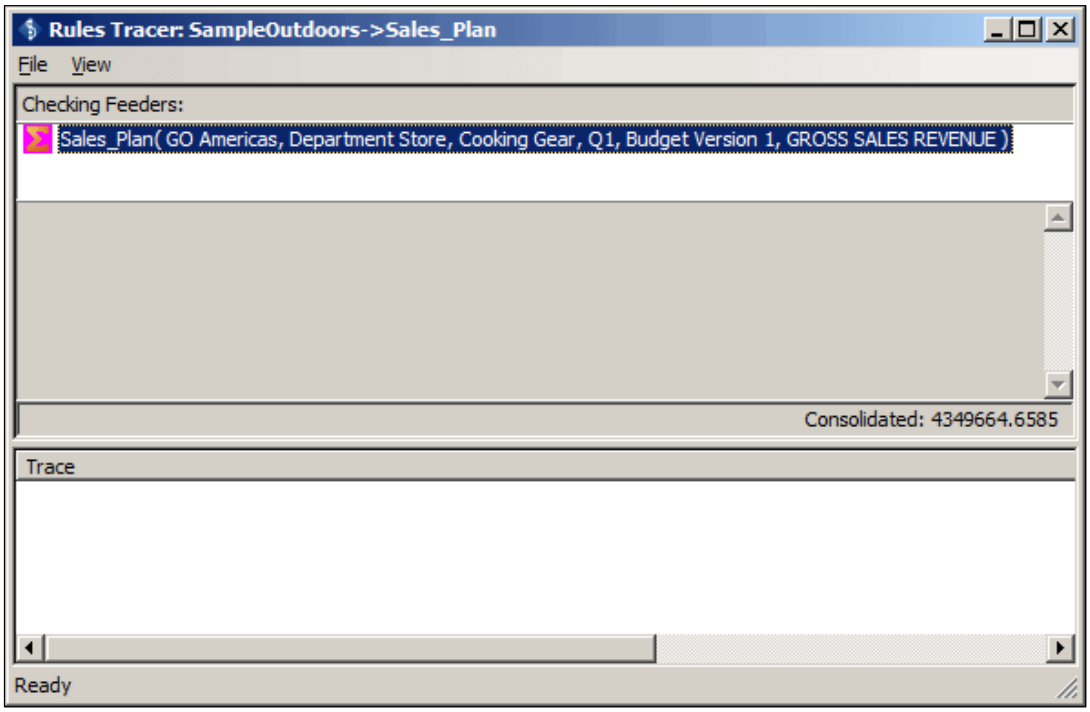
Task 4. Verify that correct data is calculated in the Sales_Plan cube.

1. Double-click the **Sales_Plan** cube.
2. From the **Select View** list, click **Revenue**.
3. Click the **Months** dimension to open the Subset Editor.
4. Click the **All** button and then click **OK**.
5. Click **Recalculate** and expand **Q1**.
6. Verify the **Q1/GROSS SALES REVENUE** displays values instead of zeros:



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7. Right-click **Q1/GROSS SALES REVENUE** and choose **Check Feeders**.
   You should not see any statements in the bottom pane indicating that the cell is not fed:



8. Close the **Rules Tracer**.
9. Close the **Cube Viewer** without saving changes.
10. Right-click **SampleOutdoors** and click **Save Data**.
11. Close **Architect** and all other open windows.

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IBM Training

Transfer data into the model

IBM Planning Analytics (v2.0)

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IBM Training

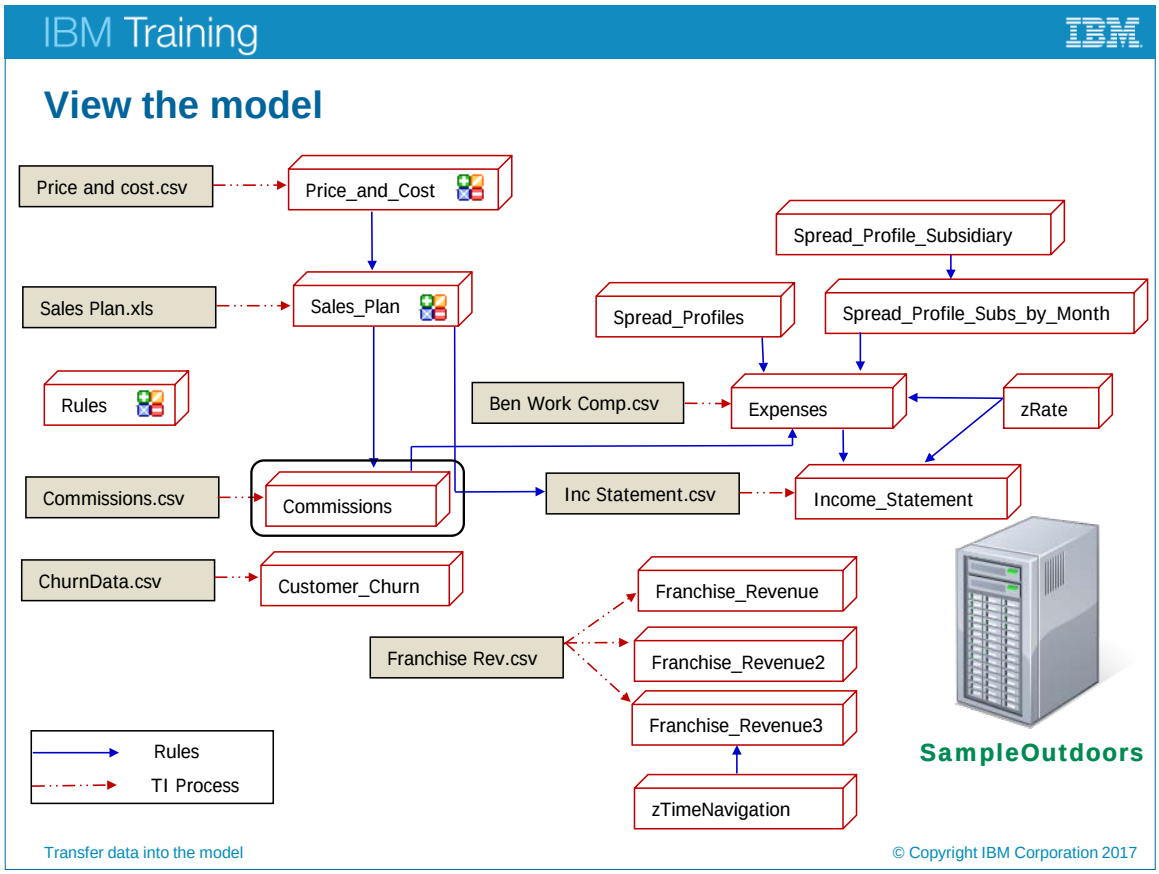
Unit objectives

- Describe the Advanced tabs in TurboIntegrator
- Load custom data into a cube
- Add a subset to a dimension
- Use logic in scripts
- Export data to ASCII
- Move data between versions
- Construct chores

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Unit objectives

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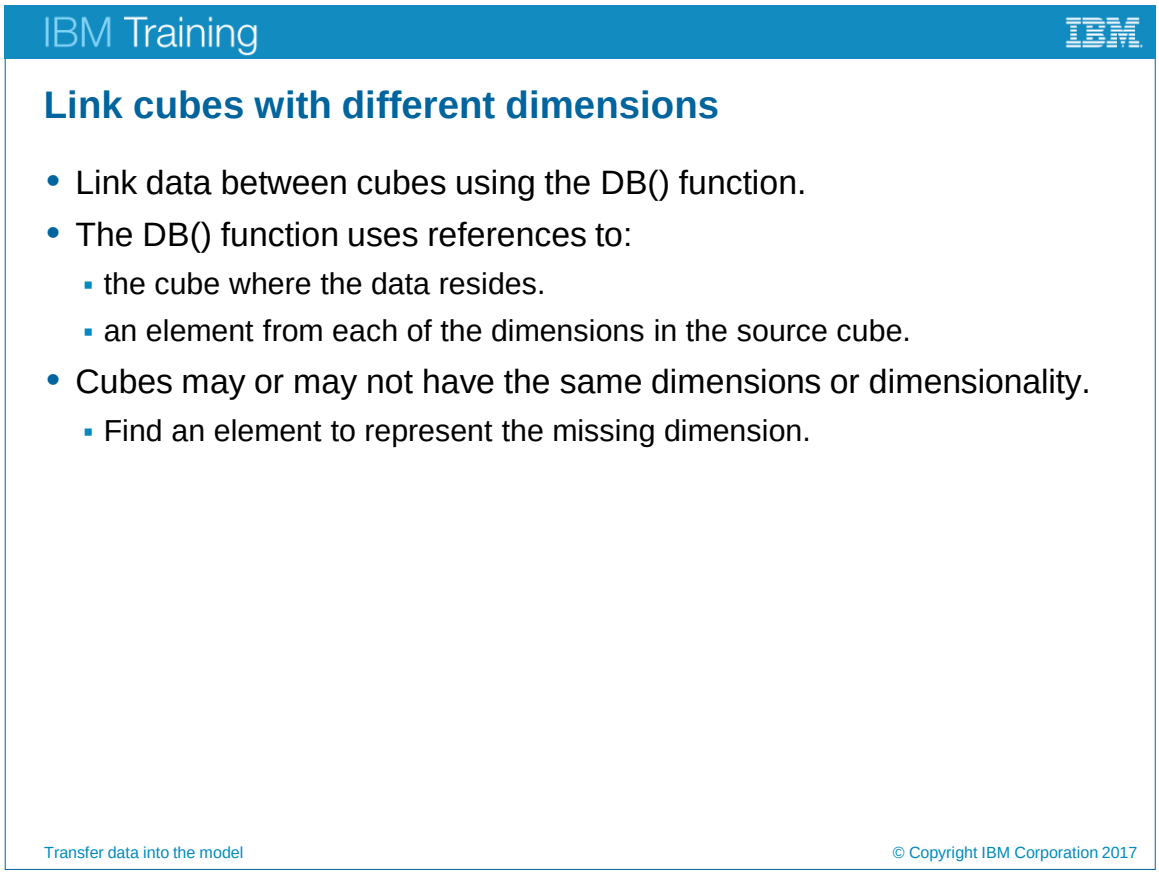
View the model

In this unit, you will create one new cube:

- Commissions

You will create a number of TurboIntegrator processes.

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Link cubes with different dimensions

If the target cube contains a dimension not found in the source, there will be a reference for it included in the area definition of the rule.

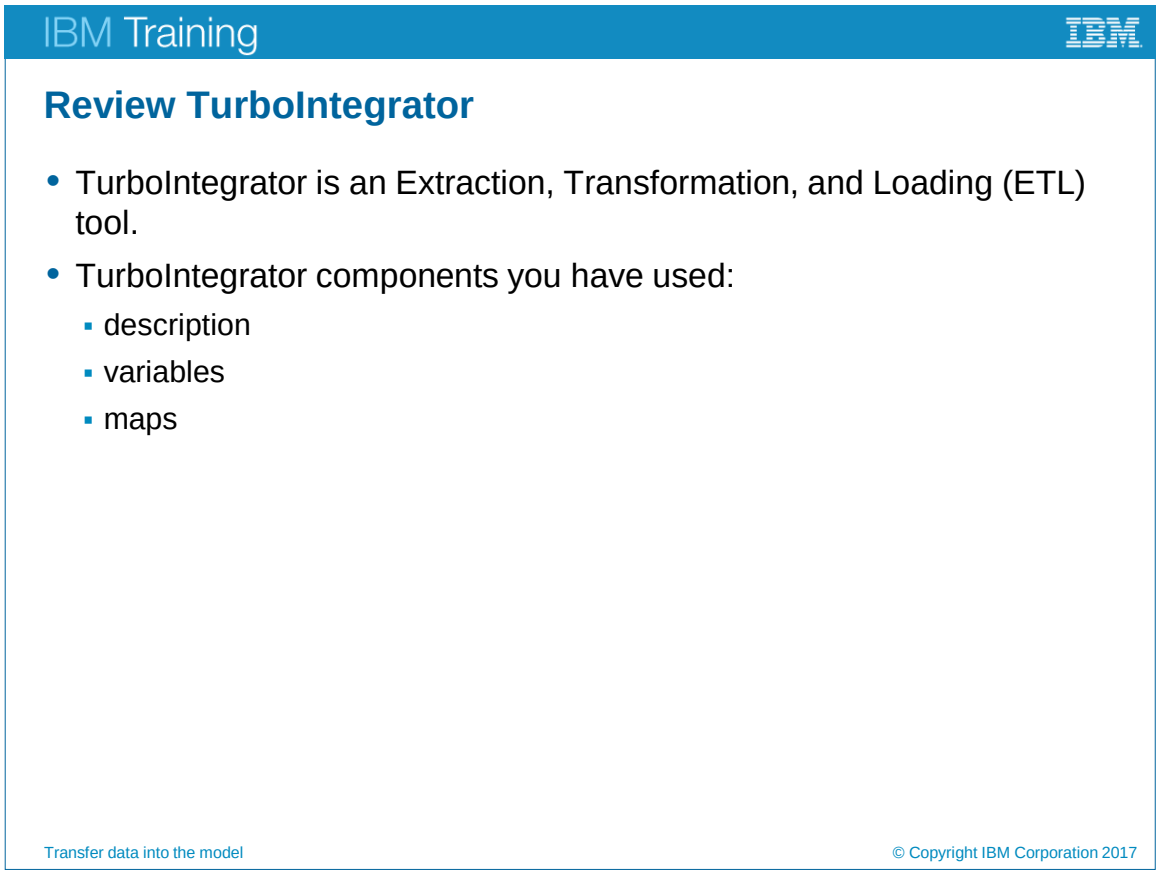
- In the following example, there is a Day dimension in the target cube that is not found in the source. The 'Jan 1' identifies the missing element:
- ['Jan 1', 'Cost']=DB('Price_and_Cost', !Products, !Date, 'Cost'];

If the source cube contains a dimension not found in the target cube, there will be a reference for it included in the DB() function dimension variables. The apex of the hierarchy is often used (i.e., Total Products, Total Channels, etc.)

- In the following example, there is a Market dimension in the source cube that does not reside in the target cube:

['Cost']=DB('Price_and_Cost', !Products, !Date, 'Total Market', 'Cost'];

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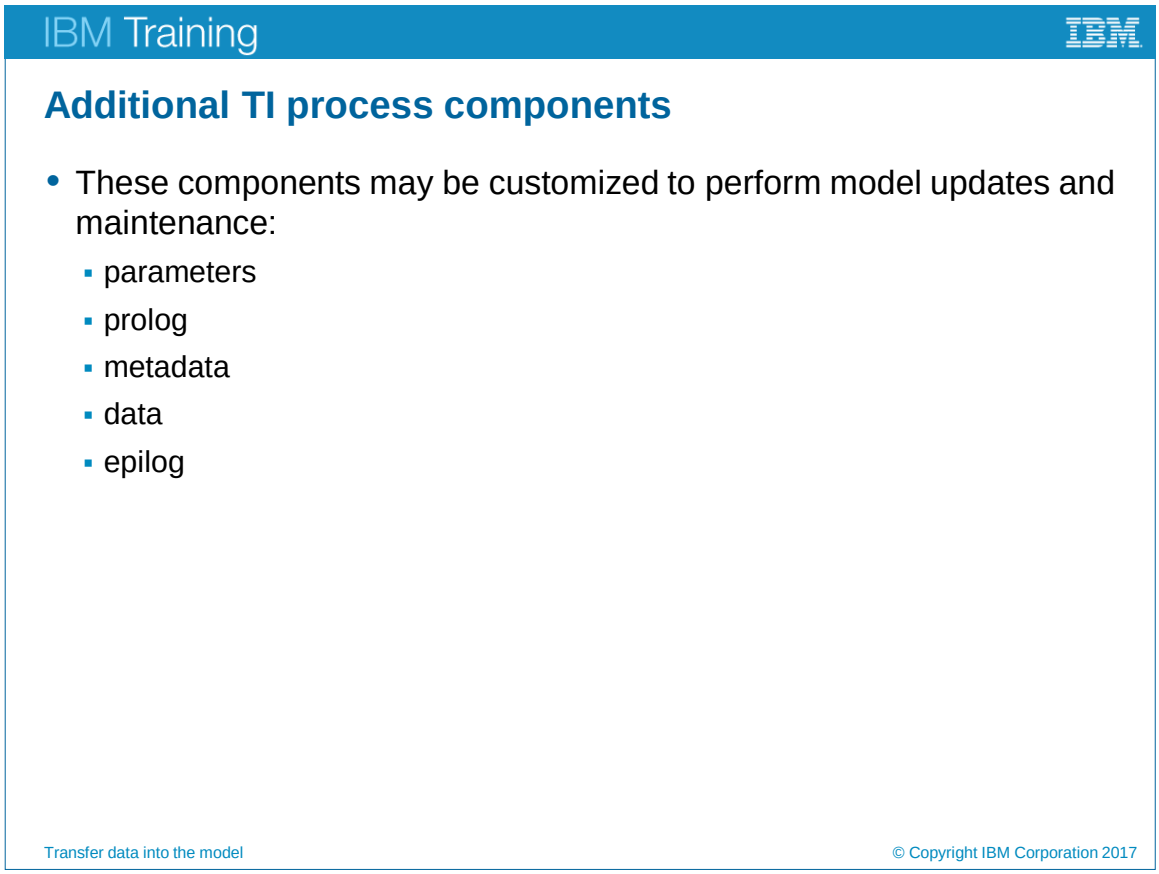
Review TurboIntegrator

TurboIntegrator (TI) can be used to define dimensions, create cubes, and populate cubes from ASCII files, TM1 cube views, and other sources. TI may also be used to write data to ODBC sources.

TI allows you to create processes that automate data importation, metadata management, and other tasks.

TI includes a complete set of functions that you can use to enhance process capabilities. You can use these functions to create scripts that export data to ASCII files and ODBC sources, or that use conditional expressions to control processing. In addition to these TI functions, you can also incorporate most TM1 rules functions.

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Additional TI process components

A set of parameters can be used to generalize a process so it can be used in multiple situations.

A prolog procedure consists of a series of actions to be executed before the data source is processed.

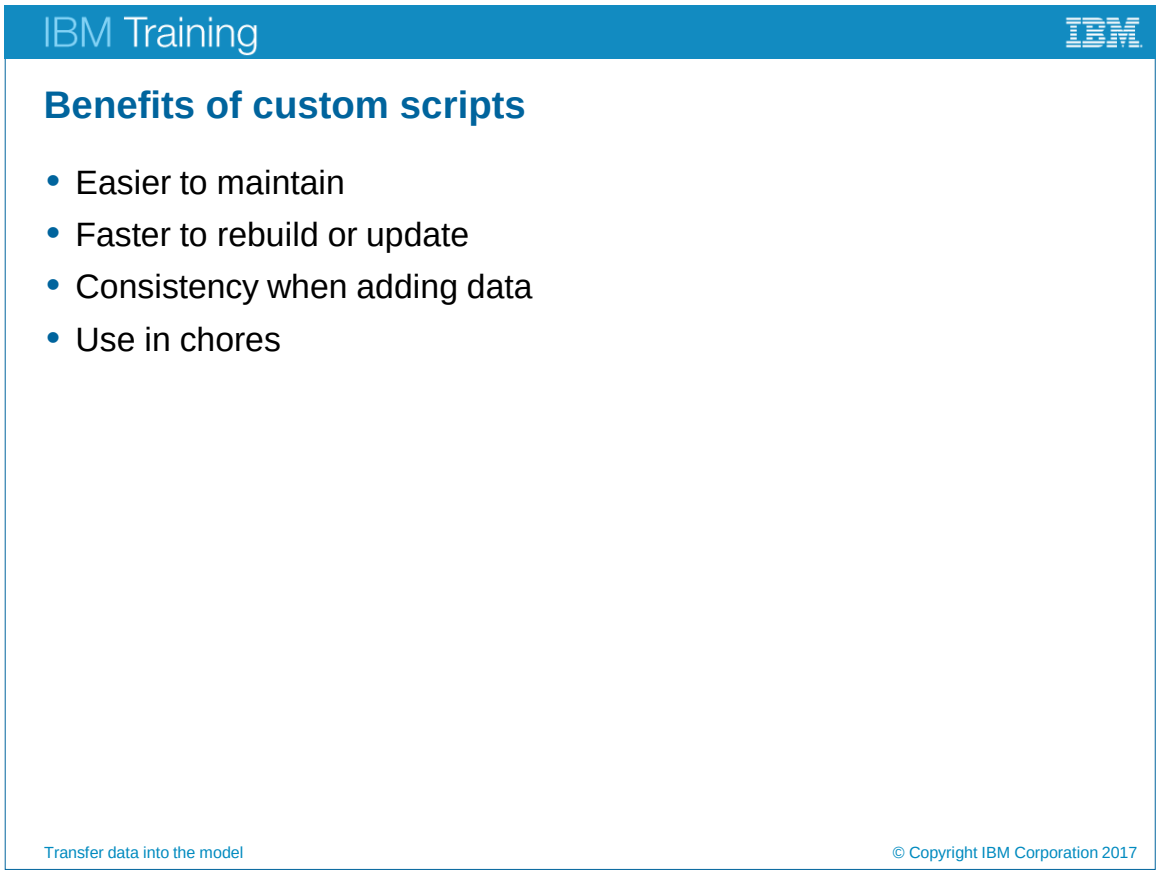
A metadata procedure consists of a series of actions that update or create cubes, dimensions, and other metadata structures.

A data procedure consists of a series of actions to be executed for each record in the data source.

An epilog procedure is to be executed after the data source is processed.

Every TI process is made up of these four procedures. They are executed in succession. The Advanced tabs show the statements generated using the TI interface and you may customize your own procedures by using TI functions. Any customization must be added outside of the reserved area.

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Benefits of custom scripts

TI is very useful as an ETL tool using only the basic steps to create variables, map data, and run the resulting processes. However, your TM1 model may be much easier to update and maintain once you become familiar with customizing and generating your own TI process scripts.

A few of the things you can do include:

- schedule processes to run periodically
- create subsets automatically
- extract data from a view and move it into another one
- archive a cube

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IBM Training

Demonstration 1

Load data into a cube from another cube with a different number of dimensions

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Demonstration 1: Load data into a cube from another cube with a different number of dimensions

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Demonstration 1: Load data into a cube from another cube with a different number of dimensions

Purpose:
You want to create and populate the Commissions cube.

First you will create the Commission_Measures dimension and the Commissions cube. Next, you will write rules that define how data will be populated, indicating that Gross Sales Revenue comes from another cube, and that it is used in another measure in the Commission_Measures dimension. Finally, you will populate the cube using a TurboIntegrator process.

Task 1. Create the Commission Measures dimension.

1. Open **Windows Explorer** and go to **C:\Program Files\IBM\cognos\tm1_64\bin**.
   Since you will be working with the Advanced Rules Editor tool, you will open the 32-bit version of Architect.
2. Double-click **tm1a.exe**.
3. Maximize the **Architect** window and then close **Windows Explorer**.
4. Expand **TM1**, double-click **SampleOutdoors**, and then log on with a user ID of **admin** and a password of **apple**.
5. Right-click **Dimensions** and then click **Create New Dimension**.
6. From the **Edit** menu, click **Insert Element**.
7. Add the following simple elements, clicking **Add** after each one:
   - **Gross Sales Revenue**
   - **Commission**
   - **Commission%**
   - **Commission_hold**
   - **Commission_count**

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8. Click **OK**.

   The result appears as shown below:



9. Save the dimension as **Commission_Measures**, and then close the **Dimension Editor**.

Task 2. Create the Commissions cube.

1. Right-click **Cubes**, and then click **Create new cube**.
2. In the **Cube Name** box, type **Commissions**.
3. In the **Available Dimensions** list, double-click the following dimensions to move them to the **Dimensions in new Cube** list:
   - **Subsidiaries**
   - **Months**
   - **Versions**
   - **Commission_Measures**
4. Click **Create Cube**.

   The Commissions cube appears in the list of cubes.

Task 3. Write rules for the Commissions cube.

When you use the Rules Editor, TM1 automatically turns off its fast consolidation algorithm, which skips empty cells. To ensure that performance is not slowed down, the first line you will add to the Rules Editor is "SKIPCHECK", which tells TM1 to skip all empty cells.

1. Under **Cubes**, right-click the **Commissions** cube and click **Create Rule**.

   The Rules Editor opens (this may take a few moments).
2. On **line 1**, type **SKIPCHECK;**.

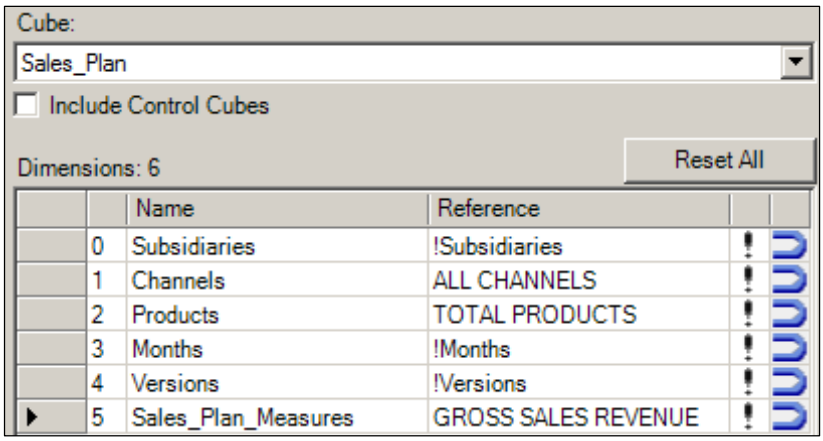
   Next you want the data from the Sales_Plan cube to be used in the Commissions cube.
3. On **line 2**, type **# Pull data from the Sales_Plan cube**.

   Lines with comments start with '#'.
4. On line 3, click **Brackets** and double-click **Commission_Measures**.
5. Click **Gross Sales Revenue** and then click **OK**.

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6. Outside the brackets, type **=**, click **Insert Cube Reference**  and then in the **Cube** list click **Sales_Plan**.
7. Click the **Subset** icon next to **Channels**, in the **Subset Editor** window, click **ALL CHANNELS**, and then click **OK**.
8. Repeat **Step 7** using the following:
   - **Products dimension**, click **TOTAL PRODUCTS**
   - **Sales_Plan_Measures** dimension, click **GROSS SALES REVENUE**

   The result appears as shown below:



   The Commissions cube does not contain all the dimensions that the Sales_Plan cube does. This is why you must select specific elements from those dimensions (Channels, Products, and Sales_Plan_Measures). You do not want to exclude any data from those dimensions so you need to choose the highest levels or the totals in the dimensions.
9. Click **OK** to close the Insert Cube Reference dialog box, and then after the bracket type **;** at the end of the line.
10. After **!Subsidiaries,**, press **Enter**, and then press **Tab**.

    This makes the expression more readable.

```
1  SKIPCHECK;
2  # Pull data from the Sales_Plan cube
3  ['Gross Sales Revenue' ] = DB('Sales_Plan', !Subsidiaries,
4        'ALL CHANNELS', 'TOTAL PRODUCTS', !Months, !Versions, 'GROSS SALES REVENUE');
```

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Task 4. Calculate average Commission% at the consolidated level.

1. On **line 6**, type the following comment:
2. **# Calculate average Commission% at the consolidated level**
3. On **line 7**, click **Brackets** and then double-click **Commission_Measures**.
4. Click **Commission_hold** and then click **OK**.
5. Click outside the brackets, click the **Insert Qualifier or Operator** icon and then click **=N:**.
6. Click **Brackets**, double-click **Commission_Measures**, and then click **Commission%**.
7. Click **OK**, and then after the bracket type **;**.
8. On **lines 8** and **9**, type the following:

   ['Commission_count']=N:IF(['Commission%']<> 0,1,0);

   ['Commission%']=C:(['Commission_hold']\['Commission_count']);

   The conditional statement on line 8 indicates that if the Commission% is not zero, the Commission_count = 1, otherwise Commission_count=0.

   The "\" on line 9 indicates that Commission% is equal to Commission_hold divided by Commission_count, when Commission_count is not equal to zero. When Commission_count is equal to zero, then Commission% is also equal to zero.

```
1  SKIPCHECK;
2  # Pull data from the Sales_Plan cube
3  ['Gross Sales Revenue' ] = DB('Sales_Plan', !Subsidiaries,
4        'ALL CHANNELS', 'TOTAL PRODUCTS', !Months, !Versions, 'GROSS SALES REVENUE');
5
6  # Calculate average Commission% at the consolidated level
7  ['Commission_hold' ]  = N: ['Commission%' ] ;
8  ['Commission_count' ]  = N:IF(['Commission%' ] <> 0,1,0);
9  ['Commission%' ]  = C: (['Commission_hold' ] \ ['Commission_count' ]) ;
```

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Task 5. Calculate commission and add feeders.

1. On **lines 11** and **12**, type the following:

   # Calculate commission

   ['Commission'] = N: (['Gross Sales Revenue']*['Commission%'])/100;

2. On **lines 14** and **15**, enter the following:

   FEEDERS;

   ['Commission%']=>['Commission'],['Commission_hold'], ['Commission_count'];

   The result appears as shown below:

```
1   SKIPCHECK;
2   # Pull data from the Sales_Plan cube
3   ['Gross Sales Revenue' ] = DB('Sales_Plan', !Subsidiaries,
4         'ALL CHANNELS', 'TOTAL PRODUCTS', !Months, !Versions, 'GROSS SALES REVENUE');
5
6   # Calculate average Commission% at the consolidated level
7   ['Commission_hold'] = N:['Commission%'];
8   ['Commission_count'] = N:IF(['Commission%']<> 0,1,0);
9   ['Commission%']=C:(['Commission_hold']\['Commission_count']);
10
11  # Calculate commission
12  ['Commission'] = N: (['Gross Sales Revenue']*['Commission%'])/100;
13
14  FEEDERS;
15  ['Commission%']=>['Commission'],['Commission_hold'],['Commission_count'];
```

3. Save and close the **Rules Editor**.

Task 6. Load data into the Commissions cube.

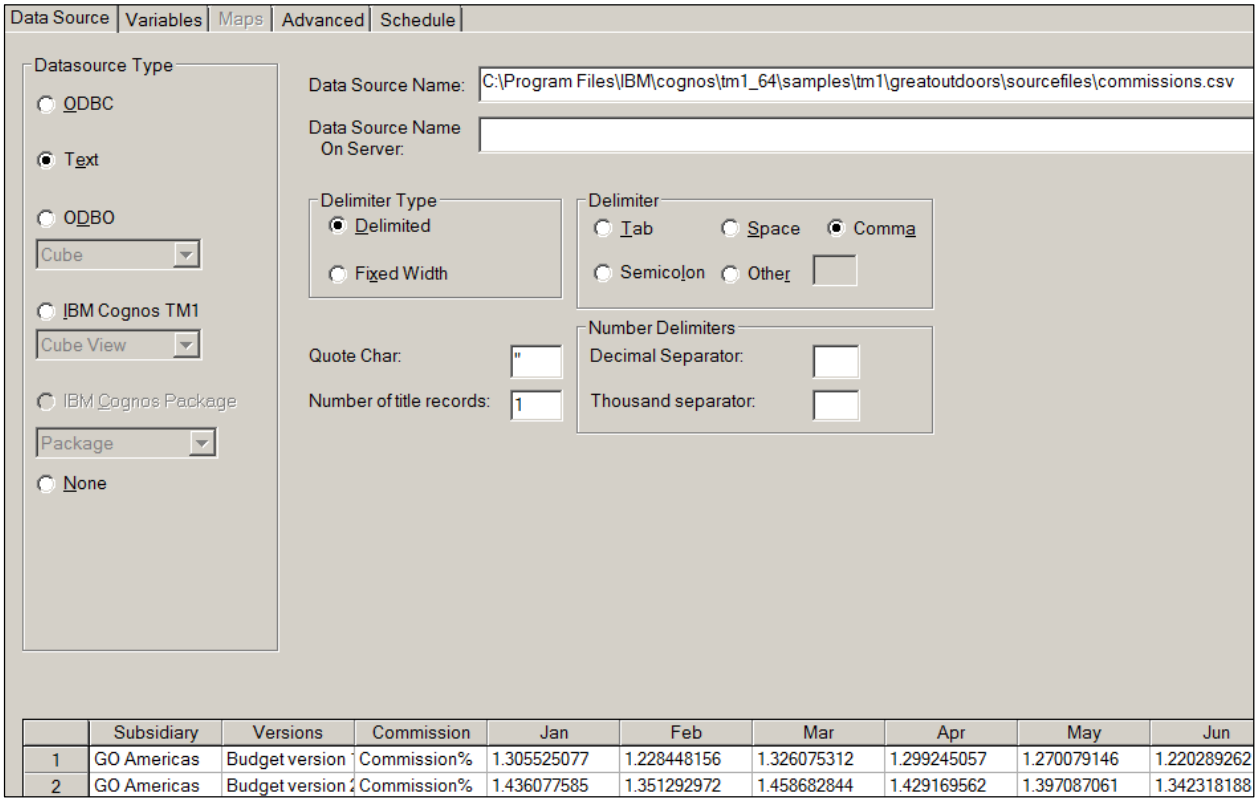
1. **Right-click Processes** and click **Create New Process**.

   - Datasource Type: **Text**
   - Data Source Name: **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\ greatoutdoors\sourcefiles\commissions.csv** (click **OK** to dismiss the warning message)
   - Number of title records: **1**

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2. Click **Preview**.

   The result appears as follows:



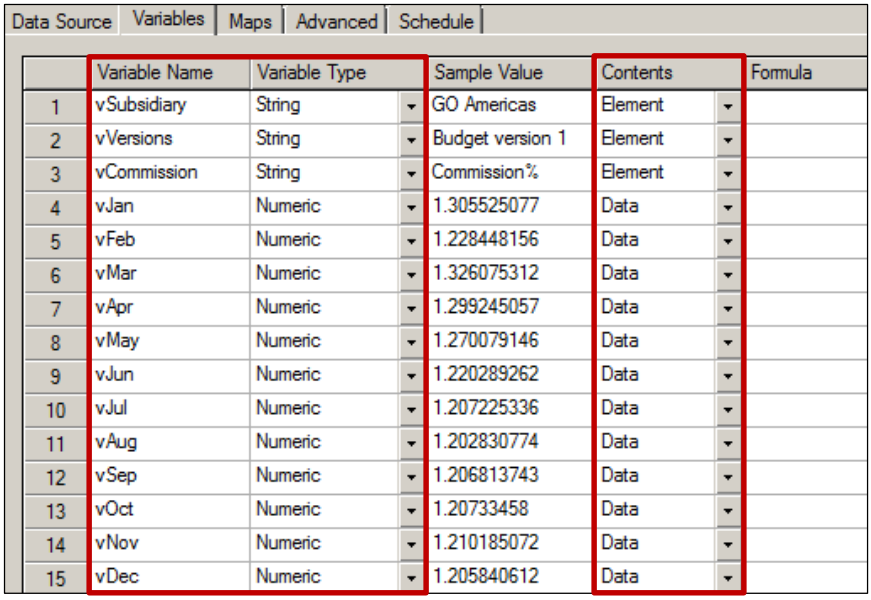
	Subsidiary	Versions	Commission	Jan	Feb	Mar	Apr	May	Jun
1	GO Americas	Budget version 1	Commission%	1.305525077	1.228448156	1.326075312	1.299245057	1.270079146	1.220289262
2	GO Americas	Budget version 2	Commission%	1.436077585	1.351292972	1.458682844	1.429169562	1.397087061	1.342318188

3. Click the **Variables** tab and add a **v** in front of each of the variable names.

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4. On the **Variables** tab, set the following:
   - For **vSubsidiary, vVersions**, and **vCommission**, in the **Contents** column, click **Element**.
   - From **vJan** to **vDec**, in the **Variable Type** column, click **Numeric** (if necessary), and then in the **Contents** column, click **Data**.

   The result appears as follows:



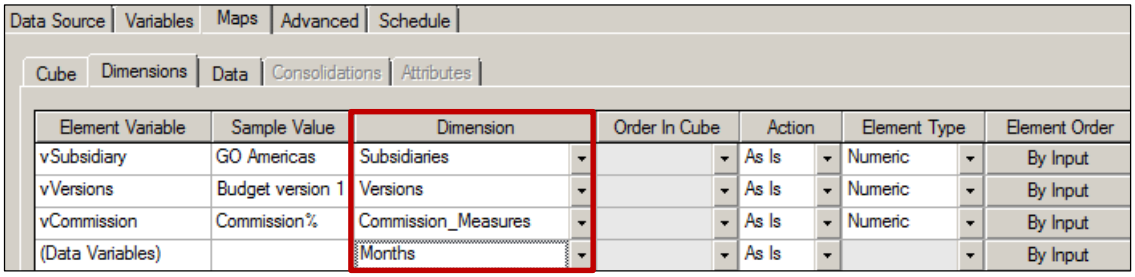
	Variable Name	Variable Type	Sample Value	Contents	Formula
1	vSubsidiary	String	GO Americas	Element	
2	vVersions	String	Budget version 1	Element	
3	vCommission	String	Commission%	Element	
4	vJan	Numeric	1.305525077	Data	
5	vFeb	Numeric	1.228448156	Data	
6	vMar	Numeric	1.326075312	Data	
7	vApr	Numeric	1.299245057	Data	
8	vMay	Numeric	1.270079146	Data	
9	vJun	Numeric	1.220289262	Data	
10	vJul	Numeric	1.207225336	Data	
11	vAug	Numeric	1.202830774	Data	
12	vSep	Numeric	1.206813743	Data	
13	vOct	Numeric	1.20733458	Data	
14	vNov	Numeric	1.210185072	Data	
15	vDec	Numeric	1.205840612	Data	

5. Click the **Maps** tab.
6. Below **Cube Action**, click **Update Cube**, and then in the **CubeName** box, click **Commissions**.
7. Click the **Dimensions** tab, and then beside each **Element Variable**, click the appropriate **Dimension**:

Element Variable	Dimension
vSubsidiary	Subsidiaries
vVersions	Versions
vCommission	Commission_Measures
(Data Variables)	Months

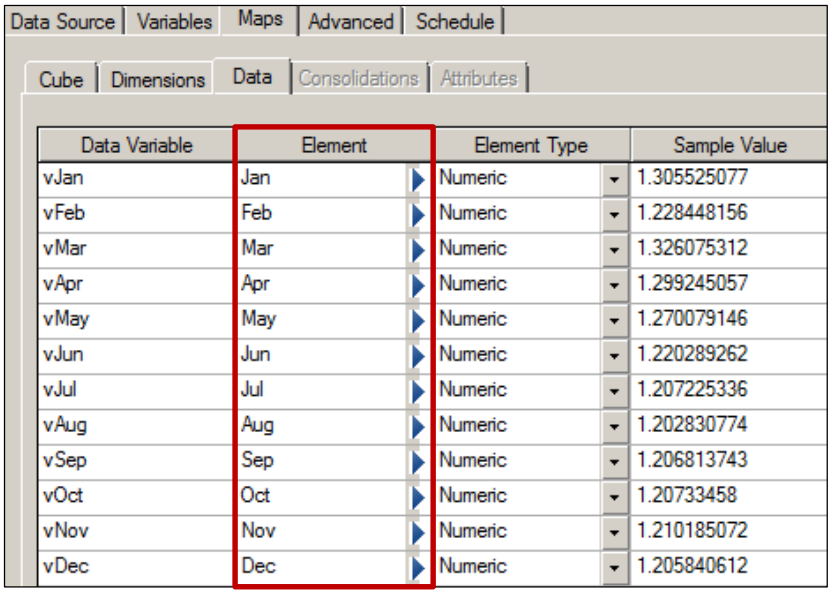
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The results appear as follows:



8. Click the **Data** tab.
9. In the **Element** column, type **Jan** to **Dec**, matching the contents of the **Data Variable** column.

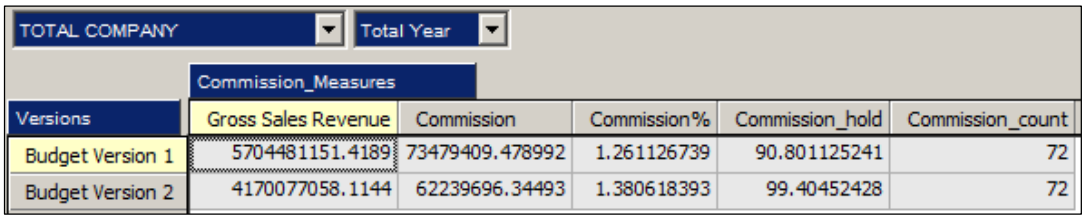
   The results appear as follows:



10. Click the **Advanced** tab, and click through the **Prolog**, **Metadata**, **Data**, and **Epilog** tabs.

    You need to scroll through the different tabs to regenerate the scripts.
11. Save the process as **LoadCommissionPctCSV**, and then click **Run**.
12. Click **OK** and then close **TurboIntegrator**.
13. Open the **Commissions** cube, and then click **Recalculate**.

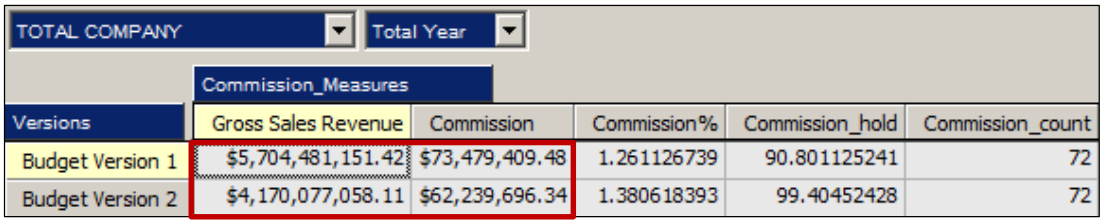
    The result appears as follows:



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14. Close the **Cube Viewer** without saving the view.
15. In Server Explorer, right click **Commission_Measures**, and then click **Edit Element Attributes**.
16. Click and drag to select the **Format (Text)** cells for **Gross Sales Revenue** and **Commission** and then click the **Format** button.
17. Click **Currency**, click **OK**, and then click **OK** again.
18. Double-click the **Commissions** cube, and then click **Recalculate**.

    The results appear as follows:

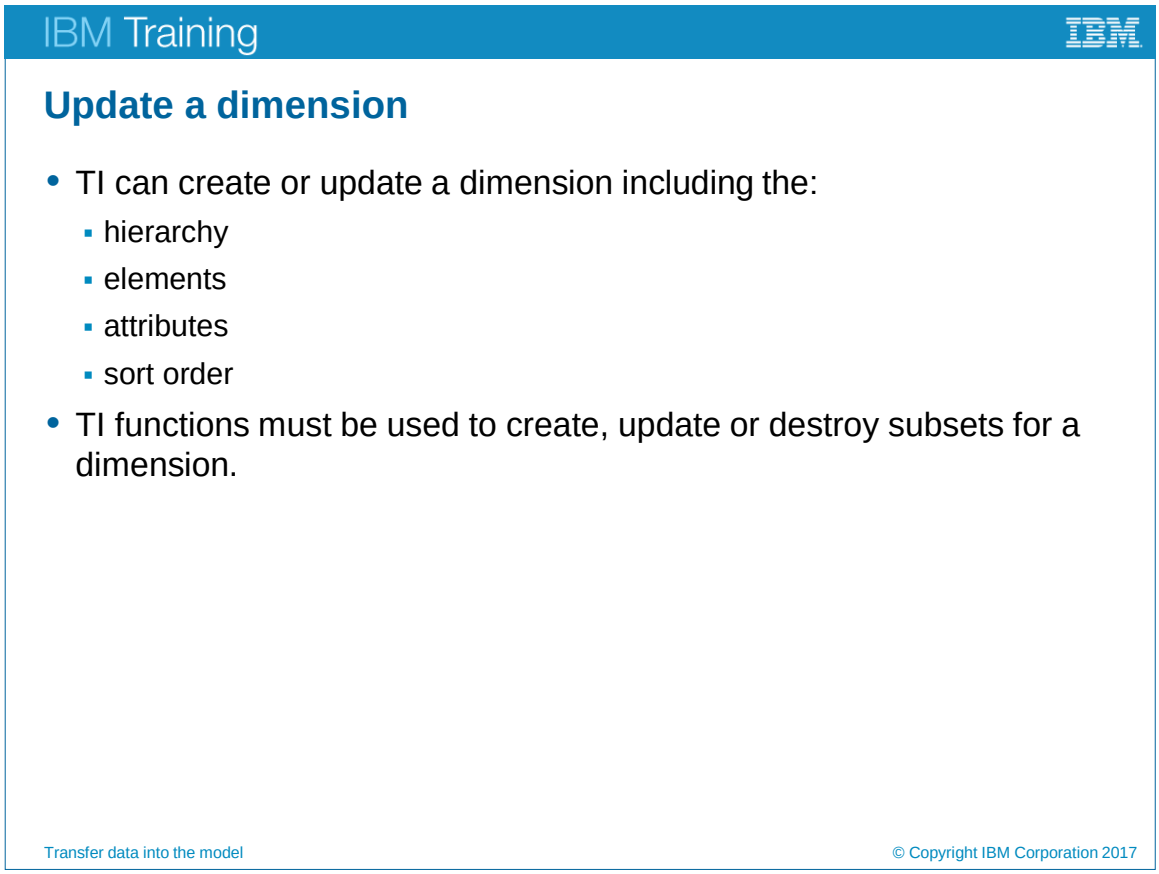


TOTAL COMPANY	Total Year				
	Commission_Measures				
Versions	Gross Sales Revenue	Commission	Commission%	Commission_hold	Commission_count
Budget Version 1	$5,704,481,151.42	$73,479,409.48	1.261126739	90.801125241	72
Budget Version 2	$4,170,077,058.11	$62,239,696.34	1.380618393	99.40452428	72

19. Close the Cube Viewer.
20. Right-click **SampleOutdoors** and click **Save Data**.
21. Leave **Architect** open for the next demonstration.

Results:
You created and populated the Commissions cube. First you created the Commission_Measures dimension, then you created the Commissions cube, and then you wrote rules that defined how the Commissions cube is to be populated with data. Lastly, you populated the cube using a TurboIntegrator process.

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Update a dimension

The tabs that guide you through building a TI process are very useful but they may only perform a set number of tasks. You can perform additional tasks using TI and functions directly in the TI process scripts.

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IBM Training

Demonstration 2

Add subsets to a dimension with TurboIntegrator

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Demonstration 2: Add subsets to a dimension with TurboIntegrator

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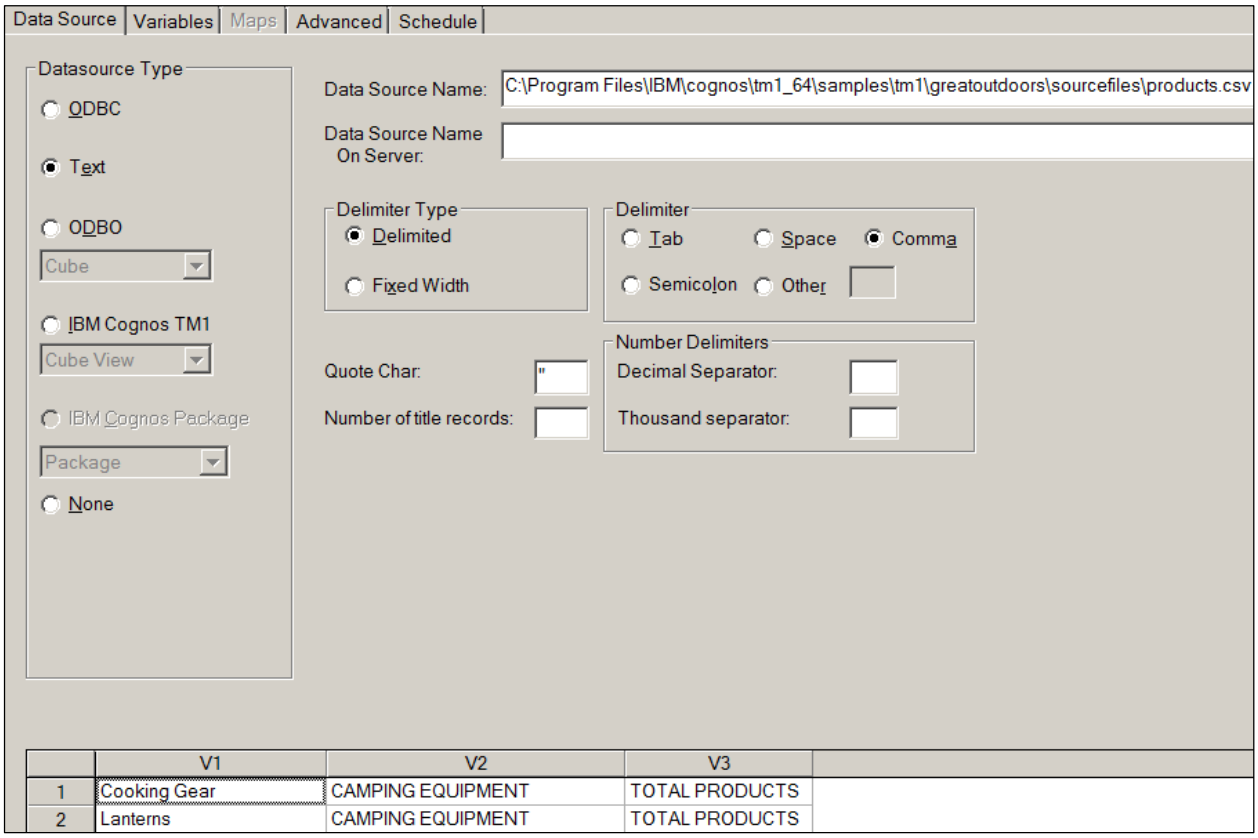
Demonstration 2: Add subsets to a dimension with TurboIntegrator

Purpose:
You want to create product type subsets that are available to users. You will use functions in TurboIntegrator to do this.

Task 1. Create a new process in TurboIntegrator.

1. Right-click **Processes**, and then click **Create New Process**.
   - Data Source Type: **Text**
   - Data Source Name: **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles\products.csv** (click **OK** to dismiss the warning message)
2. Click **Preview**.

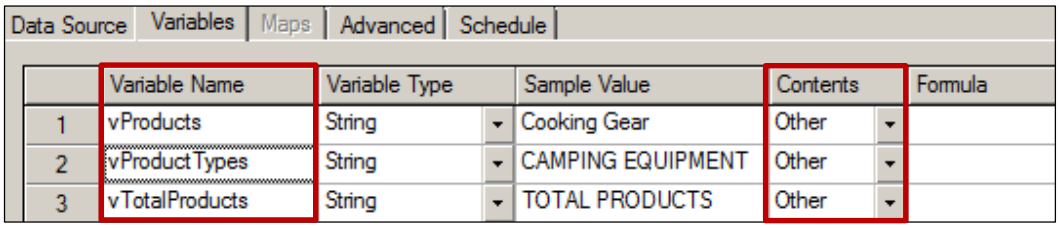
   The result appears as follows:



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3. Click the **Variables** tab and set the following:
   - Rename **V1** as **vProducts**
   - Rename **V2** as **vProductTypes**
   - Rename **V3** as **vTotalProducts**
4. Set the **Contents** for all three variables as **Other**

   The result appears as follows:



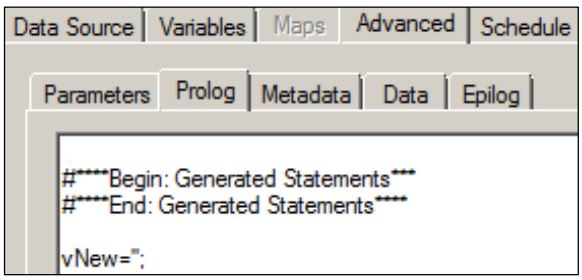
   Marking the Contents as Other allows the variables to still be available to TI processes (unlike Ignore) but does not force you to map them under the Maps tab.

   The "Other" setting allows you to use the variables to write custom scripts on the Advanced tabs.
5. Click the **Advanced** tab, click the **Prolog** sub-tab, and then after the generated statements type the following:

   vNew='';

   Use two single quotes, not a double quote.

   The results appear as follows:



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6. Click the **Metadata** sub-tab, and then after the generated statements add the following:

If subsets already exist for each product type, destroy them so that they can be recreated

IF(SubsetExists('Products',vProductTypes)<>0);

IF(vNew@<>vProductTypes);

SubsetDestroy('Products',vProductTypes);

vNew=vProductTypes;

ENDIF;

ENDIF;

Create a subset for each product type and insert elements into each subset

IF(SubsetExists('Products',vProductTypes)=0);

SubsetCreate('Products',vProductTypes);

SubsetElementInsert('Products',vProductTypes,vProductTypes,0);

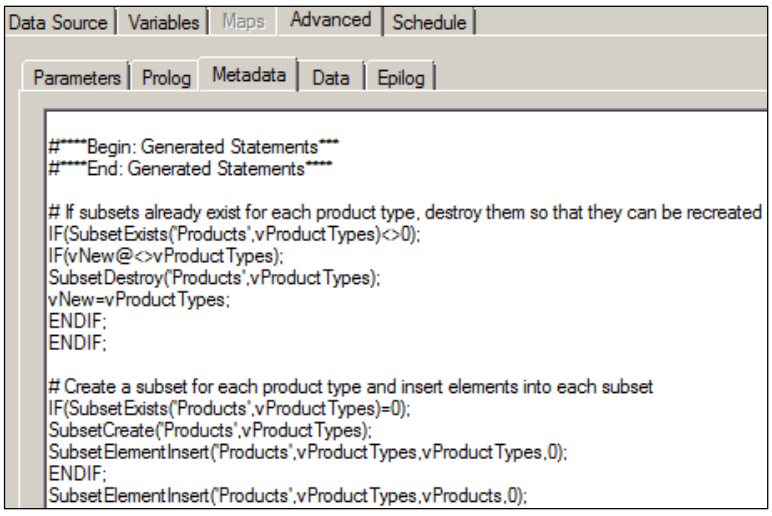
ENDIF;

SubsetElementInsert('Products',vProductTypes,vProducts,0);

Note: Statements with # before them are comments.

Note: You can copy the syntax of the different functions (such as SUBSETEXISTS) from the online help, to minimize typos. You will need to right-click, and then click Copy.

The results appear as follows:



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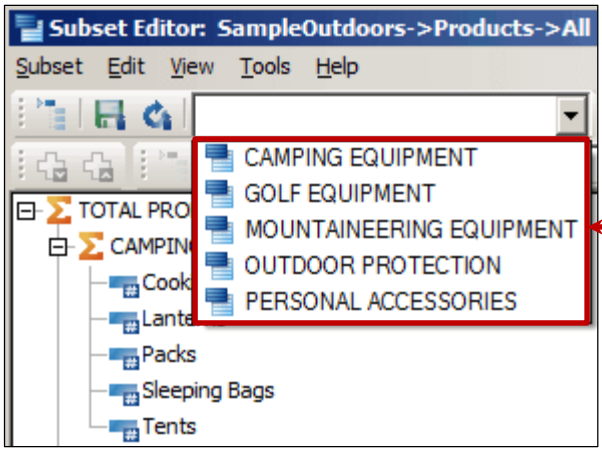
- SYNTAX: **SubsetExists(DimName, SubsetName);**
  - This checks to see if the specified Subset (SubsetName) exists for the specified dimension (DimName). You are not overwriting an existing subset.
- SYNTAX: **SubsetDestroy(DimName, SubName);**
  - This function deletes a subset (SubName) from a specified dimension (DimName). The code in the top section of the Metadata tab (under the Generated Statements) will allow this process to be run repeatedly without error to update the Products dimension with subsets based on new product types.
- SYNTAX: **SubsetCreate(DimName, SubName);**
  - This creates the specified Subset (SubName) for the specified dimension (DimName).
- **IF, ENDIF, (ELSEIF)**
  - The IF statement allows a process to execute a statement or series of statements when a given expression is true. You can use arithmetic operators, logical operators, and comparison operators to construct an expression.
  - The TurboIntegrator IF statement differs from the Rules IF function in that the TurboIntegrator statement can accept multiple ELSEIF statements to evaluate multiple expressions, while the Rules IF function can evaluate only one expression.
  - You can nest up to 20 IF statements in a TurboIntegrator process. If you exceed 20 nested IF statements, you will receive an error when attempting to save the process.
- SYNTAX: **SubsetElementInsert(DimName, SubName, ElName, Position);**
  - This function adds an element to an existing subset. Products (DimName) is the dimension to which you want to add the subsets, vProductTypes (SubName) is the variable containing the subset names, vProductTypes and vProducts (ElName) are the variables containing the elements that you are adding to the subsets (individual product types and products within product types), and 0 (Position) indicates the first position in the index.

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7. Save the process as **CreateProductType_subsets**, run the process, and then close **TurboIntegrator**.
8. Under **Dimensions**, double-click the **Products** dimension to open it in the **Subset Editor**.
9. Click the **Select Subset** box, to view the subsets.

   The result appears as follows:

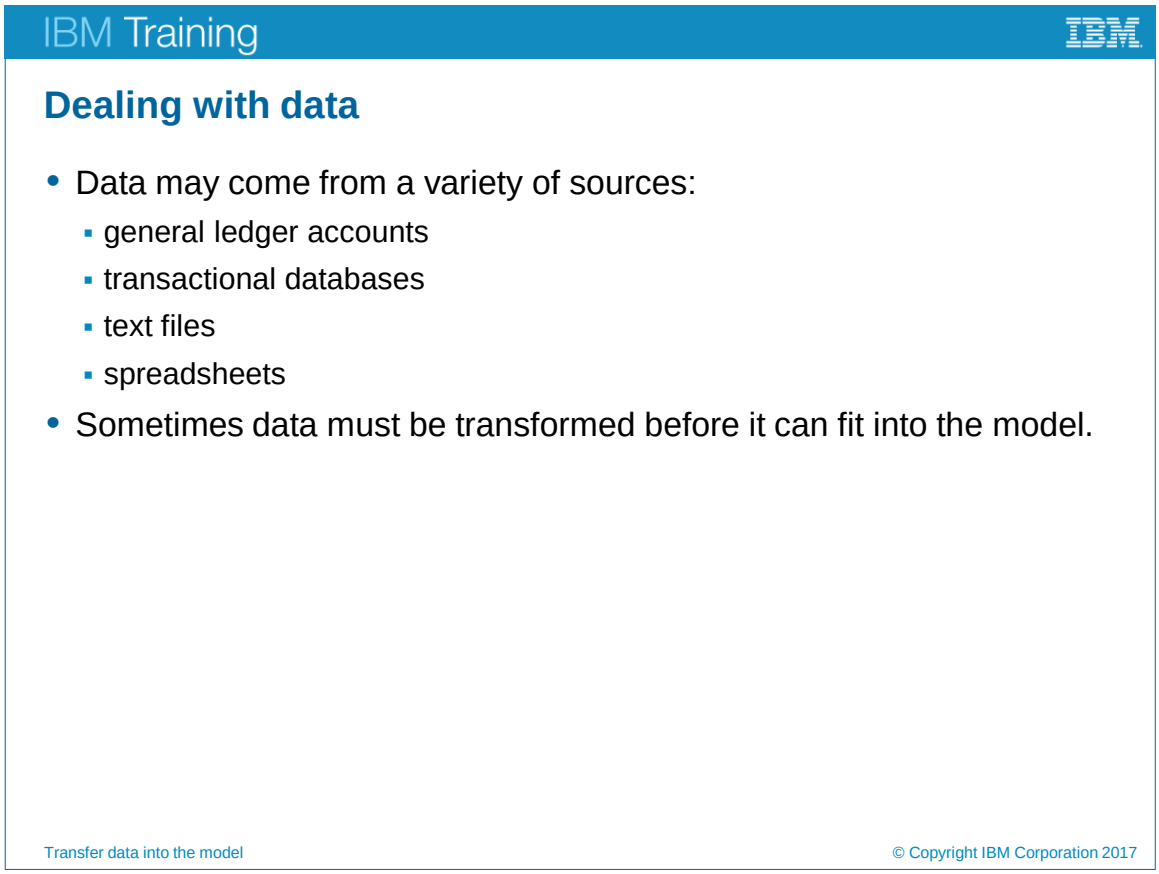


Subsets created by running the **CreateProductType_subsets process**.

10. Close the **Subset Editor**.
11. Right-click **SampleOutdoors** and click **Save Data**.
12. Leave Architect open for the next demonstration.

Results:
You ran a process to create product type subsets which are available to users. As you add new elements, you may run this process and the new elements will be added to the appropriate subset. You used functions in TurboIntegrator to create and populate the subsets.

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Dealing with data

TI functions may be used to alter the data as it comes into the model. You have used some functions to change data types (NUMBERTOSTRING() and STRINGTONUMBER())or to extract just the necessary components from the data (MONTHS()). Some functions require editing the process scripts directly.

Customizing the hierarchy to accommodate uneven levels can only be resolved by TI if the data is altered. Another option is to customize the scripts to adjust the parentage to support the uneven levels.

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IBM Training

Demonstration 3

Create a dimension with an uneven hierarchy

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Demonstration 3: Create a dimension with an uneven hierarchy

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Demonstration 3: Create a dimension with an uneven hierarchy

Purpose:
You want to create a dimension that organizes data by subsidiary and by region, so that all the subsidiaries in a region are aggregated at the region level.

In some cases, there is only one subsidiary in a region, and no region is specified. This is also known as having an uneven hierarchy because the 'parent' has no child objects.

To resolve any process errors, you will create a TurboIntegrator process that terminates the process at different levels for different hierarchies.

Task 1. Create the process and specify the data source and variables.

1. Right-click **Processes**, and then click **Create New Process**.
   - Data Source Type: **Text**
   - Data Source Name: **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles\uneven_subsidiaries.csv** (click **OK** to dismiss the warning message if necessary)
2. Click **Preview**.
   The result appears as follows:

	V1	V2	V3
1	GO Americas	GO Americas	TOTAL COMPANY
2	GO Central Europe	GO EUROPE GMBH	TOTAL COMPANY
3	GO Northern Europe	GO EUROPE GMBH	TOTAL COMPANY
4	GO Southern Europe	GO EUROPE GMBH	TOTAL COMPANY
5	GO Asia Pacific	GO Asia Pacific	TOTAL COMPANY
6	GO Accessories Gml	GO EUROPE GMBH	TOTAL COMPANY

Note: V1 and V2 have the same value for GO Americas and GO Asia Pacific. TM1 requires that elements (either leaf level or string level) be unique. Otherwise TurboIntegrator creates dimensions with parents (consolidations) that have no child objects.

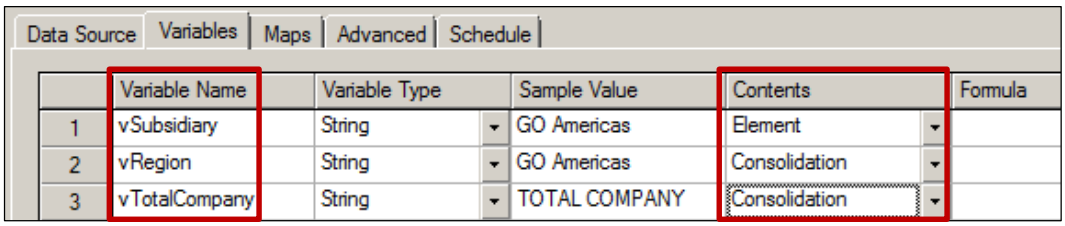
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There are two ways to resolve this:

- Alter the names of the elements which have the same name on both levels. You can see this in the **subsidiaries.csv** file that is in the **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles** directory.
- Create a TurboIntegrator process to create the uneven hierarchy.

You will first create a TurboIntegrator process that does not address the duplicates, and then later you will correct the process.

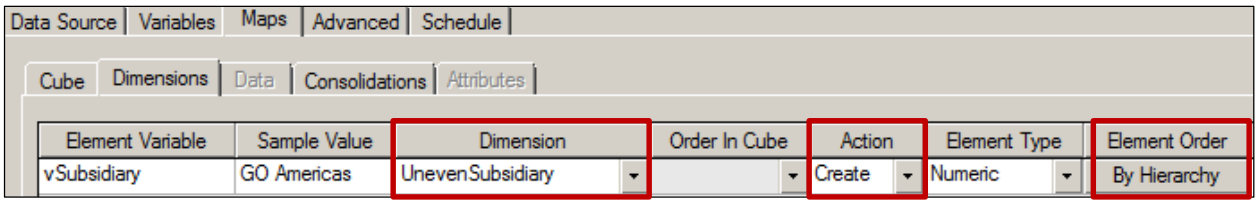
3. Click the **Variables** tab and set the following:
   - Rename **V1** as **vSubsidiary**
   - Rename **V2** as **vRegion**
   - Rename **V3** as **vTotalCompany**
4. Set the **Contents** as shown below:



Task 2. Map the variables.

1. Click **Maps**, click the **Dimensions** tab, and then in the **Dimension** column, beside **vSubsidiary**, type **UnevenSubsidiary**.

   You want to set the Element Order to "by Hierarchy".
2. In the **Element Order** column, click **By Input** to open the Dimension Element Ordering dialog box, click **Automatic**, click **Hierarchy**, and then click **OK**.



3. Click the **Consolidations** tab.

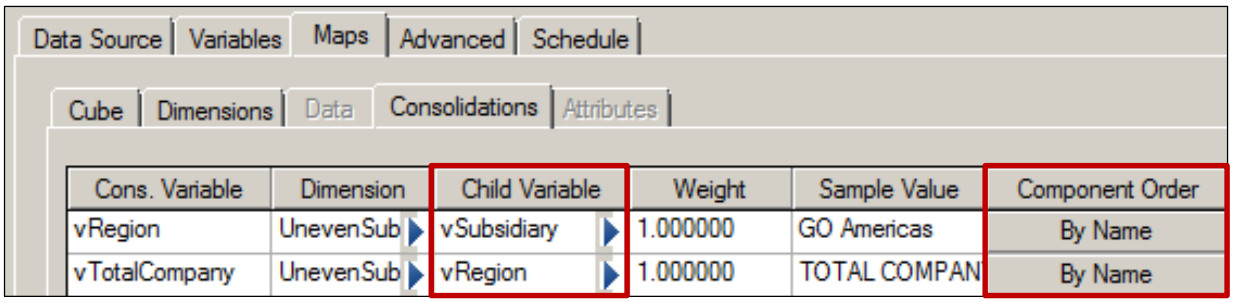
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4. Beside **vRegion**, under **Child Variable**, click **vSubsidiary**, and then beside **vTotalCompany**, click **vRegion**.

   Next you want to set each consolidation variable to order its components by name.

5. Beside **vRegion**, click **By Input** to open the **Component Element Ordering** dialog box, click **Automatic**, and click **OK**, and then repeat for **vTotalCompany**.

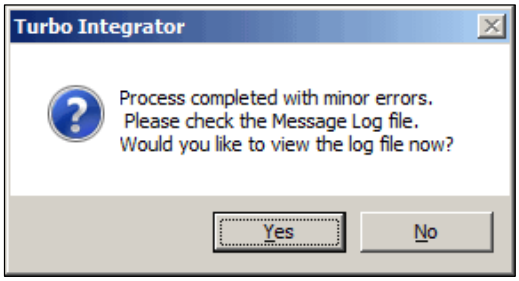
   The result appears as follows:



6. Click the **Advanced** tab, and then click through the **Prolog**, **Metadata**, **Data**, and **Epilog** tabs.

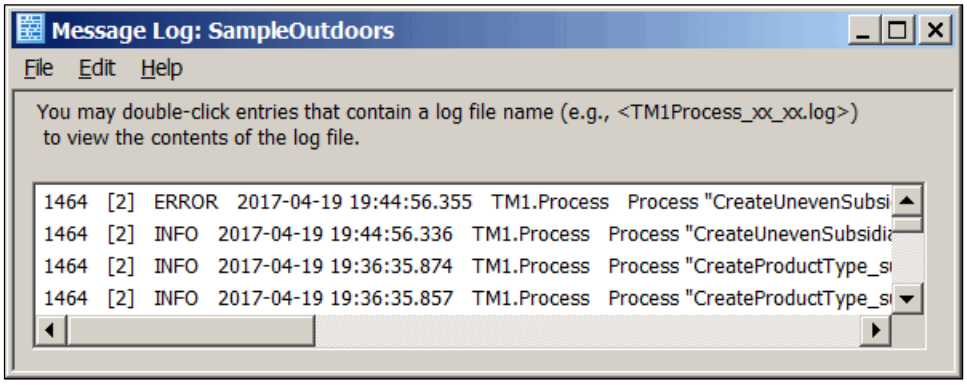
7. Save the process as **CreateUnevenSubsidiary**, and then run the process.

   You will get the message that the process completed with minor errors:



8. Click **Yes** to view the log file.

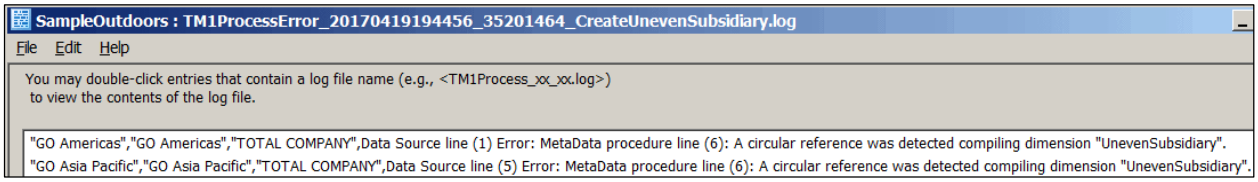
   The results appear as follows:



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9. Double-click the first row.

   The results appear as follows:



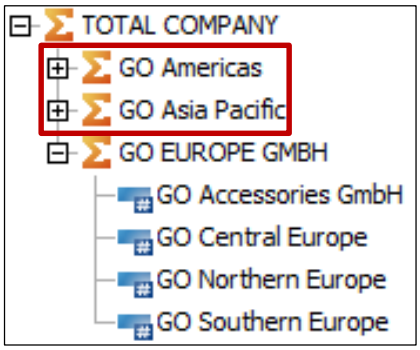
   The circular reference indicates that the items at the leaf level are not unique.

10. Close the log file windows and leave TI open.

    The log file can be found in the TM1 server data directory (**C:\Program Files\IBM\cognos\tm1_64\samples\tm1\SampleOutdoors\Logfiles** in this environment). The file name is TM1ProcessError_xx_xx.log. The first xx represents the date/timestamp and the 2nd xx represents the TI Process name (for example, TM1ProcessError_20170419194456_35201464_CreateUnevenSubsidiary.log). This file can be opened in any text editor.

11. Open the **UnevenSubsidiary** dimension, click **All**  and then click **Hierarchy Sort**  .

    The result appears as follows:



    GO Americas and GO Asia Pacific should be identified as leaf level elements since they have no children.

    Instead GO Americas and GO Asia Pacific are defined as consolidations with no children. You cannot load data into these consolidated elements.

    Next you will edit the process to handle the duplicate data and terminate the hierarchies at different levels for the different elements.

12. Save as **Subset1** and then close the **Subset Editor**.

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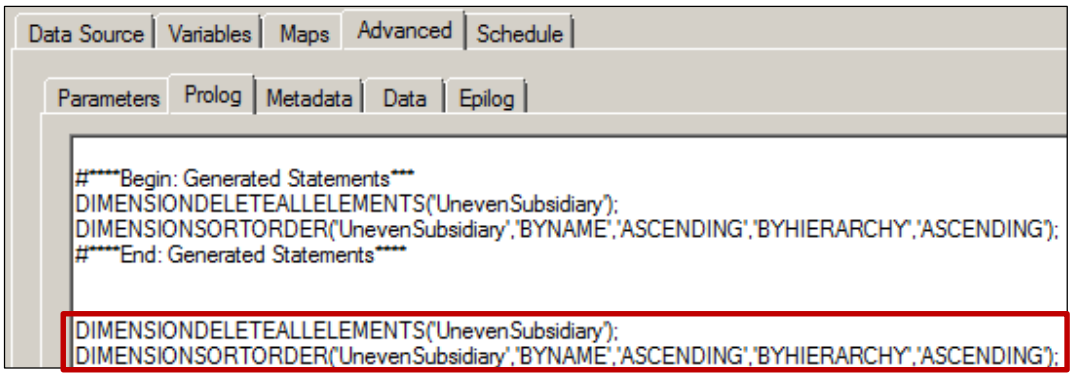
Task 3. Modify the process to terminate the hierarchies at different levels for different elements.

1. Double-click the **CreateUnevenSubsidiary** process to open it (if it is not already open).
2. Click the **Maps** tab, click the **Dimensions** sub-tab, and change the **Action** to **Recreate**.

   Because the dimension has already been created, you cannot make any other changes to the process until you change the action.
3. On the **Advanced** tab, click the **Prolog** sub-tab.
4. Copy everything between **Begin: Generated Statements** and **End: Generated Statements** and paste it beneath **End: Generated Statements**.

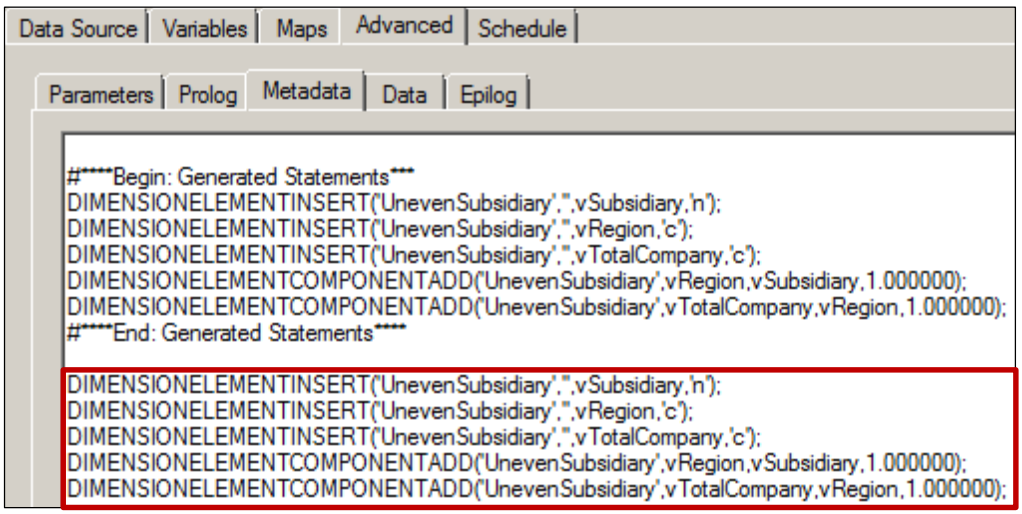
   Note: You can right-click to copy and paste in TM1.

   The result appears as follows:



5. Click the **Metadata** tab, and repeat step **4**.

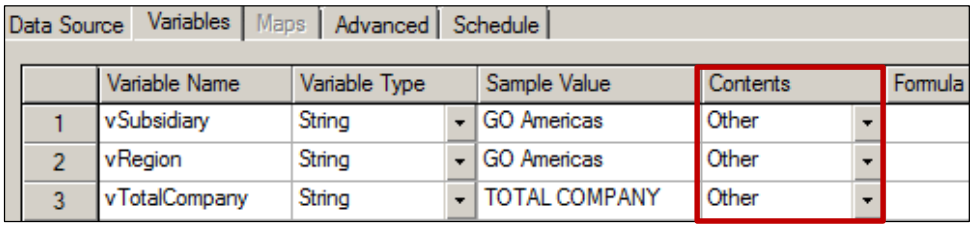
   The result appears as follows:



   There are no other statements that you will need to update on the other tabs.

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6. Click the **Variables** tab, and then change the **Contents** column for all variables to **Other**:



   You need to copy the statements before changing the Contents to Other because once you change them to Other, the variables and any statements using them are removed from the Generated Statements sections.

7. Click the **Advanced** tab, and then click the **Prolog** sub-tab.

   The Generated Statements section should be empty but everything pasted below the Generated Statements section remains.

8. Click the **Metadata** sub-tab, and ensure that the section between the Generated Statements is empty.

   Next you will make some changes to the scripts on the Metadata tab.

   You will add an IF statement to evaluate the vSubsidiary variable. If it DOES NOT match vRegion, put this element into a three-level hierarchy; if it DOES match vRegion, put it into a two-level hierarchy. Since you only have two levels with a possible conflict you can stop here. Otherwise you could add multiple ELSEIF statements.

9. After the Generated Statements, type the following:

   IF (vSubsidiary@<>vRegion);

   # create the 3 level hierarchy

10. After the **DIMENSIONELEMENTCOMPONENTADD** functions, type the following comment:

    # assigns elements to parent and weights to elements in hierarchy

    Now you will create the ELSE block and close the conditional statement.

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11. At the end of the IF block, below the comment that you just typed, type the following:

    ELSE;

    # create the 2 level hierarchy changing the vRegion to be the leaf level element

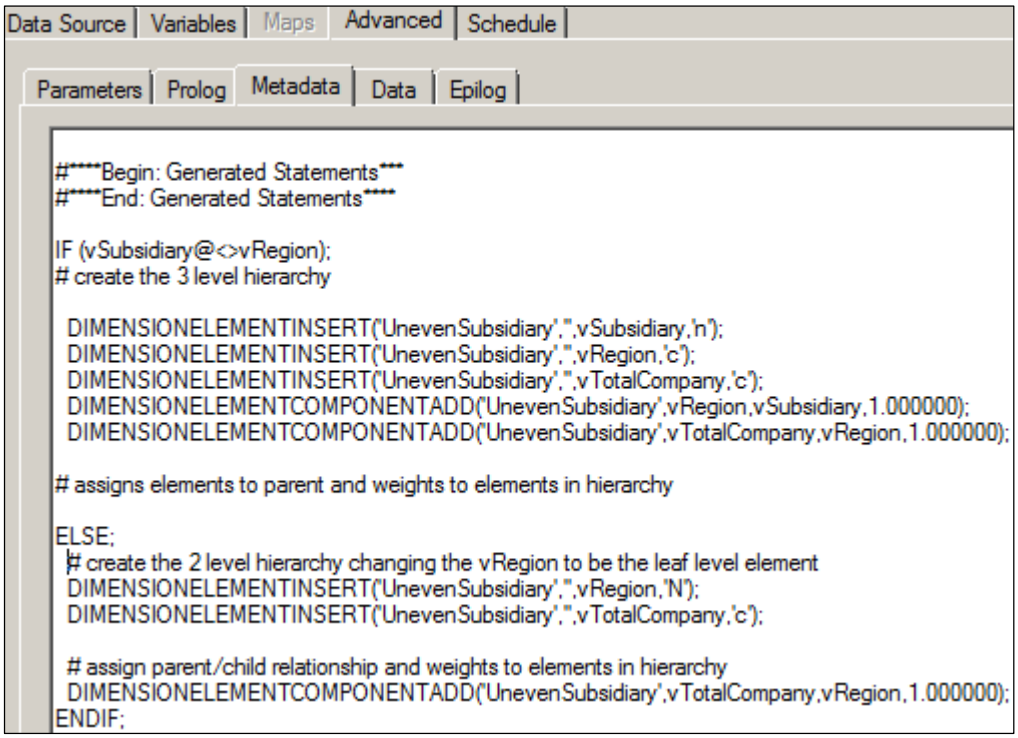
    DIMENSIONELEMENTINSERT('UnevenSubsidiary','',vRegion,'N');

    DIMENSIONELEMENTINSERT('UnevenSubsidiary','',vTotalCompany,'c');

    # assign parent/child relationship and weights to elements in hierarchy

    DIMENSIONELEMENTCOMPONENTADD('UnevenSubsidiary',vTotalCompany,vRegion,1.000000);

    ENDIF;



12. Click through the **Advanced** tabs, and then save and run the process.

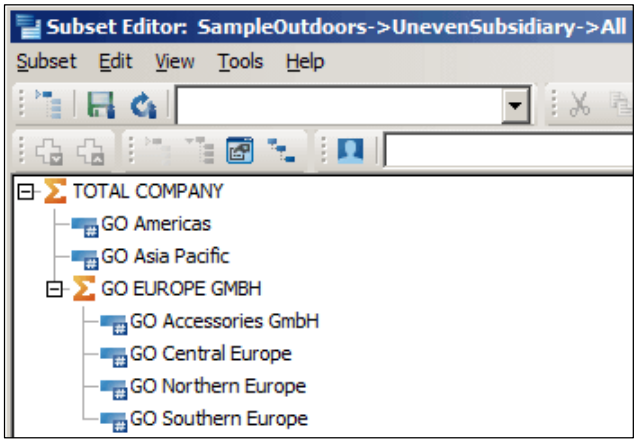
    The process runs without errors.

13. Close **TurboIntegrator**.

14. Double-click the **UnevenSubsidiary** dimension.

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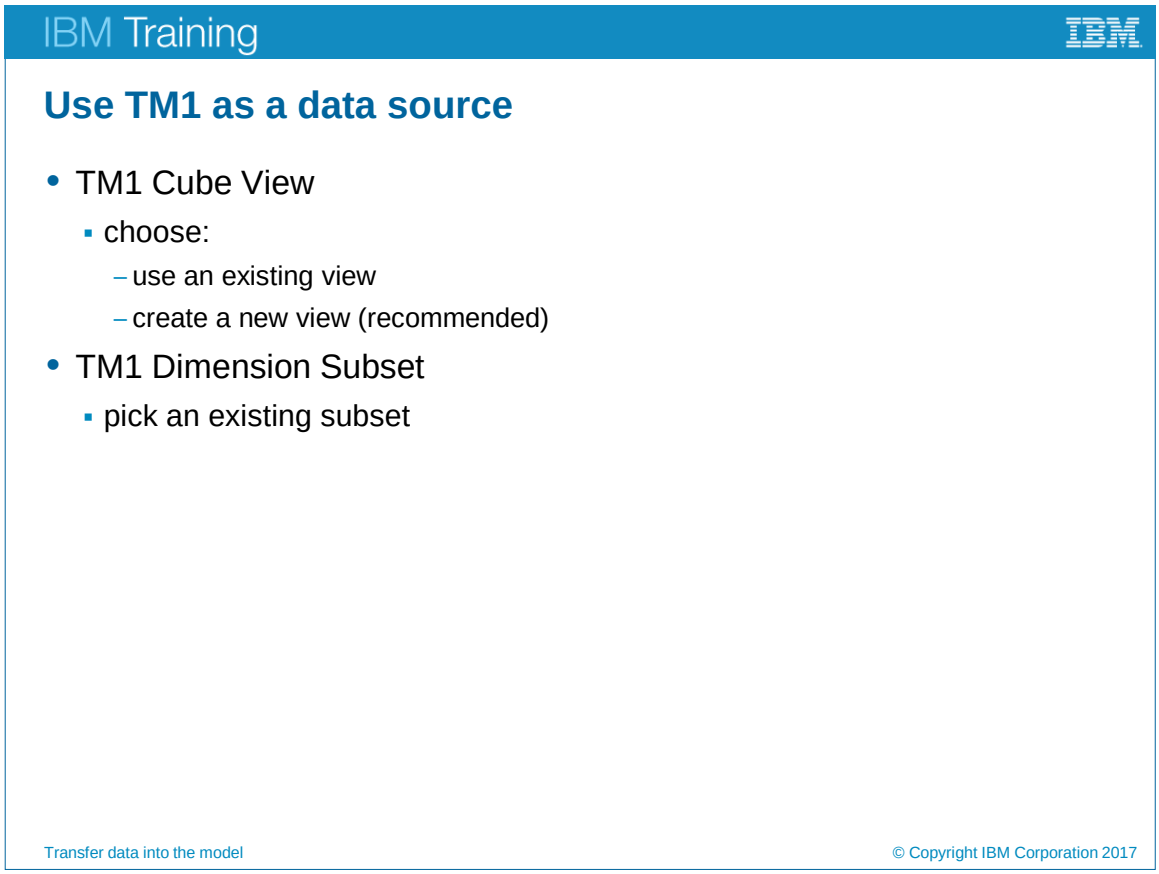
15. Click **All** and then click **Hierarchy Sort**.
    The result appears as follows:



16. Close the **Subset Editor** without saving.
17. Right-click **SampleOutdoors** and click **Save Data**.
18. Leave **Architect** open for the next demonstration.

Results:
You created a TurboIntegrator process that terminates at different levels for different hierarchies. You then ran the process and viewed the dimension that was created.

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Use TM1 as a data source

When you create the view in TurboIntegrator, the view is properly formatted for exporting. A view that has been formatted for viewing may not be properly formatted for exporting.

TM1 may be used as a source of data for external sources by exporting dimensions or cube views to text files. This allows data to be shared among a large variety of systems.

Sometimes it is useful to have one cube push data into another cube. TI can do this by using a TM1 view as a data source. This allows data to move from one cube to another in a static manner as opposed to sharing data with a rule, which will dynamically update the data between cubes.

- Archive data (and freeze rules)
- Summarize data (update on a scheduled basis; e.g., monthly)

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IBM Training

Demonstration 4

Export data to ASCII text using TurboIntegrator

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Demonstration 4: Export data to ASCII text using TurboIntegrator

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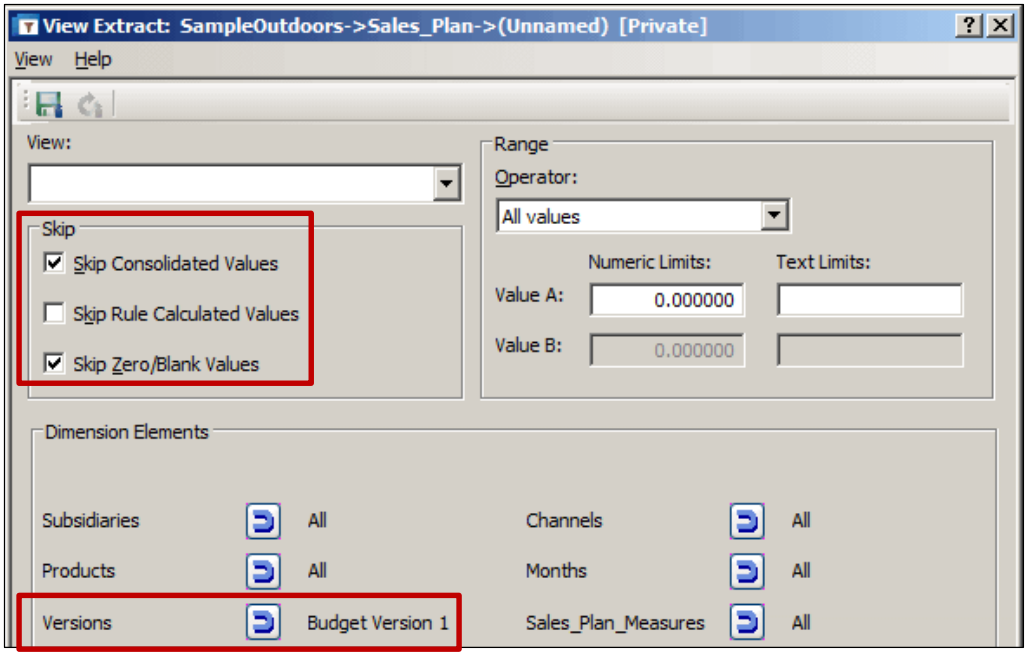
Demonstration 4:
Export data to ASCII text using TurboIntegrator

Purpose:
You want to export data for Budget Version 1 from the Sales_Plan cube to an ASCII file, so that the data can potentially be transferred to another system. You will make use of the TurboIntegrator TextOutput function to do this.

Task 1. Set the data source.

1. Create a new process with Datasource Type: **IBM CognosTM1 Cube View**.
2. Beside **Data Source Name**, click **Browse**, and then click **SampleOutdoors:Sales_Plan**.
3. Click **Create View**, and then ensure that **Skip Consolidated Values** and **Skip Zero/Blank Values** are selected.
4. Beside **Versions**, click **Subset Editor**, click **Budget Version 1**, and then click **OK**.

   The results appear as follows:



5. Click **OK**.

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6. In the **Select or Enter Named View** box, type **zExport view**, and then click **OK**.

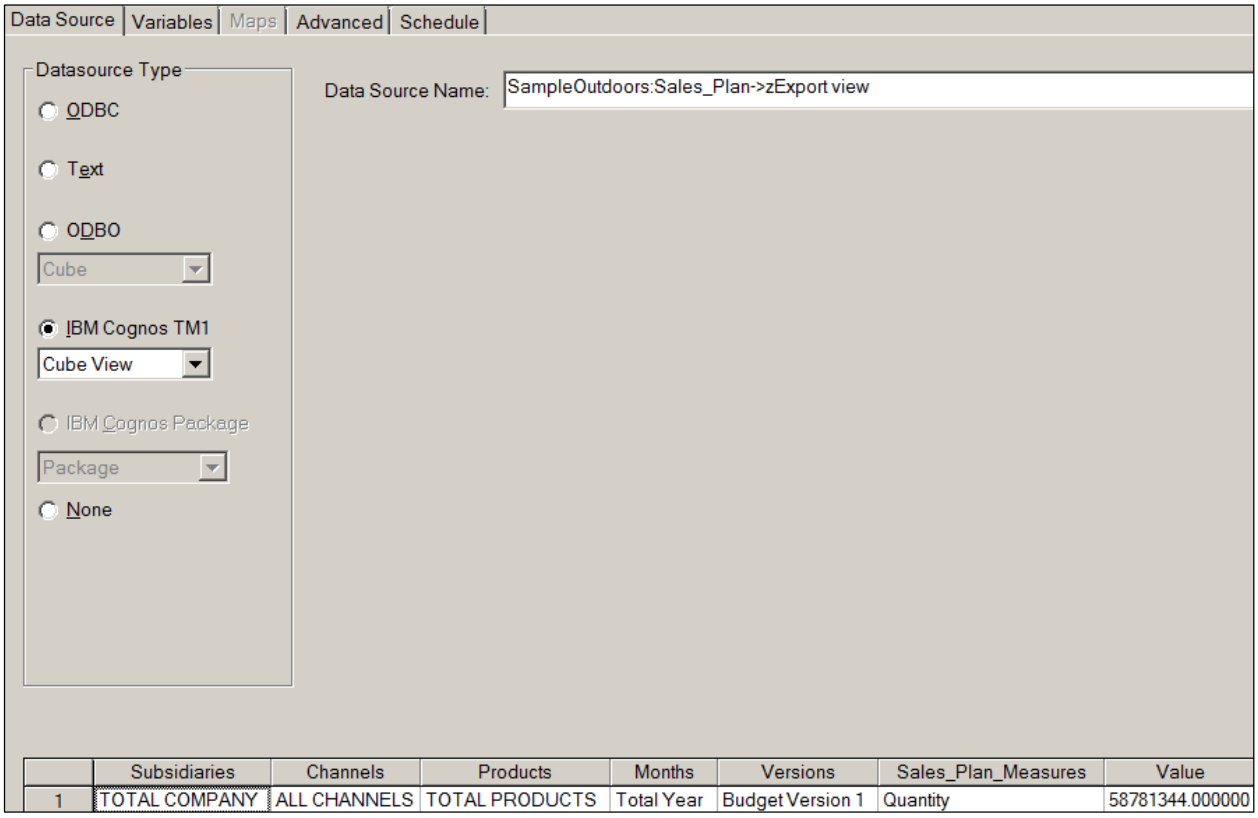
   You cannot chose to make this a private view because TI may only use public views.

   It is a good idea to give the view a name that will identify it as a view used in a process or a system view. In this example, you preceded the name with the letter z so it will be at the end of the list of views. Another option would be to name it something like 'system.ExportView' or use another predetermined prefix.

   Alternatively, you could precede the name with a } and TM1 will automatically hide it from the display as a control object.

7. Below **Select View**, click **zExport view**, and then click **OK**.
8. Click **Preview**.

   The result appears as follows:



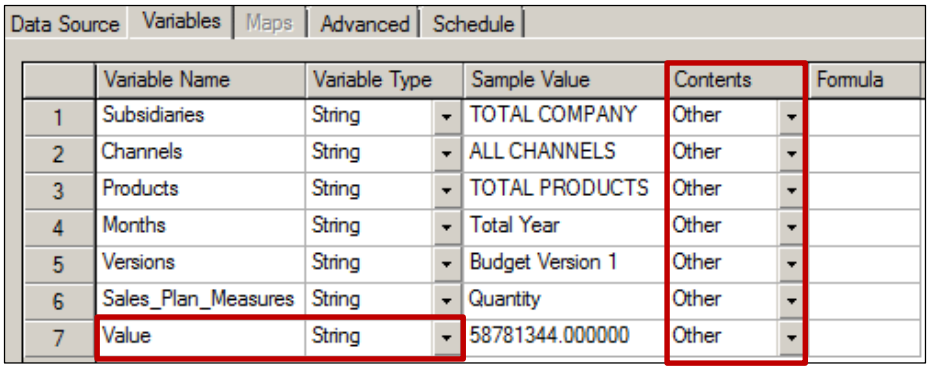
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Task 2. Set up variables, and then add a function.

1. Click the **Variables** tab.
2. Beside **Value**, set the **Variable Type** to **String**, and then for all elements set the **Contents** column to **Other**.

   When exporting data to text or ASCII files, all variables must be strings, even the data. If the value is not text, TI will generate an error message.

   The result appears as follows:



	Variable Name	Variable Type	Sample Value	Contents	Formula
1	Subsidiaries	String	TOTAL COMPANY	Other	
2	Channels	String	ALL CHANNELS	Other	
3	Products	String	TOTAL PRODUCTS	Other	
4	Months	String	Total Year	Other	
5	Versions	String	Budget Version 1	Other	
6	Sales_Plan_Measures	String	Quantity	Other	
7	Value	String	58781344.000000	Other	

3. Click the **Advanced** tab, and then click the **Metadata** sub-tab.

   You want to insert the TurboIntegrator TextOutput function.

   This function writes a comma-delimited record to a text file:

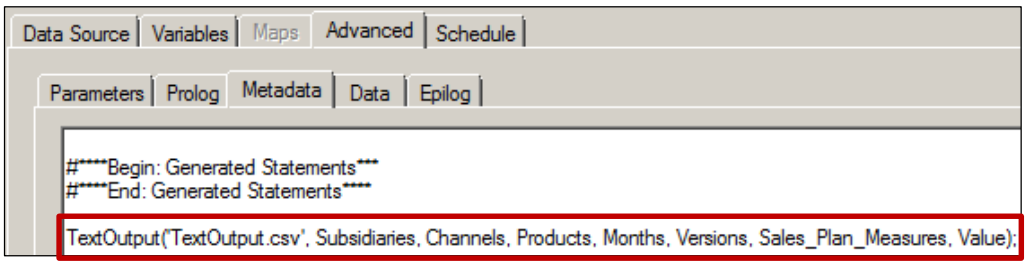
   TextOutput(FileName, String1, String2, ...Stringn);

   FileName is the full path to the text file you are writing to and **String1…Stringn** corresponds to each field you want to create in the text file.

4. Type the following on the **Metadata** tab, below the Generated Statements:

   TextOutput('TextOutput.csv', Subsidiaries, Channels, Products, Months, Versions, Sales_Plan_Measures, Value);

   The result appears as follows:



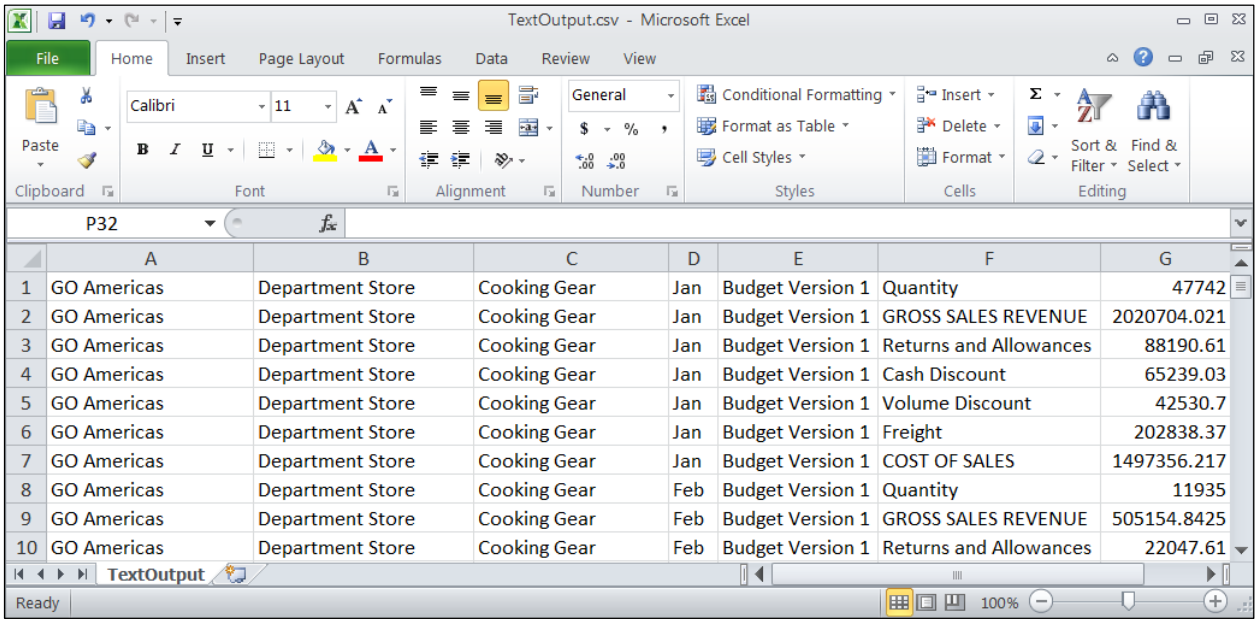
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5. Click through all the sub-tabs on the **Advanced** tab, save the process as **CreateTextOutput**, and then run the process.

   The output file (**TextOutput.csv**) is created on your TM1 data server (in this case, **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\SampleOutdoors\DataFiles\TextOutput.csv**).

6. Open **Windows Explorer**, navigate to **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\SampleOutdoors\DataFiles**, and then open **TextOutput.csv** in **Microsoft Excel** (click **Close** if you are prompted to restore any other files).

   The results appear similar to the following:

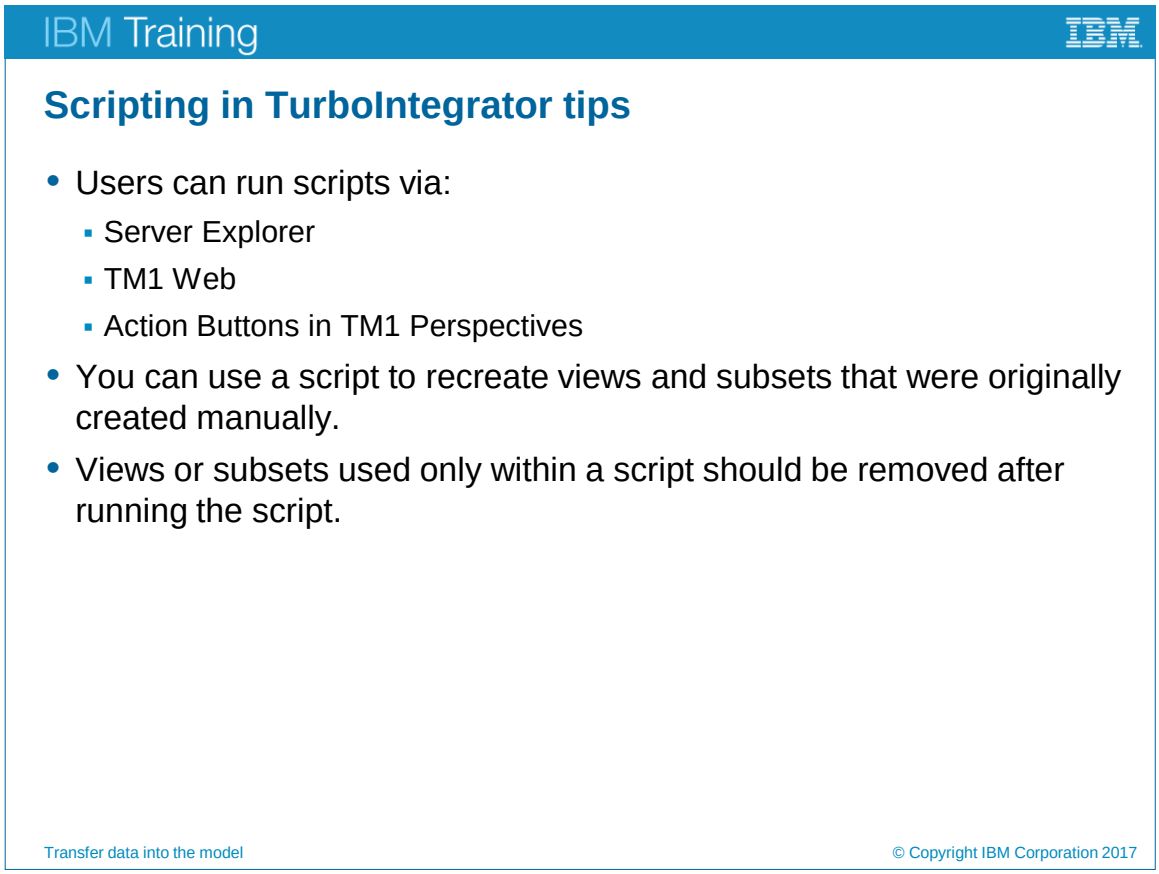


	A	B	C	D	E	F	G
1	GO Americas	Department Store	Cooking Gear	Jan	Budget Version 1	Quantity	47742
2	GO Americas	Department Store	Cooking Gear	Jan	Budget Version 1	GROSS SALES REVENUE	2020704.021
3	GO Americas	Department Store	Cooking Gear	Jan	Budget Version 1	Returns and Allowances	88190.61
4	GO Americas	Department Store	Cooking Gear	Jan	Budget Version 1	Cash Discount	65239.03
5	GO Americas	Department Store	Cooking Gear	Jan	Budget Version 1	Volume Discount	42530.7
6	GO Americas	Department Store	Cooking Gear	Jan	Budget Version 1	Freight	202838.37
7	GO Americas	Department Store	Cooking Gear	Jan	Budget Version 1	COST OF SALES	1497356.217
8	GO Americas	Department Store	Cooking Gear	Feb	Budget Version 1	Quantity	11935
9	GO Americas	Department Store	Cooking Gear	Feb	Budget Version 1	GROSS SALES REVENUE	505154.8425
10	GO Americas	Department Store	Cooking Gear	Feb	Budget Version 1	Returns and Allowances	22047.61

7. Close **TextOutput.csv** without saving and close **TurboIntegrator**.
8. Right-click **SampleOutdoors** and click **Save Data**.
9. Leave **Architect** open for the next demonstration.

Results:
You exported the data for Budget Version 1 from the Sales_Plan cube to an ASCII file, so the data can potentially be transferred to another system. You made use of the TurboIntegrator TextOutput function to do this.

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Scripting in TurboIntegrator tips

Administrators can create TI scripts and users may run them (with appropriate security settings).

Custom scripting should be planned and well documented. Doing so will make it much easier to maintain and update the model.

Even if you can create a view or subset manually, you may wish to create a TI script to recreate it. This will be easier to document and recreate as data is added or changed.

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IBM Training

Demonstration 5

Move data between versions

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Demonstration 5: Move data between versions

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Demonstration 5: Move data between versions

Purpose:
The Unit figures for Budget Version 1 in the Sales_Plan cube need to be approved. You want to move the data from Budget Version 1 to Submit Version, so users can easily identify the finalized data.

You will manually script the processes to copy the data. You will perform the following tasks:

- **Create parameters to specify the source, target and month data**
- **Create a view and subset to hold the data to copy**
- **Load the data to the new location in the cube**
- **Delete the views and subsets used to move data**

Task 1. Create a new version.

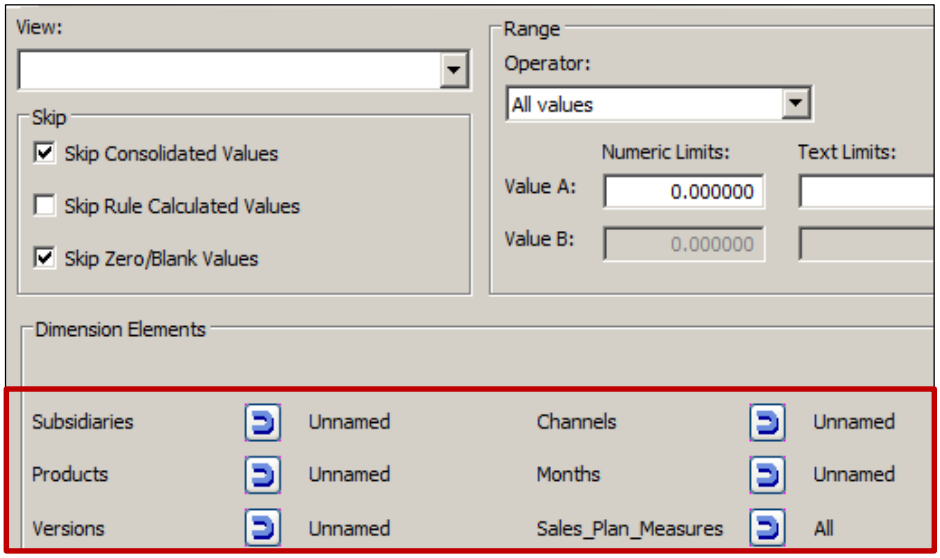
1. Right-click the **Versions** dimension, and then click **Edit Dimension Structure**. If you double-click Versions, you will open the dimension in Subset Editor.
2. Click **Budget Version 2**.
3. Click **Edit**, and then click **Insert Element**
4. Type **Submit Version**, click **Add**, and then click **OK**.
5. Save and close the **Dimension Editor**.

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Task 2. Set the data source and variables.

1. Create a new process with Datasource Type: **IBM Cognos TM1 Cube View**.
2. In the **Data Source Name** box, click **Browse**, click **sampleoutdoors:Sales_Plan**, and then click **Create View**.
3. Beside each of the following dimensions, click the **Subset Editor**  , click **All**, click **Filter by Level**  , click **0** and then click **OK** twice:
   - Subsidiaries
   - Products
   - Channels
   - Months
   - Versions

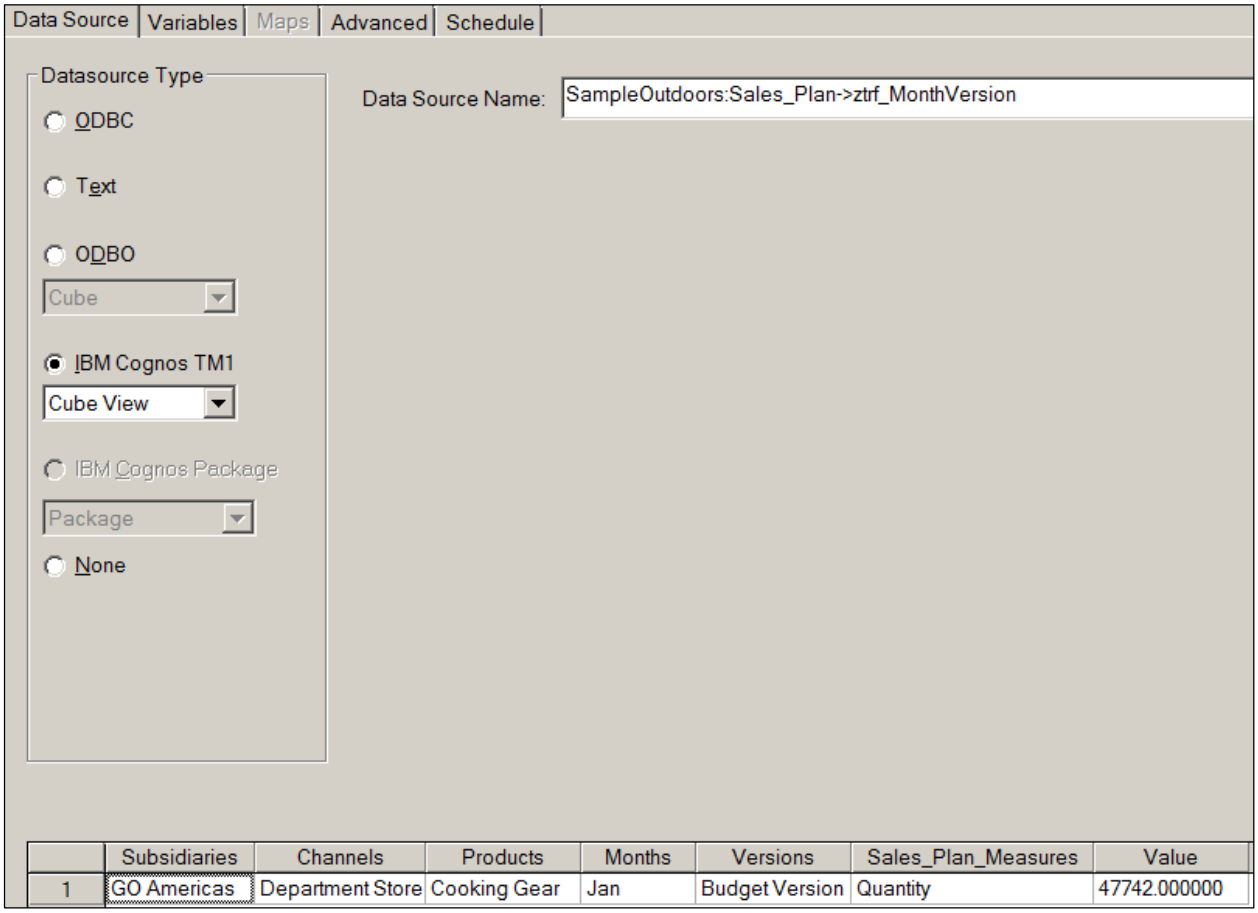
   Leave Sales_Plan_Measures as All.
   The results appear as follows:



4. Click **OK**, name the view as **ztrf_MonthVersion**, and then click **OK**.
5. Below **Select View**, click **ztrf_MonthVersion**, and then click **OK**.

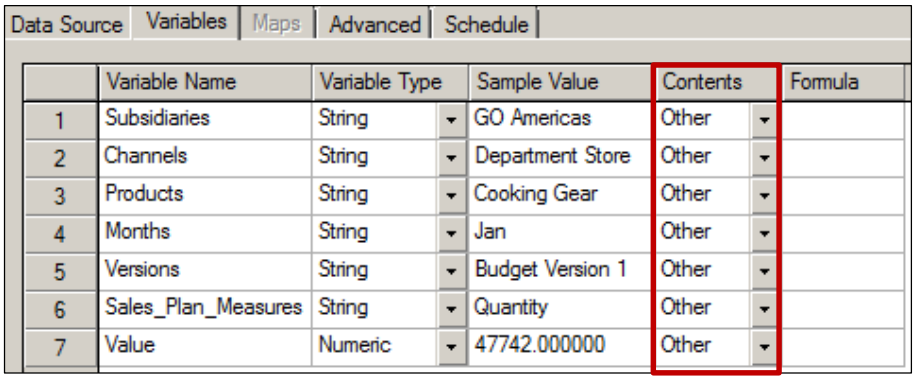
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6. Click **Preview**.
   The results appear as follows:



	Subsidiaries	Channels	Products	Months	Versions	Sales_Plan_Measures	Value
1	GO Americas	Department Store	Cooking Gear	Jan	Budget Version	Quantity	47742.000000

7. Click the **Variables** tab.
8. Change the **Contents** column for all variables to **Other**.
   The results appear as follows:



	Variable Name	Variable Type	Sample Value	Contents	Formula
1	Subsidiaries	String	GO Americas	Other	
2	Channels	String	Department Store	Other	
3	Products	String	Cooking Gear	Other	
4	Months	String	Jan	Other	
5	Versions	String	Budget Version 1	Other	
6	Sales_Plan_Measures	String	Quantity	Other	
7	Value	Numeric	47742.000000	Other	

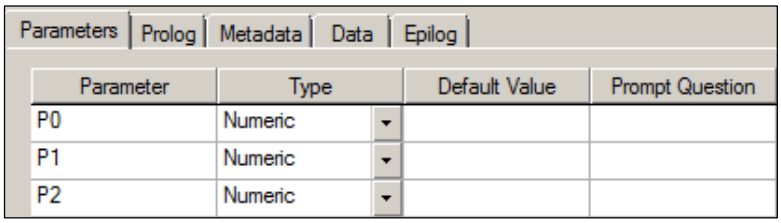
9. Click the **Advanced** tab.

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Task 3. Create the parameters.

1. Click **Insert** three times.

   The results appear as follows:



2. Click on each parameter and rename them to:

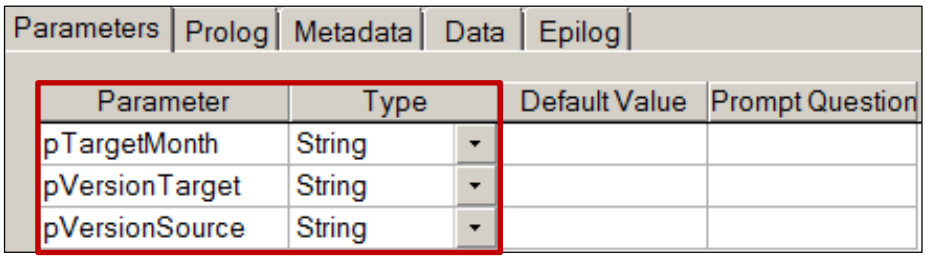
   P0 = **pTargetMonth**

   P1 = **pVersionTarget**

   P2 = **pVersionSource**

   In this example, you have used the **p** to indicate this value comes from a Parameter and not a variable defined on the Variables tab. This will make it easier to identify its purpose in the script.

3. Click **String** under **Type** for each parameter.



4. Under **Default Value** type:

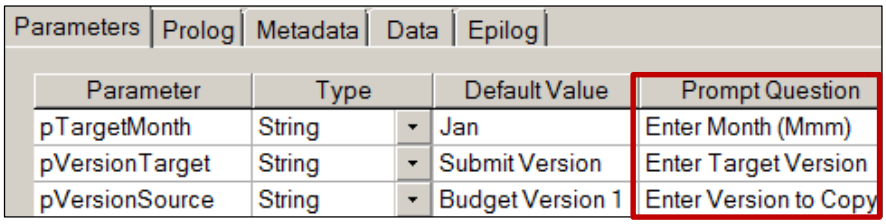
   - **Jan**
   - **Submit Version**
   - **Budget Version 1**

   These should match the elements in the Months and Versions dimensions.

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5. Under **Prompt Question**, type the following:
   - **Enter Month (Mmm)**
   - **Enter Target Version**
   - **Enter Version to Copy**

   This will be displayed in the window that is displayed to the user when they run the script.



6. Click the **Prolog** sub-tab.

Task 4. Script the Prolog process.

1. Press **Enter** twice after #****End: Generated Statements****.
2. Type the following two comment lines and then press **Enter**:

   # This TI process copies Sales_Plan Cube data from a source Month & Version to a target Month & Version.

   #**DECLARE VARIABLES******

   It is often helpful to open the Reference Guide for TI functions when scripting. In the TurboIntegrator window, click the Help menu, click Contents and Index (allow blocked content if prompted), and then click TM1 TurboIntegrator Functions in the left pane to see the list of available functions and their syntax.

   You will now create two variables to hold the name of the cube and the name of the view and subset to hold the data to move in the cube.
3. Press **Enter**, and then type the following:

   Cube = 'Sales_Plan';

   Object = '$Transfer'|Cube;

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4. Press **Enter** twice, and then type the following:

   #**TURN CUBE LOGGING ON****

   #LogFlag = CUBEGETLOGCHANGES(Cube);

   #CUBESETLOGCHANGES(Cube, 0);

   This code is optional and commented out but it may be useful if there is a problem moving the data. It turns cube logging on when you remove the # from the 2nd and 3rd lines.

   You should destroy the view before trying to create it. This way, if the process failed before removing the view, it will be removed now without generating an error (and halting the process).

5. Press **Enter** twice, and then type the following:

   #**DELETE OLD VIEW (IF ANY)****

   VIEWDESTROY(Cube, Object);

   Now you will specify that you want to destroy old subsets that will be created in this view.

6. Press **Enter** twice, and then type the following:

   #**DELETE OLD SUBSETS (IF ANY)****

   SUBSETDESTROY('Months', Object);

   SUBSETDESTROY('Versions', Object);

   Now you will create the new subsets (for the target view) and add the elements from the user input (parameters).

7. Press **Enter** twice, and then type the following:

   #**CREATE NEW SUBSETS****

   SUBSETCREATE('Months', Object);

   SUBSETELEMENTINSERT('Months', Object, pTargetMonth, 1);

   SUBSETCREATE('Versions', Object);

   SUBSETELEMENTINSERT('Versions', Object, pVersionTarget, 1);

   Now you will create the new target view using the subsets created above.

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8. Press **Enter** twice, and then type the following:
   #**CREATE VIEW****
   VIEWCREATE(Cube, Object);
   VIEWSUBSETASSIGN(Cube, Object, 'Months', Object);
   VIEWSUBSETASSIGN(Cube, Object, 'Versions', Object);
   Now you will clear the view, destroy it, and delete the subsets associated with it (as a precaution).
9. Press **Enter** twice, and then type the following:
   #**CLEAR VIEW****
   VIEWZEROOUT(Cube, Object);

   #**DELETE THE CLEARED VIEW****
   VIEWDESTROY(Cube, Object);

   #**DELETE ASSOCIATED SUBSETS****
   SUBSETDESTROY('Months', Object);
   SUBSETDESTROY('Versions', Object);
   Now you will create new views to be used as the data source.
10. Press **Enter** twice, and then type the following:
    #**CREATE NEW VIEWS TO USE AS DATA SOURCES FOR NEXT STEP****

    #**CREATE SOURCE SUBSETS****
    SUBSETCREATE('Months', Object);
    SUBSETELEMENTINSERT('Months', Object, pTargetMonth, 1);
    SUBSETCREATE('Versions', Object);
    SUBSETELEMENTINSERT('Versions', Object, pVersionSource, 1);
    Now you will create the source views and designate them as the source.

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11. Press **Enter** twice, and then type the following:

```
#****CREATE SOURCE VIEWS****
VIEWCREATE(Cube, Object);
VIEWSUBSETASSIGN(Cube, Object, 'Months', Object);
VIEWSUBSETASSIGN(Cube, Object, 'Versions', Object);

#****REESTABLISH THE NEW VIEW AS THE DATA SOURCE****
DATASOURCECUBEVIEW = Object;
```

In the above script you have:

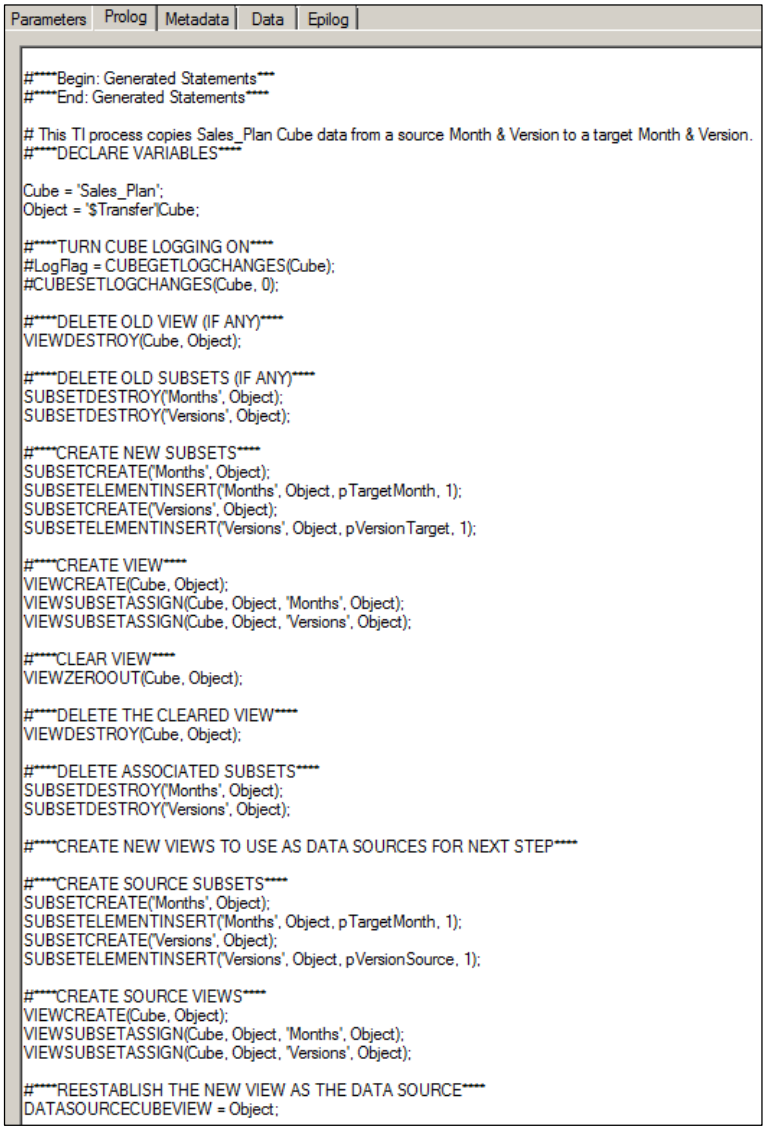
- created a target view to zero out the data (and then destroyed it and its subsets)
- created a view for the data transfer based on the source (and then destroyed it and its subsets).

You typically add a destroy subset command to:

- ensure the subset is based on the input parameter
- clean up the system so that there will not be public views left on the system for users

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The results appear as follows:



Nothing is needed on the Metadata script.

Task 5. Script the Data process.

1. Click the **Data** sub-tab.
2. Add two returns below **#****End: Generated Statements******.
3. Type the following comment line and then press **Enter**:

   # Check if the Value is a String or Number

   The following statements will check to see if you are copying numeric or string data.

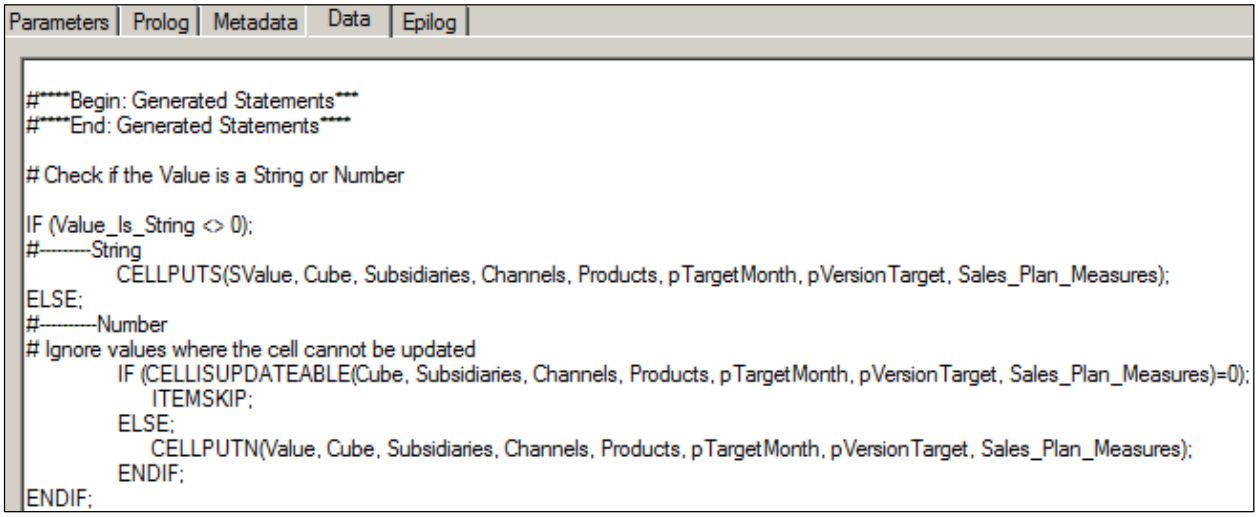
   Next you will type the following conditional check using a TI local variable called Value_Is_String.

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4. Press **Enter** twice and type the following:

```
IF (Value_Is_String <> 0);
#---------String
    CELLPUTS(SValue, Cube, Subsidiaries, Channels, Products,
pTargetMonth, pVersionTarget, Sales_Plan_Measures);
ELSE;
#----------Number
# Ignore values where the cell cannot be updated
    IF (CELLISUPDATEABLE(Cube, Subsidiaries, Channels, Products,
pTargetMonth, pVersionTarget, Sales_Plan_Measures)=0);
        ITEMSKIP;
    ELSE;
        CELLPUTN(Value, Cube, Subsidiaries, Channels, Products,
pTargetMonth, pVersionTarget, Sales_Plan_Measures);
    ENDIF;
ENDIF;
```



Note: Any functions on this sub-tab are executed for each line of data in the source. Any dimensions or cubes that should be updated with data need to be created before the functions updating them may be processed.

5. Click the **Epilog** sub-tab.

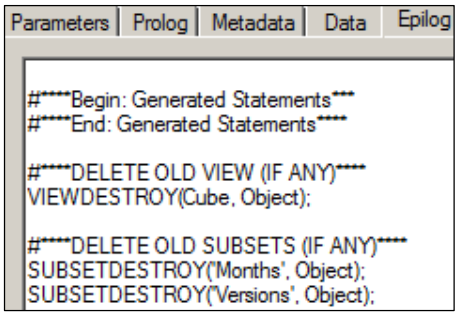
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Task 6. Script the Epilog process.

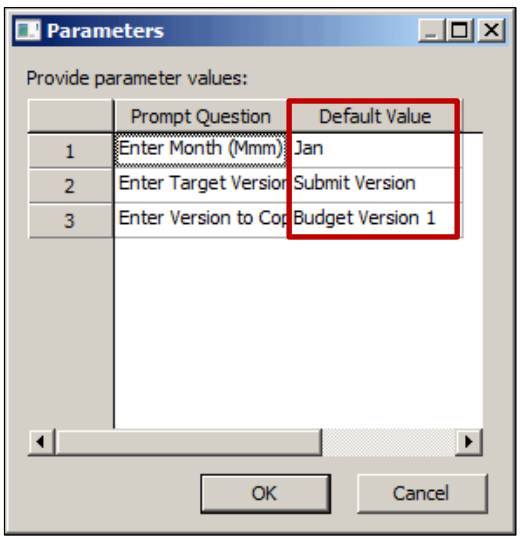
1. After **#****End: Generated Statements****, press **Enter** twice.
2. Type the following:

   #**DELETE OLD VIEW (IF ANY)****
   VIEWDESTROY(Cube, Object);

   #**DELETE OLD SUBSETS (IF ANY)****
   SUBSETDESTROY('Months', Object);
   SUBSETDESTROY('Versions', Object);



3. Save the process as **Transfer_Versions_Month**, run the process, and then enter the following values (if necessary):
   - Jan
   - Submit Version
   - Budget Version 1

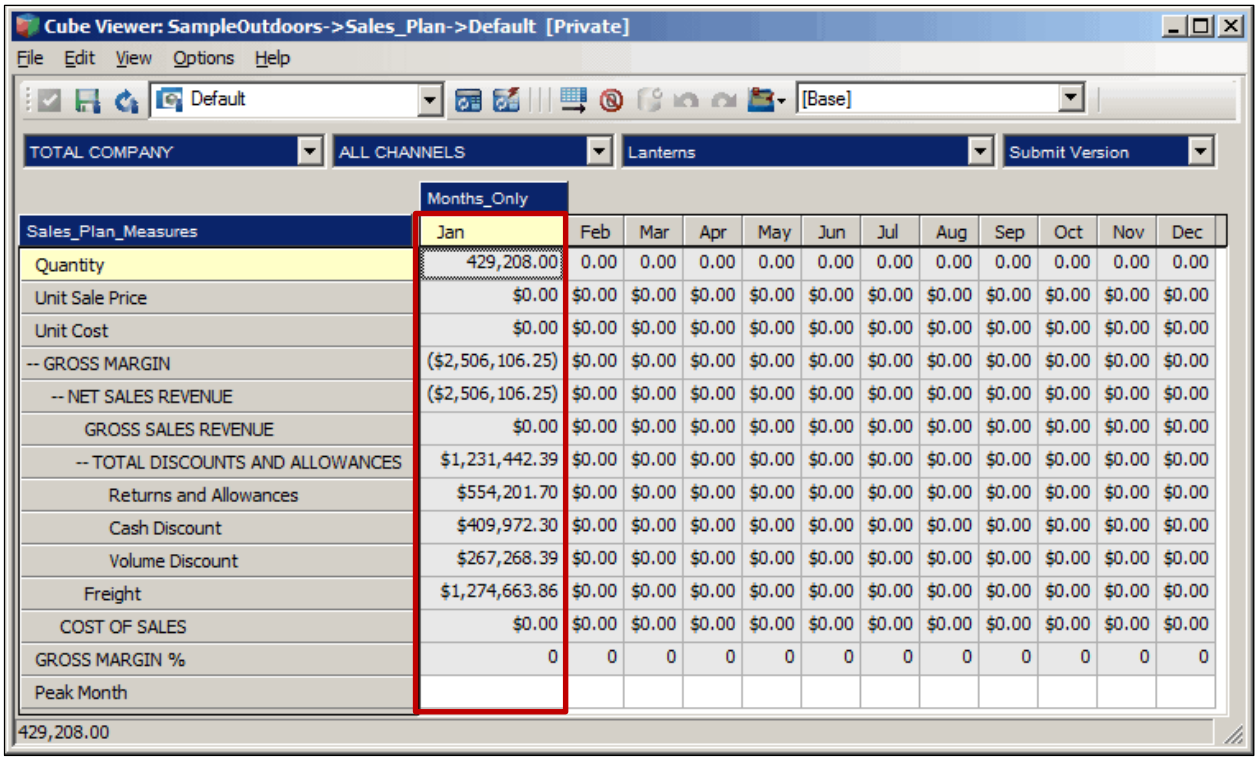


4. Click **OK**, and then click **OK** to dismiss the confirmation message.
5. Close **TurboIntegrator**.

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6. Double-click the **Sales_Plan** cube.
7. Double-click the **Versions** dimension (currently displaying **Budget Version 1**), click **All**, click **Submit Version**, and then click **OK**.
8. Recalculate the view.

   The result appears as follows (Your rows many be sorted differently, but this is acceptable):



   If your rows are sorted differently, you can click Sales_Plan_Measures to open it in the Subset Editor, click All, and then recalculate the view.

   You moved data from one portion of the cube to another by scripting.
9. Close the **Cube Viewer** without saving the changes if prompted.
10. Right-click **SampleOutdoors** and click **Save Data**.
11. Leave **Architect** open for the next demonstration.

Results:
You moved the data from Budget Version 1 to Submit Version in the Sales_Plan cube, so users can easily identify the data to be approved. This script may be run by many users as needed from Server Explorer or from an Action Button in TM1 Perspectives.

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IBM Training

Use chores with TurboIntegrator

- Chores execute:
  - one or more TurboIntegrator processes
  - at a user-defined frequency
- Chores may be:
  - run on demand
  - scheduled for automatic execution

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Use chores with TurboIntegrator

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IBM Training

Demonstration 6

Create and schedule a chore

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Demonstration 6: Create and schedule a chore

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Demonstration 6: Create and schedule a chore.

Purpose:
The grocery store part of your enterprise updates its inventory regularly. You want to create a chore that creates and populates the Products dimension and then creates the Product Type subsets. Next you want to schedule the chore to run once every week.

Since you already have processes to accomplish these tasks, you can use them in the chore.

Task 1. Create a chore.

1. In Server Explorer, right-click **Chores**, and then click **Create New Chore**.
2. In the **Available** column, double-click **CreateProductsCSV** and **CreateProductType_subsets** to add them to the **Selected** column.
3. Click **Next**.

   You want to schedule this chore to run once every 7 days.
4. Select the **Activate Chore** check box, and then beside **Days** type **7**.

   If, at some point in time, you decide that the chore no longer needs to be run, you can deactivate the chore.
5. Click **Finish**.
6. Save the chore as **CreateProducts_ProductTypeSubsets**.

   If you wish to edit a chore, you must first deactivate it. This will prevent the chore from trying to execute while someone is making changes to it. You must also deactivate a chore when editing a TurboIntegrator process that the chore will execute.
7. Leave **Architect** open for the upcoming exercise.

Results:
You created a chore that creates and populates the Products dimension and then creates the Product Type subsets. Next you scheduled the chore to run once every week.

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IBM Training

Unit summary

- Describe the Advanced tabs in TurboIntegrator
- Load custom data into a cube
- Add a subset to a dimension
- Use logic in scripts
- Export data to ASCII
- Move data between versions
- Construct chores

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Unit summary

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IBM Training

Exercise 1

Create subsets within a TurboIntegrator process

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Exercise 1: Create subsets within a TurboIntegrator process

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Exercise 1: Create subsets within a TurboIntegrator process

Your users would like to have subsets in the Subsidiaries dimension for each region. You have decided to automate this by creating a TurboIntegrator process to create the subsets. You will then couple this process and the process to create the Subsidiaries dimension into a chore and schedule it to run at 12 a.m. on the first day of every month.

To accomplish this:

- Create a TurboIntegrator process to automatically create subsets and apply elements to them. You can use an existing process as a starting point and save the modified process under a different name.
- Create a chore that will execute the processes to create the:
  - Subsidiaries dimension
  - subsets for the Subsidiaries dimension
- Schedule the process to run monthly.

For more information about where to work and the exercise results, refer to the Tasks and results section that follows. If you need more information to complete a task, refer to earlier demonstrations for detailed steps.

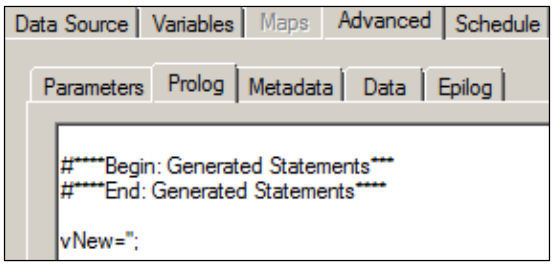
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Exercise 1:
Tasks and results

Task 1. Create a TurboIntegrator process for subsets.

- Open the CreateSubsidiariesDimCSV process.
- Click the Maps tab and then click the Dimensions sub-tab.
- Change the Action to Update.
- Click the Advanced tab, and then click all sub-tabs to regenerate the scripts.
- Click File, and then click Save As.
- In the Name box type CreateSubsidiariesRegion_Subset, and then click OK.
- Click the Variables tab and change the Contents to Other for all of the variables.
- Click the Advanced tab, and then click the Prolog sub-tab.
- Type vNew=''; (after the Generated Statements).

  Use two single quotes, not a single double quote:



- Click the **Metadata** sub-tab.
- Type the following (after the Generated Statements):

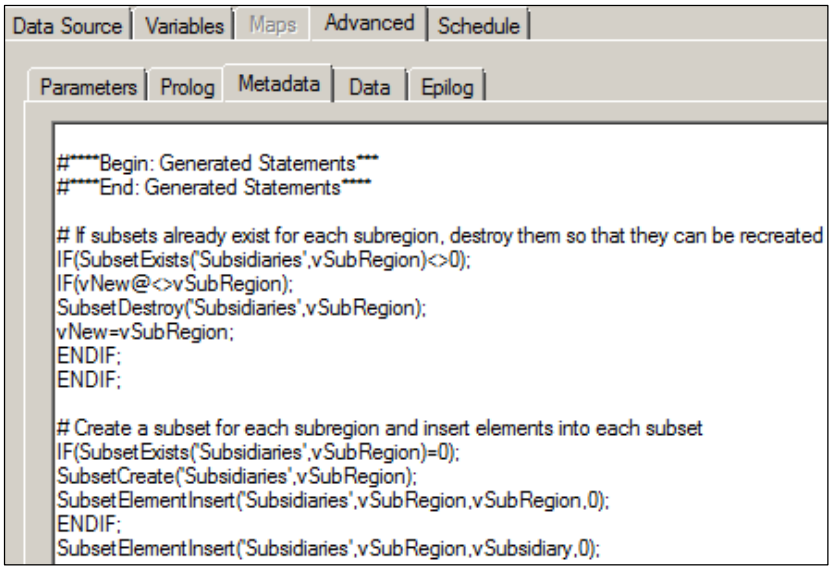
```
# If subsets already exist for each subregion, destroy them so that they can be recreated
IF(SubsetExists('Subsidiaries',vSubRegion)<>0);
IF(vNew@<>vSubRegion);
SubsetDestroy('Subsidiaries',vSubRegion);
vNew=vSubRegion;
ENDIF;
ENDIF;

# Create a subset for each subregion and insert elements into each subset
IF(SubsetExists('Subsidiaries',vSubRegion)=0);
```

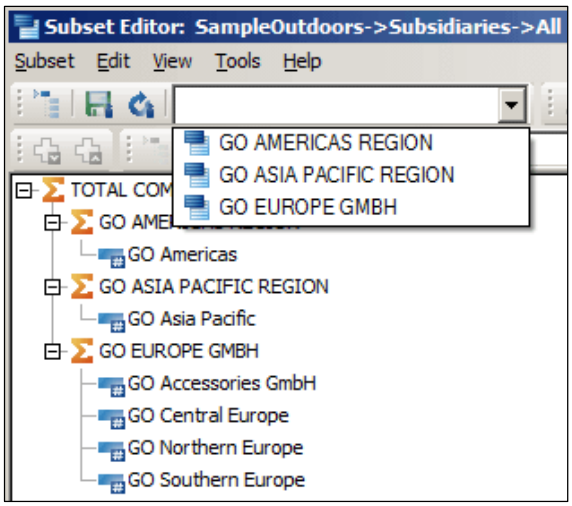
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```
SubsetCreate('Subsidiaries',vSubRegion);
SubsetElementInsert('Subsidiaries',vSubRegion,vSubRegion,0);
ENDIF;
SubsetElementInsert('Subsidiaries',vSubRegion,vSubsidiary,0);
```

The result appears as follows:



- Save and run the process.
- Click **OK** when completed successfully.
- Close the process and open the **Subsidiaries** dimension in the Subset Editor.
- Click the **Select Subset** list to see the list of subsets:



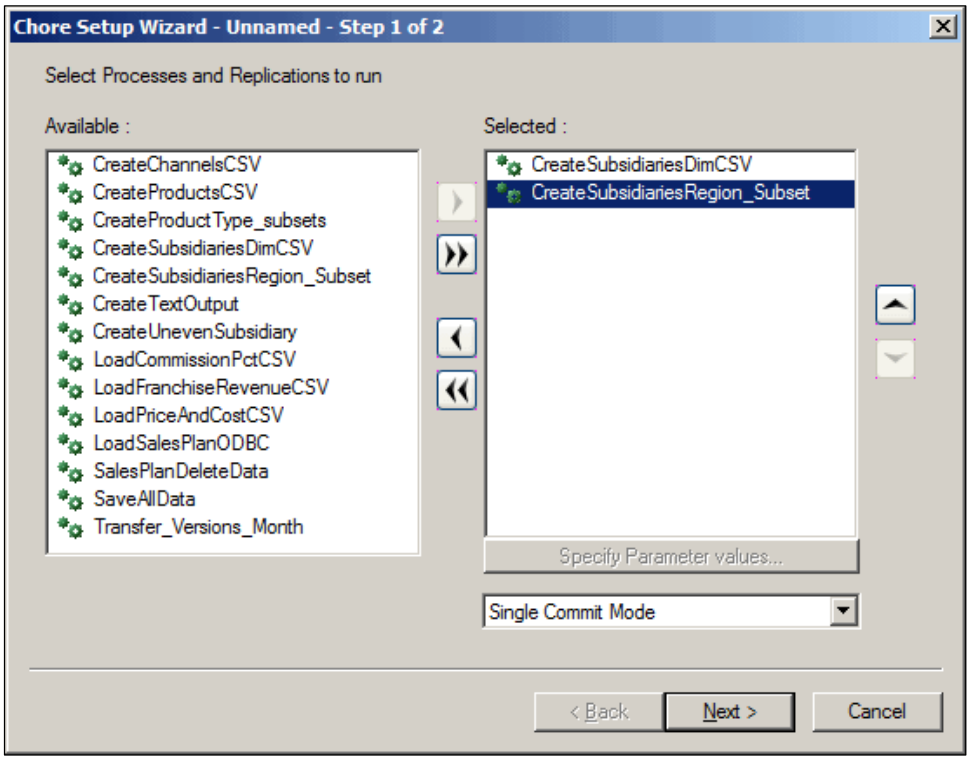
- Close the **Subset Editor**.

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Task 2. Create a chore.

- In Server Explorer, right-click **Chores** and click **Create New Chore**.
- Double-click:
  - CreateSubsidiariesDimCSV
  - CreateSubsidiariesRegion_Subset

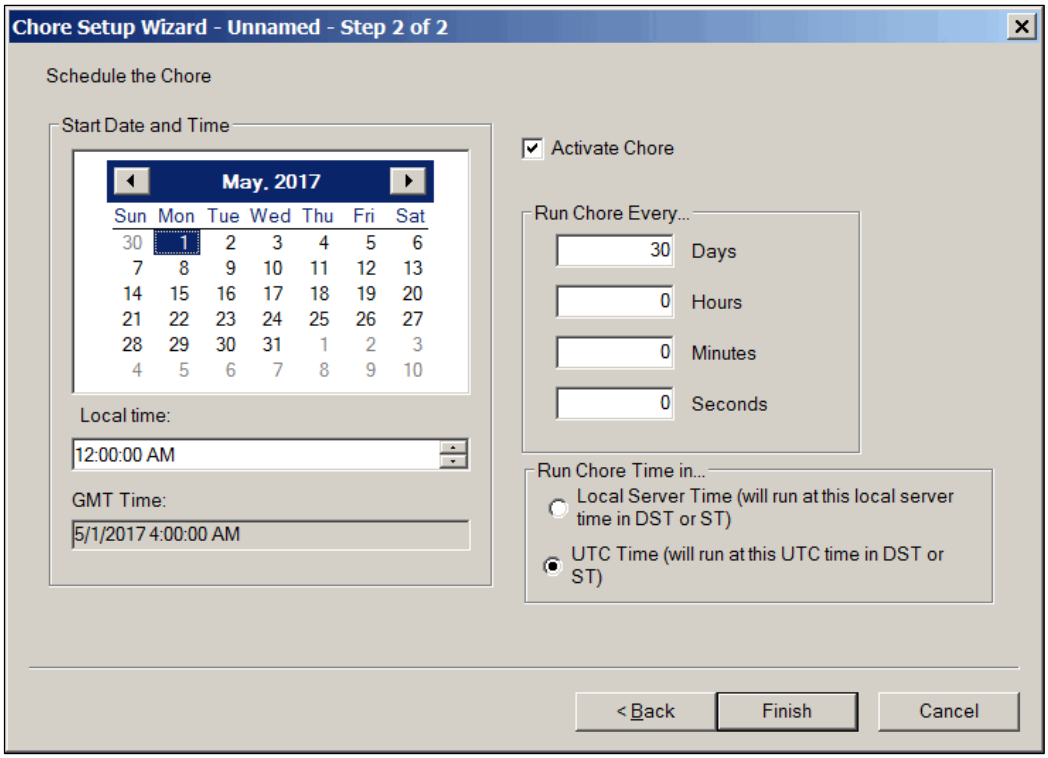
  The result appears as follows:



- Click **Next**.
- Select the **Activate Chore** check box.
- Under **Run Chore Every**, beside **Days**, type **30**.

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- Change the **Start Date and Time** section to the first day of the next month at **12:00 a.m.**:



- Click **Finish**.
- Save as **SubsidiariesChore**.
- Right-click **SampleOutdoors** and click **Save Data**.
- Close all open windows.

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IBM Training

Customize drill paths

IBM Planning Analytics (v2.0)

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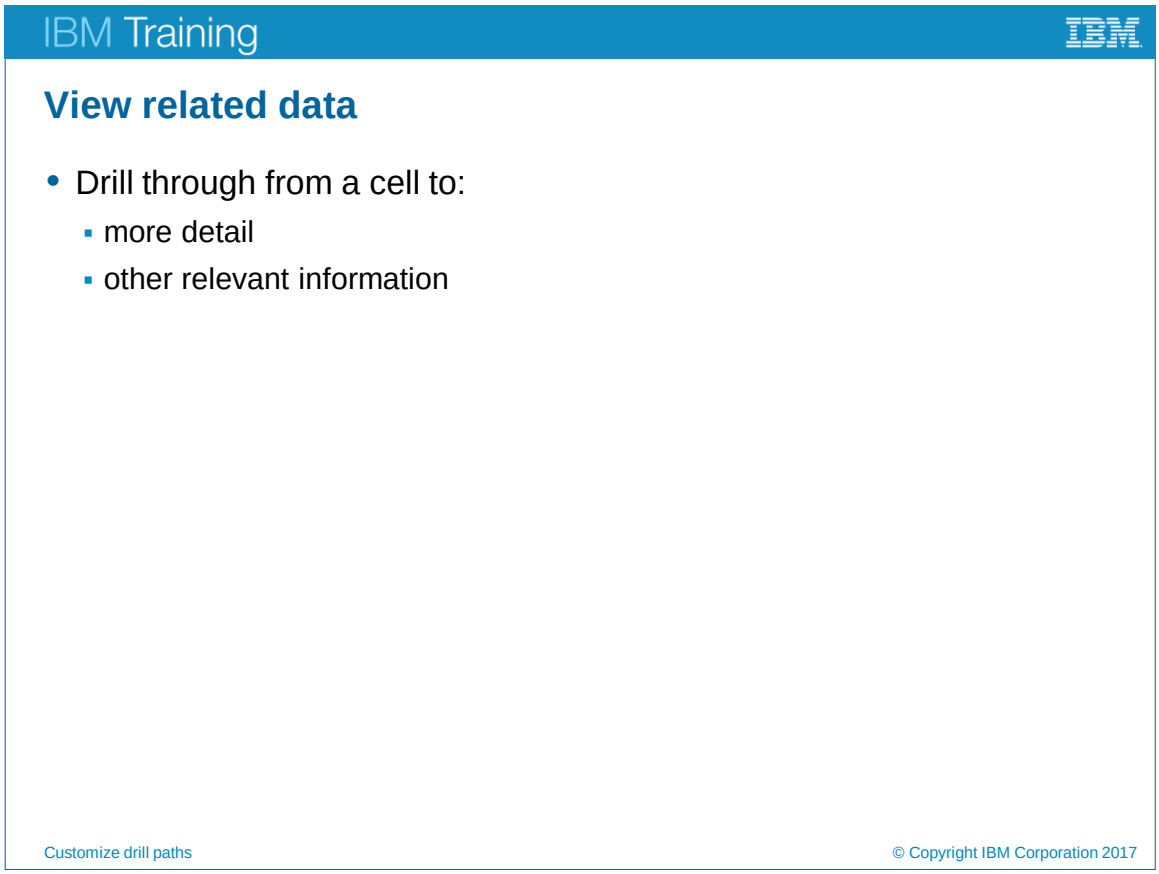
Unit objectives

- View related data
- Create a drill process
- Create a drill assignment rule
- Edit a drill process

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Unit objectives

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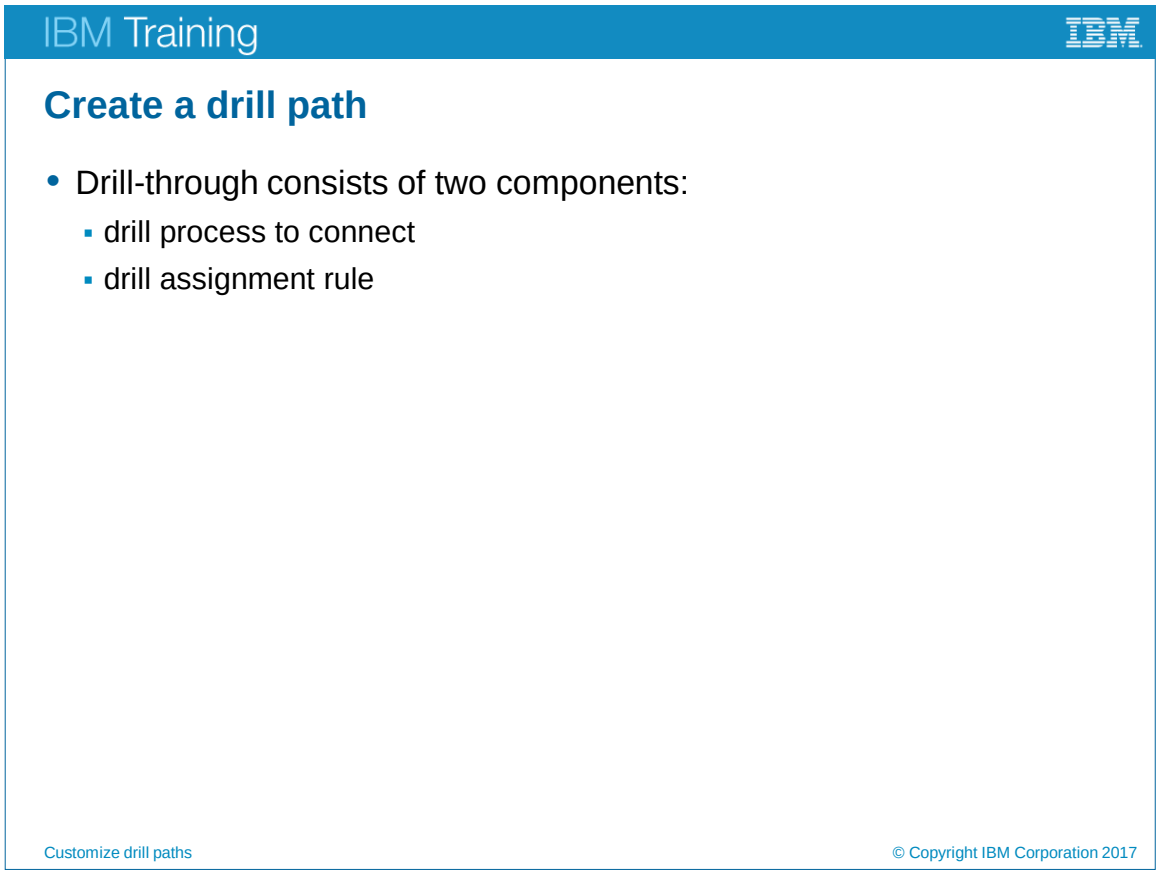
View related data

If a cell is associated with more than one drill processes, a list of available processes will appear when the drill-through process is executed.

A TM1 drill-through process can drill to a cube view, an Open DataBase Connectivity (ODBC) data source or any other data source supported by TurboIntegrator.

When drilling to an external source, you must have the connection information and any SQL or queries formatted properly for the source data. You may wish to test the ODBC connection and/or SQL statements outside of TM1 to ensure that they are valid for your data source.

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Create a drill path

The drill process defines the detailed data you want to associate with a cell.

- It populates parameters with elements from each dimension in the originating cube which may be used to limit the resulting data.
- It specifies the data source and any connection information.

The drill assignment rule defines the relationship between the cell and the detailed data.

- A string rule provides a list of drill processes.
- Place the list of processes in single quotes, separated by commas.

It is a good idea to drill through to relational data sources only from leaf-level cells. If you drill from higher level cells, the relational query could be quite large and complicated due to differences in the data structures. Use view-to-view drill through for higher level drill-through.

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IBM Training

Demonstration 1

Create a drill-through path to a view in another cube

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Demonstration 1: Create a drill-through path to a view in another cube

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Demonstration 1: Create a drill-through path to a view in another cube

Purpose:
You need to look at additional Price and Cost information when reviewing the Unit Sale Price in the Sales_Plan cube. You will create a drill path from the Sales_Plan cube to go to the default view of the Price_and_Cost cube.

Task 1. Create a public default view of the Price_and_Cost cube.

1. Open **Windows Explorer** and go to **C:\Program Files\IBM\cognos\tm1_64\bin**.

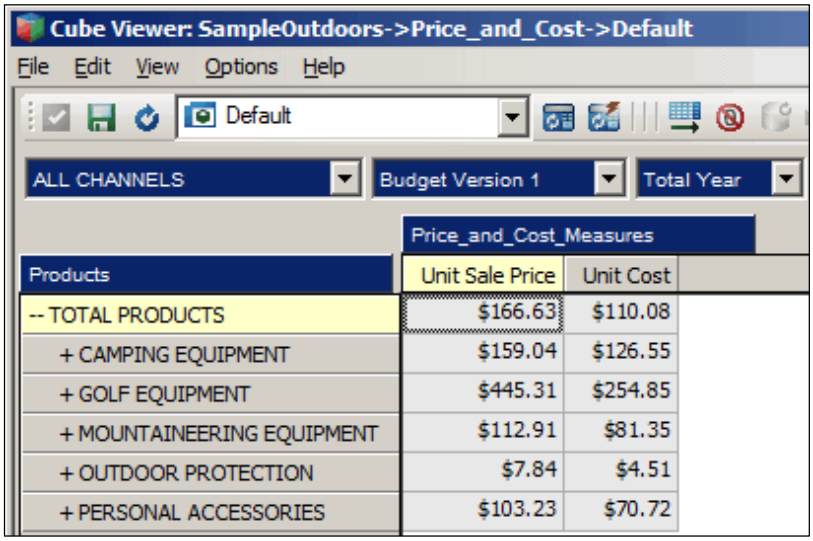
   Since you will be working with the Advanced Rules Editor later in this demonstration, you need to open the 32-bit version of Architect.
2. Double-click **tm1a.exe**.
3. Maximize the **Architect** window.
4. Expand **TM1**, double-click **SampleOutdoors**, and then log on with a user ID of **admin** and a password of **apple**.
5. Expand **Cubes**.
6. Double-click the **Price_and_Cost** cube.

   You will create a view to drill to and then map to it later.
7. Swap the **Price_and_Cost_Measures** dimension (currently displaying **Unit Sale Price**) with the **Months** dimension (currently displaying the **Months_Only** subset).
8. In the **Channels** dimension, select **ALL CHANNELS**, and then in the **Versions** dimension, select **Budget Version 1** (if necessary).
9. Double-click the **Months** dimension to open the Subset Editor.
10. Click **All**, select **Total Year**, and then click **OK**.
11. Click the **Price_and_Cost_Measures** dimension on columns, Ctrl+click the **Unit Sale Price** and **Unit Cost** elements, and then click **OK**.

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12. Click Recalculate.

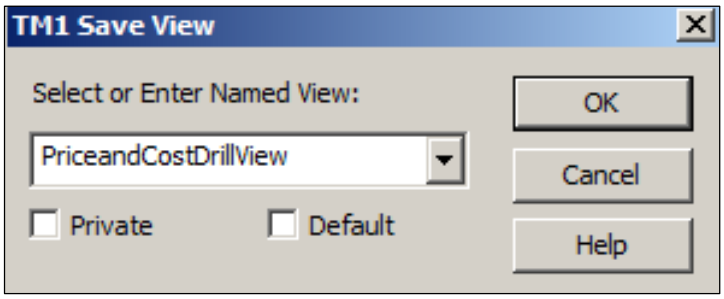
    The result appears as follows:



13. Click **File** and then click **Save As**.
14. In the **Select or Enter Named View** box, type **PriceandCostDrillView**, and then clear the **Private** check box.

    You need a public view for drill-through.

    The result appears as follows:



15. Click **OK**, and then close the view.

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Task 2. Create the drill process.

1. In Server Explorer, right-click the **Sales_Plan** cube, point to **Drill**, and then click **Create Drill Process**.

   The Drill Process Setup wizard appears.

   Ensure that your Parameter Values match the results below:

Parameter Name	Parameter Value
cubename	Sales_Plan
Subsidiaries	TOTAL COMPANY
Channels	ALL CHANNELS
Products	TOTAL PRODUCTS
Months	Total Year
Versions	Budget Version 1
Sales_Plan_Measure	Quantity

   If they do not, click the parameter value that you want to change and then click the **Select Element** button to open the Subset Editor window and choose the correct element.
2. Click **Next**.
3. Click **Cube View**, and then click **Browse**.
4. Under **Select Cube**, click **SampleOutdoors:Price_and_Cost**, and then under **Select View**, click **PriceandCostDrillView**.
5. Click **OK**, and then click **Finish**.
6. In the **Name** box, type **SalesPlanToPriceandCost**, and then click **OK**.

Task 3. Create a drill assignment rule.

1. In Server Explorer, right-click the **Sales_Plan** cube, point to **Drill**, and then click **Create Drill Assignment Rules**.
2. The Rules Editor window opens (this may take a few moments).
3. Click the **Brackets** and then double-click **Sales_Plan_Measures**.
4. Click **Unit Sale Price** and then click **OK**.

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5. To the right of the bracket, type **= S: 'SalesPlanToPriceandCost';**

   This indicates the drill process that will be run when drilling from any cell in the Unit Sale Price dimension.

   The result appears as follows:

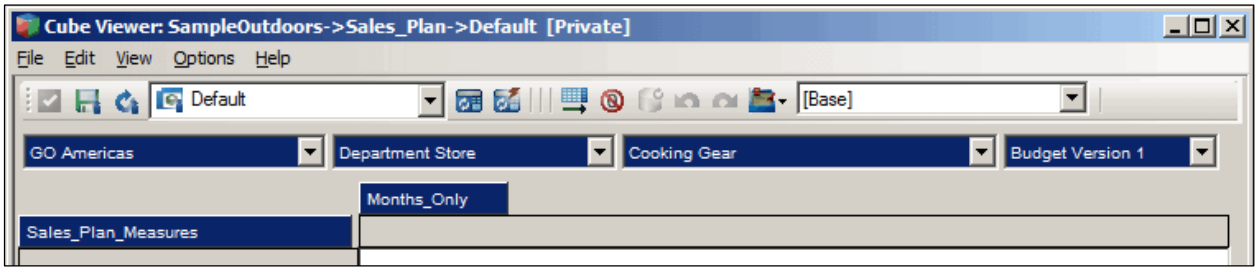
```
1   ['Unit Sale Price' ]  = S: 'SalesPlanToPriceandCost';
```

6. Click **Save**, and then close the Rules Editor.

   When you create a drill assignment rule, it places the rule on a control cube called }CubeDrill_Sales_Plan (}CubeDrill_*cubename*).

Task 4. Drill to the Price_and_Cost cube.

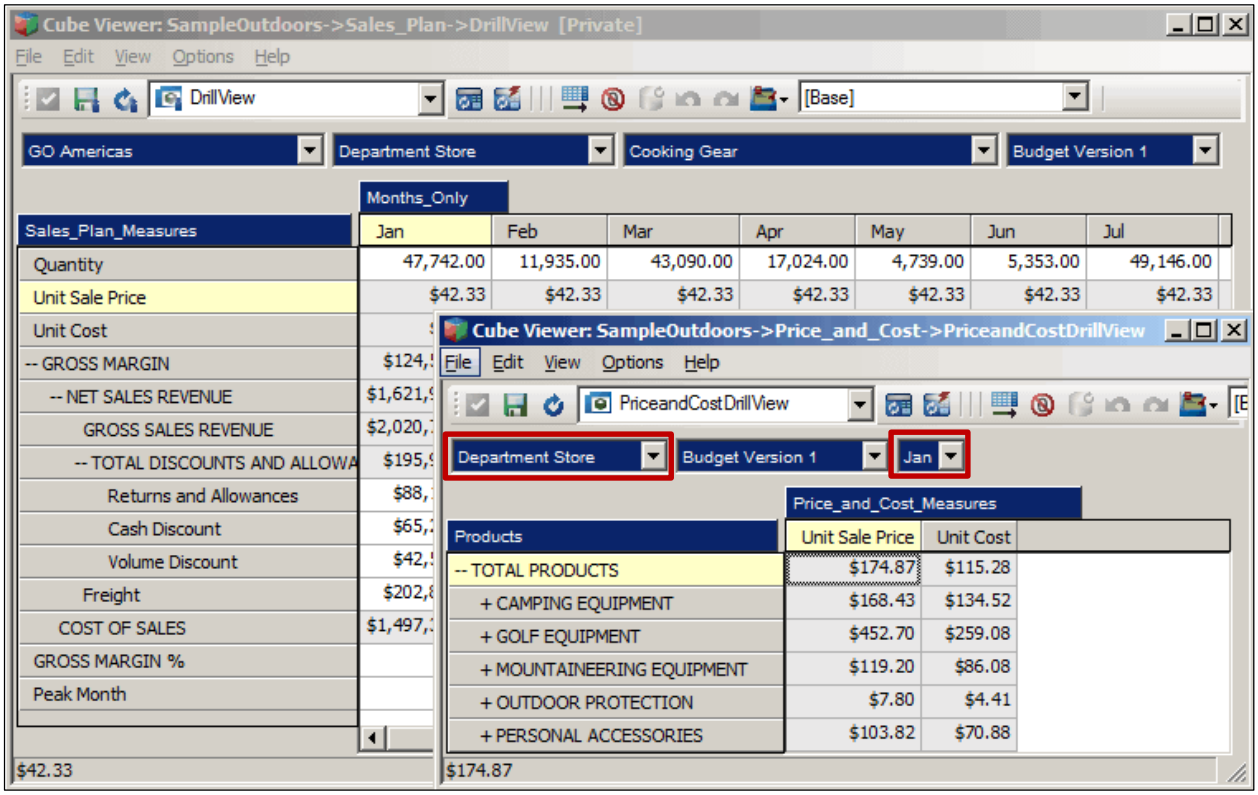
1. In Server Explorer, double-click the **Sales_Plan** cube.
2. Double-click **Subsidiaries** (currently displaying **TOTAL COMPANY**) to open the **Subset Editor**.
3. Expand **GO AMERICAS REGION**, select **GO Americas**, and then click **OK**.
4. Under **Channels**, click **Department Store**.
5. Under **Products**, click **Cooking Gear**.
6. Under **Versions**, click **Budget Version 1** (if necessary).
7. Ensure that the dimensions are arranged as follows:



8. Click **Recalculate**, and then save the view as **DrillView**.

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9. Right-click **Unit Sale Price** for **Jan**, and then click **Drill**.
   The result appears as follows:

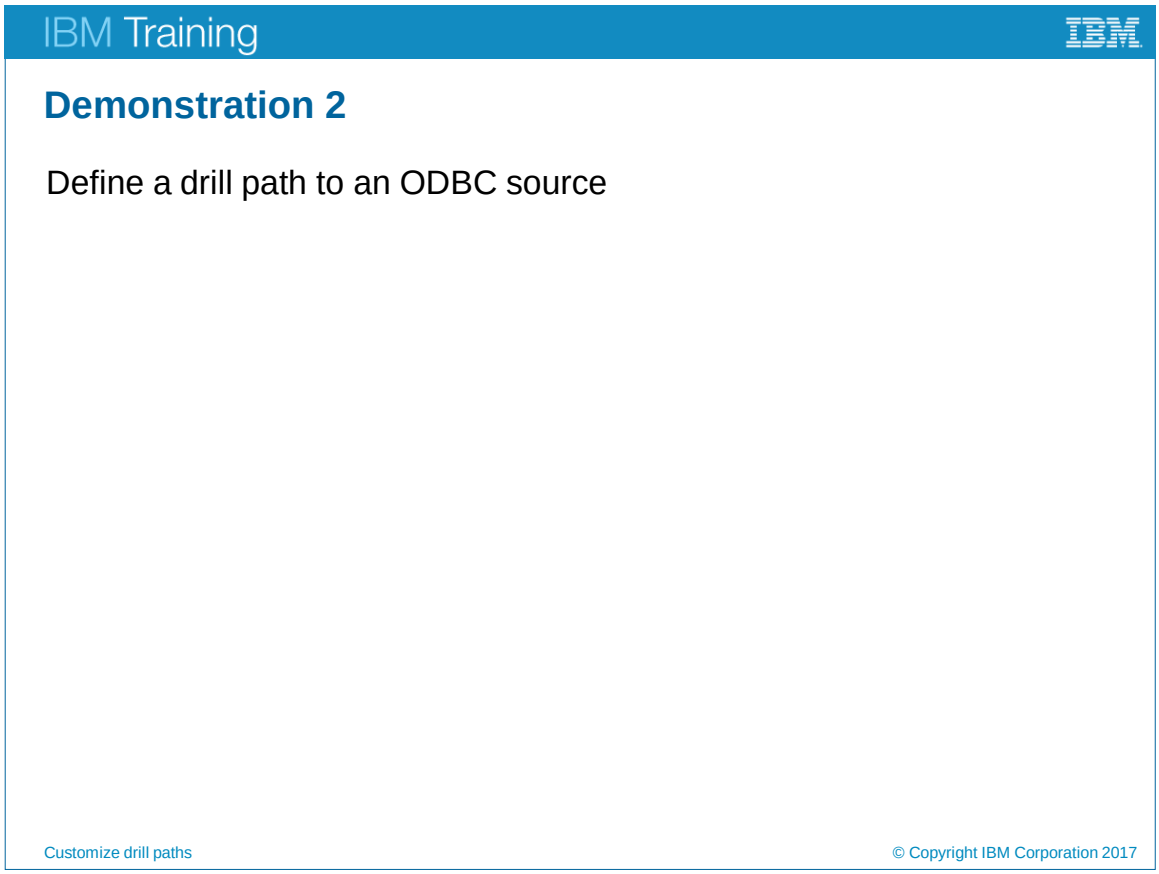


   Here you can see the Department Store and Jan data was pulled over from the Sales_Plan cube. The original PriceandCostDrillView had ALL CHANNELS and Total Year in the title.

10. Close the **PriceandCostDrillView**.
11. In the **Sales_Plan** cube, right-click in **Quantity for January**.
    You will not be able to click Drill because you have not made your drill process available to elements other than the Unit Sale Price.
12. Close **Cube Viewer** without saving changes.
13. Right-click **SampleOutdoors** and click **Save Data**.
14. Leave **Architect** open for the next demonstration.

Results:
You needed to look at additional Price and Cost information when reviewing the Unit Sale Price in the Sales_Plan cube. You created a drill path from the Sales_Plan cube to go to the default view of the Price_and_Cost cube.

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Demonstration 2: Define a drill path to an ODBC source

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Demonstration 2: Define a drill path to an ODBC source

Purpose:
In addition to looking at the data in the Price_and_Cost cube, you will need to review the source data. You will create another drill path that will be available to users at the leaf level elements to review data in the ODBC source from which you loaded the data. You will use the same ODBC connection that you created to load the data. This process will only be available to the leaf elements in the cube. You will use a function to determine the leaf elements of the cube. You will then modify the drill assignment rule so that the drill process can return data for consolidated elements as well.

Task 1. Create the drill process.

1. Right-click the **Sales_Plan** cube, point to **Drill**, and then click **Create Drill Process**.
2. In the **Drill Process Setup Wizard**, click **Next**.
3. Ensure that **ODBC** is selected, and then click **Browse**.
4. In the **ODBC Data Sources** list, click **Sales_Plan**, and then click **OK**.
5. Click **Finish**.
6. In the **Name** box, type **SalesPlanToSalesPlanXLS**, and then click **OK**.

Task 2. Edit the drill process.

1. Right-click the **Sales_Plan** cube, point to **Drill**, and then click **Edit Drill Process**.
2. In the **Select** list, click **salesplantosalesplanxls**, and then click **OK**.
3. Click **OK** to dismiss the error message related to the SQL statement.

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4. In the **Query** box, type:

   SELECT * FROM Sales
   WHERE Subsidiaries = '?Subsidiaries?'
   AND Channel = '?Channels?'
   AND "Product Type" = '?Products?'
   AND Version = '?Versions?'
   AND MONTH("Order Date") = '?Months?'

   In the query, the column names to the left of the **=** refer to the columns in the data source. The references to the right (enclosed in single quotes and question marks) refer to the TM1 dimension names in the cube.

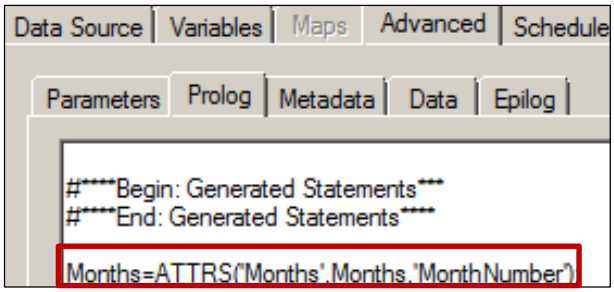
   The last line of this query extracts the month number from the source data. When the drill process runs, it will return only the data for the selected month instead of all the data.

   Since the "Month" value is a number from 1 to 12, you would need to have the MonthNumber alias turned on for the Months dimension in the Cube Viewer. Otherwise, the drill would not return any data. You will add code to the Prolog sub-tab of this process so that aliases do not have to be enabled for the drill to return data.

5. Click the **Advanced** tab, click the **Prolog** sub-tab, and then type the following under the Generated Statements:

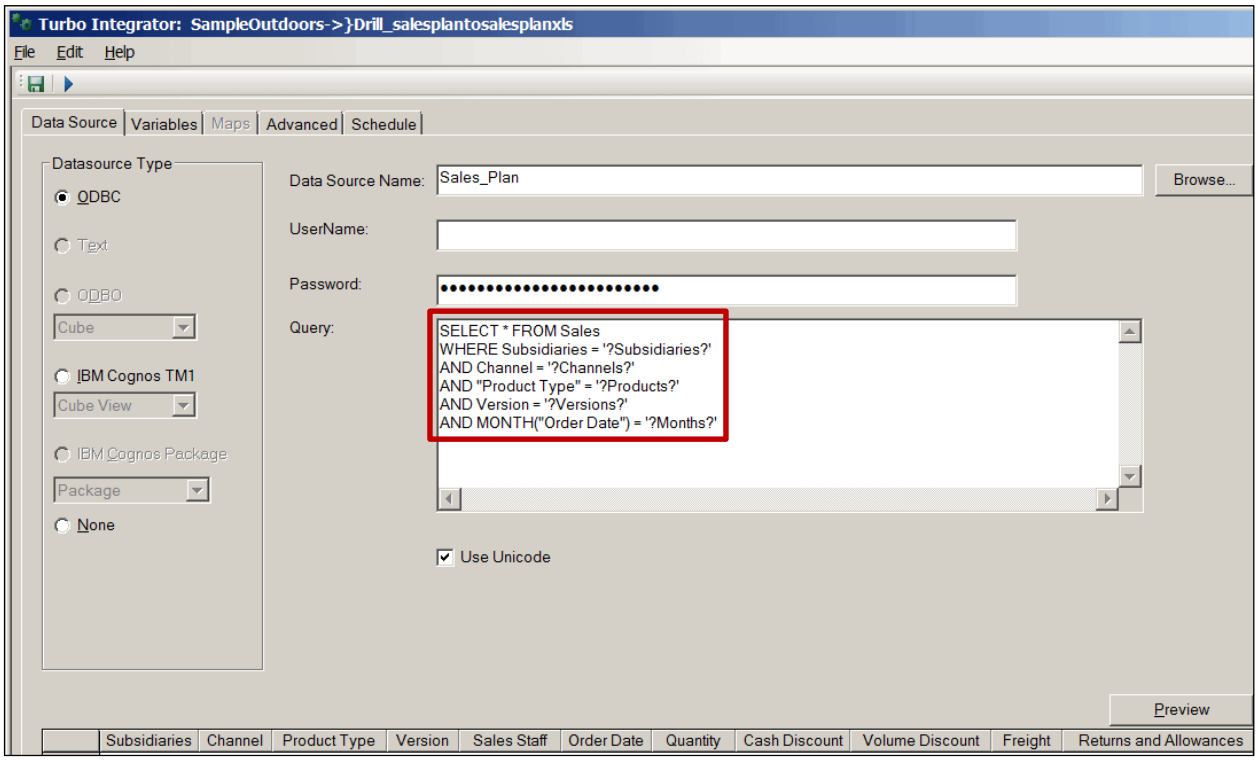
   Months=ATTRS('Months',Months,'MonthNumber');

   The results appear as follows:

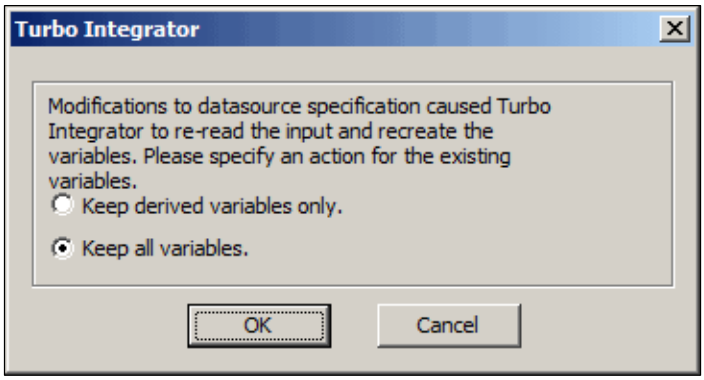


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6. Click the **Data Source** tab, and then click **Preview**.
   You will not see any results other than the column names from the data source.
   The result appears as follows:



7. Click **Save**.
   The results appear as follows:



8. Ensure that **Keep all variables** is selected, and then click **OK**.
9. Close **TurboIntegrator**.

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Task 3. Edit the drill assignment rule.

1. Right-click the **Sales_Plan** cube, point to **Drill**, and then click **Edit Drill Assignment Rules**.
2. Place your cursor at the end of the existing rule, and then press **Enter** twice.
3. On **row 3** type **[] = S:IF (ISLEAF=1, 'SalesPlanToSalesPlanXLS', '');**.

   The results appear as follows:

```
['Unit Sale Price' ]  = S: 'SalesPlanToPriceandCost';


[]  = S: IF(ISLEAF=1, 'SalesPlanToSalesPlanXLS', '');
```

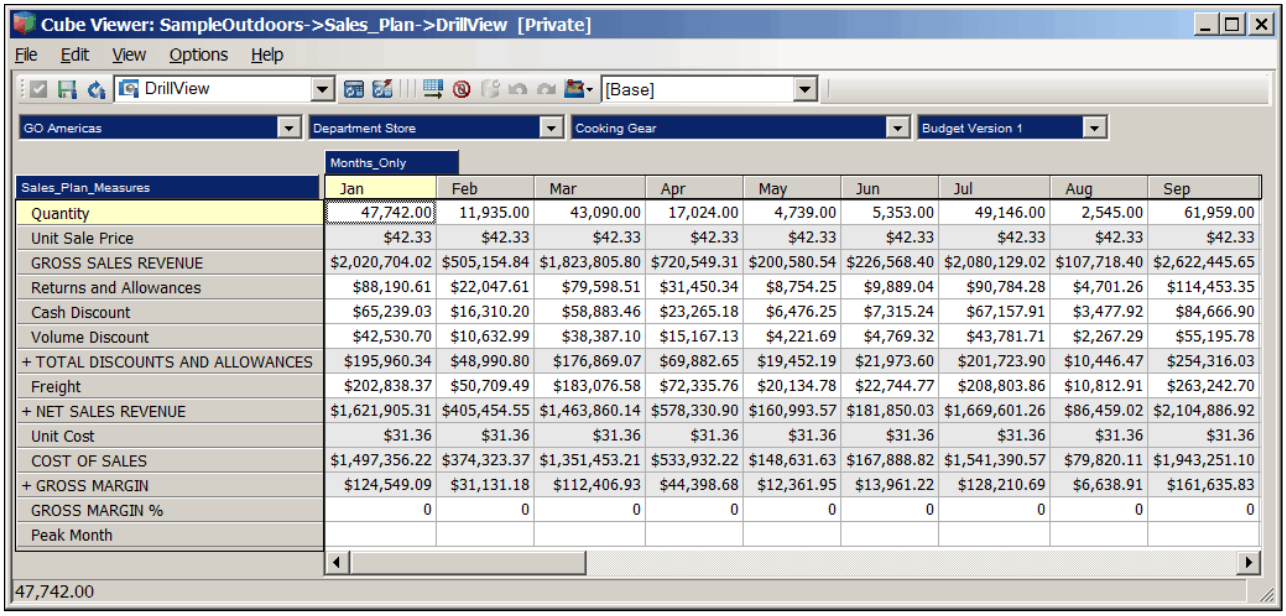
   Note: Enter two single quotes before the closing parenthesis, not a double quote.

4. Click **Save** and close the editor.

Task 4. Drill to the ODBC source.

1. Open the **DrillView** view in Cube Viewer.

   The results appear as follows:

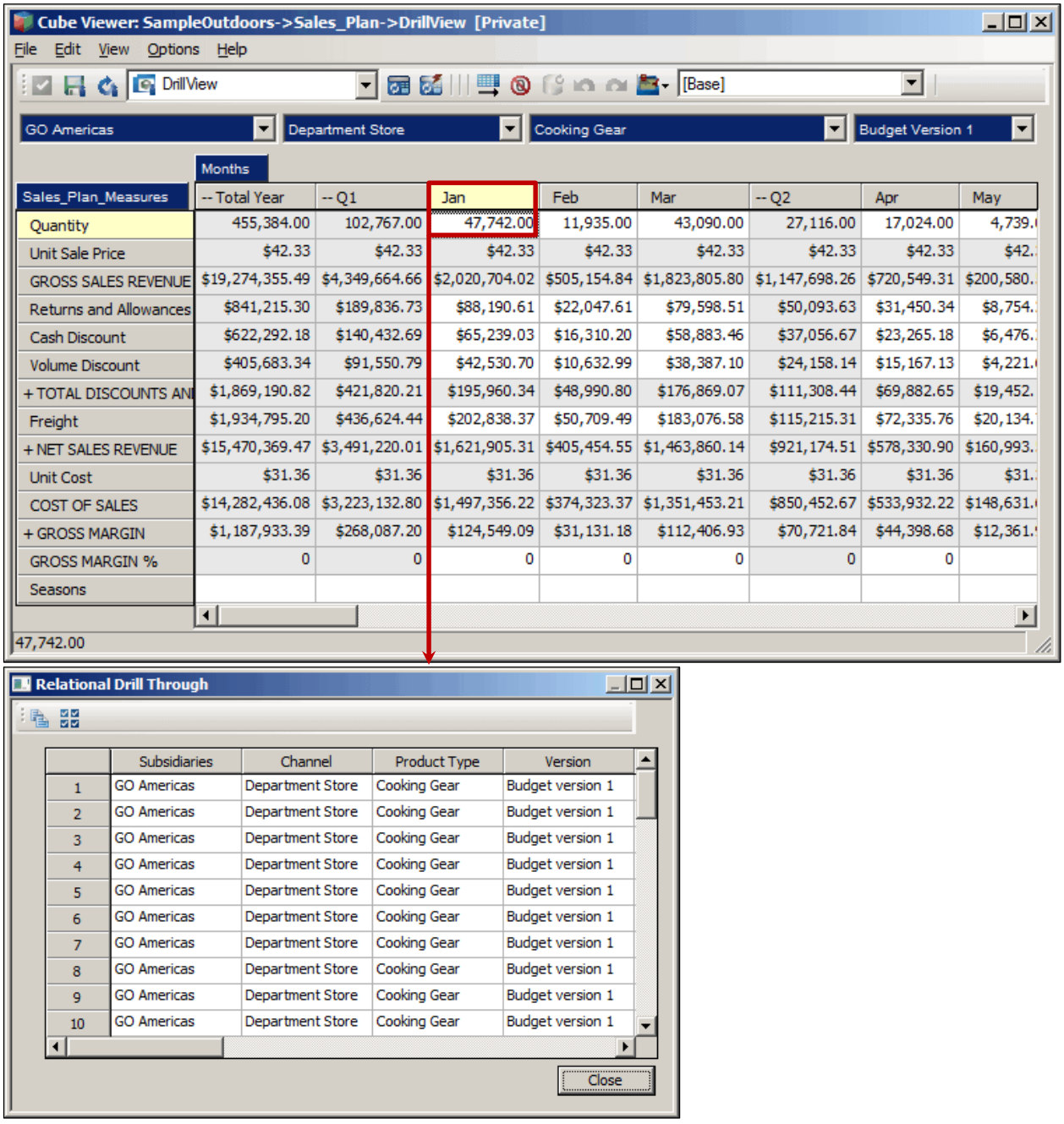


Sales_Plan_Measures	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
Quantity	47,742.00	11,935.00	43,090.00	17,024.00	4,739.00	5,353.00	49,146.00	2,545.00	61,959.00
Unit Sale Price	$42.33	$42.33	$42.33	$42.33	$42.33	$42.33	$42.33	$42.33	$42.33
GROSS SALES REVENUE	$2,020,704.02	$505,154.84	$1,823,805.80	$720,549.31	$200,580.54	$226,568.40	$2,080,129.02	$107,718.40	$2,622,445.65
Returns and Allowances	$88,190.61	$22,047.61	$79,598.51	$31,450.34	$8,754.25	$9,889.04	$90,784.28	$4,701.26	$114,453.35
Cash Discount	$65,239.03	$16,310.20	$58,883.46	$23,265.18	$6,476.25	$7,315.24	$67,157.91	$3,477.92	$84,666.90
Volume Discount	$42,530.70	$10,632.99	$38,387.10	$15,167.13	$4,221.69	$4,769.32	$43,781.71	$2,267.29	$55,195.78
+ TOTAL DISCOUNTS AND ALLOWANCES	$195,960.34	$48,990.80	$176,869.07	$69,882.65	$19,452.19	$21,973.60	$201,723.90	$10,446.47	$254,316.03
Freight	$202,838.37	$50,709.49	$183,076.58	$72,335.76	$20,134.78	$22,744.77	$208,803.86	$10,812.91	$263,242.70
+ NET SALES REVENUE	$1,621,905.31	$405,454.55	$1,463,860.14	$578,330.90	$160,993.57	$181,850.03	$1,669,601.26	$86,459.02	$2,104,886.92
Unit Cost	$31.36	$31.36	$31.36	$31.36	$31.36	$31.36	$31.36	$31.36	$31.36
COST OF SALES	$1,497,356.22	$374,323.37	$1,351,453.21	$533,932.22	$148,631.63	$167,888.82	$1,541,390.57	$79,820.11	$1,943,251.10
+ GROSS MARGIN	$124,549.09	$31,131.18	$112,406.93	$44,398.68	$12,361.95	$13,961.22	$128,210.69	$6,638.91	$161,635.83
GROSS MARGIN %	0	0	0	0	0	0	0	0	0
Peak Month									

2. Click **Months_Only** (on the columns) to open the **Subset Editor**.
3. Click **All**, expand all four quarters and then click **OK**.
4. Click **Recalculate**.

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5. Right-click **Quantity** for **Jan** and then click **Drill**.
The result appears as follows:



The Relational Drill Through window shows the data in the ODBC data source that you specified as your drill-through target.

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6. Scroll to the right until the **Order Date** column is visible.

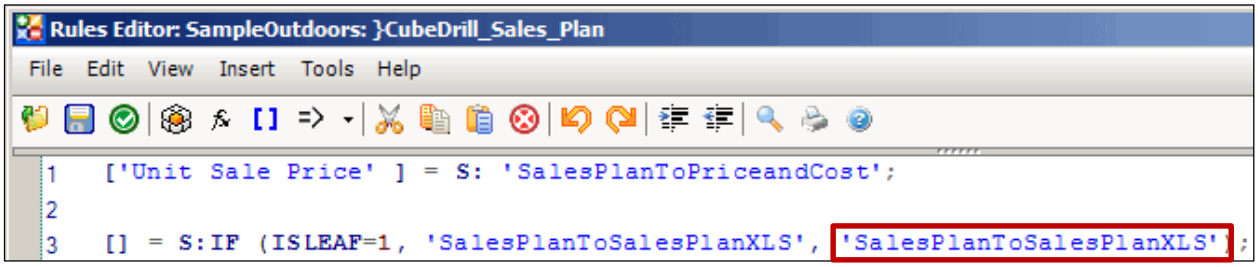
   You can see that the data being returned is from the selected month (January). If you were to drill from the intersection of Feb and Quantity, you would only see February-related data. In the drill process, you specified that you only wanted to see data from the selected month, not all of the data in the underlying source file.
7. Click **Close** to close the **Relational Drill Through** window.
8. Right-click **Unit Sale Price** for **Jan** and then click **Drill**.

   The default drill view (PriceandCostDrillView) of the Price_and_Cost cube appears. This is because the more specific rule (which applies only to the Unit Sale Price element) is on the first line of the drill assignment rules.
9. Right-click **Quantity** for **Q1**.

   The Drill menu item is not available. This is because you used the ISLEAF function in the rule to only assign the drill rule to leaf level elements. You will now modify the rule so that you can drill from consolidated elements as well as leaf level elements.
10. Leave the **Cube Viewer** window open for the next task.

Task 5. Modify the drill assignment rule so that you can drill from consolidated elements in the Sales_Plan cube.

1. In **Server Explorer**, right-click the **Sales_Plan** cube, point to **Drill**, and then click **Edit Drill Assignment Rules**.
2. Modify the rule on **line 3** as shown below:



   The rule now specifies that the SalesPlanToSalesPlanXLS process will be run regardless of whether a leaf level cell is selected or not.
3. Save and close the **Rules Editor**.
4. In the **Sales_Plan** cube, click **Recalculate**.
5. Right-click the intersection of **Q1** and **Quantity**.

   The Drill menu item is now available.

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6. Click **Drill**.

   The Relational Drill Through window opens. However, it is empty because the source file does not include quarter-related data. The file only contains individual orders made on a day-by-day basis. The Order Date column in this file does not specify the quarter in which each order was made.
7. Close all **Cube Viewer** windows without saving.
8. Right-click **SampleOutdoors** and click **Save Data**.
9. Leave **Architect** open for the upcoming exercise.

Results:
You created a drill process to display relational data from an ODBC source when requested from a leaf level element in a cube. You then modified the process so that the drill can be run from a consolidated element.

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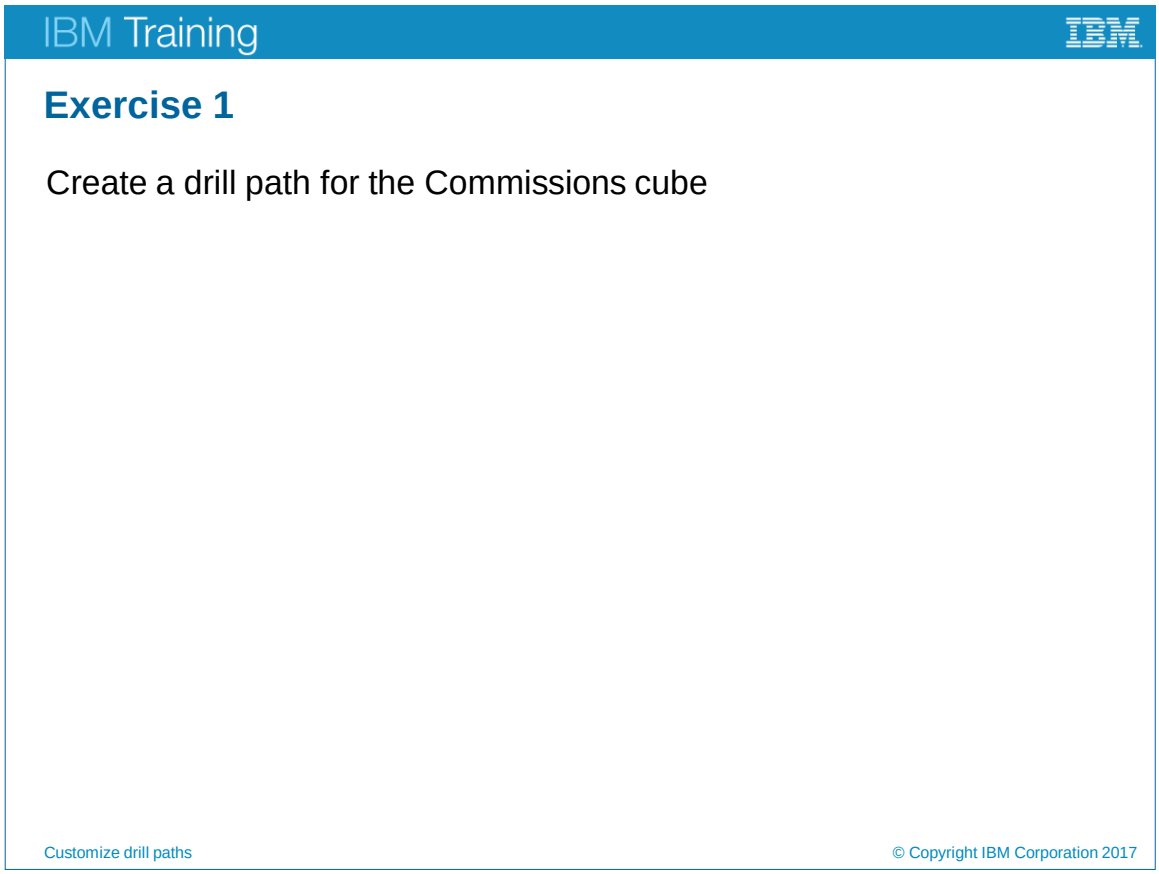
Unit summary

- View related data
- Create a drill process
- Create a drill assignment rule
- Edit a drill process

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Unit summary

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Exercise 1: Create a drill path for the Commissions cube

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Exercise 1: Create a drill path for the Commissions cube

Your users have expressed a desire to confirm data in the Sales_Plan default view as they review Commissions. You will create a public default view of the Sales_Plan cube and make it available to the Commissions cube. You will then drill from the Commissions cube to the Sales_Plan cube.

To accomplish this:

- Create a public default view of the Sales_Plan cube. In the title area, select the total company, all channels, total products, and Budget Version 1. On the rows, display all sales plan measures. On the columns, display months only.
- Create a drill process from the Commissions cube to open the default Sales_Plan view.
- Create a drill assignment rule from Commissions that is available at the leaf level cells only.
- Create a view of the Commissions cube that you can use to test the drill process.

For more information about where to work and the exercise results, refer to the Tasks and results section that follows. If you need more information to complete a task, refer to earlier demonstrations for detailed steps.

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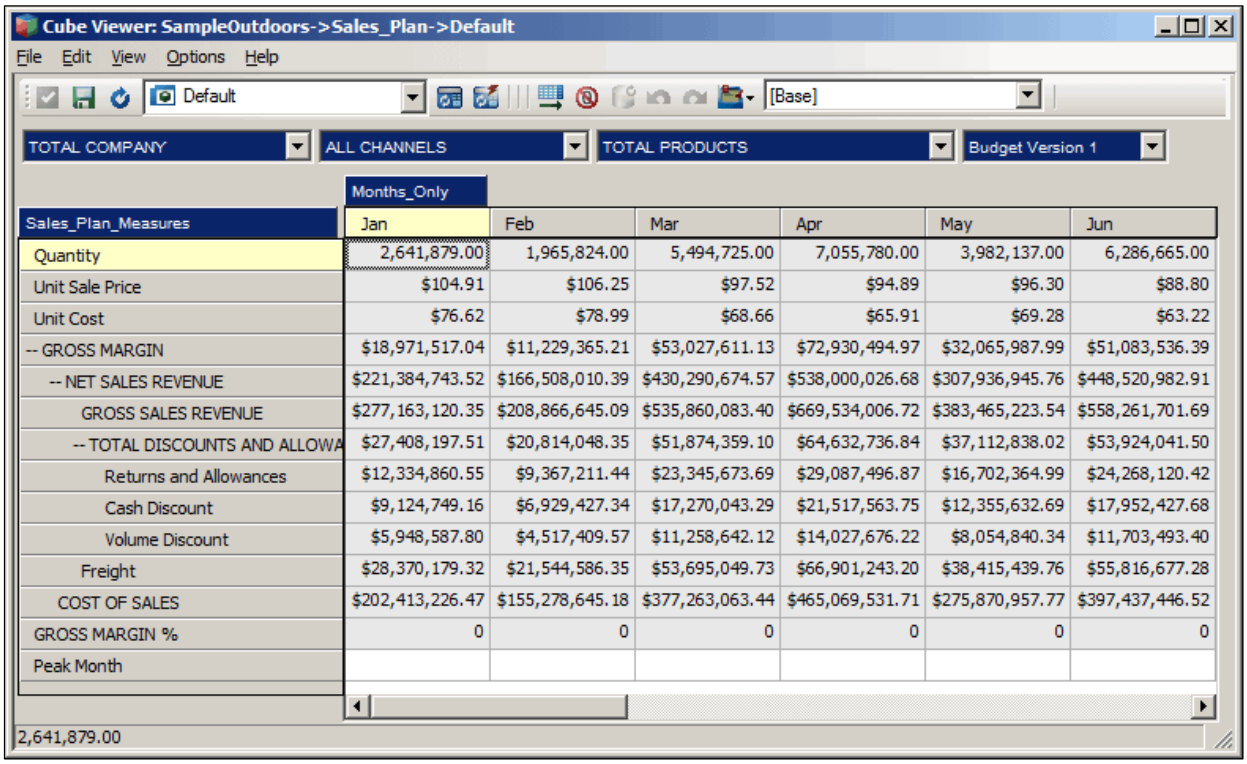
Exercise 1: Tasks and results

Task 1. Create a default view of the Sales_Plan cube.

- Double-click the **Sales_Plan** cube.
- Ensure that the title area has **TOTAL COMPANY**, **ALL CHANNELS**, **TOTAL PRODUCTS**, **Budget Version 1**.
- If necessary, click **Sales_Plan_Measures**, click **All**, and then click **OK**.
- Recalculate the cube, and then from the **File** menu, click **Save As**.
- Click **Default** and then clear the **Private** check box.

  This saves the current view as the public Default view.
- Click **OK**.

  The result appears as follows:



- Close the view.

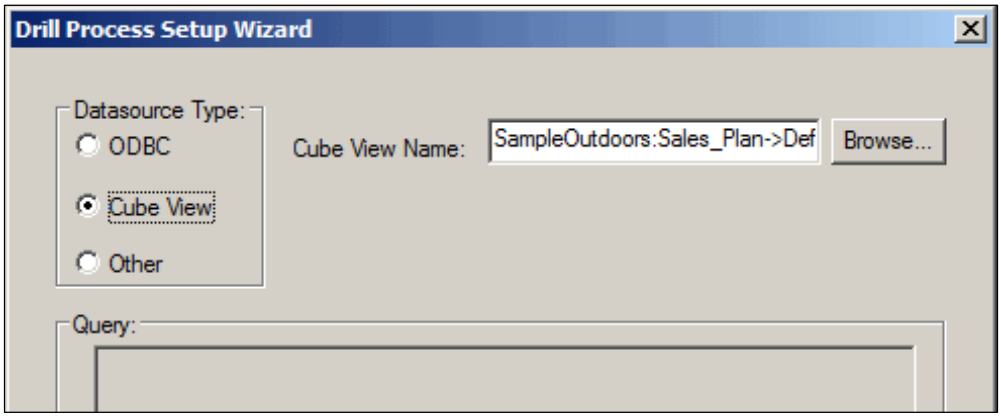
Task 2. Create a drill process to drill to the Sales_Plan default view from the Commissions cube.

- Right-click the **Commissions** cube, point to **Drill**, and then click **Create Drill Process**.

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- Click **Next**.
- Click **Cube View**, and then click **Browse**.
- Click **SampleOutdoors: Sales_Plan**, click **Default**, and then click **OK**.

  The result appears as follows:



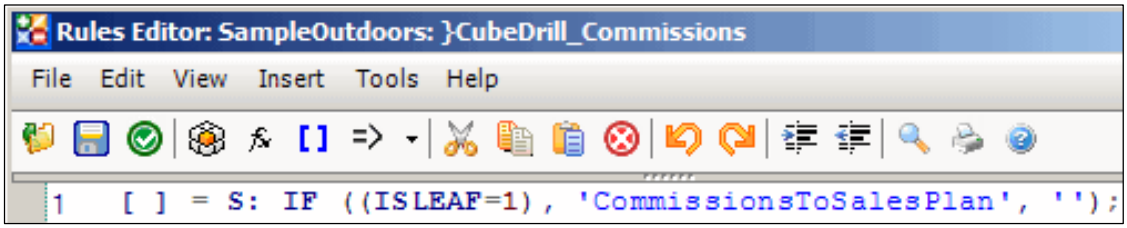
- Click **Finish**.
- Name the process **CommissionsToSalesPlan**, and then click **OK**.

Task 3. Create a drill assignment rule.

- Right-click the **Commissions** cube, point to **Drill**, and then click **Create Drill Assignment Rules**.
- Type the following on **line 1**:

  [ ] = S: IF ((ISLEAF=1), 'CommissionsToSalesPlan', '');

  Enter two single quotes before the closing parenthesis, not a double quote:



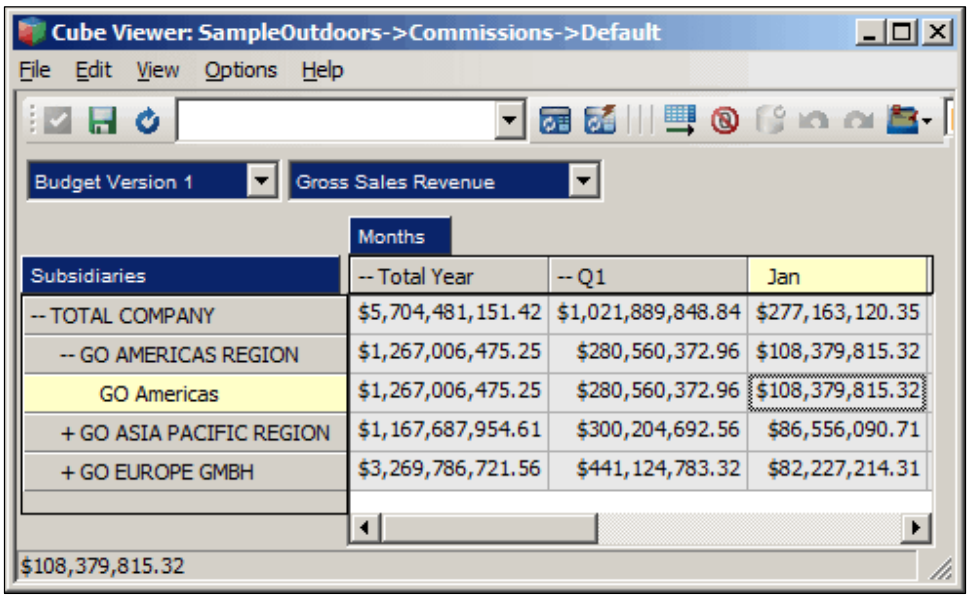
- Save the rule and close the **Rules Editor**.

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Task 4. Create a view to test the drill.

- Double-click the **Commissions** cube.
- Swap the **Commission_Measures** and **Months** dimensions so that Months is on columns.
- Swap the **Versions** and **Subsidiaries** dimensions so that Subsidiaries is on rows.
- In the title area, ensure that **Budget Version 1** is selected from the **Versions** dimension and that **Gross Sales Revenue** is selected from the **Commission Measures** dimension.
- Click **Recalculate**, and then expand **GO AMERICAS REGION** and **Q1**.

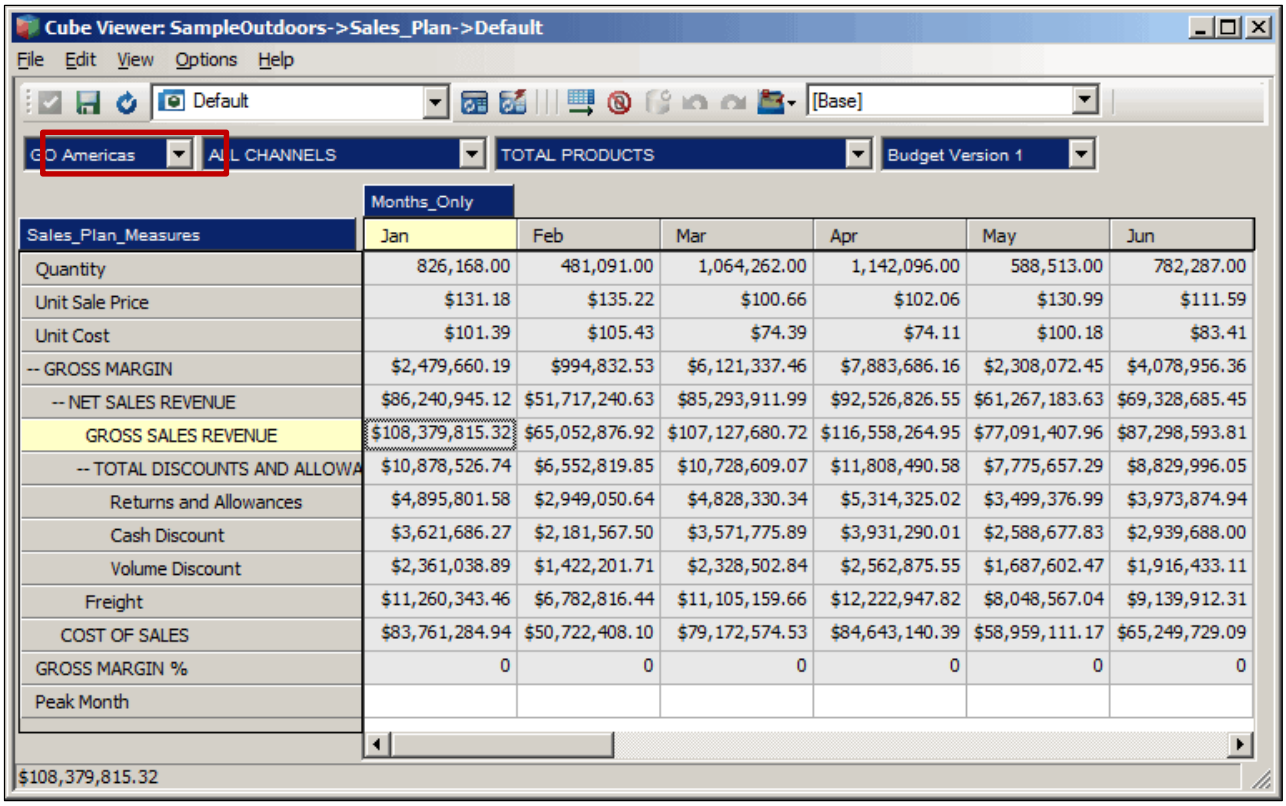
  The result appears as follows:



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- Right-click **GO Americas** for **Jan** and click **Drill**.

  The result appears as follows:



- In **Server Explorer**, right-click **SampleOutdoors** and click **Save Data**.
- Close all open windows without saving.

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IBM Training

Using rules for advanced modeling

IBM Planning Analytics (v2.0)

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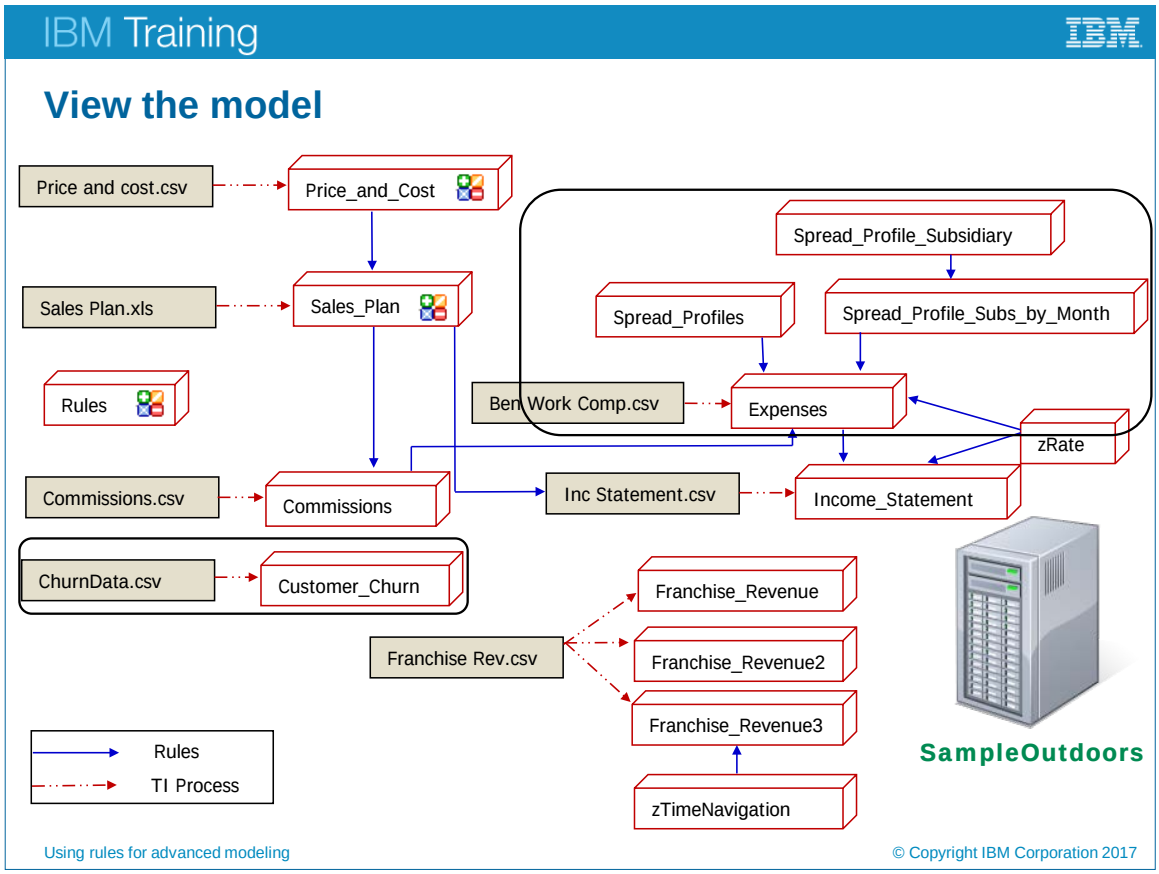
Unit objectives

- At the end of this module, you should be able to:
  - describe virtual cubes
  - list uses for lookup cubes
  - create and use a spread profiles cube
  - implement moving balances in a cube

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Unit objectives

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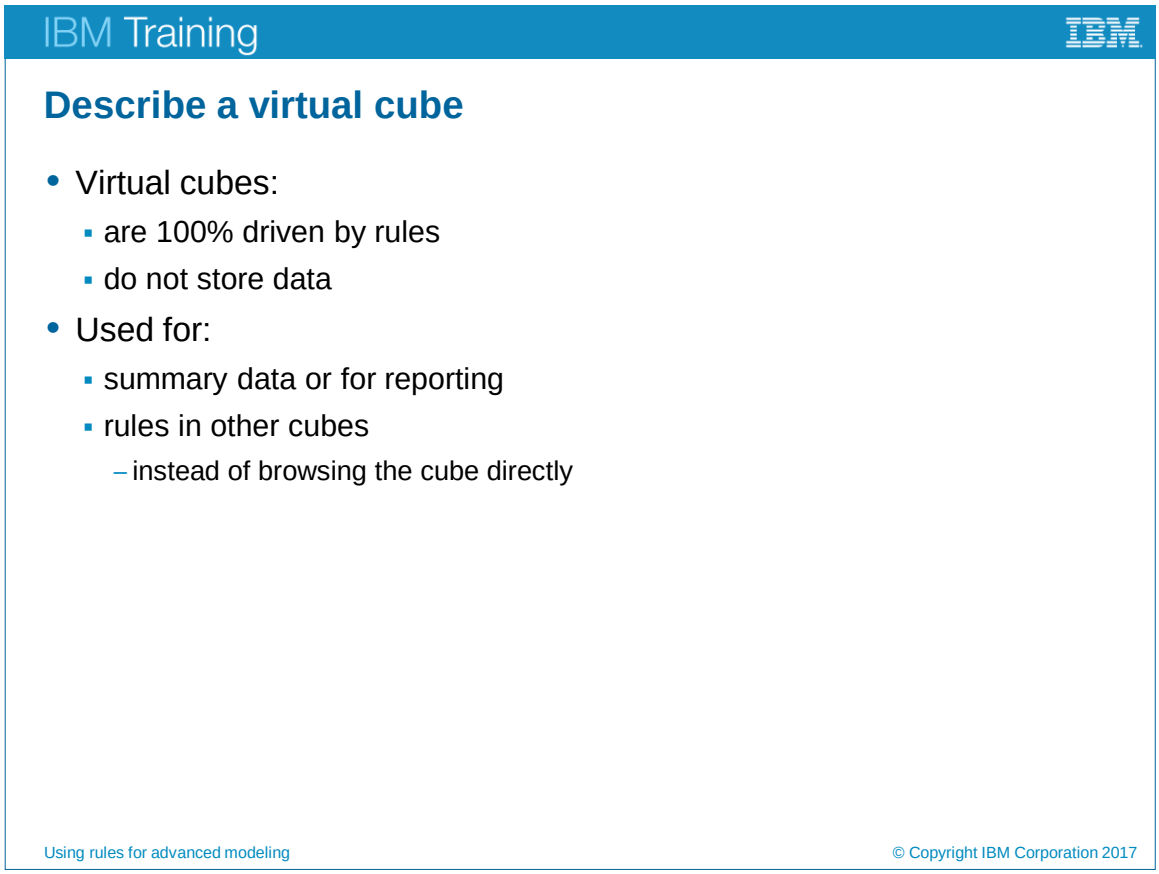


View the model

In this unit, you will create the following new cubes:

- Spread_Profiles
- Spread_Profile_Subsidiary
- Spread_Profile_Subs_by_Month
- Expenses
- Customer_Churn

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Describe a virtual cube

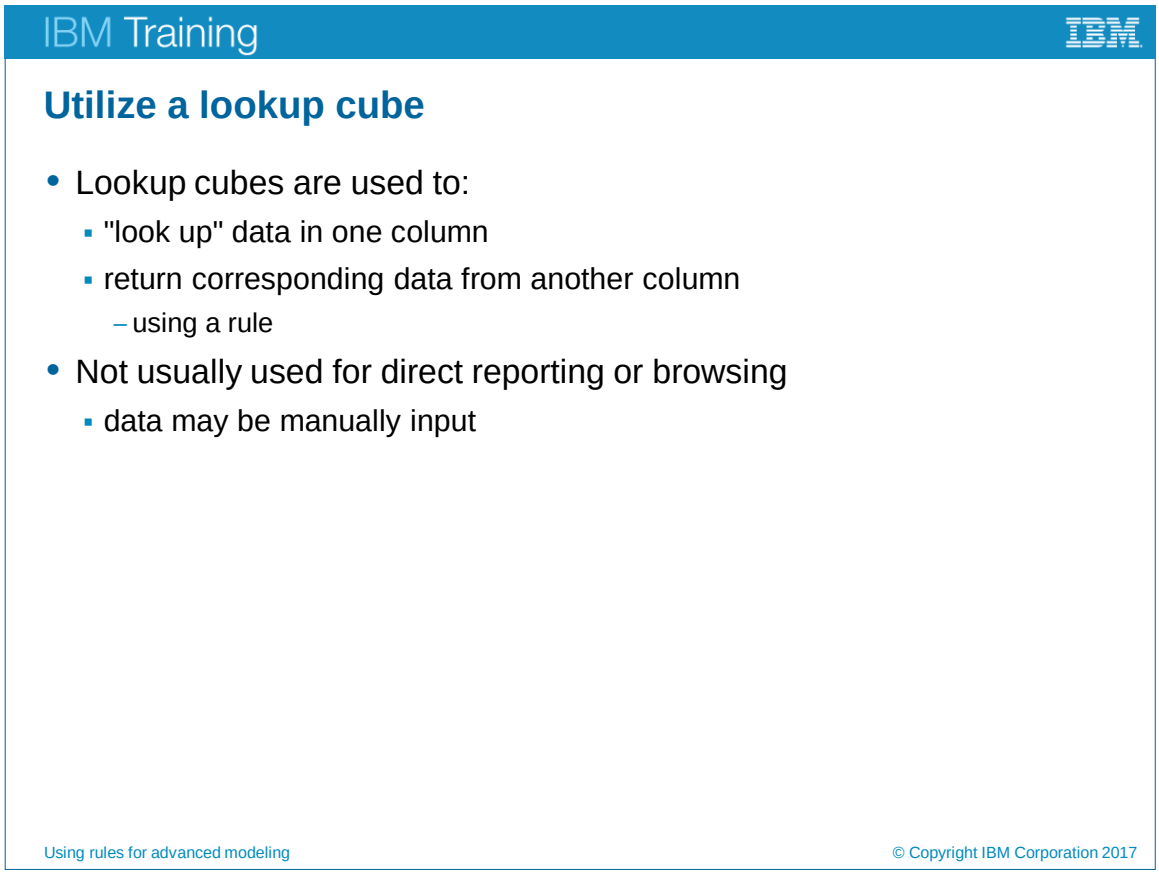
Virtual cubes are often given special names that identify them to the model developer.

You may wish to give the cube a prefix so it is sorted in the Server Explorer window with other virtual cubes.

Virtual cubes may be used simply to combine data from other cubes.

Another potential use for a virtual cube is to summarize data. You may have a large cube with a lot of data. If others need to see similar data at a higher level or minus a dimension you might use a rule to pull over the data in real time.

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Utilize a lookup cube

Lookup cubes may have a prefix such as:

- zCube
- lu_Moving_Balance

In Microsoft Excel, you might call them lookup tables.

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IBM Training

Use relative spreading

- Use spreading to distribute data among cells
  - writes data back to the cubes
  - data is inserted into the leaf-level cells
- Relative spread options
  - use other cubes' data patterns as a basis
  - must share a common dimension
    - between the source and target cubes

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Use relative spreading

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IBM Training

Use a spread profiles cube

- Create a lookup cube for the relative spread patterns.
- It should contain a dimension that also exists in the target cube.

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Use a spread profiles cube

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IBM Training

Demonstration 1

Create the Expenses cube

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Demonstration 1: Create the Expenses cube

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Demonstration 1: Create the Expenses cube

Purpose:
You need to create an Expenses cube. This cube will be populated in part by data from spreadsheets, rules, and user input. Users will need to spread data in order to project some of the Overhead expenses into different regions.

Task 1. Create the Expenses dimension.

1. Open **Windows Explorer** and go to **C:\Program Files\IBM\cognos\tm1_64\bin**.

   You will be working with the Advanced Rules Editor, so you will open the 32-bit version of Architect.
2. Double-click **tm1a.exe**.
3. Maximize the **Architect** window.
4. Expand the **TM1** node, double-click **SampleOutdoors**, and then log on to the server with a user ID of **admin** and a password of **apple**.
5. Create a new TurboIntegrator process.

   Source: **Text**

   Data Source Name: **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles\expenses.csv**
6. Click **OK** to close the warning and then click **Preview**.

   The results in the Preview pane should appear as follows:

V1	V2	V3
Salaries: direct	SALARIES	TOTAL COMPENSA
Salaries: indirect	SALARIES	TOTAL COMPENSA
Commission	SALARIES	TOTAL COMPENSA
Pension Plan %		
PENSION PLAN	BENEFITS	TOTAL COMPENSA
Social Security %		
SOCIAL SECURITY	BENEFITS	TOTAL COMPENSA
Health Insurance %		
HEALTH INSURAN(	BENEFITS	TOTAL COMPENSA
Workers Compensat	BENEFITS	TOTAL COMPENSA

7. Click the **Variables** tab.

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8. Rename the variables in the following order:
   - **vExpenses**
   - **vExpenseCategory**
   - **vTotalExpenses**
9. Assign the **Contents** as follows:
   - **vExpenses = Element**
   - **vExpenseCategory = Consolidation**
   - **vTotalExpenses = Consolidation**

   The result appears as follows:

	Variable Name	Variable Type	Sample Value	Contents
1	vExpenses	String	Salaries: direct	Element
2	vExpenseCategory	String	SALARIES	Consolidation
3	vTotalExpenses	String	TOTAL COMPENSATION	Consolidation

10. Click the **Maps** tab, and then click the **Dimensions** sub-tab.
11. In the **Dimension** column, type **Expenses**, and then click **By Input** to open the **Dimension Element Ordering** dialog box.
12. Click **Automatic**, click **Hierarchy**, and then click **OK** to close.

Cube | Dimensions | Data | Consolidations | Attributes

Element Variable	Sample Value	Dimension	Order In Cube	Action	Element Type	Element Order
vExpenses	Salaries: direct	Expenses		Create	Numeric	By Hierarchy

13. Click the **Consolidations** tab, and then in the first row under **Child Variable**, select **vExpenses**.
14. In the second row under **Child Variable** select **vExpenseCategory**.

    The result appears as follows:

Cube | Dimensions | Data | Consolidations | Attributes

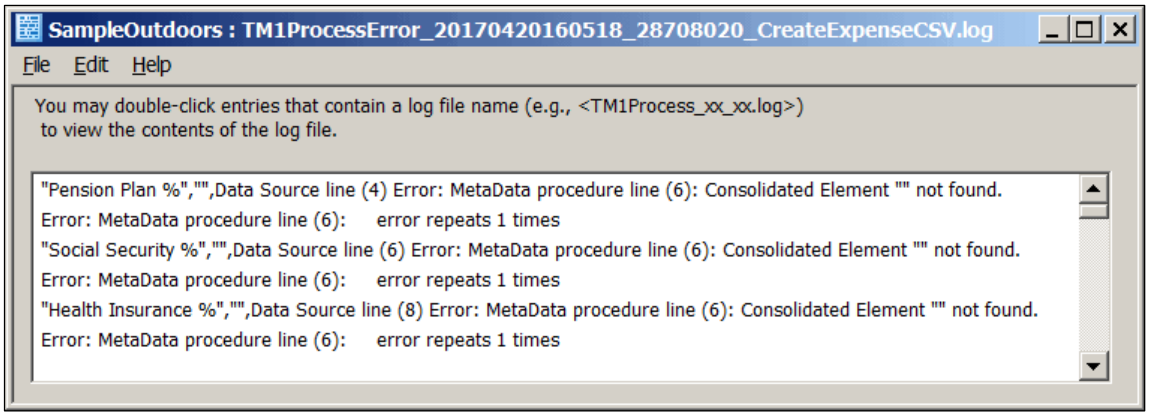
Cons. Variable	Dimension	Child Variable	Weight	Sample Value	Component Order
vExpenseCategory	Expenses	vExpenses	1.000000	SALARIES	By Input
vTotalExpenses	Expenses	vExpenseCategory	1.000000	TOTAL COMPENSATION	By Input

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15. Click the **Advanced** tab, and then click the **Prolog**, **Metadata**, **Data** and **Epilog** tabs.
16. Click **Save** and name the process **CreateExpenseCSV**.
17. Run the process.

    A message appears, indicating that the process completed with minor errors.
18. Click **Yes**, and then double-click the first line.

    This will open the log file and report that three elements were not able to load as consolidations:



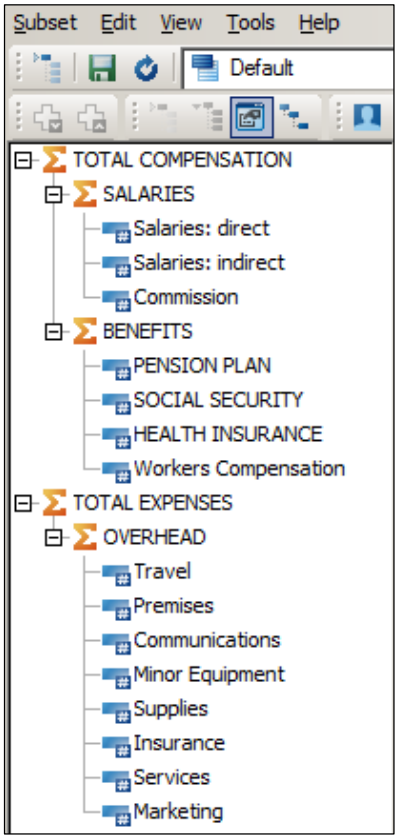
    This is fine as you will not be using these elements in your cube.
19. Close **TurboIntegrator** and the **Message Log** windows.
20. Double-click the **Expenses** dimension to open it in the **Subset Editor**.

    You will remove these elements from the subset but they will remain as part of the dimension. If you deleted them using the Dimension Editor, the elements would be permanently removed.
21. Select the **Pension Plan %**, **Social Security %**, and **Health Insurance %** elements, and then click **Delete**  .

    You are not removing these elements from the dimension, simply creating a default subset excluding these elements. They still exist in the dimension and will appear when the All button is clicked.

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22. Save this as the **public**, **Default** subset.
    The result appears as follows:



23. Close the **Subset Editor**.

Task 2. Manually create the Expense_Measures dimension.

1. In **Server Explorer**, right-click **Dimensions** and click **Create New Dimension**.
2. Insert two new simple elements, **Amount** and **Percent**.
   The result appears as follows:



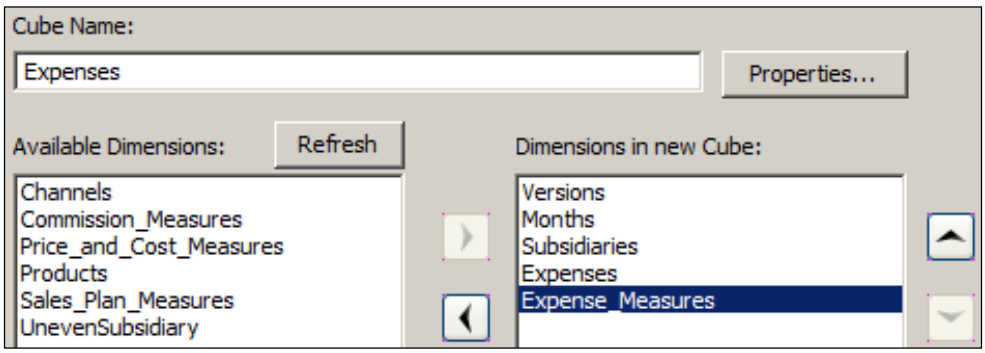
3. Save the dimension as **Expense_Measures** and close the **Dimension Editor**.

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Task 3. Create the Expenses cube.

1. In Server Explorer, right-click **Cubes**, and then click **Create new cube**.
2. In the **Cube Name** box type **Expenses**.
3. Double-click the following dimensions:
   - **Versions**
   - **Months**
   - **Subsidiaries**
   - **Expenses**
   - **Expense_Measures**

   The result appears as follows:



4. Click **Create Cube**.

   You should now see the Expenses cube under the Cubes node.

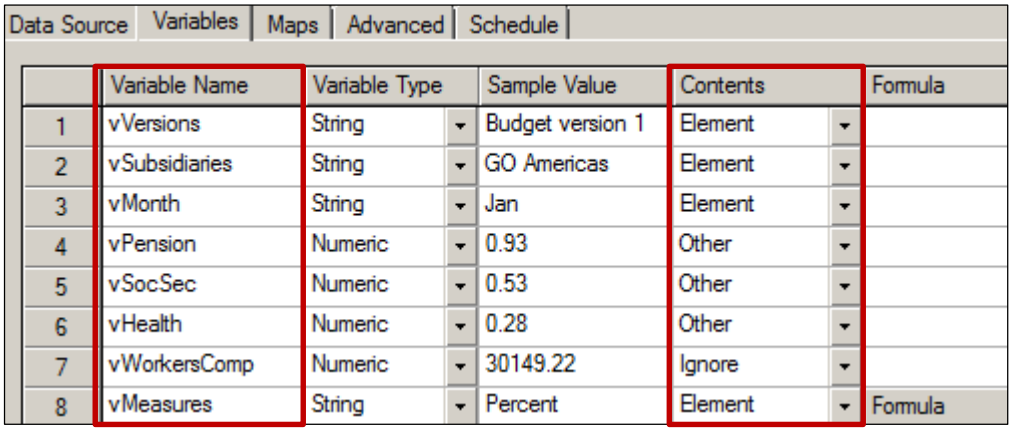
   The dimensions listed are not listed in the optimal order. After loading data into this cube, you may wish to use the Cube Optimizer utility. Simply right-click the cube and click Re-order Dimensions. Use the up and down arrows to internally reorder the dimensions and click Test to see if the Percent Change increases or decreases. This does not physically reorder the cube but it does remap dimensions internally so that functions and worksheets still work properly.

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Task 4. Create a process to load benefits data.

1. Create a new TurboIntegrator process, loading text data from **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles\ben_work_comp.csv** (click **OK** to dismiss the warning message).
2. Beside **Number of title records**, type **1**, and then click **Preview**.
3. Click the **Variables** tab.
4. Create a new variable called **vMeasures** and type the following formula:
5. **vMeasures='Percent';**

   You do not have a Percent element in the data source. Creating a variable with this value will assume that all data read into the cube will be assigned to the Percent element of the Expense_Measures dimension that you created previously.
6. Click **Evaluate**, and then click **OK**.
7. Change the **Variable Type** to **String**.
8. Rename the other variables and set the contents to the following:



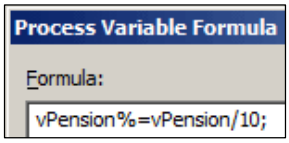
	Variable Name	Variable Type	Sample Value	Contents	Formula
1	vVersions	String	Budget version 1	Element	
2	vSubsidiaries	String	GO Americas	Element	
3	vMonth	String	Jan	Element	
4	vPension	Numeric	0.93	Other	
5	vSocSec	Numeric	0.53	Other	
6	vHealth	Numeric	0.28	Other	
7	vWorkersComp	Numeric	30149.22	Ignore	
8	vMeasures	String	Percent	Element	Formula

   You will load the Workers Compensation data in another process. It would not be loaded into the Percent element. Instead, it would be assigned to the Amount element. Because it is not used in this TI process, it may be ignored.

   In this process, you will calculate the percentage values for Pension, Social Security, and Health benefits. To get the correct numbers, you need to divide the incoming values from the source file by 10.
9. Click **New Variable** and name the new variable **vPension%**.

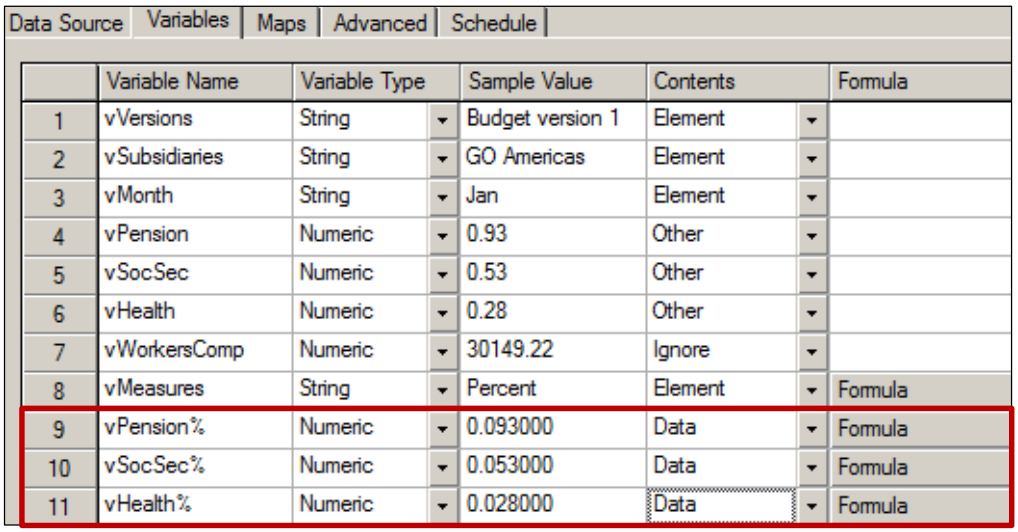
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10. For the **vPension%** variable, specify a formula of **vPension%=vPension/10;**



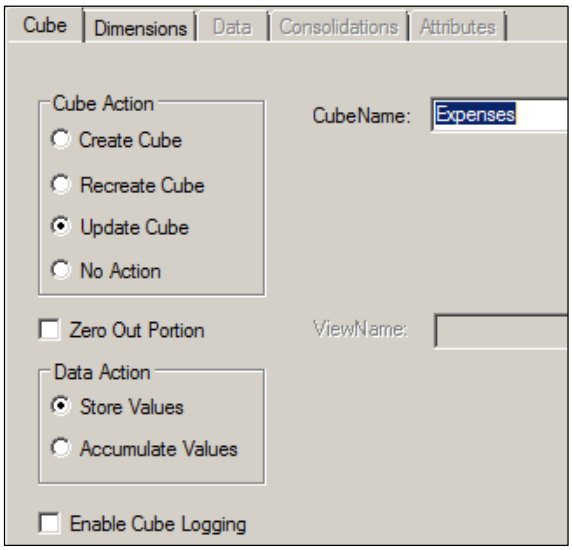
11. Click **OK**.
12. Repeat steps **8** to **10** to create two variables called **vSocSec%** and **vHealth%**. For the **vSocSec%** variable, specify a formula of **vSocSec/10**. For the **vHealth%** variable, specify a formula of **vHealth/10**.
13. In the **Contents** column, specify **Data** for the **vPension%**, **vSocSec%**, and **vHealth%** variables.

    The results appear as follows:



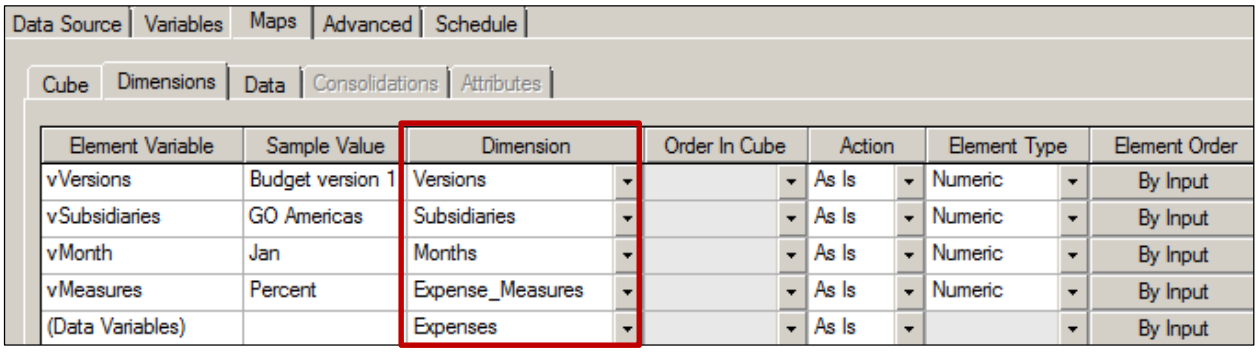
	Variable Name	Variable Type	Sample Value	Contents	Formula
1	vVersions	String	Budget version 1	Element	
2	vSubsidiaries	String	GO Americas	Element	
3	vMonth	String	Jan	Element	
4	vPension	Numeric	0.93	Other	
5	vSocSec	Numeric	0.53	Other	
6	vHealth	Numeric	0.28	Other	
7	vWorkersComp	Numeric	30149.22	Ignore	
8	vMeasures	String	Percent	Element	Formula
9	vPension%	Numeric	0.093000	Data	Formula
10	vSocSec%	Numeric	0.053000	Data	Formula
11	vHealth%	Numeric	0.028000	Data	Formula

14. Click the **Maps** tab, and then on the **Cube** tab, click **Update Cube**.
15. Click the down arrow beside **CubeName**, and then click **Expenses**.

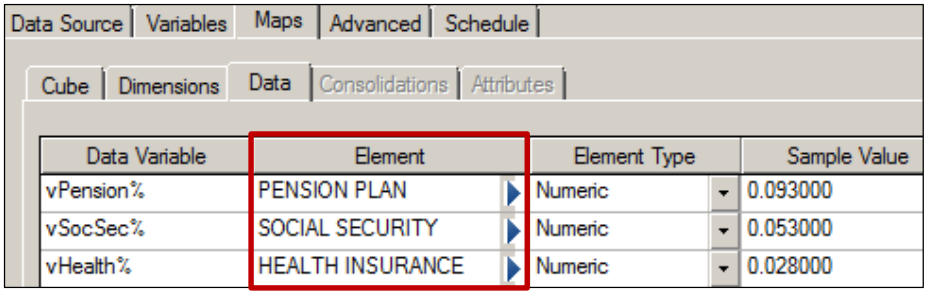


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16. Click the **Dimensions** tab and map the values in the **Dimension** column as follows:



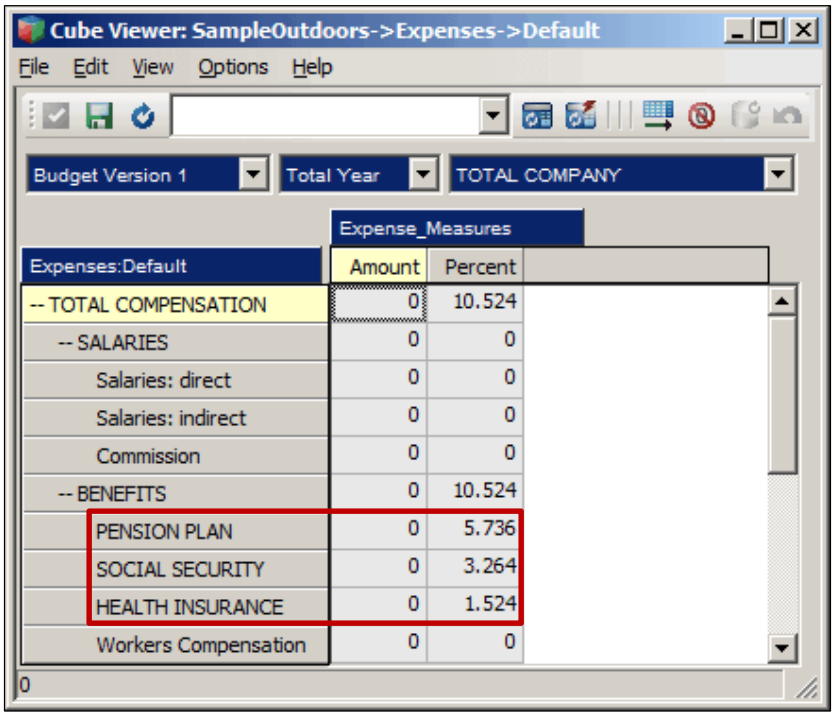
17. Click the **Data** sub-tab and map the data to elements as follows:



18. Click all the sub-tabs under the **Advanced** tab.
19. Save the process as **LoadBenWorkCompCSVPercent**.
20. Click **Run**, click **OK**, and then close **TurboIntegrator**.

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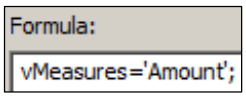
21. Double-click the **Expenses** cube, set the **Versions** dimension to **Budget Version 1** (if necessary), and then click **Recalculate**.
    The result appears as follows:



22. Leave the **Expenses** cube open.

Task 5. Create a process to load worker's compensation values.

1. Open the **LoadBenWorkCompCSVPercent** process.
2. Click **File** and then click **Save As**.
3. In the **Name** box type **LoadBenWorkCompCSVAmount**, and then click **OK**.
4. Click the **Variables** tab, and then next to **vMeasures**, click **Formula**.
5. Change **'Percent'** to **'Amount'**.
   The result appears as follows:

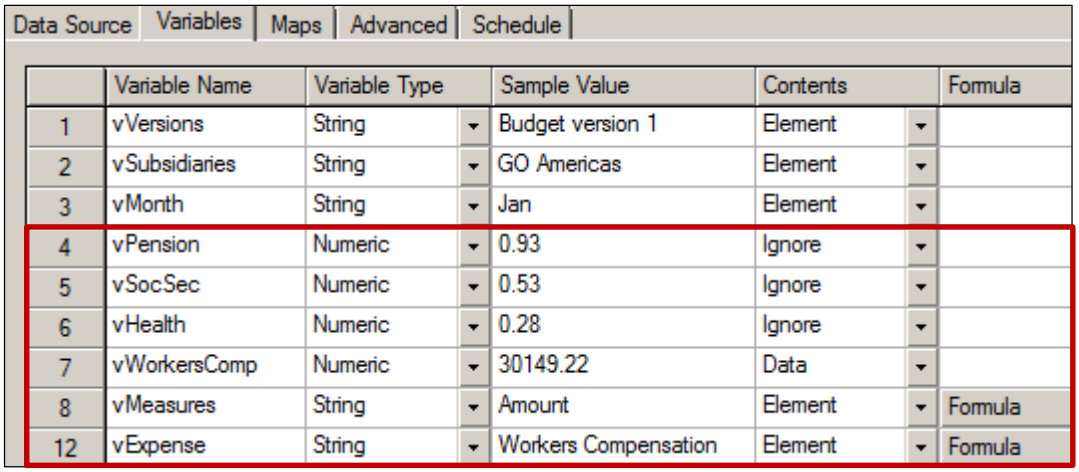


6. Click **OK**.
7. Next to **vPension**, **vSocSec**, and **vHealth** , change the **Contents** to **Ignore**.
8. Next to **vWorkersComp**, change the **Contents** to **Data**.

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9. Create a new variable called **vExpense** with a formula of **vExpense='Workers Compensation';**
10. Change the **Variable Type** to **String** and the **Contents** to **Element**.
11. Click the **vPension%** variable, and then click **Delete**.
12. Repeat step **11** for the **vSocSec%** and **vHealth%** variables.

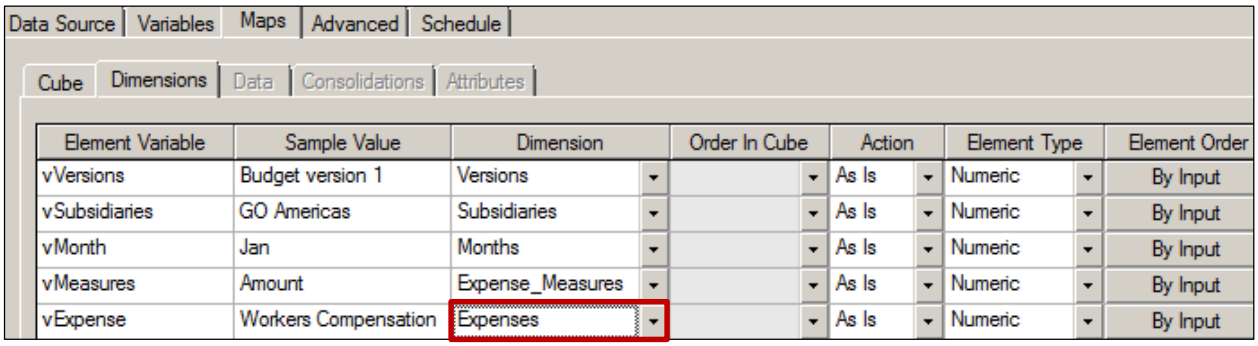
    The result appears as follows:



	Variable Name	Variable Type	Sample Value	Contents	Formula
1	vVersions	String	Budget version 1	Element	
2	vSubsidiaries	String	GO Americas	Element	
3	vMonth	String	Jan	Element	
4	vPension	Numeric	0.93	Ignore	
5	vSocSec	Numeric	0.53	Ignore	
6	vHealth	Numeric	0.28	Ignore	
7	vWorkersComp	Numeric	30149.22	Data	
8	vMeasures	String	Amount	Element	Formula
12	vExpense	String	Workers Compensation	Element	Formula

13. Click the **Maps** tab, and then click the **Dimensions** sub-tab.
14. Map **vExpense** to the **Expenses** dimension.

    The result appears as follows:



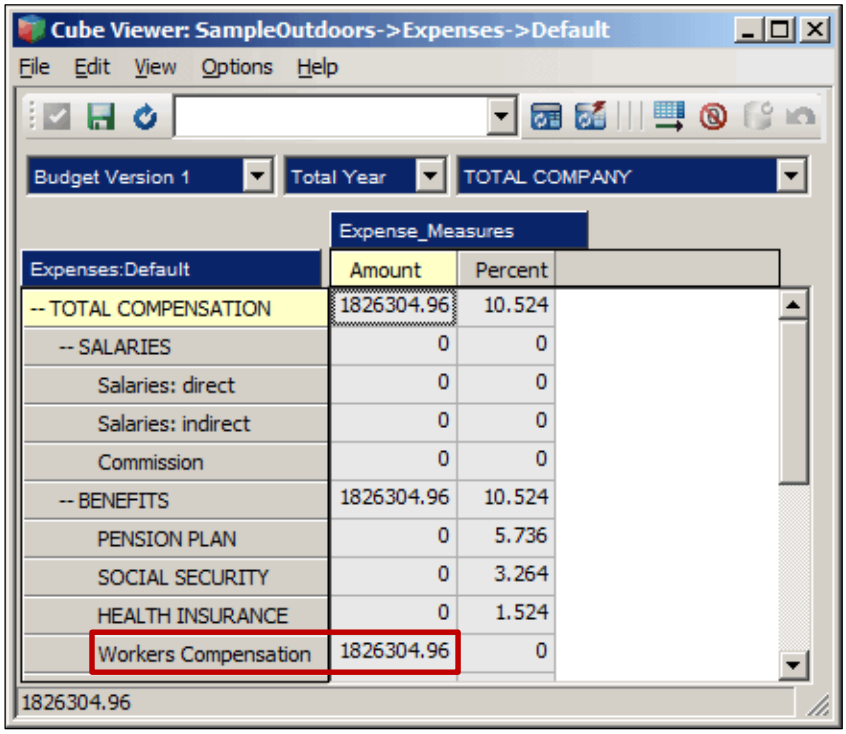
Element Variable	Sample Value	Dimension	Order In Cube	Action	Element Type	Element Order
vVersions	Budget version 1	Versions		As Is	Numeric	By Input
vSubsidiaries	GO Americas	Subsidiaries		As Is	Numeric	By Input
vMonth	Jan	Months		As Is	Numeric	By Input
vMeasures	Amount	Expense_Measures		As Is	Numeric	By Input
vExpense	Workers Compensation	Expenses		As Is	Numeric	By Input

    Because there is only one data element and it matches an element in our measures dimension, we do not have to map it manually on the Data sub-tab.
15. Click the **Advanced** tab and all sub-tabs under it.

    Be sure that the code on the Data tab now refers to 'Workers Compensation'.
16. Click **Save**, and then click **Run**.

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17. Click **OK**, close **TurboIntegrator**, and then recalculate the **Expenses** cube. The result appears as follows:



18. Save this as the public **Default** view and leave it open for the next task.

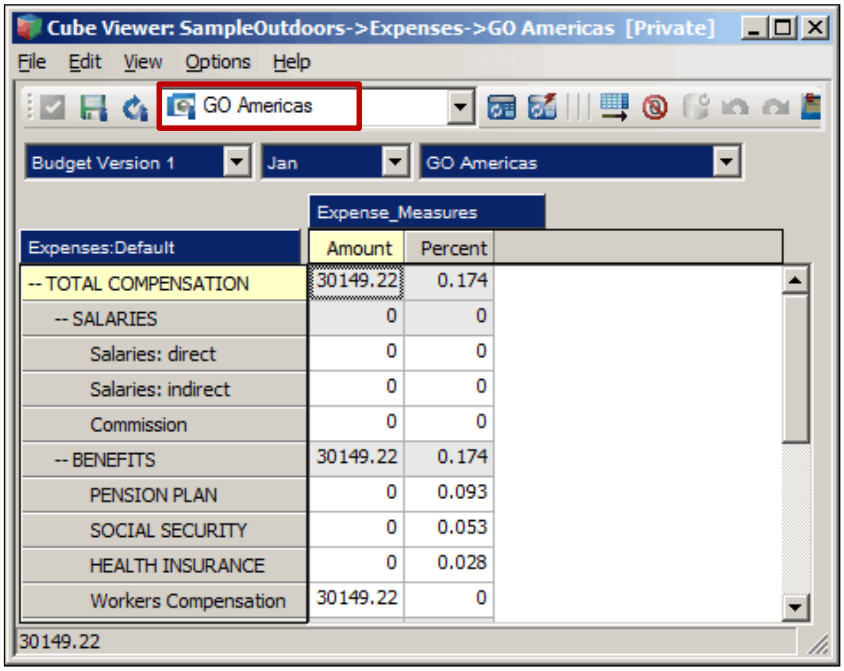
Task 6. Create a rule to populate Commissions.

This cube (Expenses) will have some stored data, loaded via TI. In addition, it will also pull data from other cubes. In this task you will create rules to pull commissions from the Commissions cube into the Expenses cube.

1. In the open view, double-click the **Months** dimension (currently displaying **Total Year**), in the **Subset Editor**, expand **Q1**, and then click **Jan**.
2. Click **OK**, and then double-click the **Subsidiaries** dimension (currently displaying **TOTAL COMPANY**).
3. Expand **GO AMERICAS REGION**, click **GO Americas**, and then click **OK**.

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4. Click **Recalculate**, and then save this as the **GO Americas** view.
   The result appears as follows:



   Leave this view open but move it aside so that you can see the list of cubes in the Server Explorer window.
5. In **Server Explorer**, right-click the **Expenses** cube and click **Create Rule**.
   The Rule Editor opens (this may take a few moments).
6. On **line 1**, type **SKIPCHECK;** and then press **Enter**.
7. Click the **Brackets** and then double-click **Expenses**.
8. Click **Commission** and then click **OK**.
9. Inside the square brackets, after **'Commission'**, type **, 'Amount'** and then press the **End** key.

```
1   SKIPCHECK;
2   ['Commission', 'Amount' ]
```

10. Type **= N:** and click **Insert Cube Reference**.

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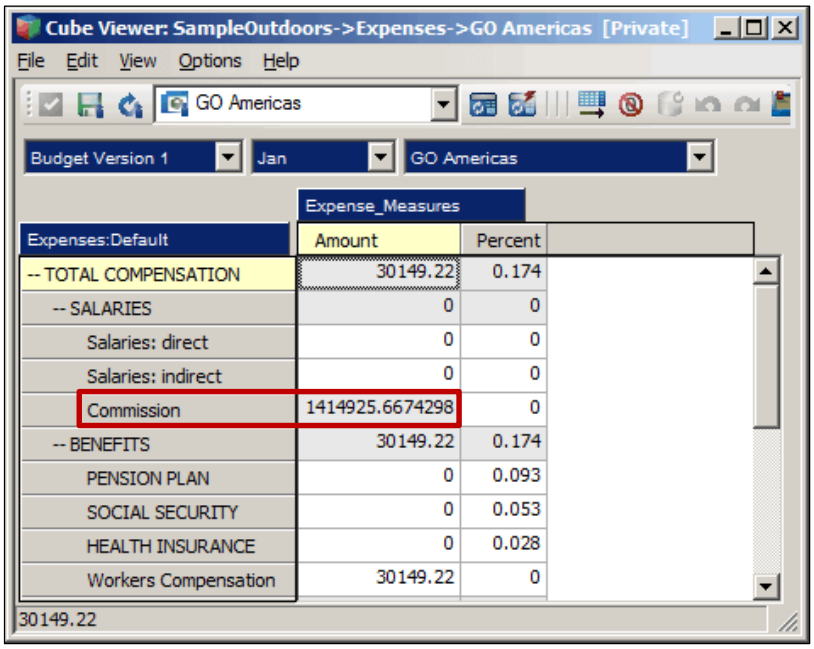
11. In the **Cube** list, select the **Commissions** cube.
12. Click the **Subset Editor** icon for **Commission_Measures**.
13. Click **Commission**, click **OK**, and then click **OK** to close the **Insert Cube Reference** dialog box.
14. Type a **;** at the end of the line.

    The result appears as follows:

```
1   SKIPCHECK;
2   ['Commission' ,'Amount']  = N: DB('Commissions', !Subsidiaries,
3                       !Months, !Versions, 'Commission');
```

15. Save the rule, close the **Rules Editor**, and then recalculate the **GO Americas** view.

    The results appear as follows:



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Task 7. Create a feeder in the Commissions cube for the commissions-related rule (in the Expenses cube).

1. Open the rules for the **Commissions** cube.



2. Add the following feeder to the rules in the **Commissions** cube under the FEEDERS; statements:

   ['Commission%'] => DB('Expenses', !Versions, !Months, !Subsidiaries, 'Commission', 'Amount') ;

   Note: Because the Commissions cube contains the source of the feeder, the feeder statement must be placed in the Commissions cube. The DB() function is not permitted on the left side of the =>.

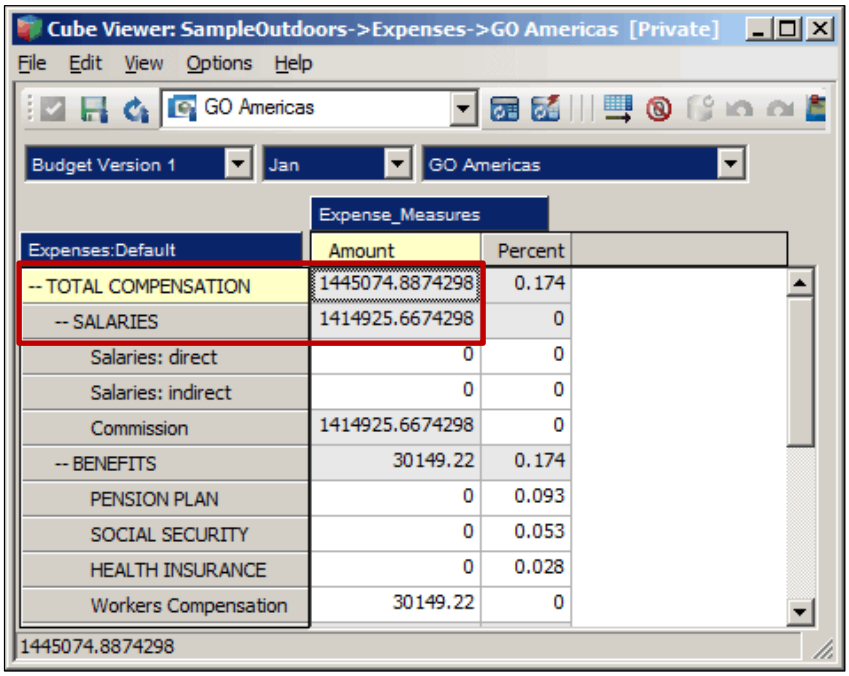
```
FEEDERS;
['Commission%' ]  => ['Commission' ] , ['Commission hold' ] , ['Commission count' ] ;
['Commission%' ]  => DB('Expenses', !Versions, !Months, !Subsidiaries, 'Commission', 'Amount') ;
```

3. Save the rule, and then close the **Rules Editor**.

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4. If necessary, open the **GO Americas** view for the **Expenses** cube and recalculate it.

   The result appears as follows:



   The values for the Commission element are consolidating properly in the Expenses cube because you added the feeder to the Commissions cube.

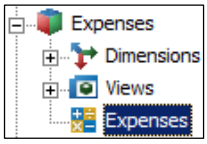
   You can assign a currency format to the Amount element and a percentage format to the Percent element in the Expense_Measures dimension to make it easier to read the values.

5. In Server Explorer, right-click **Expense_Measures** and then click **Edit Element Attributes**.
6. Click **Amount**, click **Format**, and then click **Currency**.
7. Click **OK** to close the **Number Format** dialog box.
8. Click **Percent**, click **Format**, and then click **Percentage**.
9. Click **OK** to close the **Number Format** dialog box, and then click **OK** to close the **Attributes Editor** window.
10. Leave the **Cube Viewer** open for the next task.

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Task 8. Create rules to calculate benefits.

1. Open the **Rules Editor** for the **Expenses** cube.



2. On the line after the existing rules, click the **Brackets**, and then double-click **Expenses**.
3. Click **PENSION PLAN**, and then click **OK**.
4. Inside the square brackets, after 'PENSION PLAN', type **, 'Amount'**, and then press the **End** key.

   The result appears as follows:

```
['PENSION PLAN','Amount' ]
```

5. Type the following :

   = N:['SALARIES', 'Amount'] * ['PENSION PLAN','Percent'] ;
6. Repeat steps **2** to **5** for **SOCIAL SECURITY** and **HEALTH INSURANCE**.

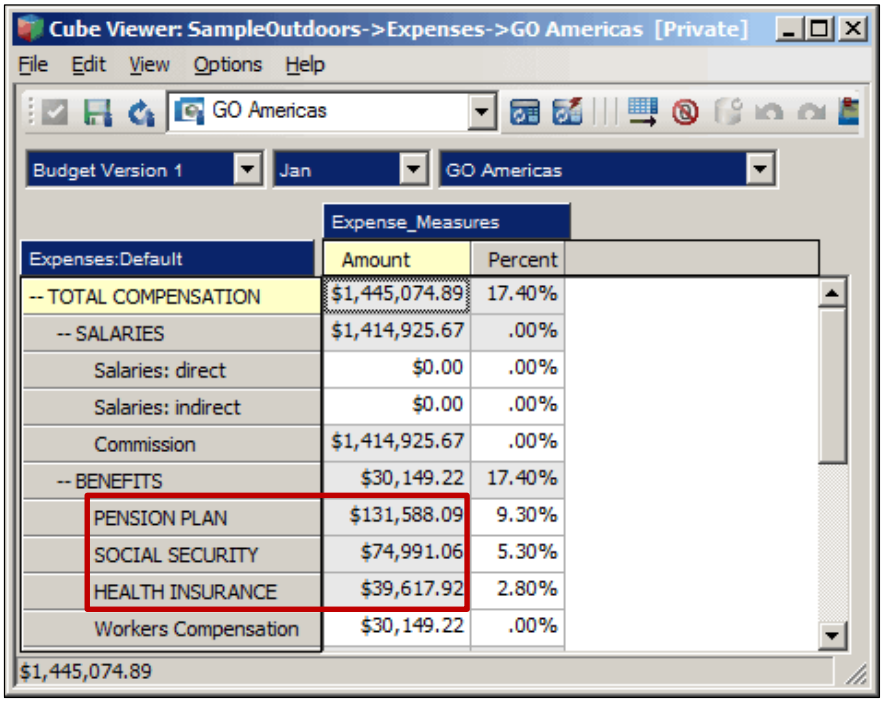
   The result appears as follows:

```
1  SKIPCHECK;
2  ['Commission' ,'Amount']  = N: DB('Commissions', !Subsidiaries,
3                    !Months, !Versions, 'Commission');
4  ['PENSION PLAN','Amount' ] = N:['SALARIES', 'Amount']  * ['PENSION PLAN','Percent'] ;
5  ['SOCIAL SECURITY','Amount' ] = N:['SALARIES', 'Amount']  * ['SOCIAL SECURITY','Percent'] ;
6  ['HEALTH INSURANCE','Amount' ] = N:['SALARIES', 'Amount']  * ['HEALTH INSURANCE','Percent'] ;
```

7. Click **Check Syntax**, click **Save**, and then close the **Rules Editor**.

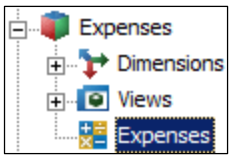
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8. In the **Cube Viewer**, click **Recalculate**.



The BENEFITS consolidation is only aggregating Workers Compensation ($30,149.22) because only that rule has feeders. Next, you will write feeders for the rest of the benefits. These calculations use elements that exist within the Expenses cube so this is where you will write the feeder.

9. On the next line, create the following feeders in the rules for the **Expenses** cube:



FEEDERS;

['SALARIES', 'Amount'] => ['HEALTH INSURANCE','Amount'], ['PENSION PLAN','Amount'],['SOCIAL SECURITY','Amount'];

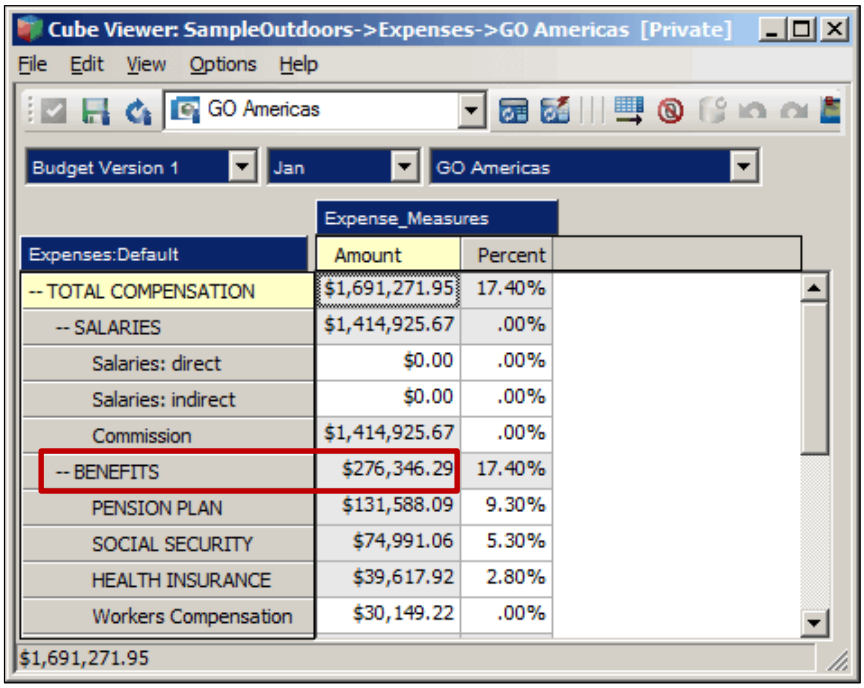
The result appears as follows:

```
1   SKIPCHECK;
2   ['Commission' ,'Amount']  = N: DB('Commissions', !Subsidiaries,
3                     !Months, !Versions, 'Commission');
4   ['PENSION PLAN','Amount' ] = N:['SALARIES', 'Amount']  * ['PENSION PLAN','Percent'] ;
5   ['SOCIAL SECURITY','Amount' ] = N:['SALARIES', 'Amount']  * ['SOCIAL SECURITY','Percent'] ;
6   ['HEALTH INSURANCE','Amount' ] = N:['SALARIES', 'Amount']  * ['HEALTH INSURANCE','Percent'] ;
7
8   FEEDERS;
9   ['SALARIES', 'Amount'] => ['HEALTH INSURANCE','Amount'],
10                    ['PENSION PLAN','Amount'],['SOCIAL SECURITY','Amount'];
```

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10. Click **Save** in the Rules Editor, and then close it.
11. Recalculate the view.

    The result appears as follows:



    All consolidations should have values if the child elements have values. If they do not, the consolidations are not being fed properly.

12. Close the **Cube Viewer**, saving and overwriting the **GO Americas** view if prompted.
13. Right-click the **SampleOutdoors** node and click **Save Data**.
14. Leave **Architect** open for the next demonstration.

Results:
You have created a new cube that contains expenses. The salary and benefit information has been populated by data in a text file and through rules. The data for the other expenses will come from user input through spreading.

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IBM Training

Demonstration 2

Create the Spread Profiles Cube

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Demonstration 2: Create the Spread Profiles cube

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Demonstration 2: Create the Spread Profiles cube

Purpose:
You just created the Expenses cube that contains salary and benefit expense information. Those using this model will also need to perform some what-if scenarios for planning purposes on the overhead expenses.

You need to create a new cube for your users so that they can employ proportional spreading by month, by subsidiary, or both. This cube will have a variety of spreading options. You will then test the proportional spreading function to ensure that the lookup cubes have been constructed properly.

Task 1. Create the SpreadMethod dimension.

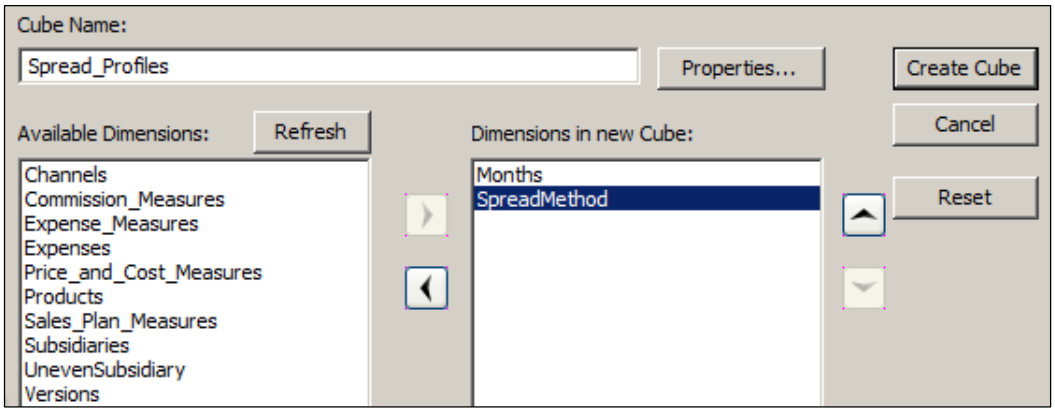
1. In Server Explorer, right-click **Dimensions** and click **Create New Dimension**.
2. Insert the following simple elements:
   - **Flat**
   - **Quarterly**
   - **4 4 5**
   - **Business**
3. Save the dimension as **SpreadMethod**, and then close the **Dimension Editor**.

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Task 2. Create the Spread_Profiles cube.

1. In Server Explorer, right-click **Cubes** and then click **Create new cube**.
2. In the **Cube Name** box type **Spread_Profiles**.
3. Move the following dimensions to the **Dimensions in new Cube** section:
   - **Months**
   - **SpreadMethod**

   The result appears as follows:



4. Click **Create Cube**.

Task 3. Populate the Spread_Profiles cube.

1. In Server Explorer, double-click the **Spread_Profiles** cube.
2. Swap the **Months** and **SpreadMethod** dimensions.
3. Spread_Method is in the row area and Months is in the column area.
4. Click the **Months** dimension, and then select only the leaf elements (Jan to Dec).
5. Click **OK**, and then click **Recalculate**.

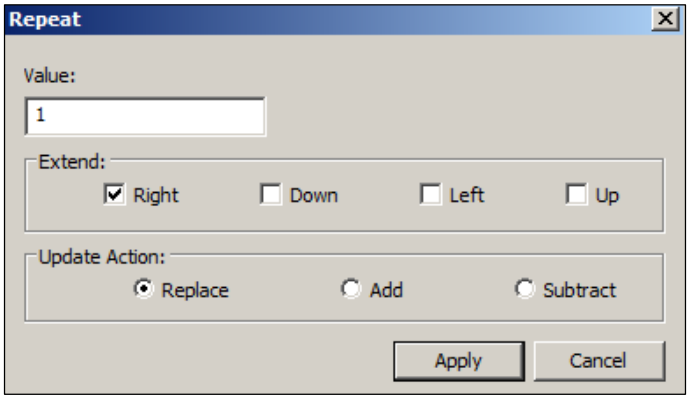
   The result appears as follows:

	Months											
SpreadMethod	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Flat	0	0	0	0	0	0	0	0	0	0	0	0
Quarterly	0	0	0	0	0	0	0	0	0	0	0	0
4 4 5	0	0	0	0	0	0	0	0	0	0	0	0
Business	0	0	0	0	0	0	0	0	0	0	0	0

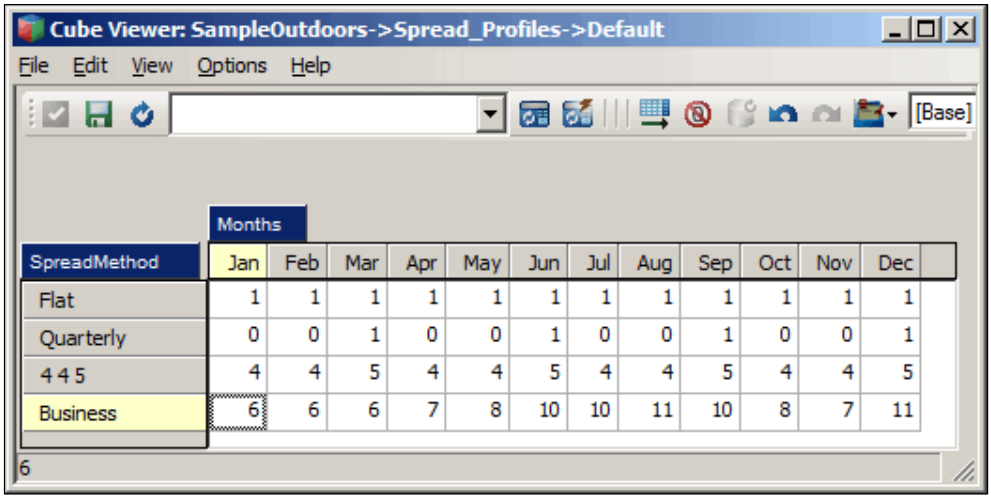
6. Right-click the **Flat**, **Jan** cell, point to **Data Spread**, and then click **Repeat**.

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7. In the **Value** box, type **1**, and then click **Right**.
   The result appears as follows:



8. Click **Apply**.
9. Enter values for the **Business**, **Quarterly**, and **4 4 5** rows so that the result appears as follows:

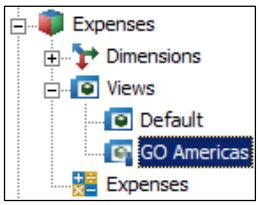


   This cube now contains a pattern of data by month, which may be used as the basis for relational proportional spreading by any other cube that contains the Months dimension.
10. Click **Recalculate**, and then click **Save**.
11. Select **Default**, deselect **Private**, and then click **OK**.
12. Close the **Cube Viewer**.

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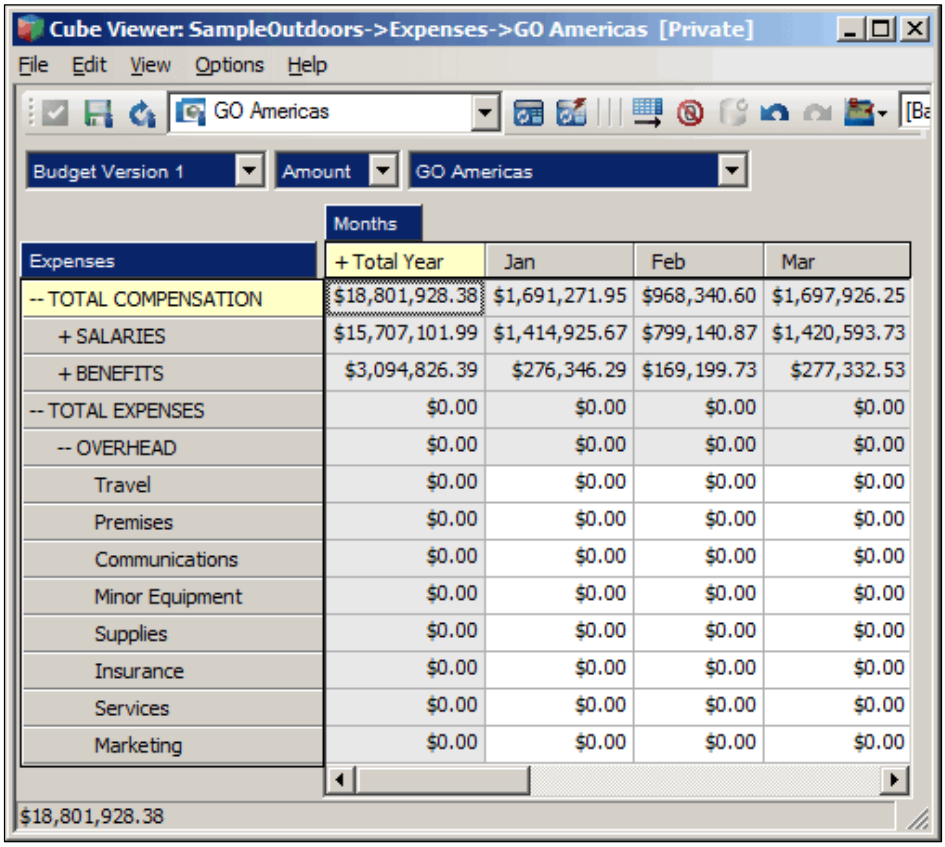
Task 4. Spread data in the Expenses cube using the values in the Spread_Profiles cube.

1. Open the **GO Americas** view of **Expenses**.



2. Swap **Expense_Measures** and **Months**.
3. Click **Recalculate,** and then collapse **SALARIES** and **BENEFITS**.
4. Click **Months** (on columns) and then click **All**.
5. Ctrl+click **Total Year** and leaf elements from **Months**, click **OK**, and then recalculate.

   The result appears as follows:



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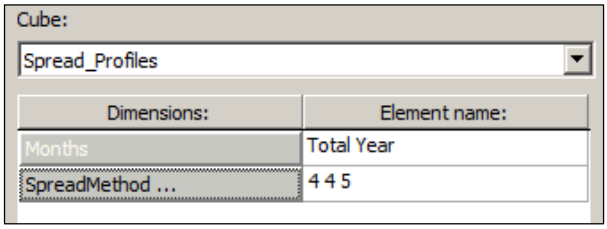
6. Save the public view as **GO Americas Expenses**.

   Now you will try spreading your Travel Expenses for the Americas. They seem to take on the 4 4 5 pattern where they are greater during the last month of the quarter. The budget for Travel Expenses is supposed to be 1,000,000 or less for the year.
7. Click **TOTAL COMPENSATION** to collapse it.
8. Right-click **Travel**, **Total Year**, point to **Data Spread**, and then click **Relative Proportional Spread**.

   Relative Proportional Spread will look at the pattern of values in the reference cells and then spread the data in the current cells, following the same pattern.

   You want to use the same pattern as the **4 4 5** cells in the **Spread_Profiles** cube.
9. Under **Cube**, click **Spread_Profiles**.
10. Under **Dimensions**, click **SpreadMethod**, click **4 4 5**, and then click **OK**.

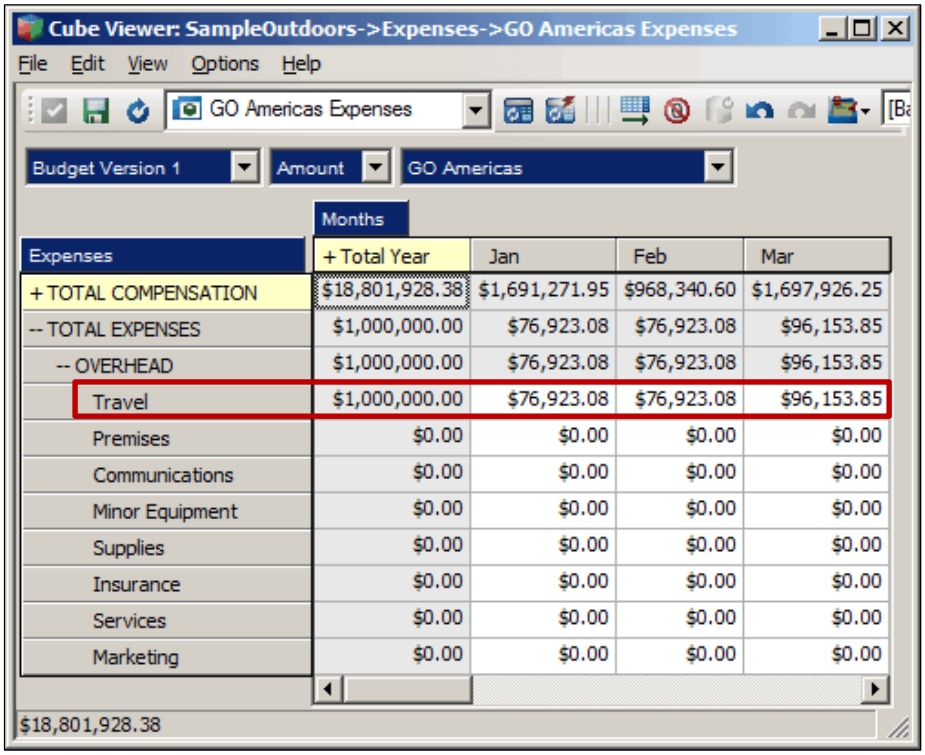
    The result appears as follows:



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11. Click **Select**, in the **Value** box, type **1000000**, and then click **Apply**.
    The result appears as follows:



    You can see that the 1,000,000 was spread across the months in the same proportions as the values in the Spread_Profiles cube. The proportions in the 4 4 5 cells of the Spread_Profiles cube were the same for Jan and Feb (4), while Mar was more (5). The result in the Expenses cube is following that same pattern.

12. Close the **GO Americas Expenses** view, saving changes and overwriting the existing view if prompted.

    It is important to note that the Months dimension exists in both the Spread_Profiles cube and the Expenses cube. Only cubes that share common dimensions can be used as a spreading basis for a relative proportional spread. Also, you don't have to create a specific cube (such as your Spread_Profiles) to do this. You can use a reporting cube as the basis as long as it contains a common dimension.

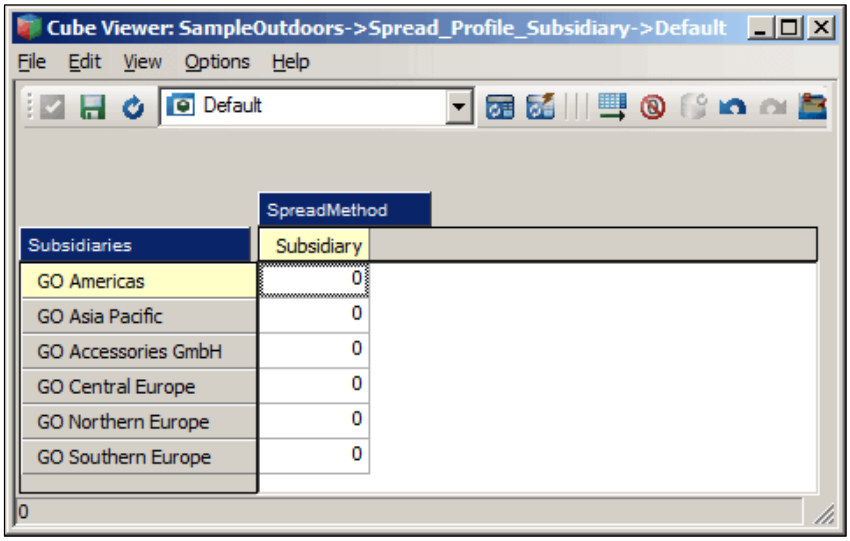
    You may wish to try this by right-clicking on Overhead (a consolidation in the Expenses dimension) for the Total Year. Click Data Spread and then click Relative Proportional Spread. You no longer can select the Spread Profiles cube because Spread Profiles does not contain the Expenses dimension, therefore you must choose a leaf level element from this dimension (like Travel).

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Task 5. Create another spread profile cube for Subsidiaries.

1. In Server Explorer, right-click the **SpreadMethod** dimension and then click **Edit Dimension Structure**.
2. Click **Business**, add a new simple element called **Subsidiary**, and then save and close the dimension.
3. Right-click **Cubes** and then click **Create new cube**.
4. Include **Subsidiaries** and **SpreadMethod** in the cube and name the cube **Spread_Profile_Subsidiary**.
5. Double-click the **Spread_Profile_Subsidiary** cube to open it in the **Cube Viewer**.
6. Click the **Subsidiaries** dimension (on rows), click **All**, and then click **Filter by Level**.
7. Click **0**, click **OK**, and then click **OK** to close the **Subset Editor** window.
8. Click the **SpreadMethod** dimension (on columns), and then click **All**.
9. Click **Subsidiary**, and then click **OK**.
10. Click **Recalculate**, and then save a **public**, **Default** view.

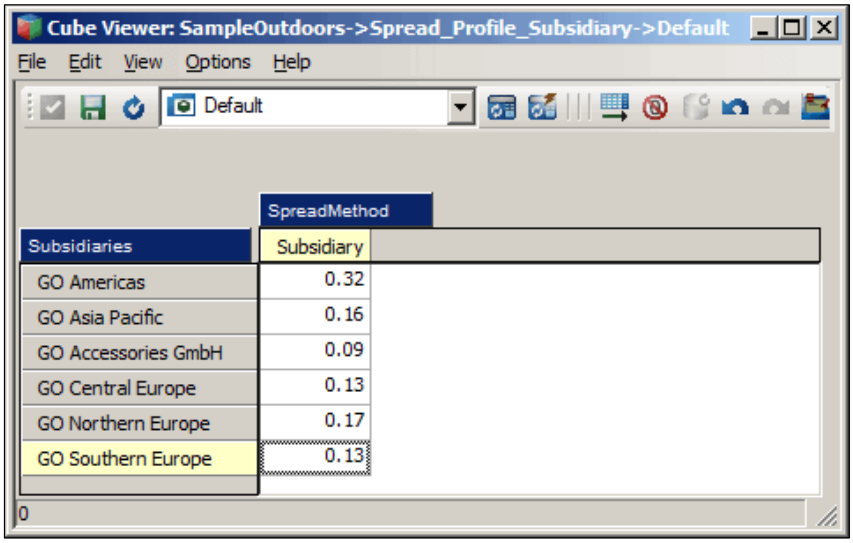
    The result appears as follows:



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11. Enter the following values for each **Subsidiary**.

    The result appears as follows:



12. Recalculate and close the view (without saving).

    You can use this view as a basis to spread data by subsidiary. However, it is common for different subsidiaries to have different monthly proportions. We will create a third spread profile cube that will combine the 4 4 5 proportion by subsidiary to accommodate this need.

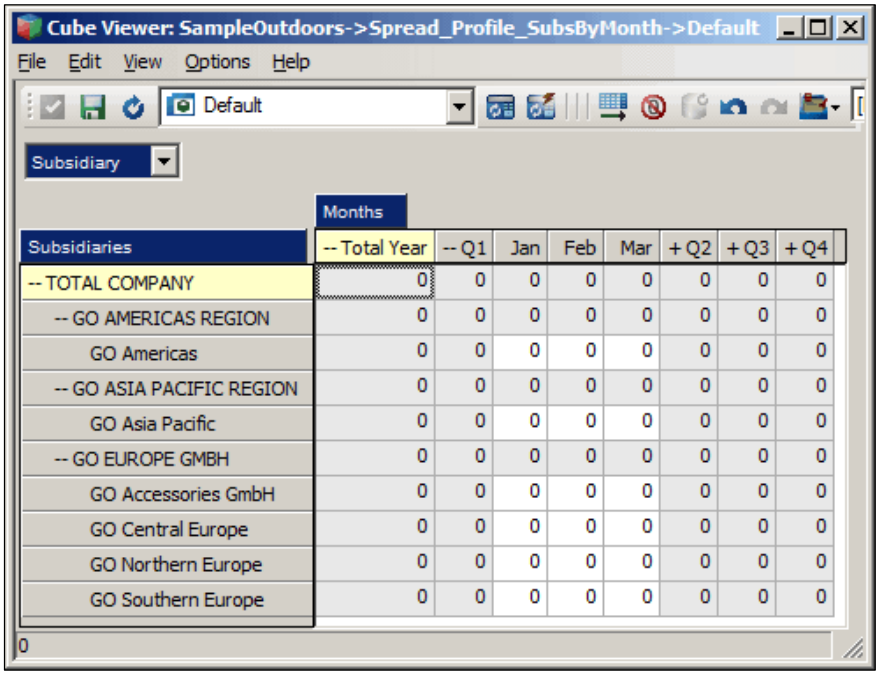
Task 6. Create the Spread_Profile_SubsByMonth cube.

1. Create a new cube called **Spread_Profile_SubsByMonth**.
2. Include three dimensions:
   - **Months**
   - **Subsidiaries**
   - **SpreadMethod**
3. Create a **public**, **Default** view of this cube.
4. Swap **Months** (currently displaying **Total Year**) and **SpreadMethod**.
5. In the **SpreadMethod** list, click **Subsidiary**.
6. Click **Recalculate**, and then click **Q1** to expand.
7. Click **Subsidiaries**, click **All**, and then click **OK**.

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8. Click **Recalculate**.

   The result appears as follows:



   You can leave this view open. Instead of typing values into this cube, you will write a rule to allocate spread proportions by month and subsidiary.

9. In Server Explorer, right-click the **Spread_Profile_SubsByMonth** cube and then click **Create Rule**.

10. Create the following rule:

    ['Subsidiary']=DB('Spread_Profile_Subsidiary', !Subsidiaries, 'Subsidiary')*DB('Spread_Profiles', !Months, '4 4 5');

    The results appear as follows:

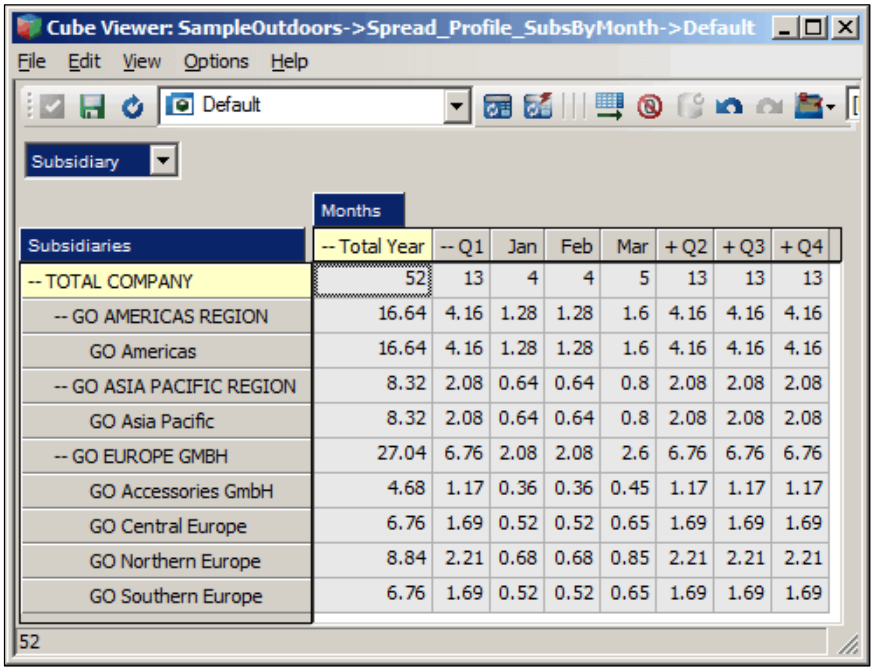
```
1   ['Subsidiary']=DB('Spread_Profile_Subsidiary', !Subsidiaries, 'Subsidiary')*
2                  DB('Spread_Profiles', !Months, '4 4 5');
```

11. Save the rule, and then close the **Rules Editor**.

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12. Recalculate the **Spread_Profile_SubsByMonth** cube.

    The result appears as follows:



Subsidiaries	-- Total Year	-- Q1	Jan	Feb	Mar	+ Q2	+ Q3	+ Q4
-- TOTAL COMPANY	52	13	4	4	5	13	13	13
-- GO AMERICAS REGION	16.64	4.16	1.28	1.28	1.6	4.16	4.16	4.16
GO Americas	16.64	4.16	1.28	1.28	1.6	4.16	4.16	4.16
-- GO ASIA PACIFIC REGION	8.32	2.08	0.64	0.64	0.8	2.08	2.08	2.08
GO Asia Pacific	8.32	2.08	0.64	0.64	0.8	2.08	2.08	2.08
-- GO EUROPE GMBH	27.04	6.76	2.08	2.08	2.6	6.76	6.76	6.76
GO Accessories GmbH	4.68	1.17	0.36	0.36	0.45	1.17	1.17	1.17
GO Central Europe	6.76	1.69	0.52	0.52	0.65	1.69	1.69	1.69
GO Northern Europe	8.84	2.21	0.68	0.68	0.85	2.21	2.21	2.21
GO Southern Europe	6.76	1.69	0.52	0.52	0.65	1.69	1.69	1.69

52

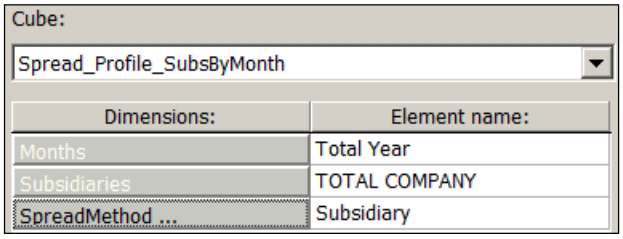
    Review the Total Year/Total Company values. They both roll up to 52. This indicates the 4 4 5 proportion was allocated properly by subsidiary. You can also see the 4 4 5 proportions along the TOTAL COMPANY for each month. This matches the Spread_Profiles (by month) cube proportions.

    Next you can spread data in the Expenses cube based on the proportions by Subsidiary and Month for 4 4 5.

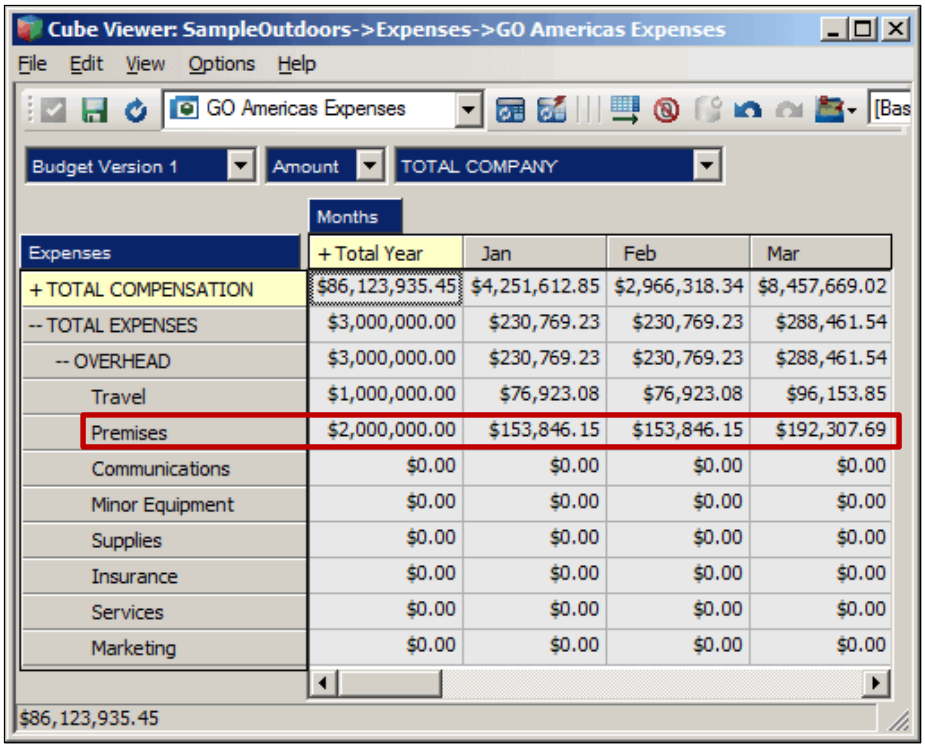
13. Close the **Spread_Profile_SubsByMonth** cube, saving and overwriting the **Default** view if prompted.
14. Open the **GO Americas Expenses** view of the **Expenses** cube.
15. Change the **Subsidiaries** dimension to display **TOTAL COMPANY**, and then click **Recalculate**.
16. Right-click **Total Year** and **Premises**, point to **Data Spread**, and then click **Relative Proportional Spread**.
17. In the **Cube** list, select **Spread_Profile_SubsByMonth**.

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18. Under **Dimensions**, click **SpreadMethod**, click **Subsidiary**, and then click **OK**.
    The results appear as follows:



19. Click **Select**.
20. In the **Value** box (currently displaying 52.00), type **2000000** and then click **Apply**.
    The result appears as follows:



The result has been allocated along two dimensions, both Months and Subsidiaries.

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21. Close all **Cube Viewer** windows without saving changes.

    You are using these cubes for spreading but they can also be used in calculations or other rules.

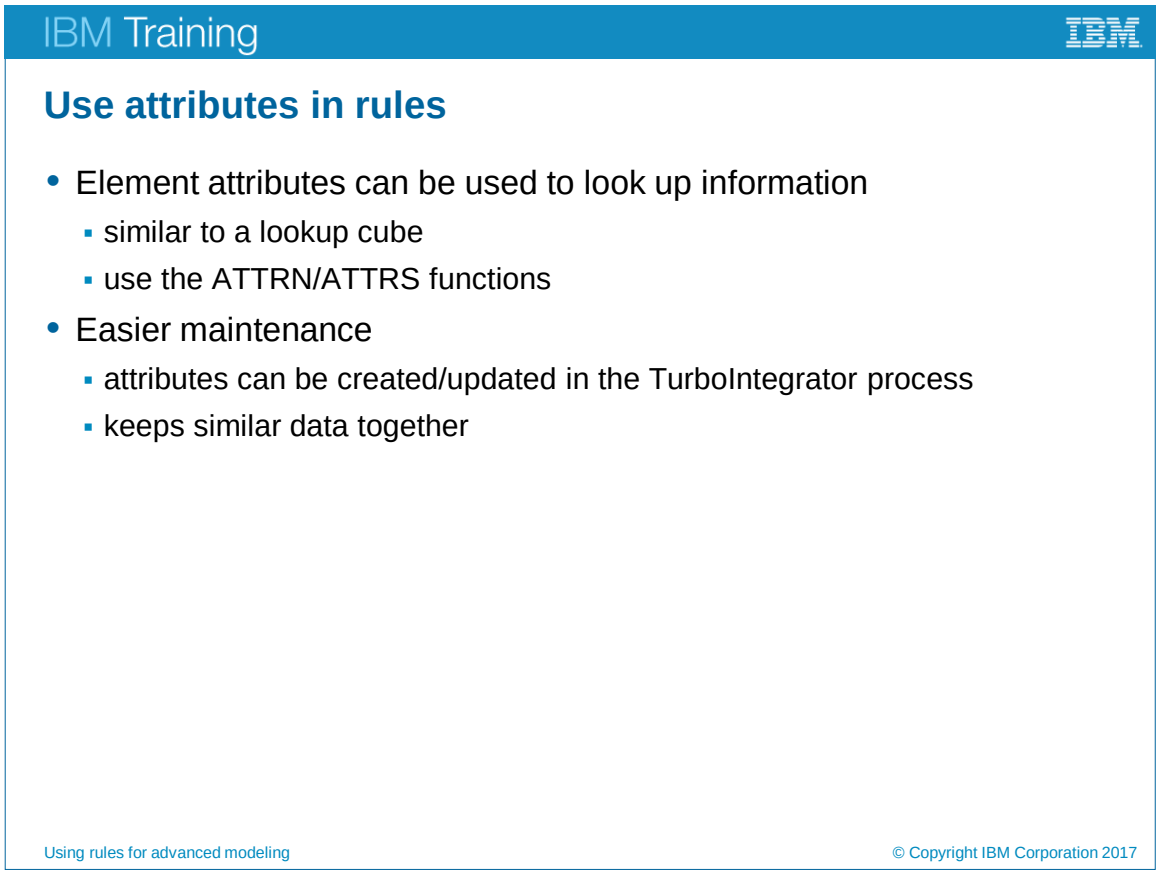
    You do not have to create feeders in this cube, mainly because there is so little data. The only time you do not create feeders is if you have very small and/or very dense cubes.

    The **Spread_Profile_SubsByMonth** cube is an example of a virtual cube. It contains no physical data but instead is populated by a calculation that allocated spread ratios among the Subsidiaries and Months dimensions.
22. Right-click the **SampleOutdoors** node, and then click **Save Data**.
23. Leave **Architect** open for the next demonstration.

Results:
You have created three spread profile cubes that can be used to spread data to other cubes. These are lookup cubes. One of these cubes was a virtual cube (Spread_Profiles_SubsByMonth) that allocates data from one cube (Spread_Profiles) by the Subsidiaries dimension in another cube (Spread_Profiles_Subsidiary).

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Use attributes in rules

Instead of using a DB function to pull data from another cube, we will pull data from the attributes of a dimension using the ATTRS function. ATTRS returns a string attribute for a specified element of a dimension. ATTRN returns a numeric attribute for a specified element.

ATTRS(dimension, element, attribute) or

ATTRN(dimension, element, attribute)

Argument	**Description**
dimension	A valid dimension name.
element	An element name in the dimension.
attribute	The attribute name to retrieve. It must be a valid attribute of the element.

These functions are frequently used to 'look up' an element based on the value of an attribute.

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IBM Training

Demonstration 3

Move balances from last month to next month

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Demonstration 3: Move balances from last month to next month

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Demonstration 3: Move balances from last month to next month

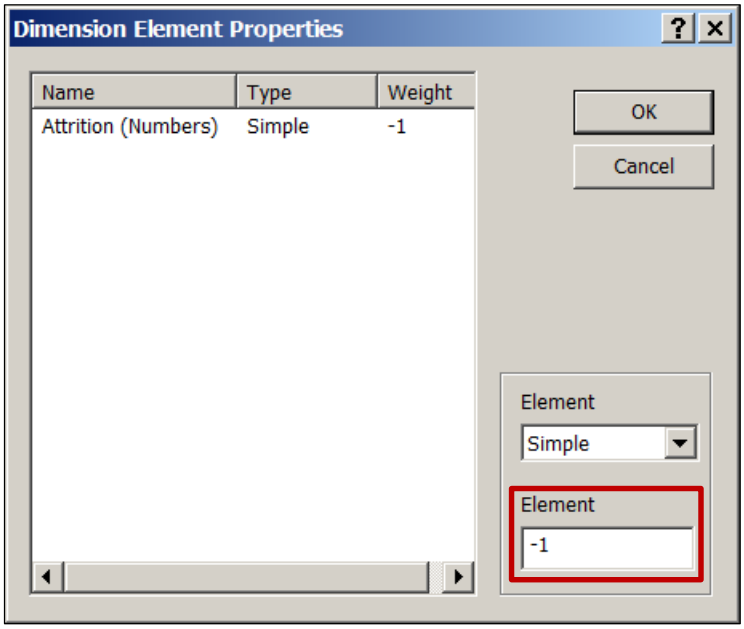
Purpose:
The Sample Outdoors Company offers its customers a subscription to an outdoors magazine that includes specially-priced merchandise. Subscribers typically take advantage of these offers (about 90% of customers). The marketing department has projected figures for new subscriptions in each period, subscriber attrition rates, and average purchase amount per subscriber. Based on this information, you want to calculate the number of active subscribers in each month and the expected revenue from special offers.

Task 1. Create the Churn_Measures dimension.

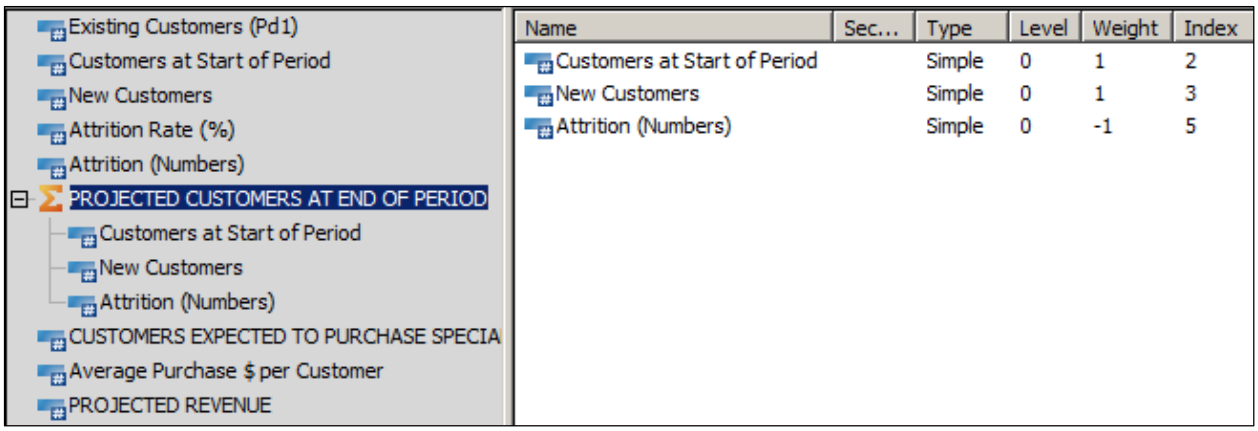
1. In **Microsoft Excel**, **open C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles\churn_calc.csv** and copy cells **A1:A9**.
2. In **Server Explorer**, create a new dimension.
3. In the **Dimension Editor**, click **Edit** and then click **Paste**.
4. Save the dimension as **Churn_Measures**.
   Next you will alter the hierarchy in this dimension.
5. Copy the following elements:
   - **Customers at Start of Period**
   - **New Customers**
   - **Attrition (Numbers)**
6. Click **PROJECTED CUSTOMERS AT END OF PERIOD**.
7. Click **Edit** and then click **Paste as Child**.
8. Right-click **Attrition (Numbers)** (child of PROJECTED CUSTOMERS AT END OF PERIOD), and then click **Element Properties**.

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9. In the **Element** box, type **-1**.
   The result appears as follows:



10. Click **OK**.
    The result appears as follows:



The elements may appear more than once but TM1 only recognizes them one time. Attrition (Numbers) will point to the same value if it is viewed in a report with or without the parent.

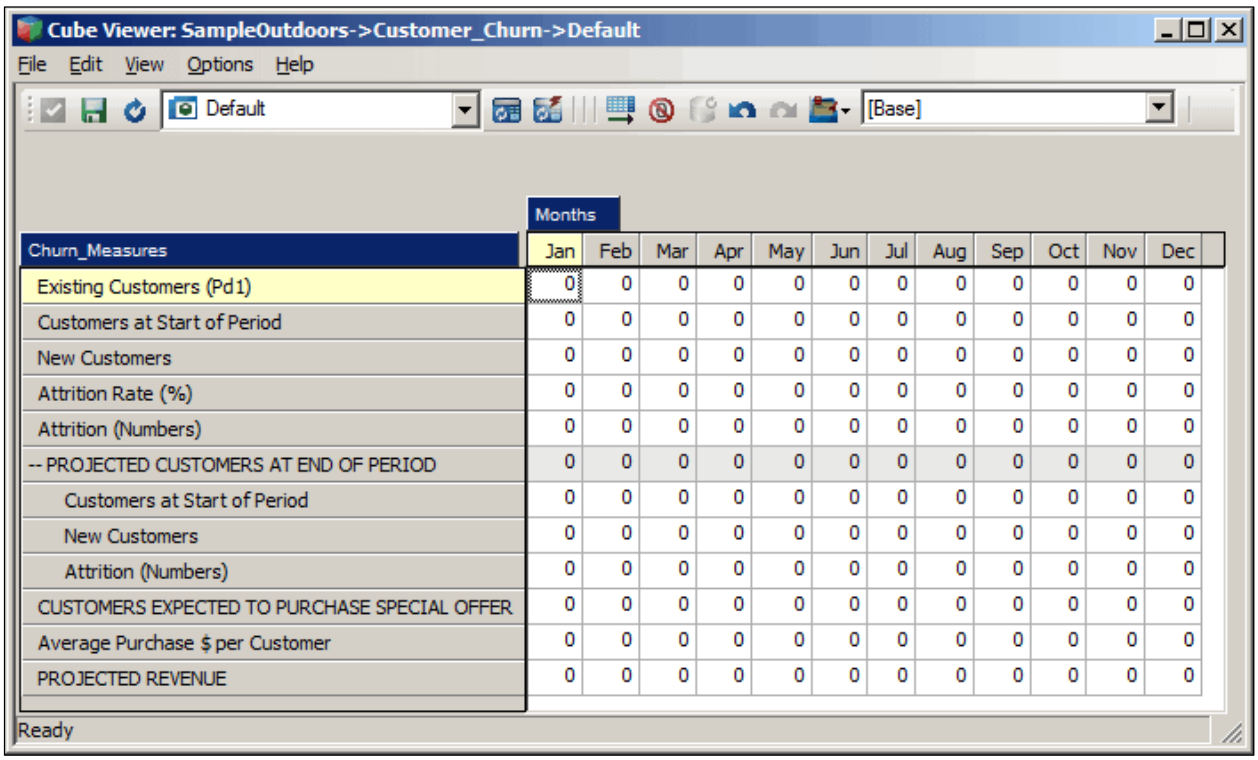
11. Click **OK** to close the **Dimension Editor** and click **Yes** when asked to save the changes to the dimension.
12. Close **Microsoft Excel** and do not save changes to the spreadsheet if prompted.

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Task 2. Create the Customer_Churn cube.

1. In **Server Explorer**, create a new cube.
2. Select the following dimensions:
   - **Months**
   - **Churn_Measures**
3. Name the cube **Customer_Churn** and then double-click the **Customer_Churn** cube to open it.
4. Swap **Churn_Measures** and **Months**
5. Click **Churn_Measures**, click **All** and then click **OK**.
6. Click **Months**, and then click **All**.
7. Select all month leaf elements (**Jan** to **Dec**) and then click **OK**.
8. Click **Recalculate** and then expand **PROJECTED CUSTOMERS AT END OF PERIOD**.
9. Save the view as the **public Default**.

   The result appears as follows:

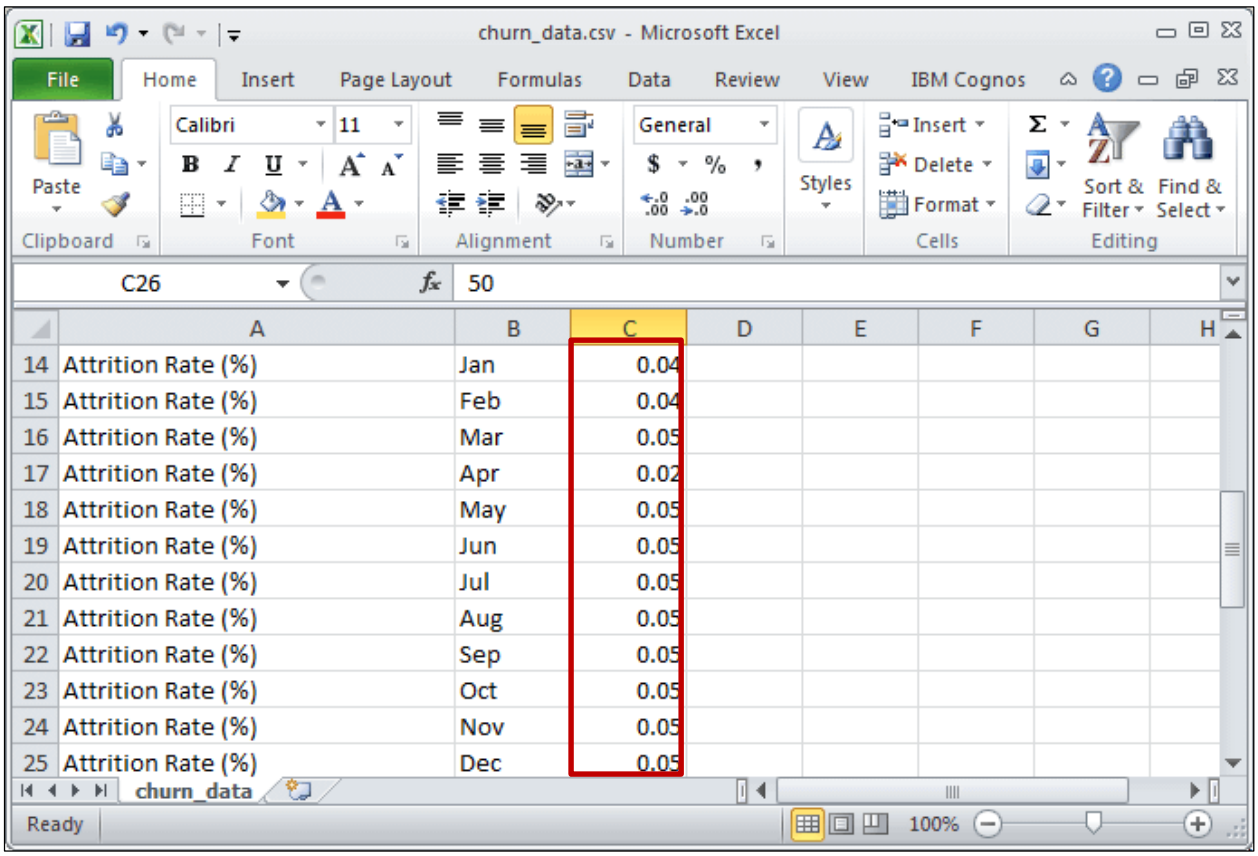


10. Leave this view open for the next task.

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Task 3. Load data from ChurnData.csv.

1. **Open C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles\churn_data.csv** in **Microsoft Excel**.
2. Modify the values in cells **C14** to **C25** as shown below:



	A	B	C
14	Attrition Rate (%)	Jan	0.04
15	Attrition Rate (%)	Feb	0.04
16	Attrition Rate (%)	Mar	0.05
17	Attrition Rate (%)	Apr	0.02
18	Attrition Rate (%)	May	0.05
19	Attrition Rate (%)	Jun	0.05
20	Attrition Rate (%)	Jul	0.05
21	Attrition Rate (%)	Aug	0.05
22	Attrition Rate (%)	Sep	0.05
23	Attrition Rate (%)	Oct	0.05
24	Attrition Rate (%)	Nov	0.05
25	Attrition Rate (%)	Dec	0.05

3. Save and close the file (click **Save** and **Yes** to any prompts that appear).
4. Create a new **TurboIntegrator** process.
5. Click **Text** and then browse to:
6. **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles\churn_data.csv**.
7. Click **OK** to dismiss the warning, click **Preview**, and then on the **Variables** tab, map the following:

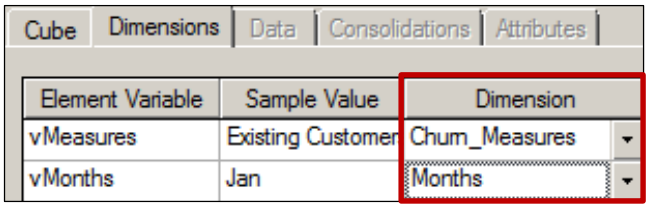
Variable Name	Variable Type	Sample Value	Contents
vMeasures	String	Existing Customers (Pd1)	Element
vMonths	String	Jan	Element
vData	Numeric	12000	Data

8. Click the **Maps** tab.

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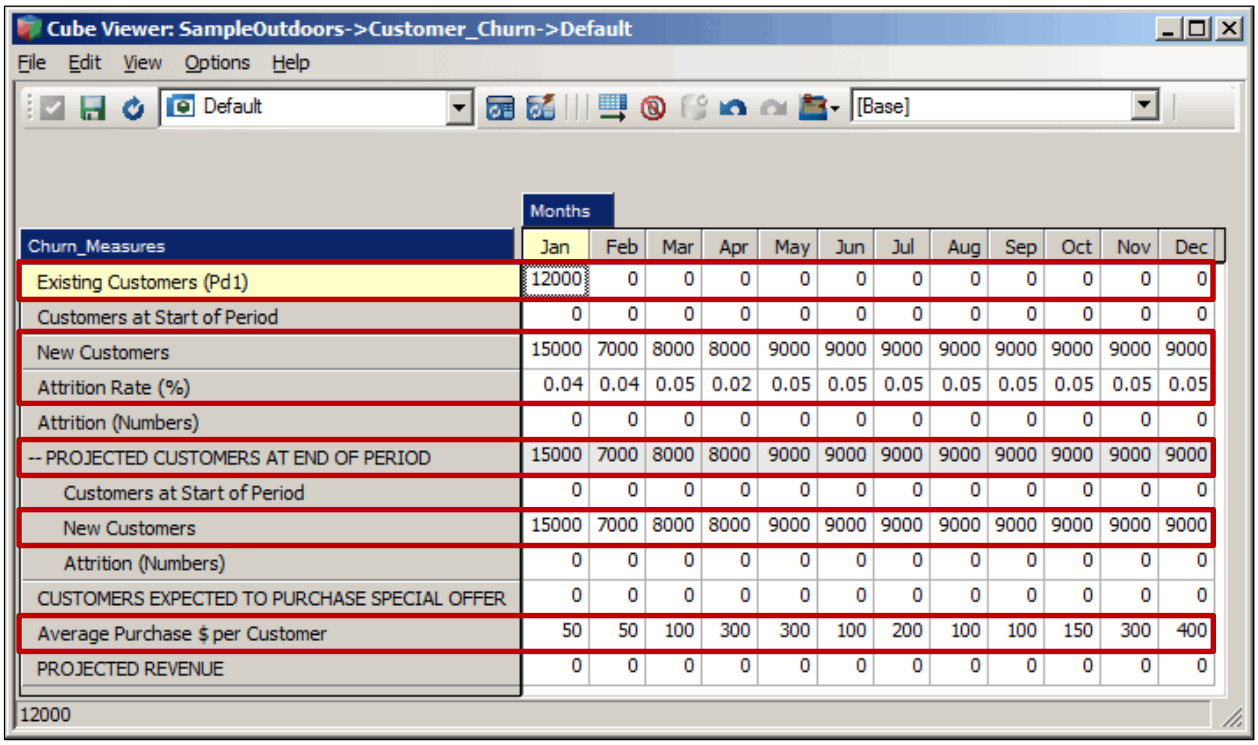
9. Click **Update Cube**, and then select **Customer_Churn** from the **CubeName** list.
10. Click the **Dimensions** tab, and then map the dimensions.

    The result appears as follows:



11. Click the **Advanced** tab and then click all the sub-tabs within.
12. Save the process as **LoadChurnDataCSV**, and then click **Run**.
13. Click **OK**, and then close **TurboIntegrator**.
14. Recalculate the **Default** view of the **Customer_Churn** cube.

    The result appears as follows:



15. Leave this view open for the next task.

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Task 4. Use rules to calculate Customer_Churn data.

You will create a number of rules to properly calculate the values for this analysis. In addition to computing Attrition (Numbers) based on rates, you will need to move the number of customers at the end of one period to the beginning of the next period.

1. In **Server Explorer**, right-click the **Customer_Churn** cube, and then click **Create Rule**.
2. Enter the following rule to move **Existing Customers (Pd1)** to **Customers at the Start of the Period**.

   Jan is our starting period. The following rule will set the value as such.

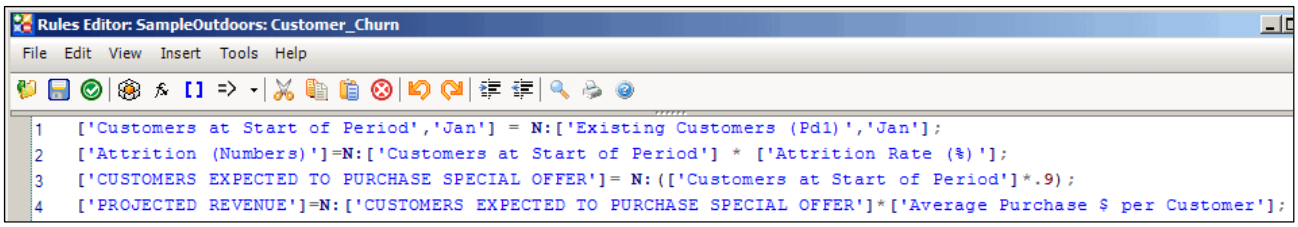
   ['Customers at Start of Period','Jan'] =N:['Existing Customers (Pd1)','Jan'];
3. Enter the following rule to multiply **Customers at the Start of Period** by **Attrition Rate (%).**

   ['Attrition (Numbers)']=N:['Customers at Start of Period']*['Attrition Rate (%)'];
4. Enter the following rule to calculate the number of customers expected to take advantage of the offer (90%).

   ['CUSTOMERS EXPECTED TO PURCHASE SPECIAL OFFER']= N:(['Customers at Start of Period']*.9);
5. Enter the following rule to calculate **PROJECTED REVENUE**:

   ['PROJECTED REVENUE']=N:['CUSTOMERS EXPECTED TO PURCHASE SPECIAL OFFER']*['Average Purchase $ per Customer'];

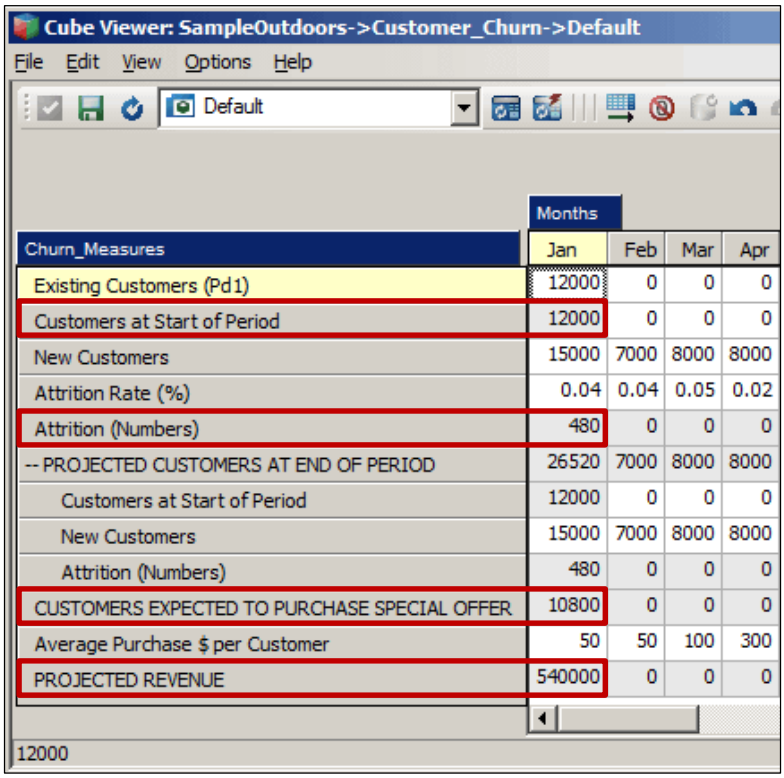
   The results appear as follows:



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6. Save the rule, save the cube view (overwriting if prompted) and then recalculate the cube view.

   The result appears as follows:



Churn_Measures	Months: Jan	Feb	Mar	Apr
Existing Customers (Pd1)	12000	0	0	0
Customers at Start of Period	12000	0	0	0
New Customers	15000	7000	8000	8000
Attrition Rate (%)	0.04	0.04	0.05	0.02
Attrition (Numbers)	480	0	0	0
-- PROJECTED CUSTOMERS AT END OF PERIOD	26520	7000	8000	8000
Customers at Start of Period	12000	0	0	0
New Customers	15000	7000	8000	8000
Attrition (Numbers)	480	0	0	0
CUSTOMERS EXPECTED TO PURCHASE SPECIAL OFFER	10800	0	0	0
Average Purchase $ per Customer	50	50	100	300
PROJECTED REVENUE	540000	0	0	0

12000

   You now need to pull the **PROJECTED CUSTOMERS AT END OF PERIOD** in **Jan** over to the **Customers at Start of Period** for **Feb**.

Task 5. Create a rule using Next and Previous attributes.

1. In the **Rules Editor** for the **Customer_Churn** cube, add a blank line at the end of the rules.
2. Type the following:

   ['Customers at Start of Period']=N:
3. Click **Insert Cube Reference**, and then select the **Customer_Churn** cube.

   In previous examples, you used the Insert Cube Reference button to point to an element in a different cube. This time you are using it to point to an element in the same cube. By using the DB () function, you are able to adjust the dimensions to meet your specific business need.

   In this example, you will alter the !Months reference to use an attribute of the Months dimension that will point to the previous month's data. This is how the value for Feb will pull the value from the Jan cell.

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4. Click the **Subset** icon for **Churn_Measures**.
5. Click **PROJECTED CUSTOMERS AT END OF PERIOD**, click **OK**, and then click **OK** to close the **Insert Cube Reference** dialog box.

   The result appears as follows:

   ['Customers at Start of Period']=N:DB('Customer_Churn', !Months, 'PROJECTED CUSTOMERS AT END OF PERIOD')

   The !Month part of this rule will retrieve the current month name for any given data point. Instead you wish to retrieve the previous Month to pull that data into the starting balance for this month. You will use the ATTRS function in place of the !Month to retrieve the Previous attribute for any given month.
6. Type a **;** at the end of the line.
7. Select **!Months** (in the DB function) and replace it with the following:

   ATTRS('Months', !Months, 'Previous')

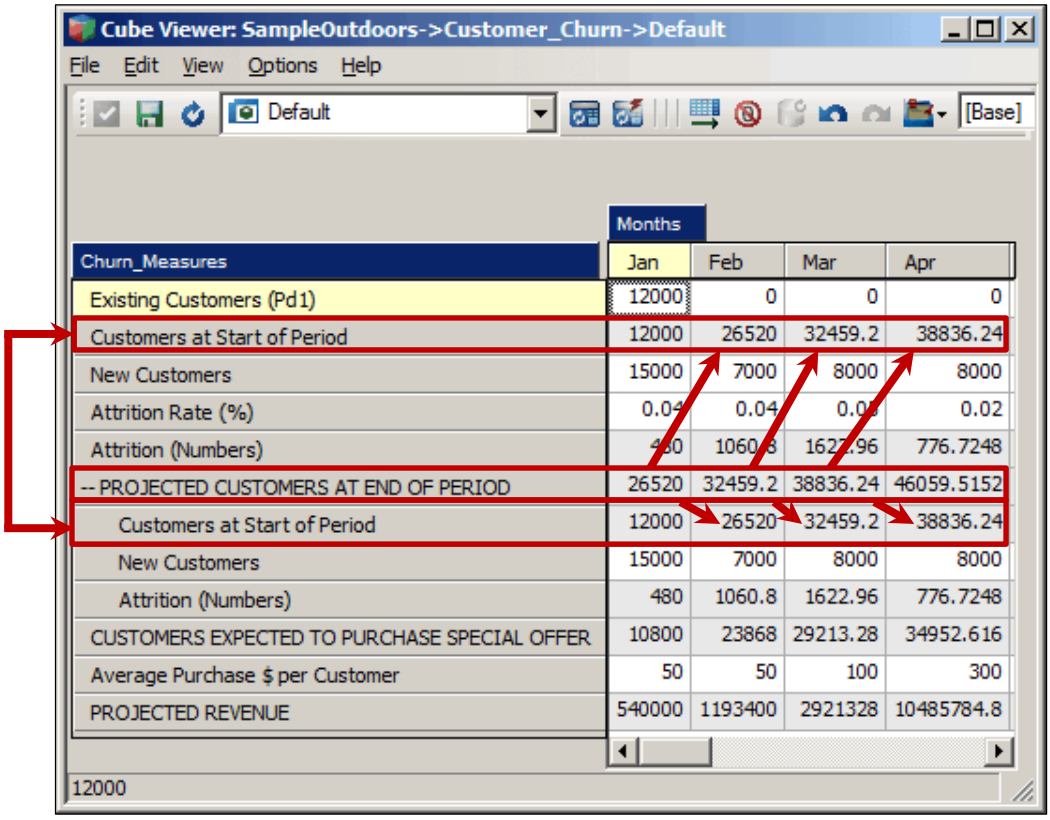
   This will retrieve the Previous attribute for the current element (!Months) in the current cube, in the Months dimension.

   The result appears as follows:

```
['Customers at Start of Period']=N:DB('Customer_Churn',
    ATTRS('Months', !Months, 'Previous') , 'PROJECTED CUSTOMERS AT END OF PERIOD');
```

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8. Save the rule and recalculate the view.
   The result appears as follows:



Task 6. Add SKIPCHECK and FEEDERS.

1. Open the rules for the **Customer_Churn** cube if they are not already open.
2. Add a blank line at the top of the rules and type **SKIPCHECK;**.
3. Enter a blank line at the bottom of the rules and type, **FEEDERS;**.
4. Add another blank line and type the following:

   ['Existing customers (Pd1)','Jan']=>['Customers at Start of Period','Jan'];
   ['Attrition Rate (%)']=>['Attrition (Numbers)'];
   ['Customers at Start of Period']=>['CUSTOMERS EXPECTED TO PURCHASE SPECIAL OFFER'];
   ['Average Purchase $ per Customer']=>['PROJECTED REVENUE'];

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5. Press **Enter** and type the following text:

   ['PROJECTED CUSTOMERS AT END OF PERIOD']=>
   DB('Customer_Churn', ATTRS('Months', !Months, 'Next'),
   'Customers at Start of Period');

   In order to feed the **Customers at Start of Period**, you need to place a marker in the **PROJECTED CUSTOMERS AT END OF PERIOD** to push into the Next month.

   Rules pull values into cells but feeders are pushed out to cells. In Feb the rule is pulling data from Jan but the marker must be pushed from Jan to the Feb cell.

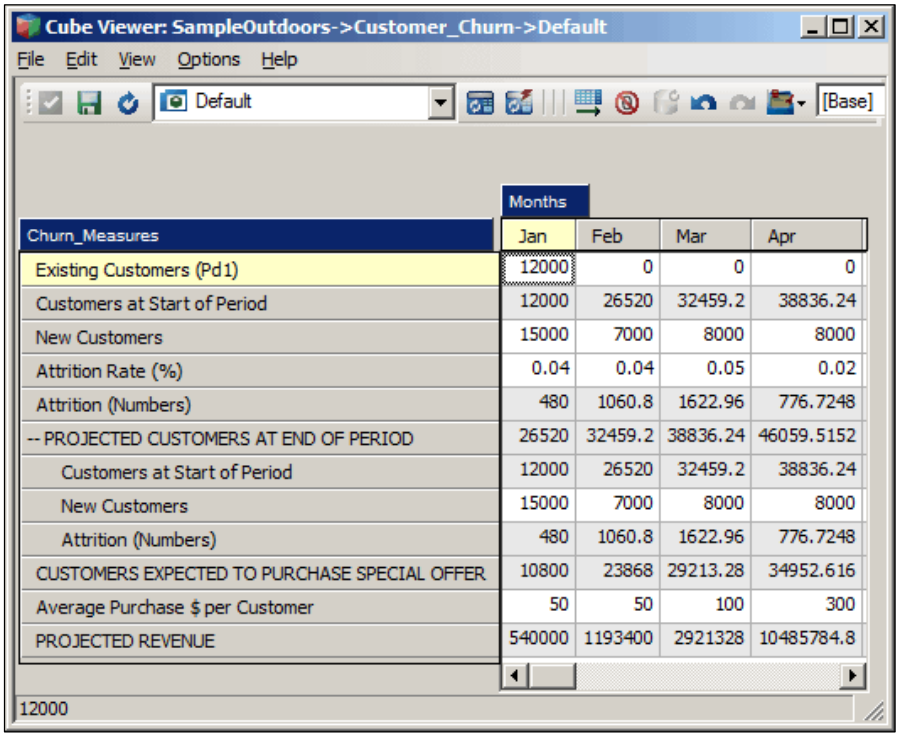
   The result appears as follows:

```
1   SKIPCHECK;
2   ['Customers at Start of Period','Jan'] =N:['Existing Customers (Pd1)','Jan'];
3   ['Attrition (Numbers)']=N:['Customers at Start of Period']* ['Attrition Rate (%)'];
4   ['CUSTOMERS EXPECTED TO PURCHASE SPECIAL OFFER']= N:(['Customers at Start of Period']*.9);
5   ['PROJECTED REVENUE']=N:['CUSTOMERS EXPECTED TO PURCHASE SPECIAL OFFER']*
6       ['Average Purchase $ per Customer'];
7
8   ['Customers at Start of Period']=N:DB('Customer_Churn',
9       ATTRS('Months', !Months, 'Previous') , 'PROJECTED CUSTOMERS AT END OF PERIOD');
10
11  FEEDERS;
12  ['Existing customers (Pd1)','Jan']=>['Customers at Start of Period','Jan'];
13  ['Attrition Rate (%)']=>['Attrition (Numbers)'];
14  ['Customers at Start of Period']=>['CUSTOMERS EXPECTED TO PURCHASE SPECIAL OFFER'];
15  ['Average Purchase $ per Customer']=>['PROJECTED REVENUE'];
16  ['PROJECTED CUSTOMERS AT END OF PERIOD']=> DB('Customer_Churn', ATTRS('Months', !Months, 'Next'),
17      'Customers at Start of Period');
```

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6. Save the rule and recalculate the view.

   The result appears as follows:



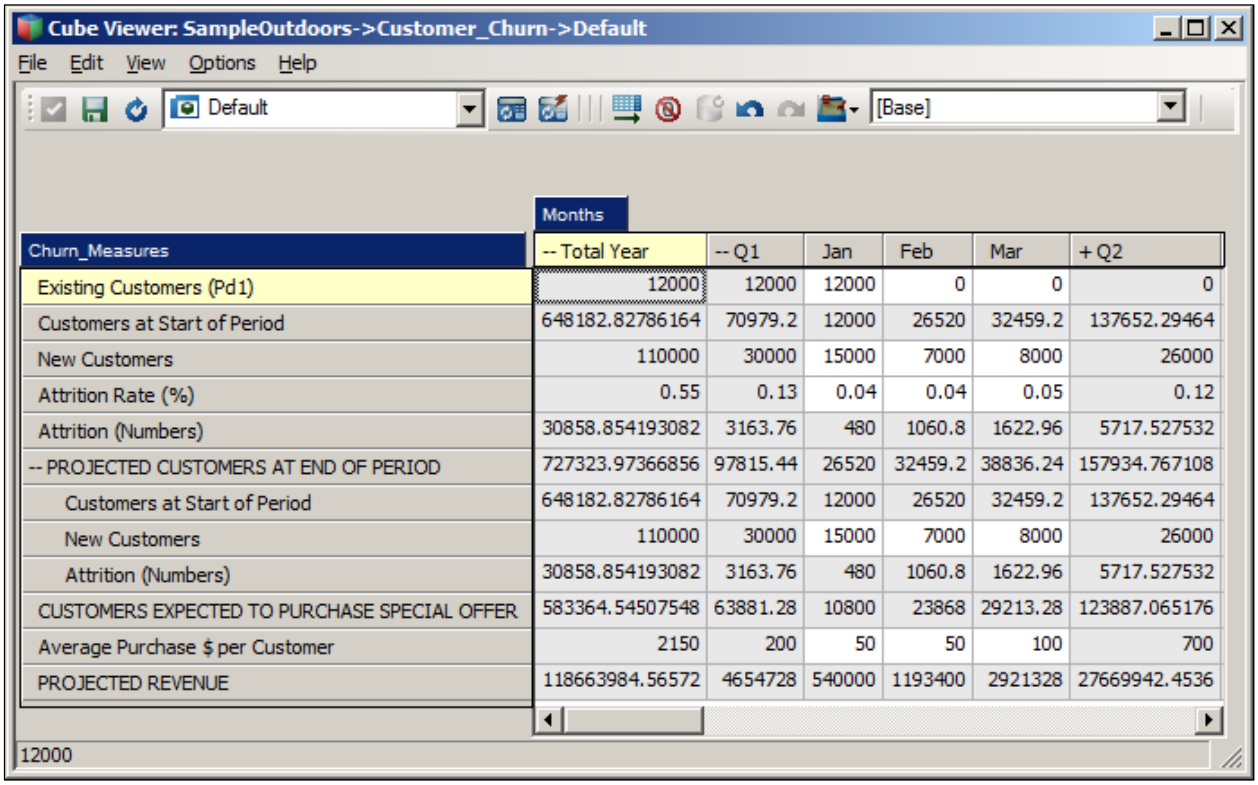
Task 7. Adjust rules so opening and closing balances do not aggregate across time.

In this example, **Customers at Start of Period** should not aggregate at the consolidated levels. Instead, the value at the end of the quarter is equal to the value at the end of the last month in the quarter. Therefore, we will write rules at the consolidated levels to override the automatic aggregation.

1. In the **Default** view of the **Customer_Churn** cube, click **Months** to open the **Subset Editor**.
2. Click **All**, click **Total Year** and then click **OK**.

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3. Click **Recalculate** and then expand **Total Year** and **Q1**.



4. Open the rules for the **Customer_Churn** cube.
5. Replace the rule for ['Customers at the Start of Period', 'Jan'] (on **line 2**) with the following:

   ['Customers at Start of Period', {'Jan','Q1','Total Year'}]=['Existing Customers (Pd1)','Jan'];

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6. Remove the **N:** from the following line:

   ['Customers at Start of Period']=N:DB('Customer_Churn', ATTRS('Months', !Months, 'Previous'), 'PROJECTED CUSTOMERS AT END OF PERIOD');

   This will override the aggregation by allowing the Customers at Start of Period to be computed at the consolidated level as well as at the base level.

   The results appear as follows:

```
1   SKIPCHECK;
2   ['Customers at Start of Period', {'Jan','Q1','Total Year'}] = ['Existing Customers (Pd1)','Jan'];
3   ['Attrition (Numbers)']=N:['Customers at Start of Period'] * ['Attrition Rate (%)'];
4   ['CUSTOMERS EXPECTED TO PURCHASE SPECIAL OFFER']= N:(['Customers at Start of Period']*.9);
5   ['PROJECTED REVENUE']=N:['CUSTOMERS EXPECTED TO PURCHASE SPECIAL OFFER']*
6       ['Average Purchase $ per Customer'];
7
8   ['Customers at Start of Period']=DB('Customer_Churn',
9       ATTRS('Months', !Months, 'Previous'), 'PROJECTED CUSTOMERS AT END OF PERIOD');
```

7. Add the following rules before the **FEEDERS;**

   ['PROJECTED CUSTOMERS AT END OF PERIOD','Q1'] = C:['Mar'];

   ['PROJECTED CUSTOMERS AT END OF PERIOD','Q2'] = C:['Jun'];

   ['PROJECTED CUSTOMERS AT END OF PERIOD','Q3'] = C:['Sep'];

   ['PROJECTED CUSTOMERS AT END OF PERIOD',{'Q4','Total Year'}] = C:['Dec'];

   These statements will set the year and quarters to the last Month closing balance.

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8. Remove the last feeder statement.

   You remove this feeder because we removed the N: from the ['Customers at Start of Period'] on line 8 (see above). Now this rule is being calculated for the consolidated levels as well as the N level which overrides the consolidation engine. Because you no longer need the consolidation engine to calculate the values, you may remove the feeder.

   The results appear as follows:

```
1   SKIPCHECK;
2   ['Customers at Start of Period', {'Jan','Q1','Total Year'}] = ['Existing Customers (Pd1)','Jan'];
3   ['Attrition (Numbers)']=N:['Customers at Start of Period'] * ['Attrition Rate (%)'];
4   ['CUSTOMERS EXPECTED TO PURCHASE SPECIAL OFFER']= N:(['Customers at Start of Period']*.9);
5   ['PROJECTED REVENUE']=N:['CUSTOMERS EXPECTED TO PURCHASE SPECIAL OFFER']*
6        ['Average Purchase $ per Customer'];
7
8   ['Customers at Start of Period']=DB('Customer_Churn',
9        ATTRS('Months', !Months, 'Previous'), 'PROJECTED CUSTOMERS AT END OF PERIOD');
10
11  ['PROJECTED CUSTOMERS AT END OF PERIOD','Q1'] = C:['Mar'];
12  ['PROJECTED CUSTOMERS AT END OF PERIOD','Q2'] = C:['Jun'];
13  ['PROJECTED CUSTOMERS AT END OF PERIOD','Q3'] = C:['Sep'];
14  ['PROJECTED CUSTOMERS AT END OF PERIOD',{'Q4','Total Year'}] = C:['Dec'];
15
16  FEEDERS;
17  ['Existing customers (Pd1)','Jan']=>['Customers at Start of Period','Jan'];
18  ['Attrition Rate (%)']=>['Attrition (Numbers)'];
19  ['Customers at Start of Period']=>['CUSTOMERS EXPECTED TO PURCHASE SPECIAL OFFER'];
20  ['Average Purchase $ per Customer']=>['PROJECTED REVENUE'];
```

   If you are using the advanced rules editor, you may use Regions to better organize your rules and feeders. You simply create a line above the lines you would like to indent and type **#Region NameOfRegion**.

9. On the line before FEEDERS; type the following:

   #Region Feeder Statements

   You should see a minus sign with a line going down to the end of the file.

```
16  #Region Feeder Statements
17  FEEDERS;
18  ['Existing customers (Pd1)','Jan']=>['Customers at Start of Period','Jan'];
19  ['Attrition Rate (%)']=>['Attrition (Numbers)'];
20  ['Customers at Start of Period']=>['CUSTOMERS EXPECTED TO PURCHASE SPECIAL OFFER'];
21  ['Average Purchase $ per Customer']=>['PROJECTED REVENUE'];
22
```

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10. After the last feeder statement type, **#endregion**

```
16  #Region Feeder Statements
17  FEEDERS;
18  ['Existing customers (Pd1)','Jan']=>['Customers at Start of Period','Jan'];
19  ['Attrition Rate (%)']=>['Attrition (Numbers)'];
20  ['Customers at Start of Period']=>['CUSTOMERS EXPECTED TO PURCHASE SPECIAL OFFER'];
21  ['Average Purchase $ per Customer']=>['PROJECTED REVENUE'];
22  #endregion
```

11. Click the minus sign to collapse the statements.

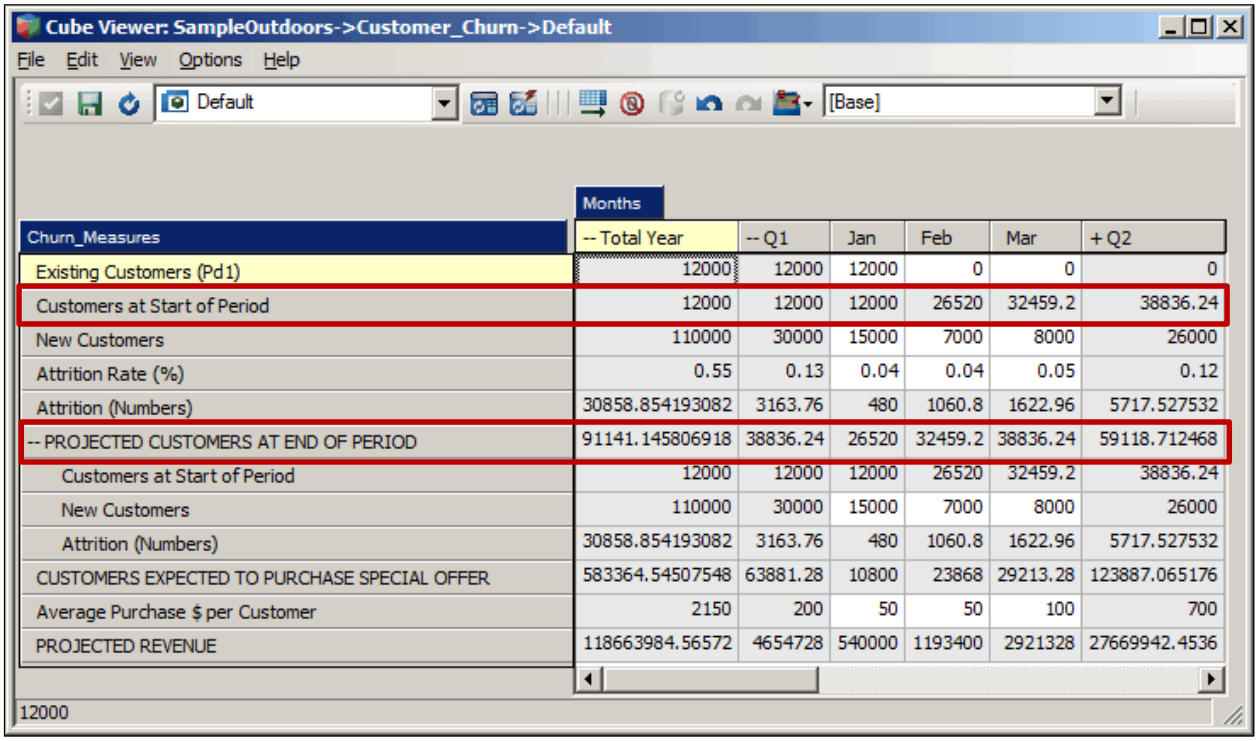
```
15
16 ⊞ Feeder Statements
```

You must be very careful when inserting and deleting lines in the advanced rules editor. It is very easy to delete a line using regions because you may not realize the whole region is selected. Use caution when pressing Enter or Delete.

12. Save and close the rules and the **Customer_Churn** cube (overwriting if prompted).
13. In Server Explorer, right-click **Customer_Churn** and then click **Unload Cube**.
14. Open the **Default** view of the **Customer_Churn** cube (if necessary).
15. If necessary, click **Months**, click **All**, and then click **OK**.

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16. Click **Recalculate**, and then expand **Q1**.
    The results appear as follows:



17. Close the **Rules Editor** and **Cube Viewer** windows, saving and overwriting if prompted.
18. Right-click the **SampleOutdoors** node and click **Save Data**.
19. Close **Architect**.

Results:
You created the customer churn cube and populated it with data from a text file and rules. You created rules that derive data from a lookup cube to determine moving balances from one time period to another.

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Unit summary

- At the end of this module, you should be able to:
  - describe virtual cubes
  - list uses for lookup cubes
  - create and use a spread profiles cube
  - implement moving balances in a cube

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Unit summary

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IBM Training

Convert currencies

IBM Planning Analytics (v2.0)

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IBM Training

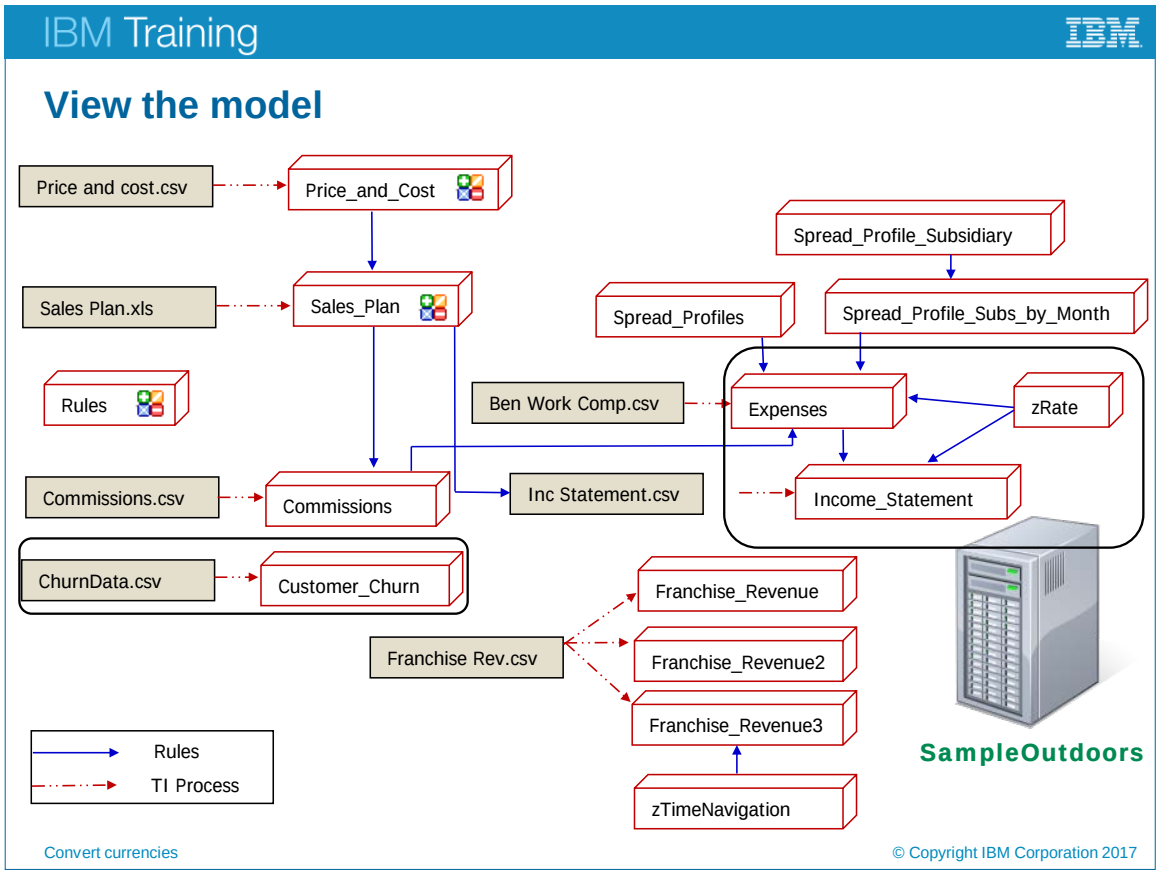
Unit objectives

- Describe currency challenges
- Create a currency dimension
- Create a currency cube
- Create rules for currency conversion

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Unit objectives

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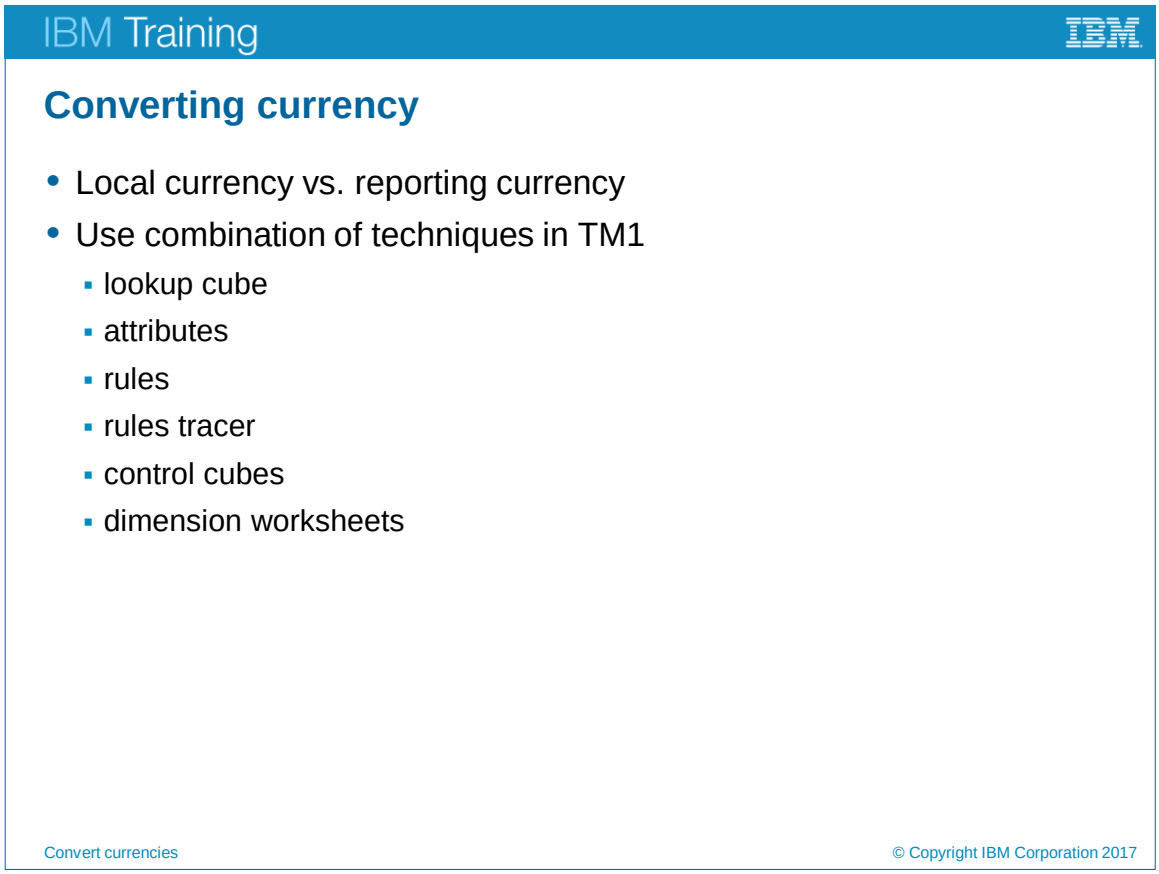


View the model

In this unit, you will create the following cubes:

- zRate
- Income_Statement

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Converting currency

In today's global markets, financial data is often coming to us in different currencies.

- Data may be input in a region's local currency.
- Reporting or analysis needs require translation.

You will want to make currency conversions easy to maintain and simple to expand.

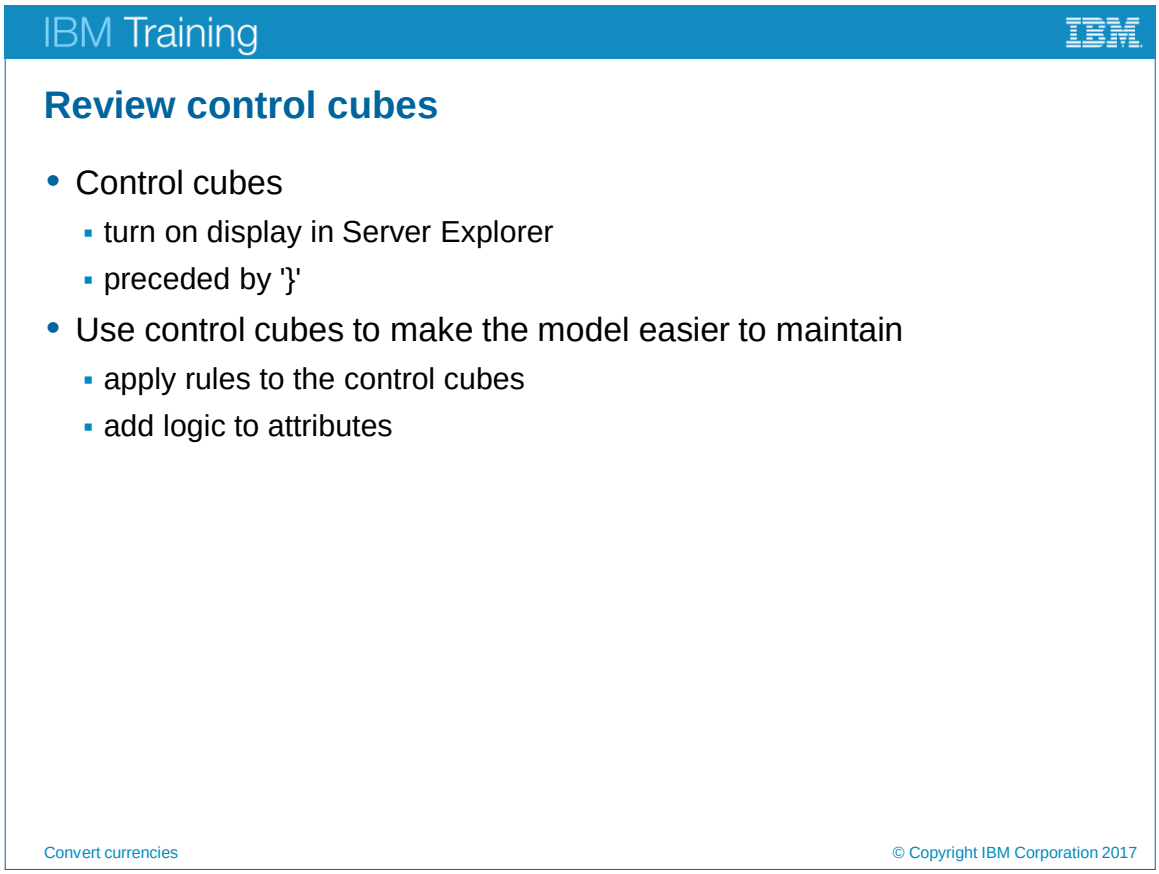
In this unit we will reuse many of the techniques from previous units (rules, attributes, logic, and lookup cubes) but we will combine them with control cubes to make our currency translation easier to expand.

This example will focus simply on currency without considering time. Currency conversion is also time dependent and your logic must reflect this as well.

Dimension worksheets are modified Microsoft Excel spreadsheets used to create and maintain elements and hierarchical relationships in dimensions.

Dimension worksheets are most often used in legacy applications but they may also make it easier to share information between models. For more information see the *TM1 for Developers* guide.

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Review control cubes

Types of data stored in control cubes include:

- element attributes
- client, cube, and server performance information
- security information

You can create views, rules, and reports on control cubes.

We will add rules containing logic to our attributes cube in order to reduce the amount of manual maintenance performed on them.

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IBM Training

Demonstration 1

Include reporting currencies in a cube

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Demonstration 1: Include reporting currencies in a cube

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Demonstration 1: Include reporting currencies in a cube

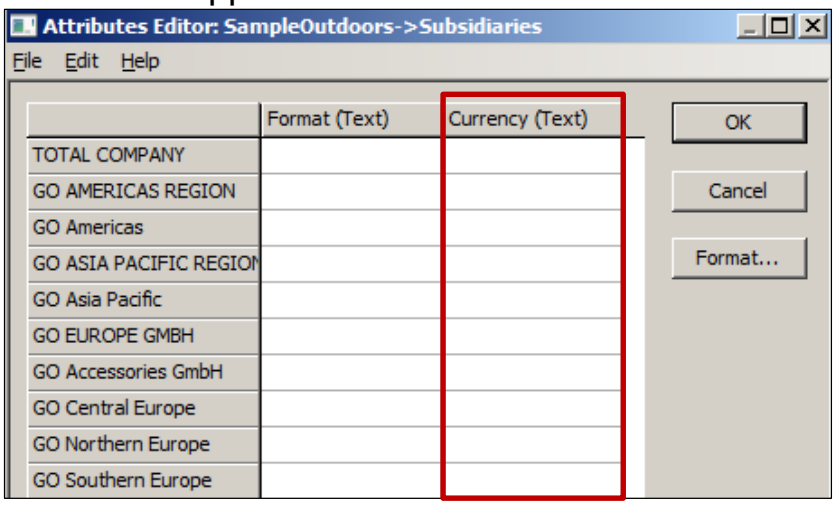
Purpose:
Sample Outdoors is a global corporation with a need to report financial results in three different currencies: U.S. Dollars, Japanese Yen, and Euros. It is assumed that each subsidiary reports sales and expenses in their local currency and then you must convert those amounts to the reporting currencies.

Task 1. Add a reporting currency attribute to the Subsidiaries and Expense_Measures dimensions.

1. Open **Windows Explorer** and go to **C:\Program Files\IBM\cognos\tm1_64\bin**.

   You will be using the Advanced Rules Editor, so you will open the 32-bit version of Architect.
2. Double-click **tm1a.exe**.
3. Maximize the **Architect** window.
4. Expand the **TM1** node, double-click **SampleOutdoors**, and then log on with a user ID of **admin** and a password of **apple**.
5. Expand **Dimensions**, right-click **Subsidiaries**, and then click **Edit Element Attributes**.
6. From the **Edit** menu, click **Add New Attribute**.
7. In the **Name** box type **Currency** and then click **OK**.

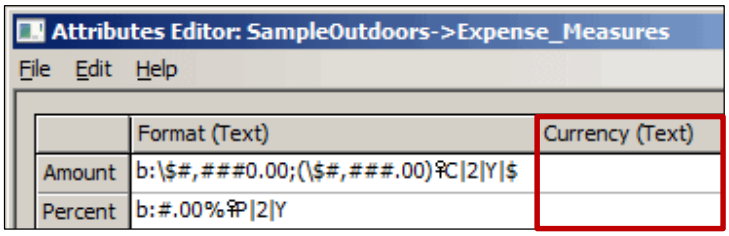
   The results appear as follows:



You will populate the Currency attribute later using a rule.

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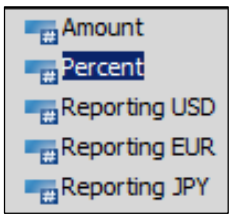
8. Click **OK**.
9. Repeat steps **3** to **5** for the **Expense_Measures** dimension.
   The results appear as follows:



10. Click **OK**.

Task 2. Add new measures and attributes to the Expense_Measures dimension.

1. In Server Explorer, right-click **Expense_Measures** and then click **Edit Dimension Structure**.
2. Click the **Percent** element.
3. From the **Edit** menu, and then click **Insert Element**.
4. Add the following simple elements:
   - **Reporting USD**
   - **Reporting EUR**
   - **Reporting JPY**
5. Click **OK**.
   The results appear as follows:



6. Save and close the **Expense_Measures** dimension.

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Task 3. Create the zRate dimensions and cube.

1. In Server Explorer, right-click **Dimensions** and then click **Create New Dimension**.

   We will create three new dimensions: **Currency**, **CurrencyType**, and **CurrencyConvertTo**.
2. From the **Edit** menu, click **Insert Element**.
3. Enter the following currency symbols as simple elements:
   - **BRL**
   - **GBP**
   - **CAD**
   - **EUR**
   - **JPY**
   - **SEK**
   - **USD**
4. Save the dimension and name it **Currency**.

   The results appear as follows:

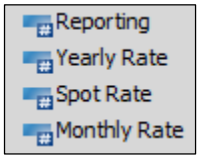
   

   You will only be using the EUR, JPY, and USD rates in this exercise, but the other currencies will be available for potential future use.
5. Close the **Dimension Editor**, and then create a new dimension.
6. Add the following simple elements:
   - Reporting
   - Yearly Rate
   - Spot Rate
   - Monthly Rate

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7. Save the dimension as **CurrencyType**.

   The results appear as follows:

   This dimension is helpful if you need to look at data or reports using different rates. In this example you will use the Reporting rate.

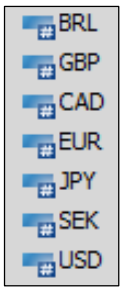
8. Close the **Dimension Editor** and create a new dimension.
9. Add the following simple elements:

   - **BRL**
   - **GBP**
   - **CAD**
   - **EUR**
   - **JPY**
   - **SEK**
   - **USD**

   You will only be using the EUR, JPY, and USD rates in this exercise, but the other currencies will be available for potential future use.

10. Save the dimension as **CurrencyConvertTo**.

    The results appear as follows:

    Models are often created for multiple regions. Data may be input into it in a variety of formats and then the data is converted. You created one dimension to store the values in the local currency (**Currency**), one dimension to store the type of rate you need to convert to (**CurrencyType**), and finally the currency the data will be converted to (**CurrencyConvertTo**).

11. Close the **Dimension Editor**.

    You will now add the **Currency**, **CurrencyType**, and **CurrencyConvertTo** dimensions to the **zRate** cube, which will store the currency rates. This cube will also contain the **Months** dimension.

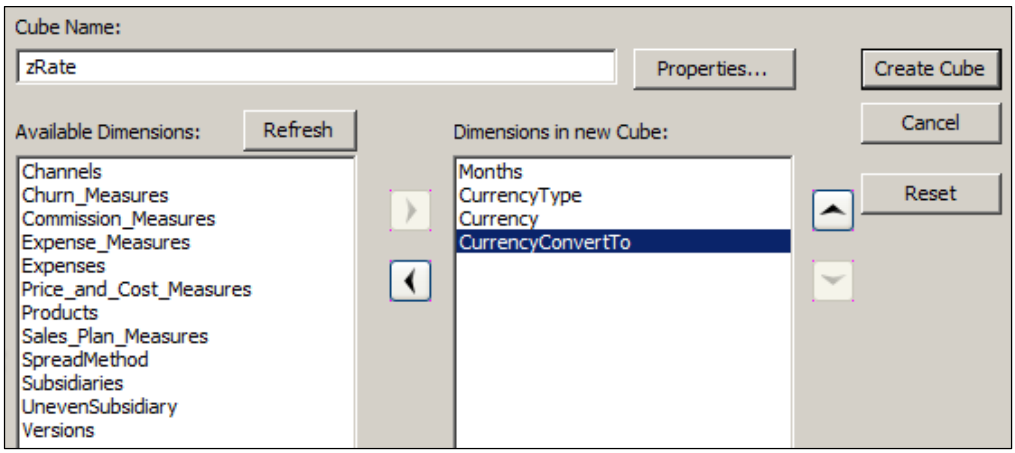
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Task 4. Create and populate the zRate cube.

1. In Server Explorer, create a new cube called **zRate**.
2. Add the following dimensions:
   - **Months**
   - **CurrencyType**
   - **Currency**
   - **CurrencyConvertTo**

   The results appear as follows:



3. Click **Create Cube**, and then double-click the **zRate** cube.
4. Double-click the **Months** dimension (currently displaying the **Total Year** element), and then click **All**.
5. Click **Jan**, click **OK**, and then click **Recalculate**.

   The results appear as follows:

Jan | Reporting

	CurrencyConvertTo						
Currency	BRL	GBP	CAD	EUR	JPY	SEK	USD
BRL	0	0	0	0	0	0	0
GBP	0	0	0	0	0	0	0
CAD	0	0	0	0	0	0	0
EUR	0	0	0	0	0	0	0
JPY	0	0	0	0	0	0	0
SEK	0	0	0	0	0	0	0
USD	0	0	0	0	0	0	0

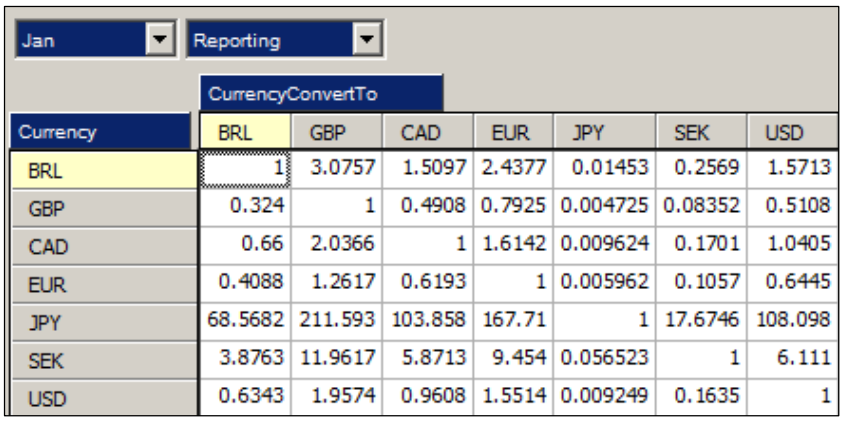
6. Save as the **public Default** view.
7. In **Microsoft Excel**, open **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles\currency_rates.xls**.
8. Select cells **B2:H8** and press **Ctrl+C** to copy.

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9. Select all of the cells in the cube view, and then press **Ctrl+V** to paste.
10. Click **Automatic Recalculate**  , and then repeat step **9** for each month (select the month from the Month dimension in the title area, where **Jan** is currently selected).

    Each month will have the same exchange rates.

    The results appear as follows:



Jan | Reporting

Currency	CurrencyConvertTo: BRL	GBP	CAD	EUR	JPY	SEK	USD
BRL	1	3.0757	1.5097	2.4377	0.01453	0.2569	1.5713
GBP	0.324	1	0.4908	0.7925	0.004725	0.08352	0.5108
CAD	0.66	2.0366	1	1.6142	0.009624	0.1701	1.0405
EUR	0.4088	1.2617	0.6193	1	0.005962	0.1057	0.6445
JPY	68.5682	211.593	103.858	167.71	1	17.6746	108.098
SEK	3.8763	11.9617	5.8713	9.454	0.056523	1	6.111
USD	0.6343	1.9574	0.9608	1.5514	0.009249	0.1635	1

11. Close **Cube Viewer** without saving changes to the **Default** view.
12. Close **Microsoft Excel** without saving changes.

Task 5. Populate the Currency attributes for the Subsidiary and Expense_Measures dimensions using control cubes.

1. In Server Explorer, from the **View** menu, click **Display Control Objects** if this option is not already selected.

   Internally TM1 uses cubes to store information about your model (including attributes).

   In order to make maintenance of our Currency attribute (in the Subsidiaries dimension) automatic, we will add a rule to apply the parent currency rate to its children.

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2. In the list of cubes, right-click **}ElementAttributes_Subsidiaries** and then click **Create Rule**.

   The Rules Editor opens (this may take a few moments).

   This rule uses the following functions:

   The ELPAR function to return the parent of an element in a specified dimension.

   SYNTAX: ELPAR(dimension, element, index)

   Note: the index refers to the index of the dimension hierarchy. It will usually be 1 unless there are multiple hierarchies in the dimension.

   IF returns one value if a logical expression you specify is TRUE and another value if it is FALSE.

   SYNTAX: IF(expression, true_value, false_value)

3. Enter the following rule:

   ['Currency']=S:IF(ELPAR('Subsidiaries',!Subsidiaries, 1)@=

   'GO AMERICAS REGION','USD', continue);

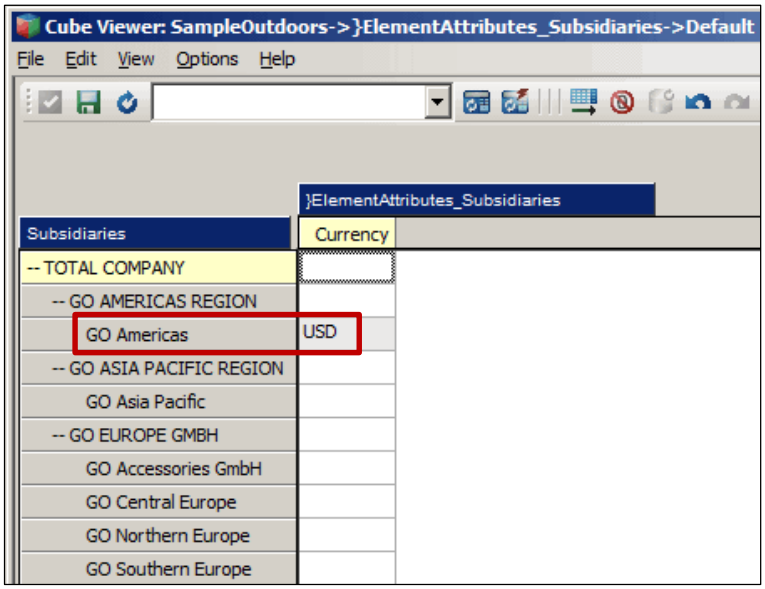
   This rule states: for the currency, look to the element's parent. If the parent is GO AMERICAS REGION, the currency will be USD; otherwise continue.

   The **continue** statement allows many rules over the same area and you will use a **STET** at the end to allow cells not covered by this function to be entered manually.

4. Save the rule.

5. Double-click the **}ElementAttributes_Subsidiaries** cube, configure it to match the graphic shown below, and then click **Recalculate**.

   The results appear as follows:



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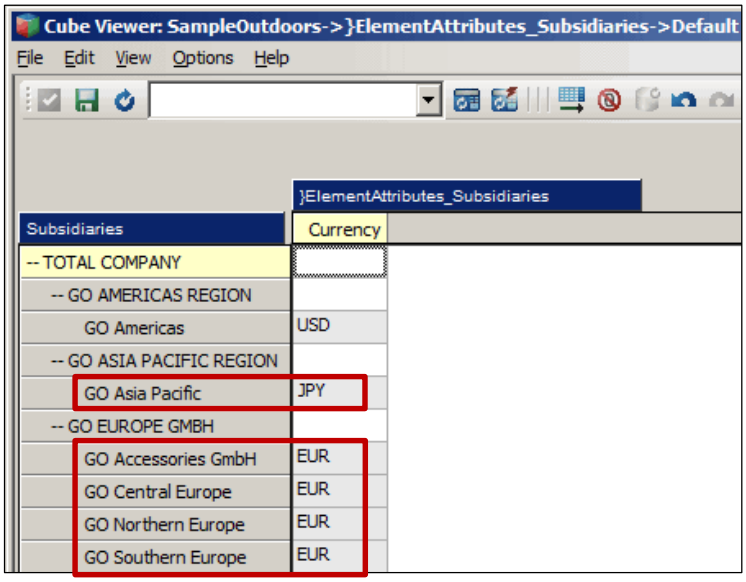
6. Add the following rules in the **Rules Editor** window:

   ['Currency']=S:IF(ELPAR('Subsidiaries',!Subsidiaries, 1)@= 'GO ASIA PACIFIC REGION','JPY', continue);

   ['Currency']=S:IF(ELPAR('Subsidiaries',!Subsidiaries, 1)@= 'GO EUROPE GMBH','EUR', STET);

7. Save the rules and recalculate the cube view.

   The results appear as follows:



8. Close the **Rules Editor** and the **Default** view of the **}ElementAttributes_Subsidiaries** cube, without saving the view.

   You will now use rules to populate the Currency attribute of the Expense_Measures dimension.

9. Double-click the **}ElementAttributes_Expense_Measures** control cube and recalculate the view.

   The Currency attribute does not have values yet.

10. Right-click the **}ElementAttributes_Expense_Measures** control cube and then click **Create Rule**.

11. Enter the following rule:

    ['Currency']=S:IF(SUBST(!Expense_Measures, 1, 9)@='Reporting', SUBST(!Expense_Measures,11, 3), STET);

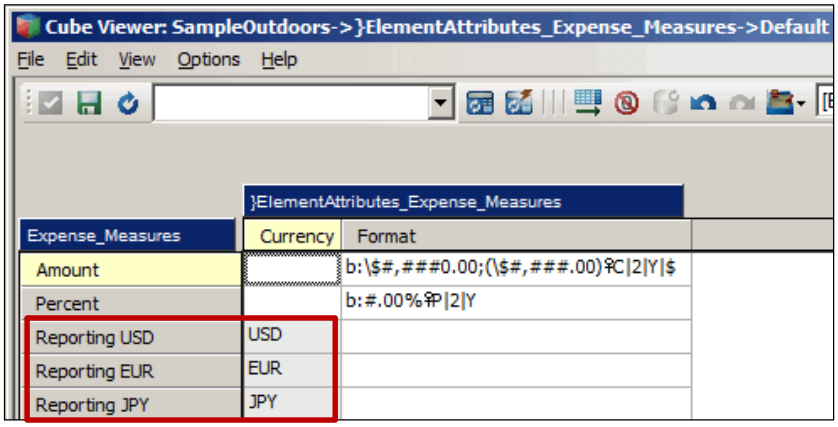
    SUBST returns a substring of a given string.

    SYNTAX: SUBST(string, beginning_position, length)

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12. Save the rule and recalculate the view.

    The results appear as follows:



    The rule looks to the element name to determine the value for the reporting currency. If you were to add a new reporting currency to the Expense_Measures dimension (for example, "Reporting GBP") you would follow the naming convention applied to the previous elements and the rule would update the currency attribute automatically (for example, it would assign "GBP" as the value of the Currency attribute for the "Reporting GBP" element).

13. Close the **Rules Editor** and the **}ElementAttributes_Expense_Measures** control cube (do not save the control cube view if prompted).

14. In Server Explorer, from the **View** menu, click **Display Control Objects** to turn off display of the control cubes.

    Because you created rules to automatically assign the currency attributes on elements in these dimensions, you will not need to manually update them. When a new reporting currency is added, simply create the new element in the Expense_Measures dimension (using the Reporting *xxx* format, such as "Reporting GBP") and it will automatically be added as an attribute.

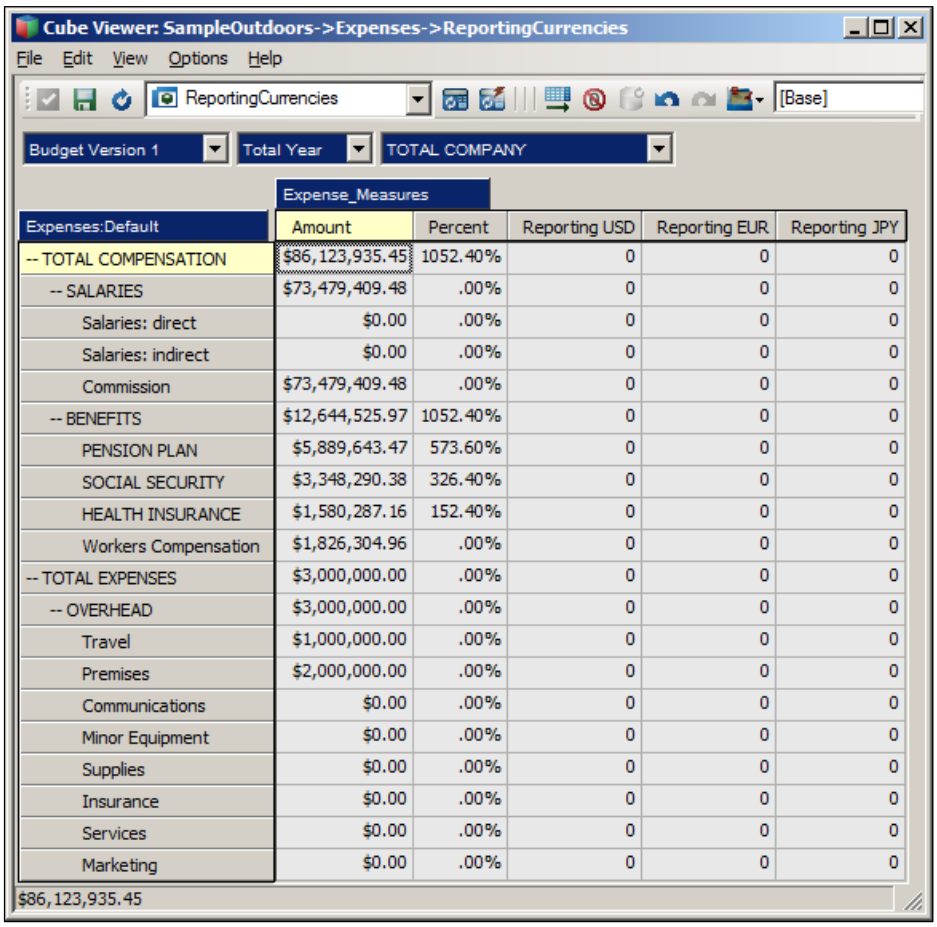
    As new subsidiaries are added to the Subsidiaries dimension, the currency attribute will also be added automatically. You would revise the rule on the }ElementAttributes_Subsidiaries control cube to supply Currency attribute values for any subsidiaries and their children.

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Task 6. Create rules to calculate the exchange rate.

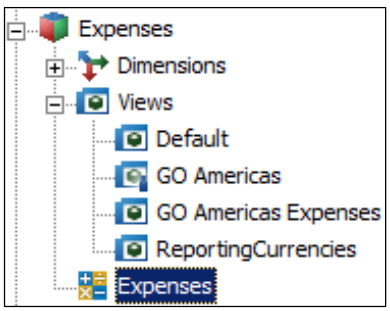
1. In Server Explorer, double-click the **Expenses** cube and create a new public view called **ReportingCurrencies**.
2. Click the **Expense_Measures** dimension (on columns) to open the **Subset Editor**, click **All**, and then click **OK**.
3. Ensure that the dimensions appear in the order shown below, and then click **Recalculate**.

   The results appear as follows:



Expenses:Default	Amount	Percent	Reporting USD	Reporting EUR	Reporting JPY
-- TOTAL COMPENSATION	$86,123,935.45	1052.40%	0	0	0
-- SALARIES	$73,479,409.48	.00%	0	0	0
Salaries: direct	$0.00	.00%	0	0	0
Salaries: indirect	$0.00	.00%	0	0	0
Commission	$73,479,409.48	.00%	0	0	0
-- BENEFITS	$12,644,525.97	1052.40%	0	0	0
PENSION PLAN	$5,889,643.47	573.60%	0	0	0
SOCIAL SECURITY	$3,348,290.38	326.40%	0	0	0
HEALTH INSURANCE	$1,580,287.16	152.40%	0	0	0
Workers Compensation	$1,826,304.96	.00%	0	0	0
-- TOTAL EXPENSES	$3,000,000.00	.00%	0	0	0
-- OVERHEAD	$3,000,000.00	.00%	0	0	0
Travel	$1,000,000.00	.00%	0	0	0
Premises	$2,000,000.00	.00%	0	0	0
Communications	$0.00	.00%	0	0	0
Minor Equipment	$0.00	.00%	0	0	0
Supplies	$0.00	.00%	0	0	0
Insurance	$0.00	.00%	0	0	0
Services	$0.00	.00%	0	0	0
Marketing	$0.00	.00%	0	0	0

4. Open the rules for the **Expenses** cube.



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5. Insert two blank lines before the **FEEDERS;** statement.

   You will be multiplying the dollar values in the **Amount** column by the exchange rate. The exchange rate will be pulled from the **zRate** cube using the FROM currency specified in the attribute in the **Subsidiaries** dimension and the TO currency specified in the attribute in the **Expense_Measures** dimension. You will use the ATTRS function to get the attributes of these dimensions.

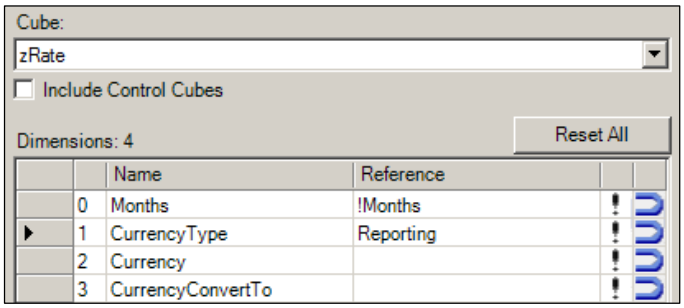
6. Enter the following directly above the **FEEDERS;** statement:

   ['Reporting USD']=N:['Amount']*

7. Click **Insert Cube Reference**, and then in the **Cube** list, select the **zRate** cube.

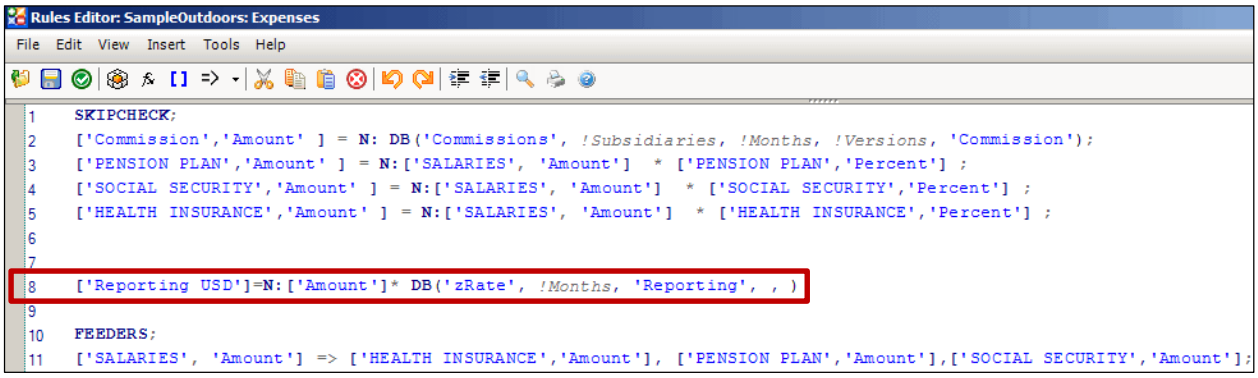
8. Click the **Subset** icon next to **CurrencyType**, click **Reporting**, and then click **OK**.

   The results appear as follows:



9. Click **OK**.

   The results appear as follows (your lines may be indented differently, but this is acceptable):

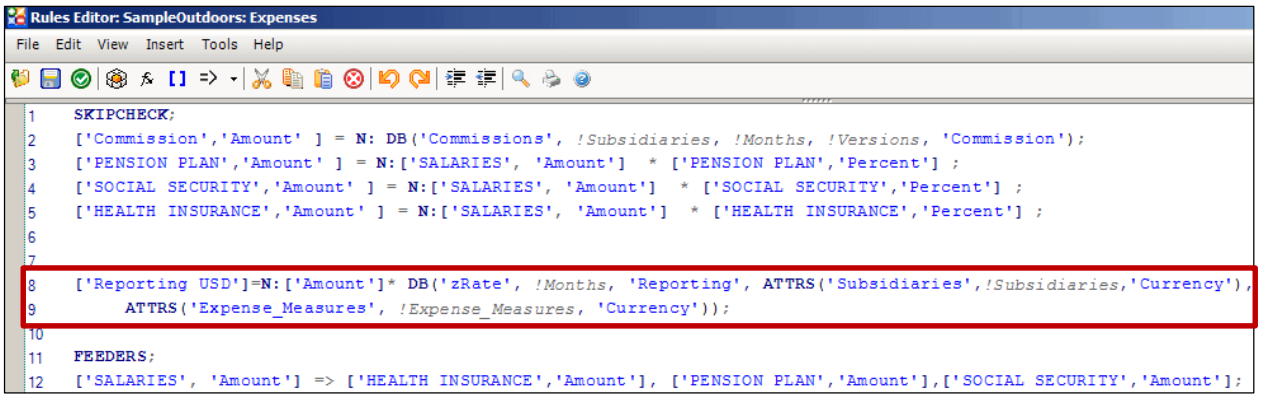


   You need to use ATTRS functions to point to the Currency and CurrencyConvertTo dimensions.

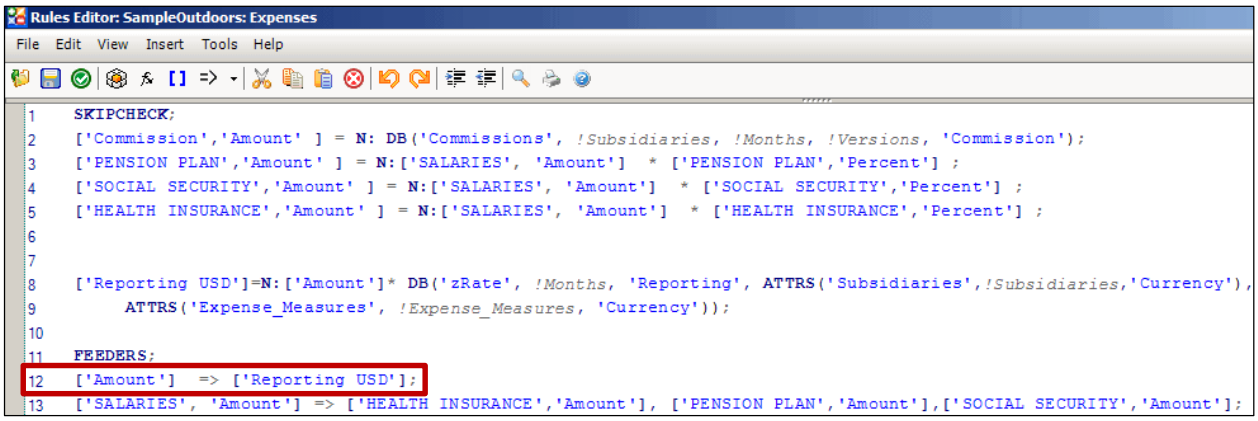
   In between the two commas will be ATTRS('Subsidiaries', !Subsidiaries, 'Currency').

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10. Click in between the last two commas and type
    ATTRS('Subsidiaries',!Subsidiaries,'Currency')
11. Click after the last comma and type:
    ATTRS('Expense_Measures', !Expense_Measures, 'Currency')
12. Add a **;** to the end of the line and save the rule.
    The results appear as follows:



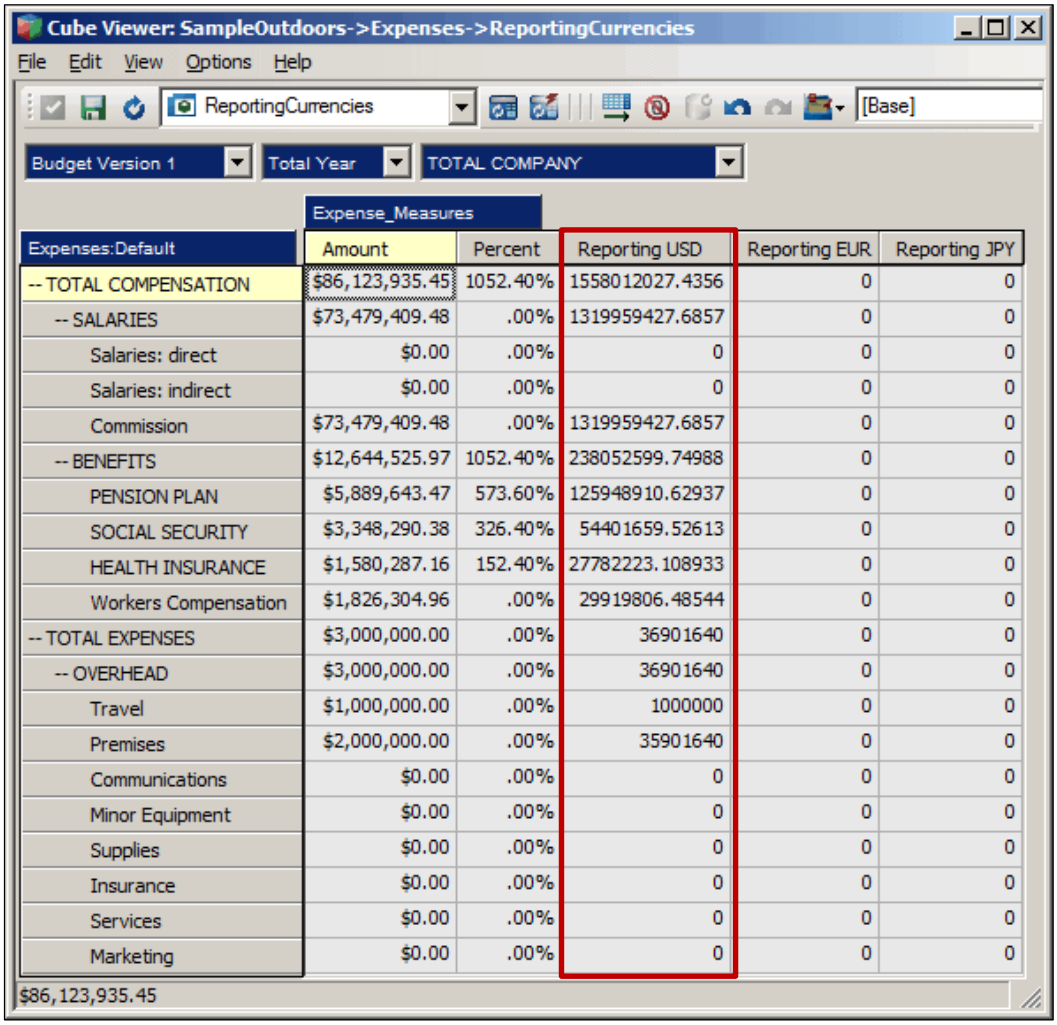
13. Add the feeder:
    ['Amount'] => ['Reporting USD'];
    The results appear as follows:



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14. Save the rule and recalculate the view.

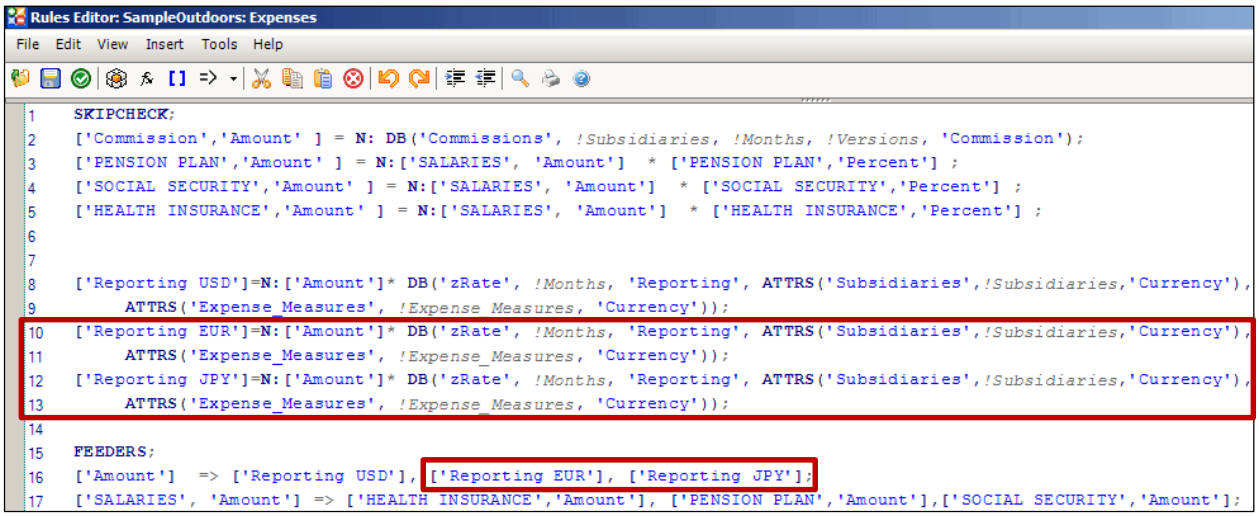
    The results appear as follows:



Expenses:Default	Amount	Percent	Reporting USD	Reporting EUR	Reporting JPY
-- TOTAL COMPENSATION	$86,123,935.45	1052.40%	1558012027.4356	0	0
-- SALARIES	$73,479,409.48	.00%	1319959427.6857	0	0
Salaries: direct	$0.00	.00%	0	0	0
Salaries: indirect	$0.00	.00%	0	0	0
Commission	$73,479,409.48	.00%	1319959427.6857	0	0
-- BENEFITS	$12,644,525.97	1052.40%	238052599.74988	0	0
PENSION PLAN	$5,889,643.47	573.60%	125948910.62937	0	0
SOCIAL SECURITY	$3,348,290.38	326.40%	54401659.52613	0	0
HEALTH INSURANCE	$1,580,287.16	152.40%	27782223.108933	0	0
Workers Compensation	$1,826,304.96	.00%	29919806.48544	0	0
-- TOTAL EXPENSES	$3,000,000.00	.00%	36901640	0	0
-- OVERHEAD	$3,000,000.00	.00%	36901640	0	0
Travel	$1,000,000.00	.00%	1000000	0	0
Premises	$2,000,000.00	.00%	35901640	0	0
Communications	$0.00	.00%	0	0	0
Minor Equipment	$0.00	.00%	0	0	0
Supplies	$0.00	.00%	0	0	0
Insurance	$0.00	.00%	0	0	0
Services	$0.00	.00%	0	0	0
Marketing	$0.00	.00%	0	0	0

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15. Modify the feeder:
    ['Amount'] => ['Reporting USD']**, ['Reporting EUR'],**
    ['Reporting JPY'];
16. Copy the rule shown on line 8 above, and paste on the next blank line.
17. In the copied line, change **['Reporting USD']** to **['Reporting EUR']**.
18. Repeat steps **16** and **17** for **Reporting JPY**.
    The results appear as follows:



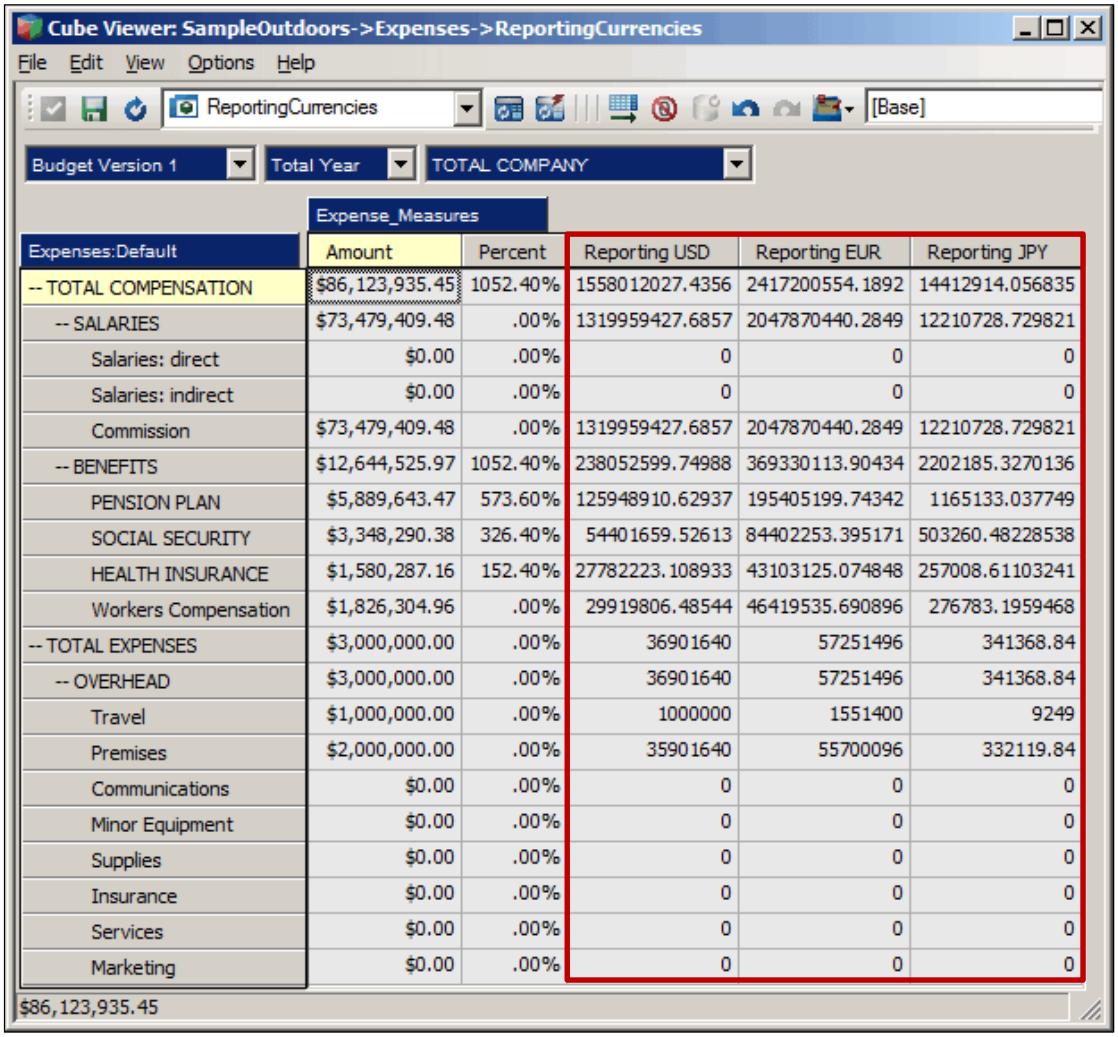
Instead of having three separate rules for the Reporting USD, Reporting EUR and Reporting JPY, you can just change the area of the first rule from **['Reporting USD']=** to:

[{'Reporting USD', 'Reporting EUR', 'Reporting JPY'}]=

The curly braces {} indicate a list of elements within the same dimension.

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19. Save the rule and recalculate the view:
    The results appear as follows:



Expenses:Default	Amount	Percent	Reporting USD	Reporting EUR	Reporting JPY
-- TOTAL COMPENSATION	$86,123,935.45	1052.40%	1558012027.4356	2417200554.1892	14412914.056835
-- SALARIES	$73,479,409.48	.00%	1319959427.6857	2047870440.2849	12210728.729821
Salaries: direct	$0.00	.00%	0	0	0
Salaries: indirect	$0.00	.00%	0	0	0
Commission	$73,479,409.48	.00%	1319959427.6857	2047870440.2849	12210728.729821
-- BENEFITS	$12,644,525.97	1052.40%	238052599.74988	369330113.90434	2202185.3270136
PENSION PLAN	$5,889,643.47	573.60%	125948910.62937	195405199.74342	1165133.037749
SOCIAL SECURITY	$3,348,290.38	326.40%	54401659.52613	84402253.395171	503260.48228538
HEALTH INSURANCE	$1,580,287.16	152.40%	27782223.108933	43103125.074848	257008.61103241
Workers Compensation	$1,826,304.96	.00%	29919806.48544	46419535.690896	276783.1959468
-- TOTAL EXPENSES	$3,000,000.00	.00%	36901640	57251496	341368.84
-- OVERHEAD	$3,000,000.00	.00%	36901640	57251496	341368.84
Travel	$1,000,000.00	.00%	1000000	1551400	9249
Premises	$2,000,000.00	.00%	35901640	55700096	332119.84
Communications	$0.00	.00%	0	0	0
Minor Equipment	$0.00	.00%	0	0	0
Supplies	$0.00	.00%	0	0	0
Insurance	$0.00	.00%	0	0	0
Services	$0.00	.00%	0	0	0
Marketing	$0.00	.00%	0	0	0

    If you want to see Trace Calculations, navigate to the leaf level (such as **GO Americas** for **January**), select a cell, and then click **Trace Calculation**. This will show the values and the exchange rates applied.
20. Close all open views without saving, and then close the **Rules Editor**.
21. Right-click the **SampleOutdoors** node, and then click **Save Data**.
22. Close **Architect**.

Results:
Each subsidiary reports sales and expenses in their local currency. You converted those amounts to the reporting currencies by using control cubes and rules.

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IBM Training

Unit summary

- Describe currency challenges
- Create a currency dimension
- Create a currency cube
- Create rules for currency conversion

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Unit summary

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IBM Training

Exercise 1

Build the Income_Statement Cube with reporting currencies

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Exercise 1: Build the Income_Statement cube with reporting currencies

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Exercise 1: Build the Income_Statement cube with reporting currencies

You need to create a cube reflecting the corporation's Income Statement. You also need to ensure that this cube will consider the finance department's requirement to submit reports in three currencies (USD, EUR, and JPY).

To accomplish this:

- Modify the DatabaseDirectory parameter in the tmp1.ini file to reference the full folder path of the SampleOutdoors server. Then restart the TM1 Admin server.
- In TM1 Perspectives, use income_statement_accounts.cma (from the TM1 sample files) to create the Income_Statement_Accounts dimension. You will need to modify rows 3 and 31 of the file for the DEPRECIATION, AMORTIZATION, INTEREST, TAXES element. Copy the cells to the Income Statement Accounts.xdi file and save the dimension to the SampleOutdoors server.
- In the 32-bit version of Architect, create the Income_Statement_Measures dimension. Include elements for Amount and the three reporting currencies (USD, EUR, and JPY).
- Format the Amount and Reporting USD elements of the Income_Statement_Measures dimension as currency and add a Currency attribute to this dimension. Create a rule on the element attributes control cube for the Income_Statement_Measures dimension. This rule should populate the Currency attribute for the "Reporting" elements of the Income_Statement_Measures dimension.
- Create the Income_Statement cube from the Versions, Months, Subsidiaries, Income_Statement_Accounts, and Income_Statement_Measures dimensions.
- Create a TurboIntegrator process to load data from the inc_statement.csv file into the Income_Statement cube.
- Add rules to the Income_Statement cube to pull data from the Sales_Plan cube. Focus on the Amount element for Gross Sales Revenue, Total Discounts and Allowances, Freight, and Cost of Sales. Add corresponding feeders to the Sales_Plan cube that reference the Income_Statement cube. View the results in the Income_Statement cube.

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- Create rules and feeders to pull data into the Income_Statement cube from the Expenses cube. Focus on the Overheads, Salaries, and Benefits elements.
- Add a rule to the Income_Statement cube to convert data by referencing the zRate cube. Focus on the Reporting USD, Reporting EUR, and Reporting JPY elements. Use the ATTRS function to reference values from the Currency attributes in the Subsidiaries and Income_Statement_Measures dimensions. View the results in the Income_Statement cube.

For more information about where to work and the exercise results, refer to the Tasks and results section that follows. If you need more information to complete a task, refer to earlier demonstrations for detailed steps.

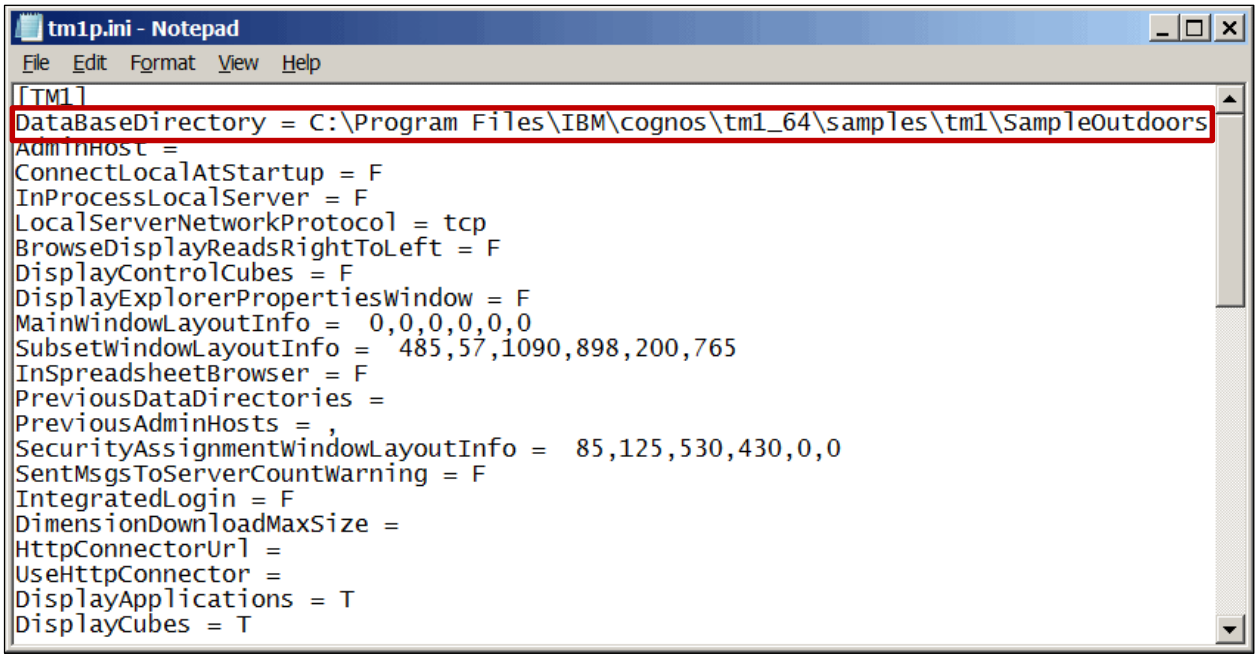
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Exercise 1: Tasks and results

Task 1. Modify the tm1p.ini file to reference the SampleOutdoors TM1 server.

1. Open **C:\Users\Student\AppData\Roaming\Applix\TM1\tm1p.ini** in **Notepad**.
2. Change the value of the DatabaseDirectory parameter to **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\SampleOutdoors**.
   The results appear as follows:



3. Save and close the file.
4. In the **taskbar**, click the **Services** button  , right-click the **IBM Cognos TM1 Admin x64** service, and then click **Restart**.
5. Click **Yes** when prompted to restart the other services, and then after the restart process is complete, close the **Services** window.

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Task 2. Create the Income_Statement Accounts dimension using IBM Cognos TM1 Perspectives.

1. On the **Desktop**, double-click the **Perspectives for MS Excel** icon.
2. Click the **TM1** tab (if necessary), and then in the top left corner, click **Connect**.
   You are prompted to connect to a TM1 server.
3. In the **Server ID** list, select **SampleOutdoors**, in the **Client ID** box, type **admin**, and then in the **Password** box, type **apple**.
4. Click **OK**, and then click **OK** to dismiss the confirmation message.
5. In the top left corner, click **Developer** 

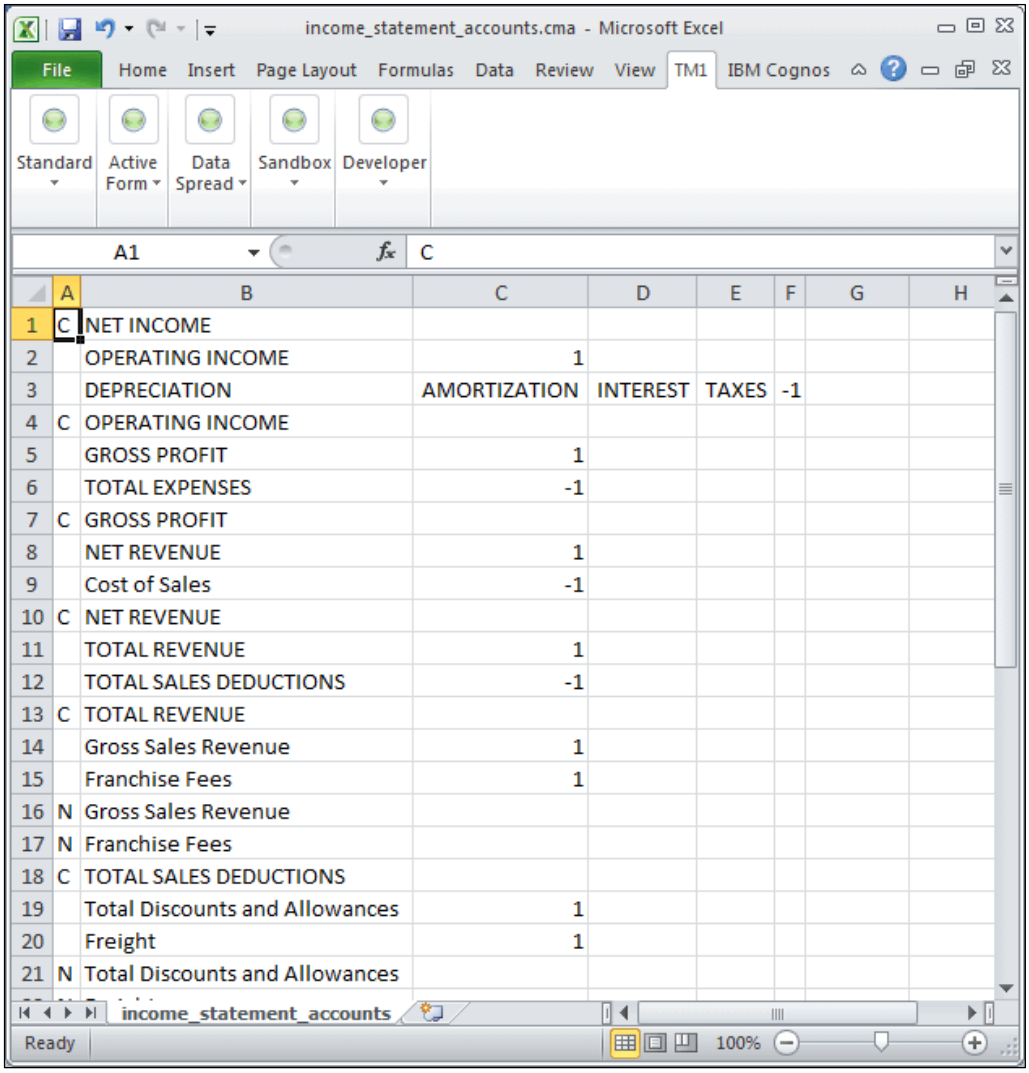
, and then click **Dimension New**.
   The Create a dimension dialog box appears.
6. In the top box, type **Income_Statement_Accounts**, and then click **OK**.
   The Income_Statement_Accounts.xdi file opens in a separate Microsoft Excel window.
7. In the original Microsoft Excel window, click the File tab, click Open, and then open the C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles\income_statement_accounts.cma file.
   In the Open dialog box, in the bottom right corner, select **All Files (*.*)** so that you can select the income_statement_accounts.cma file.
   The Text Import wizard appears.
8. In the **Original data type** section, ensure that the **Delimited** button is selected, in the **Start import at row** box, ensure that **1** is selected, and then in the **File origin** list, ensure that **Windows (ANSI)** is selected.
9. Click **Next**.
10. In the **Delimiters** area, ensure that only the **Comma** check box is selected, ensure that the **Treat consecutive delimiters as one** check box is NOT selected, and then click **Next**.

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11. In the **Column data format** section, ensure that the **General** button is selected, click **Finish**, and then resize the columns in the **income_statement_accounts.cma** Microsoft Excel window as necessary.

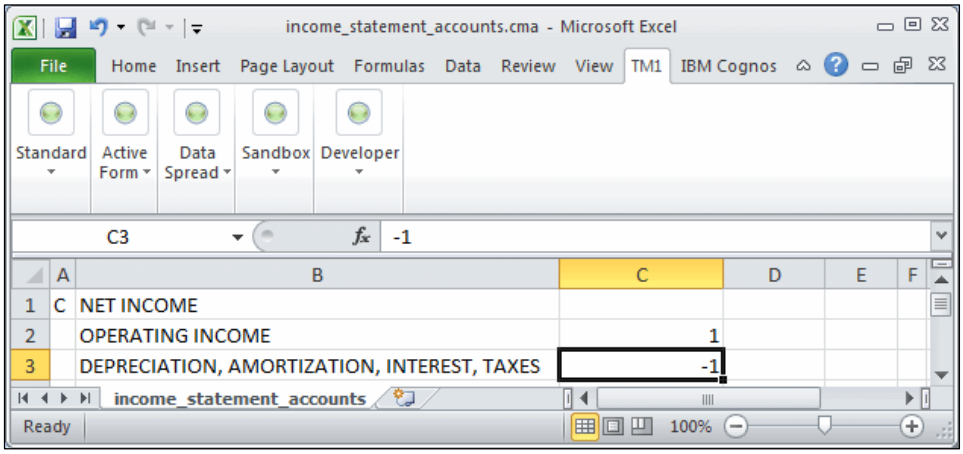
    The results appear as follows:

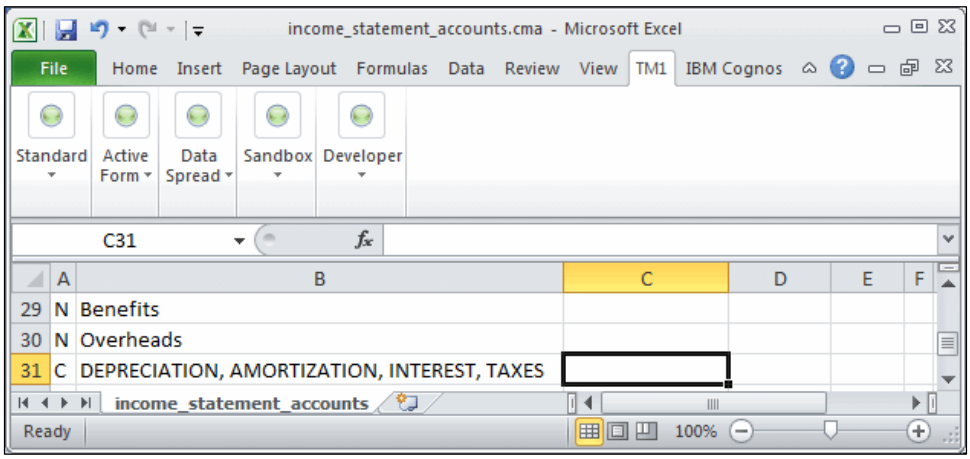


	A	B	C	D	E	F	G	H
1	C	NET INCOME						
2		OPERATING INCOME	1					
3		DEPRECIATION	AMORTIZATION	INTEREST	TAXES	-1		
4	C	OPERATING INCOME						
5		GROSS PROFIT	1					
6		TOTAL EXPENSES	-1					
7	C	GROSS PROFIT						
8		NET REVENUE	1					
9		Cost of Sales	-1					
10	C	NET REVENUE						
11		TOTAL REVENUE	1					
12		TOTAL SALES DEDUCTIONS	-1					
13	C	TOTAL REVENUE						
14		Gross Sales Revenue	1					
15		Franchise Fees	1					
16	N	Gross Sales Revenue						
17	N	Franchise Fees						
18	C	TOTAL SALES DEDUCTIONS						
19		Total Discounts and Allowances	1					
20		Freight	1					
21	N	Total Discounts and Allowances						

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12. Modify rows **3** and **31** so that they appear as shown below:





With your two modifications, this dimension worksheet has the elements and the hierarchy for the Income_Statement_Accounts dimension. All you need to do is copy it into the new dimension worksheet and upload it to the server. This will save you from having to manually create the hierarchy and weights in the Dimension Editor.

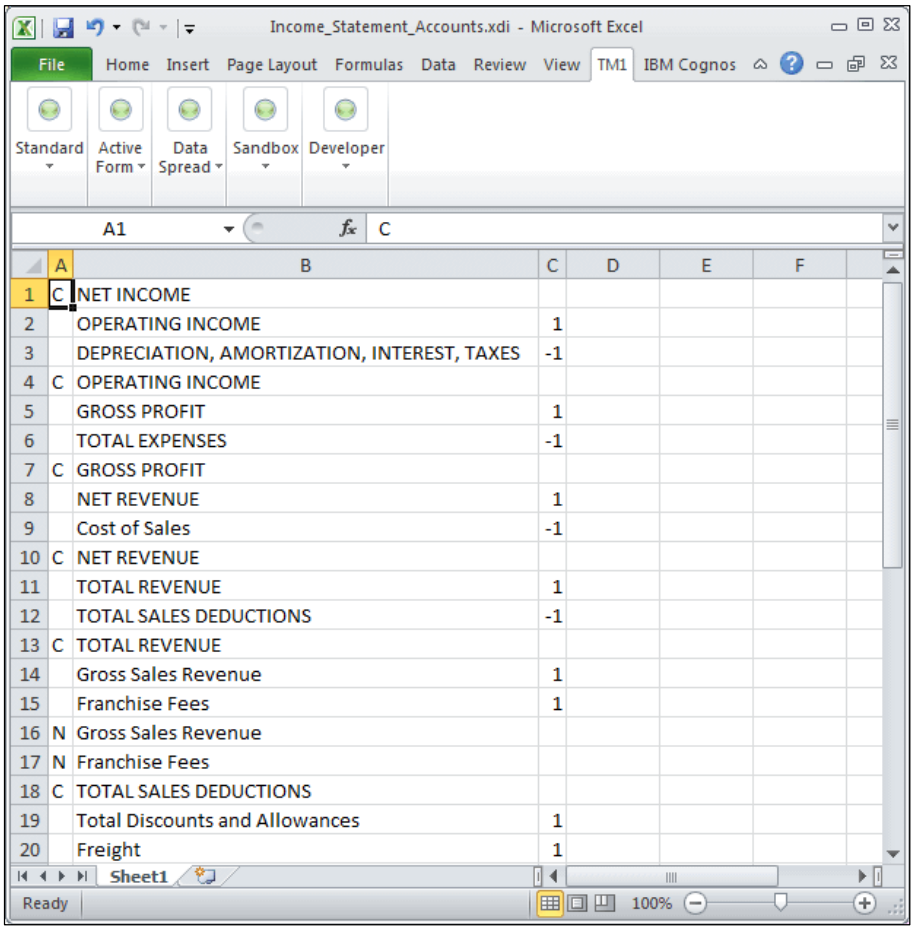
Column A contains the hierarchy, column B contains the element names, and column C contains the element weights. This dimension was originally exported from a TM1 server by right-clicking the dimension and then clicking Export.

13. Select cells **A1:C39**, and click **Ctrl+C** to copy.
14. From the **taskbar**, click **Income Statement Accounts.xdi**.

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15. With your cursor in cell **A1**, click **Ctrl+V** to paste, and then resize the columns as necessary.

    The results appear as follows:



	A	B	C	D	E	F
1	C	NET INCOME				
2		OPERATING INCOME	1			
3		DEPRECIATION, AMORTIZATION, INTEREST, TAXES	-1			
4	C	OPERATING INCOME				
5		GROSS PROFIT	1			
6		TOTAL EXPENSES	-1			
7	C	GROSS PROFIT				
8		NET REVENUE	1			
9		Cost of Sales	-1			
10	C	NET REVENUE				
11		TOTAL REVENUE	1			
12		TOTAL SALES DEDUCTIONS	-1			
13	C	TOTAL REVENUE				
14		Gross Sales Revenue	1			
15		Franchise Fees	1			
16	N	Gross Sales Revenue				
17	N	Franchise Fees				
18	C	TOTAL SALES DEDUCTIONS				
19		Total Discounts and Allowances	1			
20		Freight	1			

    The exported cma file and the xdi file use the same column formats.

16. In the top right corner of the **Income_Statement_Accounts.xdi** file, click **Developer**, and then click **Dimension Save**.

    The Select server name dialog box appears.

17. Select **SampleOutdoors** from the list of servers, and then click **OK**.

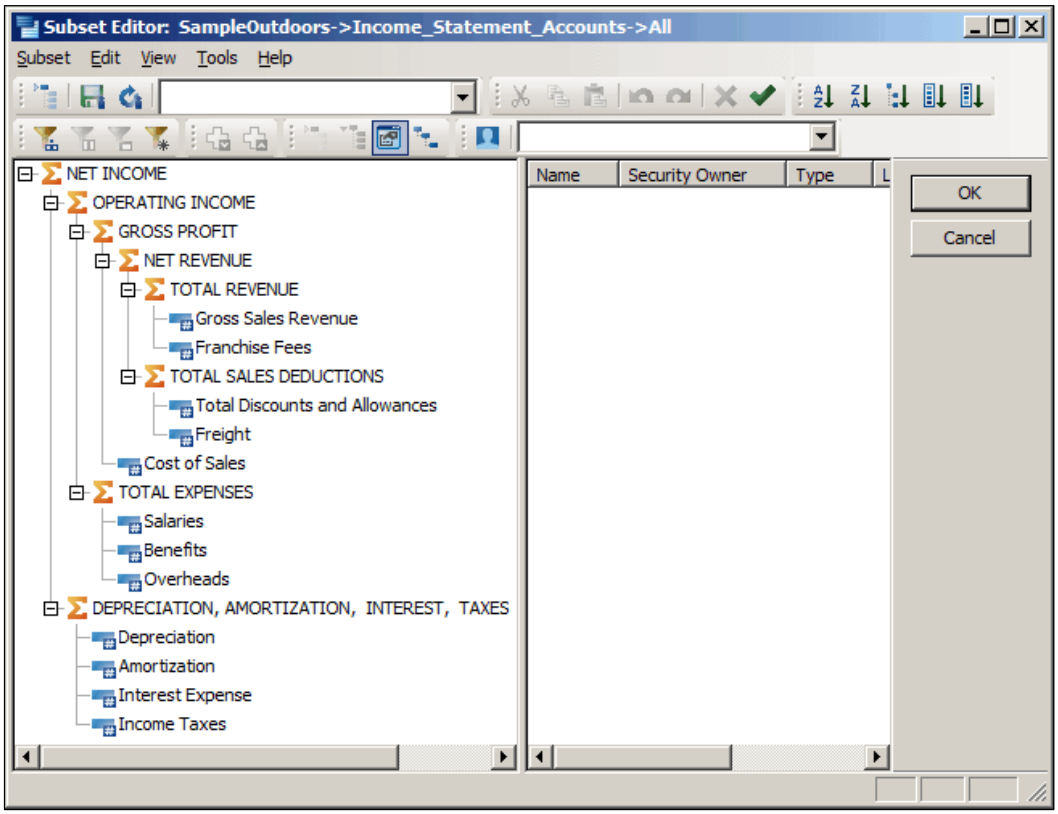
18. Close all **Microsoft Excel** windows without saving, open **tm1a.exe** from **C:\Program Files\IBM\cognos\tm1_64\bin**, and then maximize the **Architect** window.

19. Expand the **TM1** node, double-click **SampleOutdoors**, and then log on with a user ID of **admin** and a password of **apple**.

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20. Expand **Dimensions** and double-click the **Income_Statement_Accounts** dimension.

    You can see the hierarchical list of dimension elements in the Subset Editor:



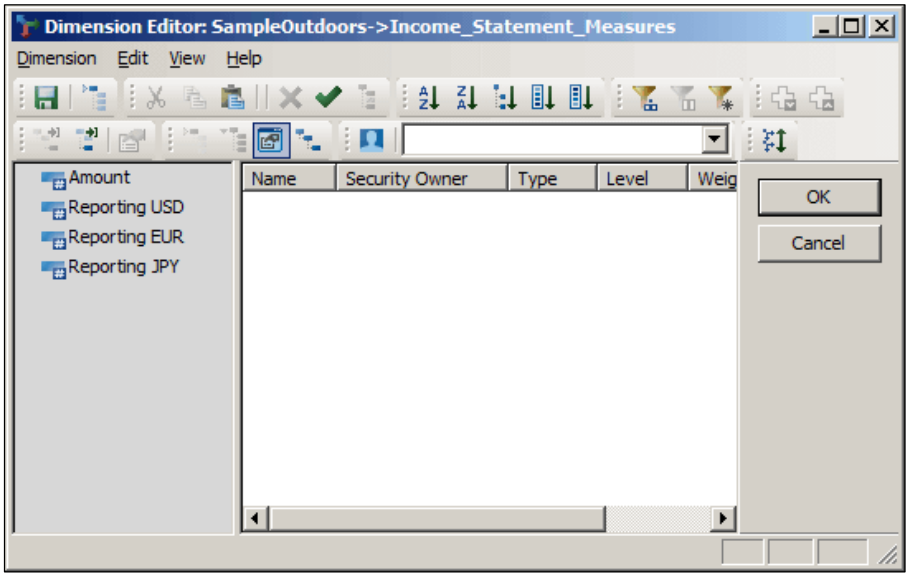
21. Close the **Subset Editor** window.

Task 3. Create the Income_Statement_Measures dimension.

1. Create a new dimension.
2. Insert the following simple elements:
   - **Amount**
   - **Reporting USD**
   - **Reporting EUR**
   - **Reporting JPY**

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3. Save the dimension as **Income_Statement_Measures**.
   The result appears as follows:

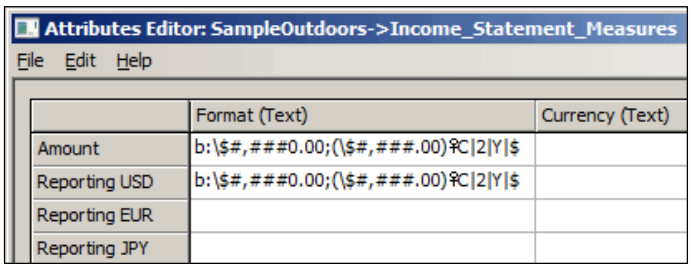


4. Close the **Dimension Editor**.

Task 4. Format elements in the Income_Statement_Measures dimension, add an attribute, and then populate the attribute using a rule on a control cube.

These attributes will be used to format data and currency.

1. Right-click the Income_Statement_Measures dimension and then click Edit Element Attributes.
2. Shift+click Amount and Reporting USD, click Format, click Currency and then click OK.
3. Add a new text attribute named Currency, and then click OK to close the New Attribute dialog box.
   The result appears as follows:



4. Click **OK** to close the **Attributes Editor**.
5. From the **View** menu, click **Display Control Objects**.

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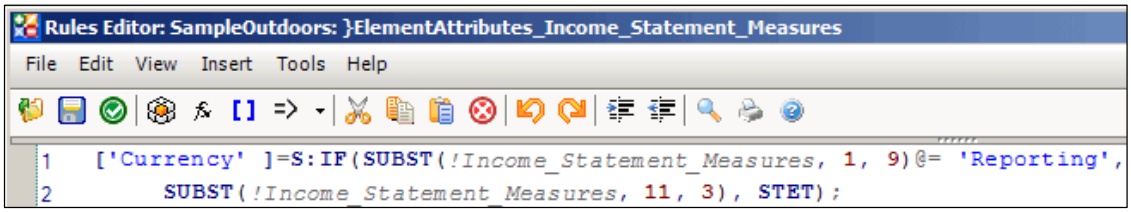
6. Right-click the **}ElementAttributes_Income_Statement_Measures** cube, and then click **Create Rule**.

   The Rules Editor window opens (this may take a few moments).

7. On **line 1**, enter the following rule:

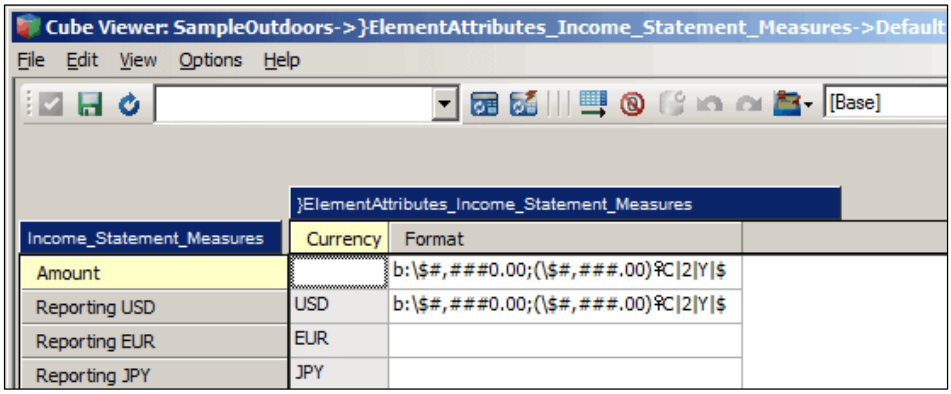
   ['Currency']=S:IF(SUBST(!Income_Statement_Measures, 1, 9)@= 'Reporting', SUBST(!Income_Statement_Measures, 11, 3), STET);

8. The result appears as follows:



9. Save the rule, close the **Rules Editor** and then open and recalculate the **}ElementAttributes_Income_Statement_Measures** cube.

   The result appears as follows (the columns may be in a different order, but that is acceptable):

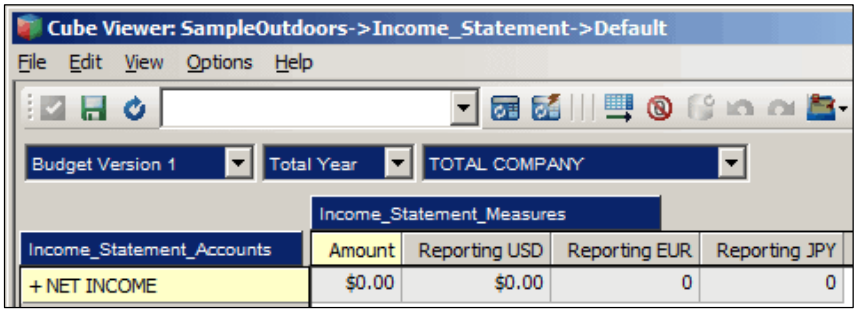


10. Close the **Cube Viewer** without saving.
11. From the **View** menu, deselect **Display Control Objects**.

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Task 5. Create the Income_Statement cube.

1. Create a new cube.
2. Double-click the following dimensions:
   - **Versions**
   - **Months**
   - **Subsidiaries**
   - **Income_Statement_Accounts**
   - **Income_Statement_Measures**
3. In the **Cube Name** box, type **Income_Statement**.
4. Click **Create Cube**.
5. Double-click the **Income_Statement** cube and click **Recalculate**.



6. Close the **Income_Statement** cube without saving.

Task 6. Load data into the Income_Statement cube.

1. Create a new TurboIntegrator process using **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\sourcefiles\inc_statement.csv** as the source (click **OK** to dismiss the warning message).
2. In the **Number of title records** box, type **1**, and then click **Preview**.
3. Click the **Variables** tab and add a **v** to the **Subsidiaries**, **Expense**, and **Version** variable names.
4. Rename **V3** to **vDate** and **V5** to **vData**.
5. In the **Contents** column for **vSubsidiaries**, **vExpense**, and **vVersion**, select **Element**.

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6. In **Contents** for **vDate**, click **Other**.
7. In **Contents** for **vData** click **Data**.
8. Create a new variable called **vMonth**:
   vMonth=SUBST(vDate, 3, 3);
9. For **vMonth** change the **Variable Type** to **String** and the **Contents** to **Element**.
10. Create a new variable called **vMeasures**:
    vMeasures='Amount';
11. Change the **Variable Type** to **String** and the **Contents** to **Element**.
    Ensure that the result appears as follows:

Data Source | Variables | Maps | Advanced | Schedule

	Variable Name	Variable Type	Sample Value	Contents	Formula
1	vSubsidiaries	String	GO Americas	Element	
2	vExpense	String	Amortization	Element	
3	vDate	String	7-Jan	Other	
4	vVersion	String	Budget version 1	Element	
5	vData	Numeric	497745.81	Data	
6	vMonth	String	Jan	Element	Formula
7	vMeasures	String	Amount	Element	Formula

12. Click the **Maps** tab.
13. On the **Cube** tab click **Update Cube**, and then beside **CubeName** select **Income_Statement**.
14. Click the **Dimensions** tab and map to the dimensions:

Data Source | Variables | Maps | Advanced | Schedule

Cube | Dimensions | Data | Consolidations | Attributes

Element Variable	Sample Value	Dimension	Order In Cube	Action	Element Type	Element Order
vSubsidiaries	GO Americas	Subsidiaries		As Is	Numeric	By Input
vExpense	Amortization	Income_Statement_Accounts		As Is	Numeric	By Input
vVersion	Budget version 1	Versions		As Is	Numeric	By Input
vMonth	Jan	Months		As Is	Numeric	By Input
vMeasures	Amount	Income_Statement_Measures		As Is	Numeric	By Input

15. Click all the **Advanced** tabs.
16. Save the process as **LoadIncomeStatementCSV**.
17. Run the process, click **OK**, and then close **TurboIntegrator**.

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Task 7. Create rules to pull data from the Sales_Plan cube.

1. Right-click the **Income_Statement** cube and then click **Create Rule**.
2. On the first line, type **SKIPCHECK;**.
3. Add the following rule for **Gross Sales Revenue**:

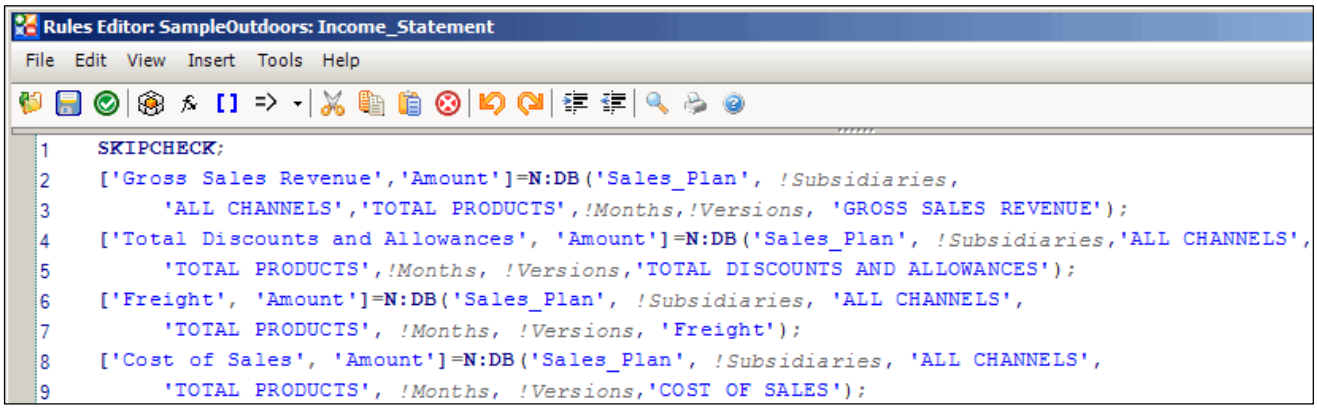
   ['Gross Sales Revenue','Amount']=N:DB('Sales_Plan', !Subsidiaries, 'ALL CHANNELS','TOTAL PRODUCTS',!Months,!Versions, 'GROSS SALES REVENUE');
4. Add the following rule for Total Discounts and Allowances:

   ['Total Discounts and Allowances', 'Amount']=N:DB('Sales_Plan', !Subsidiaries,'ALL CHANNELS','TOTAL PRODUCTS',!Months, !Versions,'TOTAL DISCOUNTS AND ALLOWANCES');
5. Add the following rule for Freight:

   ['Freight', 'Amount']=N:DB('Sales_Plan', !Subsidiaries, 'ALL CHANNELS','TOTAL PRODUCTS', !Months, !Versions, 'Freight');
6. Add the following rule for Cost of Sales:

   ['Cost of Sales', 'Amount']=N:DB('Sales_Plan', !Subsidiaries, 'ALL CHANNELS','TOTAL PRODUCTS', !Months, !Versions,'COST OF SALES');

   The result appears as follows:



7. Click **Save**, and then close the **Rules Editor**.
8. Right-click the **Sales_Plan** cube, click **Edit Rule**, and then add the following under the **FEEDERS;** statement:

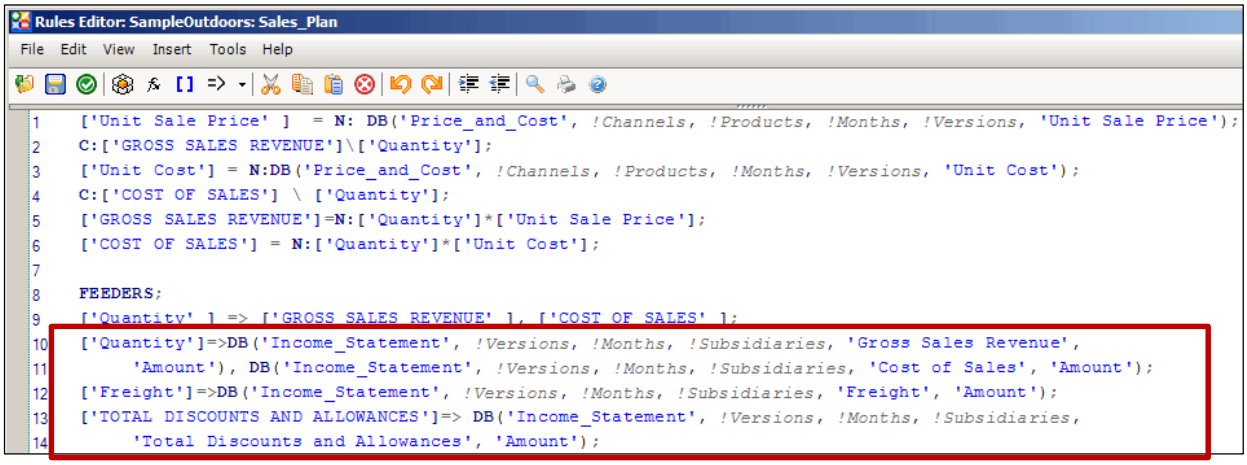
   ['Quantity']=>DB('Income_Statement', !Versions, !Months, !Subsidiaries, 'Gross Sales Revenue', 'Amount'), DB('Income_Statement', !Versions, !Months, !Subsidiaries, 'Cost of Sales', 'Amount');

   ['Freight']=>DB('Income_Statement', !Versions, !Months, !Subsidiaries, 'Freight', 'Amount');

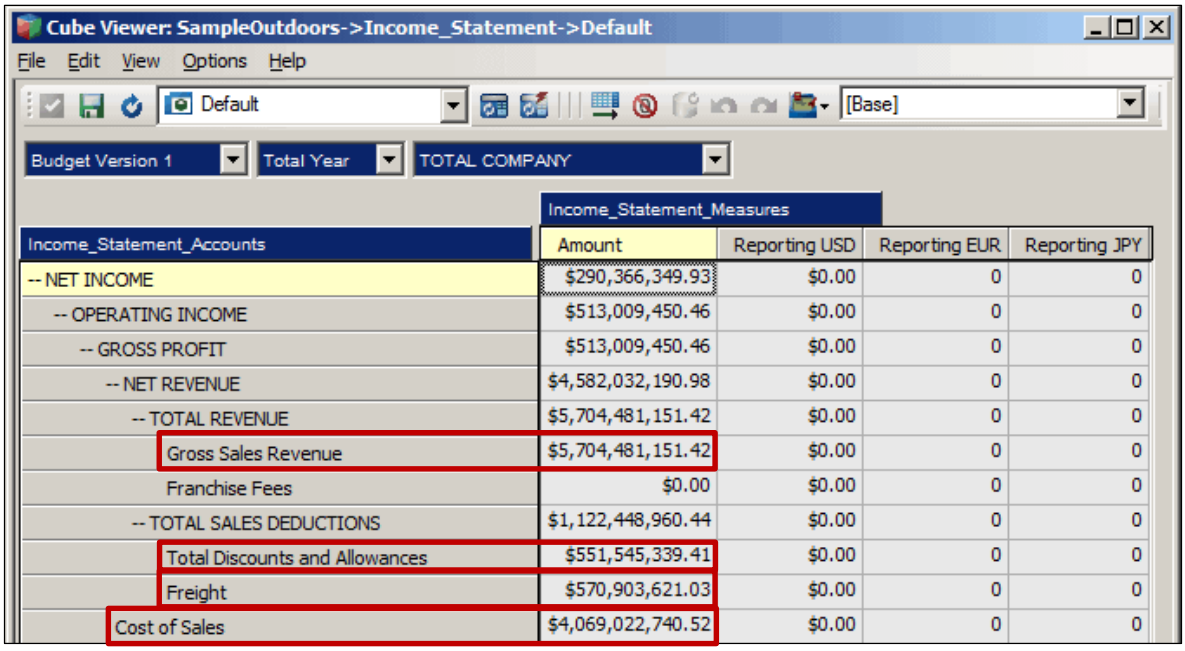
   ['TOTAL DISCOUNTS AND ALLOWANCES']=> DB('Income_Statement', !Versions, !Months, !Subsidiaries, 'Total Discounts and Allowances', 'Amount');

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The result appears as follows:



9. Click **Save**, and then close the **Rules Editor**.
10. Open the **Income Statement** cube and set **Versions** to **Budget Version 1** (if necessary).
11. Click **Income_Statement_Accounts** (on rows), click **All** and then click **OK**.
12. Recalculate the **Income_Statement** cube:



13. Click **Save**, select **Default**, deselect **Private**, and then click **OK**.
14. Close **Cube Viewer**.

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Task 8. Create rules and feeders to pull data from the Expenses cube.

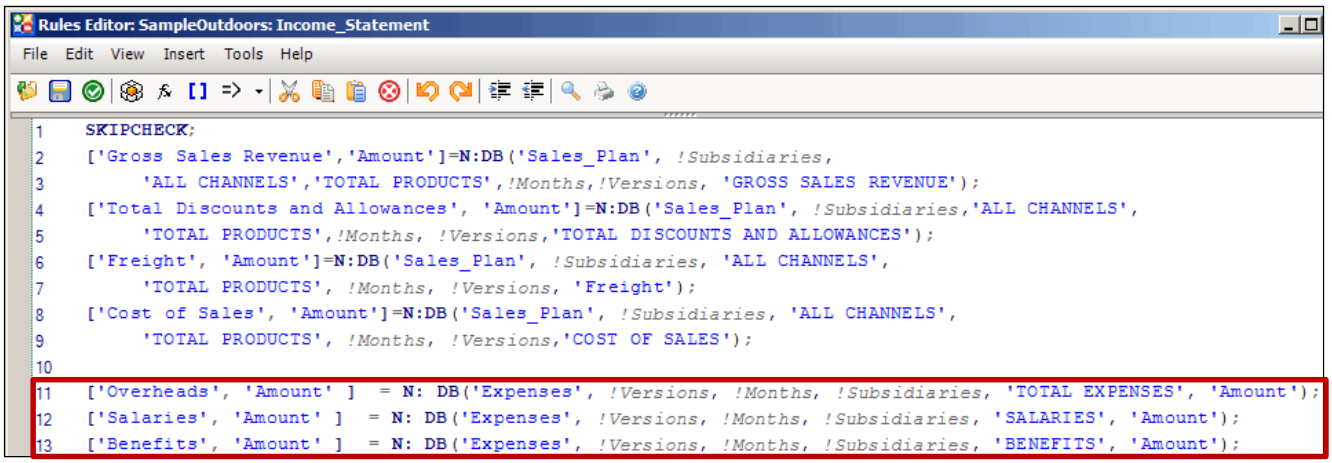
1. Right-click the **Income_Statement** cube and then click **Edit Rule**.
2. Insert a return to create a blank line after the last rule.
3. Insert the following rules:

   ['Overheads', 'Amount'] = N: DB('Expenses', !Versions, !Months, !Subsidiaries, 'TOTAL EXPENSES', 'Amount');

   ['Salaries', 'Amount'] = N: DB('Expenses', !Versions, !Months, !Subsidiaries, 'SALARIES', 'Amount');

   ['Benefits', 'Amount'] = N: DB('Expenses', !Versions, !Months, !Subsidiaries, 'BENEFITS', 'Amount');
4. Click **Check Syntax**, and then click **Save**.

   The result appears as follows:



5. Close the **Rules Editor**.
6. Right-click the **Expenses cube**, and then click **Edit Rule**.
7. Insert a blank line after the last feeder statement.
8. Insert the following feeder statements:

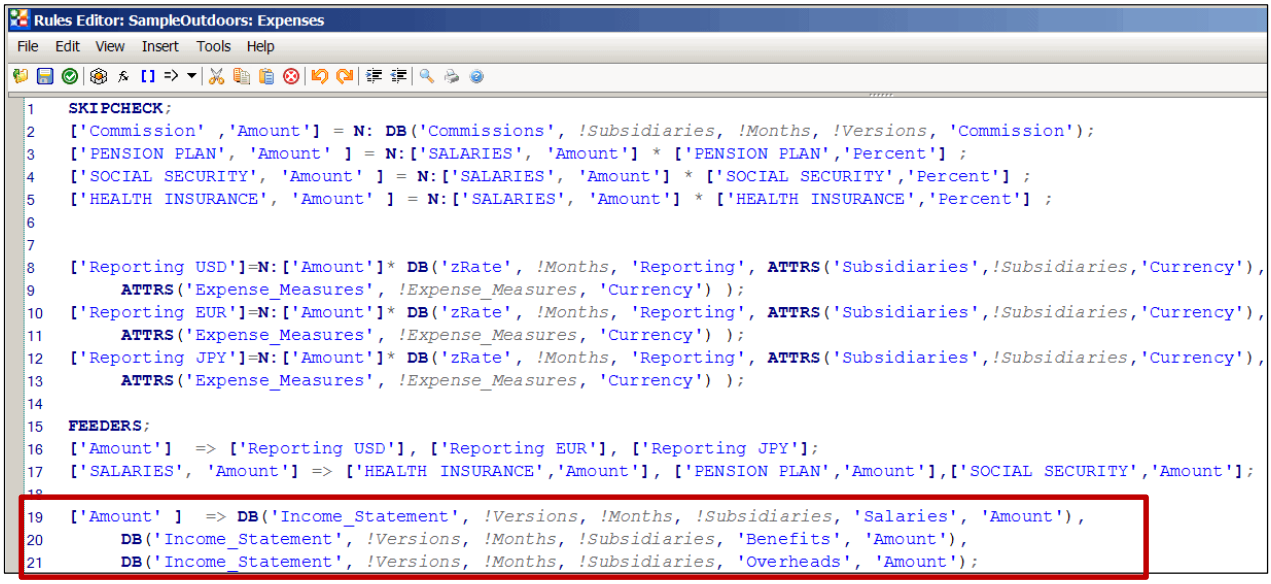
   ['Amount'] => DB('Income_Statement', !Versions, !Months, !Subsidiaries, 'Salaries', 'Amount'),

   DB('Income_Statement', !Versions, !Months, !Subsidiaries, 'Benefits', 'Amount'),

   DB('Income_Statement', !Versions, !Months, !Subsidiaries, 'Overheads', 'Amount');

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9. Click **Check Syntax**, and then click Save.
   The result appears as follows:



10. Close the **Rules Editor**.

Task 9. Create rules in the Income_Statement cube to calculate reporting currencies.

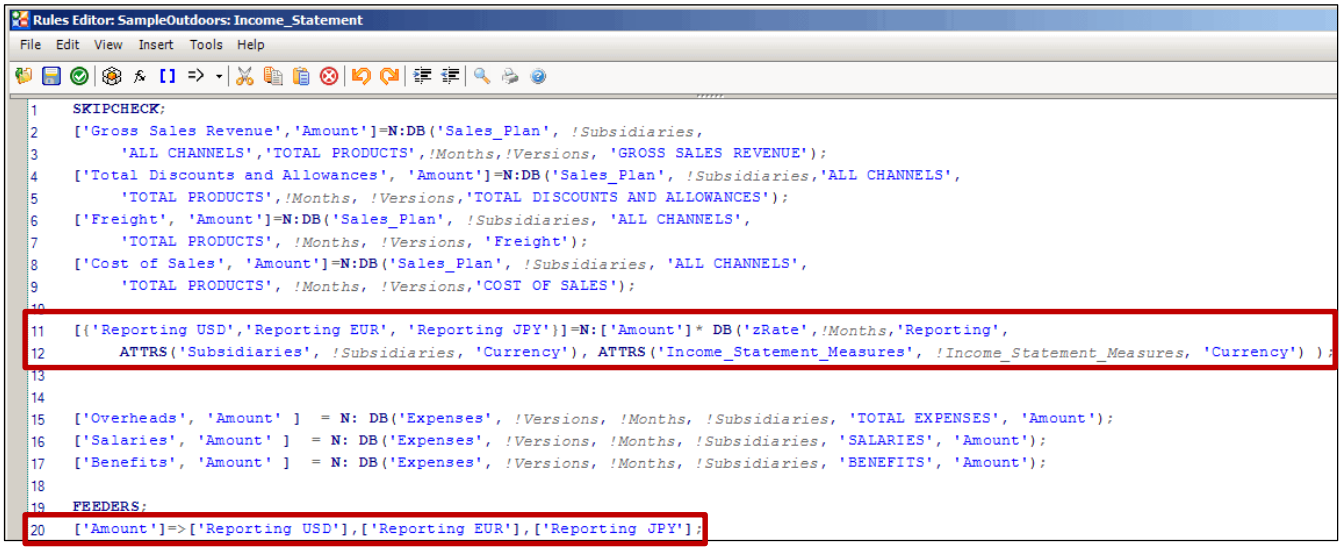
1. Right-click the **Income_Statement** cube and click **Edit Rule**.
2. After the last rule, type:
   [{'Reporting USD','Reporting EUR', 'Reporting JPY'}]=N:['Amount']*
3. Insert references to the **zRate** cube.
   DB('zRate',!Months,'Reporting',
4. Use the ATTRS function to retrieve the currencies from the Subsidiaries and Income_Statement_Measures dimensions.

   ATTRS('Subsidiaries', !Subsidiaries, 'Currency'),
   ATTRS('Income_Statement_Measures',
   !Income_Statement_Measures, 'Currency'));
5. Add **FEEDERS;** after the last rule.
6. Add the following feeder statement:
   ['Amount']=>['Reporting USD'],['Reporting EUR'],['Reporting JPY'];

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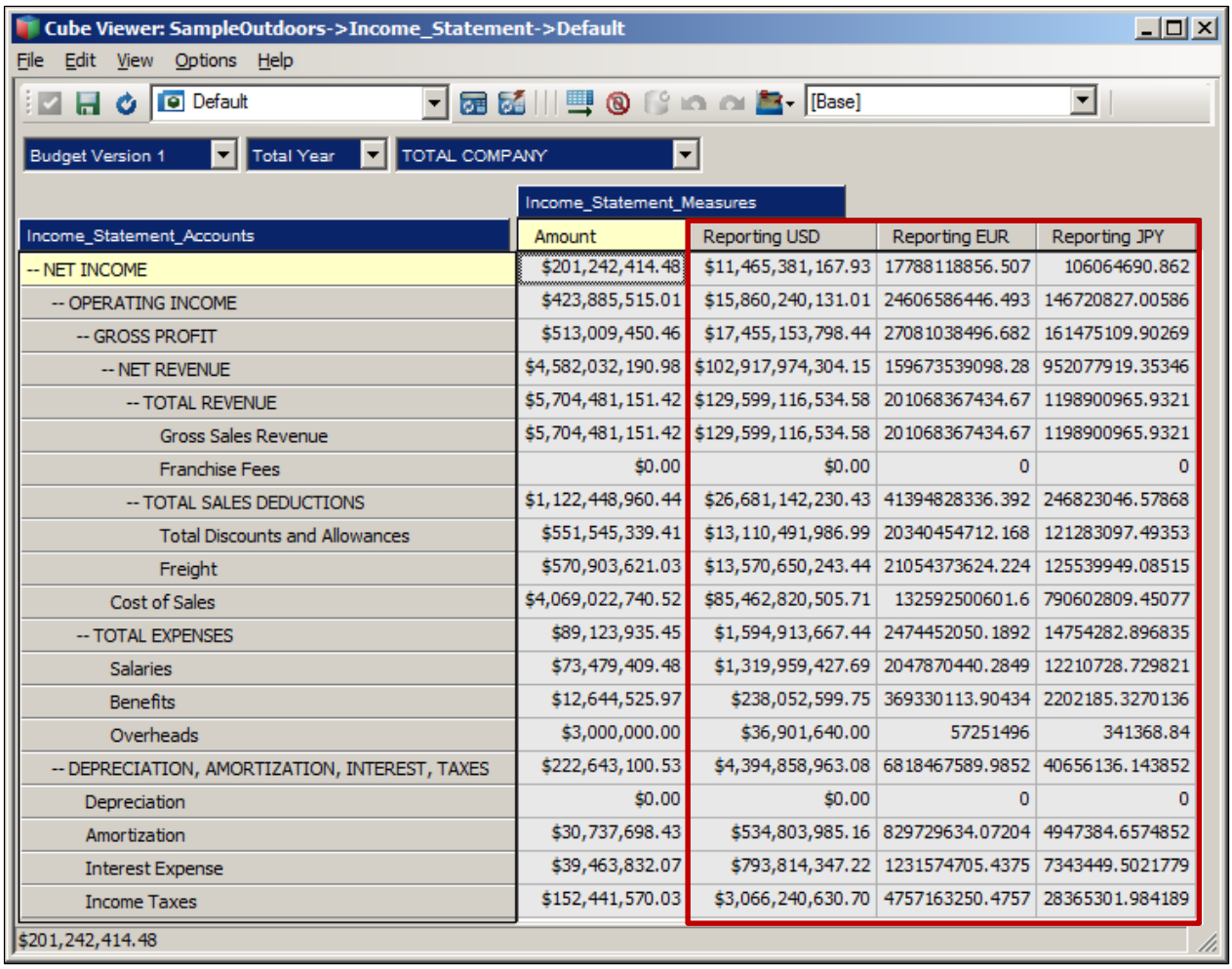
7. Save the rule.

   The result appears as follows:



8. Close the **Rules Editor**.
9. Open the **Income_Statement** cube and recalculate the view (if necessary).

   The result appears as follows:



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10. Close the **Cube Viewer** without saving.
11. Right-click the **SampleOutdoors** node and click **Save Data**.
12. Close all open windows.

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IBM Training

Model for different fiscal requirements

IBM Planning Analytics (v2.0)

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IBM Training

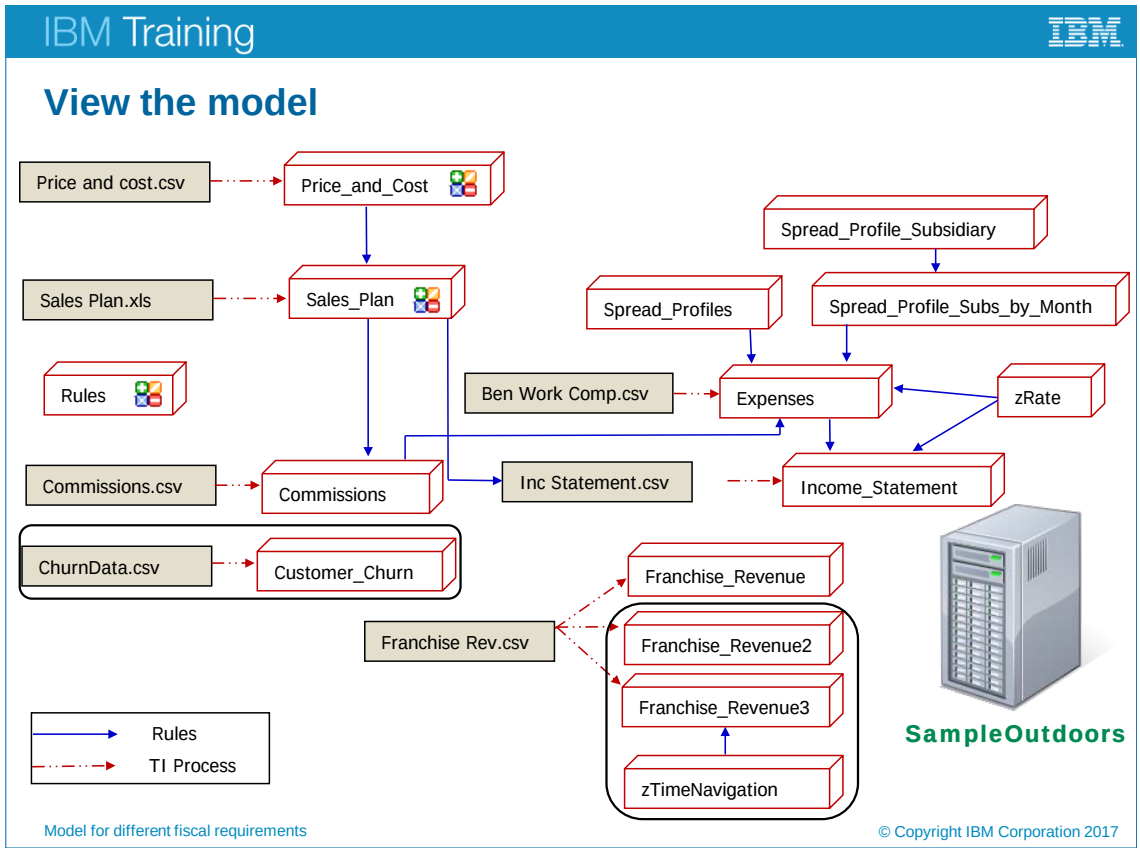
Unit objectives

- Identify time considerations
- Use discrete time dimensions
- Implement a continuous time dimension model

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Unit objectives

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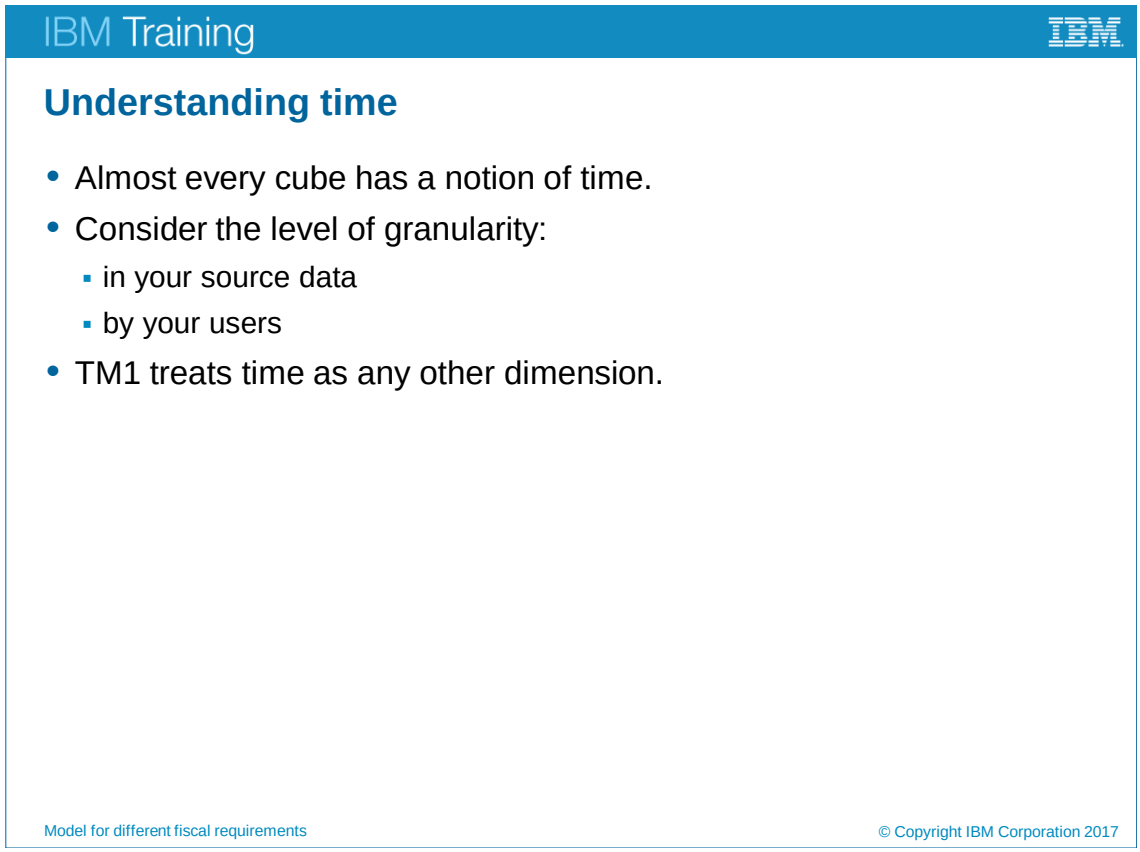


View the model

In this unit you will create the following cubes:

- Franchise_Revenue2
- Franchise_Revenue3
- zTimeNavigation

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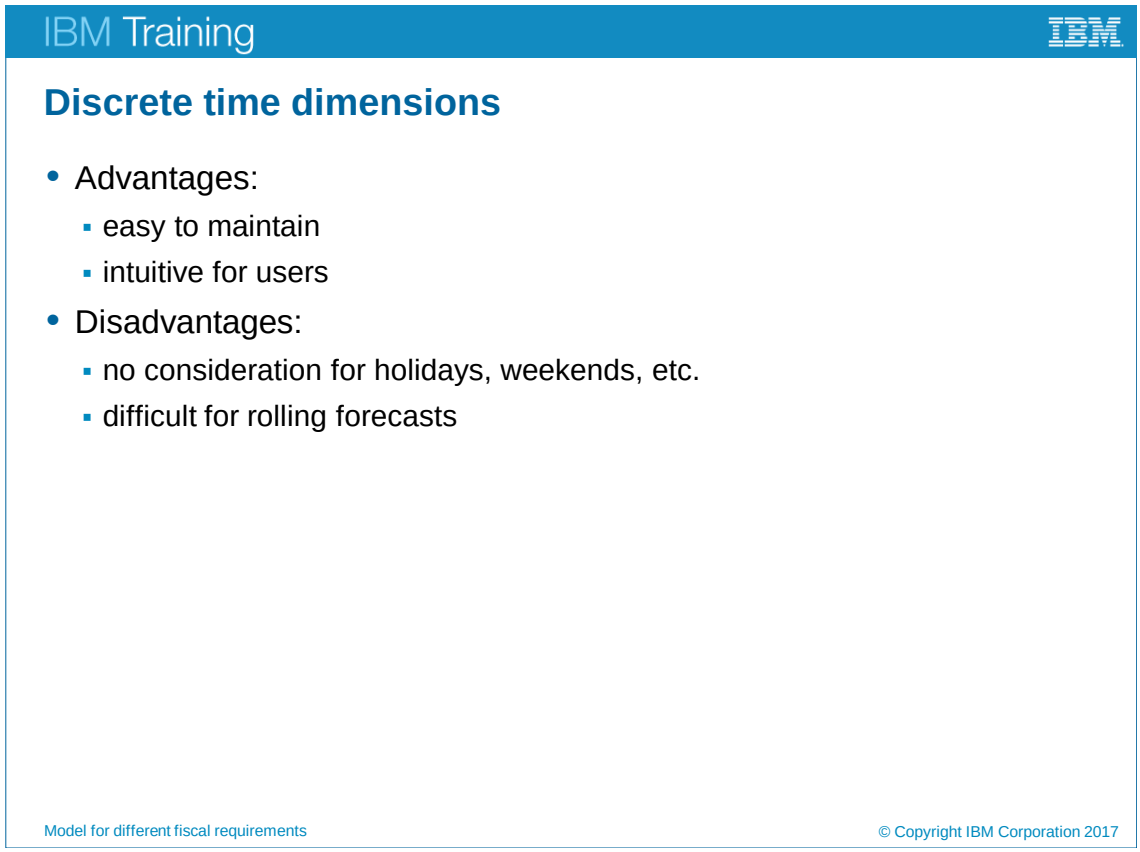


Understanding time

In the cube properties, you can specify the measures and time dimensions in a cube. This has no impact in TM1, but if you are using your TM1 model with other systems (such as IBM Cognos Analytics) they may require these settings.

To access cube properties, right-click a cube and click Properties.

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Discrete time dimensions

Discrete time dimensions are typically better for backward-looking financial reporting cubes.

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Demonstration 1

Add a Year dimension

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Demonstration 1: Add a Year dimension

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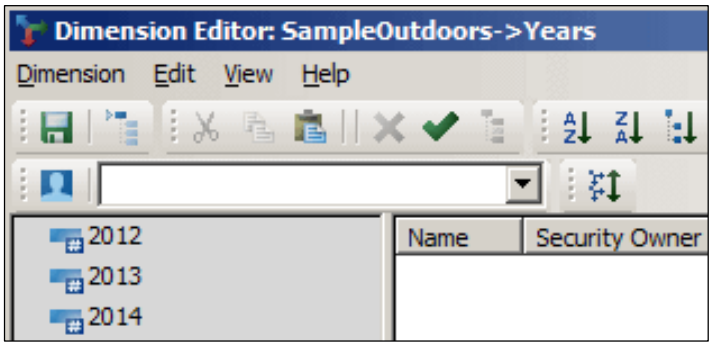
Demonstration 1: Add a Year dimension

Purpose:
You must decide how to present time-sensitive information to your users.

Create a new version of the Franchise_Revenue cube that contains a Year dimension as well as the Months. The Year should contain three years starting with 2012. Next, load data into the year 2012.

Task 1. Create the Year dimension.

1. Open **Windows Explorer** and go to **C:\Program Files\IBM\cognos\tm1_64\bin**.
   You will be working with the Advanced Rules Editor in a later demonstration, so you will open the 32-bit version of Architect.
2. Double-click **tm1a.exe** and maximize the **Architect** window.
3. Expand the **TM1** node, double-click **SampleOutdoors**, and then log on with a user ID of **admin** and a password of **apple**.
4. Create a new dimension with the following simple elements:
   - **2012**
   - **2013**
   - **2014**
5. Save the dimension as **Years**.
   The results appear as follows:

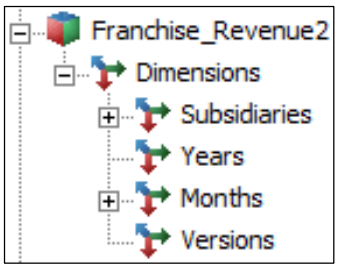


6. Close the **Dimension Editor**.

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Task 2. Create the Franchise_Revenue2 cube.

1. Create a new cube named **Franchise_Revenue2**.
2. Include the following dimensions:
   - **Subsidiaries**
   - **Years**
   - **Months**
   - **Versions**
3. Click **Create Cube**.
   The results appear as follows:

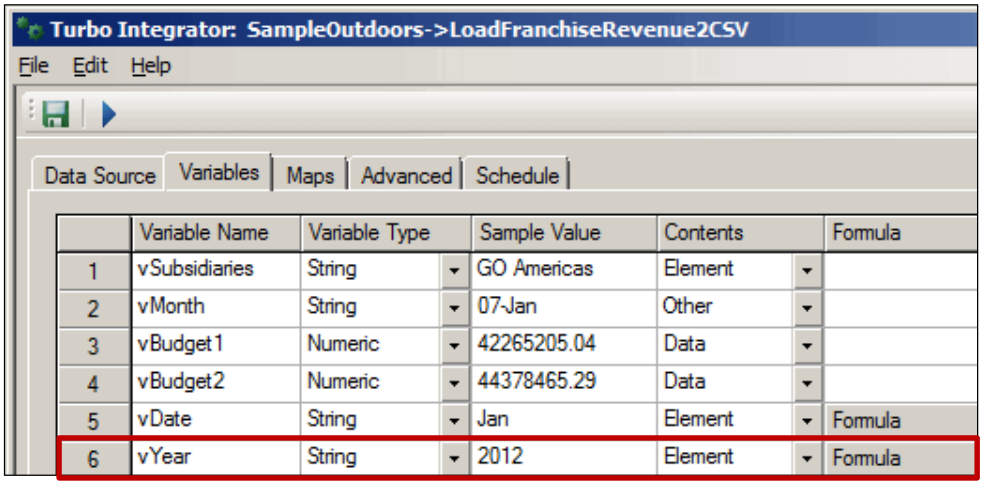


Task 3. Load data into the Franchise_Revenue2 cube.

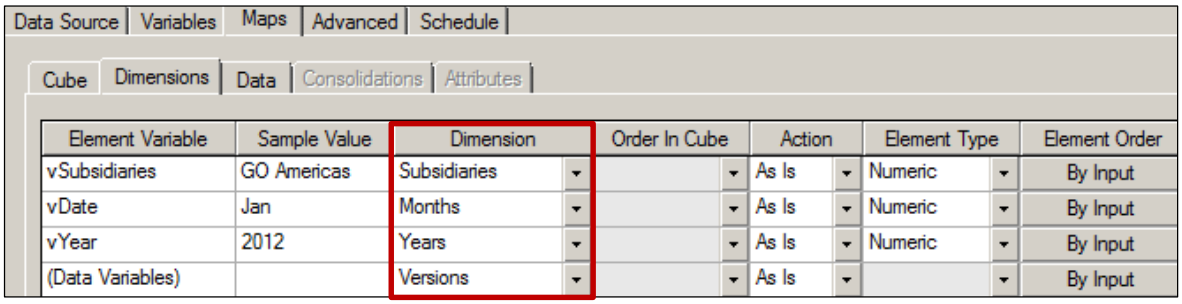
1. Open the **LoadFranchiseRevenueCSV** process.
   You will save this process under a different name, modify it as necessary, and then run it.
2. From the **File** menu, click **Save As**.
3. Name the new process **LoadFranchiseRevenue2CSV**.
4. On the **Variables** tab, create a new variable called **vYear** and give it the following formula:
   vYear='2012';

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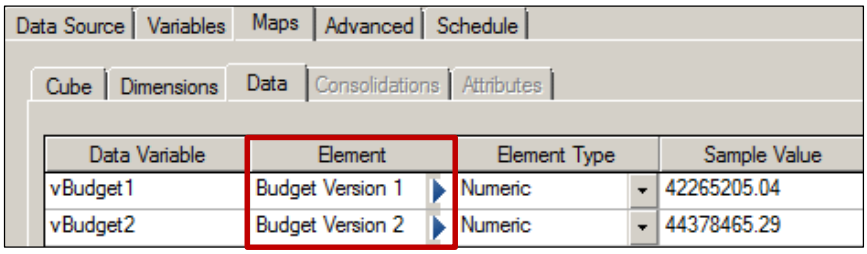
5. Under **Variable Type** click **String**, and then under **Contents** click **Element**.
   The results appear as follows:



6. On the **Maps** tab, ensure that **Update Cube** is selected, and then in the **CubeName** list, select **Franchise_Revenue2**.
7. On the **Dimensions** sub-tab map the dimensions as follows:
   vSubsidiaries = Subsidiaries
   vDate = Months
   vYear = Years
   (Data Variables) = Versions
   The results appear as follows:



8. Click the **Data** sub-tab and map the **Version** dimension elements.
   The results appear as follows:

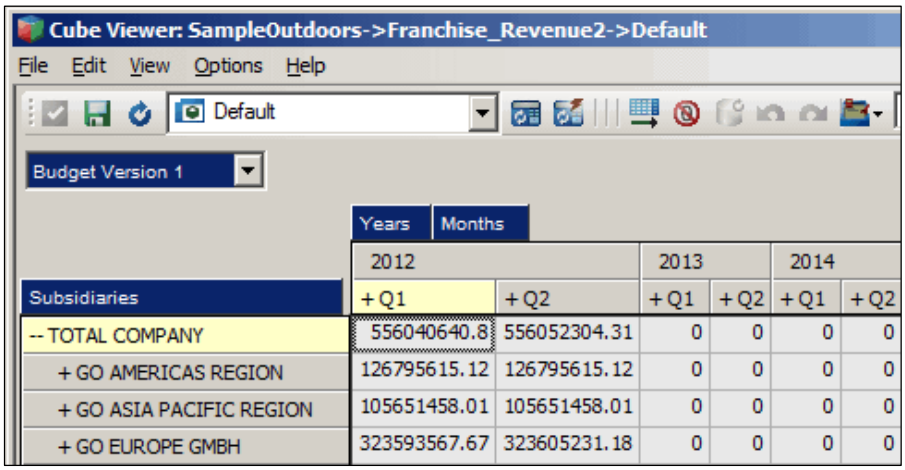


9. Click the **Advanced** tabs and all sub-tabs within.

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10. Save and run the process, and then close **TurboIntegrator**.
11. Create a **public**, **Default** view of the **Franchise_Revenue2** cube.
12. Swap **Months** and **Subsidiaries** (currently displaying **TOTAL COMPANY**), and then swap **Months** and **Versions**.
13. Click **Months** (on columns), click **All**, Ctrl+click **Q1** and **Q2**, and then click **OK**.
14. Drag **Years** (currently displaying 2012) to the column area (to the left of Months) to stack the dimensions.
15. In the **Versions** list, select **Budget Version 1** (if necessary), and then click **Recalculate**.

    The result appears as follows:



16. Close the **Cube Viewer**, saving and overwriting the view when prompted.
17. Right-click the **SampleOutdoors** node and click **Save Data**.
18. Leave **Architect** open for the next demonstration.

Results:
The resulting cube can easily be manipulated to compare the same month or quarter over time.

This type of view works well for reporting scenarios where the time period does not need to be more flexible.

If users need greater granularity or more time-specific information (comparing weekends to weekdays, etc.) a continuous time dimension may be a better option.

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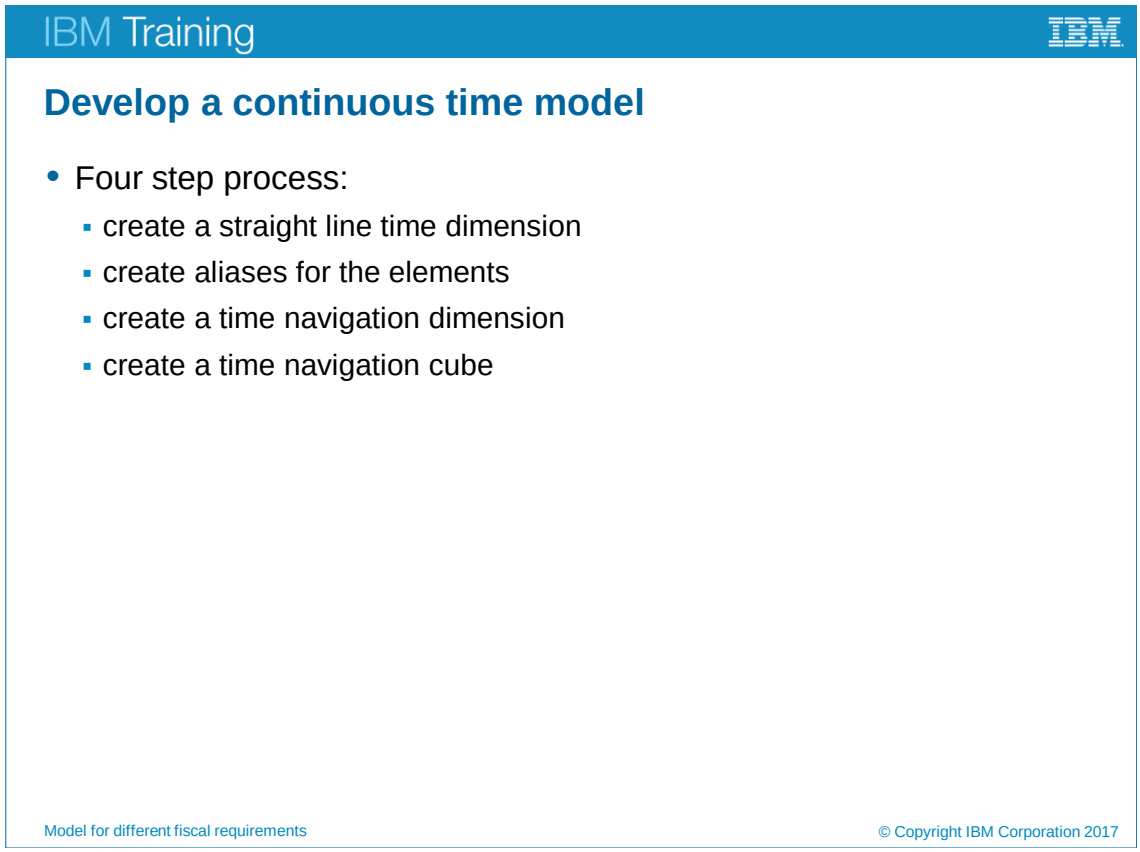
Continuous time dimensions

- Continuous time dimensions:
  - provide more flexibility
  - require more maintenance
- Better for:
  - forecasting applications
  - spanning year boundaries

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Continuous time dimensions

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Develop a continuous time model

When working with a continuous time dimension, you should construct a time cube to provide more flexibility in determining periods like:

- Next and Previous Periods
- Next and Previous Periods Last Year

This will allow for more flexibility in reporting (as the multiple time dimensions do).

Elements in this dimension should have generic names like W1 for Week 1, W2 for Week 2, M1 for January, etc. As the periods of one year roll into the next one, the numbering continues. Use aliases to make navigation easier for users.

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Demonstration 2

Create a continuous time model

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Demonstration 2: Create a continuous time model

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Demonstration 2: Create a continuous time model

Purpose:
You are constructing your model going back to January 1 of last year. You want to build a model that reflects three years.

However, your users want to see the Franchise_Revenue cube with a continuous time period that spans three years beginning with January 1, 2012. They also want to be able to analyze data by comparing current period values to the previous periods both this year and last year.

You will create another version of the Franchise_Revenue cube using a straight line time dimension that will calculate the next and previous time periods using a lookup cube (zTime_Navigation).

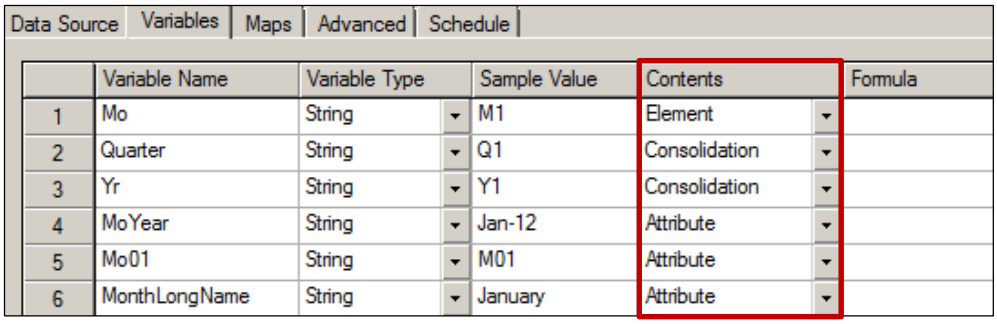
You will also correct the problem of having no measures dimension in this cube. The Franchise_Revenue cubes, as they now exist, would not be able to do the continuous time analysis or convert currency in the manner described.

1. Create a new **TurboIntegrator** process.
2. Under **Datasource Type**, click **Text**, and then browse to **C:\Training\P8361\SLTime.csv** (click **OK** to dismiss the warning message).
3. Next to **Number of title records**, type **1** and then click **Preview**.
4. Click the **Variables** tab.

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5. Under **Contents**, select the following:
   - Mo: **Element**
   - Quarter: **Consolidation**
   - Yr: **Consolidation**
   - MoYear: **Attribute**
   - Mo01: **Attribute**
   - MonthLongName: **Attribute**

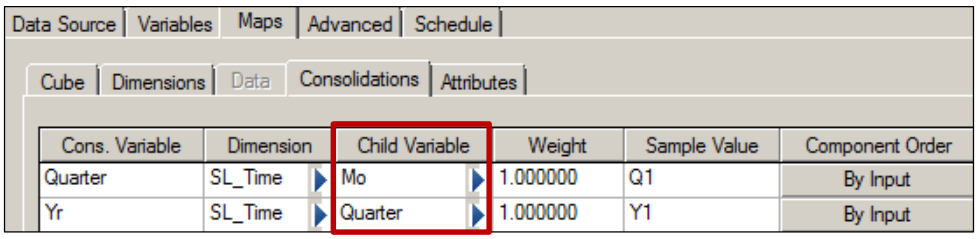
   The results appear as follows:



	Variable Name	Variable Type	Sample Value	Contents	Formula
1	Mo	String	M1	Element	
2	Quarter	String	Q1	Consolidation	
3	Yr	String	Y1	Consolidation	
4	MoYear	String	Jan-12	Attribute	
5	Mo01	String	M01	Attribute	
6	MonthLongName	String	January	Attribute	

   MoYear will be an alias but Mo01 and Month Long Name will be text attributes.
6. Click the **Maps** tab, and then click the **Dimensions** sub-tab.
7. In the **Dimension** column for **Mo**, type **SL_Time** and then click **By Input**.
8. Click **Automatic**, click **Hierarchy**, and then click **OK** to close the **Dimension Element Ordering** dialog box.
9. Click the **Consolidations** tab.
10. Click the arrow in the **Child Variable** cell for **Quarter**, select **Mo**, and then click **OK**.
11. Click the arrow in the **Child Variable** cell for **Yr**, select **Quarter**, and then click **OK**.

    The results appear as follows:



Cons. Variable	Dimension	Child Variable	Weight	Sample Value	Component Order
Quarter	SL_Time	Mo	1.000000	Q1	By Input
Yr	SL_Time	Quarter	1.000000	Y1	By Input

12. Click the **Attributes** tab.

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13. Set the **Element Variable** to **Mo** for all three attributes.
14. Under **Attribute**, type the following:
    - MoYear: **MonthYear**
    - Mo01: **M00**
    - MonthLongName: **Full Name**
15. Under **Attribute Type** select the following:
    - MoYear: **Alias**
    - Mo01: **Text**
    - MonthLongName: **Text**

    The results appear as follows:

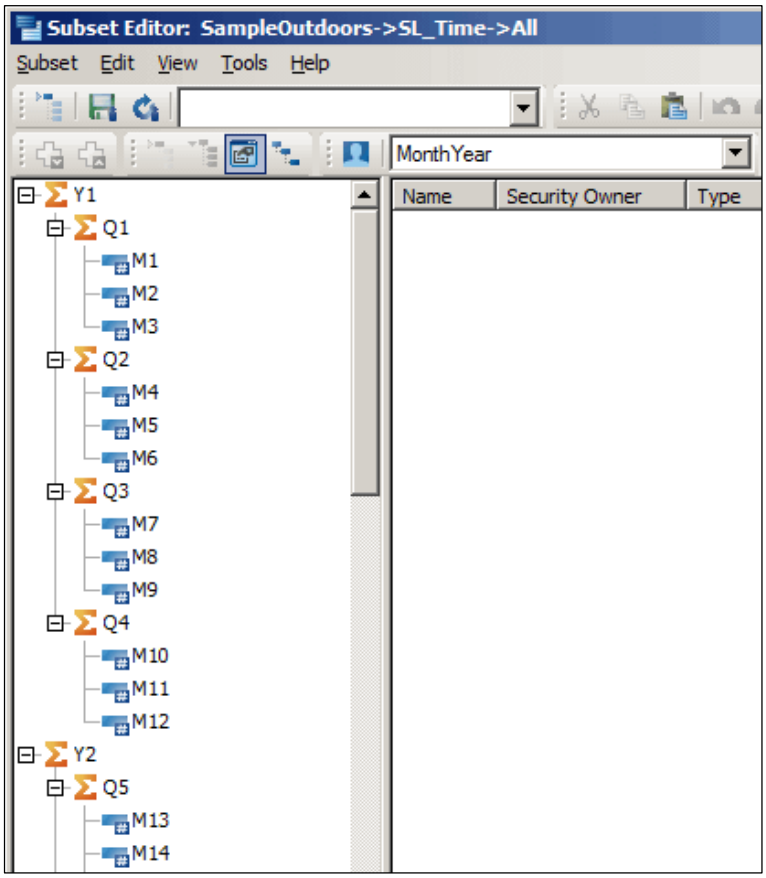
Attribute Variable	Sample Value	Dimension	Element Variable	Attribute	Action	Attribute Type
MoYear	Jan-12	SL_Time	Mo	MonthYear	Create	Alias
Mo01	M01	SL_Time	Mo	M00	Create	Text
MonthLongName	January	SL_Time	Mo	Full Name	Create	Text

16. Click the **Advanced** tab, and then click all the sub-tabs within it.
17. Save the process as **CreateSLTimeCSV**, and then run the process.

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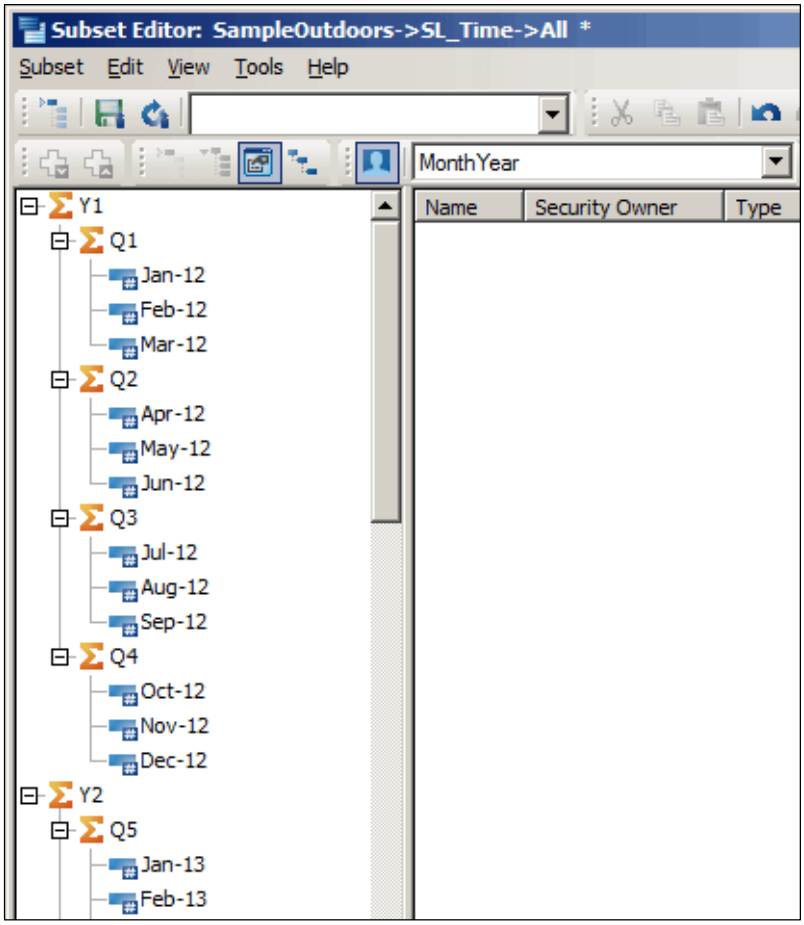
18. Click **OK** to dismiss the confirmation message, close **TurboIntegrator**, and then double-click the **SL_Time** dimension.

    The results appear as follows:



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19. Click **Use Aliases** (currently off)  .
    The results appear as follows:



20. Close the **Subset Editor** without saving.

Task 2. Create a time navigation dimension.

1. Create a new dimension.

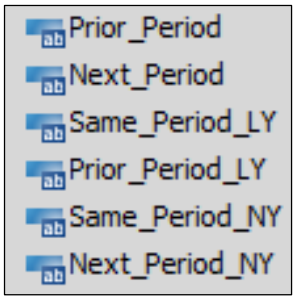
   You will create a cube similar to the zRate cube called **zTime_Navigation**.

   The zTime_Navigation cube will be used to automatically determine the next and previous periods for a given time period. This functionality will allow you to create a multitude of time-based reports and comparisons from your model.

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2. Insert the following **String** elements:
   - **Prior_Period**
   - **Next_Period**
   - **Same_Period_LY**
   - **Prior_Period_LY**
   - **Same_Period_NY**
   - **Next_Period_NY**

   The results appear as follows:

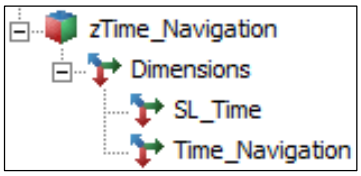


3. Save the dimension as **Time_Navigation**, and then close the **Dimension Editor**.

Task 3. Create and populate a time navigation cube.

1. Create a new cube named **zTime_Navigation**.
2. Double-click the following dimensions.
   - **SL_Time**
   - **Time_Navigation**
3. Click **Create Cube**.

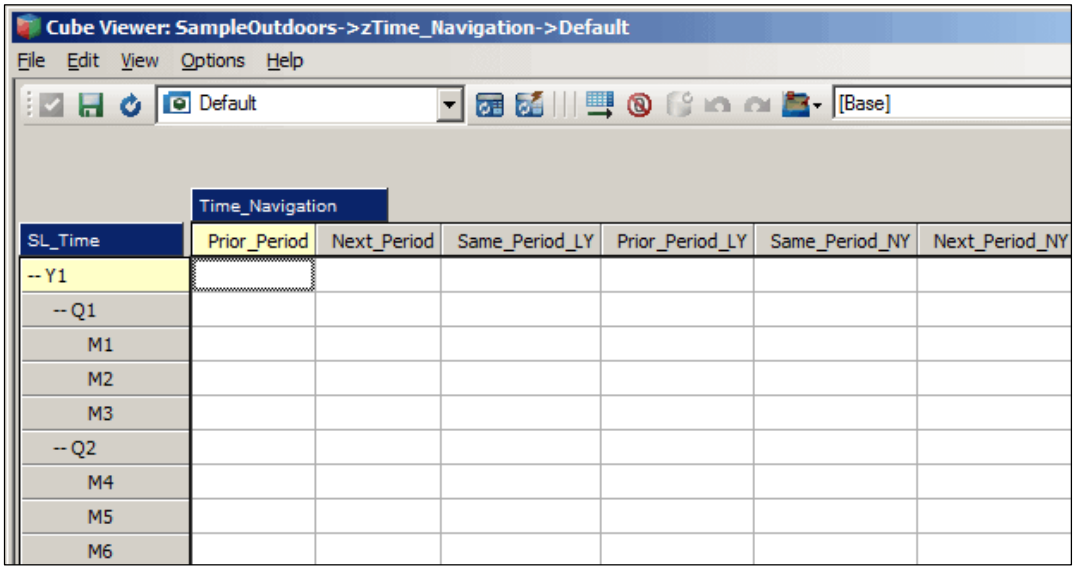
   The results appear as follows:



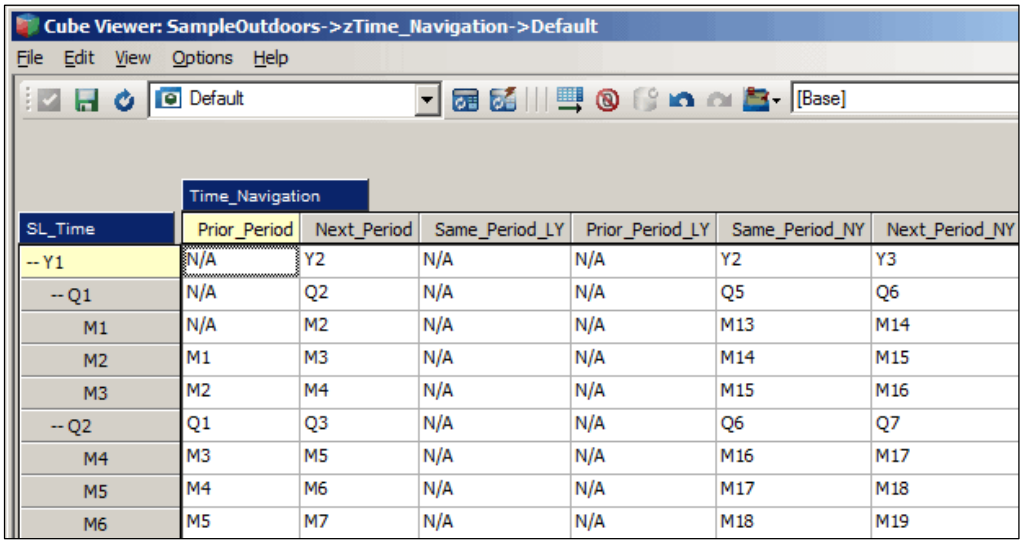
4. Double click **zTime_Navigation** to open the default view.
5. Click the **SL_Time** dimension (on rows), click **All**, and then click **OK**.

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6. Recalculate the cube, and then save a **public**, **Default** view of the cube.
   The results appear as follows:



7. In **Microsoft Excel**, open **C:\Training\P8361\TimeData.csv**.
8. Copy cells **B2:G52**.
9. Click the first cell in the cube view, click **Edit** and then click **Paste**.
10. Recalculate the view.
    The results appear as follows:



11. Close the **Cube Viewer** and **TimeData.csv** without saving changes.

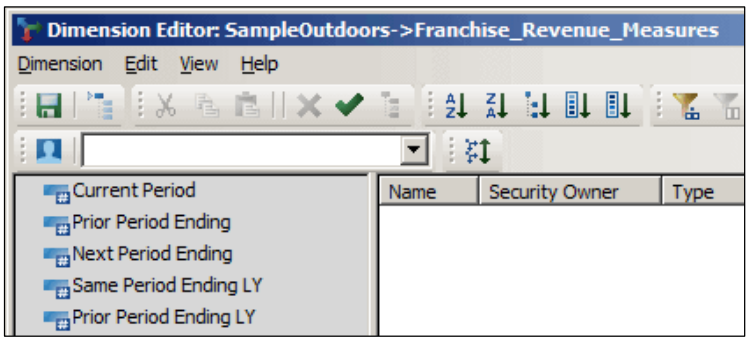
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Task 4. Create a Franchise_Revenue_Measures dimension.

The Franchise_Revenue cubes have no designated measures dimension. This makes modifying either one of them to accommodate different currencies or time dimensions very difficult. You will create a measures dimension to fix this problem.

1. Create a new dimension, and then insert the following **Simple** elements:
   - **Current Period**
   - **Prior Period Ending**
   - **Next Period Ending**
   - **Same Period Ending LY**
   - **Prior Period Ending LY**
2. Save the dimension as **Franchise_Revenue_Measures**.

   The results appear as follows:



3. Close the **Dimension Editor**.

   Here you can see why it is important to have a clearly defined measures dimension in a cube. The original version of the Franchise_Revenue cube that you created did not have a measures dimension and it was unable to accommodate the new measures.

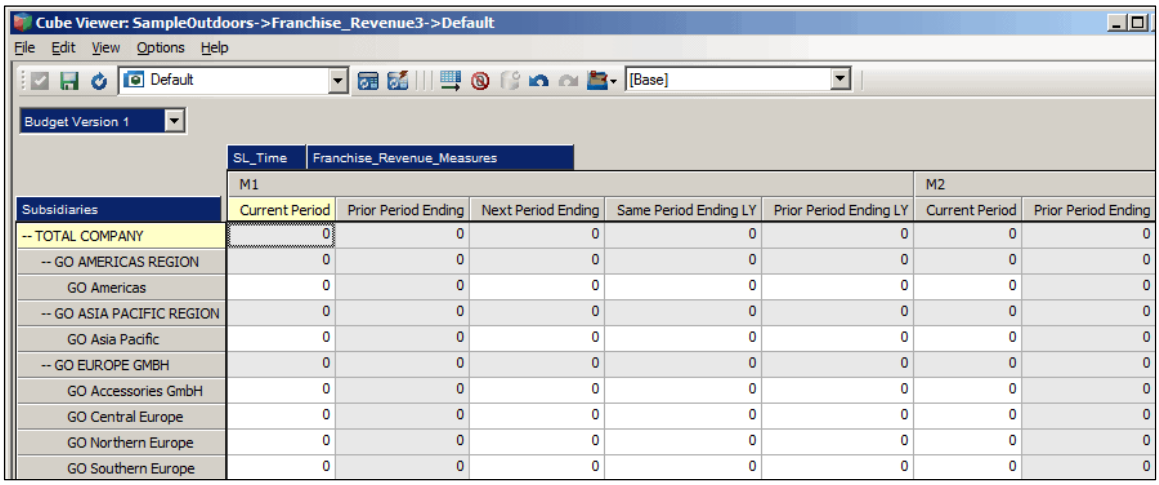
Task 5. Create and populate the reporting cube (Franchise_Revenue3).

1. Create a new cube called **Franchise_Revenue3**.
2. Double-click the following dimensions:
   - **Subsidiaries**
   - **SL_Time**
   - **Versions**
   - **Franchise_Revenue_Measures**

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3. Click **Create Cube**.
4. Open the **Franchise_Revenue3** cube.
5. Swap the **Versions** and **Subsidiaries** dimensions.
6. Select **Budget Version 1** from the **Versions** dimension (if necessary).
7. Move the **SL_Time** dimension to the left of **Franchise_Revenue_Measures** on columns.
8. Click **SL_Time** and expand the **Q1** element.
9. From the **View** menu, click **Expand Above**, select **Q1, M1, M2, M3**, and then click **OK**.
10. Recalculate the view and expand all of the subsidiaries on the rows.

    The results appear as follows:



11. Save the view as **public**, **Default** view.
12. Open the **LoadFranchiseRevenueCSV** process.

    You will save this process under a different name and modify the process.
13. From the **File** menu, click **Save As**, type **LoadFranchiseRevenue3CSV**, and then click **OK**.
14. Click the **Variables** tab, and then click **New Variable**.
15. Change **V6** to **vMeasures**, click **Formula**, and then specify the following:

    vMeasures='Current Period';
16. Click **OK** to close the **Process Variable Formula** dialog box, under **Variable Type** for **vMeasures**, click **String**, and then under **Contents**, click **Element**.

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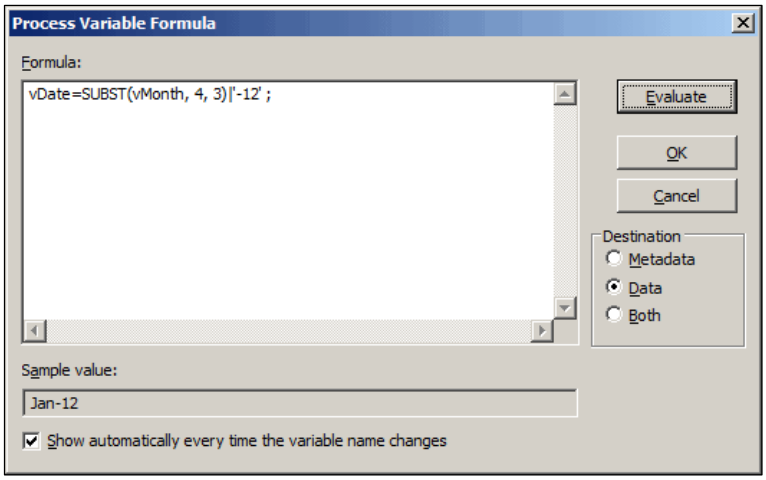
17. Click the **Formula** button for the **vDate** variable.

    The original function extracted just the month name from the data. Now you want to specify the Mon-Year with the year being 2012. You will concatenate the -12 to the existing function using a pipe symbol (|).

18. After the **)** but before the semi-colon, add the following, **|'-12'** and then click **Evaluate**.

    Ensure the dash is typed before 12.

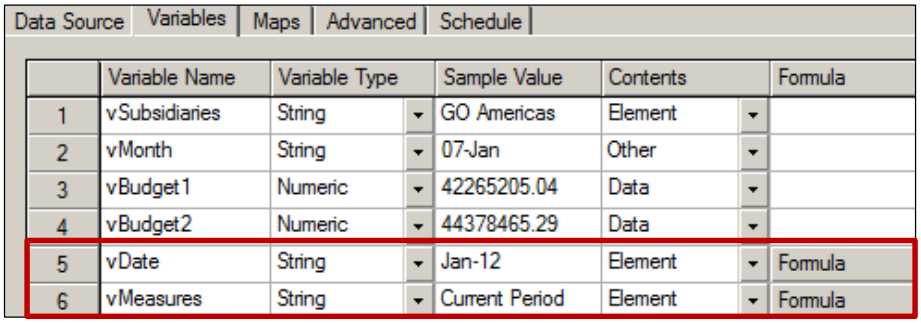
    The results appear as follows:



19. Click **OK** to close the **Process Variable Formula** dialog box.

    You do not need to change the Variable Type or the Contents because they are still correct.

    The results appear as follows:

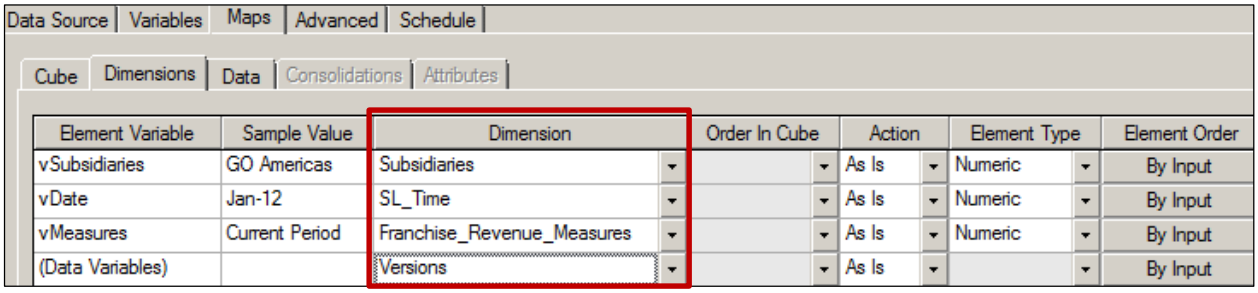


	Variable Name	Variable Type	Sample Value	Contents	Formula
1	vSubsidiaries	String	GO Americas	Element	
2	vMonth	String	07-Jan	Other	
3	vBudget1	Numeric	42265205.04	Data	
4	vBudget2	Numeric	44378465.29	Data	
5	vDate	String	Jan-12	Element	Formula
6	vMeasures	String	Current Period	Element	Formula

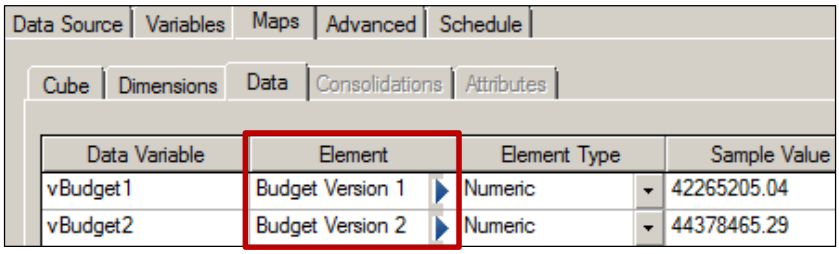
20. Click the **Maps** tab.
21. In the **CubeName** list, select **Franchise_Revenue3**.

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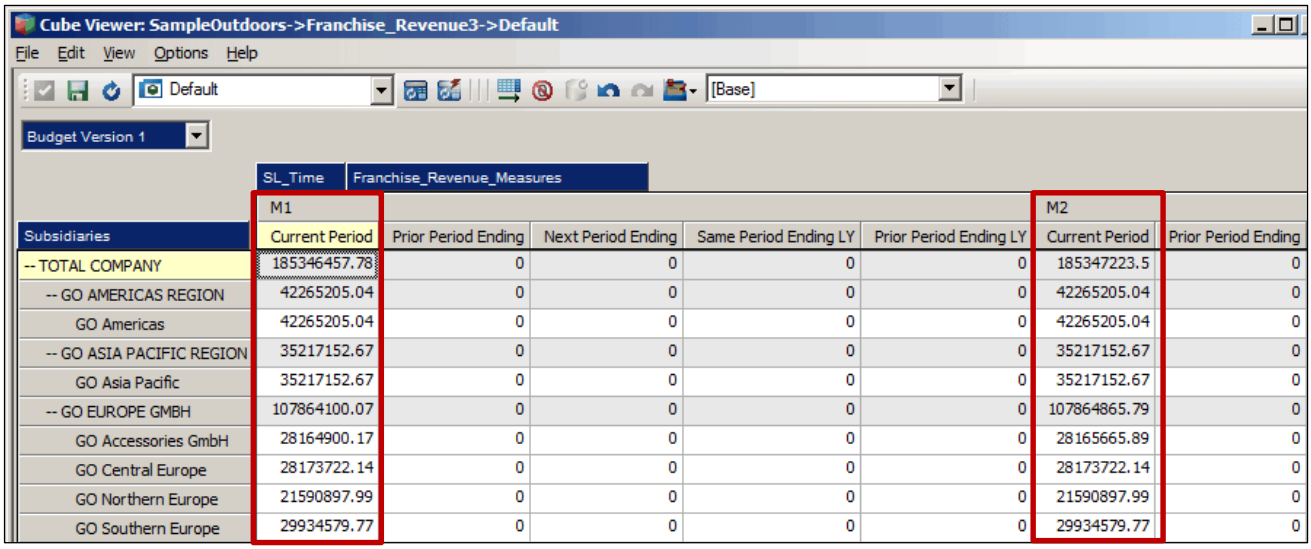
22. Click the **Dimensions** sub-tab, and then map the dimensions as follows:



23. Click the Data sub-tab and map to Budget Version 1 and Budget Version 2.
    The results appear as follows:



24. Click the **Advanced** tab and all the sub-tabs within.
25. Save and run the process, click **OK**, and then close **TurboIntegrator**.
26. Recalculate the default view of **Franchise_Revenue3** to verify that the **Current Period** columns contain values.
    The results appear as follows:



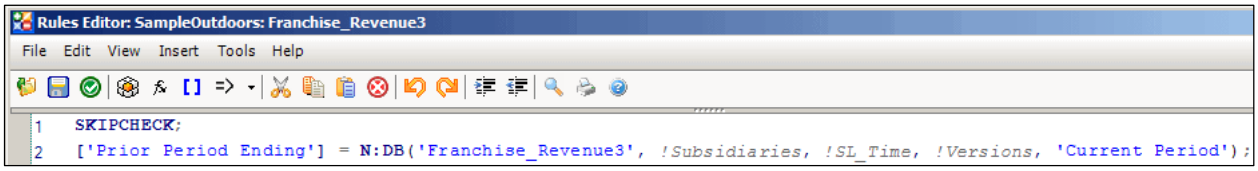
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Task 6. Create rules to calculate the time-related measures in the Franchise_Revenue_Measures dimension.

1. Right-click the **Franchise_Revenue3** cube, and then click **Create Rule**.
2. The Rules Editor window opens (this may take a few moments).
3. On the first line, type **SKIPCHECK;** and then press **Enter**.
4. Enter the following:

   ['Prior Period Ending'] = N:
5. Click **Insert Cube Reference** and then in the **Cube** list, select **Franchise_Revenue3** (if necessary).
6. Next to **Franchise_Revenue_Measures**, click the **Subset** icon, click **Current Period**, and then click **OK**.
7. Click **OK** to close the **Insert Cube Reference dialog** box.
8. Type a semicolon (**;**) at the end of the line.

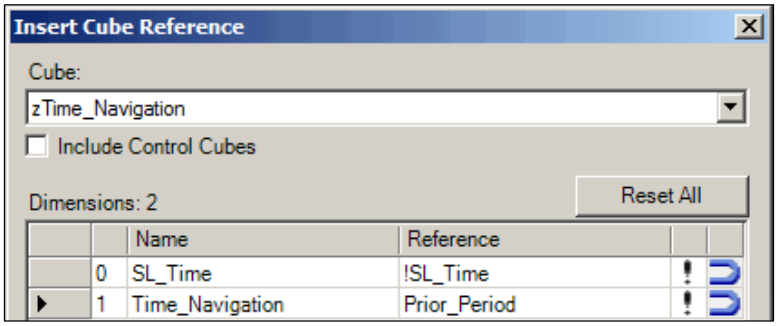
   The results appear as follows:



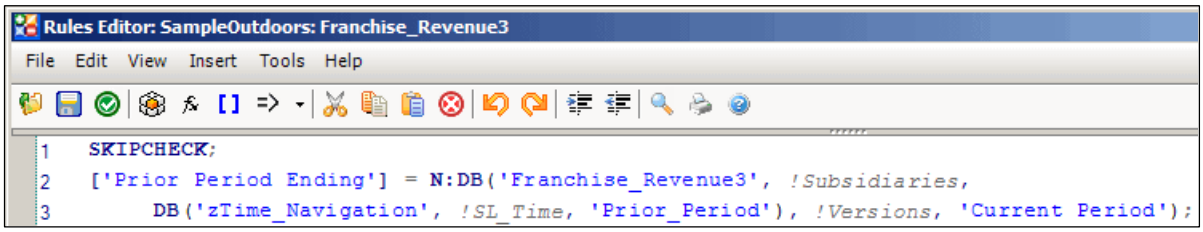
   You will now replace the **!SL_Time** part of the rule with a reference to the corresponding value for Prior Period in the zTime_Navigation cube. This way, no matter what the time period is, the rule can look to the zTime_Navigation cube and get the prior period for it. This makes for a much more flexible rule (with much less maintenance).
9. Select **!SL_Time** and press **Delete**, and then click **Insert Cube Reference**.
10. In the **Cube** list, **zTime_Navigation**, and then click the **Subset** icon next to **Time_Navigation**.

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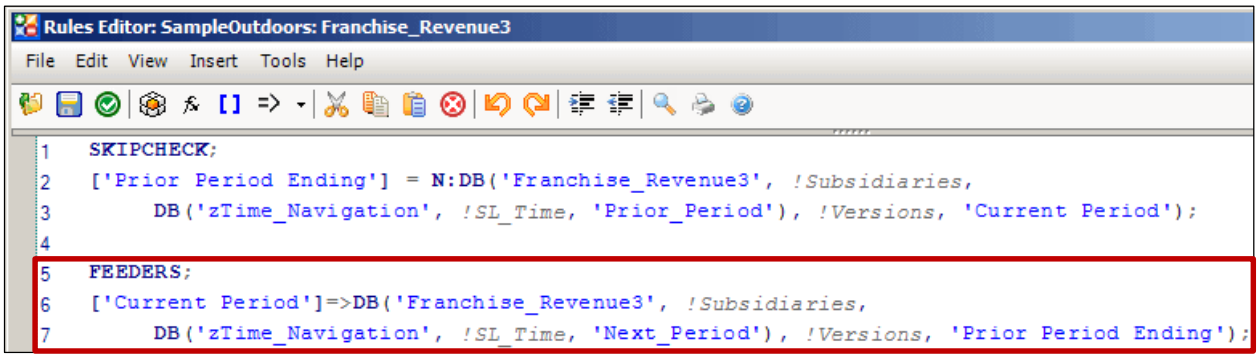
11. Click **Prior_Period**, and then click **OK**.
    The results appear as follows:



12. Click **OK** to close the **Insert Cube Reference** dialog box.
    The results appear as follows:

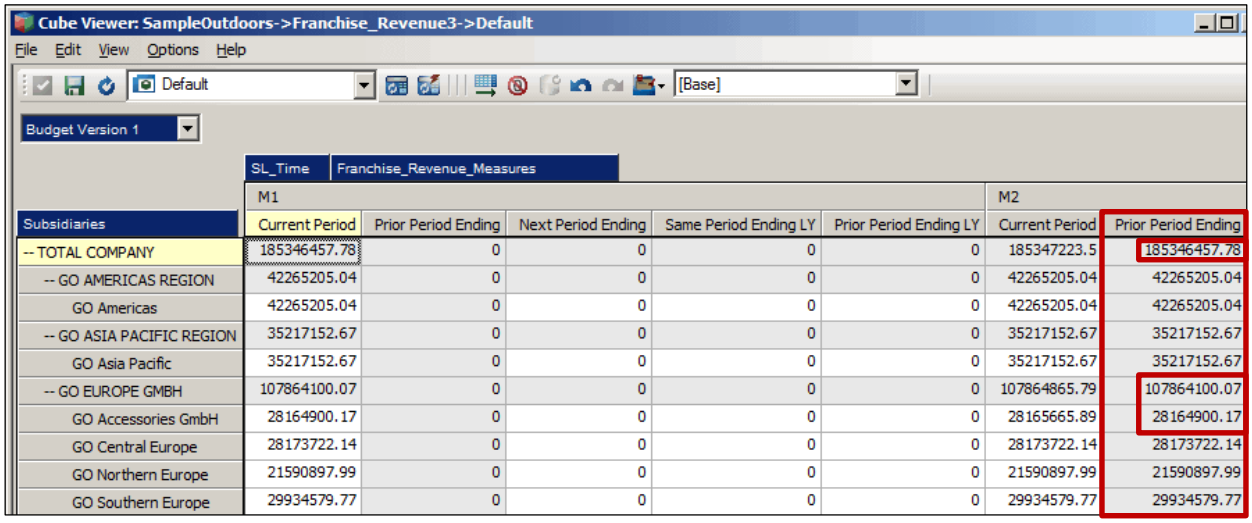


13. Press **Enter** twice, and then type **FEEDERS;**
14. Enter the following feeder:
    ['Current Period']=>DB('Franchise_Revenue3', !Subsidiaries, DB('zTime_Navigation', !SL_Time, 'Next_Period'), !Versions, 'Prior Period Ending');
    The results appear as follows:



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15. Save the rule, close the **Rules Editor** and then recalculate the view.
    The results appear as follows:



	M1					M2	
Subsidiaries	Current Period	Prior Period Ending	Next Period Ending	Same Period Ending LY	Prior Period Ending LY	Current Period	Prior Period Ending
-- TOTAL COMPANY	185346457.78	0	0	0	0	185347223.5	185346457.78
-- GO AMERICAS REGION	42265205.04	0	0	0	0	42265205.04	42265205.04
GO Americas	42265205.04	0	0	0	0	42265205.04	42265205.04
-- GO ASIA PACIFIC REGION	35217152.67	0	0	0	0	35217152.67	35217152.67
GO Asia Pacific	35217152.67	0	0	0	0	35217152.67	35217152.67
-- GO EUROPE GMBH	107864100.07	0	0	0	0	107864865.79	107864100.07
GO Accessories GmbH	28164900.17	0	0	0	0	28165665.89	28164900.17
GO Central Europe	28173722.14	0	0	0	0	28173722.14	28173722.14
GO Northern Europe	21590897.99	0	0	0	0	21590897.99	21590897.99
GO Southern Europe	29934579.77	0	0	0	0	29934579.77	29934579.77

16. The view now retrieves the prior period (shown in the Prior Period Ending column) as specified in the zTime_Navigation cube.

    Notice that the values in the **Prior Period Ending** cells for **M2** are different for the **TOTAL COMPANY**, **GO EUROPE GMBH** and **GO Accessories GmbH** subsidiaries. All of the values in this column come from the Current Period values for M1.
17. Close the **Cube Viewer** without saving changes.
18. Right-click the **SampleOutdoors** node and click **Save Data**.
19. Leave **Architect** open for the upcoming exercise.

Results:
You created another version of the Franchise_Revenue cube called Franchise_Revenue3. This new cube uses a straight line time dimension that calculates the next and previous time periods using a lookup cube.

You also corrected the problem of having no measures dimension in this cube. Unlike the other Franchise_Revenue cubes, this new cube can display continuous time analysis and converted currency data because it includes the Franchise_Revenue_Measures dimension.

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IBM Training

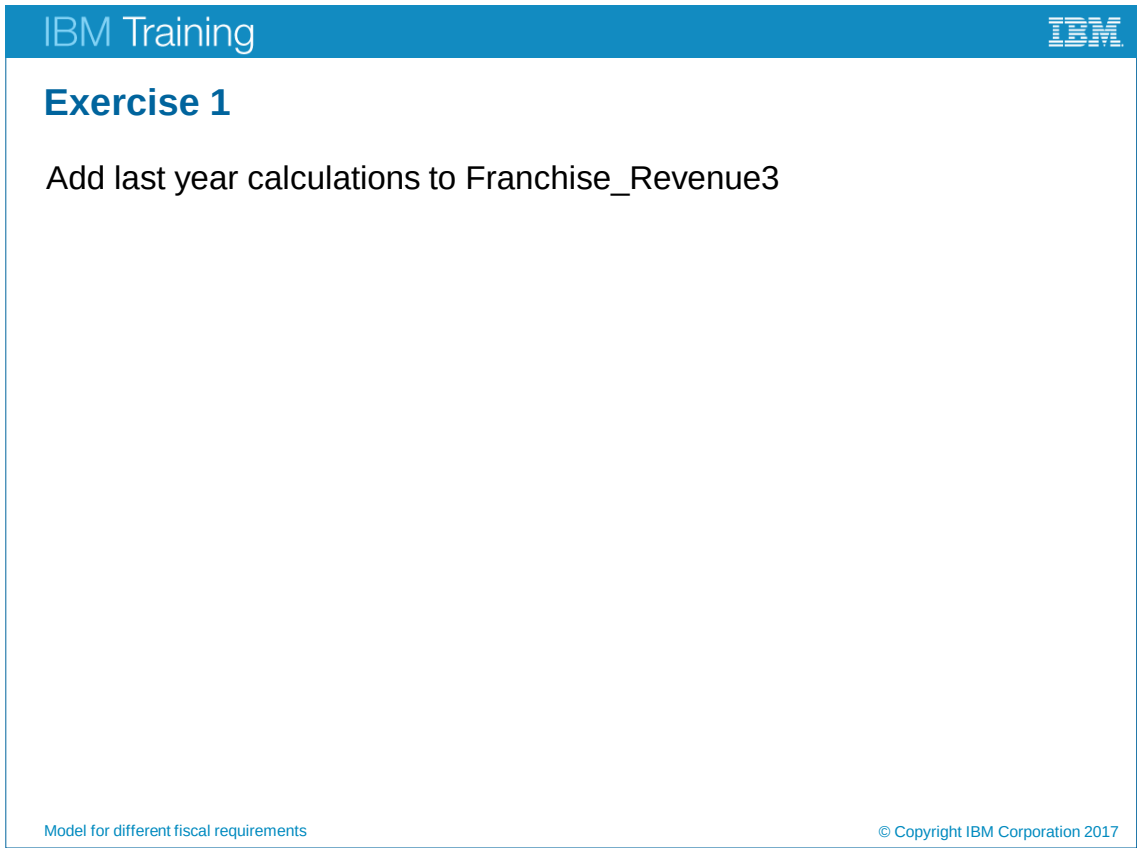
Unit summary

- Identify time considerations
- Use discrete time dimensions
- Implement a continuous time dimension model

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Unit summary

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Exercise 1: Add last year calculations to Franchise_Revenue3

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Exercise 1: Add last year calculations to Franchise_Revenue3

Your users would like to make comparisons between this year's and last year's data. You will need to add rules and feeders to calculate the same (or current) period last year balances. You have already created the time navigation cube for these calculations.

To accomplish this:

- Add rules to the Franchise_Revenue3 cube to pull in data for the same period ending last year and the prior period ending last year.
- Modify the feeder of the Franchise_Revenue3 cube to push data to the Same Period Ending LY and Prior Period Ending LY elements. Then open the Franchise_Revenue3 cube and view the data for the Current Period, Prior Period Ending, and Same Period Ending LY elements for January 2013. Save data to the SampleOutdoors server.

For more information about where to work and the exercise results, refer to the Tasks and results section that follows. If you need more information to complete a task, refer to earlier demonstrations for detailed steps.

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Exercise 1:
Tasks and results

Task 1. Add rules for last year calculations.

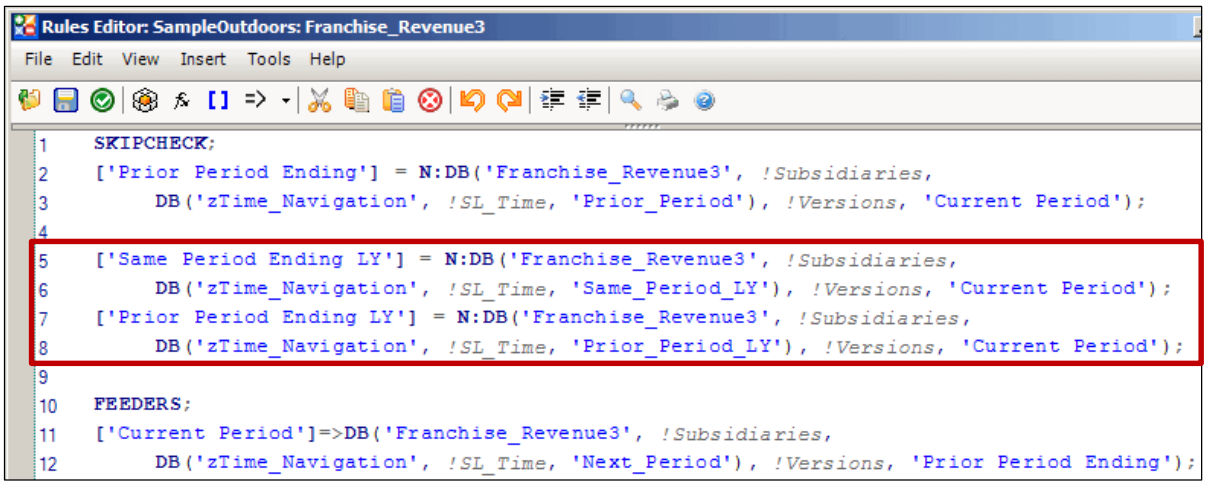
1. Right-click the **Franchise_Revenue3** cube and click **Edit Rule**.
2. Add a blank line above the **FEEDERS;** statement and then type the following:
   ['Same Period Ending LY'] = N:
3. Insert a cube reference to the Current Period measure:
   DB('Franchise_Revenue3', !Subsidiaries, !SL_Time, !Versions, 'Current Period');
4. Replace the **!SL_Time** part of the rule with the following:
   DB('zTime_Navigation', !SL_Time, 'Same_Period_LY')
5. Ensure that there is a semi-colon at the end of the line.
   The results appear as follows:

```
['Same Period Ending LY'] = N:DB('Franchise_Revenue3', !Subsidiaries,
     DB('zTime_Navigation', !SL_Time, 'Same_Period_LY'), !Versions, 'Current Period');
```

6. Repeat Steps **2** to **5** to add a rule for **Prior Period Ending LY**:

```
['Same Period Ending LY'] = N:DB('Franchise_Revenue3', !Subsidiaries,
     DB('zTime_Navigation', !SL_Time, 'Same_Period_LY'), !Versions, 'Current Period');
['Prior Period Ending LY'] = N:DB('Franchise_Revenue3', !Subsidiaries,
     DB('zTime_Navigation', !SL_Time, 'Prior_Period_LY'), !Versions, 'Current Period');
```

7. Click **Save**.
   The results appear as follows:



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Task 2. Modify the feeder and view the results in the Franchise_Revenue3 cube.

1. Place a comma after the first feeder (before the ;), and then press **Enter**.
2. Enter the following:

   DB('Franchise_Revenue3', !Subsidiaries, DB('zTime_Navigation', !SL_Time, 'Same_Period_NY'), !Versions, 'Same Period Ending LY'), DB('Franchise_Revenue3', !Subsidiaries, DB('zTime_Navigation', !SL_Time, 'Next_Period_NY'), !Versions, 'Prior Period Ending LY')

   Ensure that the feeder ends with a single semi-colon.

   Note: This feeder statement feeds both the Same Period Ending LY and Prior Period Ending LY rules.

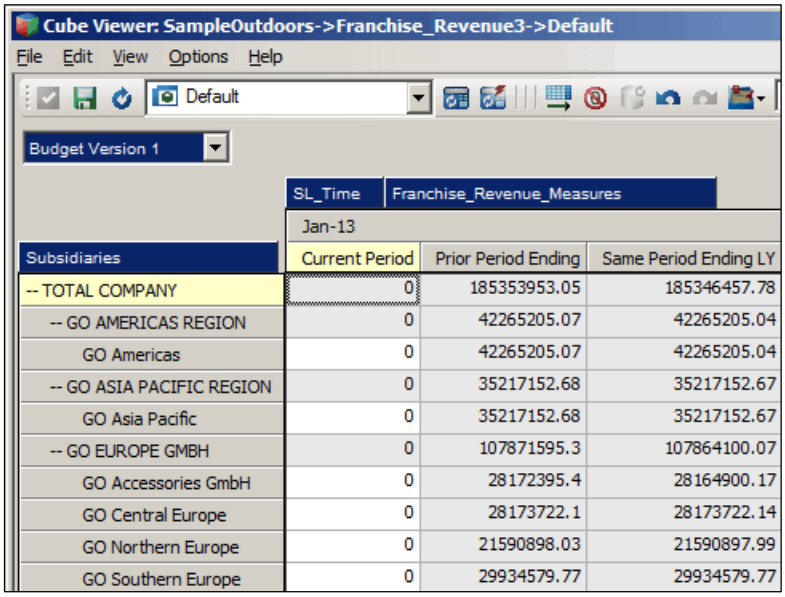
   The results appear as follows:

```
FEEDERS;
['Current Period']=>DB('Franchise_Revenue3', !Subsidiaries,
     DB('zTime_Navigation', !SL_Time, 'Next_Period'), !Versions, 'Prior Period Ending'),
     DB('Franchise_Revenue3', !Subsidiaries, DB('zTime_Navigation', !SL_Time, 'Same_Period_NY'),
         !Versions, 'Same Period Ending LY'),
     DB('Franchise Revenue3', !Subsidiaries, DB('zTime_Navigation', !SL_Time, 'Next_Period_NY'),
         !Versions, 'Prior Period Ending LY');
```

3. Click **Save**, and then close the **Rules Editor**.
4. Open the **Franchise_Revenue3** cube, and then click the **SL_Time** dimension (on columns).
5. Click the **Use Aliases (currently off)** button.
6. Click **All**, click **Jan-13**, and then click **OK**.
7. Click **Franchise_Revenue_Measures** (on columns).
8. Click **All**, Ctrl+click **Current Period**, **Prior Period Ending**, and **Same Period Ending LY**, and then click **OK**.

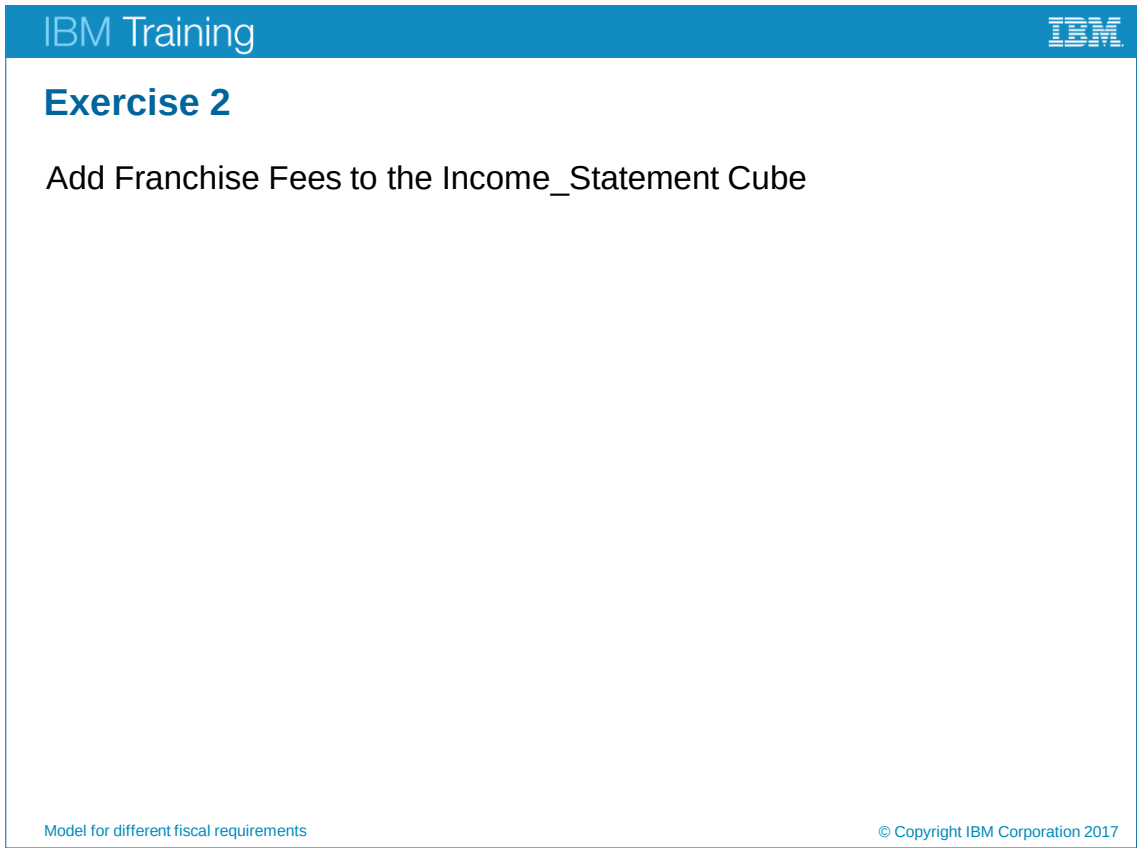
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9. Click **Recalculate** and expand all the subsidiaries on the rows (if necessary). The results appear as follows:



10. Close **Cube Viewer** without saving.
11. Right-click the **SampleOutdoors** node and click **Save Data**.
12. Leave **Architect** open for the next exercise.

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Exercise 2: Add Franchise Fees to the Income_Statement Cube

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Exercise 2: Add Franchise Fees to the Income_Statement Cube

Now that you have completed the cube containing franchise revenue, you need to add this data to the Income_Statement cube.

To accomplish this:

- Create a view of the Income_Statement cube that includes the Franchise Fees element from the Income_Statement_Accounts dimension. Put this dimension on rows and the Income_Statement_Measures dimension on columns. Then add a rule to the Income_Statement cube to pull only the relevant data from the Franchise_Revenue cube.
- Add a feeder to the Franchise_Revenue cube that pushes the relevant data to the Income_Statement cube. You only need to reference Budget Version 1. Then recalculate the view of the Income_Statement cube to see the results of the rule and feeder.

For more information about where to work and the exercise results, refer to the Tasks and results section that follows. If you need more information to complete a task, refer to earlier demonstrations for detailed steps.

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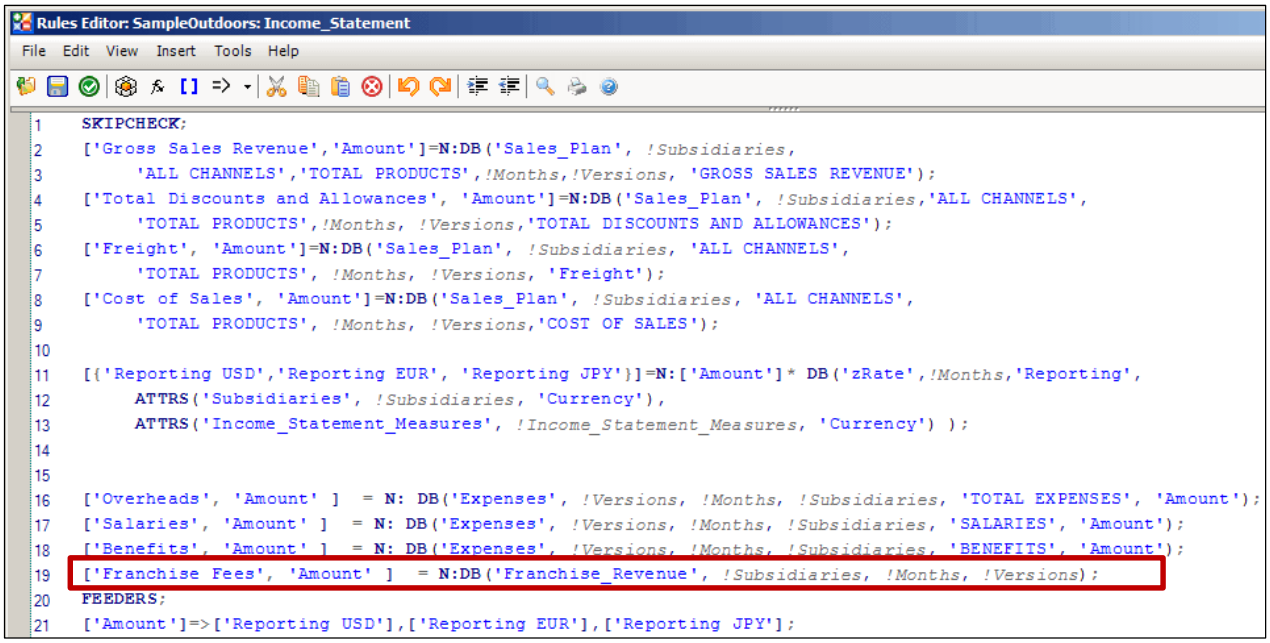
Exercise 2: Tasks and results

Task 1. Add a rule to pull data from the Franchise_Revenue cube into the Income_Statement cube.

1. Open the default view for the **Income_Statement** cube.
2. Arrange the view as follows:
   - Title area: **Versions** (with **Budget Version 1** selected), **Months** (with **Jan** selected), **Subsidiaries** (with **GO Americas** selected)
   - Rows: **Income Statement Accounts** (with all elements expanded)
   - Columns: **Income Statement Measures**
3. Save the view as **Final**.
4. In **Architect**, right-click the **Income_Statement** cube, and then click **Edit Rule**.
5. On the line above the **FEEDERS;** statement type the following:

   ['Franchise Fees', 'Amount'] = N:DB('Franchise_Revenue', !Subsidiaries, !Months, !Versions);

   The results appear as follows:



6. Click **Save** and then close the **Rules Editor**.

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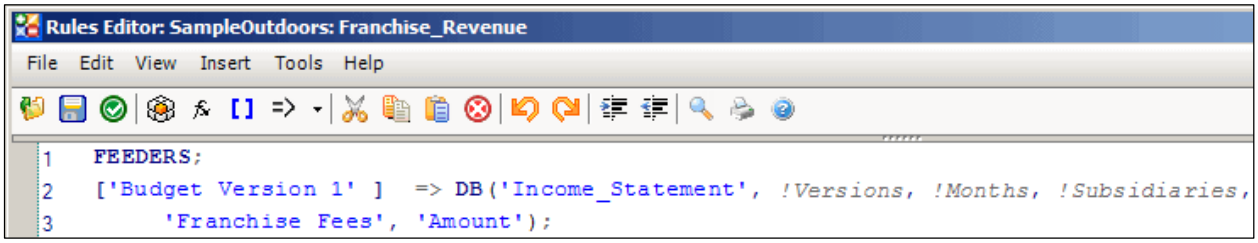
Task 2. Add a feeder to push data from the Franchise_Revenue cube into the Income_Statement cube.

1. Right click the **Franchise_Revenue** cube and click **Create Rule**.
2. Type the following:

   FEEDERS;

   ['Budget Version 1'] => DB('Income_Statement', !Versions, !Months, !Subsidiaries, 'Franchise Fees', 'Amount');

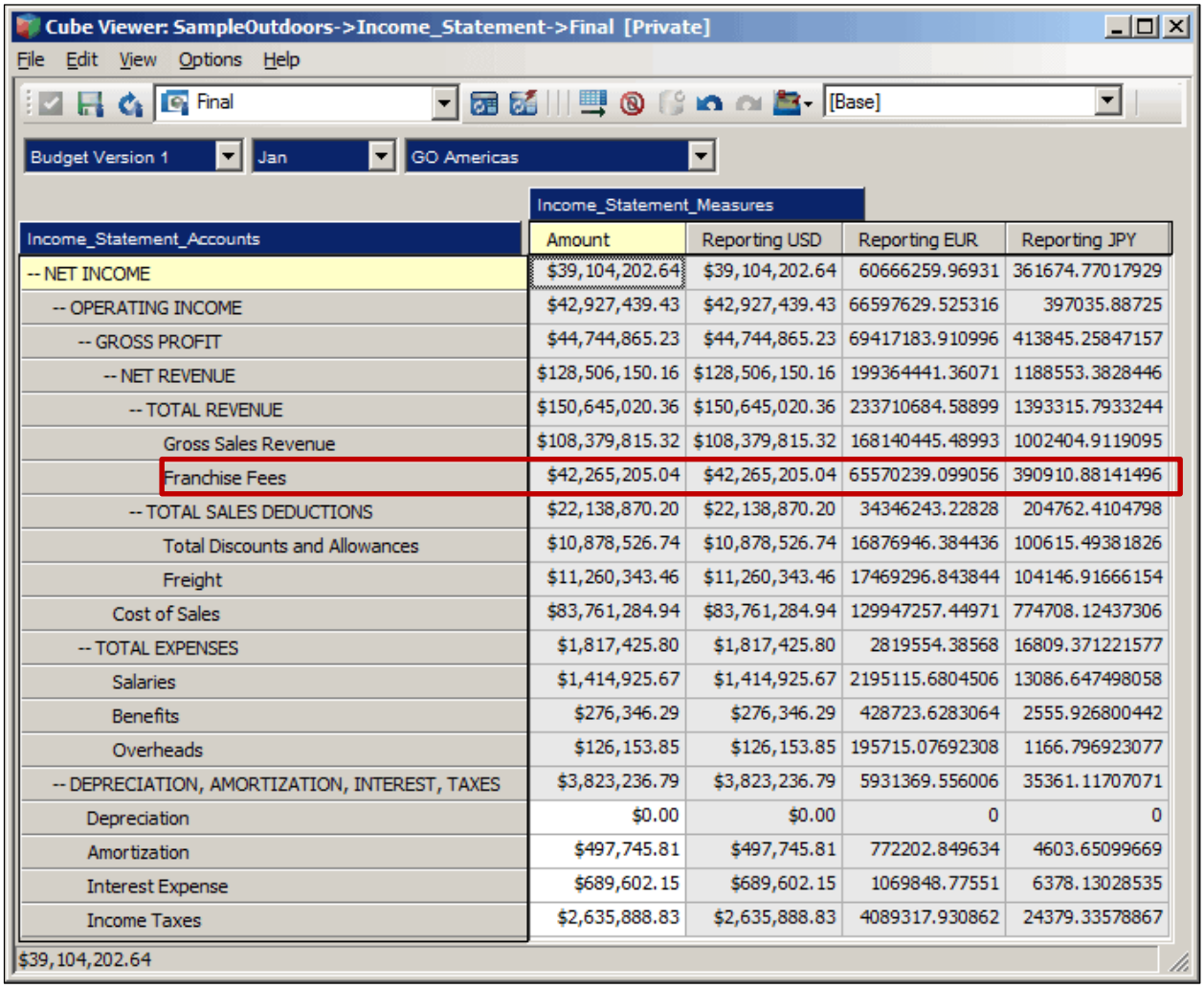
   The results appear as follows:



3. Click **Save** and then close the Rules Editor.
4. Click the **Final** view of the **Income_Statement** and recalculate.

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The results appear as follows:



Cube Viewer: SampleOutdoors->Income_Statement->Final [Private]

File Edit View Options Help

Final | [Base]

Budget Version 1 | Jan | GO Americas

Income_Statement_Accounts	Income_Statement_Measures: Amount	Reporting USD	Reporting EUR	Reporting JPY
-- NET INCOME	$39,104,202.64	$39,104,202.64	60666259.96931	361674.77017929
-- OPERATING INCOME	$42,927,439.43	$42,927,439.43	66597629.525316	397035.88725
-- GROSS PROFIT	$44,744,865.23	$44,744,865.23	69417183.910996	413845.25847157
-- NET REVENUE	$128,506,150.16	$128,506,150.16	199364441.36071	1188553.3828446
-- TOTAL REVENUE	$150,645,020.36	$150,645,020.36	233710684.58899	1393315.7933244
Gross Sales Revenue	$108,379,815.32	$108,379,815.32	168140445.48993	1002404.9119095
Franchise Fees	$42,265,205.04	$42,265,205.04	65570239.099056	390910.88141496
-- TOTAL SALES DEDUCTIONS	$22,138,870.20	$22,138,870.20	34346243.22828	204762.4104798
Total Discounts and Allowances	$10,878,526.74	$10,878,526.74	16876946.384436	100615.49381826
Freight	$11,260,343.46	$11,260,343.46	17469296.843844	104146.91666154
Cost of Sales	$83,761,284.94	$83,761,284.94	129947257.44971	774708.12437306
-- TOTAL EXPENSES	$1,817,425.80	$1,817,425.80	2819554.38568	16809.371221577
Salaries	$1,414,925.67	$1,414,925.67	2195115.6804506	13086.647498058
Benefits	$276,346.29	$276,346.29	428723.6283064	2555.926800442
Overheads	$126,153.85	$126,153.85	195715.07692308	1166.796923077
-- DEPRECIATION, AMORTIZATION, INTEREST, TAXES	$3,823,236.79	$3,823,236.79	5931369.556006	35361.11707071
Depreciation	$0.00	$0.00	0	0
Amortization	$497,745.81	$497,745.81	772202.849634	4603.65099669
Interest Expense	$689,602.15	$689,602.15	1069848.77551	6378.13028535
Income Taxes	$2,635,888.83	$2,635,888.83	4089317.930862	24379.33578867

$39,104,202.64

You may choose to pull values from the Franchise_Revenue3 cube. This will be more challenging because you would need a more complex rule. You could also update the zTime_Navigation cube to contain additional values to match the attributes or elements in the Months dimension.

5. Close **Cube Viewer**.
6. Right-click the **SampleOutdoors** node and click **Save Data**.
7. Close all open windows.

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IBM Training

Deploy IBM Planning Analytics applications

IBM Planning Analytics (v2.0)

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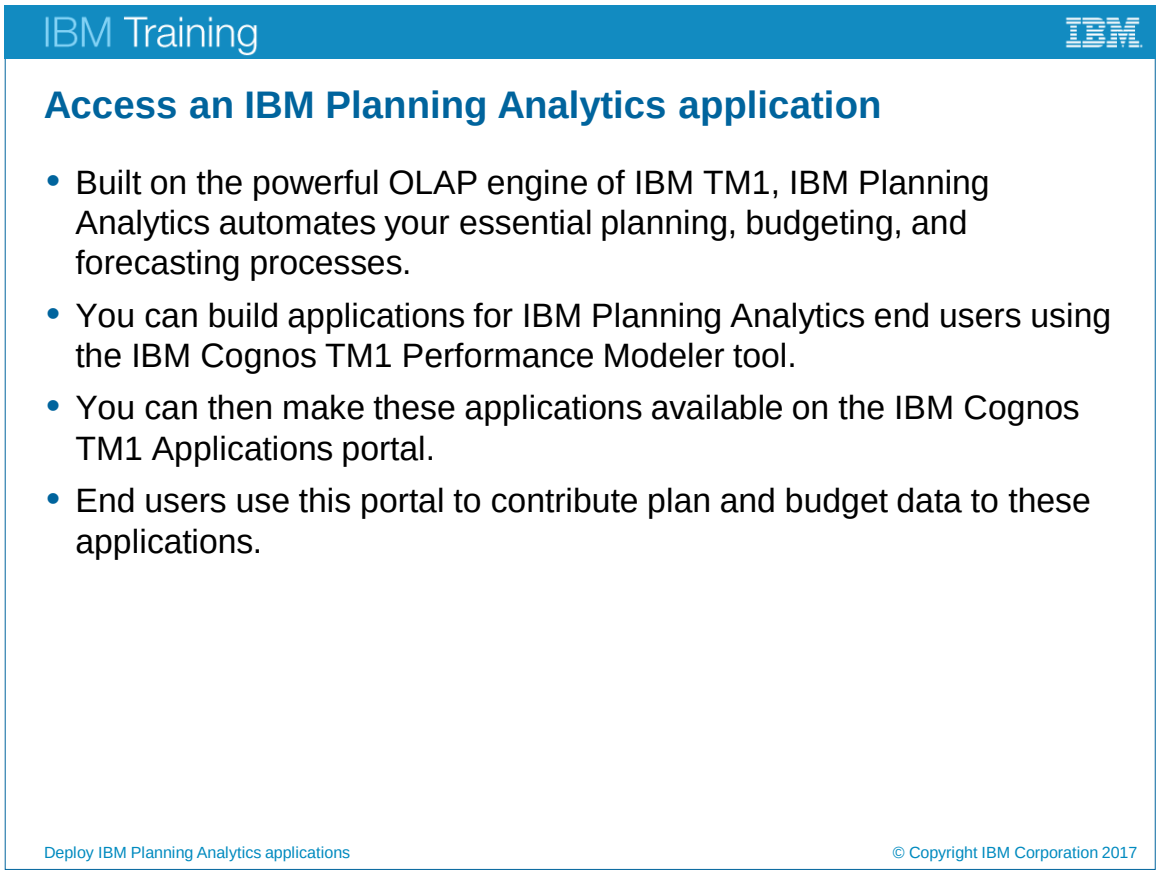
Unit objectives

- Explain the application types
- Access an IBM Planning Analytics application
- Create a new application
- Set the available clients
- Apply security in the application
- Activate and deactivate an application

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Unit objectives

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Access an IBM Planning Analytics application

In this unit, you will use the IBM Cognos TM1 Applications portal to access various plans.

TM1 application administrators use this portal to:

- create and maintain applications
- manage rights to secure applications
- import and export applications
- activate and deactivate applications

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IBM Training

What is an application?

- A TM1 application is a series of TM1 cube views designed to collect and calculate data for a specific business plan, budget, or forecast.
- Some applications have defined completion dates, while others are ongoing.
- Depending on how the application is designed by the administrator, users may download and work with the entire application, or only the part for which they are responsible.

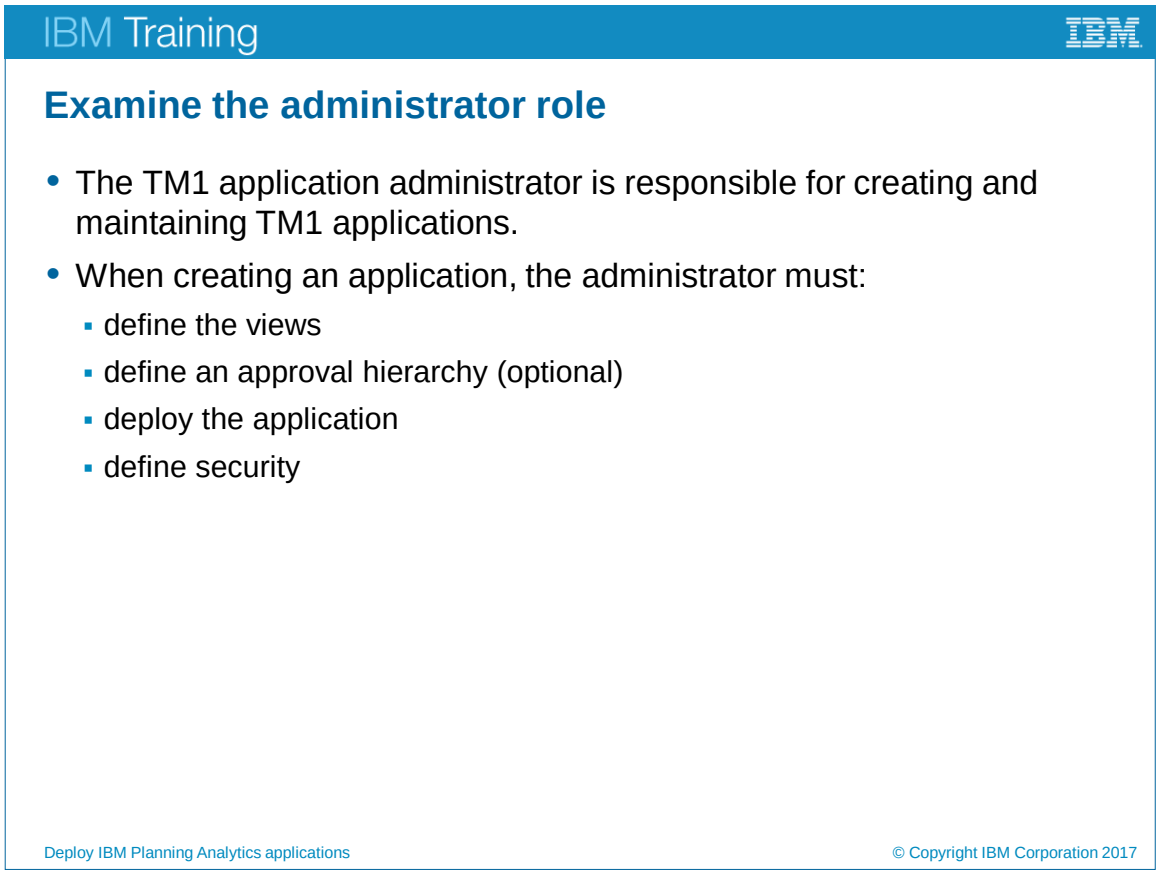
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What is an application?

If an application has been created with an approval hierarchy, it will be sectioned into parts, such as a separate part for each Sales Division or each Department.

Each part is called a node.

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Examine the administrator role

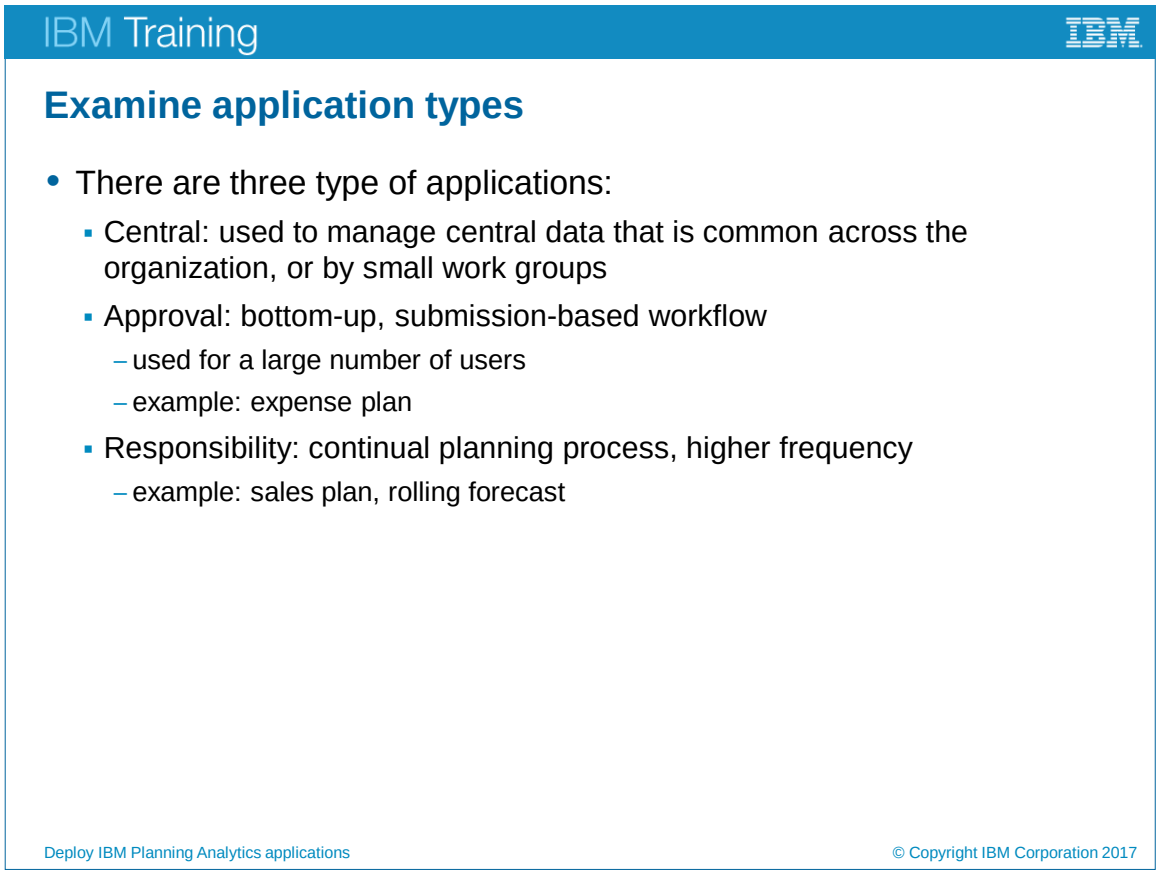
Define views: select the public views to be included in the application and set their properties.

Define an approval hierarchy (optional): choose the dimension subset that reflects the structure of your business. This defines the workflow for the application.

Deploy the application: makes the application available on the IBM Cognos TM1 Applications portal (for administrators only at this point).

Define security: after the application is deployed, define security to determine the rights users have to the application and to make the application available to users in the portal.

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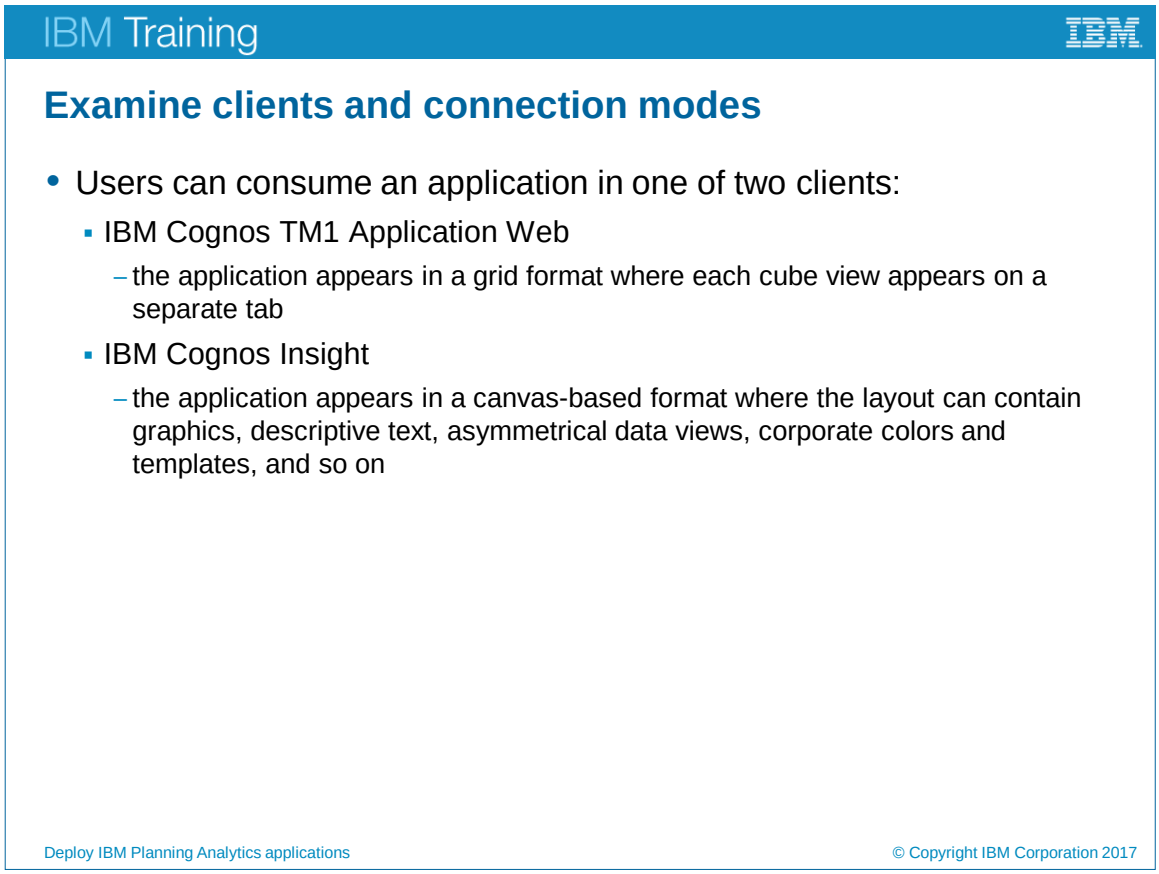
Examine application types

Central: No hierarchy. When you connect to a Central application, you download the entire plan and are directly connected to the server. Used by a smaller number of users for top-down budgeting, managing assumption data, model maintenance, and so on.

Approval: Contains hierarchy. You can only download and work with the part of the plan for which you are responsible. Contributors must submit their data, and reviewers can either accept the data by submitting it to the next level of approval, or reject the data so it can be adjusted. Often these plans have fixed end dates.

Responsibility: Contains hierarchy. You can only download and work with the part of the plan for which you are responsible. These applications have no submit and reject capabilities, and no fixed end dates. The data is adjusted on an ongoing basis.

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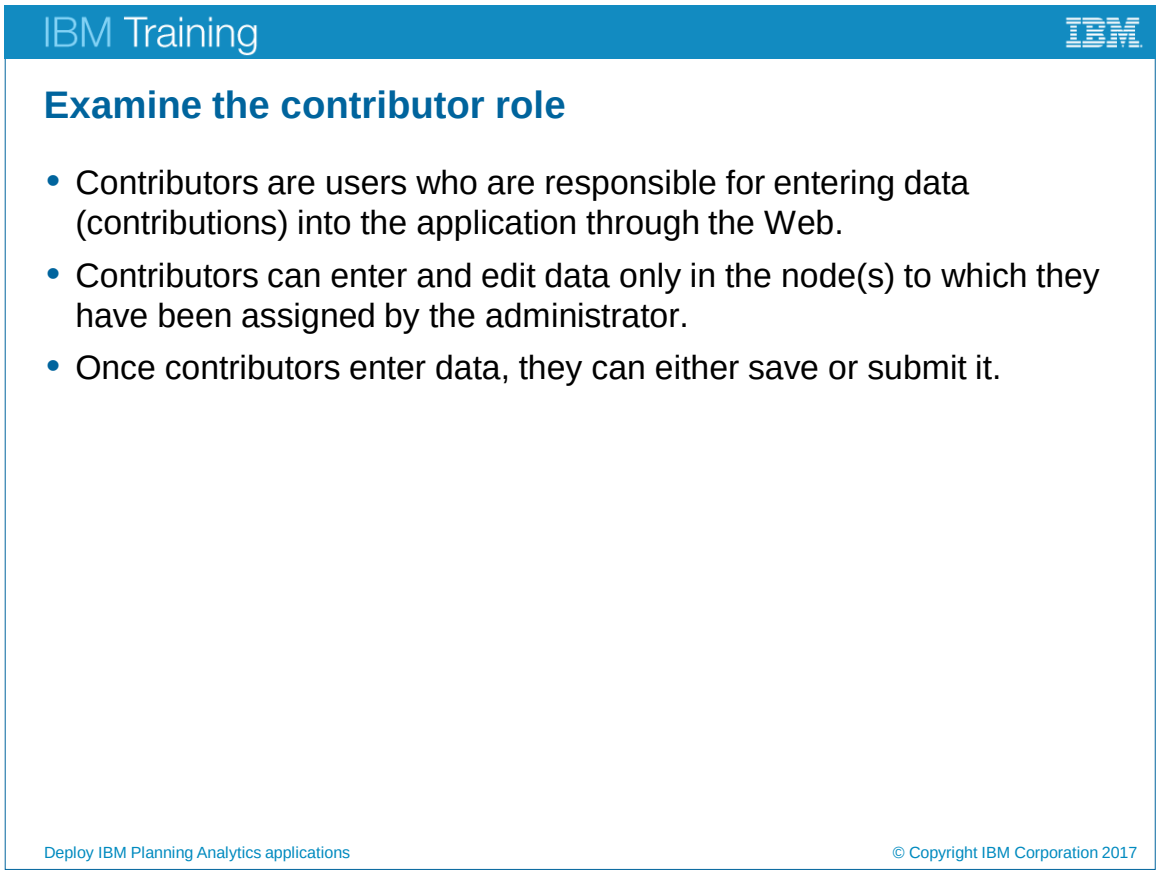
Examine clients and connection modes

Users can open an application in Cognos Insight in one of two modes, depending on how the administrator has set up the application:

- Distributed: Users select their node, and that part of the plan is downloaded to their computer. Processing is done on their computer for increased responsiveness. In this mode, you can take a node offline.

Connected: Users select their node, and work directly with the data on the server. Useful for working with extremely large data that may not fit within the memory available on the user's computer.

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Examine the contributor role

When a contributor saves data, it is saved to the server, and can be viewed and edited by planners who have access to that node.

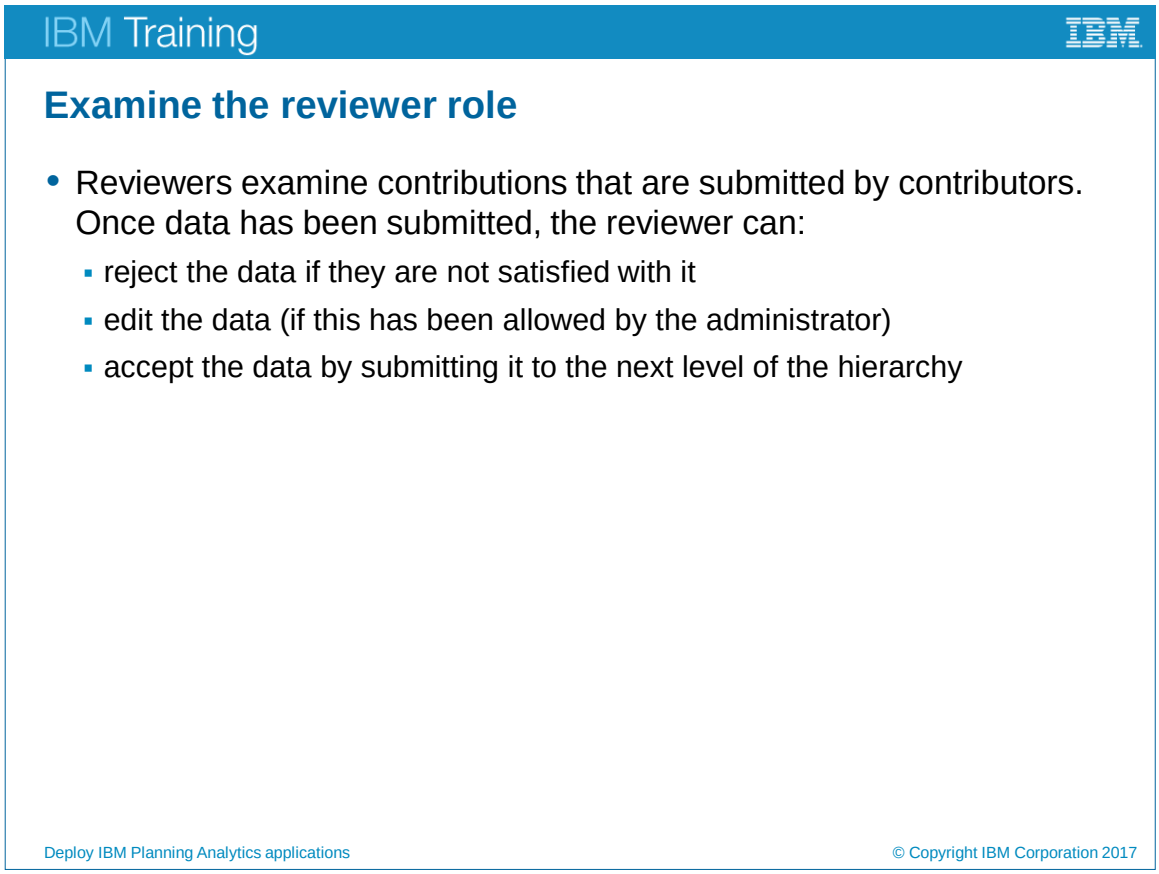
When a contributor submits data, the data is forwarded to a reviewer and cannot be edited further by planners unless the reviewer rejects it.

A contributor may be responsible for entering data for more than one node.

If a contributor is responsible for multiple nodes under the same parent, they can view each contribution node individually, or see all contribution nodes in a single view.

If they have different parents, you may only view them separately.

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Examine the reviewer role

At any time during the planner/reviewer cycle, reviewers can view contributions that they are responsible for managing.

A user can be both a planner and a reviewer for the same node.

If a complete set of submitted data has been viewed by the reviewer and is satisfactory, the data can then be submitted to the next reviewer in the hierarchy.

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IBM Training

Identify Workflow States

	**Available**. The node has not been opened and the data has not been changed or saved.
	**Reserved**. The user has taken ownership of the node and data in this state can be submitted for review.
	**Locked**. The data was submitted and the node is locked. Data in this state is read only. If this node is rejected, its state returns to Reserved.
	**Incomplete**. At least one child node of this parent node is Available, and at least one other child node is Reserved, Locked, or Ready.
	**Work in Progress**. No nodes belonging to this parent node are Available, but not all are locked.
	**Ready**. All child nodes of this parent node are locked. This parent node is ready to be submitted to the next level in the hierarchy

Deploy IBM Planning Analytics applications

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Identify Workflow States

Icons below the dotted line refer to states that apply only to parent nodes (such as North America).

Icons above the dotted line refer to states that apply to child (leaf level) nodes (such as Canada). Parent nodes can also be Available or Locked.

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IBM Training

Demonstration 1

Examine three types of applications

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Demonstration 1: Examine three types of applications

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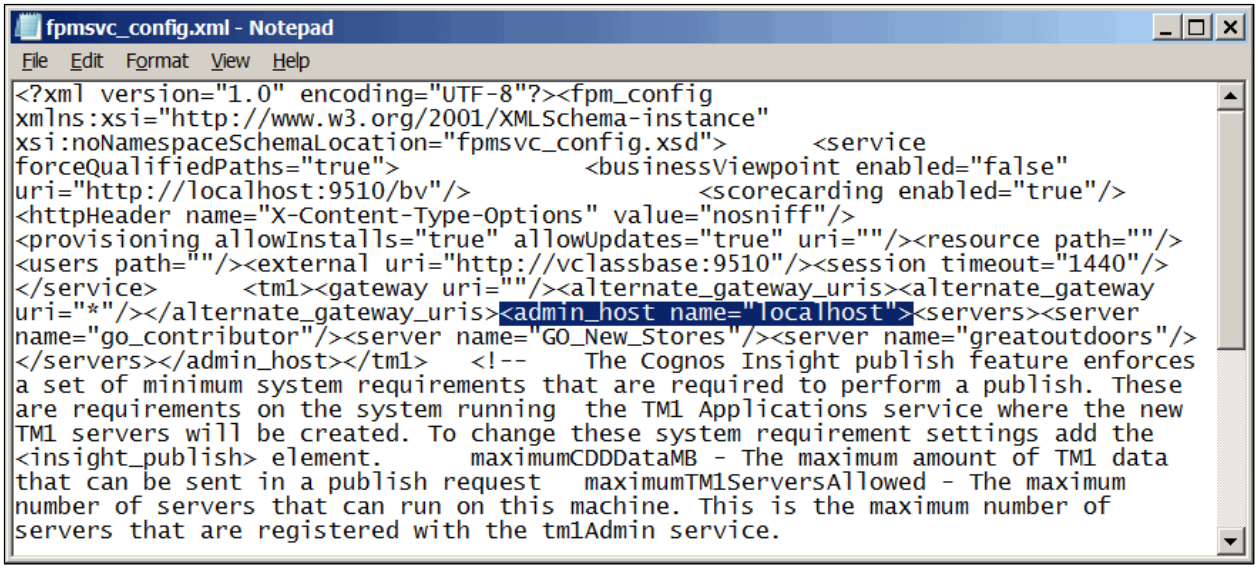
Demonstration 1: Examine three types of applications

Purpose:
You want to familiarize yourself with the various types of TM1 applications. You will log on to the Web portal and open and explore three applications that have just been created by another administrator.

Task 1. Adjust the configuration file.

1. Open **Windows Explorer**, and then navigate to **C:\Program Files\IBM\cognos\tm1_64\webapps\pmpsvc\WEB-INF\configuration**.
2. Open **fpmsvc_config.xml** in **Notepad**, and then from the **Format** menu, select **Word Wrap** to enable it (if necessary).
3. If necessary, change the value of the **admin_host** name property from the existing value to **localhost**.

   The results appear similar to the following:



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4. Save and close the file, and then close **Windows Explorer**.

   Verifying this setting will prevent any errors from appearing when you open an application node in IBM Cognos TM1 Application Web. It should also decrease the amount of time that it takes for the IBM Cognos TM1 Applications portal to open in your environment.

   If you find that it takes a long time for the portal to open (for example, after you have restarted the operating system), verify that the admin_host name property in the fpmsvc_config.xml file still has a value of "localhost" (instead of, for example, an IP address). If you have to change this value, ensure that you save the file and restart the IBM Cognos TM1 service again.

5. In the **taskbar**, click the **Services** button  , and then stop the **IBM Cognos TM1** service.
6. Start the **IBM Cognos TM1** service, and then close the **Services** window.

Task 2. Explore an Approval application.

1. Open **Internet Explorer**, and then go to **http://localhost:9510/pmpsvc**.
2. Log in as **admin** with the password **apple**.

   The IBM Cognos TM1 Applications Web portal appears (this may take a few moments):



   There are three applications available to you. The order of the applications may vary. You will start by exploring the Income_Statement application, which is an Approval application.

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If the Income_Statement application is not visible, perform the following steps:

a) Open the **Services** window and check to see if the **IBM Cognos TM1 Server - greatoutdoors** service is running. If it is already running, restart this service and proceed to step **f)**. If it is not already running, proceed to step **b)**.

b) In **Windows Explorer**, go to **C:\Program Files\IBM\cognos\tm1_64\samples\tm1\greatoutdoors\logfiles**.

c) Send all the files in this folder to the **Recycle Bin**.

d) On the **taskbar**, click the **Services** button to open the **Services** window.

e) Start the **IBM Cognos TM1 Server - greatoutdoors** service.

f) Close the **Services** window.

g) In **Internet Explorer**, from the **View** menu, click **Refresh**.

3. Click **Income_Statement**.

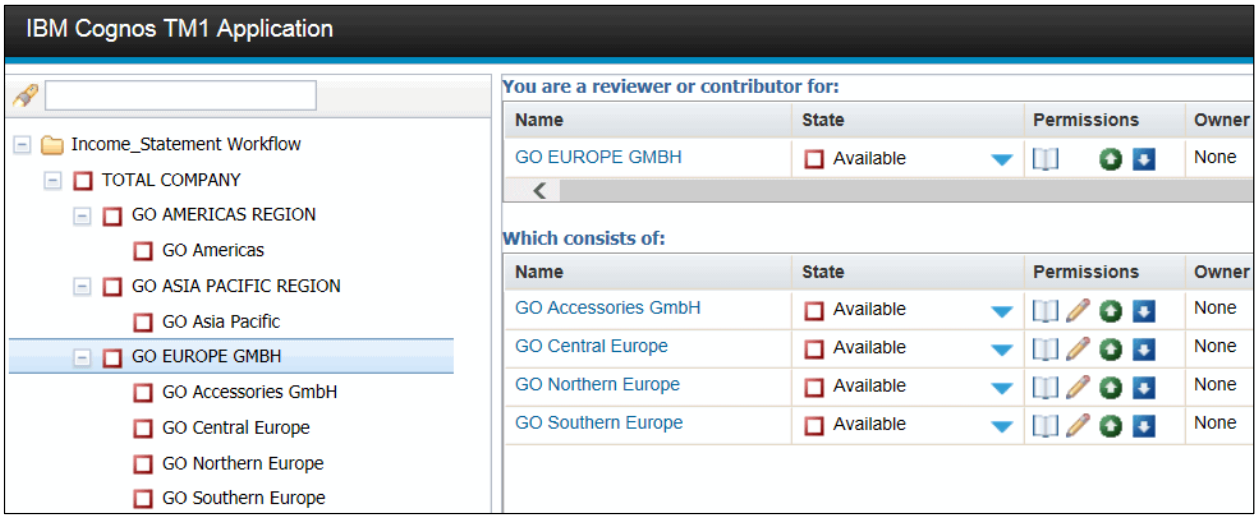
   The workflow page appears in IBM Cognos TM1 Application. Because this is an Approval application, you can see a hierarchy of nodes in the left pane.

4. In the left pane, expand **TOTAL COMPANY**, **GO AMERICAS REGION**, **GO ASIA PACIFIC REGION**, and **GO EUROPE GMBH**.

   This application is distributed by Subsidiary, and security has been defined so that users that belong to the appropriate security group can contribute to or review data for a specific subsidiary.

5. In the tree on the left, ensure that **GO EUROPE GMBH** is selected.

   The results appear as follows:



The top of the table displays details for the parent node selected in the tree. The bottom of the table displays details for the child nodes that belong to the parent node.

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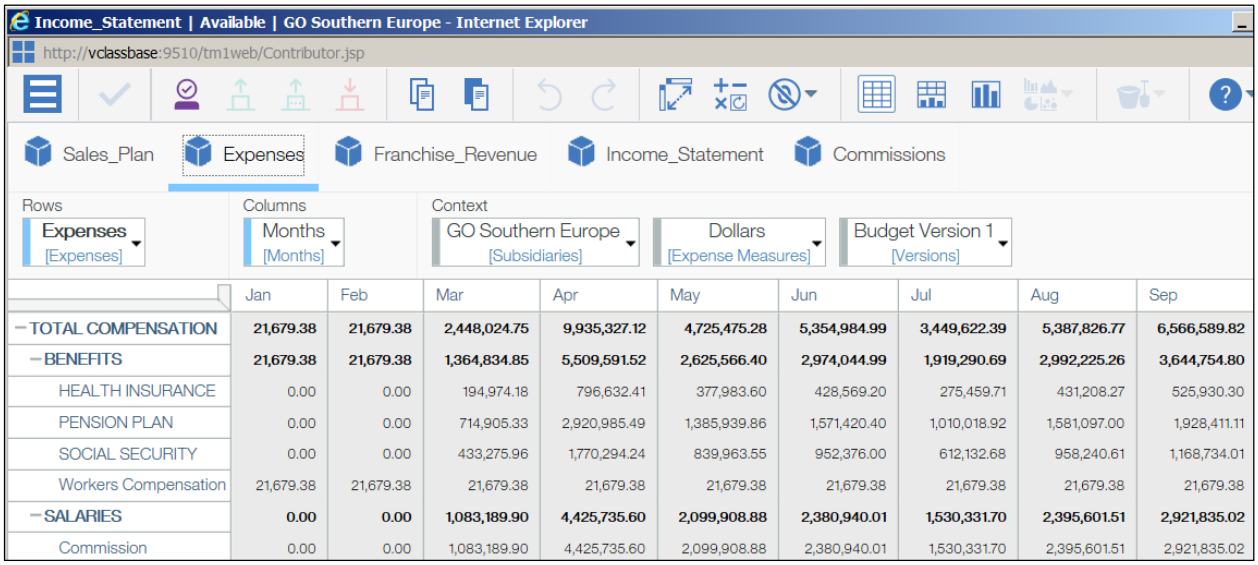
You can see that all the nodes are Available, which means that no one has taken ownership of these nodes and saved changes to the data. Because this is an Approval application, you can see in the Permissions column that all nodes can be Submitted or Rejected .

6. In the bottom table, right-click **GO Southern Europe**.

   You can open this node in TM1 Application Web, or in Cognos Insight in either Distributed or Connected mode. The administrator determines which clients will be made available when creating the application.

7. Click **Open TM1 Application Web**, maximize the window, and then when the node opens (it may take a few moments to appear), click the **Expenses** tab.

   The result appears as shown below:



	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
TOTAL COMPENSATION	21,679.38	21,679.38	2,448,024.75	9,935,327.12	4,725,475.28	5,354,984.99	3,449,622.39	5,387,826.77	6,566,589.82
BENEFITS	21,679.38	21,679.38	1,364,834.85	5,509,591.52	2,625,566.40	2,974,044.99	1,919,290.69	2,992,225.26	3,644,754.80
HEALTH INSURANCE	0.00	0.00	194,974.18	796,632.41	377,983.60	428,569.20	275,459.71	431,208.27	525,930.30
PENSION PLAN	0.00	0.00	714,905.33	2,920,985.49	1,385,939.86	1,571,420.40	1,010,018.92	1,581,097.00	1,928,411.11
SOCIAL SECURITY	0.00	0.00	433,275.96	1,770,294.24	839,963.55	952,376.00	612,132.68	958,240.61	1,168,734.01
Workers Compensation	21,679.38	21,679.38	21,679.38	21,679.38	21,679.38	21,679.38	21,679.38	21,679.38	21,679.38
SALARIES	0.00	0.00	1,083,189.90	4,425,735.60	2,099,908.88	2,380,940.01	1,530,331.70	2,395,601.51	2,921,835.02
Commission	0.00	0.00	1,083,189.90	4,425,735.60	2,099,908.88	2,380,940.01	1,530,331.70	2,395,601.51	2,921,835.02

   This application collects budgeted expense and revenue data for each subsidiary in order to calculate a forecasted income statement.

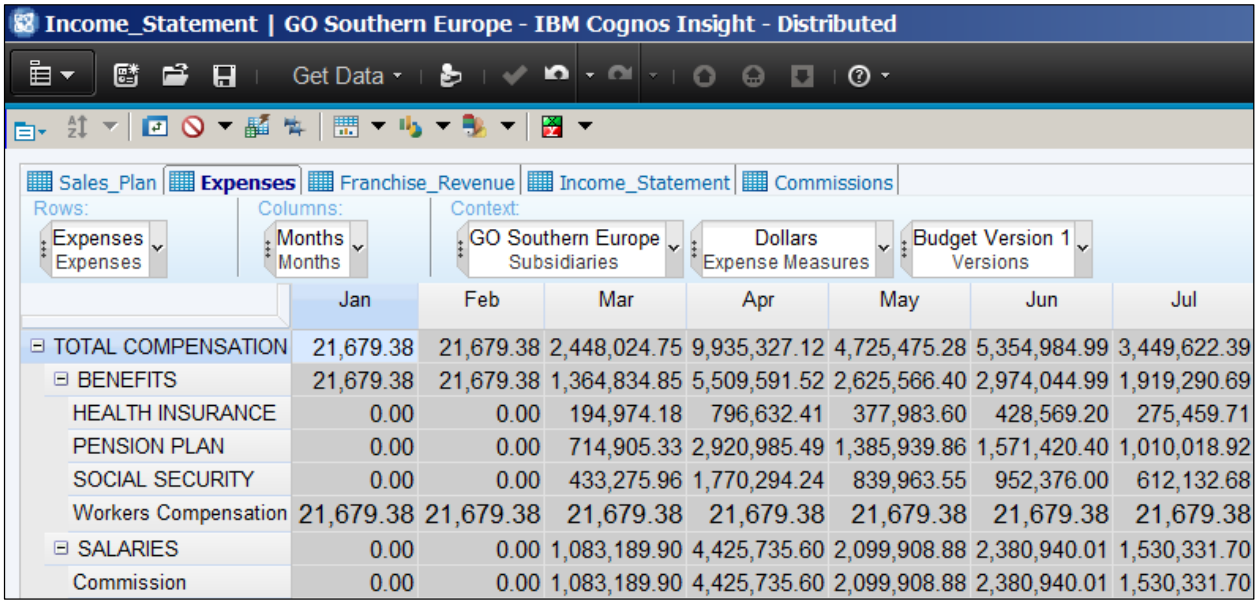
   You now want to open this node in Cognos Insight - Distributed mode.

8. Close the **Web** grid.
9. On the workflow page, in the bottom table, right-click **GO Southern Europe**, and then click **Open Cognos Insight - Distributed**.
10. If you are prompted to open the **provagent.cogrcp_insight** file, click the **Open** button at the bottom.
11. Wait for the application to open (it will take a few moments), maximize the window, and then click the **Expenses** tab.

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12. Expand all of the rows on the left side (if necessary).

    The result appears as shown below:



	Jan	Feb	Mar	Apr	May	Jun	Jul
TOTAL COMPENSATION	21,679.38	21,679.38	2,448,024.75	9,935,327.12	4,725,475.28	5,354,984.99	3,449,622.39
BENEFITS	21,679.38	21,679.38	1,364,834.85	5,509,591.52	2,625,566.40	2,974,044.99	1,919,290.69
HEALTH INSURANCE	0.00	0.00	194,974.18	796,632.41	377,983.60	428,569.20	275,459.71
PENSION PLAN	0.00	0.00	714,905.33	2,920,985.49	1,385,939.86	1,571,420.40	1,010,018.92
SOCIAL SECURITY	0.00	0.00	433,275.96	1,770,294.24	839,963.55	952,376.00	612,132.68
Workers Compensation	21,679.38	21,679.38	21,679.38	21,679.38	21,679.38	21,679.38	21,679.38
SALARIES	0.00	0.00	1,083,189.90	4,425,735.60	2,099,908.88	2,380,940.01	1,530,331.70
Commission	0.00	0.00	1,083,189.90	4,425,735.60	2,099,908.88	2,380,940.01	1,530,331.70

    This application has not had a layout created for it in Cognos Insight, so this is the default appearance of an application when opened with this client. It looks similar to TM1 Application Web, and you can perform many of the same tasks in both clients, such as save, submit, reject, and so on.

    You have been told that because these applications are all new, none of them have had layouts created in Cognos Insight yet. Therefore you will explore the remaining applications in TM1 Application Web only.

    You now want to explore a Responsibility application.

13. Close **IBM Cognos Insight**, and then click **OK**.

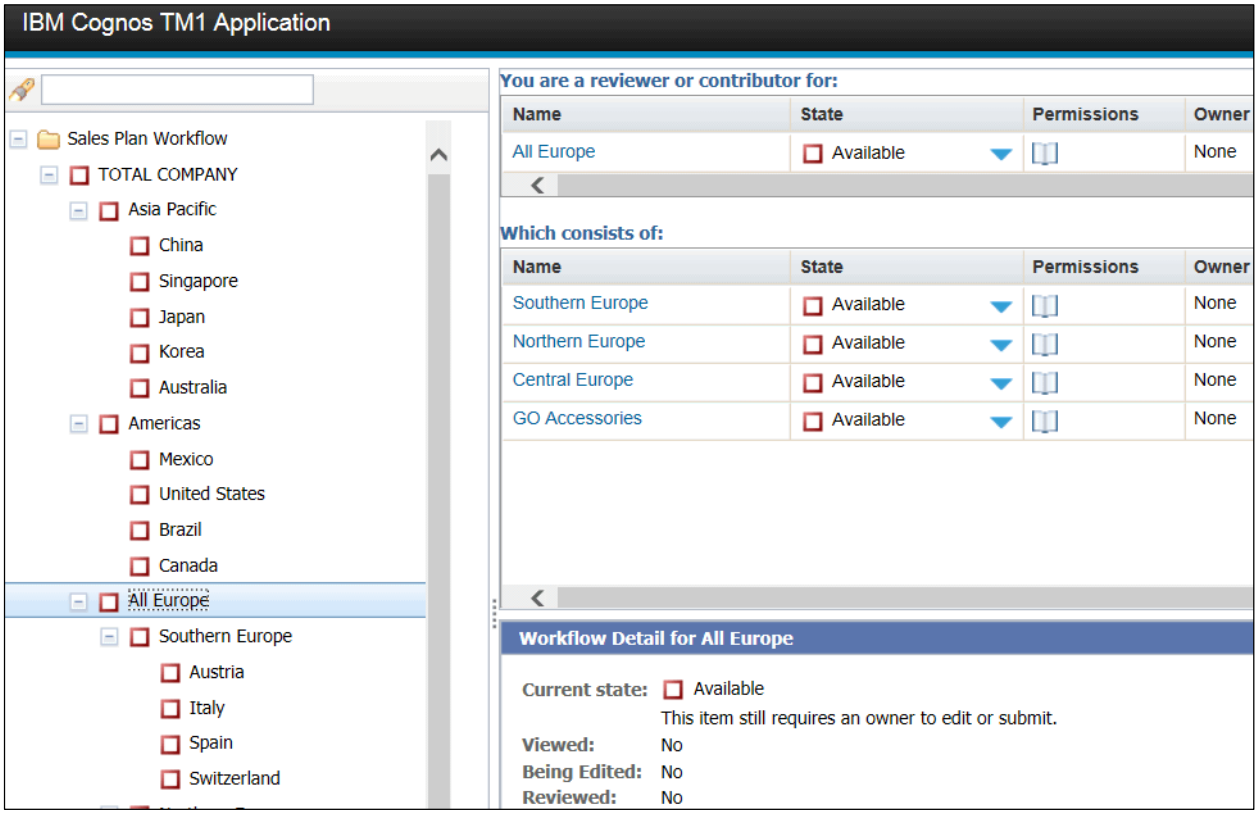
Task 3. Explore a Responsibility application.

1. On the toolbar of the workflow page, click **Return**  to return to the portal.
2. In the **IBM Cognos TM1 Applications** portal, click **Sales Plan**.

   The workflow page appears. Because this is a Responsibility application, you can see a hierarchy of nodes in the left pane.

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3. In the left pane, expand **all nodes** and then click the **All Europe** node.
   The result appears as shown below:



   This application is distributed by Country, a different hierarchy than the Income_Statement application. This is a new application so all nodes are Available.

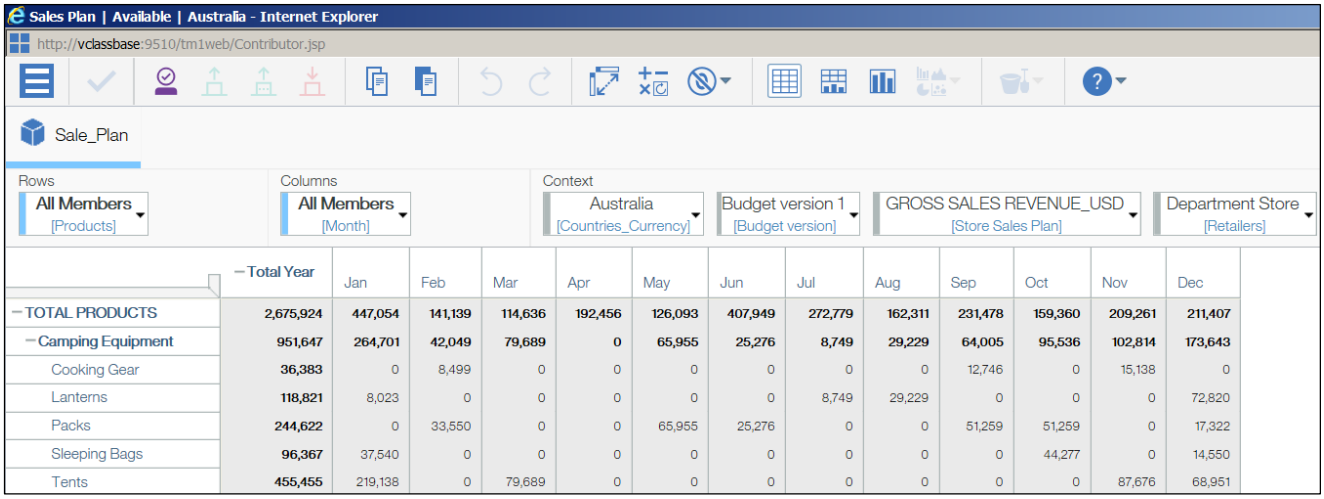
   Because this is a Responsibility application, you can see in the Permissions column that you can view or edit data, but you cannot submit or reject data.

4. In the tree, click **Asia Pacific**.

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5. In the bottom table, right-click **Australia**, click **Open TM1 Application Web**, and then maximize the window. You may need to resize the Workflow Detail for Asia Pacific pane at the bottom in order to see the Australia link.

   The result appears as shown below:



	Total Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
TOTAL PRODUCTS	2,675,924	447,054	141,139	114,636	192,456	126,093	407,949	272,779	162,311	231,478	159,360	209,261	211,407
Camping Equipment	951,647	264,701	42,049	79,689	0	65,955	25,276	8,749	29,229	64,005	95,536	102,814	173,643
Cooking Gear	36,383	0	8,499	0	0	0	0	0	0	12,746	0	15,138	0
Lanterns	118,821	8,023	0	0	0	0	0	8,749	29,229	0	0	0	72,820
Packs	244,622	0	33,550	0	0	65,955	25,276	0	0	51,259	51,259	0	17,322
Sleeping Bags	96,367	37,540	0	0	0	0	0	0	0	0	44,277	0	14,550
Tents	455,455	219,138	0	79,689	0	0	0	0	0	0	0	87,676	68,951

   This application is a rolling sales forecast for each country, forecasting sales metrics by product, month, and retailer.

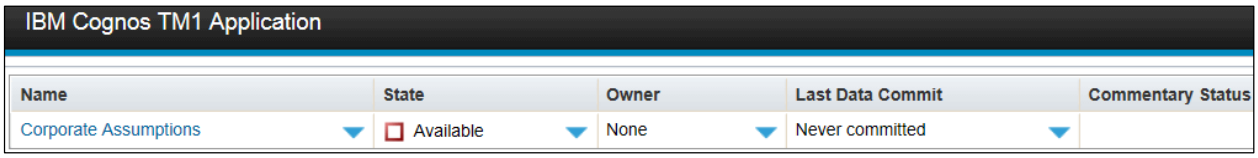
   You now want to examine a Central application.

6. Close **IBM Cognos TM1 Application Web**.

Task 4. Explore a Central application.

1. On the toolbar of the workflow page, click **Return** to return to the portal.
2. In the **IBM Cognos TM1 Applications** portal, click **Corporate Assumptions**.

   The workflow page appears in IBM Cognos TM1 Application:



   Because this is a Central application, there is no hierarchy of nodes in the left pane. You only see a table showing details about the entire application. There is no distinction between contributors and reviewers, and data cannot be submitted or rejected. If you have access to the application, you can edit the data and commit it back to the database.

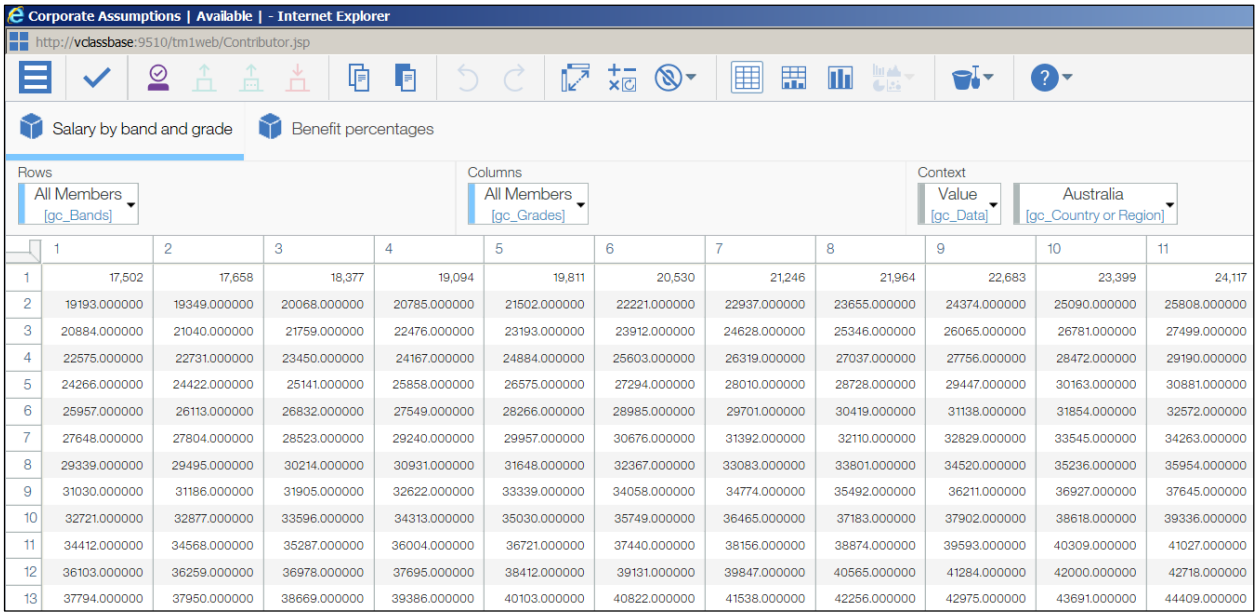
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3. In the table, right-click **Corporate Assumptions**.

   Currently, you can only open this application in TM1 Application Web or Cognos Insight - Connected mode. In either client, you access the entire application and work with data directly on the server. However, the administrator can choose to enable Cognos Insight - Distributed mode for a Central application.

4. Click **Open TM1 Application Web**, and then maximize the window.

   The result appears as shown below:



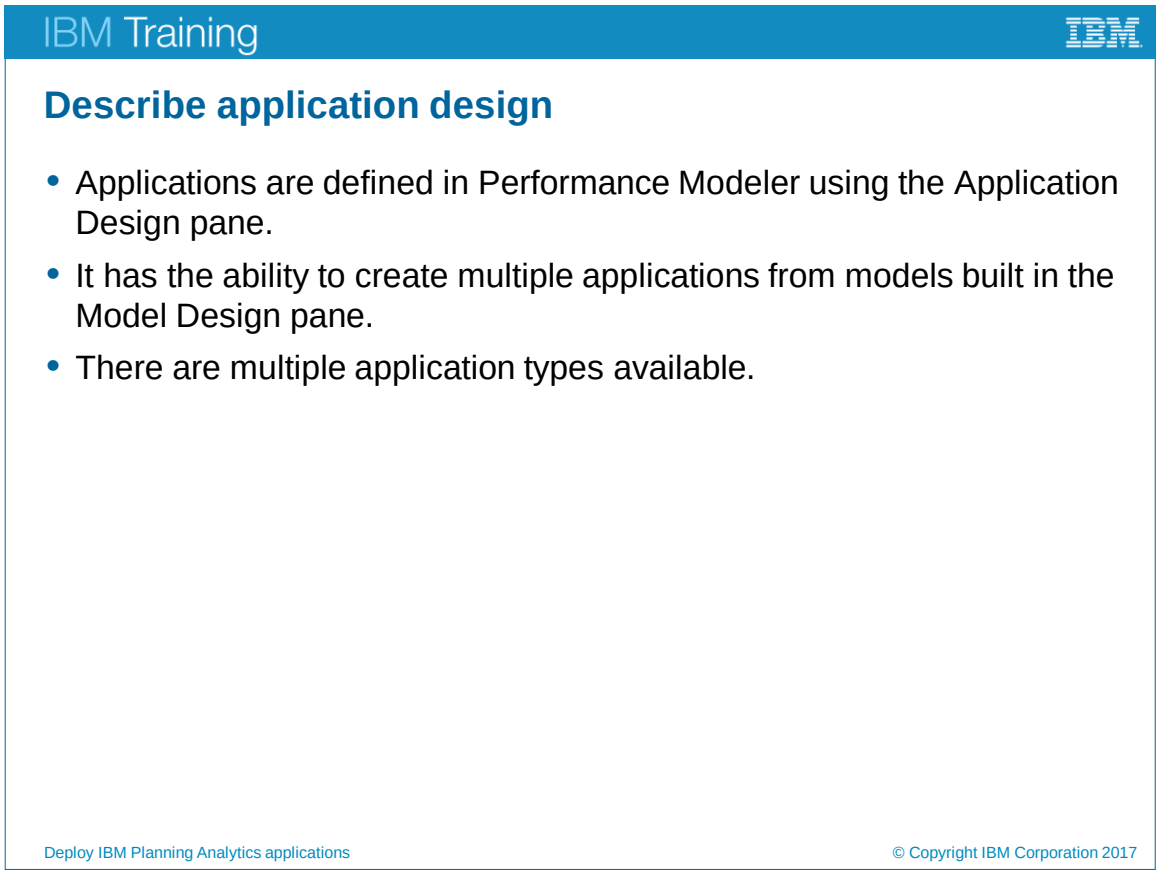
	1	2	3	4	5	6	7	8	9	10	11
1	17,502	17,658	18,377	19,094	19,811	20,530	21,246	21,964	22,683	23,399	24,117
2	19193.000000	19349.000000	20068.000000	20785.000000	21502.000000	22221.000000	22937.000000	23655.000000	24374.000000	25090.000000	25808.000000
3	20884.000000	21040.000000	21759.000000	22476.000000	23193.000000	23912.000000	24628.000000	25346.000000	26065.000000	26781.000000	27499.000000
4	22575.000000	22731.000000	23450.000000	24167.000000	24884.000000	25603.000000	26319.000000	27037.000000	27756.000000	28472.000000	29190.000000
5	24266.000000	24422.000000	25141.000000	25858.000000	26575.000000	27294.000000	28010.000000	28728.000000	29447.000000	30163.000000	30881.000000
6	25957.000000	26113.000000	26832.000000	27549.000000	28266.000000	28985.000000	29701.000000	30419.000000	31138.000000	31854.000000	32572.000000
7	27648.000000	27804.000000	28523.000000	29240.000000	29957.000000	30676.000000	31392.000000	32110.000000	32829.000000	33545.000000	34263.000000
8	29339.000000	29495.000000	30214.000000	30931.000000	31648.000000	32367.000000	33083.000000	33801.000000	34520.000000	35236.000000	35954.000000
9	31030.000000	31186.000000	31905.000000	32622.000000	33339.000000	34058.000000	34774.000000	35492.000000	36211.000000	36927.000000	37645.000000
10	32721.000000	32877.000000	33596.000000	34313.000000	35030.000000	35749.000000	36465.000000	37183.000000	37902.000000	38618.000000	39336.000000
11	34412.000000	34568.000000	35287.000000	36004.000000	36721.000000	37440.000000	38156.000000	38874.000000	39593.000000	40309.000000	41027.000000
12	36103.000000	36259.000000	36978.000000	37695.000000	38412.000000	39131.000000	39847.000000	40565.000000	41284.000000	42000.000000	42718.000000
13	37794.000000	37950.000000	38669.000000	39386.000000	40103.000000	40822.000000	41538.000000	42256.000000	42975.000000	43691.000000	44409.000000

   This application collects data for the entire enterprise regarding base salaries and benefit percentages in each country. Only members of the Human Resources department can access this application.

5. Close **IBM Cognos TM1 Application Web**, click **Return** in the top right corner, and then leave **IBM Cognos TM1 Applications** open for the next demonstration.

Results:
You have familiarized yourself with the various types of planning applications. You logged on to the Web portal and explored three applications created by another administrator.

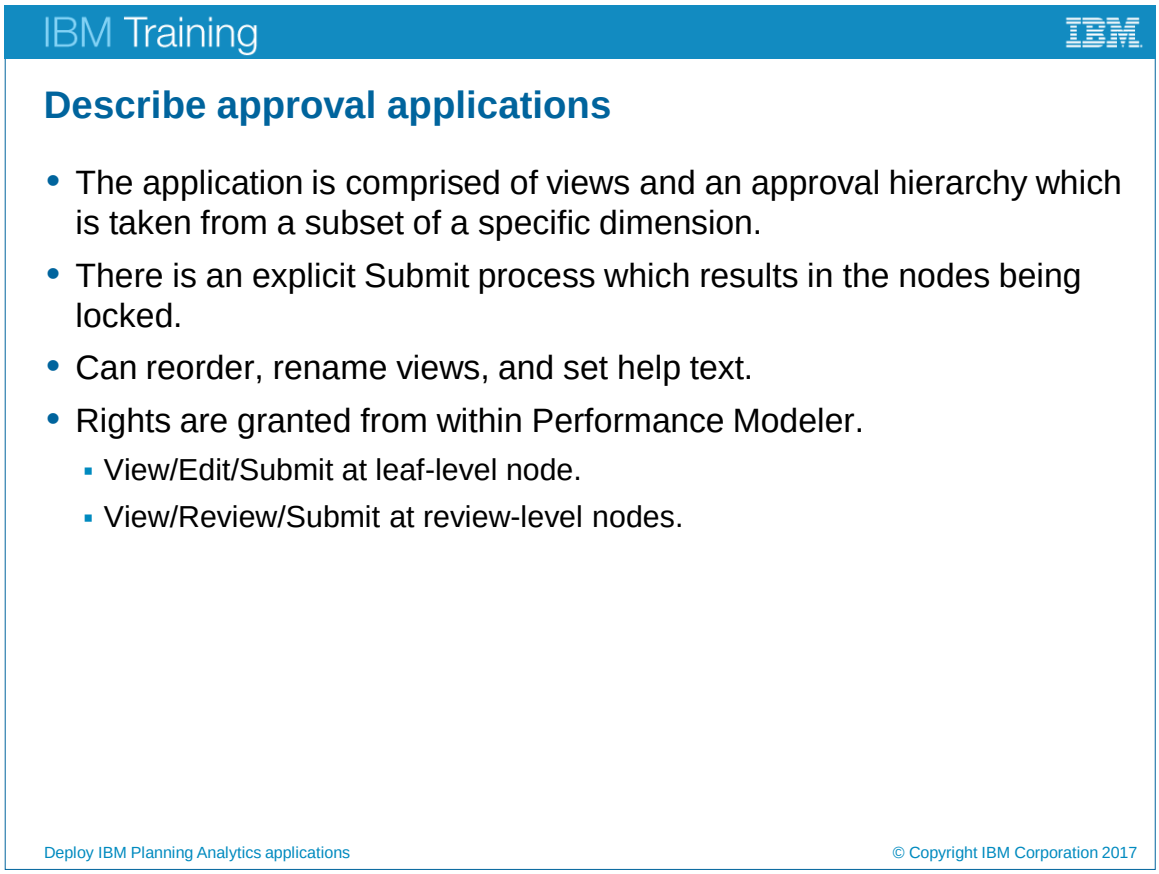
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Describe application design

IBM Cognos TM1 Performance Modeler is an alternate tool for creating TM1 models. You also use Performance Modeler to create applications for use in the IBM Cognos TM1 Applications portal.

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Describe approval applications

Rights can be set from within Performance Modeler or through the portal.

View, Edit, and Submit are available at the lowest level of the Approval hierarchy and View, Review, and Submit are available at the Review level. The Reviewer Edit check box is also available (and checked by default so that reviewers can take ownership of a node below them and edit the data if required).

All client types can be selected if they have been made available by the Administrator.

For both Approval and Responsibility application types, ownership must be taken before the cubes with the approval hierarchy can be written to. Any "assumption" cubes which do not have the approval hierarchy are writable by default and should be secured by the modeler before the application is deployed.

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Describe responsibility applications

- Rights are granted from within Performance Modeler.
  - Only View/Edit at leaf-level node.
  - Only View/Review at review nodes.
- An approval hierarchy is selected.
- No concept of submit.

Deploy IBM Planning Analytics applications

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Describe responsibility applications

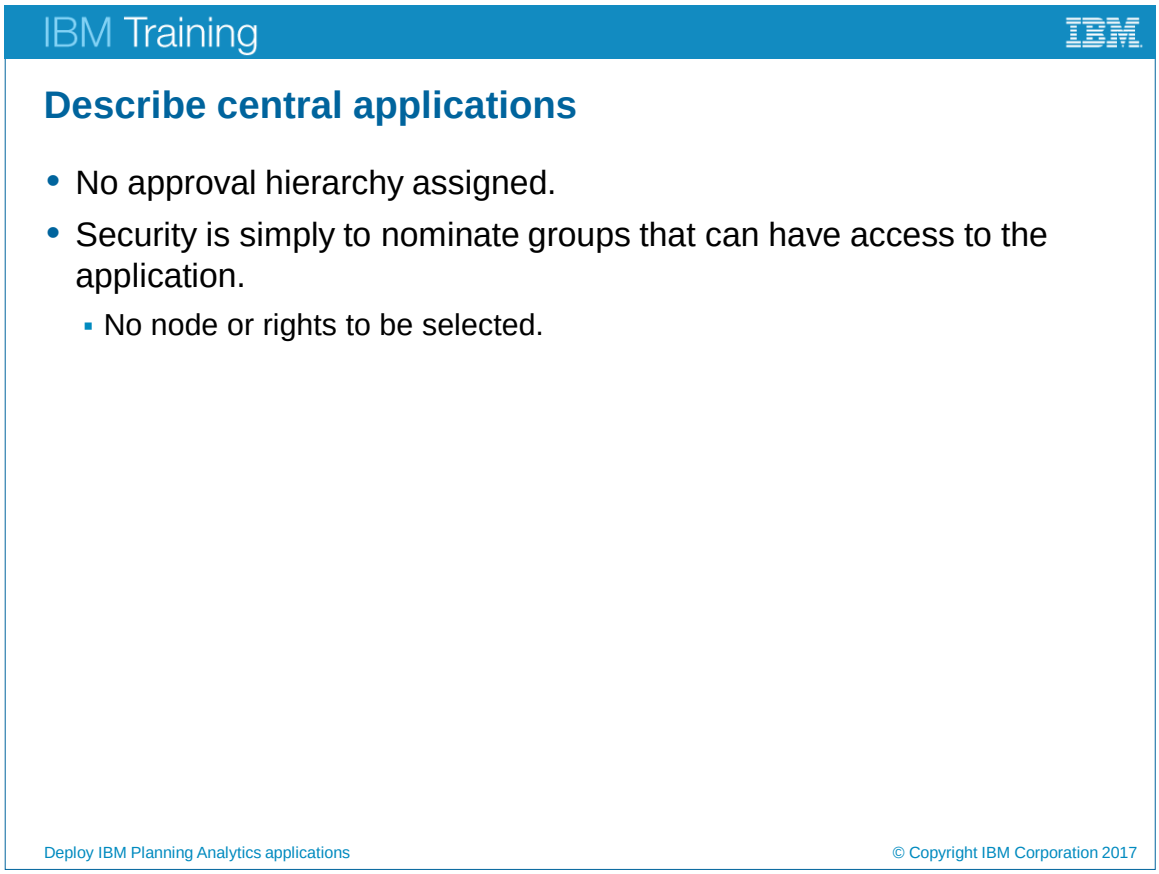
No Submit - process ongoing so always in edit state.

Reviewer edit still available.

Can reorder, rename views, and set help text.

All client types can be selected if made available by the Administrator.

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Describe central applications

No different types of access - it is simply if a group is given access all users in that group have access to the application.

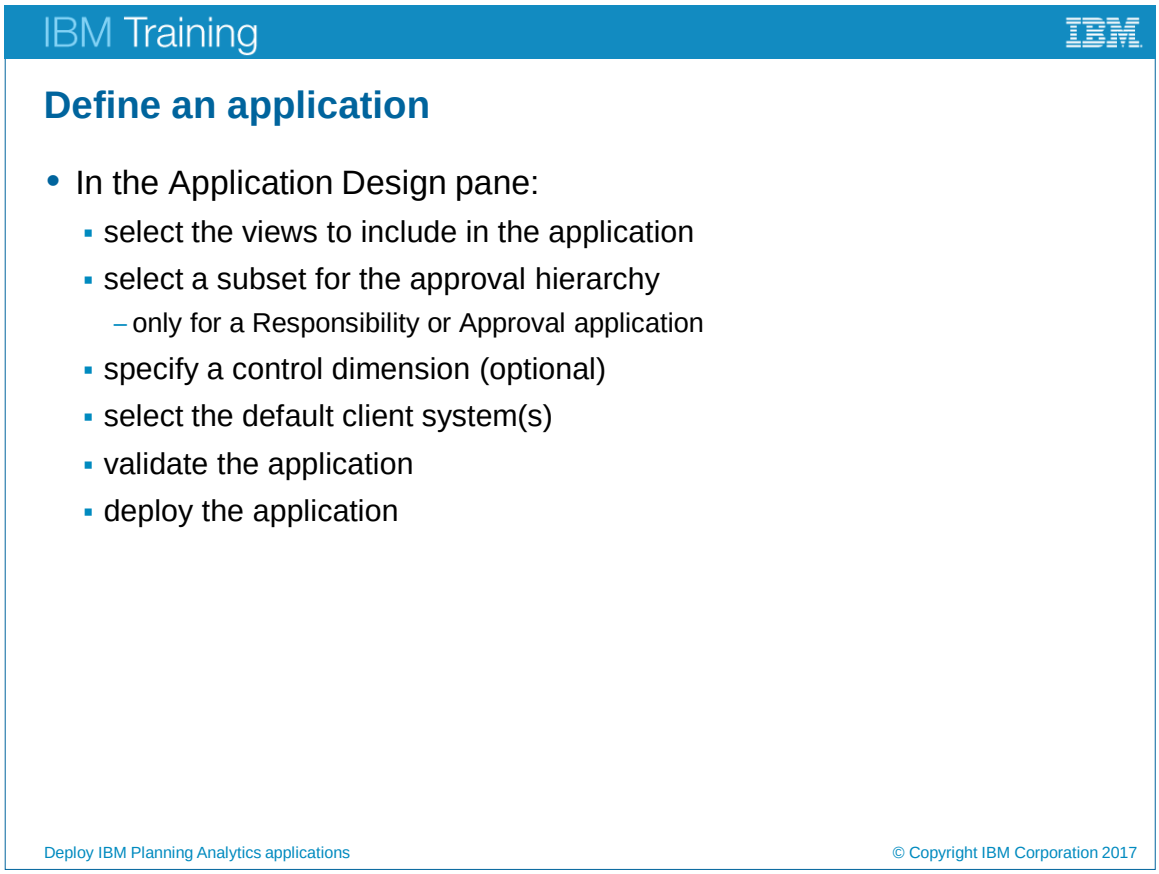
Taking ownership is optional. Multiple users can work simultaneously to make minor changes. The normal TM1 paradigm of "last entry wins" is in place.

However, if there are major changes to be made throughout the application it is recommended that the user takes ownership which will make all views reserved for that user.

You can open central applications in TM1 Application Web, or either Connected or Distributed modes of IBM Cognos Insight. However, if you are going to use IBM Cognos Insight, Connected mode is recommended for performance reasons. In Connected mode, users work directly with the data on the server. This mode is useful for working with extremely large data that may not fit within the memory available on the user' computer.

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Define an application

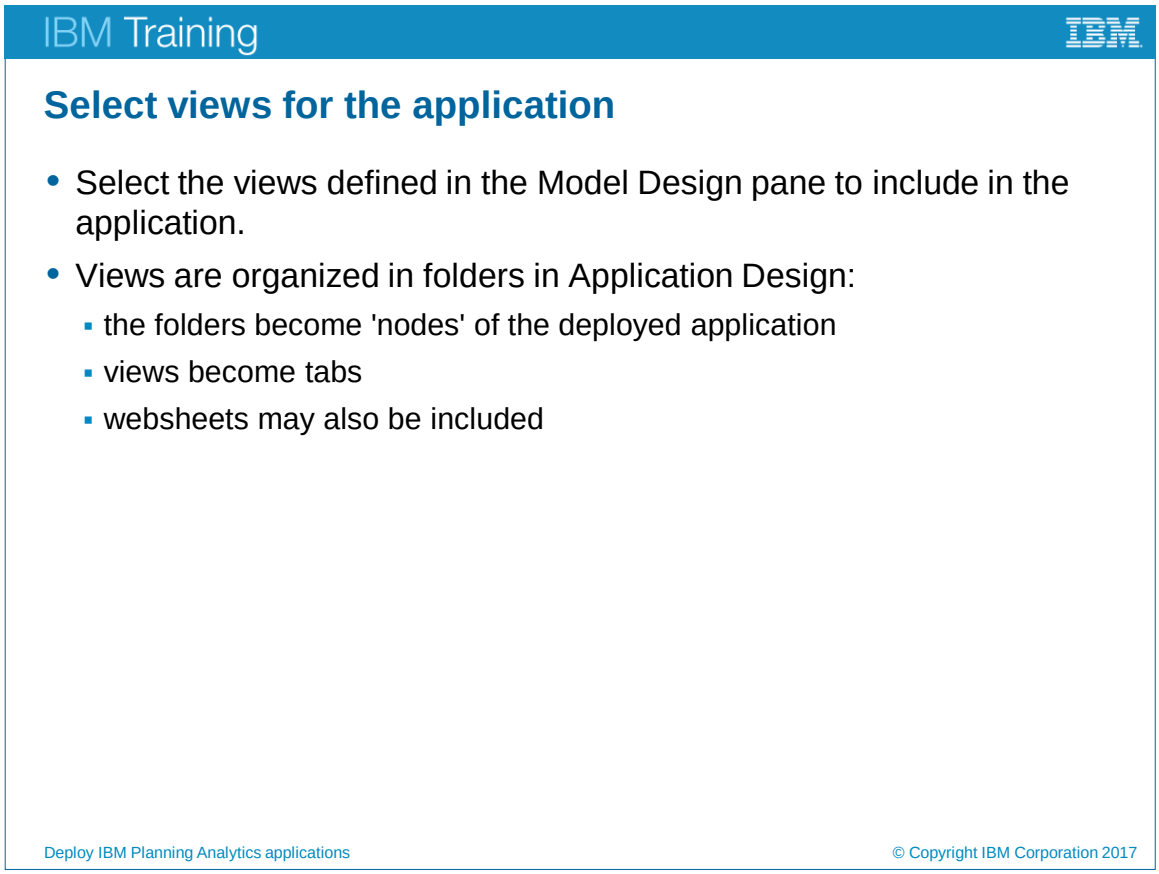
You can create and edit more than one application at a time, and you can create more than one type of application.

To build an application, you must have a valid cube.

When considering which client is best for an application, consider these points:

- Both IBM Cognos Insight (in distributed or connected mode), and IBM Cognos TM1 Application Web can be used interchangeably if the application uses cube views alone and has multiple sandboxes disabled. The user can select which client they want to use on these applications.
- Applications that use a canvas layout show a simple multi-tabbed view when used with IBM Cognos TM1 Application Web.
- Applications that contain a websheet can only be opened by IBM Cognos TM1 Application Web.

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Select views for the application

The views used in your application must exist in the Model Design pane before you can include them in your application.

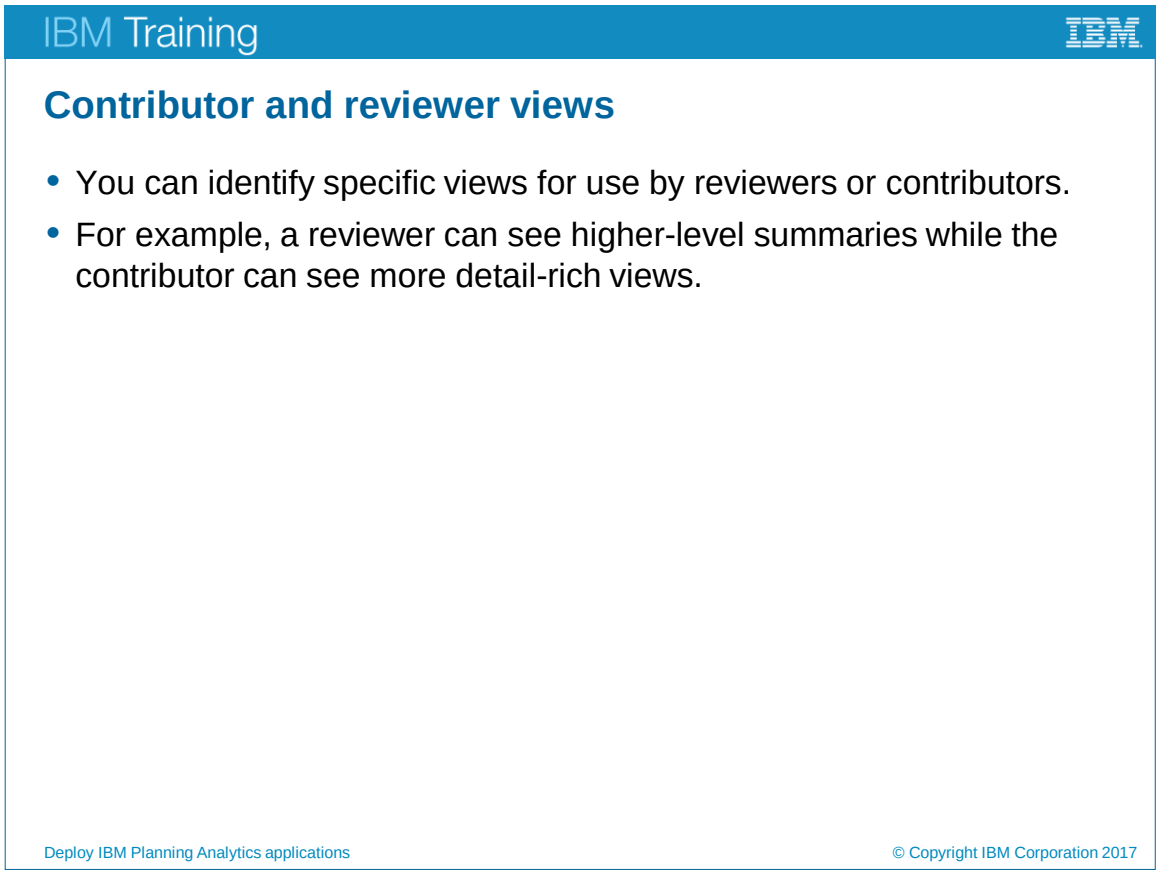
Each view has two properties to help the user:

- **Help Text**: shows when a user clicks the View > Help option when working with a view in IBM Cognos TM1 Application Web. This property displays instructions or information to assist users in entering data in the view. The help text you enter is also available in the workflow page.
- **Tab Name**: the name displayed on the view tab in the client. Modify this property if you want the tab to display text other than the name of the view.

Applications containing websheets are only deployed to IBM Cognos TM1 Application Web.

In order to include websheets, Advanced Modeling must be enabled for the application.

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Contributor and reviewer views

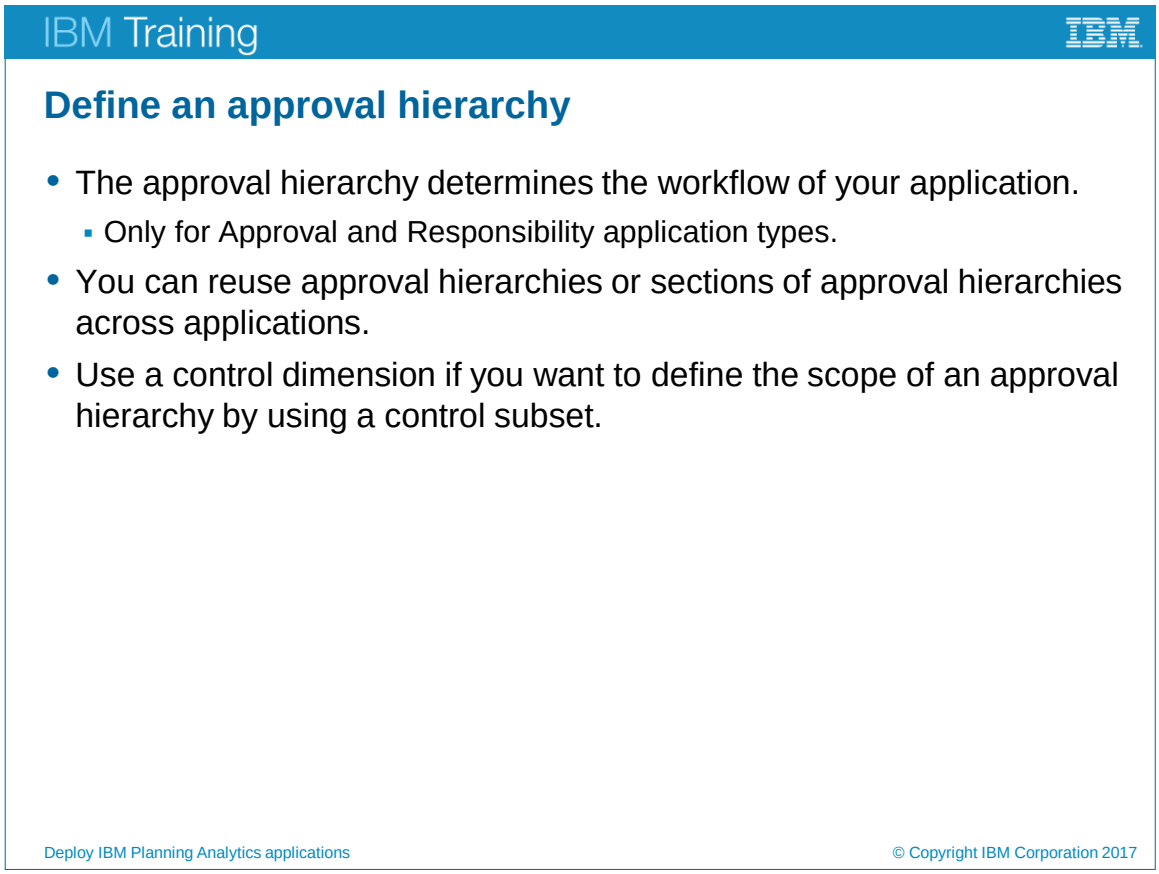
On the Application Design tab, you can specify which views are designed as Contributor or Reviewer views.

Targeted views also facilitate the deployment of reporting cubes. Reporting cubes can deliver improved performance when large numbers of nodes need to be reviewed. To improve performance, a TurboIntegrator process based on a cube with many rules can report into a cube with only a few rules for review.

If there are no Reviewer views specified, then all users with Review access at the consolidated level also have access to the views specified for contributors at the leaf level.

If a view is defined as for both Reviewers and Contributors, the user can choose which view to use.

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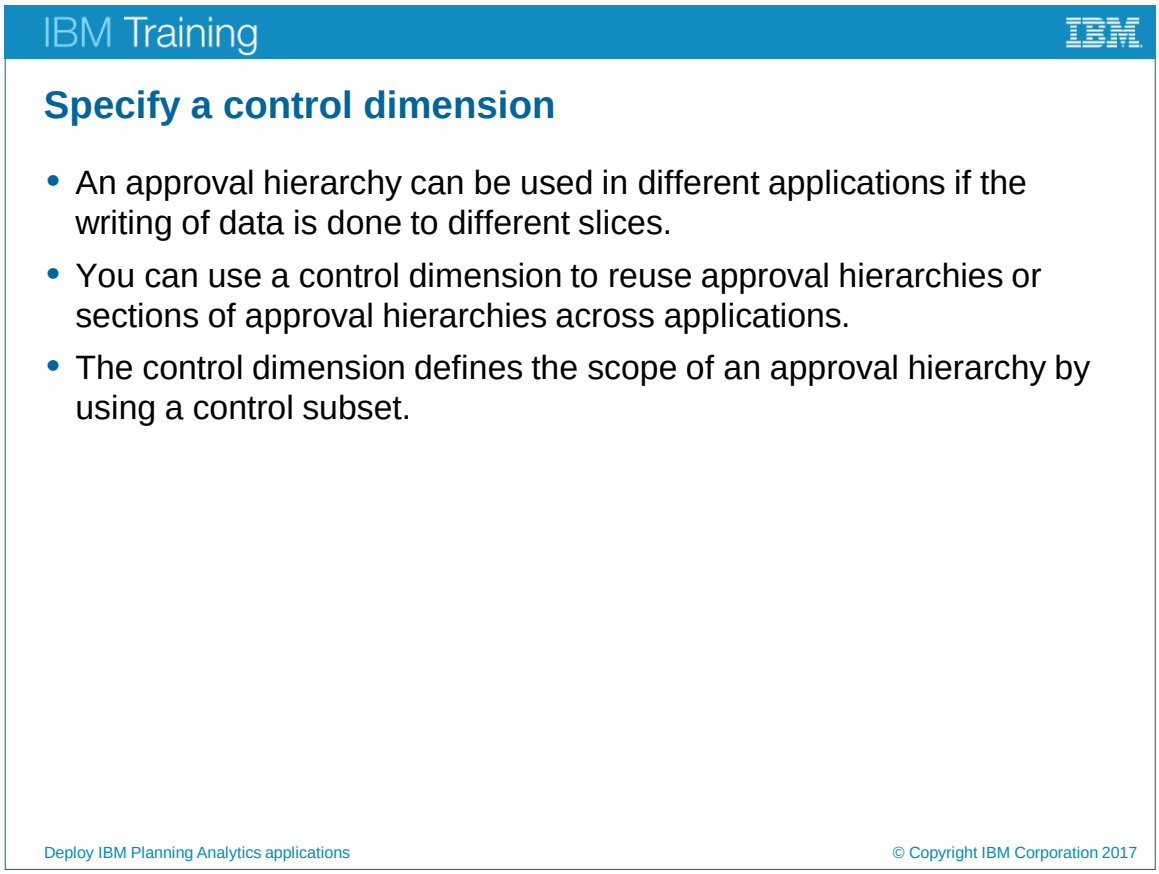


Define an approval hierarchy

The application types Approval and Responsibility need to have an approval hierarchy defined. An approval hierarchy is a dimension subset on your IBM Cognos TM1 server. Each member in a subset is referred to as a "node" in the approval hierarchy. An approval hierarchy has these requirements:

- An approval hierarchy can only be used once for each IBM Cognos TM1 server.
- At least one view in your application must include the dimension that contains your approval hierarchy subset.
- The approval hierarchy subset must only contain one top level member. If the subset contains multiple top-level members, you will receive an error.
- The approval hierarchy subset cannot contain any string members.
- When a subset is designated as an approval hierarchy, all security for the parent dimension of the subset is controlled by IBM Cognos TM1 Performance Modeler.

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Specify a control dimension

You can reuse approval hierarchies across applications or sections of applications if the data does not overlap.

For example, you can create a budget application and a forecast application that use data from the same cube but operate on different schedules or that use different rollups. Different kinds of rollups include geographic rollups versus market maturity rollups.

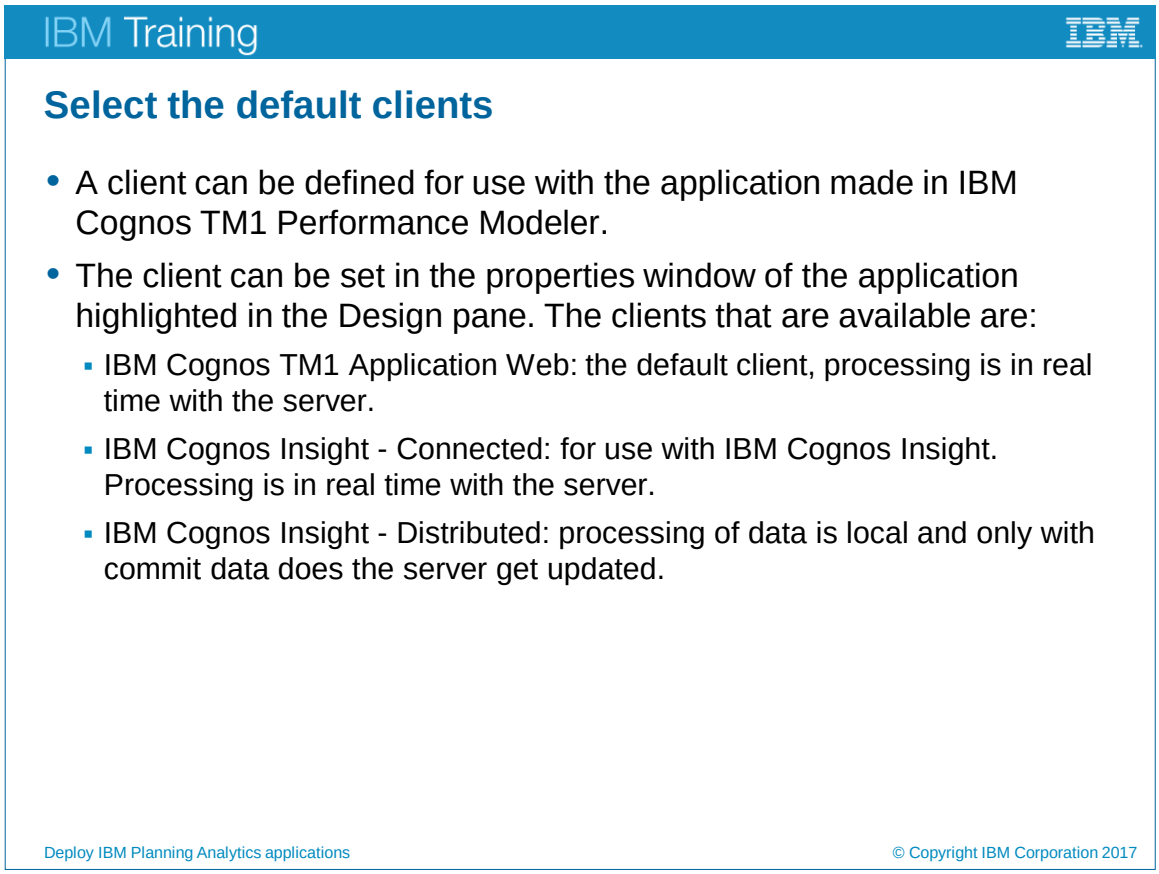
Similarly, the same approval hierarchy can be used when the forecast application writes to the forecast slice and the budget application writes to a budget slice. You can also share hierarchies when, for example, the application uses the Europe portion of the hierarchy while the other uses the North American hierarchy.

To share an approval hierarchy, define a control dimension and subset in the application to control the scope within the context of another dimension (such as plan_version).

The control dimension determines the maximum access allowed by any user of the application. For any writeable member of the control dimension, the user's access is determined by their approval hierarchy rights.

The control dimension creates a subset that defines the visible slices of the control dimension. Do not alter or delete the control subset.

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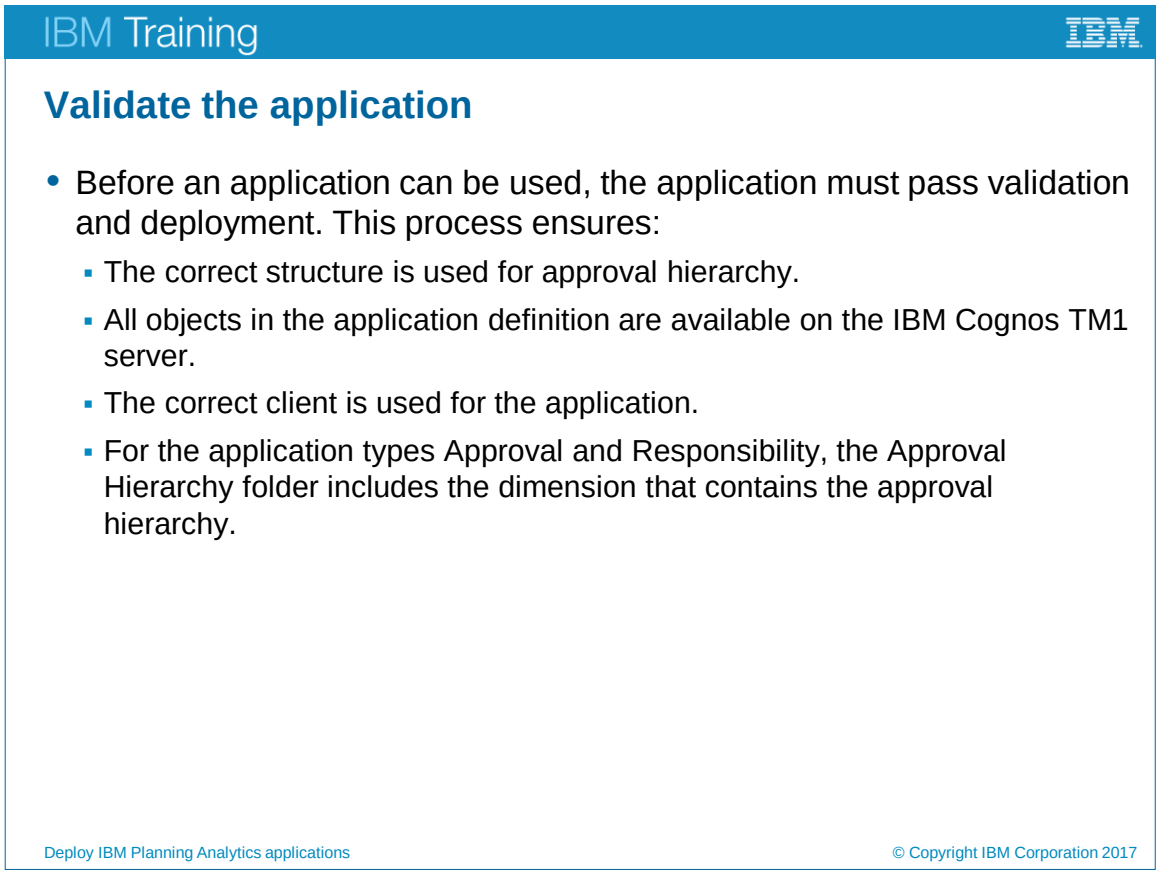


Select the default clients

The available clients are saved as a property of the application and can be changed by the Administrator.

- When considering which client is best for an application, consider these points:
  - Both IBM Cognos Insight (in distributed or connected mode) and IBM Cognos TM1 Application Web can be used interchangeably if the application uses cube views alone and has multiple sandboxes disabled. The user can select which client they want to use on these applications.
  - Applications that use a canvas layout show a simple multi-tabbed view when used with IBM Cognos TM1 Application Web.
  - Applications that contain a websheet can only be opened by IBM Cognos TM1 Application Web.

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Validate the application

The validation process ensures that all conditions required to deploy the application are in place.

To validate an application, open it in the Application Pane, right-click the application itself or any node beneath it and then click Validate Application. You can also click the Validate Application button on the toolbar.

Before you can edit the user rights for the application, you must first validate and deploy the application.

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IBM Training

Deploy an application

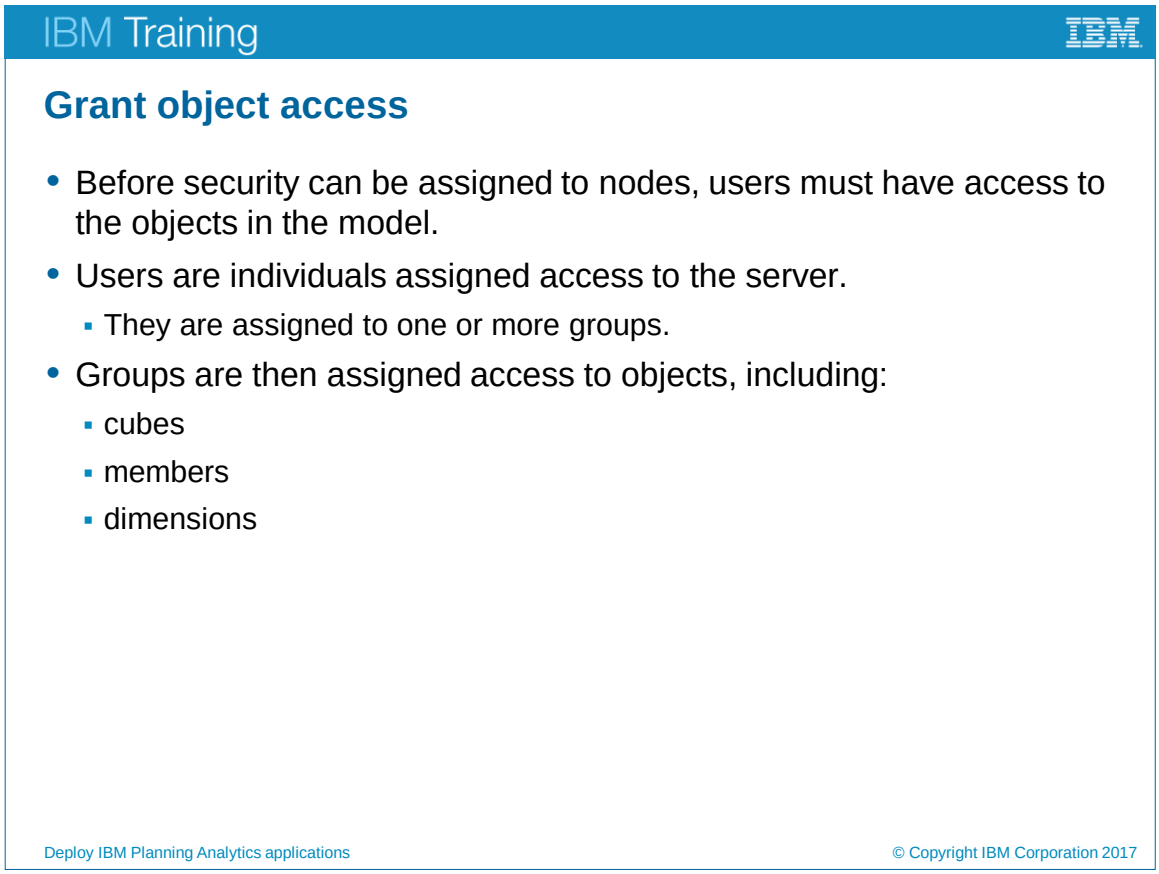
- All application types are deployed to the portal.
- All are deployed in inactive state and must be activated in the portal.
- Once deployed, an application will behave differently within IBM Cognos Performance Modeler.
  - Approval - no data entry allowed, need to take ownership in application through portal.
  - Centralized - can enter data if in correct security group.

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Deploy an application

Administrators can activate applications from the portal by clicking the Activate icon for each application.

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Grant object access

Users and groups can be defined using native and non-native security. However, in this course you are using native security to define the users and groups for the SampleOutdoors model.

Once users are assigned to groups, groups are then assigned one of the following privileges to TM1 objects:

- READ (see it, not modify it)
- WRITE (write to the data)
- LOCK (permanently lock it until the user unlocks it)
- ADMIN (may do anything, edit, create, delete, and so forth)
- NONE (may not do anything, cannot see the object)
- RESERVE (temporarily lock it until the server is reset or the admin or user unlocks it)

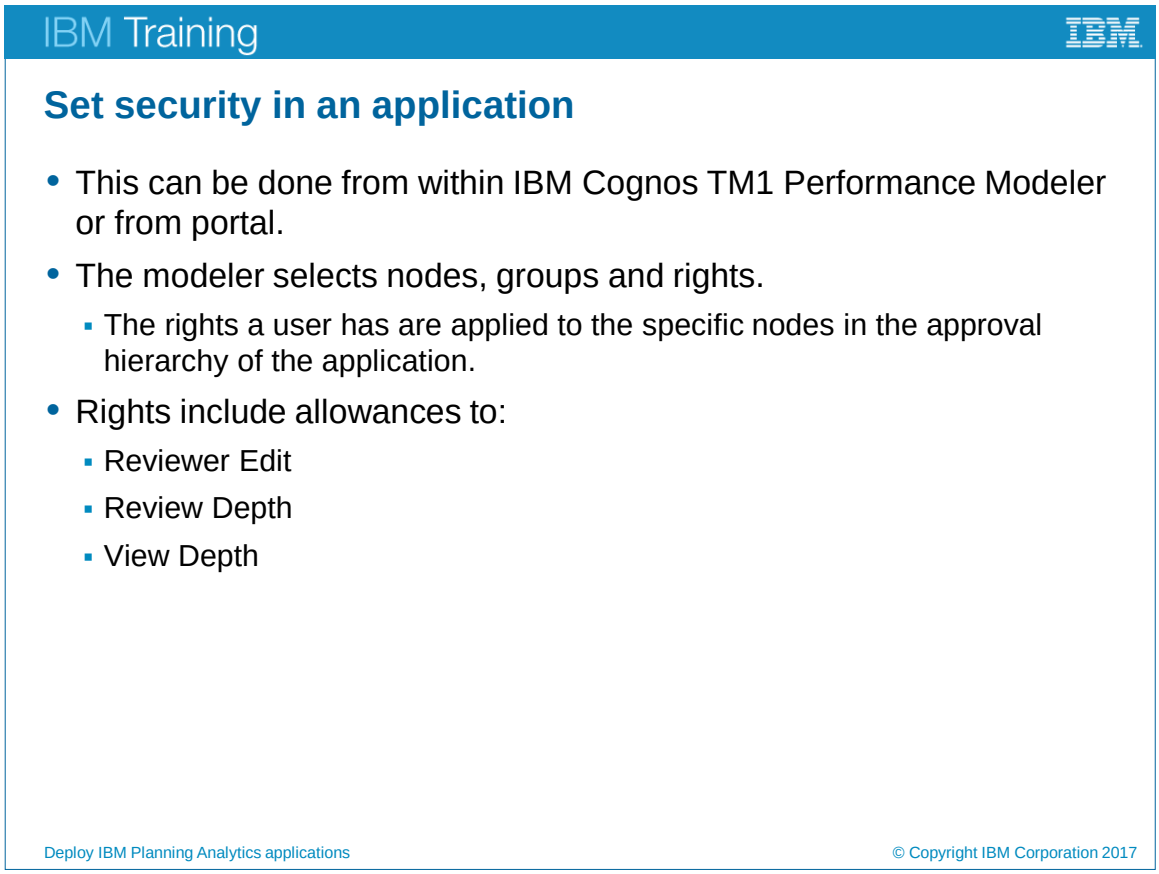
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Users and groups must be brought into TM1. Instead of adding them manually, you may create TI scripts to define the clients and groups to the control objects from a text file (exported from the original source).

For more information on controlling access to TM1 objects, see the *TM1 for Developers* guide.

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Set security in an application

The rights a user has are applied to the specific nodes in the approval hierarchy of the application. Different users can access a node but at any given point in time only one user has exclusive write access or ownership of a specific node.

If rights are explicitly applied to a parent node they are implicitly applied to the children of the node. The children can have different rights from the parent nodes but cannot exceed the rights of the parent.

For more information on managing rights for an application, including some scenarios, please see the *IBM Cognos TM1 Performance Modeler* guide.

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Authorized user Access Rights

View	View the data in the node. Cannot take ownership of a node.
Edit	Take ownership of the node. Can edit values in the node.
Submit	Take ownership, edit values, and submit the node for review. Can reject submissions from child nodes.
Review	Can edit or submit their own nodes. Can review and reject child nodes which have been submitted. If Allow Reviewer Edit is enabled, users can take ownership of any child node which has not been submitted, edit the values and then submit the node.

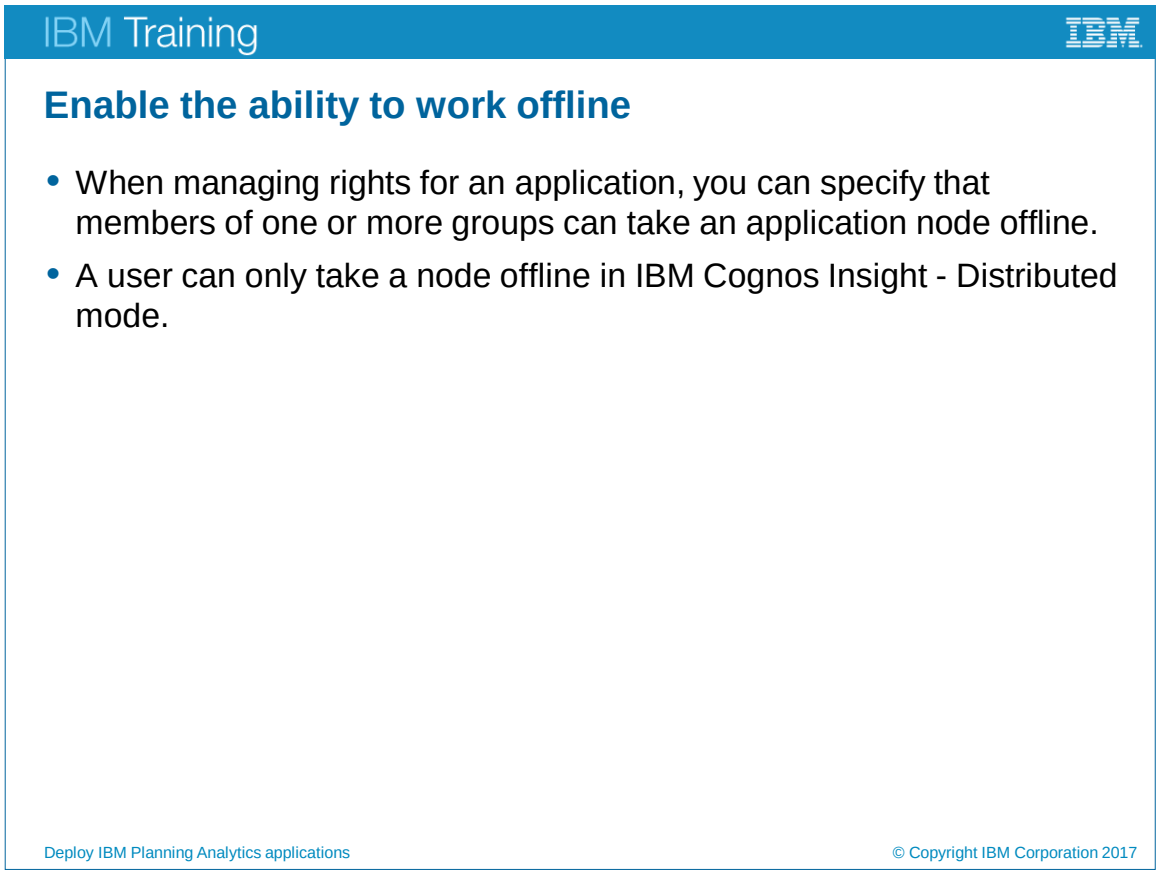
The Review Depth option is enabled only when the Review or Submit right is selected. The Review Depth determines how many levels beneath the current node will inherit the selected access right. If you want only the immediate children of the current node to inherit rights, select 1. If you want the two levels beneath the current node to inherit rights, select 2, and so on.

The Review Depth setting can never be greater than the View Depth setting.

The View Depth determines how many levels below the current node are visible to the user group. If you want members of the user group to be able to view only the current node and its immediate children, select 1. If you want members of the user group to be able to view only the current node and the next two levels of children, select 2, and so on.

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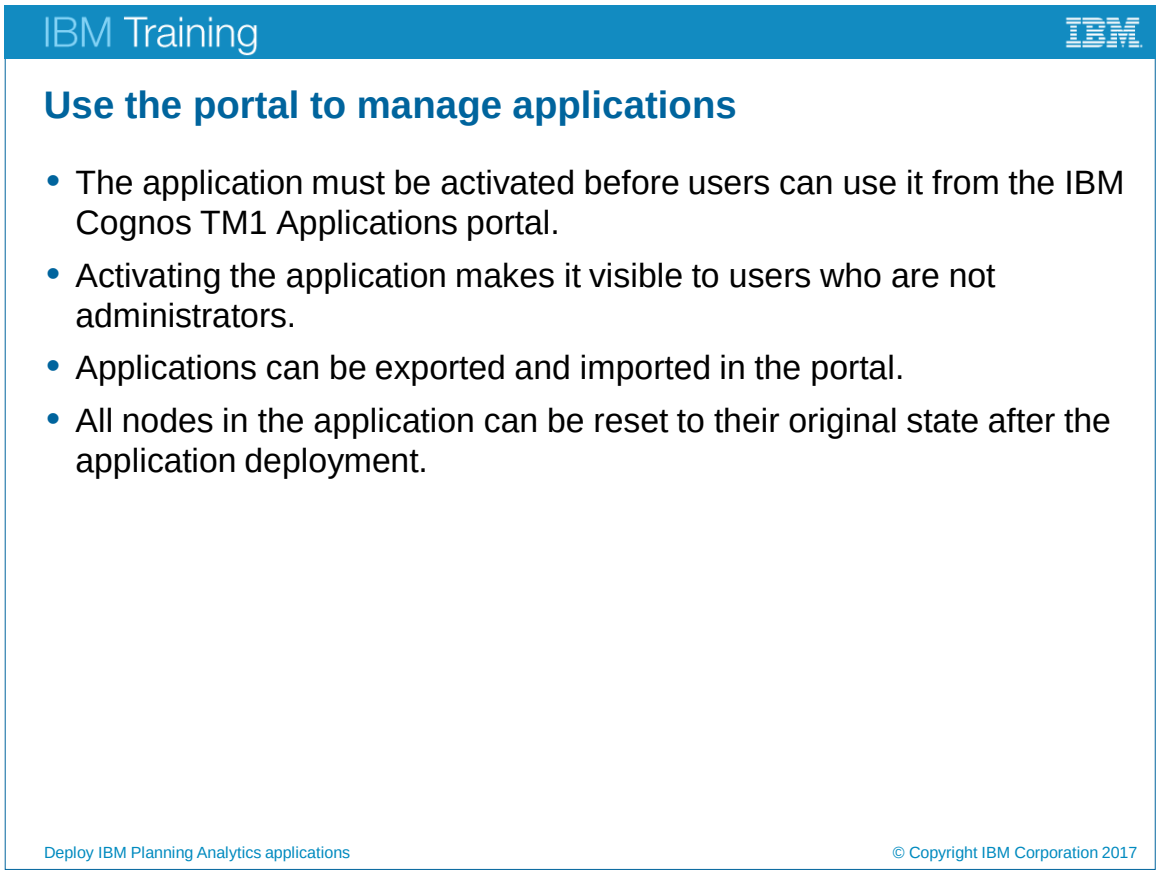
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Enable the ability to work offline

If an administrator configures it, a user can take a node offline in a Central, Approval, or Responsibility application.

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Use the portal to manage applications

In the portal, an application is automatically visible to an administrator.

An application should only be exported from an IBM Cognos TM1 server to a server that does not have that application or uses a different dimension for the approval hierarchy of the exported application.

An archive is created and contains the XML files that describe the structure and security of your application.

Resetting an application discards all progress made in the planning process so that you can restart the planning process. Resetting an application does not reset or discard any data changes.

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IBM Training

Demonstration 2

Create and deploy an Approval application

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Demonstration 2: Create and deploy an Approval application

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Demonstration 2: Create and deploy an Approval application

Purpose:
You will now create a managed planning application for the SampleOutdoors TM1 server. Users need to be able to create and contribute to the expense budget. Some will need to contribute to it and others will review it.

Task 1. Add the SampleOutdoors TM1 server to the IBM Cognos TM1 Applications portal.

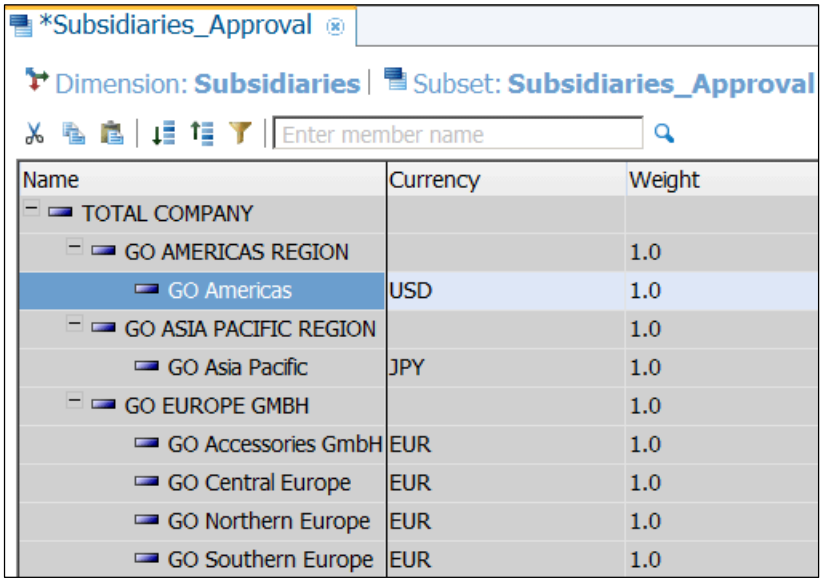
1. In the top right corner, click **Administer IBM Cognos TM1 Applications**  .
2. Under the **Server Name** section, click **Add**.
3. In the **Admin Host** box, type l**ocalhost**, and then click **Refresh**  .
4. In the **Server Name** list, select **SampleOutdoors**, and then click **OK**.
5. Click **OK** to close the **IBM Cognos TM1 Applications Configuration** page.
6. Click **No** if prompted to use **AutoComplete** to remember web form entries.

Task 2. Create a subset to use in an approval hierarchy.

1. On the top right side, click **Open IBM Cognos TM1 Performance Modeler**  .
2. If you are prompted to open the **provagent.cogrcp_modeler** file, click the **Open** button at the bottom.
3. In the **TM1 Server** list, select the **SampleOutdoors** server and click **OK**.
   The SampleOutdoors TM1 server opens in Performance Modeler (this will take a few moments).
4. Maximize the **Performance Modeler** window, and then close the **Model Design** tab.
5. In the **Model Design** pane, expand the **Dimensions** node.
6. Right-click the **Subsidiaries** dimension, point to **New**, and then click **Subset**.
7. Name it **Subsidiaries_Approval**.

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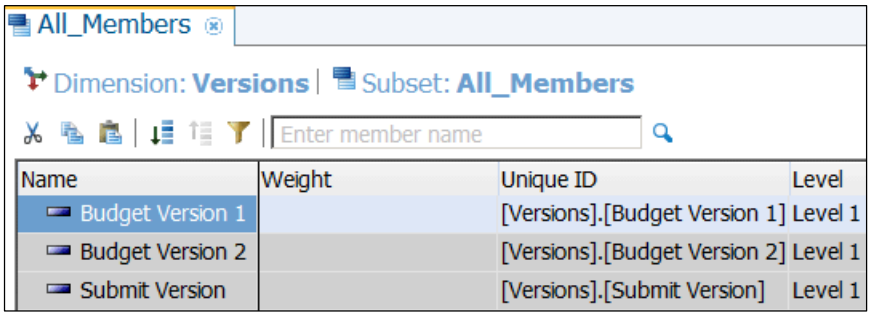
8. Right-click **GO Americas**, and then click **Hierarchy sort**.
   The results appear as follows:



9. Click **Save** and then close the **Subset Editor** tab on the right side.

Task 3. Create a subset for use in a control dimension.

1. Right-click the **Versions** dimension, point to **New**, and then click **Subset**.
2. Name it **All_Members**.
   The results appear as follows:



3. Close the **Subset Editor** tab on the right side.

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Task 4. Create users and groups to allow access to the server and cubes.

1. In the **Model Design** pane, right-click the **SampleOutdoors** TM1 server, point to **Configure Security** and **Define Users and Groups**, and then click **Users and Groups**.

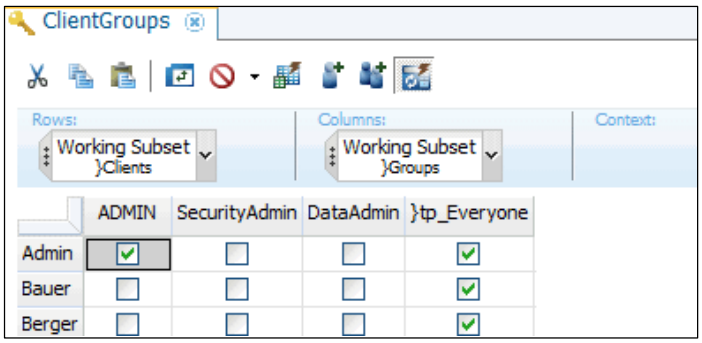
   The ClientGroups cube opens.

   You can also create these users and groups in Architect.

2. Click **New User**  , type each of the following **Usernames (no passwords)** and then click **OK**:

   - **Bauer**
   - **Berger**

   The results appear as follows:



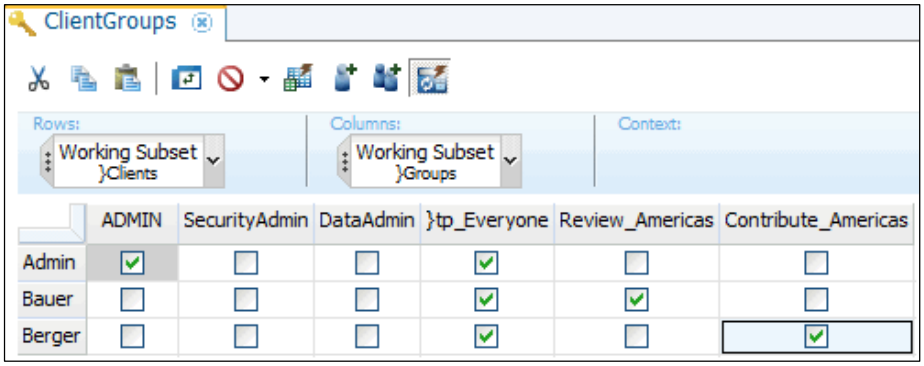
3. Click **New Group**  , and then add each of the following groups:

   - **Review_Americas**
   - **Contribute_Americas**

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4. Select the boxes at the intersections of the following:
   - **Bauer/Review_Americas**
   - **Berger/Contribute_Americas**

   The results appear as follows:



5. Close the **ClientGroups** tab on the right side.

Task 5. Set access permissions for a cube.

Now that users have been assigned to groups, the groups need to be assigned access to the objects. You will grant everyone write access to the Expenses cube and in the application further define the group roles.

You can also set access permissions for a cube in Architect.

1. In the **Model Design** pane on the left side, expand the **Model Security** folder, and then double-click **CubeSecurity**.

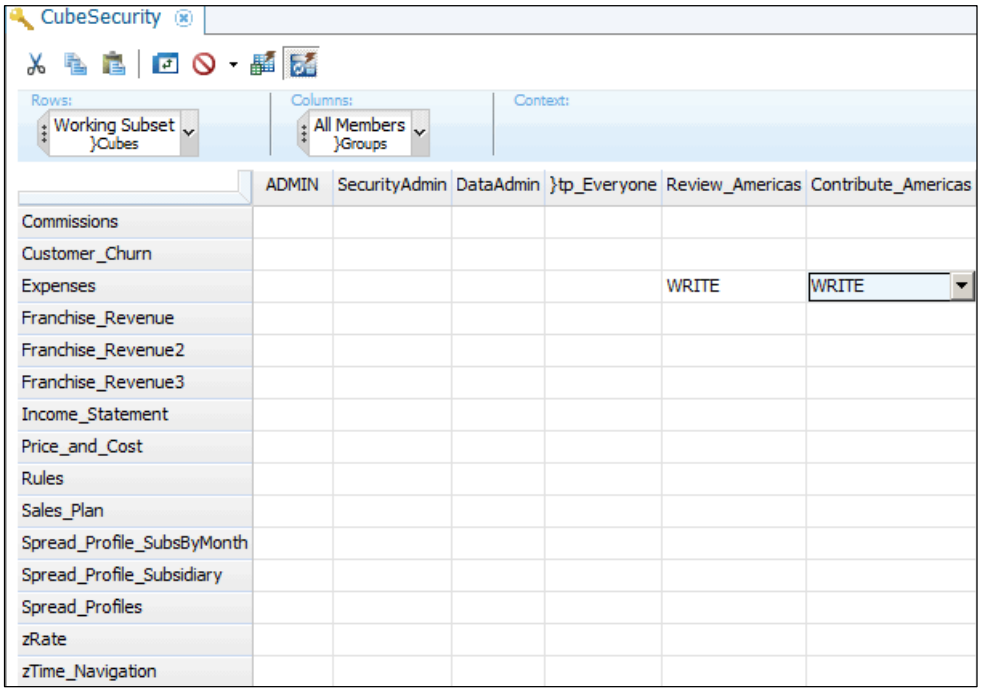
   The CubeSecurity cube opens. This cube contains a list of cubes on the rows and the groups defined in Performance Modeler on the columns.

   Privileges are assigned by clicking in intersecting cells and selecting READ, WRITE, LOCK, NONE, or RESERVE.
2. Double-click the intersection of **Expenses** and **Review_Americas**, and then in the list, click **WRITE**.
3. Press **Enter**, and then press **CTRL+C** to copy.

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4. Click the intersection of **Expenses** and **Contribute_Americas** and then press **CTRL+V** to paste.

   The results appear as follows:



   You can select multiple cells but only one cell can be updated at a time.

5. Close the **CubeSecurity** cube.

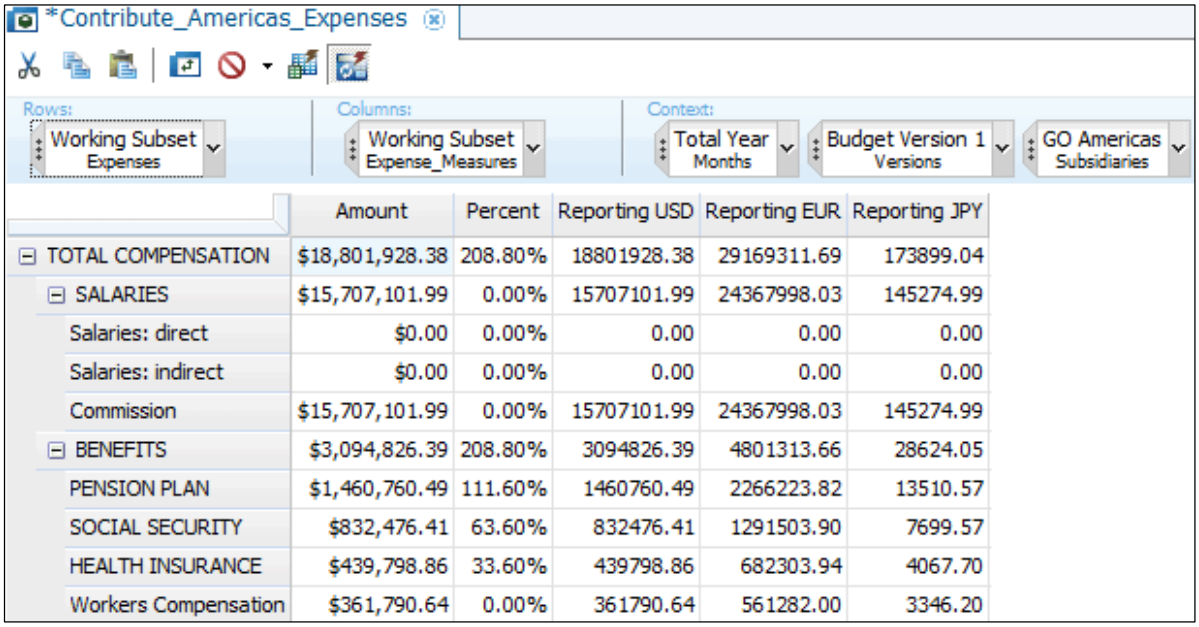
   There is no need to save this cube.

Task 6. Create a contributor view for use in an application.

1. In the **Model Design** pane, expand the **Cubes** folder.
2. Right-click the **Expenses** cube, point to **New**, and then click **View**.
3. Type **Contribute_Americas_Expenses**, and then click **OK**.

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4. Modify the view so that the result appears as shown below:



	Amount	Percent	Reporting USD	Reporting EUR	Reporting JPY
TOTAL COMPENSATION	$18,801,928.38	208.80%	18801928.38	29169311.69	173899.04
SALARIES	$15,707,101.99	0.00%	15707101.99	24367998.03	145274.99
Salaries: direct	$0.00	0.00%	0.00	0.00	0.00
Salaries: indirect	$0.00	0.00%	0.00	0.00	0.00
Commission	$15,707,101.99	0.00%	15707101.99	24367998.03	145274.99
BENEFITS	$3,094,826.39	208.80%	3094826.39	4801313.66	28624.05
PENSION PLAN	$1,460,760.49	111.60%	1460760.49	2266223.82	13510.57
SOCIAL SECURITY	$832,476.41	63.60%	832476.41	1291503.90	7699.57
HEALTH INSURANCE	$439,798.86	33.60%	439798.86	682303.94	4067.70
Workers Compensation	$361,790.64	0.00%	361790.64	561282.00	3346.20

The values in the Percent column currently sum up over the Months and Subsidiaries dimensions. This can be adjusted in the rules for the cube.

5. Save and close the view.

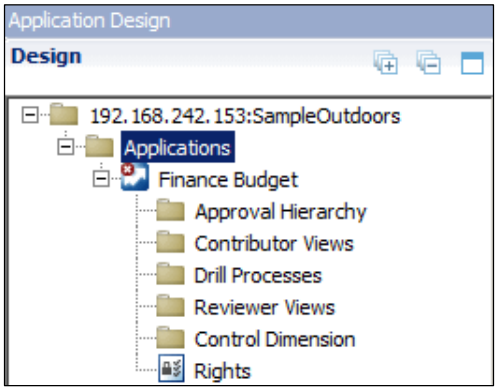
Task 7. Use the Application Design pane to create the approval application.

1. In the lower left corner of the window, click **Application Design**, and then on the right side, close the **Application Design** tab.
2. Right-click the **Applications** folder, point to **New**, and then click **Application**.
   The Create new application dialog box appears.
3. In the **Enter name for new application** box, type **Finance Budget**.
4. In the **Application Type** list, ensure that **Approval** is selected, and then click **OK**.

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5. In the **Application Design** pane on the left side, expand **Finance Budget**.

   The results appear as follows:



   You can build the application by dragging views and/or subsets from the model (displayed under the TM1 Objects section) to the appropriate folders or to the boxes on the right side. Once finished, you will apply rights, validate, and then deploy the application.

6. Under **TM1 Objects**, expand the **SampleOutdoors** TM1 server and the **Cubes** folder (if necessary).
7. Under the **Expenses** cube, drag the **Contribute_Americas_Expenses** view to the **Contributor views** box on the right side.
8. Under the **Expenses** cube, drag the **Default** view to the **Reviewer views** box on the right side.
9. In the **Application Design** pane on the left side, under the **Reviewer Views** folder, right-click the **Default** view and then click **Rename**.
10. Type **Overall Expenses**, and then click **OK**.

    You are creating an approval application which will have a review and submit workflow structure applied to it. Now you will need to specify a subset of a dimension that reflects this submit/reject process.

    This is also known as the approval hierarchy. This application will use the Subsidiaries_Approval subset created earlier.

11. Under **TM1 Objects**, expand **Dimensions** (if necessary), and then expand the **Subsidiaries** dimension (if necessary).

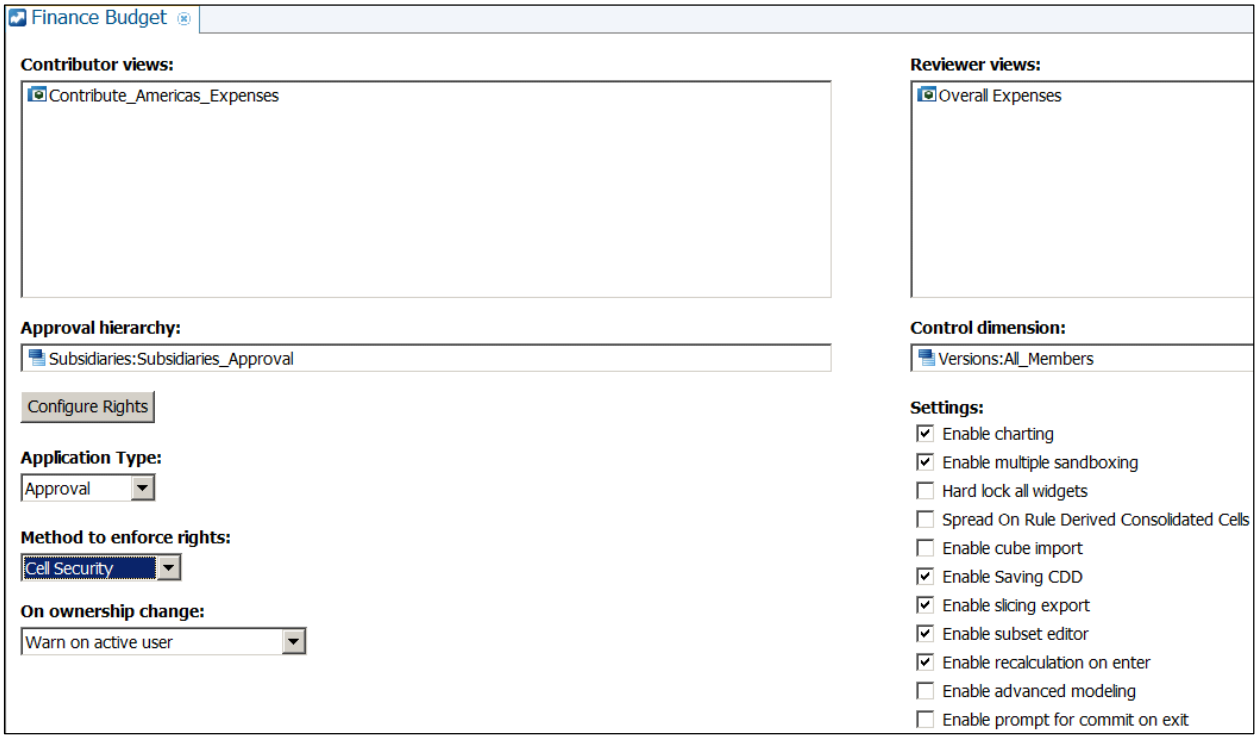
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12. Click the **Subsidiaries_Approval** subset and drag it to the **Approval Hierarchy** box on the right side.

    You want to be able to deploy applications to different slices of the Expenses cube. You will use a control dimension to split things up by version. Users of the Finance Budget application will only be working with Budget Version 1 data.

13. In the **Model Design** pane, expand the **Versions** dimension (if necessary), and then drag the **All_Members** subset to the **Control Dimension** box on the right side.
14. In the **Method to enforce rights** list, select **Cell Security**.
15. Click **Save** and then click **OK**.

    The results appear as follows:



    Since the view name is not what the application users wish to see, you changed the view's tab name by right-clicking the view in the Application Design pane. You can also select the view and modify its tab name by changing the Tab Name property in the Properties section on the bottom right side. By default, the tab name is the same as the view name.

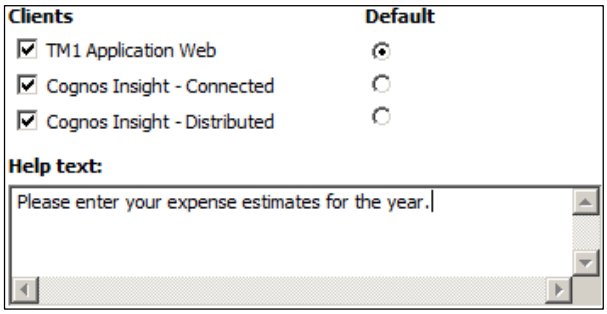
    You can also apply some instructions in the Help Text property for a view, as well as for the application as a whole.

16. Under **Design** on the top left side, under the **Contributor Views** folder, click **Contribute_Americas_Expenses**, and then click in the **Help Text** box.

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17. Type the following: **Please enter your expense estimates for the year.**, and then press **Enter**.

    The results appear as follows:

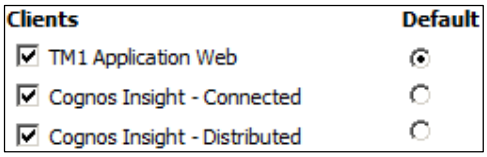


18. Click **Save**, and then click **OK**.

Task 8. Review the default clients for the application.

1. In the right pane, scroll down to the bottom (if necessary).

   The available clients are visible:



   By default, all the clients are enabled. To disable a client, deselect the applicable check box. You can also change the default client.

2. Leave all clients enabled, and then select the **Default** button for **Cognos Insight - Distributed**.

Task 9. Validate, deploy and activate the application.

1. On the toolbar, click **Validate Application**  .

   The warning messages indicates that the approval hierarchy dimension (Subsidiaries) is lower than the first or second position in the dimension order of the Expenses cube. This positioning can have an impact on performance in Cognos Insight in Distributed mode. You will leave the dimension order as-is for now, although you can change it later if necessary.

2. Click **OK** to dismiss the warning messages, and then click **OK** to dismiss the confirmation message.

3. On the toolbar, click **Save**, and then click **OK**.

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4. On the toolbar, click **Deploy Application**  .

   The warning messages related to dimension order appear again. You can disregard these for now.
5. Click **OK** to dismiss the warning messages, and then click **OK** to dismiss the confirmation message (it may take a few moments for the confirmation message to appear).
6. Close the **Application Design** tab on the right side.

   Now you can apply security to the application. This can be done in Performance Modeler or in the IBM Cognos TM1 Applications portal.

Task 10. Apply security to the nodes in the application.

1. In the right pane, click the **Configure Rights** button.

   The Rights for Finance Budget tab opens and the warning messages related to dimension order appear. You can disregard these for now.
2. Click **Close**.

   In the Rights Editor, you see the approval hierarchy (if applicable), the security groups, and the privileges that can be assigned. You can also configure the control dimension and offline users.

   For applications with an approval hierarchy, select a node in the hierarchy, then the corresponding group, and finally the rights to apply. Click Add and that setting will be added to the list at the top.

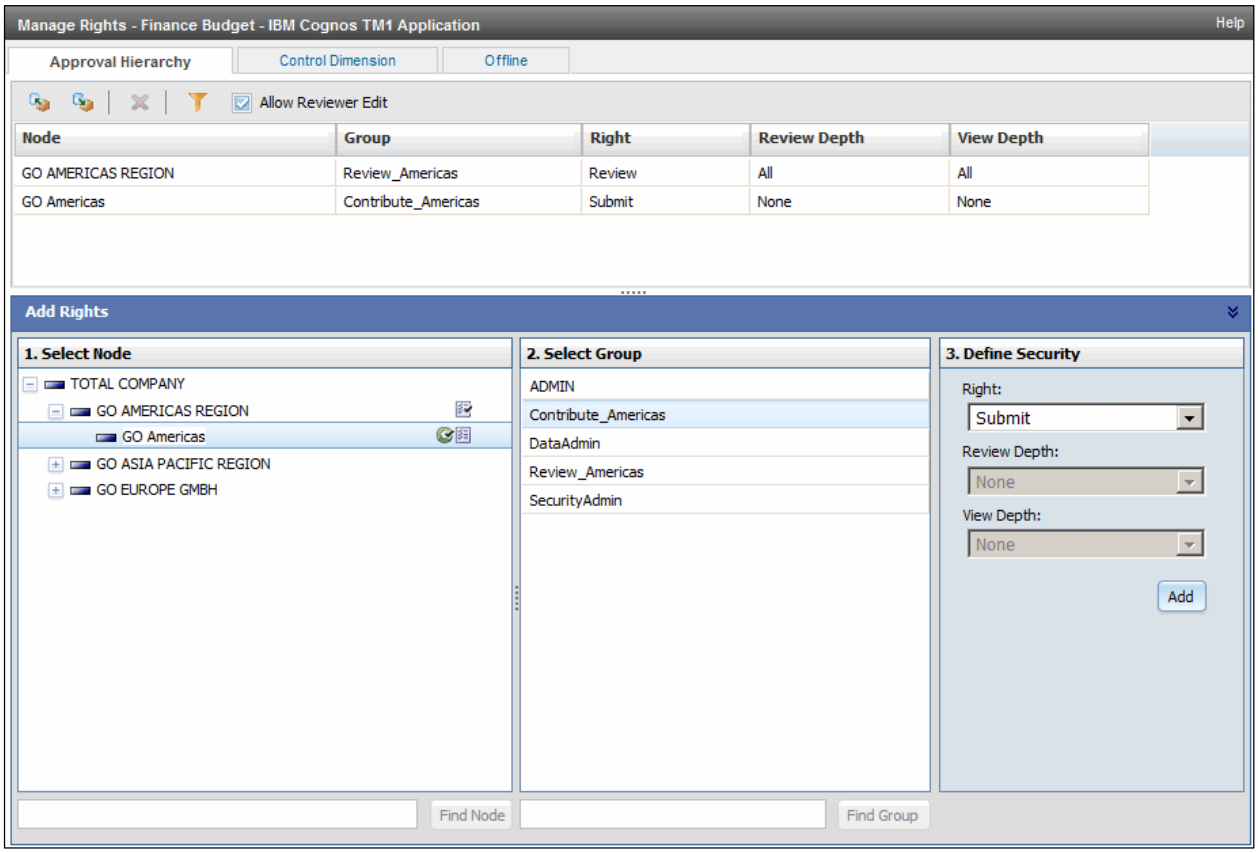
   The Find Node and Find Group boxes allow you to type in values to help you locate the appropriate node or group.
3. From the **Select Node** pane of the **Add Rights** section, click **GO AMERICAS REGION**.
4. In the box under **Select Group**, click **Review_Americas**.
5. Under **Define Security**, select **Review** from the **Right** drop down list.
6. Leave the defaults for **Review Depth** and **View Depth**, and then click **Add**.
7. Under **Select Node**, click **GO Americas**.

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8. In **Select Group**, click **Contribute_Americas**, in the **Right** list, click **Submit**, and then click **Add**.

   The results appear as follows:



   Rights can be exported from one application and imported into another one using tools on the toolbar.

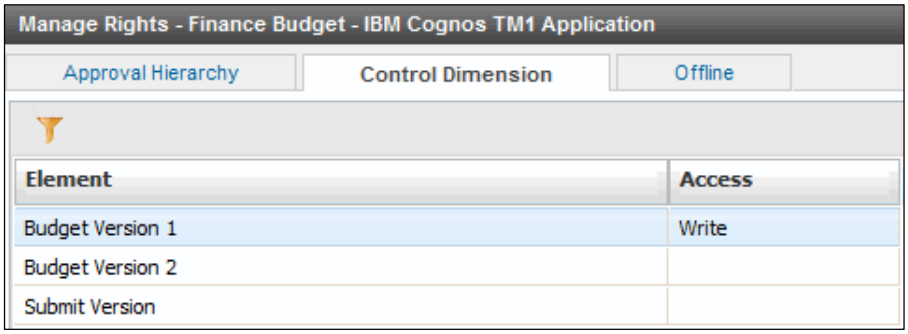
   You will now configure the control dimension. You want this application to be applicable to Budget Version 1 only. Other applications that use this approval hierarchy will apply to other versions of the budget.

9. Click the **Control Dimension** tab, and then click the **Access** cell for **Budget Version 1**.

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10. Select **Write**, and then press **Enter**.

    The results appear as follows:



    These settings indicate that users will be able to write to Budget Version 1. They will not see the other elements at all.

11. Click the **Offline** tab.

    By default, all users can take nodes offline when working in IBM Cognos Insight - Distributed mode. This is acceptable for now.

12. Click **Apply**, and then click **Close**.

13. Close **IBM Cognos TM1 Performance Modeler**.

14. In the **IBM Cognos TM1 Applications** portal, click **Refresh**  .

    The results appear as follows:



    The Finance Budget application is currently only visible to administrators. You need to activate it before anyone else can see it.

15. To the right of the **Finance Budget** application, click **Activate the IBM Cognos TM1 Application - Finance Budget**  , and then click **Close** to dismiss the warning messages.

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16. To the right of the **Finance Budget** application, click **Open IBM Cognos TM1 Application - Finance Budget**  .

    To the left is the list of nodes in the applications and the details appear to the right. If you had not activated the application, you would have been able to open it but you would have received a warning message indicating that the application was inactive.

17. In the top right corner, click the **Return** button, and then click **Log Off**.

    You will now access the application as a contributor. You will log in as Berger, who is a member of the Contribute_Americas group.

Task 11. Open the approval application as a contributor.

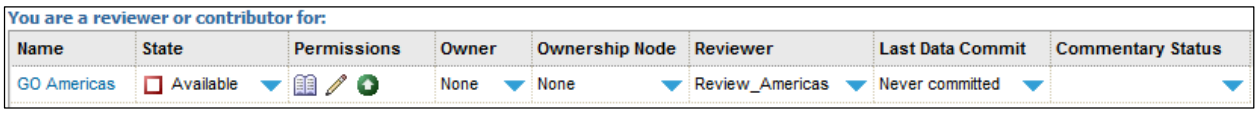
1. Log on as **Berger** with no password.

   Berger is a member of the Contribute_Americas group.

   Only the Finance Budget application is visible in the portal.

2. Click the **Finance Budget** link, and then under **Finance Budget Workflow** on the left side, click **GO Americas**.

   Here you can see information about the state of the selected node:



   And below are the workflow details for the node:



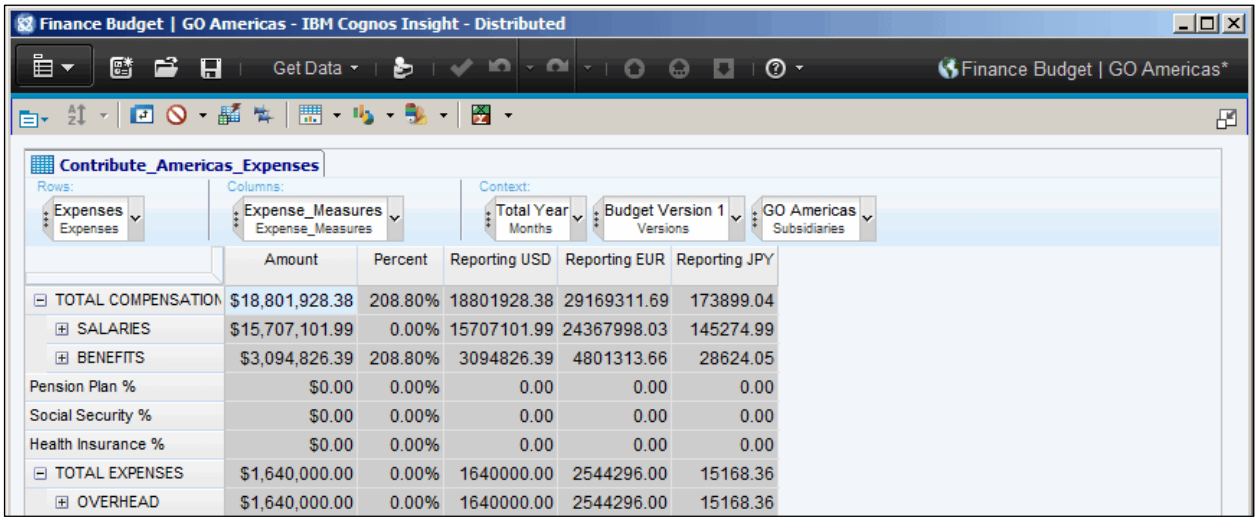
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3. In the table to the right, click the **GO Americas** link.

   If you are prompted to open the **provagent.cogrcp_insight** file, click the **Open** button at the bottom.

   By default the application opens in IBM Cognos Insight - Distributed mode (this will take a few moments).

   You will see the view that has been designated for the contributors, which is Contribute_Americas_Expenses:



	Amount	Percent	Reporting USD	Reporting EUR	Reporting JPY
TOTAL COMPENSATION	$18,801,928.38	208.80%	18801928.38	29169311.69	173899.04
SALARIES	$15,707,101.99	0.00%	15707101.99	24367998.03	145274.99
BENEFITS	$3,094,826.39	208.80%	3094826.39	4801313.66	28624.05
Pension Plan %	$0.00	0.00%	0.00	0.00	0.00
Social Security %	$0.00	0.00%	0.00	0.00	0.00
Health Insurance %	$0.00	0.00%	0.00	0.00	0.00
TOTAL EXPENSES	$1,640,000.00	0.00%	1640000.00	2544296.00	15168.36
OVERHEAD	$1,640,000.00	0.00%	1640000.00	2544296.00	15168.36

4. Maximize the window, and then on the toolbar, click **Take Ownership**  .

   The data can now be edited.

5. In the **Context** area, click the **Months** dimension, expand **Q1**, and then select **Jan**.

   Now that you are at a leaf level, you can enter data into the writeable cells of the application.

6. In the **Context** area, click the **Versions** dimension.

   Due to the control dimension, you can see that only Budget Version 1 is available for data entry.

7. In the **Context** area, click the **Subsidiaries** dimension.

   Because of the security that you applied, you can only enter data for the GO Americas subsidiary.

8. In the top left corner, click the **Actions** menu  , and then click **Work Offline**.

   The title bar of the window indicates that you are now working offline. You can now enter data into your slice of the plan without being connected to the network.

9. From the **Actions** menu, deselect the **Work Offline** option.

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10. Close **IBM Cognos Insight**, and then click **OK**.
11. Under **Name**, right-click the **GO_Americas** link, click **Open TM1 Application Web**.

    The application opens in IBM Cognos TM1 Application Web (this may take a few moments). You can use either this tool or IBM Cognos Insight to take ownership of nodes and use the application. However, you cannot take nodes offline in IBM Cognos TM1 Application Web or IBM Cognos Insight - Connected mode.
12. Close **IBM Cognos TM1 Application Web**.

    You will now open the application as a reviewer.

Task 12. Open the approval application as a reviewer.

1. In the top right corner, click **Log Off**, and then log on as **Bauer** with no password.

   Bauer is a member of the Review_Americas group.
2. Click the **Finance Budget** link.

   The application opens. Due to the security you set, Bauer can see both the GO AMERICAS REGION node and the GO Americas node underneath it.
3. In the left pane, click **GO AMERICAS REGION**.

   The Workflow Detail for GO AMERICAS REGION pane at the bottom indicates that data in this node is aggregated, and that the node underneath it (GO Americas) is not available. GO AMERICAS REGION itself has a workflow status of Work In Progress, while GO Americas has a status of Reserved.
4. Right-click the **GO AMERICAS REGION** link and point to **Open IBM Cognos Insight - Distributed**.

   Because of the security you set, and because you have designated both contributor and reviewer views for this application, you can open this node as a Reviewer or a Contributor. Bauer is a member of the Review_Americas group, which has Review rights to the GO AMERICAS REGION node and all the nodes underneath GO AMERICAS REGION. Members of this group can also edit nodes, because the Allow Reviewer Edit check box was selected on the Manage Rights screen.
5. Click **As Reviewer**.
6. If you are prompted to open the **provagent.cogrcp_insight** file, click the **Open** button at the bottom.

   IBM Cognos Insight opens in distributed mode (this may take a few moments). You can see the Overall Expenses tab, which you had designated earlier as a reviewer view.
7. Close **IBM Cognos Insight**, and then click **OK**.

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8. Right-click the **GO AMERICAS REGION** link, point to **Open IBM Cognos Insight - Distributed**, and then click **As Contributor**.
9. If you are prompted to open the **provagent.cogrcp_insight** file, click the **Open** button at the bottom.

   IBM Cognos Insight opens in distributed mode (this may take a few moments). You can see the Contribute_Americas_Expenses tab, which you had designated earlier as a contributor view.
10. Close **IBM Cognos Insight**, and then click **OK**.
11. Click **Log Off**, and then close **Internet Explorer**.

Results:
You created and deployed a managed planning application for the SampleOutdoors TM1 server.

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IBM Training

Unit summary

- Explain the application types
- Access an IBM Planning Analytics application
- Create a new application
- Set the available clients
- Apply security in the application
- Activate and deactivate an application

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Unit summary

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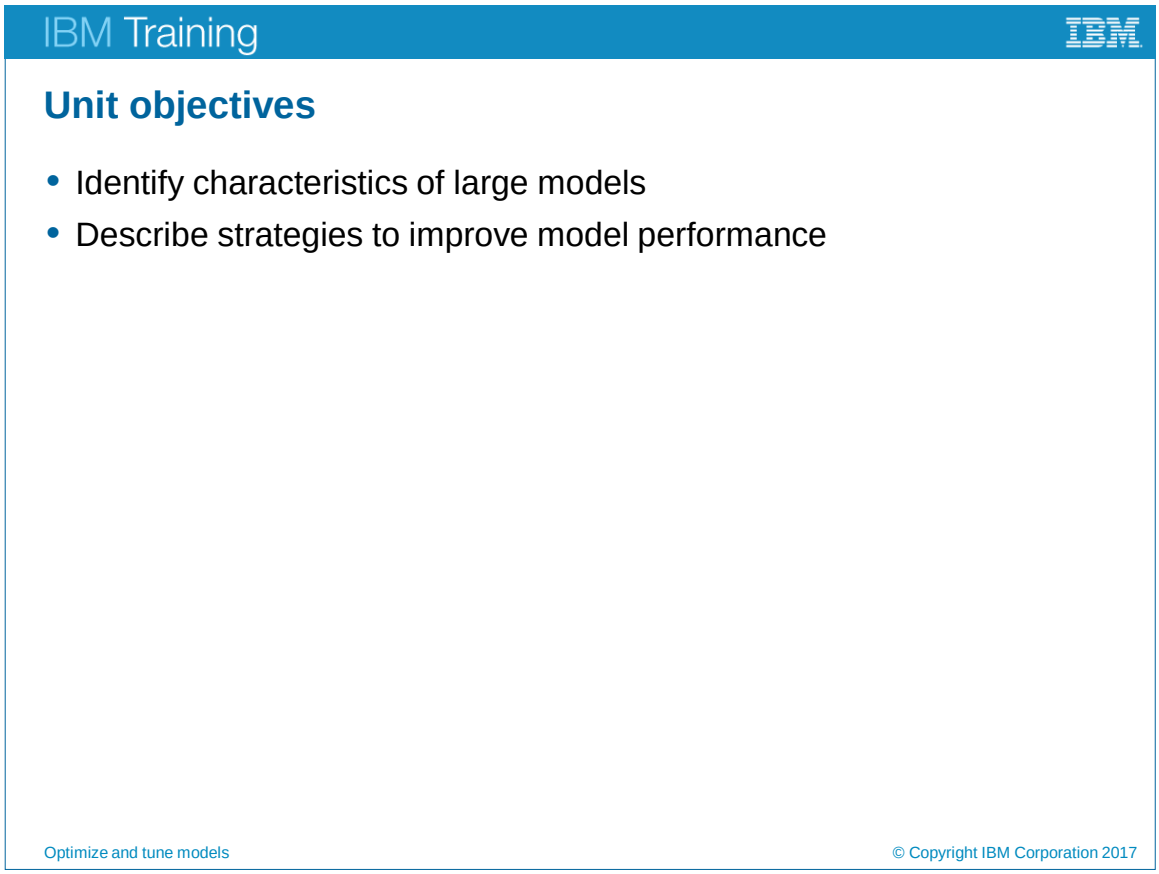
IBM Training

Optimize and tune models

IBM Planning Analytics (v2.0)

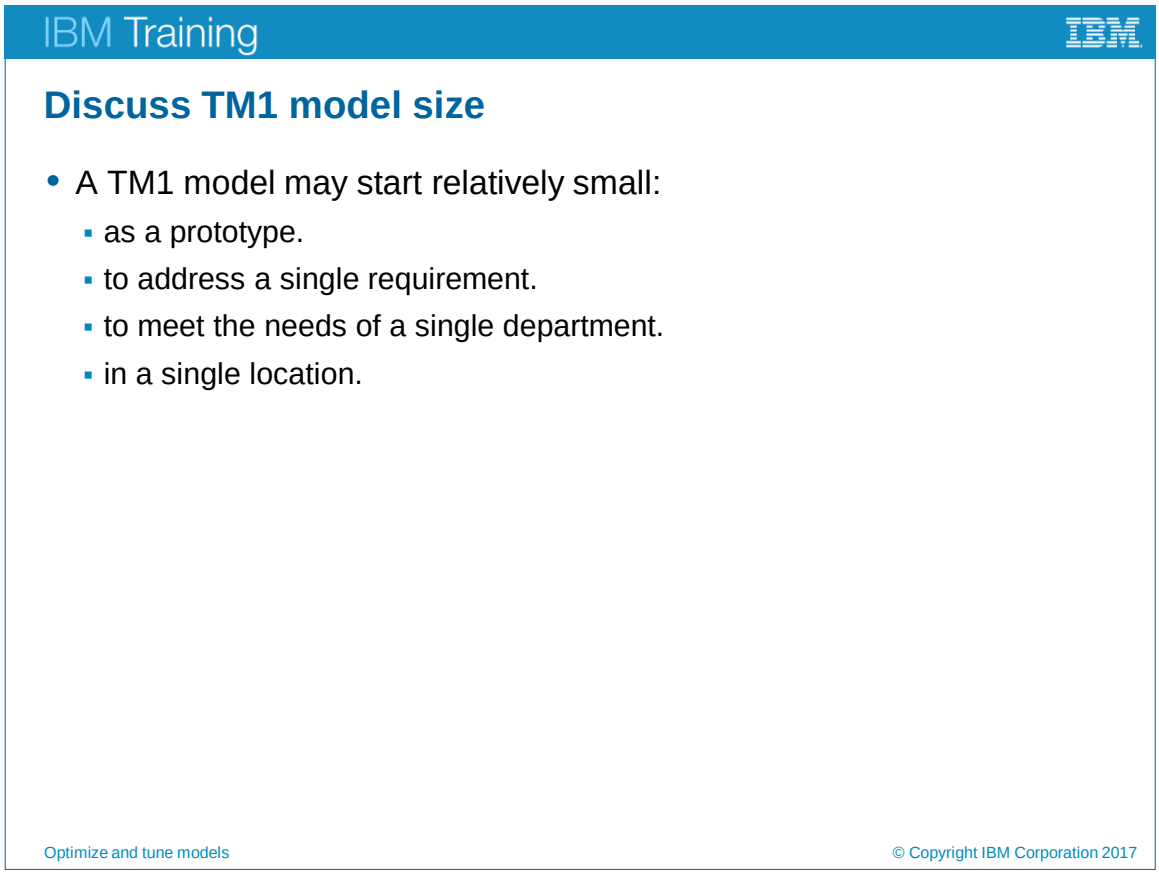
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Unit objectives

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Discuss TM1 model size

Model size can be measured in terms of size of data in memory or number of users.

In terms of the size of a model in memory, you can use the following as guidelines.

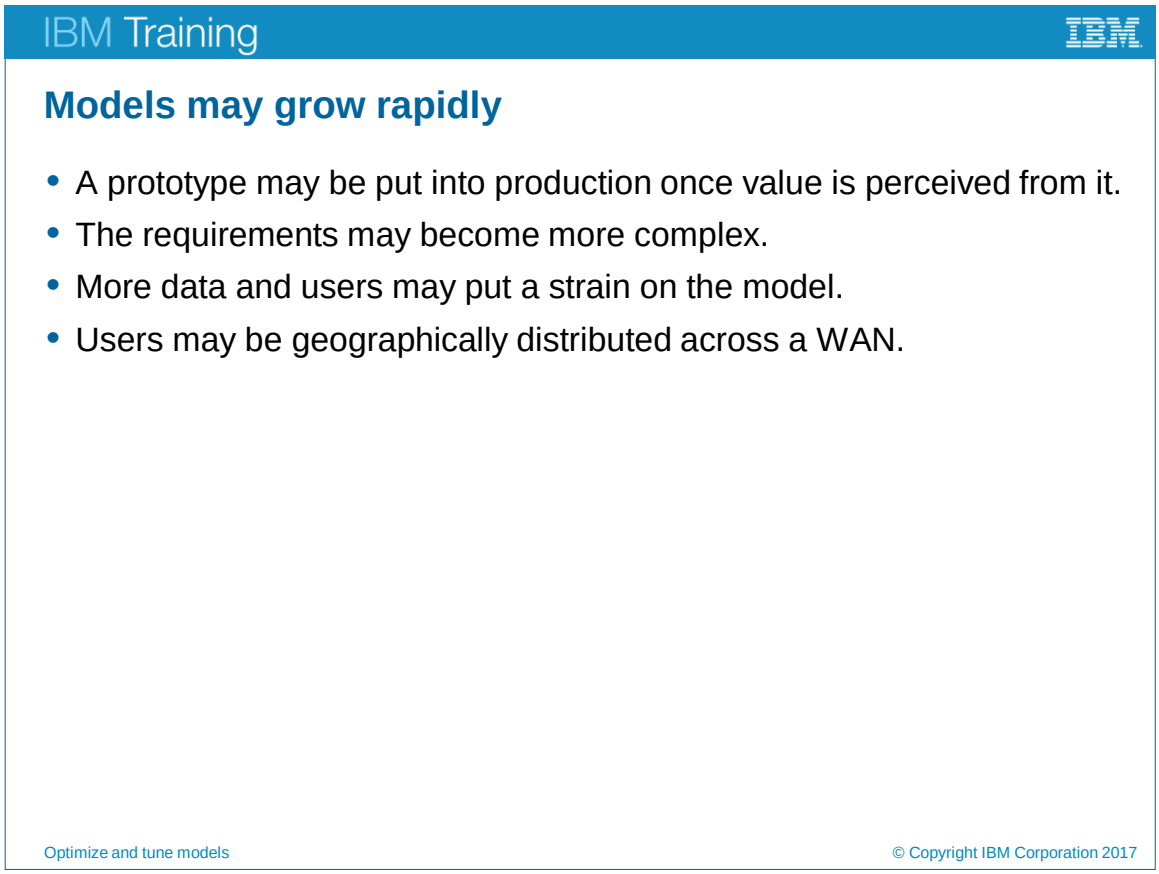
- A smaller model may be about 1 GB in memory.
- A medium or average model may be about 10 GB in memory.
- A large model may be about 100 GB in memory.

A similar scale may be used for number of users:

- A small model may have about 10 users.
- A medium or average model may have about 100 users.
- A large model may have over 1,000 users.

These are estimates and every model and environment is unique.

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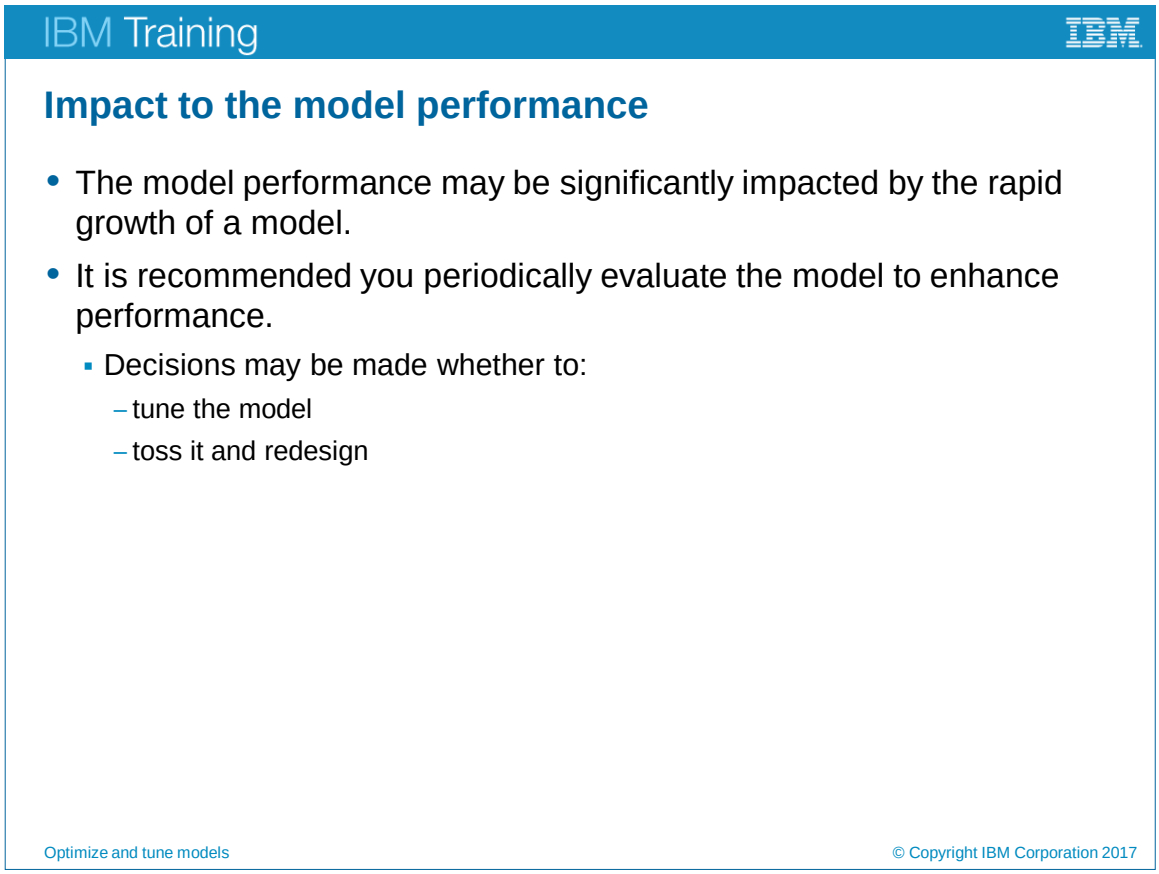
Models may grow rapidly

It is relatively easy to create a simple prototype. However, it should only be used as a starting point in order to identify requirements or early user adoption. A prototype may only have been designed with a subset of data and requirements in mind. The prototype should be used to help determine the requirements for the production model.

It may not utilize best practices for design.

Best practices for a small model may not be the same as for a large one.

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Impact to model performance

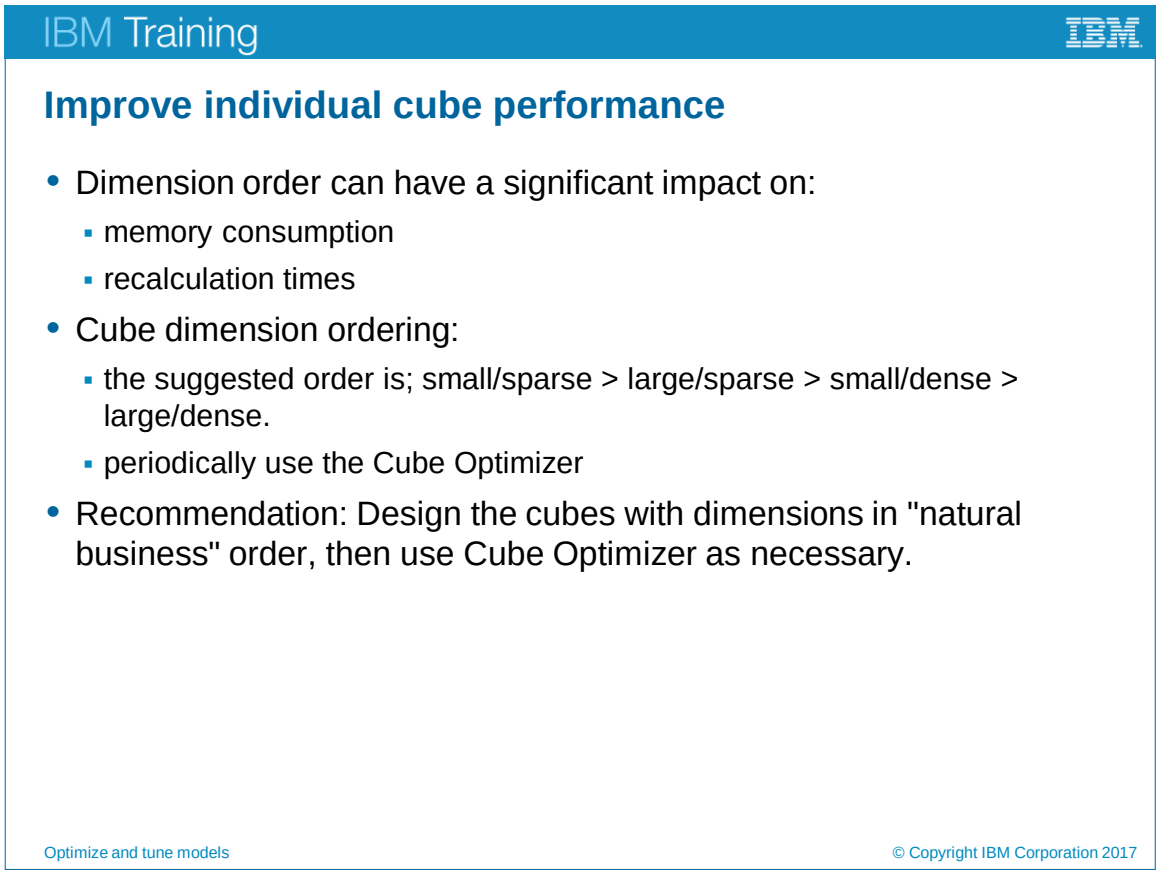
If a model was not intended to move into a production environment, there may be enhancements that can be made to improve performance. Over time, users may request enhancements or develop new requirements.

Continually ask yourself:

- Why does this object exist?
- Can I make a more general version of this object?

If you're not 100% sure why something exists, get rid of it.

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Improve individual cube performance

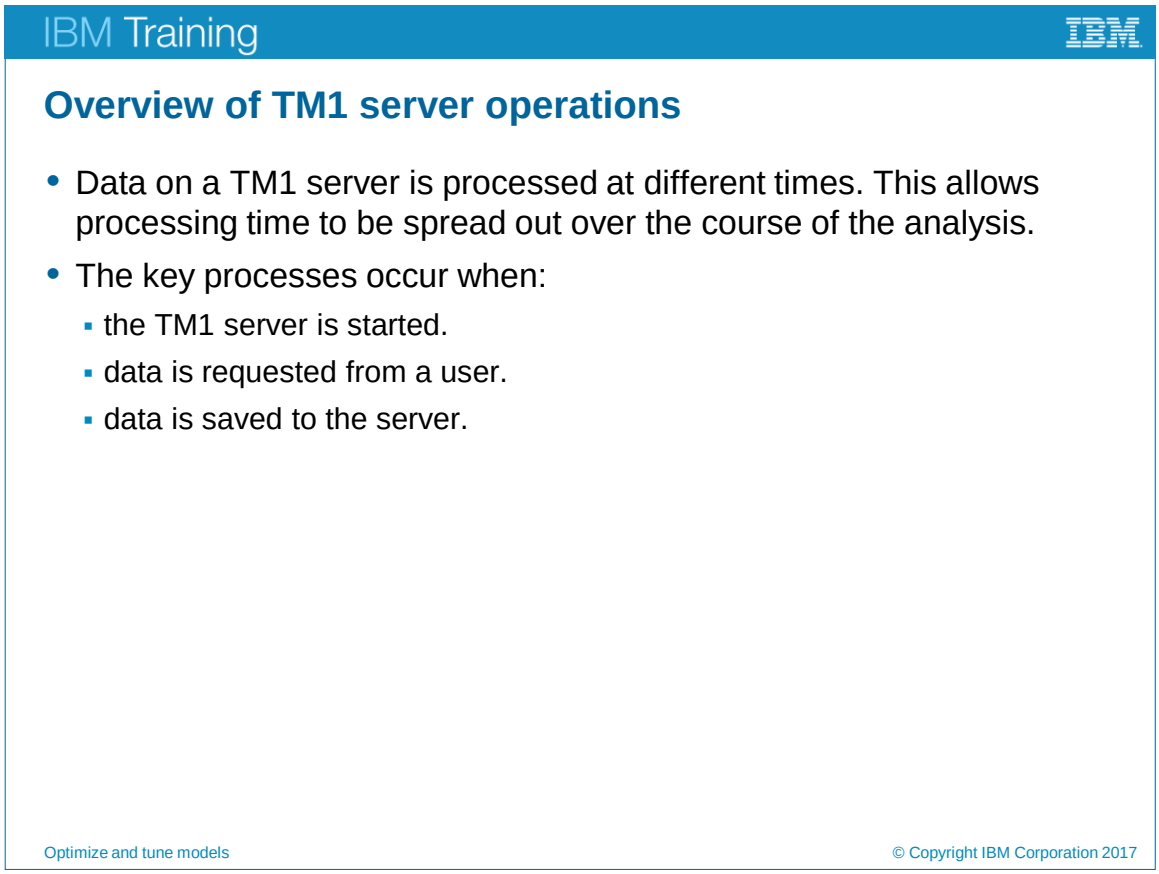
TM1 functions in spreadsheets and rules may be referencing dimensions in a specific order, so it is important not to physically change the dimension order. The Cube Optimizer allows internal re-ordering of cube dimensions to decrease storage requirements and increase recalculation performance.

Dimension order is important to TM1, because the recalculation engine is RAM-based (on-the-fly).

The sparsity algorithm needs to "prune" dimension branches as early as possible for best performance.

It requires 2X of cube memory footprint to be available (for example, 1GB must be available to reorder a 500MB cube).

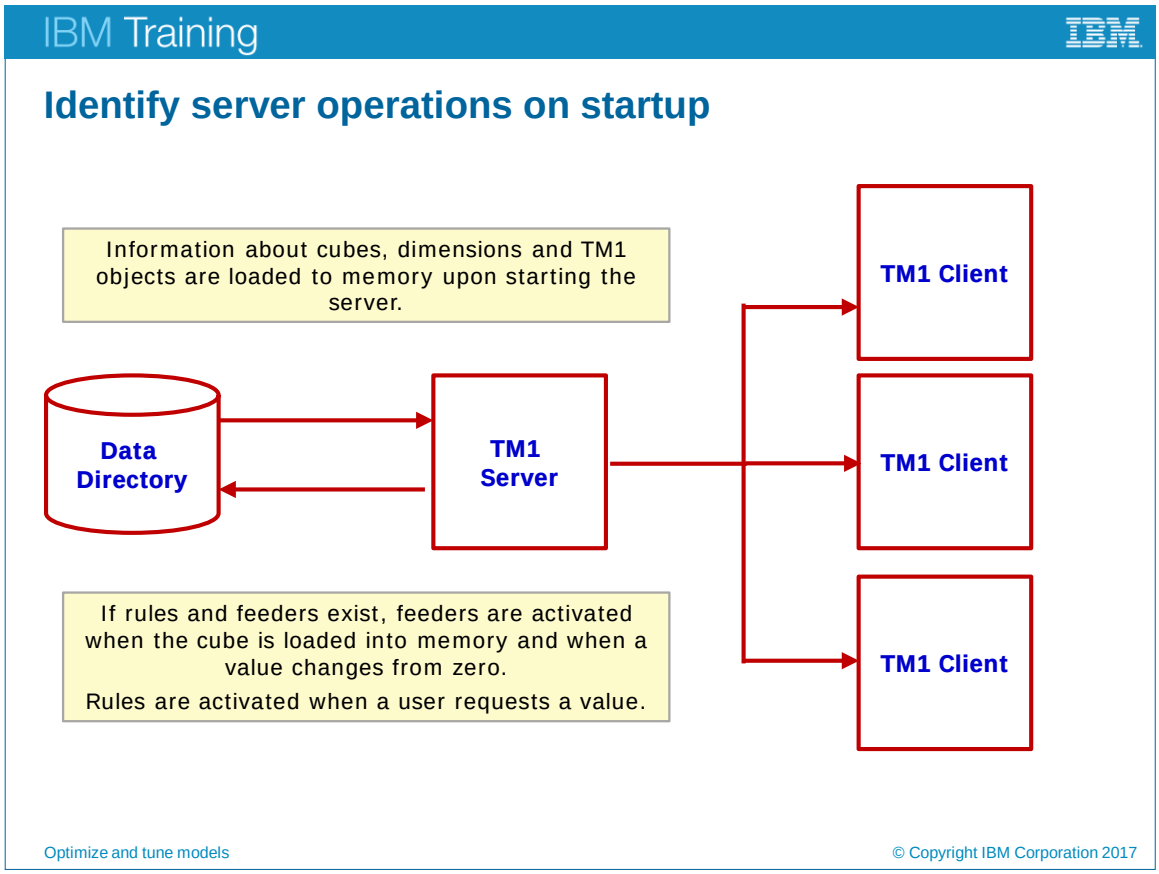
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Overview of TM1 server operations

Understanding what type of process occurs at what time can alter the way the model is designed.

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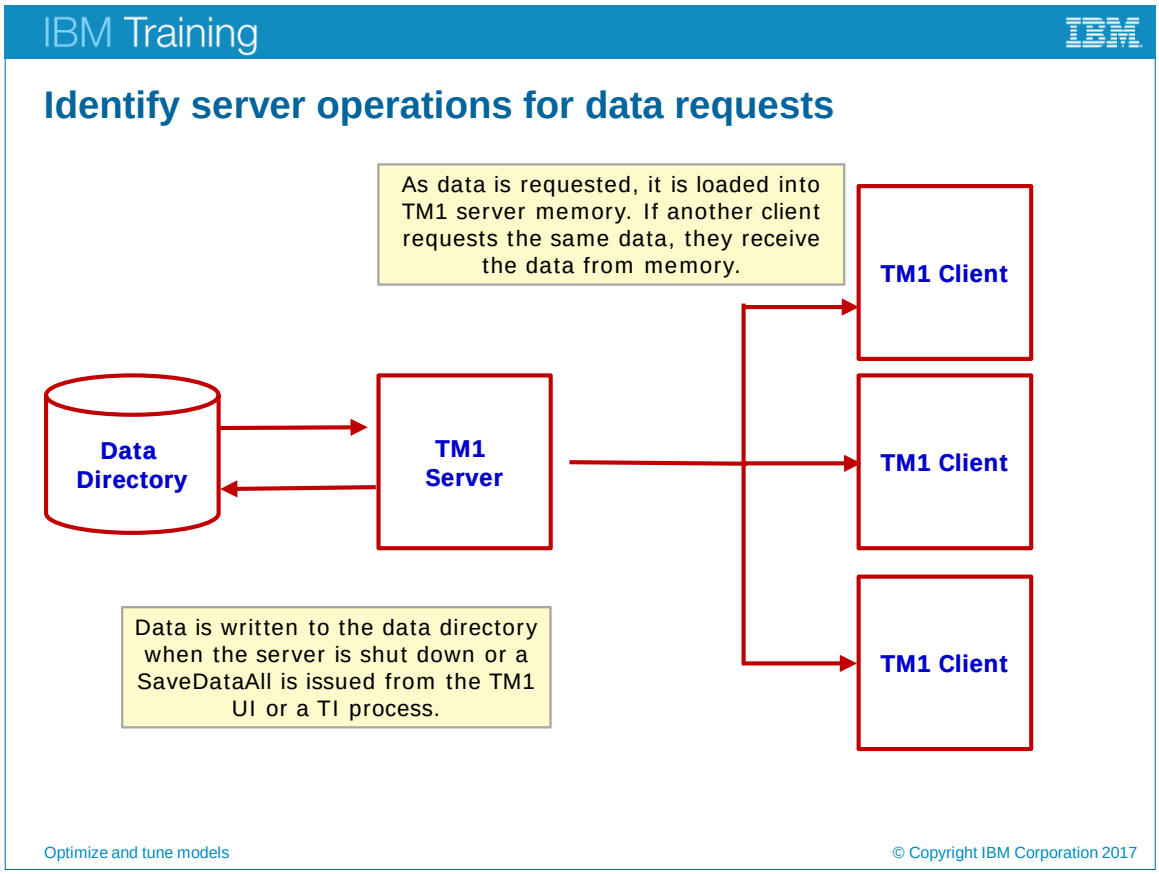


Identify server operations on startup

As a TM1 server is started, information about TM1 objects is loaded into memory. When a server is started from a desktop icon, the TM1s.log file is displayed. This displays the name of each object as it is loaded into memory.

Information about dimensions, cubes and other TM1 structures are loaded into memory. Data stored in the cubes is not yet loaded.

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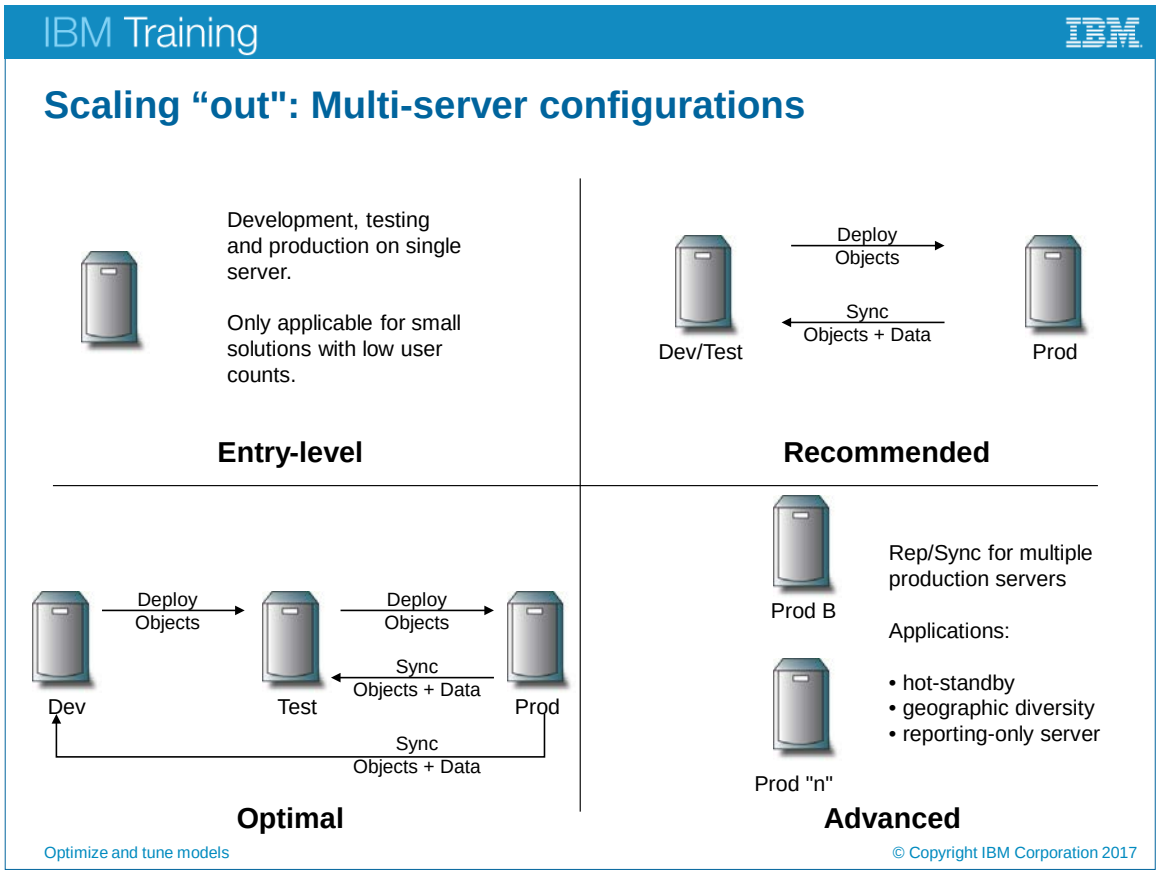
Identify server operations for data requests

As data in a cube is requested, it is held in the server memory. If the same data is later requested, the data in memory is presented. This enables TM1 to respond very rapidly to the second request.

Data remains in memory until it is explicitly written to the cubes using one of the following methods:

- from Server Explorer, click Save Data.
- from a TI process or chore using SaveDataAll.
- shutting down the server.

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Scaling "out": Multi-server configurations

Scaling "out" refers to the act of increasing the number of host machines or TM1 server instances. Scaling "up" is adding additional memory and/or CPUs to an existing host machine.

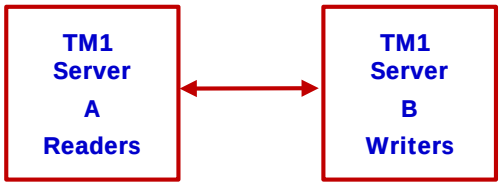
As the number of users grow, it may be necessary to have separate servers based on activity (reader/writer), geography, or other functional area. This is not required for TM1 to function but could improve server performance.

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IBM Training

Mirrored servers

- All cubes, dimensions, and other objects are synchronized.
- Addresses problem of contention between 'Readers' and 'Writers' seeking to access the same data.



Optimize and tune models

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Mirrored servers

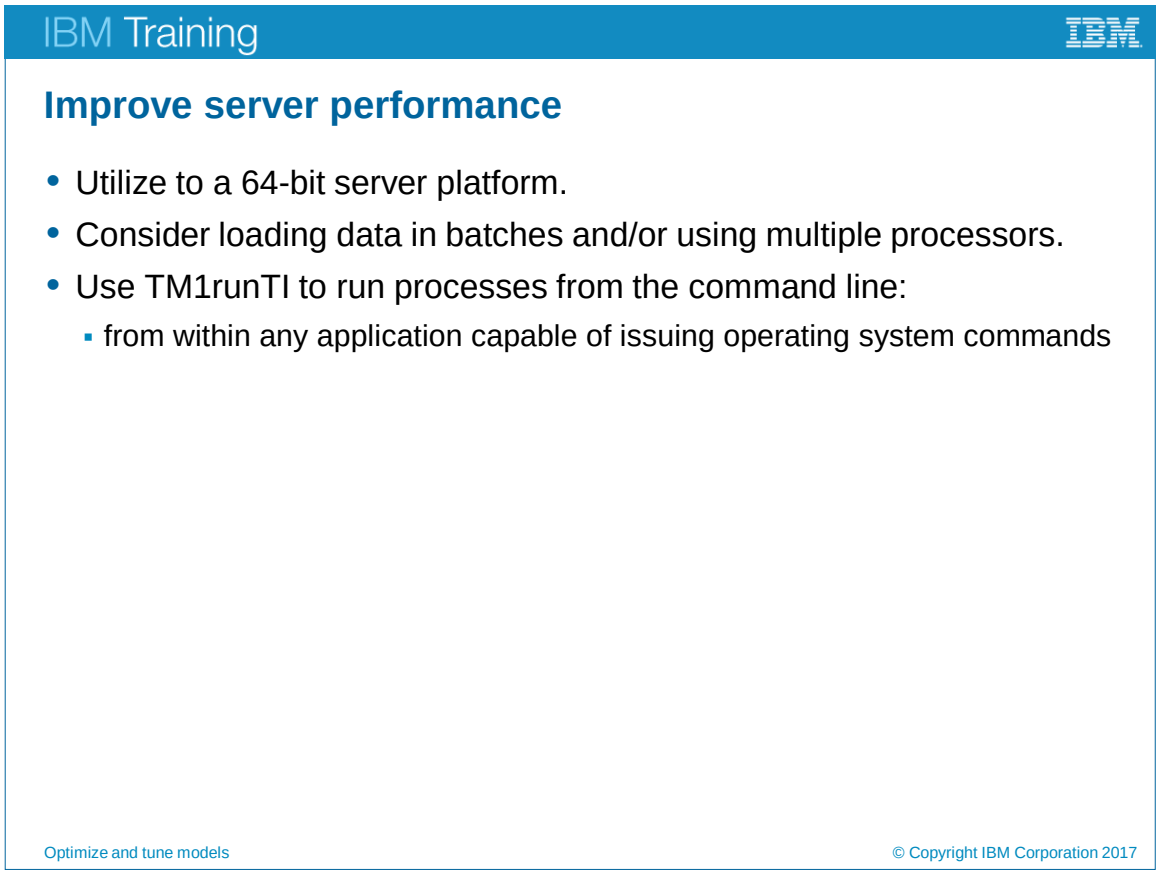
The worst case scenario for multi-user performance occurs when there are users performing updates while other users are performing long calculations.

Even though write operations generally take very little time, because of the queuing rules, if there is a writer waiting while a long calculation is taking place, all users will queue behind the writer. If, in turn, there are other readers requesting long calculations and there are writers interspersed between them, the result is that all requests are processed sequentially, and the average waiting times may get unacceptably long.

The idea is to segregate readers and writers into separate "mirrored servers". By mirrored servers we mean servers where all cubes, dimensions and other objects are synchronized. The synchronization may be done on a regular interval or on demand.

The effect is that between synchronizations, the reader's server acts as a read-only server and will deliver all the benefits of stored calculations.

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Improve server performance

All Win32 systems are bound by an immutable limit: 2^32 = approximately 4GB, meaning that 32-bit processors, and Win32, can only access 4GB of RAM.

By default, Win32 operating systems reserve 2GB of this 4GB space for the O/S, leaving 2GB for all applications. On "advanced" versions of Windows Server, a "3GB switch" can be set to force the O/S to work within 1GB, and leave 3GB available for all applications.

One of the most attractive features of the x64 platform is that it natively supports both 32-bit and 64-bit applications and device drivers, allowing IT administrators to gracefully transition from 32-bit to 64-bit operation over time.

For TM1, one immediate benefit is that 32-bit applications that are "Large Address Aware" under x64 receive one (1) extra GB of address space, up to a full 4GB. This can offer dramatic improvements in performance and stability, as it adds a significant "safety buffer" for large TM1 views and calculations.

True 64-bit applications executing under x64 can address as much physical RAM as is available on the host machine that has not been reserved by the O/S.

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IBM Training

Unit summary

- Identify characteristics of large models
- Describe strategies to improve model performance

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Unit summary

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IBM Training

Identify common data structures

IBM Planning Analytics (v2.0)

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IBM Training

Unit objectives

- Identify characteristics of operational and reporting databases
- Discuss Online Analytical Processing (OLAP)
- Examine core model design principles
- Identify the basic tasks required to create a TM1 model and deploy it

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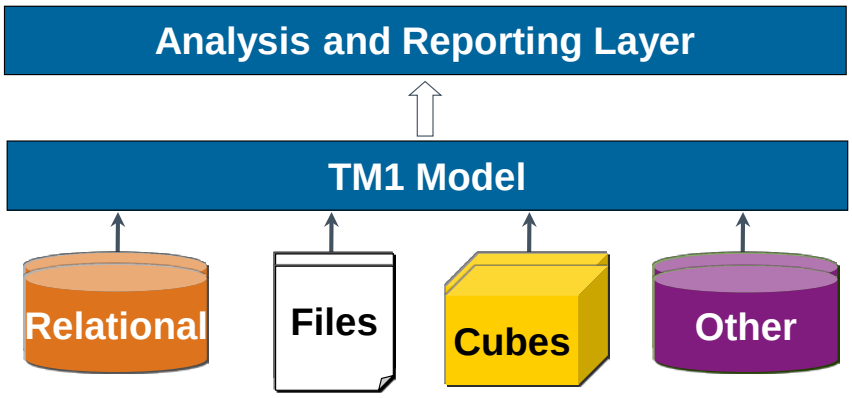
Unit objectives

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IBM Training

Examine the role of a TM1 metadata model

- A TM1 model provides a business presentation view of an organization's data sources.
- TM1 consumers use the model to analyze and report on their data sources.



Identify common data structures

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Examine the role of a TM1 metadata model

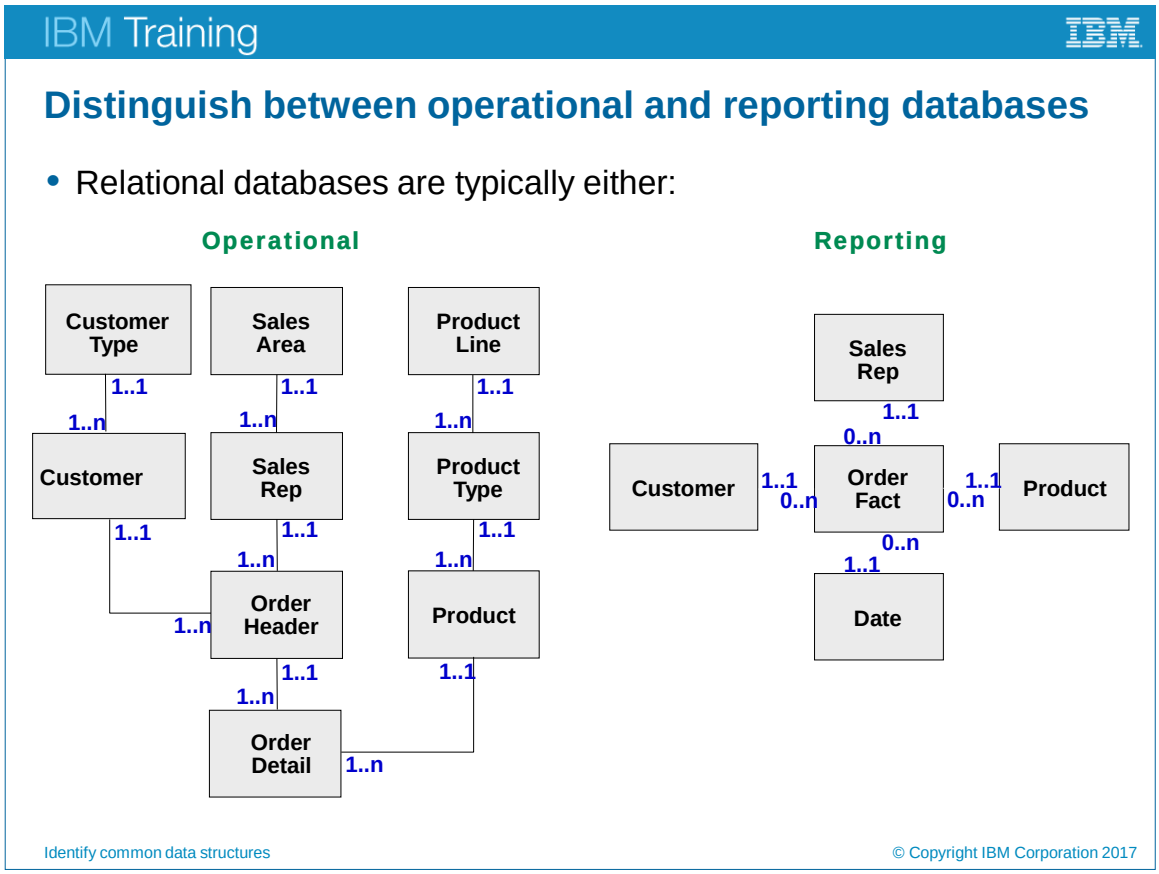
A TM1 model can hide the structural complexity of your underlying data sources.

By creating a TM1 model, you have more control over how your data is presented to end users. You can also choose which data to display to your end users and how it will be organized. The overall goal of modeling the metadata is to create a model that provides predictable results as well as an easy-to-use view of the metadata for authors and analysts.

Your underlying data sources may be very diverse. For example, you may have operational or reporting data in one or more relational databases. You may also have legacy data in various file formats, such as text, comma separated values (.csv), and spreadsheets (.xlsx).

This unit will focus on some industry standard data structures. This knowledge is essential before beginning a metadata modeling project.

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Distinguish between operational and reporting databases

Operational databases are:

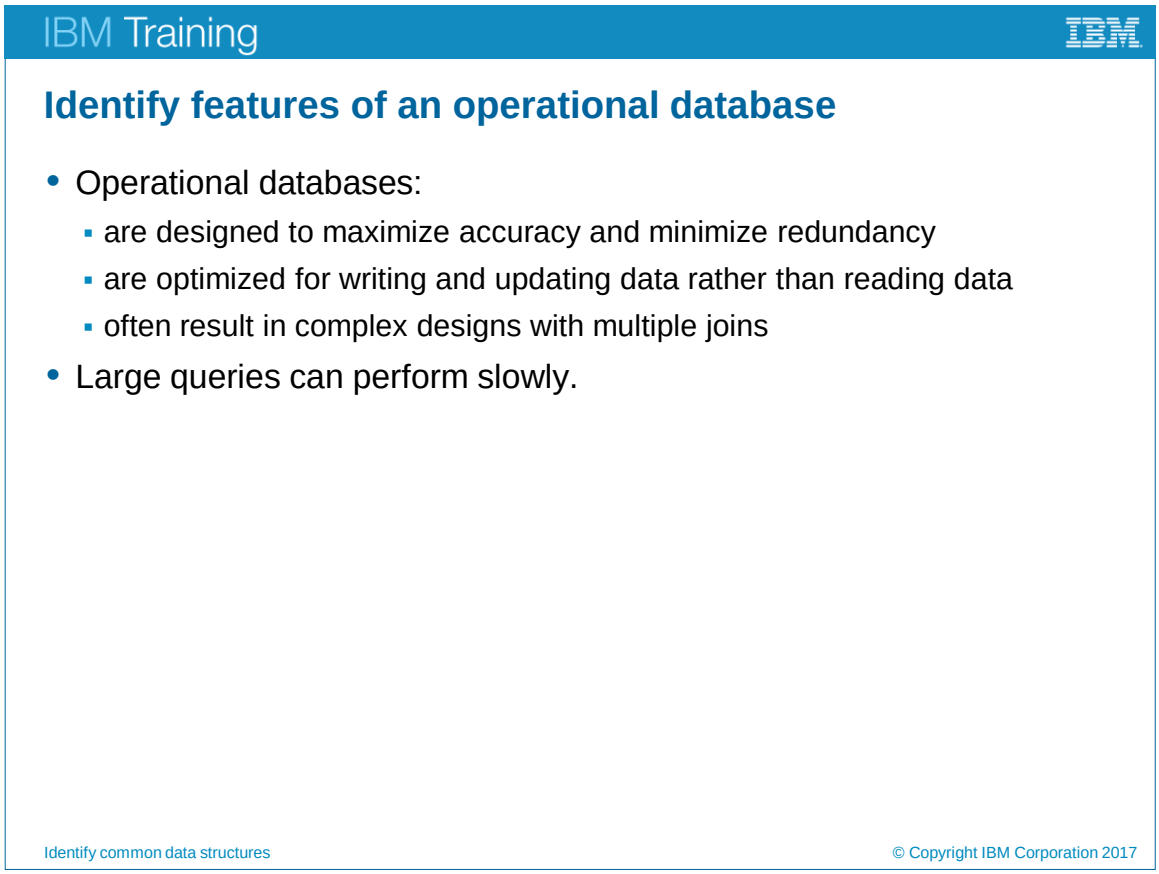
- used to track the day-to-day operations of a business
- usually normalized or part of an enterprise resource planning (ERP) vendor package

Reporting databases are:

- typically a copy of the operational data
- structured differently from operational databases to make reporting faster and easier
- usually dimensional, taking the form of a star schema design

TM1 does not require a star schema but a cube structure uses many of the same modeling principles.

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Identify features of an operational database

It can be difficult to report from an operational database because:

- reading the data can impact the performance of the system
- the number of tables that must be joined to satisfy a business question can be prohibitively large
- a more complex database requires more metadata modeling efforts to make it consumable by business users

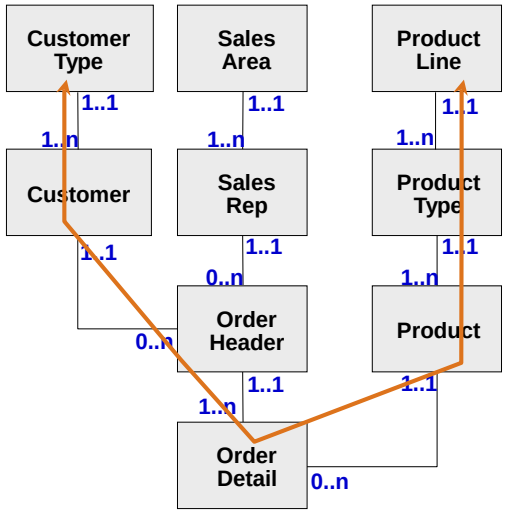
Operational systems are designed with one goal in mind: to get data into the database quickly. These databases are normalized to reduce redundancy. Having little to no redundancy ensures that there is data integrity and that database triggers function properly, so that the right data is captured.

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IBM Training

Identify issues with operational databases

- "Show all customer types that bought from a product line."
- The query must check data in seven tables before returning a result set.



Query requires 7 tables

Identify common data structures

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Identify issues with operational databases

Operational databases may be too complex for general reporting. As a result, reports that are generated against these structures may take a long time to run. They may also have unpredictable results.

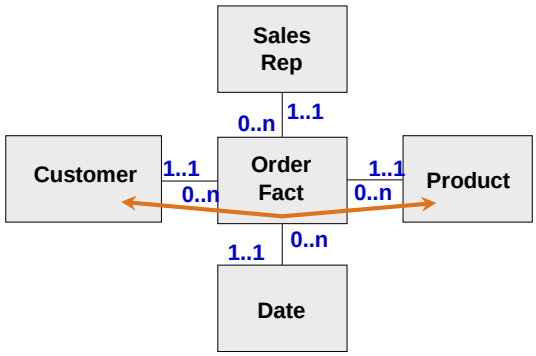
As shown in the slide example, a report may need to access many tables to retrieve all the necessary data.

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IBM Training

Examine reporting databases (star schema design)

- Transactional data is stored in a fact table
- Reference data is stored in separate dimension tables



Query only requires 3 tables

Identify common data structures

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Examine reporting databases (star schema design)

A star schema is a design with two basic components:

- a central table, known as the fact table (typically numeric data)
- satellite tables, known as the dimension tables

Because a star schema database contains fewer tables than a fully normalized database, query performance is much faster.

- A typical query against a star schema database focuses on the central fact table and makes integrity checks against the related dimension tables.

If the query is to retrieve information about a specific subject area only, such as all the products that belong to a particular product line, then the query will be even faster.

When building a TM1 cube, instead of creating a 'fact table' you will identify the numeric data for your cube. This data will probably be identified in your 'Measures' dimension.

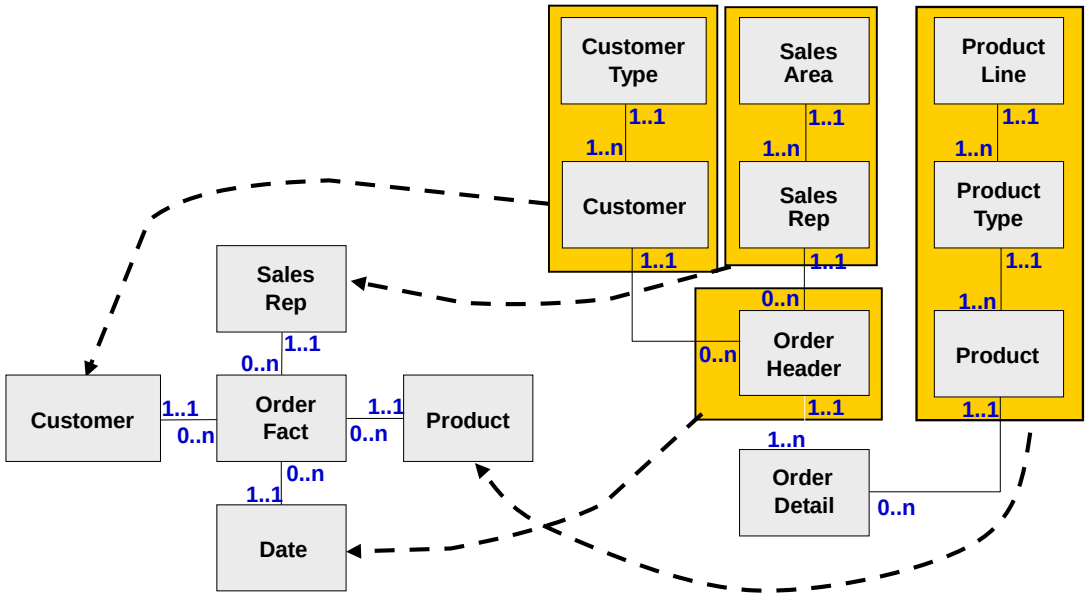
Instead of creating dimension 'tables', in TM1 you will build dimensions and define a hierarchy.

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IBM Training

Create a star schema

- Collapse the relationships to form dimensions (perspectives).



Identify common data structures

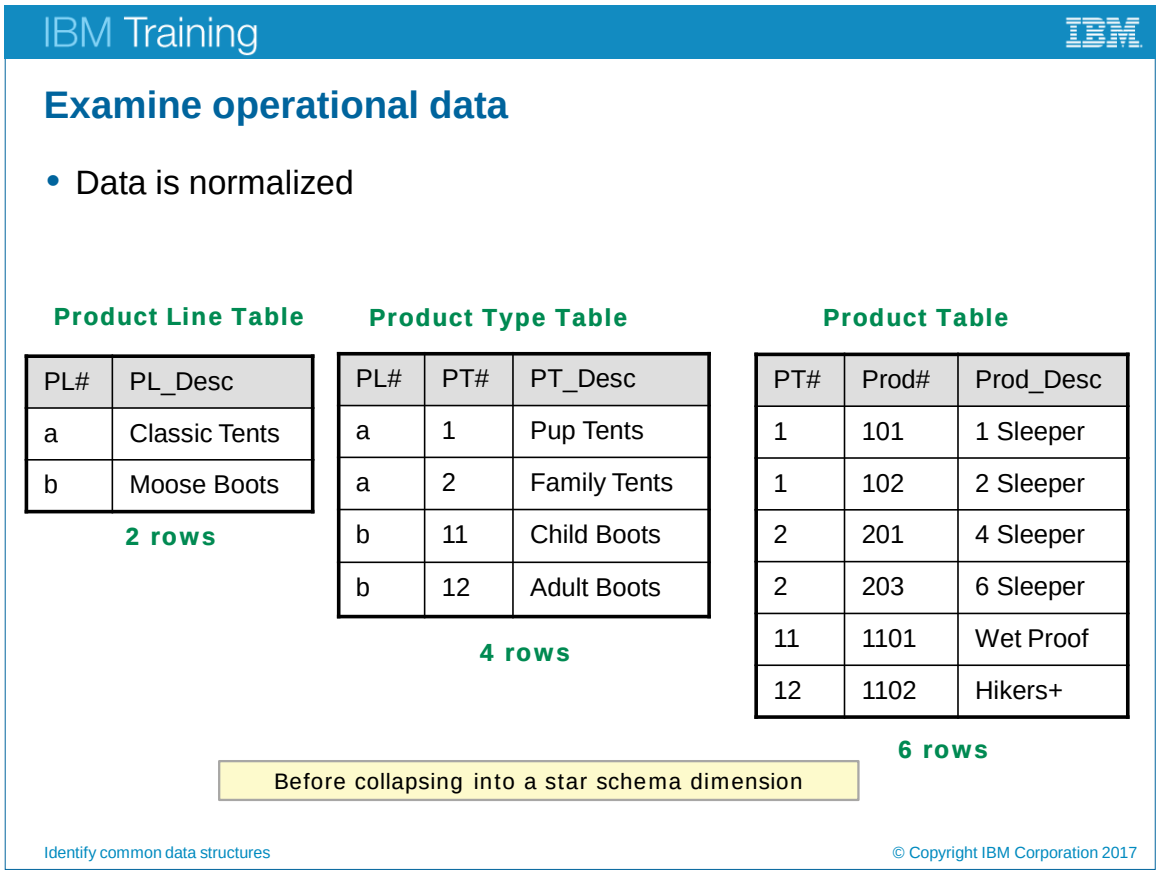
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Create a star schema

Each table in a star schema database will contain an expanded set of data.

Extract Transform and Load (ETL) tools can be used to create a star schema data warehouse, or you may use a metadata modeling tool to emulate a star schema structure by generating the appropriate SQL at report design time.

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Examine operational data

The slide example shows three normalized tables that represent three hierarchical levels. Products roll up into product types, and product types roll up into product lines.

The Product Line table has two rows that indicate two lines of products sold by the company.

The Product Type table contains four rows to indicate the four types of products that fall under the previous two product lines (two types per product line).

The Product table contains the greatest level of detail. It holds 6 rows to represent the 6 products that fall under the four product types.

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IBM Training

Examine reporting data

- Data is de-normalized

Product Dimension Table

PL#	PL_Desc	PT#	PT_Desc	Prod#	Prod_Desc
A	Classic Tents	1	Pup Tents	101	1 Sleeper
A	Classic Tents	1	Pup Tents	102	2 Sleeper
A	Classic Tents	2	Family Tents	201	4 Sleeper
A	Classic Tents	2	Family Tents	203	6 Sleeper
B	Moose Boots	11	Child Boots	1101	Wet Proof
B	Moose Boots	12	Adult Boots	1102	Hikers

6 rows

After collapsing into a star schema dimension

Identify common data structures

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Examine reporting data

The slide example shows a de-normalized dimension table created from the three normalized tables shown on the previous slide.

The Product Line table forms the first two columns of the new dimension table (PL# and PL_Desc), the Product Type table forms the next two columns (PT# and PT_Desc), and the Product table forms the last two columns (Prod# and Prod_Desc).

The main characteristic of this table is its redundancy. Note that each product line (Classic Tents and Moose Boots) is repeated, once for each product that the product line contains. The same applies for product type.

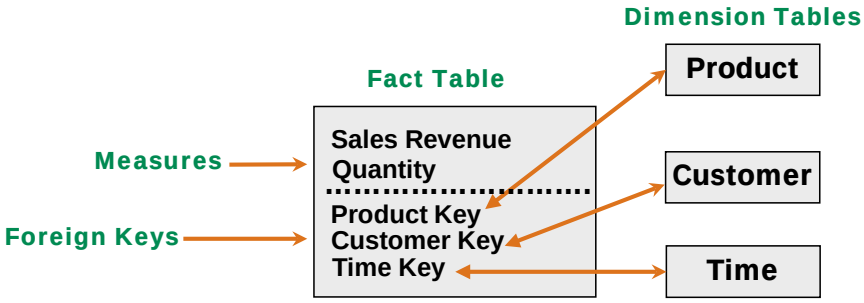
This type of table is unsuitable for a normalized system, but is ideal for a reporting and analysis structure.

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Examine fact tables

- Fact tables contain the (usually additive) values by which a company measures itself:
  - Standard Selling Price - not additive
  - Sale Amount - additive



Identify common data structures

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Examine fact tables

Fact tables are the focal point of any star schema, and typically contain the most rows. There are typically no descriptive attributes in a fact table. Instead, there are foreign keys that relate to the dimension tables, which contain descriptive attributes.

Facts in a fact table are also known as metrics, measures, or key performance indicators.

In TM1, you will often identify the metrics or measures as calculations. TM1 does not require the identification of a specific measures dimension. However, the modeler should be aware of the measures or metrics of the model. These will usually be additive but not always (as in the example above).

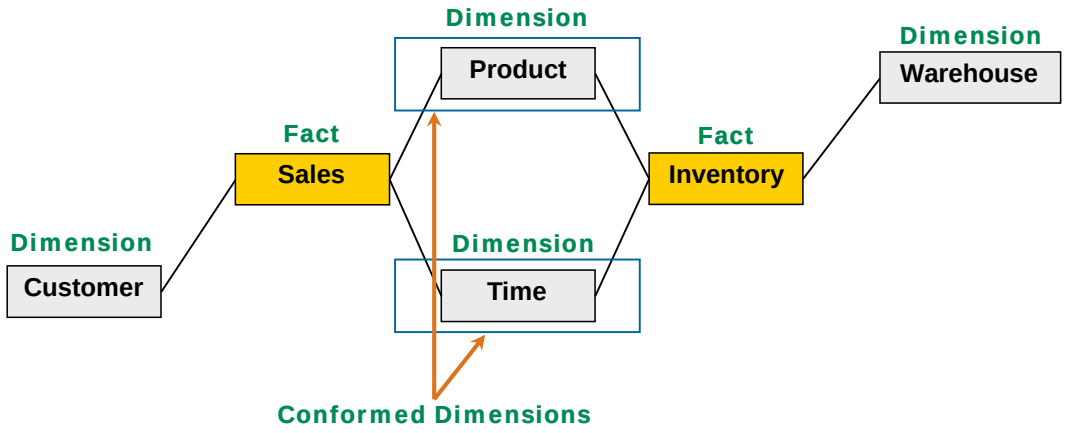
When you look at data to load into a cube that data will need to be similar to a "fact table" where elements from each dimension in the cube identify where to place the data into a given cube.

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Examine dimension tables

- Dimension tables provide descriptive information.
- Dimension tables may be "conformed" so that they are applicable to multiple fact tables across the business.



Identify common data structures

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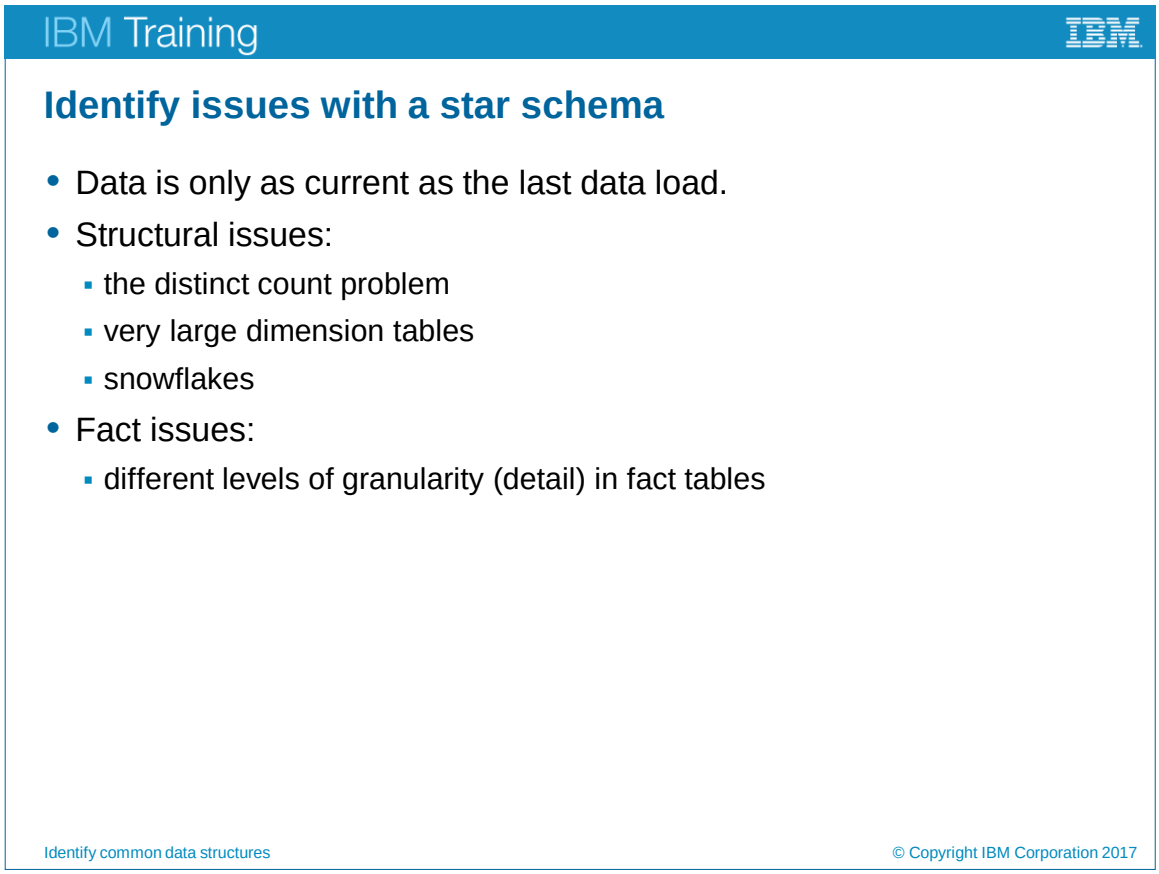
Examine dimension tables

A conformed dimension is one which is common to several fact tables; it has the same meaning and content when being referred to from several fact tables.

In the slide example, you can query either sales or inventory data through the Product dimension table, the Time dimension table, or both. For example, Product acts as a context when you want to compare quantity sold with stock count.

In TM1, dimensions can easily be shared among cubes, such as Sales and Inventory. TM1 dimensions support multiple hierarchies to support multiple cube structures. Additionally, dimension attributes can be used to support different dimension format requirements across cubes.

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Identify issues with a star schema

Data is periodically loaded from an operational system into the reporting database (star schema layout). Therefore the data is only as current as the last time it was loaded.

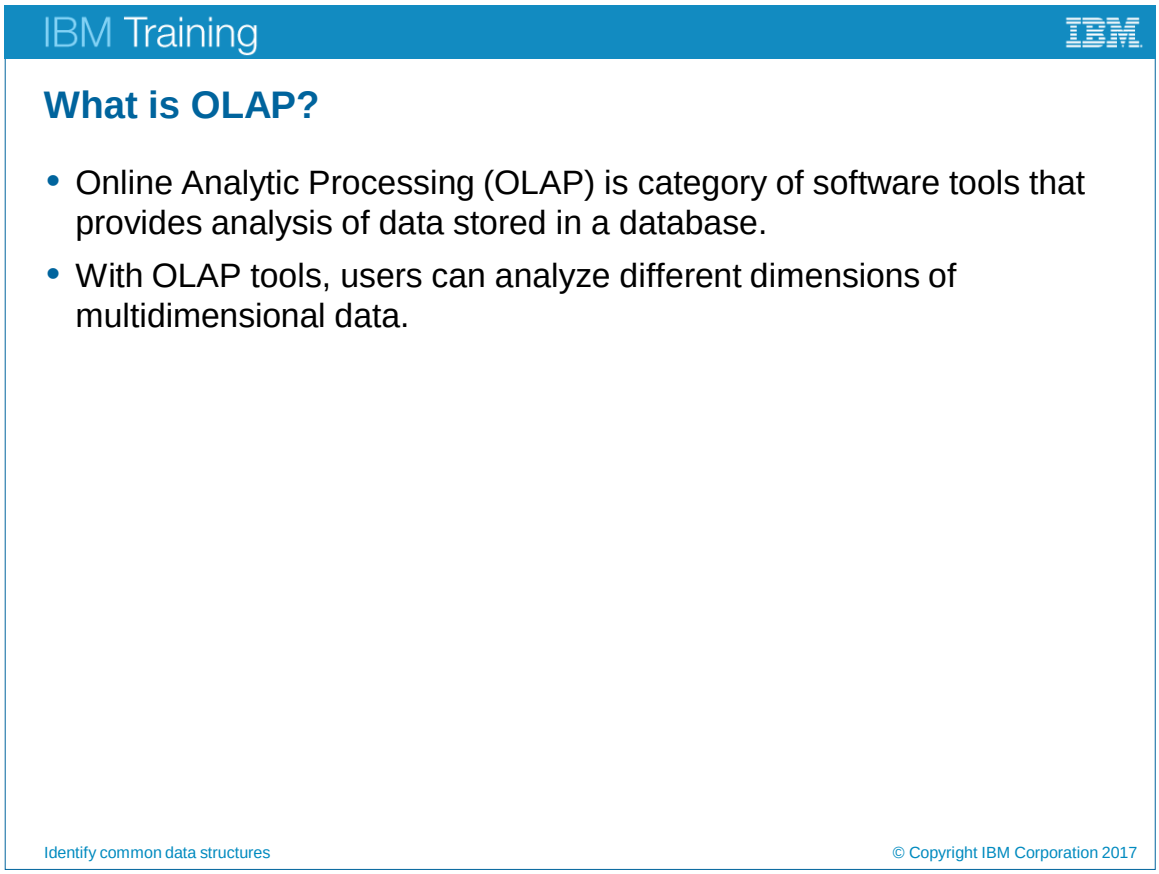
When reporting against a star schema database, you may encounter difficulties in counting the exact number of items in the dimension, such as separate products.

The star schema may also include large dimension tables that result in reports that run too slowly. These tables may be broken out into smaller tables through a normalization process, which in turn creates snowflake tables.

In some cases, fact tables can have different levels of granularity. For example the data values may either be associated with the day level or the month level. The granularity may change based on the dimensions referenced in the fact table.

TM1 is a multidimensional database employing Online Analytical Processing (OLAP) technology to provide faster access to data.

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What is OLAP?

With OLAP, data is stored in cubes, which are made of two or more dimensions. Each dimension has a hierarchical structure.

You can drill down (or drill up) on a dimension to view data at the level of detail that you choose. For example, you can drill down on the products dimension to view information about camping equipment. You can drill down another level to tents, or even more to view information about a specific type of tent. With OLAP, you can also slice and dice the data to view the dimensions that are of interest to you.

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OLAP versus OLTP (relational)

	OLAP	OLTP
Definition	Analytical Processing against Multi-Dimensional Data	Transaction Processing against Relational Data
Purpose	Optimize the business	Automate the business
Use	Planning, Modeling, Analysis	Transaction Processing
Schema	Multi-dimensional, Hierarchical	Two-dimensional, Normalized
Reporting	Crosstab, Interactive, speed of thought analysis from multiple perspectives	Query-based, batch run, banded transactional reporting

Identify common data structures

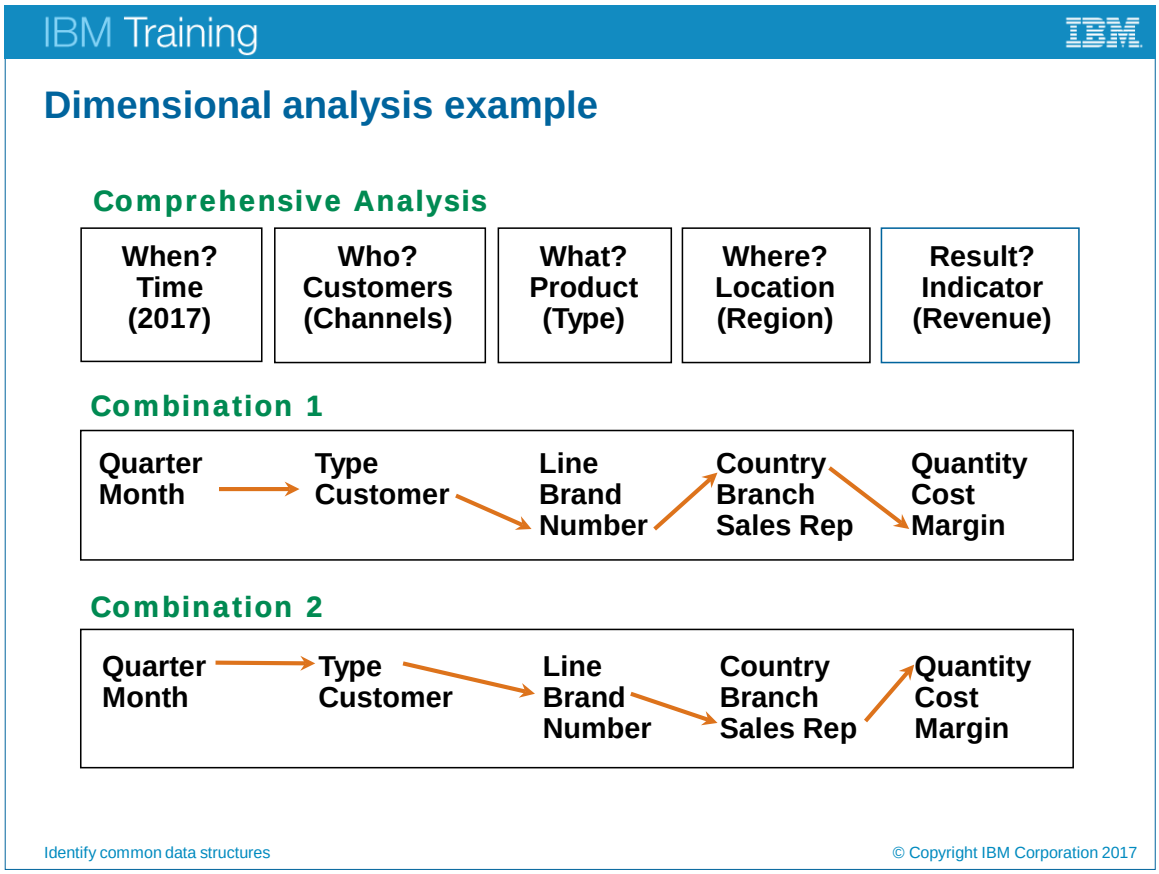
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OLAP versus OLTP (relational)

OLAP - Online Analytical Processing:

- designed for ad hoc data access and analysis
- fast aggregation of hierarchies / fast computations
- ideal for management of data with multiple dimensions

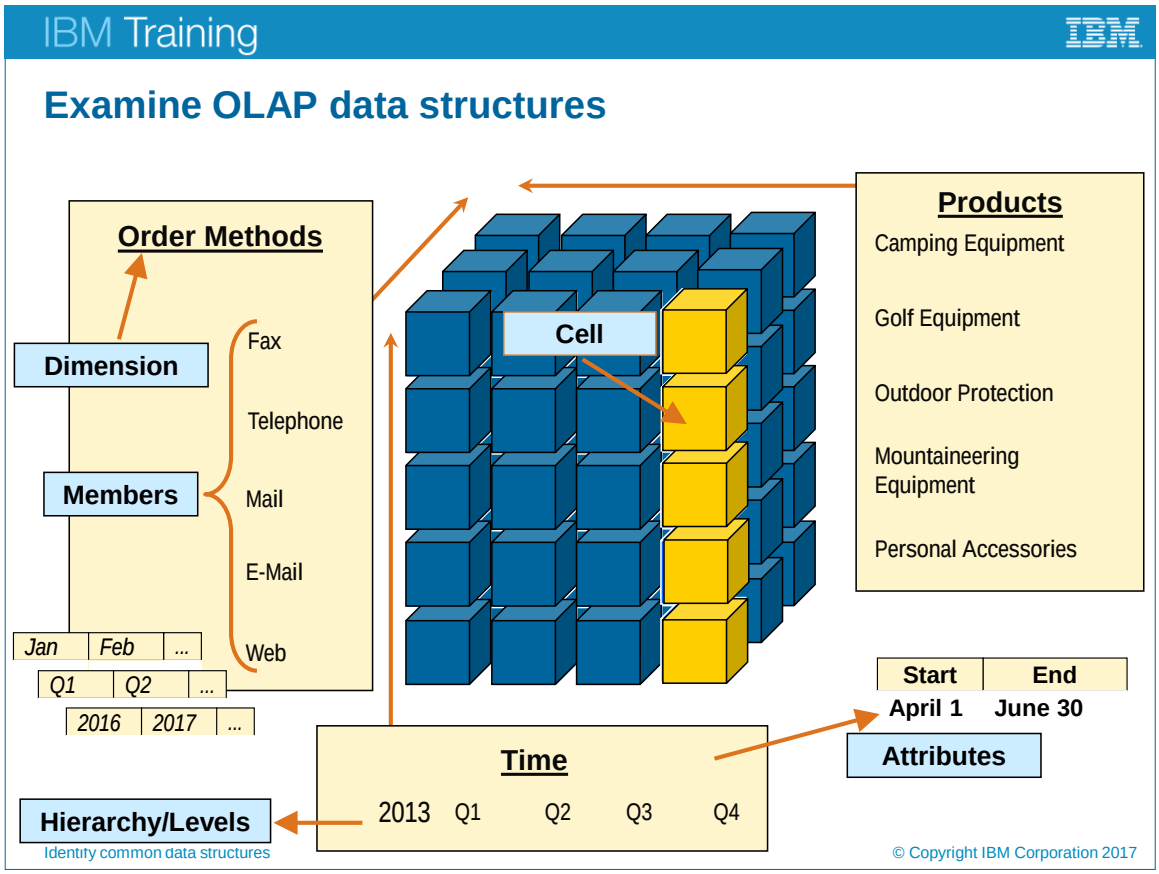
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Dimensional analysis example

This example shows that each dimension answers a specific business question. The data you view depends on the choices you make in each dimension.

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Examine OLAP data structures

The OLAP structure consists of the following:

- Dimensions - contain members that may be structured into hierarchies and levels
- Elements/Members - are data entities that provide context to cell values
- Hierarchies - provide context to the level structures they contain
- Levels - provide structure for the members of a hierarchy
- Attributes - provide additional information for members
- Cells - are intersection points containing values (measures) for various members from different dimensions (also referred to as tuples)
- Slice - is a two dimensional view of multidimensional data.

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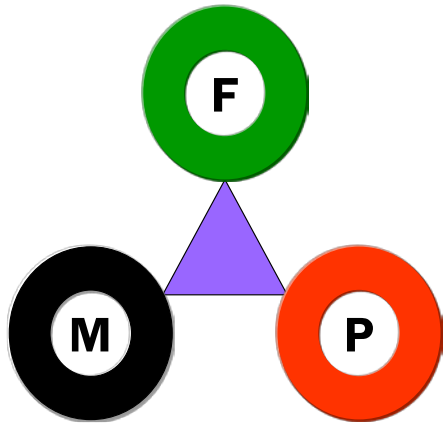
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The three core design principles

DFF: Design For Functionality
- The solution must fulfill its design intent
- Make the solution intuitive and "discoverable"

DFM: Design For Maintainability
- Think beyond the immediate requirements
- Adhere to coherent design standards

DFP: Design For Performance
- Leverage TM1's unique strengths
- Configure the most efficient model for today and in the future



Identify common data structures

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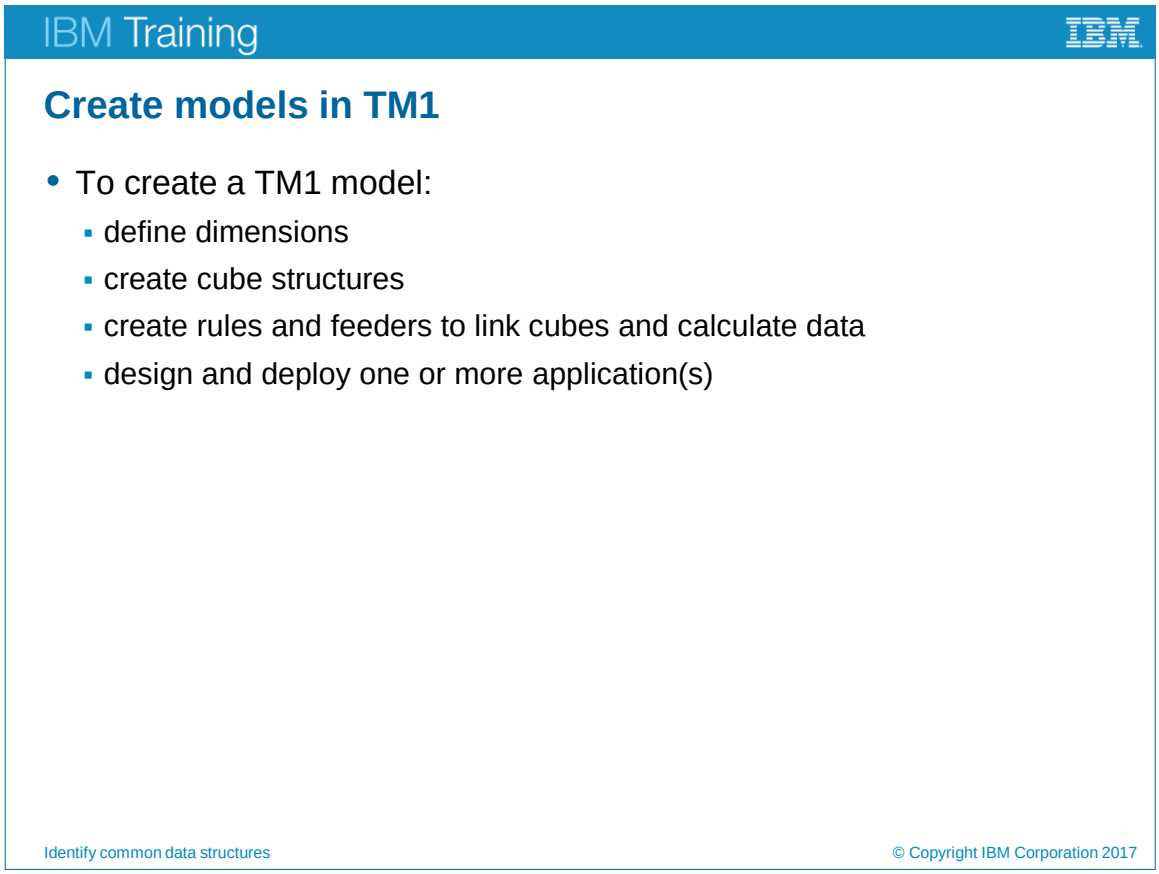
The three core design principles

Functionality, Maintainability, and Performance are often in conflict with each other.

Principles	Effect
Functionality versus Maintainability	If the design does not meet the intent, it does not matter how maintainable it is.
Maintainability versus Performance	In a BPM application, it is usually more important to make the solution maintainable than to wring out every last ounce of performance.

Every solution is different. As the Modeler, it is your job to make effective choices based on your user requirements and the environment.

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Create models in TM1

Populate the dimensions manually or using data from an external source such as a text file or a database.

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Unit summary

- Identify characteristics of operational and reporting databases
- Discuss Online Analytical Processing (OLAP)
- Examine core model design principles
- Identify the basic tasks required to create a TM1 model and deploy it

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Unit summary

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